FHIR-STU3 — Administration Module (sections 8 – 8.24.9)

Source: HL7 FHIR STU3 (v3.0.2). Text-only edition for AI ingestion (NotebookLM); pages follow the source order (its table of contents when one exists).

8 Administration Module

URL: https://hl7.org/fhir/STU3/administration-module.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

Work Group Patient Administration | Ballot Status: Informative

8.0 Administration Module

8.0.1 Introduction

The Administrative module covers the base data that is then linked into the other modules for clinical content, finance/billing, workflow, etc. This of course is built on the FHIR technology platform modules.

Before any clinical data can be recorded, the basic information on the patient must be recorded, and then often the basis of the interaction (such as an encounter)

8.0.2 Index

Patient RelatedPerson Group Practitioner PractitionerRole | Organization Location HealthcareService Endpoint Schedule Slot | EpisodeOfCare Encounter Appointment AppointmentResponse Account Flag | Device DeviceComponent DeviceMetric Substance

8.0.2.1 Patient Registers

Track people involved in receiving healthcare, the basics nearly everything else references back to

Name | Aliases | Description

Patient | SubjectOfCare Client Resident | Demographics and other administrative information about an individual or animal receiving care or other health-related services.

RelatedPerson | | Information about a person that is involved in the care for a patient, but who is not the target of healthcare, nor has a formal responsibility in the care process.

Person | | Demographics and administrative information about a person independent of a specific health-related context.

Group | | Represents a defined collection of entities that may be discussed or acted upon collectively but which are not expected to act collectively and are not formally or legally recognized; i.e. a collection of entities that isn't an Organization.

Note: Patient linking should also be considered when evaluating searches with references to other resources. e.g. Searching for conditions for a patient. At present the specification does not define if the links should be also followed to include conditions that reference the linked patients too. We are currently seeking feedback on this.

Note: The Person resource may be used as a centralized register of people that may eventually be involved in healthcare, and could be used as the central core demographics register. However the fields/values in Person are duplicated in the other resources, and in many cases the Person resource will be hosted on external systems.

8.0.2.2 Clinical Categorization Resources

Most clinical activities occur grouped in some way. Long term care is typically covered by an EpisodeOfCare, where short term care by encounters. Account associates the tracking of transactions back to a Patient (or other resource). Flag is just used to highlight a warning or other notification about a patient (or other resource)

Name | Aliases | Description

EpisodeOfCare | Case Program Problem | An association between a patient and an organization / healthcare provider(s) during which time encounters may occur. The managing organization assumes a level of responsibility for the patient during this time.

Encounter | Visit | An interaction between a patient and healthcare provider(s) for the purpose of providing healthcare service(s) or assessing the health status of a patient.

Account | Cost center | A financial tool for tracking value accrued for a particular purpose. In the healthcare field, used to track charges for a patient, cost centers, etc.

Flag | Barriers to Care, Alert | Prospective warnings of potential issues when providing care to the patient.

Note: Resources shown with a dotted box are described in other sections of the specification: Coverage and Claim are from the section on Finance.

8.0.2.3 Service Provider Directory Resources

Service Directory resources are usually stored in the administration section of applications, and may even be synchronized from external systems.

Name | Aliases | Description

Organization | | A formally or informally recognized grouping of people or organizations formed for the purpose of achieving some form of collective action. Includes companies, institutions, corporations, departments, community groups, healthcare practice groups, etc.

Location | | Details and position information for a physical place where services are provided and resources and participants may be stored, found, contained or accommodated.

Practitioner | | A person who is directly or indirectly involved in the provisioning of healthcare.

PractitionerRole | | A specific set of Roles/Locations/specialties/services that a practitioner may perform at an organization for a period of time.

HealthcareService | | The details of a healthcare service available at a location.

Endpoint | | The technical details of an endpoint that can be used for electronic services, such as for web services providing XDS.b or a REST endpoint for another FHIR server. This may include any security context information.

8.0.2.4 Scheduling and Appointments

The Scheduling/Appointments resources permit the planning of encounters to occur and follow on with other clinical activities.

Name | Aliases | Description
Schedule | Availability | A container for slots of time that may be available for booking appointments.
Slot | | A slot of time on a schedule that may be available for booking appointments.
Appointment | | A booking of a healthcare event among patient(s), practitioner(s), related person(s) and/or device(s) for a specific date/time. This may result in one or more Encounter(s).
AppointmentResponse | | A reply to an appointment request for a patient and/or practitioner(s), such as a confirmation or rejection.

8.0.2.5 Devices and Substances

Other assets are often registered in the administration system, and maintained as master files.

Name | Aliases | Description
Device | | This resource identifies an instance or a type of a manufactured item that is used in the provision of healthcare without being substantially changed through that activity. The device may be a medical or non-medical device. Medical devices include durable (reusable) medical equipment, implantable devices, as well as disposable equipment used for diagnostic, treatment, and research for healthcare and public health. Non-medical devices may include items such as a machine, cellphone, computer, application, etc.
DeviceComponent | | The characteristics, operational status and capabilities of a medical-related component of a medical device.
DeviceMetric | | Describes a measurement, calculation or setting capability of a medical device.
Substance | | A homogeneous material with a definite composition.

8.0.3 Security and Privacy

Patient privacy is handled with security labels and tags in the Resource Meta property. This is the standard way that the FHIR specification provides this supporting information to a sub-system that implements it (which is not defined by FHIR).

One of the more common use cases is for marking a patient as being a celebrity.

Note that privacy considerations apply to Person, Practitioner and RelatedPerson records in addition to Patient's.

While Organization, Location, Device and other non-person-identifying records are generally subject to less stringent security precautions, such data must still be protected to avoid safety issues (e.g. someone maliciously changing the ingredients associated with a drug to cause/fail to cause alerts)

Devices can be linked to Patients. If this occurs, they must be protected as any other patient-linked element

For more general considerations, see the Security and Privacy module.

8.0.4 Common Use Cases

Administration Resources are cornerstone resources that are used by clinical and other domains of the FHIR Standard.

* Managing a Master Record of a Patient and a Person (e.g. MPI) A Patient resource is used to describe patient demographic and visit information and any updates to it. It can be used to communicate Patient information to other systems (e.g. other registries, clinical, ancillary and financial systems). Some systems distinguish the Patient Registry (or Client Registry) from the Person Registry. A Person resource is a base for the Person Registry system. The Patient/Person Management use case includes creation, update, as well as merge/unmerge and link/unlink scenarios.

* Managing a Master Record of a Provider and Service Catalogue (e.g. Provider Registry, Service Directory) A Practitioner resource is a base resource for enabling the registry of individuals, related to providing health care services. Other resources, such as Organization, Location, HealthcareService, are creating a complete picture of where, how and by whom the care services are enabled to a patient. The resources can be used for managing the master record or as a reference in clinical resources to inform about participants and places for various clinical resources.

* Managing Other Administrative Records The Administration domain of the FHIR standard includes creation and update of Device and Substance records. Resources can be used for managing a master record or communicating its information to other systems.

* Enabling Patient Profiles, Clinical Reporting and Connecting Clinical Records Administration Resources are referred by almost all clinical resources. Querying systems, using the references to Administration Resources enables the creation of profiles and reports of various complexities.

* Enabling Clinical Grouping and Financial Reporting Other use cases are included in the roadmap of resources, developed by the Patient Administration group. The roadmap section lists plans and updates of the current work.

8.0.5 Developmental Roadmap

The Patient Administration is currently working through resources that support:

* Encounters and Scheduling (enhance maturity of encounters and further develop in/outpatient scheduling)

* Service Provider Directory (in co-ordination with the Argonaut Provider Directory group)

* Financial Management interactions (account/coverage, then charge item, which links administration to billing)

Many of the administrative resources are part of the core resources that most systems use first and have formed the basis for most people's first experiences with FHIR. However this limited exposure has still to be proven in all contexts, such as veterinary, public health and clinical research.

8.1 Resource Patient - Content

URL: https://hl7.org/fhir/STU3/patient.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

Patient Administration Work Group | Maturity Level: 5 | Trial Use | Compartments: Patient, Practitioner, RelatedPerson

Demographics and other administrative information about an individual or animal receiving care or other health-related services.

8.1.1 Scope and Usage

This Resource covers data about patients and animals involved in a wide range of health-related activities, including:

* Curative activities
* Psychiatric care
* Social services
* Pregnancy care
* Nursing and assisted living
* Dietary services
* Tracking of personal health and exercise data

The data in the Resource covers the "who" information about the patient: its attributes are focused on the demographic information necessary to support the administrative, financial and logistic procedures. A Patient record is generally created and maintained by each organization providing care for a patient. A patient or animal receiving care at multiple organizations may therefore have its information present in multiple Patient Resources.

Not all concepts are included within the base resource (such as race, ethnicity, organ donor status, nationalilty, etc.), but may be found in profiles defined for specific jurisdictions (e.g., US Meaningful Use Program) or standard extensions. Such fields vary widely between jurisdictions and often have different names and valuesets for the similar concepts, but they are not similar enough to be able to map and exchange

This resource is referenced by Account, AdverseEvent, AllergyIntolerance, Appointment, AppointmentResponse, AuditEvent, Basic, BodySite, CarePlan, CareTeam, ChargeItem, Claim, ClaimResponse, ClinicalImpression, Communication, CommunicationRequest, Composition, Condition, Consent, Contract, Coverage, DetectedIssue, Device, DeviceRequest, DeviceUseStatement, DiagnosticReport, DocumentManifest, DocumentReference, EligibilityRequest, Encounter, EnrollmentRequest, EpisodeOfCare, ExplanationOfBenefit, FamilyMemberHistory, Flag, Goal, Group, GuidanceResponse, ImagingManifest, ImagingStudy, Immunization, ImmunizationRecommendation, List, MeasureReport, Media, MedicationAdministration, MedicationDispense, MedicationRequest, MedicationStatement, NutritionOrder, Observation, Person, Procedure, ProcedureRequest, Provenance, QuestionnaireResponse, ReferralRequest, RelatedPerson, RequestGroup, ResearchSubject, RiskAssessment, Schedule, Sequence, Specimen, SupplyDelivery, SupplyRequest, Task and VisionPrescription

8.1.2 Resource Content

* Structure
* UML
* XML
* JSON
* Turtle
* R2 Diff
* All

Structure

Name | Flags | Card. | Type | Description & Constraints
Patient | | | DomainResource | Information about an individual or animal receiving health care services Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | ? | 0..* | Identifier | An identifier for this patient
active | ?!? | 0..1 | boolean | Whether this patient's record is in active use
name | ? | 0..* | HumanName | A name associated with the patient
telecom | ? | 0..* | ContactPoint | A contact detail for the individual
gender | ? | 0..1 | code | male | female | other | unknown AdministrativeGender (Required)
birthDate | ? | 0..1 | date | The date of birth for the individual
deceased[x] | ?!? | 0..1 | | Indicates if the individual is deceased or not
deceasedBoolean | | | boolean
deceasedDateTime | | | dateTime
address | ? | 0..* | Address | Addresses for the individual
maritalStatus | | 0..1 | CodeableConcept | Marital (civil) status of a patient Marital Status Codes (Extensible)
multipleBirth[x] | | 0..1 | | Whether patient is part of a multiple birth
multipleBirthBoolean | | | boolean

multipleBirthInteger | | | integer
photo | | 0..* | Attachment | Image of the patient
contact | I | 0..* | BackboneElement | A contact party (e.g. guardian, partner, friend) for the patient + SHALL at least contain a contact's details or a reference to an organization
relationship | | 0..* | CodeableConcept | The kind of relationship v2 Contact Role (Extensible)
name | | 0..1 | HumanName | A name associated with the contact person
telecom | | 0..* | ContactPoint | A contact detail for the person
address | | 0..1 | Address | Address for the contact person
gender | | 0..1 | code | male | female | other | unknown AdministrativeGender (Required)
organization | I | 0..1 | Reference(Organization) | Organization that is associated with the contact
period | | 0..1 | Period | The period during which this contact person or organization is valid to be contacted relating to this patient
animal | ?!? | 0..1 | BackboneElement | This patient is known to be an animal (non-human)
species | ? | 1..1 | CodeableConcept | E.g. Dog, Cow AnimalSpecies (Example)
breed | ? | 0..1 | CodeableConcept | E.g. Poodle, Angus AnimalBreeds (Example)
genderStatus | ? | 0..1 | CodeableConcept | E.g. Neutered, Intact GenderStatus (Example)
communication | | 0..* | BackboneElement | A list of Languages which may be used to communicate with the patient about his or her health
language | | 1..1 | CodeableConcept | The language which can be used to communicate with the patient about his or her health Common Languages (Extensible but limited to All Languages)
preferred | | 0..1 | boolean | Language preference indicator
generalPractitioner | | 0..* | Reference(Organization | Practitioner) | Patient's nominated primary care provider
managingOrganization | ? | 0..1 | Reference(Organization) | Organization that is the custodian of the patient record
link | ?!? | 0..* | BackboneElement | Link to another patient resource that concerns the same actual person
other | ? | 1..1 | Reference(Patient | RelatedPerson) | The other patient or related person resource that the link refers to
type | ? | 1..1 | code | replaced-by | replaces | refer | seealso - type of link LinkType (Required)
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Patient xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier An identifier for this patient --></identifier>
 <active value="[boolean]"/><!-- 0..1 Whether this patient's record is in active use -->
 <name><!-- 0..* HumanName A name associated with the patient --></name>
 <telecom><!-- 0..* ContactPoint A contact detail for the individual --></telecom>
 <gender value="[code]"/><!-- 0..1 male | female | other | unknown -->
 <birthDate value="[date]"/><!-- 0..1 The date of birth for the individual -->
 <deceased[x]><!-- 0..1 boolean|dateTime Indicates if the individual is deceased or not --></deceased[x]>
 <address><!-- 0..* Address Addresses for the individual --></address>
 <maritalStatus><!-- 0..1 CodeableConcept Marital (civil) status of a patient --></maritalStatus>
 <multipleBirth[x]><!-- 0..1 boolean|integer Whether patient is part of a multiple birth --></multipleBirth[x]>
 <photo><!-- 0..* Attachment Image of the patient --></photo>
 <contact>  <!-- 0..* A contact party (e.g. guardian, partner, friend) for the patient -->
  <relationship><!-- 0..* CodeableConcept The kind of relationship --></relationship>
  <name><!-- 0..1 HumanName A name associated with the contact person --></name>
  <telecom><!-- 0..* ContactPoint A contact detail for the person --></telecom>
  <address><!-- 0..1 Address Address for the contact person --></address>
  <gender value="[code]"/><!-- 0..1 male | female | other | unknown -->
  <organization><!--  0..1 Reference(Organization) Organization that is associated with the contact --></organization>
  <period><!-- 0..1 Period The period during which this contact person or organization is valid to be contacted relating
     to this patient --></period>
 </contact>
 <animal>  <!-- 0..1 This patient is known to be an animal (non-human) -->
  <species><!-- 1..1 CodeableConcept E.g. Dog, Cow --></species>
  <breed><!-- 0..1 CodeableConcept E.g. Poodle, Angus --></breed>
  <genderStatus><!-- 0..1 CodeableConcept E.g. Neutered, Intact --></genderStatus>
 </animal>
 <communication>  <!-- 0..* A list of Languages which may be used to communicate with the patient about his or her
    health -->
  <language><!-- 1..1 CodeableConcept The language which can be used to communicate with the patient about his or her
    health --></language>
  <preferred value="[boolean]"/><!-- 0..1 Language preference indicator -->
 </communication>
 <generalPractitioner><!-- 0..* Reference(Organization|Practitioner) Patient's nominated primary care provider
    --></generalPractitioner>
 <managingOrganization><!-- 0..1 Reference(Organization) Organization that is the custodian of the patient record
    --></managingOrganization>
 <link>  <!-- 0..* Link to another patient resource that concerns the same actual person -->
  <other><!-- 1..1 Reference(Patient|RelatedPerson) The other patient or related person resource that the link refers to
     --></other>
  <type value="[code]"/><!-- 1..1 replaced-by | replaces | refer | seealso - type of link -->
 </link>
</Patient>
```

JSON Template

```
{
  "resourceType" : "Patient",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // An identifier for this patient
  "active" : <boolean>, // Whether this patient's record is in active use
  "name" : [{ HumanName }], // A name associated with the patient
  "telecom" : [{ ContactPoint }], // A contact detail for the individual
```

```
  "gender" : "<code>", // male | female | other | unknown
  "birthDate" : "<date>", // The date of birth for the individual
  // deceased[x]: Indicates if the individual is deceased or not. One of these 2:
  "deceasedBoolean" : <boolean>,
  "deceasedDateTime" : "<dateTime>",
  "address" : [{ Address }], // Addresses for the individual
  "maritalStatus" : { CodeableConcept }, // Marital (civil) status of a patient
  // multipleBirth[x]: Whether patient is part of a multiple birth. One of these 2:
  "multipleBirthBoolean" : <boolean>,
  "multipleBirthInteger" : <integer>,
  "photo" : [{ Attachment }], // Image of the patient
  "contact" : [{ // A contact party (e.g. guardian, partner, friend) for the patient
    "relationship" : [{ CodeableConcept }], // The kind of relationship
    "name" : { HumanName }, // A name associated with the contact person
    "telecom" : [{ ContactPoint }], // A contact detail for the person
    "address" : { Address }, // Address for the contact person
    "gender" : "<code>", // male | female | other | unknown
    "organization" : { Reference(Organization) }, // C? Organization that is associated with the contact
    "period" : { Period } // The period during which this contact person or organization is valid to be contacted
    relating to this patient
  }],
  "animal" : { // This patient is known to be an animal (non-human)
    "species" : { CodeableConcept }, // R!  E.g. Dog, Cow
    "breed" : { CodeableConcept }, // E.g. Poodle, Angus
    "genderStatus" : { CodeableConcept } // E.g. Neutered, Intact
  },
  "communication" : [{ // A list of Languages which may be used to communicate with the patient about his or her health
    "language" : { CodeableConcept }, // R!  The language which can be used to communicate with the patient about his or
     her health
    "preferred" : <boolean> // Language preference indicator
  }],
  "generalPractitioner" : [{ Reference(Organization|Practitioner) }], // Patient's nominated primary care provider
  "managingOrganization" : { Reference(Organization) }, // Organization that is the custodian of the patient record
  "link" : [{ // Link to another patient resource that concerns the same actual person
    "other" : { Reference(Patient|RelatedPerson) }, // R!  The other patient or related person resource that the link
    refers to
    "type" : "<code>" // R!  replaced-by | replaces | refer | seealso - type of link
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Patient;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Patient.identifier [ Identifier ], ... ; # 0..* An identifier for this patient
  fhir:Patient.active [ boolean ]; # 0..1 Whether this patient's record is in active use
  fhir:Patient.name [ HumanName ], ... ; # 0..* A name associated with the patient
  fhir:Patient.telecom [ ContactPoint ], ... ; # 0..* A contact detail for the individual
  fhir:Patient.gender [ code ]; # 0..1 male | female | other | unknown
  fhir:Patient.birthDate [ date ]; # 0..1 The date of birth for the individual
  # Patient.deceased[x] : 0..1 Indicates if the individual is deceased or not. One of these 2
    fhir:Patient.deceasedBoolean [ boolean ]
    fhir:Patient.deceasedDateTime [ dateTime ]
  fhir:Patient.address [ Address ], ... ; # 0..* Addresses for the individual
  fhir:Patient.maritalStatus [ CodeableConcept ]; # 0..1 Marital (civil) status of a patient
  # Patient.multipleBirth[x] : 0..1 Whether patient is part of a multiple birth. One of these 2
    fhir:Patient.multipleBirthBoolean [ boolean ]
    fhir:Patient.multipleBirthInteger [ integer ]
  fhir:Patient.photo [ Attachment ], ... ; # 0..* Image of the patient
  fhir:Patient.contact [ # 0..* A contact party (e.g. guardian, partner, friend) for the patient
    fhir:Patient.contact.relationship [ CodeableConcept ], ... ; # 0..* The kind of relationship
    fhir:Patient.contact.name [ HumanName ]; # 0..1 A name associated with the contact person
    fhir:Patient.contact.telecom [ ContactPoint ], ... ; # 0..* A contact detail for the person
    fhir:Patient.contact.address [ Address ]; # 0..1 Address for the contact person
    fhir:Patient.contact.gender [ code ]; # 0..1 male | female | other | unknown
    fhir:Patient.contact.organization [ Reference(Organization) ]; # 0..1 Organization that is associated with the
    contact
    fhir:Patient.contact.period [ Period ]; # 0..1 The period during which this contact person or organization is valid
    to be contacted relating to this patient
  ], ...;
  fhir:Patient.animal [ # 0..1 This patient is known to be an animal (non-human)
    fhir:Patient.animal.species [ CodeableConcept ]; # 1..1 E.g. Dog, Cow
    fhir:Patient.animal.breed [ CodeableConcept ]; # 0..1 E.g. Poodle, Angus
    fhir:Patient.animal.genderStatus [ CodeableConcept ]; # 0..1 E.g. Neutered, Intact
  ];
  fhir:Patient.communication [ # 0..* A list of Languages which may be used to communicate with the patient about his or
     her health
    fhir:Patient.communication.language [ CodeableConcept ]; # 1..1 The language which can be used to communicate with
    the patient about his or her health
    fhir:Patient.communication.preferred [ boolean ]; # 0..1 Language preference indicator
  ], ...;
  fhir:Patient.generalPractitioner [ Reference(Organization|Practitioner) ], ... ; # 0..* Patient's nominated primary
    care provider
  fhir:Patient.managingOrganization [ Reference(Organization) ]; # 0..1 Organization that is the custodian of the
```

```
    patient record
  fhir:Patient.link [ # 0..* Link to another patient resource that concerns the same actual person
    fhir:Patient.link.other [ Reference(Patient|RelatedPerson) ]; # 1..1 The other patient or related person resource
    that the link refers to
    fhir:Patient.link.type [ code ]; # 1..1 replaced-by | replaces | refer | seealso - type of link
  ], ...;
]
```

Changes since DSTU2

Patient

Patient.maritalStatus | Change binding strength from required to extensible

Patient.contact.relationship | Change value set from http://hl7.org/fhir/ValueSet/patient-contact-relationship to http://hl7.org/fhir/ValueSet/v2-0131

Patient.communication.language | Change binding strength from required to extensible Change value set from http://tools.ietf.org/html/bcp47 to http://hl7.org/fhir/ValueSet/languages

Patient.generalPractitioner | Renamed from careProvider to generalPractitioner

Patient.link.other | Add Reference(RelatedPerson) No longer marked as Modifier

Patient.link.type | No longer marked as Modifier

See the Full Difference for further information

This analysis is available as XML or JSON.

See R2 <--> R3 Conversion Maps (status = 18 tests that all execute ok. All tests pass round-trip testing and 6 r3 resources are invalid (12 errors).).

Structure

Name | Flags | Card. | Type | Description & Constraints

Patient | | | DomainResource | Information about an individual or animal receiving health care services Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension

identifier | ? | 0..* | Identifier | An identifier for this patient

active | ?!? | 0..1 | boolean | Whether this patient's record is in active use

name | ? | 0..* | HumanName | A name associated with the patient

telecom | ? | 0..* | ContactPoint | A contact detail for the individual

gender | ? | 0..1 | code | male | female | other | unknown AdministrativeGender (Required)

birthDate | ? | 0..1 | date | The date of birth for the individual

deceased[x] | ?!? | 0..1 | | Indicates if the individual is deceased or not

deceasedBoolean | | | boolean

deceasedDateTime | | | dateTime

address | ? | 0..* | Address | Addresses for the individual

maritalStatus | | 0..1 | CodeableConcept | Marital (civil) status of a patient Marital Status Codes (Extensible)

multipleBirth[x] | | 0..1 | | Whether patient is part of a multiple birth

multipleBirthBoolean | | | boolean

multipleBirthInteger | | | integer

photo | | 0..* | Attachment | Image of the patient

contact | I | 0..* | BackboneElement | A contact party (e.g. guardian, partner, friend) for the patient + SHALL at least contain a contact's details or a reference to an organization

relationship | | 0..* | CodeableConcept | The kind of relationship v2 Contact Role (Extensible)

name | | 0..1 | HumanName | A name associated with the contact person

telecom | | 0..* | ContactPoint | A contact detail for the person

address | | 0..1 | Address | Address for the contact person

gender | | 0..1 | code | male | female | other | unknown AdministrativeGender (Required)

organization | I | 0..1 | Reference(Organization) | Organization that is associated with the contact

period | | 0..1 | Period | The period during which this contact person or organization is valid to be contacted relating to this patient

animal | ?!? | 0..1 | BackboneElement | This patient is known to be an animal (non-human)

species | ? | 1..1 | CodeableConcept | E.g. Dog, Cow AnimalSpecies (Example)

breed | ? | 0..1 | CodeableConcept | E.g. Poodle, Angus AnimalBreeds (Example)

genderStatus | ? | 0..1 | CodeableConcept | E.g. Neutered, Intact GenderStatus (Example)

communication | | 0..* | BackboneElement | A list of Languages which may be used to communicate with the patient about his or her health

language | | 1..1 | CodeableConcept | The language which can be used to communicate with the patient about his or her health Common Languages (Extensible but limited to All Languages)

preferred | | 0..1 | boolean | Language preference indicator

generalPractitioner | | 0..* | Reference(Organization | Practitioner) | Patient's nominated primary care provider

managingOrganization | ? | 0..1 | Reference(Organization) | Organization that is the custodian of the patient record

link | ?!? | 0..* | BackboneElement | Link to another patient resource that concerns the same actual person

other | ? | 1..1 | Reference(Patient | RelatedPerson) | The other patient or related person resource that the link refers to

type | ? | 1..1 | code | replaced-by | replaces | refer | seealso - type of link LinkType (Required)

Documentation for this format

UML Diagram (Legend)

XML Template

```
<Patient xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier An identifier for this patient --></identifier>
 <active value="[boolean]"/><!-- 0..1 Whether this patient's record is in active use -->
 <name><!-- 0..* HumanName A name associated with the patient --></name>
 <telecom><!-- 0..* ContactPoint A contact detail for the individual --></telecom>
 <gender value="[code]"/><!-- 0..1 male | female | other | unknown -->
```

```
 <birthDate value="[date]"/><!-- 0..1 The date of birth for the individual -->
 <deceased[x]><!-- 0..1 boolean|dateTime Indicates if the individual is deceased or not --></deceased[x]>
 <address><!-- 0..* Address Addresses for the individual --></address>
 <maritalStatus><!-- 0..1 CodeableConcept Marital (civil) status of a patient --></maritalStatus>
 <multipleBirth[x]><!-- 0..1 boolean|integer Whether patient is part of a multiple birth --></multipleBirth[x]>
 <photo><!-- 0..* Attachment Image of the patient --></photo>
 <contact>  <!-- 0..* A contact party (e.g. guardian, partner, friend) for the patient -->
  <relationship><!-- 0..* CodeableConcept The kind of relationship --></relationship>
  <name><!-- 0..1 HumanName A name associated with the contact person --></name>
  <telecom><!-- 0..* ContactPoint A contact detail for the person --></telecom>
  <address><!-- 0..1 Address Address for the contact person --></address>
  <gender value="[code]"/><!-- 0..1 male | female | other | unknown -->
  <organization><!--  0..1 Reference(Organization) Organization that is associated with the contact --></organization>
  <period><!-- 0..1 Period The period during which this contact person or organization is valid to be contacted relating
     to this patient --></period>
 </contact>
 <animal>  <!-- 0..1 This patient is known to be an animal (non-human) -->
  <species><!-- 1..1 CodeableConcept E.g. Dog, Cow --></species>
  <breed><!-- 0..1 CodeableConcept E.g. Poodle, Angus --></breed>
  <genderStatus><!-- 0..1 CodeableConcept E.g. Neutered, Intact --></genderStatus>
 </animal>
 <communication>  <!-- 0..* A list of Languages which may be used to communicate with the patient about his or her
    health -->
  <language><!-- 1..1 CodeableConcept The language which can be used to communicate with the patient about his or her
    health --></language>
  <preferred value="[boolean]"/><!-- 0..1 Language preference indicator -->
 </communication>
 <generalPractitioner><!-- 0..* Reference(Organization|Practitioner) Patient's nominated primary care provider
    --></generalPractitioner>
 <managingOrganization><!-- 0..1 Reference(Organization) Organization that is the custodian of the patient record
    --></managingOrganization>
 <link>  <!-- 0..* Link to another patient resource that concerns the same actual person -->
  <other><!-- 1..1 Reference(Patient|RelatedPerson) The other patient or related person resource that the link refers to
     --></other>
  <type value="[code]"/><!-- 1..1 replaced-by | replaces | refer | seealso - type of link -->
 </link>
</Patient>
```

JSON Template

```
{
  "resourceType" : "Patient",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // An identifier for this patient
  "active" : <boolean>, // Whether this patient's record is in active use
  "name" : [{ HumanName }], // A name associated with the patient
  "telecom" : [{ ContactPoint }], // A contact detail for the individual
  "gender" : "<code>", // male | female | other | unknown
  "birthDate" : "<date>", // The date of birth for the individual
  // deceased[x]: Indicates if the individual is deceased or not. One of these 2:
  "deceasedBoolean" : <boolean>,
  "deceasedDateTime" : "<dateTime>",
  "address" : [{ Address }], // Addresses for the individual
  "maritalStatus" : { CodeableConcept }, // Marital (civil) status of a patient
  // multipleBirth[x]: Whether patient is part of a multiple birth. One of these 2:
  "multipleBirthBoolean" : <boolean>,
  "multipleBirthInteger" : <integer>,
  "photo" : [{ Attachment }], // Image of the patient
  "contact" : [{ // A contact party (e.g. guardian, partner, friend) for the patient
    "relationship" : [{ CodeableConcept }], // The kind of relationship
    "name" : { HumanName }, // A name associated with the contact person
    "telecom" : [{ ContactPoint }], // A contact detail for the person
    "address" : { Address }, // Address for the contact person
    "gender" : "<code>", // male | female | other | unknown
    "organization" : { Reference(Organization) }, // C? Organization that is associated with the contact
    "period" : { Period } // The period during which this contact person or organization is valid to be contacted
    relating to this patient
  }],
  "animal" : { // This patient is known to be an animal (non-human)
    "species" : { CodeableConcept }, // R!  E.g. Dog, Cow
    "breed" : { CodeableConcept }, // E.g. Poodle, Angus
    "genderStatus" : { CodeableConcept } // E.g. Neutered, Intact
  },
  "communication" : [{ // A list of Languages which may be used to communicate with the patient about his or her health
    "language" : { CodeableConcept }, // R!  The language which can be used to communicate with the patient about his or
     her health
    "preferred" : <boolean> // Language preference indicator
  }],
  "generalPractitioner" : [{ Reference(Organization|Practitioner) }], // Patient's nominated primary care provider
  "managingOrganization" : { Reference(Organization) }, // Organization that is the custodian of the patient record
  "link" : [{ // Link to another patient resource that concerns the same actual person
    "other" : { Reference(Patient|RelatedPerson) }, // R!  The other patient or related person resource that the link
    refers to
    "type" : "<code>" // R!  replaced-by | replaces | refer | seealso - type of link
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .
```

```
[ a fhir:Patient;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Patient.identifier [ Identifier ], ... ; # 0..* An identifier for this patient
  fhir:Patient.active [ boolean ]; # 0..1 Whether this patient's record is in active use
  fhir:Patient.name [ HumanName ], ... ; # 0..* A name associated with the patient
  fhir:Patient.telecom [ ContactPoint ], ... ; # 0..* A contact detail for the individual
  fhir:Patient.gender [ code ]; # 0..1 male | female | other | unknown
  fhir:Patient.birthDate [ date ]; # 0..1 The date of birth for the individual
  # Patient.deceased[x] : 0..1 Indicates if the individual is deceased or not. One of these 2
    fhir:Patient.deceasedBoolean [ boolean ]
    fhir:Patient.deceasedDateTime [ dateTime ]
  fhir:Patient.address [ Address ], ... ; # 0..* Addresses for the individual
  fhir:Patient.maritalStatus [ CodeableConcept ]; # 0..1 Marital (civil) status of a patient
  # Patient.multipleBirth[x] : 0..1 Whether patient is part of a multiple birth. One of these 2
    fhir:Patient.multipleBirthBoolean [ boolean ]
    fhir:Patient.multipleBirthInteger [ integer ]
  fhir:Patient.photo [ Attachment ], ... ; # 0..* Image of the patient
  fhir:Patient.contact [ # 0..* A contact party (e.g. guardian, partner, friend) for the patient
    fhir:Patient.contact.relationship [ CodeableConcept ], ... ; # 0..* The kind of relationship
    fhir:Patient.contact.name [ HumanName ]; # 0..1 A name associated with the contact person
    fhir:Patient.contact.telecom [ ContactPoint ], ... ; # 0..* A contact detail for the person
    fhir:Patient.contact.address [ Address ]; # 0..1 Address for the contact person
    fhir:Patient.contact.gender [ code ]; # 0..1 male | female | other | unknown
    fhir:Patient.contact.organization [ Reference(Organization) ]; # 0..1 Organization that is associated with the
    contact
    fhir:Patient.contact.period [ Period ]; # 0..1 The period during which this contact person or organization is valid
    to be contacted relating to this patient
  ], ...;
  fhir:Patient.animal [ # 0..1 This patient is known to be an animal (non-human)
    fhir:Patient.animal.species [ CodeableConcept ]; # 1..1 E.g. Dog, Cow
    fhir:Patient.animal.breed [ CodeableConcept ]; # 0..1 E.g. Poodle, Angus
    fhir:Patient.animal.genderStatus [ CodeableConcept ]; # 0..1 E.g. Neutered, Intact
  ];
  fhir:Patient.communication [ # 0..* A list of Languages which may be used to communicate with the patient about his or
     her health
    fhir:Patient.communication.language [ CodeableConcept ]; # 1..1 The language which can be used to communicate with
    the patient about his or her health
    fhir:Patient.communication.preferred [ boolean ]; # 0..1 Language preference indicator
  ], ...;
  fhir:Patient.generalPractitioner [ Reference(Organization|Practitioner) ], ... ; # 0..* Patient's nominated primary
    care provider
  fhir:Patient.managingOrganization [ Reference(Organization) ]; # 0..1 Organization that is the custodian of the
    patient record
  fhir:Patient.link [ # 0..* Link to another patient resource that concerns the same actual person
    fhir:Patient.link.other [ Reference(Patient|RelatedPerson) ]; # 1..1 The other patient or related person resource
    that the link refers to
    fhir:Patient.link.type [ code ]; # 1..1 replaced-by | replaces | refer | seealso - type of link
  ], ...;
]
```

Changes since DSTU2

Patient

Patient.maritalStatus | Change binding strength from required to extensible

Patient.contact.relationship | Change value set from http://hl7.org/fhir/ValueSet/patient-contact-relationship to http://hl7.org/fhir/ValueSet/v2-0131

Patient.communication.language | Change binding strength from required to extensible Change value set from http://tools.ietf.org/html/bcp47 to http://hl7.org/fhir/ValueSet/languages

Patient.generalPractitioner | Renamed from careProvider to generalPractitioner

Patient.link.other | Add Reference(RelatedPerson) No longer marked as Modifier

Patient.link.type | No longer marked as Modifier

See the Full Difference for further information

This analysis is available as XML or JSON.

See R2 <--> R3 Conversion Maps (status = 18 tests that all execute ok. All tests pass round-trip testing and 6 r3 resources are invalid (12 errors).).

Alternate definitions: Master Definition (XML, JSON), XML Schema/Schematron (for) + JSON Schema, ShEx (for Turtle)

8.1.2.1 Terminology Bindings

Path | Definition | Type | Reference

Patient.gender Patient.contact.gender | The gender of a person used for administrative purposes. | Required | AdministrativeGender

Patient.maritalStatus | The domestic partnership status of a person. | Extensible | Marital Status Codes

Patient.contact.relationship | The nature of the relationship between a patient and a contact person for that patient. | Extensible | v2 Contact Role

Patient.animal.species | The species of an animal. | Example | AnimalSpecies

Patient.animal.breed | The breed of an animal. | Example | AnimalBreeds

Patient.animal.genderStatus | The state of the animal's reproductive organs. | Example | GenderStatus

Patient.communication.language | A human language. | Extensible, but limited to All Languages | Common Languages

Patient.link.type | The type of link between this patient resource and another patient resource. | Required | LinkType

8.1.2.2 Constraints

* pat-1: On Patient.contact: SHALL at least contain a contact's details or a reference to an organization (expression on Patient.contact: name.exists() or telecom.exists() or address.exists() or organization.exists())

Notes:

* multipleBirth can be either expressed as a Boolean (just indicating whether the patient is part of a multiple birth) or as an integer, indicating the actual birth order.

* Patient records may only be in one of two statuses: in use (active=true) and not in use (active=false). A normal record is active, i.e. it is in use. Active is set to 'false' when a record is created as a duplicate or in error. A record does not need to be linked to be inactivated.

* The link element is used to assert that two or more Patient resources are both about the same actual patient. See below for further discussion

* There should be only one preferred language (Language.preference = true) per mode of expression.

* The Contact for a Patient has an element organization, this is for use with guardians or business related contacts where just the organization is relevant.

8.1.3 Patient id's and Patient resource id's

A Patient record's Resource Id can never change. For this reason the identifiers with which humans are concerned (often called MRN - Medical Record Number, or UR - Unit Record) should not be used for the resource's id, since MRN's may change, i.e. as a result of having duplicate records of the same patient. Instead they should be represented in the Patient.identifier list where they can be managed. This is also useful for the case of institutions that have acquired multiple numbers because of mergers of patient record systems over time.

8.1.4 Linking Patients

The link element is used to assert that patient resources refer to the same patient. This element is used to support the following scenarios where multiple patient records exist:

8.1.4.1 Duplicate Patient records

Managing Patient registration is a well known difficult problem. Around 2% of registrations are in error, mostly duplicate records. Sometimes the duplicate record is caught fairly quickly and retired before much data is accumulated. In other cases, substantial amounts of data may accumulate. By using a link of type 'replaced-by', the record containing such a link is marked as a duplicate and the link points forward to a record that should be used instead. Note that the record pointed to may in its turn have been identified as created in error and forward to yet another Patient resource. Records that replace another record may use a link type of 'replaces' pointing to the old record.

8.1.4.2 Patient record in a Patient index

A Patient record may be present in a system that acts as a Patient Index: it maintains a (summary of) patient data and a list of one or more servers that it are known to hold a more comprehensive and/or authorative record of the same patient. The link type 'refer' is used denote such a link. Note that linked records may contain contradictory information. The record referred to does not point back to the referring record.

8.1.4.3 Distributed Patient record

In a distributed architecture, multiple systems keep separate patient records concerning the same patient. These records are not considered duplicates, but contain a distributed, potentially overlapping view of the patient's data. Each such record may have its own focus or maintaining organization and there need not be a sense of one record being more complete or more authorative than another. In such cases, links of type 'see also' can be used to point to other patient records. It is not a requirement that such links are bilateral.

8.1.5 Patient vs. Person vs. Patient.Link vs. Linkage

The Person resource on the surface appears to be very similar to the Patient resource, and the usage for it is very similar to using the Patient.Link capability. The intention of the Person resource is to be able to link instances of resources together that are believed to be the same individual. This includes across resource types, such as RelatedPerson, Practitioner, Patient and even other Person resources. The Patient Link however is only intended to be used for Patient resources.

The primary use case for the Person resource is to be able to support person registries that do not necessarily have a healthcare context, and are able to identify and quantify confidence levels that this is the same person. This could include consumer portals where the maintainer of the person information is the actual person themselves. A system could use the Person entry to cross check changes to information applied to one part of a record to values in another system; e.g., when moving, a consumer updates his contact numbers and address in his person record, and then a Patient Administration system is able to see that this data is changed and prompt the organization to follow up with the patient that was linked to the person record if they want their details updated, or if they no longer need services and they should be cancelled, as they've moved from the area.

The Linkage resource and the Patient.link property conceptually perform similar functions in FHIR, both provide an assertion of linkages between multiple resource instances that are referring to the same underlying individual. When a Patient resource is linked/merged then it needs to have an internal indication that there is another patient resource that should be considered when referencing other records, which is achieved using the patient.link property. Not detecting/checking for a potential

linkage could mean that related clinical records are not discovered, potentially impacting patient safety. (which is why using the Linkage resource is not appropriate, as its use in this manner would force the use of either another query to potentially locate other patient resources to consider, or use _revInclude)

8.1.6 Patient.contact vs. RelatedPerson

The contact element on the Patient Resource should be used for storing people to contact information. Where a system has a separate record for other people for purposes other than just the contact details, the RelatedPerson resource should be used. This includes cases where these related people are actually contributing to the record, and need to be referenced individually (e.g. CarePlan.Participant, Encounter, DocumentReference, Appointment) where the Patient.Contact component cannot be used.

It is not expected that these records will be used for recording the primary care provider; this information should be stored in the Patient.generalPractitioner field.

8.1.7 Patient Gender

Tracking a patient's gender presents a number of challenges due to biological variations, differing cultural expectations and legal restrictions, and the availability of various kinds of gender re-assignment. The basic gender included in Patient.gender has a limited use, that of the administrative gender: the gender that the patient is considered to have for administration and record keeping purposes. In addition, to this gender, other kinds of gender may be represented:

* Birth Sex - the sex assigned at birth / on the birth registration. Some countries allow variations such as not yet determined, unknown, or undifferentiated, while others do not. The US realm defines a US Specific extension for this for Meaningful Use

* Clinical Gender - an observation about the patient, typically using the LOINC code 76691-5). LOINC also provides a set of possible codes , or SNOMED CT has the descendents of 285116001 : Gender identity finding

Other related Observations may describe the Karyotypic/Genetic, Gonadal, Ductal, Phenotypic status of a patient.

For veterinary use the animal component also includes the genderStatus which indicates sterility information.

8.1.8 Mother and newborn relationships

There are several ways to represent the relationship between a mother and a child. This is due to the when it is recorded and the purpose for which it is recorded:

* To express the family relationship and legal responsibility thereof for administrative purposes: use the Patient/RelatedPerson structure. This structure is consistent over time.

* To relate the encounter of a mother and her baby for a maternity encounter, for administrative and billing purposes: use the encounter.partof property

* To collect information about the patient's relatives that might be relevant to the patient's medical condition: use the FamilyMemberHistory resource

During a maternity encounter, the Patient and Encounter resources for the mother will be present. After the child is born, new Patient, Encounter and RelatedPerson (for the mother) records will be created. The Child's encounter should reference the Mother's encounter using the partOf property. The Patient/RelatedPerson structure should also be created for ongoing usage, as shown in this example:

```
<Patient>
  <id value="child"/>
  <!-- The details of the child -->
</Patient>
<RelatedPerson>
  <id value="rp-mom"/>
  <patient>
    <reference value="Patient/child"/>
  </patient>
</RelatedPerson>
<Patient>
  <id value="pat-mom"/>
  <!-- The details of the mom -->
  <link>
    <other value="rp-mom"/>
    <type value="see-also"/>
  </link>
</Patient>
<Encounter>
  <id value="mom-enc"/>
  <status value="in-progress"/>
  <class value="inpatient"/>
  <patient>
    <reference value="Patient/mom"/>
  </patient>
</Encounter>
<Encounter>
  <id value="child-enc"/>
  <status value="in-progress"/>
  <class value="inpatient"/>
  <patient>
    <reference value="Patient/child"/>
  </patient>
```

```
  <partOf>
    <reference value="Encounter/mom-enc"/>
  </partOf>
</Encounter>
```

8.1.9 Merging records

This specification does not specify merge functionality: if multiple patient records are found to be duplicates, they can be linked together, as described above. These links merely express the relationship between records, and in the case of a replacement link, indicate a "master" record. This specification does not mandate that FHIR servers migrate information between such records on finding such a link. Note:

* Health information administrators may call the process "merging", but it is often implemented as "linking" at the record level

* Servers are allowed to implement merging/record migration even though it is not mandated.

STU Note: We are seeking input from the implementer community on what effect linking/merging/unlinking should have on other functionality such as the GET operation (where the result is the old version of the Patient), searching, reverse includes, etc.; e.g., should observation resources from all linked/merged patients be returned when querying for one of them? How should an unlink behavior be done? (Assuming that no data was "re-allocated" as part of merge) These suggested updated behaviors could be the subject of a future connectathon.

Feedback here .

8.1.10 Patient Matching using an MPI

A Master Patient Index (MPI) is a service used to manage patient identification in a context where multiple patient databases exist. Healthcare applications and middleware use the MPI to match patients between the databases, and as new patient details are encountered. MPIs are highly specialized applications, often tailored extensively to the institution's particular mix of patients. MPIs can also be run on a regional and national basis.

To ask an MPI to match a patient, clients call the patient $match operation, which processes a parameters resource containing a complete or fragment of a patient resource, along with some other control parameters. This provided patient resource does not need to pass full validation (mandatory fields, or invariants) as the resource will not be stored, it does however need to be a parsable instance. The MPI can then use the properties of the resource as MPI inputs, and processed using an internal MPI algorithm of some kind to determine the most appropriate matches in the patient set. It does not have to use all the properties provided, and may ignore others provided quietly. A specific profile (with the required fields/invariants) can be used to define what parameters the MPI algorithm requires.

```
POST [base]/Patient/$match
[some headers including content-type xml or json]
[parameters body with patient resource inside]
```

The response from an "mpi" $match operation is a set of patient records, ordered from most likely to least likely. If there are not patient matches, the MPI SHALL return an empty search set with no error, but may include an operation outcome with further advice. All patient records SHALL have a score from 0 to 1, where 1 is the most certain match, along with an extension "match-grade" that indicates the MPI's position on the match quality:

```
<entry>
  <resource>
    <Patient>
      <!-- patient details -->
    </Patient>
  </resource>
  <search>
    <extension url="http://hl7.org/fhir/StructureDefinition/match-grade">
      <valueCode value="probable"/>
    </extension>
    <score value="0.80"/>
  </search>
</entry>
```

The match-grade extension has one of the following codes:

certain | This record meets the matching criteria to be automatically considered as a full match.
probable | This record is a close match, but not a certain match. Additional review (e.g. by a human) may be required before using this as a match.
possible | This record may be a matching one. Additional review (e.g. by a human) SHOULD be performed before using this as a match.
certainly-not | This record is known not to be a match. Note that usually non-matching records are not returned, but in some cases records previously or likely considered as a match may specifically be negated by the matching engine

The purpose of using an MPI search versus a regular search is that the MPI search is really intended to target and find a specific single patient for recording information about reducing errors through incorrectly selecting the wrong patient. Often MPIs won't return data if there is insuffience search parameter data, such as a partial surname. This compares to a regular search which can be used for finding lists of patients, such as to locate a group of patients that share a property in common, such as live in a specific location, or fit within an age range for performing population analysis.

A formal definition for the MPI $match operation is published.

8.1.11 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
active | token | Whether the patient record is active | Patient.active
address | string | A server defined search that may match any of the string fields in the Address, including line, city, state, country, postalCode, and/or text | Patient.address | 3 Resources
address-city | string | A city specified in an address | Patient.address.city | 3 Resources
address-country | string | A country specified in an address | Patient.address.country | 3 Resources
address-postalcode | string | A postalCode specified in an address | Patient.address.postalCode | 3 Resources
address-state | string | A state specified in an address | Patient.address.state | 3 Resources
address-use | token | A use code specified in an address | Patient.address.use | 3 Resources
animal-breed | token | The breed for animal patients | Patient.animal.breed
animal-species | token | The species for animal patients | Patient.animal.species
birthdate | date | The patient's date of birth | Patient.birthDate | 2 Resources
death-date | date | The date of death has been provided and satisfies this search value | Patient.deceased.as(DateTime)
deceased | token | This patient has been marked as deceased, or as a death date entered | Patient.deceased.exists()
email | token | A value in an email contact | Patient.telecom.where(system='email') | 4 Resources
family | string | A portion of the family name of the patient | Patient.name.family | 1 Resources
gender | token | Gender of the patient | Patient.gender | 3 Resources
general-practitioner | reference | Patient's nominated general practitioner, not the organization that manages the record | Patient.generalPractitioner (Practitioner, Organization)
given | string | A portion of the given name of the patient | Patient.name.given | 1 Resources
identifier | token | A patient identifier | Patient.identifier
language | token | Language code (irrespective of use value) | Patient.communication.language
link | reference | All patients linked to the given patient | Patient.link.other (Patient, RelatedPerson)
name | string | A server defined search that may match any of the string fields in the HumanName, including family, give, prefix, suffix, suffix, and/or text | Patient.name
organization | reference | The organization at which this person is a patient | Patient.managingOrganization (Organization)
phone | token | A value in a phone contact | Patient.telecom.where(system='phone') | 4 Resources
phonetic | string | A portion of either family or given name using some kind of phonetic matching algorithm | Patient.name | 3 Resources
telecom | token | The value in any kind of telecom details of the patient | Patient.telecom | 4 Resources

8.1.12 Resource Patient - Examples

URL: https://hl7.org/fhir/STU3/patient-examples.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.1.12 Resource Patient - Examples

Patient Administration Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Patient, Practitioner, RelatedPerson

Example List:

General Person Example | example | XML | JSON | Turtle
Patient 1 for linking | pat1 | XML | JSON | Turtle
Patient 2 for linking | pat2 | XML | JSON | Turtle
Deceased patient (using time) | pat3 | XML | JSON | Turtle
Deceased patient (using boolean) | pat4 | XML | JSON | Turtle
Stock people (defined by HL7 publishing) | b248b1b2-1686-4b94-9936-37d7a5f94b51 | XML | JSON | Turtle
Example People from cypress project | b0a5e4277-83c4-4adb-87e2-e3efe3369b6f | XML | JSON | Turtle
2nd person example | xcda | XML | JSON | Turtle
XDS Patient | xds | XML | JSON | Turtle
An example of an animal | animal | XML | JSON | Turtle
Taken from a DICOM sample | dicom | XML | JSON | Turtle
Example from IHE-PCD example | ihe-pcd | XML | JSON | Turtle
Real-world patient example (anonymized) | f001 | XML | JSON | Turtle
Real-world patient example (anonymized) | f201 | XML | JSON | Turtle
Example for glossy | glossy | XML | JSON | Turtle
Genetic Risk Assessment Person | proband | XML | JSON | Turtle
Additional Genetics Example | genetics-example1 | XML | JSON | Turtle
Example Patient resource with Chinese content | ch-example | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

8.1.13 Resource Patient - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/patient-definitions.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.1.13 Resource Patient - Detailed Descriptions

Patient Administration Work Group | Maturity Level: 5 | Trial Use | Compartments: Patient, Practitioner, RelatedPerson

Detailed Descriptions for the elements in the Patient resource.

Patient
Definition | Demographics and other administrative information about an individual or animal receiving care or other health-related services.
Control | 1..1
Requirements | Tracking patient is the center of the healthcare process.
Alternate Names | SubjectOfCare Client Resident
Patient.identifier
Definition | An identifier for this patient.
Note | This is a business identifer, not a resource identifier (see discussion)
Control | 0..*
Type | Identifier
Requirements | Patients are almost always assigned specific numerical identifiers.
Summary | true
Patient.active
Definition | Whether this patient record is in active use.
Control | 0..1
Type | boolean
Is Modifier | true
Default Value | true
Requirements | Need to be able to mark a patient record as not to be used because it was created in error.
Summary | true
Comments | Default is true. If a record is inactive, and linked to an active record, then future patient/record updates should occur on the other patient This element is labeled as a modifier because when the patient record is marked as not active it is not expected to be used/referenced without being changed back to active.
Patient.name
Definition | A name associated with the individual.
Control | 0..*
Type | HumanName
Requirements | Need to be able to track the patient by multiple names. Examples are your official name and a partner name.
Summary | true
Comments | A patient may have multiple names with different uses or applicable periods. For animals, the name is a "HumanName" in the sense that is assigned and used by humans and has the same patterns.
Patient.telecom
Definition | A contact detail (e.g. a telephone number or an email address) by which the individual may be contacted.
Control | 0..*
Type | ContactPoint
Requirements | People have (primary) ways to contact them in some way such as phone, email.
Summary | true
Comments | A Patient may have multiple ways to be contacted with different uses or applicable periods. May need to have options for contacting the person urgently and also to help with identification. The address may not go directly to the individual, but may reach another party that is able to proxy for the patient (i.e. home phone, or pet owner's phone).
Patient.gender
Definition | Administrative Gender - the gender that the patient is considered to have for administration and record keeping purposes.
Control | 0..1
Terminology Binding | AdministrativeGender (Required)
Type | code
Requirements | Needed for identification of the individual, in combination with (at least) name and birth date. Gender of individual drives many clinical processes.
Summary | true
Comments | The gender may not match the biological sex as determined by genetics, or the individual's preferred identification. Note that for both humans and particularly animals, there are other legitimate possibilities than M and F, though the vast majority of systems and contexts only support M and F. Systems providing decision support or enforcing business rules should ideally do this on the basis of Observations dealing with the specific gender aspect of interest (anatomical, chromosonal, social, etc.) However, because these observations are infrequently recorded, defaulting to the administrative gender is common practice. Where such defaulting occurs, rule enforcement should allow for the variation between administrative and biological, chromosonal and other gender aspects. For example, an alert about a hysterectomy on a male should be handled as a warning or overrideable error, not a "hard" error.
Patient.birthDate
Definition | The date of birth for the individual.
Control | 0..1
Type | date
Requirements | Age of the individual drives many clinical processes.
Summary | true
Comments | At least an estimated year should be provided as a guess if the real DOB is unknown There is a standard extension "patient-birthTime" available that should be used where Time is required (such as in maternaty/infant care systems).
LOINC Code | 21112-8
Patient.deceased[x]
Definition | Indicates if the individual is deceased or not.
Control | 0..1
Type | boolean|dateTime
[x] Note | See Choice of Data Types for further information about how to use [x]
Is Modifier | true
Requirements | The fact that a patient is deceased influences the clinical process. Also, in human communication and relation management it is necessary to know whether the person is alive.
Summary | true

Comments | If there's no value in the instance it means there is no statement on whether or not the individual is deceased. Most systems will interpret the absence of a value as a sign of the person being alive. This element is labeled as a modifier because once a patient is marked as deceased, the actions that are appropriate to perform on the patient may be significantly different.
Patient.address
Definition | Addresses for the individual.
Control | 0..*
Type | Address
Requirements | May need to keep track of patient addresses for contacting, billing or reporting requirements and also to help with identification.
Summary | true
Comments | Patient may have multiple addresses with different uses or applicable periods.
Patient.maritalStatus
Definition | This field contains a patient's most recent marital (civil) status.
Control | 0..1
Terminology Binding | Marital Status Codes (Extensible)
Type | CodeableConcept
Requirements | Most, if not all systems capture it.
Patient.multipleBirth[x]
Definition | Indicates whether the patient is part of a multiple (bool) or indicates the actual birth order (integer).
Control | 0..1
Type | boolean|integer
[x] Note | See Choice of Data Types for further information about how to use [x]
Requirements | For disambiguation of multiple-birth children, especially relevant where the care provider doesn't meet the patient, such as labs.
Comments | Where the valueInteger is provided, the number is the birth number in the sequence. E.g. The middle birth in tripplets would be valueInteger=2 and the third born would have valueInteger=3 If a bool value was provided for this tripplets examle, then all 3 patient records would have valueBool=true (the ordering is not indicated).
Patient.photo
Definition | Image of the patient.
Control | 0..*
Type | Attachment
Requirements | Many EHR systems have the capability to capture an image of the patient. Fits with newer social media usage too.
Patient.contact
Definition | A contact party (e.g. guardian, partner, friend) for the patient.
Control | 0..*
Requirements | Need to track people you can contact about the patient.
Comments | Contact covers all kinds of contact parties: family members, business contacts, guardians, caregivers. Not applicable to register pedigree and family ties beyond use of having contact.
Invariants | Defined on this element pat-1: SHALL at least contain a contact's details or a reference to an organization (expression : name.exists() or telecom.exists() or address.exists() or organization.exists(), xpath: exists(f:name) or exists(f:telecom) or exists(f:address) or exists(f:organization))
Patient.contact.relationship
Definition | The nature of the relationship between the patient and the contact person.
Control | 0..*
Terminology Binding | v2 Contact Role (Extensible)
Type | CodeableConcept
Requirements | Used to determine which contact person is the most relevant to approach, depending on circumstances.
Patient.contact.name
Definition | A name associated with the contact person.
Control | 0..1
Type | HumanName
Requirements | Contact persons need to be identified by name, but it is uncommon to need details about multiple other names for that contact person.
Patient.contact.telecom
Definition | A contact detail for the person, e.g. a telephone number or an email address.
Control | 0..*
Type | ContactPoint
Requirements | People have (primary) ways to contact them in some way such as phone, email.
Comments | Contact may have multiple ways to be contacted with different uses or applicable periods. May need to have options for contacting the person urgently, and also to help with identification.
Patient.contact.address
Definition | Address for the contact person.
Control | 0..1
Type | Address
Requirements | Need to keep track where the contact person can be contacted per postal mail or visited.
Patient.contact.gender
Definition | Administrative Gender - the gender that the contact person is considered to have for administration and record keeping purposes.
Control | 0..1
Terminology Binding | AdministrativeGender (Required)
Type | code
Requirements | Needed to address the person correctly.
Patient.contact.organization
Definition | Organization on behalf of which the contact is acting or for which the contact is working.
Control | 0..1
Type | Reference(Organization)
Requirements | For guardians or business related contacts, the organization is relevant.

Invariants | Affect this element pat-1: SHALL at least contain a contact's details or a reference to an organization (expression : name.exists() or telecom.exists() or address.exists() or organization.exists(), xpath: exists(f:name) or exists(f:telecom) or exists(f:address) or exists(f:organization))
Patient.contact.period
Definition | The period during which this contact person or organization is valid to be contacted relating to this patient.
Control | 0..1
Type | Period
Patient.animal
Definition | This patient is known to be an animal.
Control | 0..1
Is Modifier | true
Requirements | Many clinical systems are extended to care for animal patients as well as human.
Summary | true
Comments | The animal element is labeled "Is Modifier" since patients may be non-human. Systems SHALL either handle patient details appropriately (e.g. inform users patient is not human) or reject declared animal records. The absense of the animal element does not imply that the patient is a human. If a system requires such a positive assertion that the patient is human, an extension will be required. (Do not use a species of homo-sapiens in animal species, as this would incorrectly infer that the patient is an animal).
Patient.animal.species
Definition | Identifies the high level taxonomic categorization of the kind of animal.
Control | 1..1
Terminology Binding | AnimalSpecies (Example)
Type | CodeableConcept
Requirements | Need to know what kind of animal.
Summary | true
Comments | If the patient is non-human, at least a species SHALL be specified. Species SHALL be a widely recognised taxonomic classification. It may or may not be Linnaean taxonomy and may or may not be at the level of species. If the level is finer than species--such as a breed code--the code system used SHALL allow inference of the species. (The common example is that the word "Hereford" does not allow inference of the species Bos taurus, because there is a Hereford pig breed, but the SNOMED CT code for "Hereford Cattle Breed" does.).
Patient.animal.breed
Definition | Identifies the detailed categorization of the kind of animal.
Control | 0..1
Terminology Binding | AnimalBreeds (Example)
Type | CodeableConcept
Requirements | May need to know the specific kind within the species.
Summary | true
Comments | Breed MAY be used to provide further taxonomic or non-taxonomic classification. It may involve local or proprietary designation--such as commercial strain--and/or additional information such as production type.
Patient.animal.genderStatus
Definition | Indicates the current state of the animal's reproductive organs.
Control | 0..1
Terminology Binding | GenderStatus (Example)
Type | CodeableConcept
Requirements | Gender status can affect housing and animal behavior.
Summary | true
Patient.communication
Definition | Languages which may be used to communicate with the patient about his or her health.
Control | 0..*
Requirements | If a patient does not speak the local language, interpreters may be required, so languages spoken and proficiency is an important things to keep track of both for patient and other persons of interest.
Comments | If no language is specified, this implies that the default local language is spoken. If you need to convey proficiency for multiple modes then you need multiple Patient.Communication associations. For animals, language is not a relevant field, and should be absent from the instance. If the Patient does not speak the default local language, then the Interpreter Required Standard can be used to explicitly declare that an interpreter is required.
Patient.communication.language
Definition | The ISO-639-1 alpha 2 code in lower case for the language, optionally followed by a hyphen and the ISO-3166-1 alpha 2 code for the region in upper case; e.g. "en" for English, or "en-US" for American English versus "en-EN" for England English.
Control | 1..1
Terminology Binding | Common Languages (Extensible but limited to ??)
Type | CodeableConcept
Requirements | Most systems in multilingual countries will want to convey language. Not all systems actually need the regional dialect.
Comments | The structure aa-BB with this exact casing is one the most widely used notations for locale. However not all systems actually code this but instead have it as free text. Hence CodeableConcept instead of code as the data type.
Patient.communication.preferred
Definition | Indicates whether or not the patient prefers this language (over other languages he masters up a certain level).
Control | 0..1
Type | boolean
Requirements | People that master multiple languages up to certain level may prefer one or more, i.e. feel more confident in communicating in a particular language making other languages sort of a fall back method.
Comments | This language is specifically identified for communicating healthcare information.
Patient.generalPractitioner
Definition | Patient's nominated care provider.
Control | 0..*
Type | Reference(Organization | Practitioner)
Alternate Names | careProvider

Comments | This may be the primary care provider (in a GP context), or it may be a patient nominated care manager in a community/disablity setting, or even organization that will provide people to perform the care provider roles. It is not to be used to record Care Teams, these should be in a CareTeam resource that may be linked to the CarePlan or EpisodeOfCare resources.
Patient.managingOrganization
Definition | Organization that is the custodian of the patient record.
Control | 0..1
Type | Reference(Organization)
Requirements | Need to know who recognizes this patient record, manages and updates it.
Summary | true
Comments | There is only one managing organization for a specific patient record. Other organizations will have their own Patient record, and may use the Link property to join the records together (or a Person resource which can include confidence ratings for the association).
Patient.link
Definition | Link to another patient resource that concerns the same actual patient.
Control | 0..*
Is Modifier | true
Requirements | There are multiple usecases: Duplicate patient records due to the clerical errors associated with the difficulties of identifying humans consistently, and * Distribution of patient information across multiple servers.
Summary | true
Comments | There is no assumption that linked patient records have mutual links. This element is labelled as a modifier because it may not be the main Patient resource, and the referenced patient should be used instead of this Patient record. This is when the link.type value is 'replaced-by'.
Patient.link.other
Definition | The other patient resource that the link refers to.
Control | 1..1
Type | Reference(Patient | RelatedPerson)
Summary | true
Comments | Referencing a RelatedPerson here removes the need to use a Person record to associate a Patient and RelatedPerson as the same individual.
Patient.link.type
Definition | The type of link between this patient resource and another patient resource.
Control | 1..1
Terminology Binding | LinkType (Required)
Type | code
Summary | true

8.1.14 Resource Patient - Mappings

URL: https://hl7.org/fhir/STU3/patient-mappings.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.1.14 Resource Patient - Mappings

Patient Administration Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Patient, Practitioner, RelatedPerson

Mappings:

RIM Mapping

LOINC code for the element

CDA (R2)

HL7 v2 Mapping

Mappings for the patient resource.

8.1.14.1 Mappings for RIM Mapping (http://hl7.org/v3)

Patient | Patient[classCode=PAT]
identifier | id
active | statusCode
name | name
telecom | telecom
gender | player[classCode=PSN|ANM and determinerCode=INSTANCE]/administrativeGender
birthDate | player[classCode=PSN|ANM and determinerCode=INSTANCE]/birthTime
deceased[x] | player[classCode=PSN|ANM and determinerCode=INSTANCE]/deceasedInd, player[classCode=PSN|ANM and determinerCode=INSTANCE]/deceasedTime
address | addr
maritalStatus | player[classCode=PSN]/maritalStatusCode
multipleBirth[x] | player[classCode=PSN|ANM and determinerCode=INSTANCE]/multipleBirthInd, player[classCode=PSN|ANM and determinerCode=INSTANCE]/multipleBirthOrderNumber
photo | player[classCode=PSN|ANM and determinerCode=INSTANCE]/desc
contact | player[classCode=PSN|ANM and determinerCode=INSTANCE]/scopedRole[classCode=CON]
relationship | code
name | name
telecom | telecom
address | addr

gender | player[classCode=PSN|ANM and determinerCode=INSTANCE]/administrativeGender
organization | scoper
period | effectiveTime
animal | player[classCode=ANM]
species | code
breed | playedRole[classCode=GEN]/scoper[classCode=ANM, determinerCode=KIND]/code
genderStatus | genderStatusCode
communication | LanguageCommunication
language | player[classCode=PSN|ANM and determinerCode=INSTANCE]/languageCommunication/code
preferred | preferenceInd
generalPractitioner | subjectOf.CareEvent.performer.AssignedEntity
managingOrganization | scoper
link | outboundLink
other | id
type | typeCode

8.1.14.2 Mappings for LOINC code for the element (http://loinc.org)

Patient
identifier
active
name
telecom
gender
birthDate | 21112-8
deceased[x]
address
maritalStatus
multipleBirth[x]
photo
contact
relationship
name
telecom
address
gender
organization
period
animal
species
breed
genderStatus
communication
language
preferred
generalPractitioner
managingOrganization
link
other
type

8.1.14.3 Mappings for CDA (R2) (http://hl7.org/v3/cda)

Patient | ClinicalDocument.recordTarget.patientRole
identifier | .id
active | n/a
name | .patient.name
telecom | .telecom
gender | .patient.administrativeGenderCode
birthDate | .patient.birthTime
deceased[x] | n/a
address | .addr
maritalStatus | .patient.maritalStatusCode
multipleBirth[x] | n/a
photo | n/a
contact | n/a
relationship | n/a
name | n/a
telecom | n/a
address | n/a
gender | n/a
organization | n/a
period | n/a

animal | n/a
species | n/a
breed | n/a
genderStatus | n/a
communication | patient.languageCommunication
language | .languageCode
preferred | .preferenceInd
generalPractitioner | n/a
managingOrganization | .providerOrganization
link | n/a
other | n/a
type | n/a

8.1.14.4 Mappings for HL7 v2 Mapping (http://hl7.org/v2)

Patient
identifier | PID-3
active
name | PID-5, PID-9
telecom | PID-13, PID-14, PID-40
gender | PID-8
birthDate | PID-7
deceased[x] | PID-30 (bool) and PID-29 (datetime)
address | PID-11
maritalStatus | PID-16
multipleBirth[x] | PID-24 (bool), PID-25 (integer)
photo | OBX-5 - needs a profile
contact
relationship | NK1-7, NK1-3
name | NK1-2
telecom | NK1-5, NK1-6, NK1-40
address | NK1-4
gender | NK1-15
organization | NK1-13, NK1-30, NK1-31, NK1-32, NK1-41
period
animal
species | PID-35
breed | PID-37
genderStatus | N/A
communication
language | PID-15, LAN-2
preferred | PID-15
generalPractitioner | PD1-4
managingOrganization
link
other | PID-3, MRG-1
type

8.1.15 Resource Patient - Design Notes

URL: https://hl7.org/fhir/STU3/patient-explanations.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.1.15 Resource Patient - Design Notes

to do

8.1.16 Resource Patient - Extensions & Profiles

URL: https://hl7.org/fhir/STU3/patient-profiles.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.1.16 Resource Patient - Extensions & Profiles

Patient Administration Work Group | Maturity Level: N/A | Ballot Status: Informative

This table lists profiles and extensions for the Patient resource. For background information, see Profiling Resources and Extensibility. Additional profiles and resources may be found in published Implementation Guides , or in the Conformance resource registry .

8.1.16.1 Profiles

No Profiles defined for this resource

8.1.16.2 Extensions

birthPlace | Birth Place | for General Extensions for use by FHIR Implementers
patient-adoptionInfo | adoptionInfo | for Patient HL7 Extensions
patient-birthTime | birthTime | for Patient HL7 Extensions
patient-cadavericDonor | cadavericDonor | for Patient HL7 Extensions
patient-citizenship | citizenship | for Patient HL7 Extensions
patient-clinicalTrial | clinicalTrial | for Patient HL7 Extensions
patient-congregation | congregation | for Patient HL7 Extensions
patient-disability | disability | for Patient HL7 Extensions
patient-importance | importance | for Patient HL7 Extensions
patient-interpreterRequired | interpreterRequired | for Patient HL7 Extensions
patient-mothersMaidenName | mothersMaidenName | for Patient HL7 Extensions
patient-nationality | nationality | for Patient HL7 Extensions
patient-religion | religion | for Patient HL7 Extensions

Extensions that reference this resource:

communicationrequest-orderedBy | orderedBy | for CommunicationRequest HL7 Extensions
consent-Witness | Witness | for Consent HL7 Extensions
cqif-guidanceinitiatingPerson | initiatingPerson | for CQIF Guidance Extensions
cqif-guidancereceivingPerson | receivingPerson | for CQIF Guidance Extensions
familymemberhistory-patient-record | patient-record | for Family Member History HL7 Extensions
goal-acceptance | acceptance | for FHIR Core Goal Profile
questionnaireresponse-author | author | for Core extensions for QuestionnaireResponse
task-candidateList | candidateList | for Task HL7 Extensions

8.1.16.3 Search Extensions

No Search Extensions defined for this resource

8.1.17 Patient HL7 Extensions

URL: https://hl7.org/fhir/STU3/patient-extensions.html

Defines common extensions used with or related to the Patient resource

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.1.17 Patient HL7 Extensions

(No assigned work group) | Maturity Level: N/A | Ballot Status: Informative

8.1.17.1 Content

Extensions:
patient-mothersMaidenName | mothersMaidenName : Mother's maiden (unmarried) name, commonly collected to help verify patient identity.
patient-birthTime | birthTime : The time of day that the Patient was born. This includes the date to ensure that the timezone information can be communicated effectively.
patient-nationality | nationality : The nationality of the patient.
patient-citizenship | citizenship : The patient's legal status as citizen of a country.
patient-cadavericDonor | cadavericDonor : Flag indicating whether the patient authorized the donation of body parts after death.
patient-congregation | congregation : A group or place of religious practice that may provide services to the patient.
patient-adoptionInfo | adoptionInfo : Code indication the adoption status of the patient.
patient-disability | disability : Value(s) identifying physical or mental condition(s) that limits a person's movements, senses, or activities.
patient-importance | importance : The importance of the patient (e.g. VIP).
patient-clinicalTrial | clinicalTrial : The clinical trials this patient has or is participating in.
patient-interpreterRequired | interpreterRequired : This Patient requires an interpreter to communicate healthcare information to the practitioner.
patient-religion | religion : The patient's professed religious affiliations.

8.1.17.2 Search Parameters

Search parameters defined by this package. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Paths | Source
age | number | Searches for patients based on age as calculated based on current date and date of birth. Deceased patients are excluded from the search. | f:Patient/f:birthDate | XML / JSON
birthOrderBoolean | token | Search based on whether a patient was part of a multiple birth or not. | f:Patient/f:multipleBirthBoolean | f:Patient/f:multipleBirthInteger | XML / JSON
mothersMaidenName | string | Search based on patient's mother's maiden name | | XML / JSON

8.1.17.3 Patient-extensions-Patient-age

URL: https://hl7.org/fhir/STU3/patient-extensions-Patient-age.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.1.17.3 Patient-extensions-Patient-age

Search Parameter age (type: number) defined as part of the patient-extensions Profile.

Searches for patients based on age as calculated based on current date and date of birth. Deceased patients are excluded from the search.

Other Details:

* Full URL : http://hl7.org/fhir/SearchParameter/patient-extensions-Patient-age

* Paths : f:Patient/f:birthDate

* Targets :

8.1.17.4 Patient-extensions-Patient-birthOrderBoolean

URL: https://hl7.org/fhir/STU3/patient-extensions-Patient-birthOrderBoolean.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.1.17.4 Patient-extensions-Patient-birthOrderBoolean

Search Parameter birthOrderBoolean (type: token) defined as part of the patient-extensions Profile.

Search based on whether a patient was part of a multiple birth or not.

Other Details:

* Full URL : http://hl7.org/fhir/SearchParameter/patient-extensions-Patient-birthOrderBoolean

* Paths : f:Patient/f:multipleBirthBoolean | f:Patient/f:multipleBirthInteger

* Targets :

8.1.17.5 Patient-extensions-Patient-mothersMaidenName

URL: https://hl7.org/fhir/STU3/patient-extensions-Patient-mothersMaidenName.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.1.17.5 Patient-extensions-Patient-mothersMaidenName

Search Parameter mothersMaidenName (type: string) defined as part of the patient-extensions Profile.

Search based on patient's mother's maiden name

Other Details:

* Full URL : http://hl7.org/fhir/SearchParameter/patient-extensions-Patient-mothersMaidenName

* Paths :

* Targets :

8.1.18 Resource Patient - Operations

URL: https://hl7.org/fhir/STU3/patient-operations.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.1.18 Resource Patient - Operations

Patient Administration Work Group | Maturity Level: 5 | Trial Use | Compartments: Patient, Practitioner, RelatedPerson

This resource has 2 operations associated with it:

$match | Find patient matches using MPI based logic
$everything | Fetch Patient Record

For more information about operations, including how they are invoked, see Operations.

8.1.18.1 Find patient matches using MPI based logic

A Master Patient Index (MPI) is a service used to manage patient identification in a context where multiple patient databases exist. Healthcare applications and middleware use the MPI to match patients between the databases, and as new patient details are encountered. MPIs are highly specialized applications, often tailored extensively to the institution's particular mix of patients. MPIs can also be run on a regional and national basis.\n\nTo ask an MPI to match a patient, clients use the \"$match\" operation, which accepts a patient resource which may be only partially complete. The data provided is interpreted as an MPI input and passed processed by an algorithm of some kind that uses the data to determine the most appropriate matches in the patient set. \n\nNote that different MPI matching algorithms have different required inputs. The generic $match operation does not specify any particular algorithm, nor a minimum set of information that must be provided when asking for

an MPI match operation to be performed, but may implementations will have a set of minimum information, which may be declared in their definition of the $match operation by specifying a profile on the resource parameter, indicating which properties are required in the search.\n\nThe patient resource coming into the operation does not have to be complete, nor does it need to pass validation (i.e. Mandatory fields don't need to be populated), but it does have to be a valid instance. This is due to the resource being used for reference data to match from, and not being stored.

The official URL for this operation definition is

`http://hl7.org/fhir/OperationDefinition/Patient-match`

Formal Definition (as a OperationDefinition).

URL: [base]/Patient/$match

This is an idempotent operation

In Parameters:

Name | Cardinality | Type | Binding | Profile | Documentation

resource | 1..1 | Resource | | | Use this to provide an entire set of patient details for the MPI to match against (e.g. POST a patient record to Patient/$match). If a patient record is not provided, then one or more of the other parameters must be provided

onlyCertainMatches | 0..1 | boolean | | | If there are multiple potential matches, then the match should not return the results with this flag set to true. When false, the server may return multiple results with each result graded accordingly.

count | 0..1 | integer | | | The maximum number of records to return. If no value is provided, the server decides how many matches to return. Note that clients should be careful when using this, as it may prevent probable - and valid - matches from being returned

Out Parameters:

Name | Cardinality | Type | Binding | Profile | Documentation

return | 1..1 | Bundle | | | A bundle contain a set of Patient records that represent possible matches, optionally it may also contain an OperationOutcome with further information about the search results (such as warnings or information messages, such as a count of records that were close but eliminated) If the operation was unsuccessful, then an OperationOutcome may be returned along with a BadRequest status Code (e.g. security issue, or insufficient properties in patient fragment - check against profile) Note: as this the only out parameter, it is a resource, and it has the name 'return', the result of this operation is returned directly as a resource

The response from an "mpi" query is a bundle containing patient records, ordered from most likely to least likely. If there are no patient matches, the MPI SHALL return an empty search set with no error, but may include an operation outcome with further advice regarding patient selection. All patient records SHALL have a search score from 0 to 1, where 1 is the most certain match, along with an extension "match-grade" that indicates the MPI's position on the match quality.

8.1.18.1.1 Examples

Patient match using patient resource: (Request):

```
POST /open/Patient/$match
[some headers]

{
  "resourceType": "Parameters",
  "id": "example",
  "parameter": [
    {
      "name": "resource",
      "resource": {
        "resourceType": "Patient",
        "identifier": [
          {
            "use": "usual",
            "type": {
              "coding": [
                {
                  "system": "http://hl7.org/fhir/v2/0203",
                  "code": "MR"
                }
              ]
            },
            "system": "urn:oid:1.2.36.146.595.217.0.1",
            "value": "12345"
          }
        ],
        "name": [
          {
            "family": [
              "Chalmers"
            ],
            "given": [
              "Peter"
            ]
          }
        ],
        "gender": "male",
        "birthDate": "1974-12-25"
      }
    },
    {
      "name": "count",
      "valueInteger": "3"
    },
    {
      "name": "onlyCertainMatches",
```

```
      "valueBoolean": "false"
    }
  ]
}
```

Results from imaginary MPI algorithm (Response):

```
HTTP/1.1 200 OK
[other headers]

{
  "resourceType": "Bundle",
  "id": "26419249-18b3-45de-b10e-dca0b2e72b",
  "meta": {
    "lastUpdated": "2016-03-18T03:28:49Z"
  },
  "type": "searchset",
  "total": 2,
  "entry": [{
    "fullUrl": "http://server/path/Patient/example",
    "resource": {
      "resourceType": "Patient",
      "id": "example",
      .. snip ...
    },
    "search": {
      "extension": [{
        "url": "http://hl7.org/fhir/StructureDefinition/match-grade",
        "valueCode": "certain"
      }],
      "mode": "match",
      "score": 0.9
    }
  },{
    "fullUrl": "http://server/path/Patient/292",
    "resource": {
      "resourceType": "Patient",
      "id": "292",
      .. snip ...
    },
    "search": {
      "extension": [{
        "url": "http://hl7.org/fhir/StructureDefinition/match-grade",
        "valueCode": "possible"
      }],
      "mode": "match",
      "score": 0.2
    }
  }]
}
```

8.1.18.2 Fetch Patient Record

This operation is used to return all the information related to the patient described in the resource on which this operation is invoked. The response is a bundle of type "searchset". At a minimum, the patient resource itself is returned, along with any other resources that the server has that are related to the patient, and that are available for the given user. The server also returns whatever resources are needed to support the records - e.g. linked practitioners, medications, locations, organizations etc. The principle intended use for this operation is to provide a patient with access to their entire record (e.g. "Blue Button"). The server SHOULD return at least all resources that it has that are in the patient compartment for the identified patient, and any resource referenced from those, including binaries and attachments. In the US Realm, at a mimimum, the resources returned SHALL include all the data covered by the meaningful use common data elements as defined in DAF . Other applicable implementation guides may make additional rules about how much information that is returned

The official URL for this operation definition is

```
http://hl7.org/fhir/OperationDefinition/Patient-everything
```

Formal Definition (as a OperationDefinition).

URL: [base]/Patient/$everything

URL: [base]/Patient/[id]/$everything

This is an idempotent operation

In Parameters:

Name | Cardinality | Type | Binding | Profile | Documentation

start | 0..1 | date | | | The date range relates to care dates, not record currency dates - e.g. all records relating to care provided in a certain date range. If no start date is provided, all records prior to the end date are in scope.

end | 0..1 | date | | | The date range relates to care dates, not record currency dates - e.g. all records relating to care provided in a certain date range. If no end date is provided, all records subsequent to the start date are in scope.

Out Parameters:

Name | Cardinality | Type | Binding | Profile | Documentation

return | 1..1 | Bundle | | | The bundle type is "searchset" Note: as this the only out parameter, it is a resource, and it has the name 'return', the result of this operation is returned directly as a resource

The key differences between this operation and simply searching the patient compartment are:

* unless the client requests otherwise, the server returns the entire result set in a single bundle (rather than using paging)

* the server is responsible for determining what resources to return as included resources (rather than the client specifying which ones). This frees the client from needing to determine what it could or should ask for

It is assumed that the server has identified and secured the context appropriately, and can either associate the authorization context with a single patient, or determine whether the context has the rights to the nominated patient, if there is one. If there is no nominated patient (e.g. the operation is invoked at the system level) and the context is not associated with a single patient record, then the server should return an error. Specifying the relationship between the context, a user and patient records is outside the scope of this specification.

8.1.18.2.1 Examples

Get all resources linked to a single patient (Request):

```
GET /open/Patient/f001/$everything
[some headers]
```

Everything linked to the nominated patient (in this case, built automatically by the publication tooling) (Response):

```
HTTP/1.1 200 OK
[other headers]

<Bundle xml="http://hl7.org/fhir">
  <id value="8993b629-b67d-4c9a-8f2a-4dae84383054"/>
  <type value="searchset"/>
  <entry>
    <fullUrl value="http://hl7.org/fhir/Patient/f001"/>
    <resource>
     <Patient xmlns="http://hl7.org/fhir">
       <id value="f001"/>
       <identifier>
         <use value="usual"/>
         <system value="urn:oid:2.16.840.1.113883.2.4.6.3"/>
       <!--  BSN identification system  -->
         <value value="738472983"/>
       </identifier>
       <identifier>
         <use value="usual"/>
         <system value="urn:oid:2.16.840.1.113883.2.4.6.3"/>
       <!--  BSN identification system  -->
       </identifier>
       <active value="true"/>
       <name>
         <use value="usual"/>
         <family value="van de Heuvel"/>
         <given value="Pieter"/>
         <suffix value="MSc"/>
       </name>
       <telecom>
         <system value="phone"/>
         <value value="0648352638"/>
         <use value="mobile"/>
       </telecom>
       <telecom>
         <system value="email"/>
         <value value="p.heuvel@gmail.com"/>
         <use value="home"/>
       </telecom>
       <gender value="male"/>
       <birthDate value="1944-11-17"/>
       <deceasedBoolean value="false"/>
       <address>
         <use value="home"/>
         <line value="Van Egmondkade 23"/>
         <city value="Amsterdam"/>
         <postalCode value="1024 RJ"/>
         <country value="NLD"/>
       <!--  ISO 3166 Codes (Countries)  -->
       </address>
       <maritalStatus>
         <coding>
           <system value="http://hl7.org/fhir/v3/MaritalStatus"/>
           <code value="M"/>
           <display value="Married"/>
         </coding>
         <text value="Getrouwd"/>
       </maritalStatus>
       <multipleBirthBoolean value="true"/>
       <contact>
         <relationship>
           <coding>
             <system value="http://hl7.org/fhir/v2/0131"/>
             <code value="C"/>
           </coding>
         </relationship>
         <name>
           <use value="usual"/>
           <family value="Abels"/>
```

```
        <given value="Sarah"/>
      </name>
      <telecom>
        <system value="phone"/>
        <value value="0690383372"/>
        <use value="mobile"/>
      </telecom>
    </contact>
    <communication>
      <language>
        <coding>
          <system value="urn:ietf:bcp:47"/>
        <!--  IETF language tag  -->
          <code value="nl"/>
          <display value="Dutch"/>
        </coding>
        <text value="Nederlands"/>
      </language>
      <preferred value="true"/>
    </communication>
    <managingOrganization>
      <reference value="Organization/f001"/>
      <display value="Burgers University Medical Centre"/>
    </managingOrganization>
   </Patient>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/Observation/f002"/>
  <resource>
   <Observation xmlns="http://hl7.org/fhir">
    <id value="f002"/>
   <!--  urn:oid:2.16.840.1.113883.4.642.1.7  -->
   <!--  2.16.840.1.113883.4.642.1.118  -->
    <identifier>
      <use value="official"/>
      <system value="http://www.bmc.nl/zorgportal/identifiers/observations"/>
      <value value="6324"/>
    </identifier>
    <status value="final"/>
    <code>
      <coding>
        <system value="http://loinc.org"/>
        <code value="11555-0"/>
        <display value="Base excess in Blood by calculation"/>
      </coding>
    </code>
    <subject>
      <reference value="Patient/f001"/>
      <display value="P. van de Heuvel"/>
    </subject>
    <effectivePeriod>
      <start value="2013-04-02T10:30:10+01:00"/>
      <end value="2013-04-05T10:30:10+01:00"/>
    </effectivePeriod>
    <issued value="2013-04-03T15:30:10+01:00"/>
    <performer>
      <reference value="Practitioner/f005"/>
      <display value="A. Langeveld"/>
    </performer>
    <valueQuantity>
      <value value="12.6"/>
      <unit value="mmol/l"/>
      <system value="http://unitsofmeasure.org"/>
      <code value="mmol/L"/>
    </valueQuantity>
    <interpretation>
      <coding>
        <system value="http://hl7.org/fhir/v2/0078"/>
        <code value="H"/>
        <display value="High"/>
      </coding>
    </interpretation>
    <referenceRange>
      <low>
        <value value="7.1"/>
        <unit value="mmol/l"/>
        <system value="http://unitsofmeasure.org"/>
        <code value="mmol/L"/>
      </low>
      <high>
        <value value="11.2"/>
        <unit value="mmol/l"/>
        <system value="http://unitsofmeasure.org"/>
        <code value="mmol/L"/>
      </high>
    </referenceRange>
   </Observation>
  </resource>
```

```
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/Consent/consent-example-notTime"/>
  <resource>
   <Consent xmlns="http://hl7.org/fhir">
     <id value="consent-example-notTime"/>

     <text>
       <status value="generated"/>
       <div xmlns="http://www.w3.org/1999/xhtml">The following scenario is based on existing jurisdictional policy and
   are realized in existing systems in Canada.
    The default policy is one of implied consent for the provision of care, so these scenarios all deal with
  withdrawal or withholding consent for that purpose.
    In other jurisdictions, where an express consent model is used (Opt-In), these would examples would contain the
  phrase "consent to" rather
    than "withhold" or "withdraw" consent for.
       <p>
    specific to use-case 6. Withhold or withdraw consent for disclosure of records related published during a
  timeframe
    </p><p>
       Patient "P. van de Heuvel" wishes to withhold disclosure of all data from a timeframe to any provider.
      </p>
      </div>
     </text>

     <status value="active"/>

     <patient>
       <reference value="Patient/f001"/>
       <display value="P. van de Heuvel"/>
     </patient>

     <dateTime value="2015-11-18"/>
      <!-- not bound by a timeframe - Consent.period -->

     <!-- I assume the example given is Canada Infoway wide???  AND I assume it is desired to state that in the
  Consent.authority element -->
     <organization>
       <reference value="Organization/Infoway"/>
      <display value="Canada Infoway"/>
     </organization>

     <!-- the text terms of the consent in lawyer speak -->
     <sourceAttachment>
       <title value="The terms of the consent in lawyer speak."/>
       <!-- likely use url pointer to common text -->
    </sourceAttachment>

     <!-- this is opt-in - e.g. everything approved unless otherwise stated -->
     <policyRule value="http://hl7.org/fhir/ConsentPolicy/opt-in"/>

     <except>
       <type value="deny"/>
        <period>
        <start value="2015-01-01"/>
        <end value="2015-02-01"/>
      </period>
    </except>
   </Consent>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/Condition/f001"/>
  <resource>
   <Condition xmlns="http://hl7.org/fhir">
     <id value="f001"/>
     <clinicalStatus value="active"/>
     <verificationStatus value="confirmed"/>
     <category>
       <coding>
         <system value="http://snomed.info/sct"/>
         <code value="439401001"/>
         <display value="diagnosis"/>
       </coding>
     </category>
     <severity>
       <coding>
         <system value="http://snomed.info/sct"/>
         <code value="6736007"/>
         <display value="Moderate"/>
       </coding>
     </severity>
     <code>
       <coding>
         <system value="http://snomed.info/sct"/>
         <code value="368009"/>
         <display value="Heart valve disorder"/>
       </coding>
     </code>
```

```
      <bodySite>
        <coding>
          <system value="http://snomed.info/sct"/>
          <code value="40768004"/>
          <display value="Left thorax"/>
        </coding>
        <text value="heart structure"/>
      </bodySite>
      <subject>
        <reference value="Patient/f001"/>
        <display value="P. van de Heuvel"/>
      </subject>
      <context>
        <reference value="Encounter/f001"/>
      </context>
      <onsetDateTime value="2011-08-05"/>
      <assertedDate value="2011-10-05"/>
      <asserter>
        <reference value="Patient/f001"/>
        <display value="P. van de Heuvel"/>
      </asserter>
      <evidence>
        <code>
          <coding>
            <system value="http://snomed.info/sct"/>
            <code value="426396005"/>
            <display value="Cardiac chest pain"/>
          </coding>
        </code>
      </evidence>
    </Condition>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/Contract/pcd-example-notAuthor"/>
  <resource>
   <Contract xmlns="http://hl7.org/fhir">
     <id value="pcd-example-notAuthor"/>

     <text>
       <status value="generated"/>
       <div xmlns="http://www.w3.org/1999/xhtml">The following scenario is based on existing jurisdictional policy and
    are realized in existing systems in Canada.
     The default policy is one of implied consent for the provision of care, so these scenarios all deal with
  withdrawal or withholding consent for that purpose.
     In other jurisdictions, where an express consent model is used (Opt-In), these would examples would contain the
  phrase "consent to" rather
     than "withhold" or "withdraw" consent for.
        <p>
     specific to use-case 5) Withhold or withdraw consent for disclosure of records that were authored by a specific
  organization (or service delivery location).
     </p><p>
        Patient "P. van de Heuvel" wishes to have all of the PHI collected at the Good Health Psychiatric Hospital
      restricted from disclosure to every provider.
      </p>
      </div>
     </text>

     <issued value="2015-11-18"/>
     <!-- not bound by a timeframe - Contract.applies -->

     <subject>
       <reference value="Patient/f001"/>
       <display value="P. van de Heuvel"/>
     </subject>

     <!-- I assume the example given is Canada Infoway wide???  AND I assume it is desired to state that in the
  Contract.authority element -->
     <authority>
         <reference value="Organization/Infoway"/>
      <display value="Canada Infoway"/>
      </authority>

      <!-- and/or would this [also] go into Contract.domain as a Jurisdiction?? (see the example on UK Pharamacy
  Juristiction) -->
     <domain>
         <reference value="Location/Infoway"/>
      <display value="Canada Infoway"/>
      </domain>

     <type>
        <coding>
       <system value="http://loinc.org"/>
       <code value="57016-8"/>
     </coding>
     </type>

     <!-- made up code-system to represent the set of privacy consent sub-types known and published by Canada Infoway
  -->
```

```
      <subType>
         <coding>
        <system value="http://www.infoway-inforoute.ca.org/Consent-subtype-codes"/>
        <code value="Opt-In"/>
        <display value="Default Authorization with exceptions."/>
      </coding>
      </subType>

      <term>
         <type>
           <coding>
          <!-- made up code system -->
          <system value="http://example.org/fhir/consent-term-type-codes"/>
          <code value="withhold-authored-by"/>
        <display value="Withhold all data authored by specified actor entity."/>
        </coding>
      </type>
      <topic>
            <reference value="Organization/2.16.840.1.113883.19.5"/>
               <display value="Good Health Clinic"/>
      </topic>
      <text value="Withhold all data authored by Good Health provider."/>
       </term>
       <!-- the terms of the consent in friendly consumer speak -->
       <friendly>
         <contentAttachment>
        <title value="The terms of the consent in friendly consumer speak."/>
        <!-- likely use url pointer to common text -->
      </contentAttachment>
       </friendly>
       <!-- the legal terms of the consent in lawyer speak -->
       <legal>
         <contentAttachment>
        <title value="The terms of the consent in lawyer speak."/>
        <!-- likely use url pointer to common text -->
        <!-- I think some think that this should be the pointer to the law by which this consent is derived under. Such
    as http://lois-laws.justice.gc.ca/eng/acts/P-21/index.html -->
      </contentAttachment>
       </legal>
    </Contract>
   </resource>
 </entry>
 <entry>
   <fullUrl value="http://hl7.org/fhir/Encounter/f001"/>
   <resource>
    <Encounter xmlns="http://hl7.org/fhir">
      <id value="f001"/>
        <identifier>
            <use value="official"/>
            <system value="http://www.amc.nl/zorgportal/identifiers/visits"/>
            <value value="v1451"/>
        </identifier>
        <status value="finished"/>
        <class>
            <system value="http://hl7.org/fhir/v3/ActCode"/>
            <code value="AMB"/> <!-- outpatient -->
            <display value="ambulatory"/>
        </class>
        <type>
            <coding>
                <system value="http://snomed.info/sct"/>
                <code value="270427003"/>
                <display value="Patient-initiated encounter"/>
            </coding>
        </type>
        <priority>
            <coding>
                <system value="http://snomed.info/sct"/>
                <code value="310361003"/>
                <display value="Non-urgent cardiological admission"/>
            </coding>
        </priority>
     <subject>
            <reference value="Patient/f001"/>
            <display value="P. van de Heuvel"/>
        </subject>
        <participant>
            <individual>
                <reference value="Practitioner/f002"/>
                <display value="P. Voigt"/>
            </individual>
        </participant>
        <length>
            <value value="140"/>
            <unit value="min"/>
            <system value="http://unitsofmeasure.org"/>
            <code value="min"/>
        </length>
        <reason>
```

```
            <coding>
                <system value="http://snomed.info/sct"/>
                <code value="34068001"/>
                <display value="Heart valve replacement"/>
            </coding>
        </reason>
        <hospitalization>
            <preAdmissionIdentifier>
                <use value="official"/>
                <system value="http://www.amc.nl/zorgportal/identifiers/pre-admissions"/>
                <value value="93042"/>
            </preAdmissionIdentifier>
      <!--      <preAdmissionTest>
                <coding>
                    <system value="http://snomed.info/sct"/>
                    <code value="164847006"/>
                    <display value="Standard ECG"/>
                </coding>
                <coding>
                    <system value="http://snomed.info/sct"/>
                    <code value="396550006"/>
                    <display value="Blood test"/>
                </coding>
            </preAdmissionTest> -->
            <admitSource>
                <coding>
                    <system value="http://snomed.info/sct"/>
                    <code value="305956004"/>
                    <display value="Referral by physician"/>
                </coding>
            </admitSource>
            <dischargeDisposition>
                <coding>
                    <system value="http://snomed.info/sct"/>
                    <code value="306689006"/>
                    <display value="Discharge to home"/>
                </coding>
            </dischargeDisposition>
        </hospitalization>
        <serviceProvider>
            <reference value="Organization/f001"/>
            <display value="Burgers University Medical Center"/>
        </serviceProvider>
    </Encounter>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/Consent/consent-example-notAuthor"/>
  <resource>
   <Consent xmlns="http://hl7.org/fhir">
     <id value="consent-example-notAuthor"/>

     <text>
       <status value="generated"/>
       <div xmlns="http://www.w3.org/1999/xhtml">The following scenario is based on existing jurisdictional policy and
   are realized in existing systems in Canada.
     The default policy is one of implied consent for the provision of care, so these scenarios all deal with
  withdrawal or withholding consent for that purpose.
     In other jurisdictions, where an express consent model is used (Opt-In), these would examples would contain the
  phrase "consent to" rather
     than "withhold" or "withdraw" consent for.
        <p>
     specific to use-case 5) Withhold or withdraw consent for disclosure of records that were authored by a specific
  organization (or service delivery location).
     </p><p>
        Patient "P. van de Heuvel" wishes to have all of the PHI collected at the Good Health Psychiatric Hospital
      restricted from disclosure to every provider.
      </p>
      </div>
     </text>

     <status value="active"/>

     <patient>
       <reference value="Patient/f001"/>
       <display value="P. van de Heuvel"/>
     </patient>

     <dateTime value="2015-11-18"/>
      <!-- not bound by a timeframe - Consent.period -->

     <!-- I assume the example given is Canada Infoway wide???  AND I assume it is desired to state that in the
  Consent.authority element -->
     <organization>
       <reference value="Organization/Infoway"/>
      <display value="Canada Infoway"/>
     </organization>

     <!-- the text terms of the consent in lawyer speak -->
     <sourceAttachment>
```

```
      <title value="The terms of the consent in lawyer speak."/>
      <!-- likely use url pointer to common text -->
    </sourceAttachment>

    <!-- this is opt-in - e.g. everything approved unless otherwise stated -->
    <policyRule value="http://hl7.org/fhir/ConsentPolicy/opt-in"/>

    <except>
      <type value="deny"/>
     <actor>
        <role>
          <coding>
            <system value="http://hl7.org/fhir/v3/ParticipationType"/>
            <code value="CST"/>
          </coding>
        </role>
        <reference>
          <reference value="Organization/2.16.840.1.113883.19.5"/>
           <display value="Good Health Clinic"/>
        </reference>
     </actor>
    </except>

  </Consent>
 </resource>
</entry>
<entry>
 <fullUrl value="http://hl7.org/fhir/Observation/unsat"/>
 <resource>
  <Observation xmlns="http://hl7.org/fhir">
    <id value="unsat"/>
  <!--  urn:oid:2.16.840.1.113883.4.642.1.7  -->
  <!--  2.16.840.1.113883.4.642.1.118  -->
    <identifier>
      <use value="official"/>
      <system value="http://www.bmc.nl/zorgportal/identifiers/observations"/>
      <value value="6323"/>
    </identifier>
    <status value="cancelled"/>
    <code>
      <coding>
        <system value="http://loinc.org"/>
        <code value="15074-8"/>
        <display value="Glucose [Moles/volume] in Blood"/>
      </coding>
    </code>
    <subject>
      <reference value="Patient/f001"/>
      <display value="P. van de Heuvel"/>
    </subject>
    <effectivePeriod>
      <start value="2013-04-02T09:30:10+01:00"/>
      <end value="2013-04-05T09:30:10+01:00"/>
    </effectivePeriod>
    <issued value="2013-04-03T15:30:10+01:00"/>
    <performer>
      <reference value="Practitioner/f005"/>
      <display value="A. Langeveld"/>
    </performer>
    <dataAbsentReason>
      <coding>
        <system value="http://snomed.info/sct"/>
        <code value="125154007"/>
        <display value="Specimen unsatisfactory for evaluation"/>
      </coding>
    </dataAbsentReason>
    <comment value="Tube broken in transit and sample leaked"/>
    <referenceRange>
      <low>
        <value value="3.1"/>
        <unit value="mmol/l"/>
        <system value="http://unitsofmeasure.org"/>
        <code value="mmol/L"/>
      </low>
      <high>
        <value value="6.2"/>
        <unit value="mmol/l"/>
        <system value="http://unitsofmeasure.org"/>
        <code value="mmol/L"/>
      </high>
    </referenceRange>
  </Observation>
 </resource>
</entry>
<entry>
 <fullUrl value="http://hl7.org/fhir/Contract/pcd-example-notLabs"/>
 <resource>
  <Contract xmlns="http://hl7.org/fhir">
    <id value="pcd-example-notLabs"/>
```

```
   <text>
     <status value="generated"/>
     <div xmlns="http://www.w3.org/1999/xhtml">The following scenario is based on existing jurisdictional policy and
 are realized in existing systems in Canada.
  The default policy is one of implied consent for the provision of care, so these scenarios all deal with
withdrawal or withholding consent for that purpose.
  In other jurisdictions, where an express consent model is used (Opt-In), these would examples would contain the
phrase "consent to" rather
  than "withhold" or "withdraw" consent for.
     <p>
  specific to use-case 1. Withhold or withdraw consent for disclosure of records related to specific domain (e.g.
DI, LAB, etc.)
  </p><p>
     Patient "P. van de Heuvel" wishes to withhold disclosure of all of her lab orders and results to any provider.
    </p>
    </div>
   </text>

   <issued value="2014-08-17"/>
  <!-- not bound by a timeframe - Contract.applies -->

   <subject>
     <reference value="Patient/f001"/>
     <display value="P. van de Heuvel"/>
   </subject>

   <!-- I assume the example given is Canada Infoway wide???  AND I assume it is desired to state that in the
Contract.authority element -->
   <authority>
       <reference value="Organization/Infoway"/>
    <display value="Canada Infoway"/>
    </authority>

    <!-- and/or would this [also] go into Contract.domain as a Jurisdiction?? (see the example on UK Pharamacy
Juristiction) -->
   <domain>
       <reference value="Location/Infoway"/>
    <display value="Canada Infoway"/>
    </domain>

   <type>
      <coding>
     <system value="http://loinc.org"/>
     <code value="57016-8"/>
   </coding>
   </type>

   <!-- made up code-system to represent the set of privacy consent sub-types known and published by Canada Infoway
-->
   <subType>
      <coding>
     <system value="http://www.infoway-inforoute.ca.org/Consent-subtype-codes"/>
     <code value="Opt-In"/>
     <display value="Default Authorization with exceptions."/>
   </coding>
   </subType>

  <!-- I can easily say to exclude a type of FHIR Resource, but lab-orders is not a direct Resource type. Might we
put in FHIR based query parameters? -->
    <term>
      <type>
        <coding>
       <!-- made up code system -->
       <system value="http://example.org/fhir/consent-term-type-codes"/>
       <code value="withhold-object-type"/>
     </coding>
   </type>
   <subType>
        <coding>
       <system value="http://hl7.org/fhir/resource-types"/>
       <code value="ProcedureRequest"/>
     </coding>
   </subType>
      <!-- by not specifying term.actor I assume this means everyone? -->
   <text value="Withhold orders from any provider."/>
    </term>
    <term>
      <type>
        <coding>
       <!-- made up code system -->
       <system value="http://example.org/fhir/consent-term-type-codes"/>
       <code value="withhold-object-type"/>
     </coding>
   </type>
   <subType>
        <coding>
       <system value="http://hl7.org/fhir/resource-types"/>
       <code value="DiagnosticReport"/>
     </coding>
```

```
    </subType>
       <!-- by not specifying term.actor I assume this means everyone? -->
     <text value="Withhold order results from any provider."/>
      </term>
      <!-- the terms of the consent in friendly consumer speak -->
      <friendly>
        <contentAttachment>
       <title value="The terms of the consent in friendly consumer speak."/>
       <!-- likely use url pointer to common text -->
     </contentAttachment>
      </friendly>
      <!-- the legal terms of the consent in lawyer speak -->
      <legal>
        <contentAttachment>
       <title value="The terms of the consent in lawyer speak."/>
       <!-- likely use url pointer to common text -->
       <!-- I think some think that this should be the pointer to the law by which this consent is derived under. Such
   as http://lois-laws.justice.gc.ca/eng/acts/P-21/index.html -->
     </contentAttachment>
      </legal>
   </Contract>
  </resource>
</entry>

<entry>
  <fullUrl value="http://hl7.org/fhir/Observation/ekg"/>
  <resource>
   <!-- Sampled ECG potentials (Leads I, II, II). There are 2 chunks (numbered as 0 and 1) for each lead. The voltage
  of each lead is near zero. --><Observation xmlns="http://hl7.org/fhir">
    <id value="ekg"/>
    <status value="final"/>
    <category>
      <coding>
        <system value="http://hl7.org/fhir/observation-category"/>
        <code value="procedure"/>
        <display value="Procedure"/>
      </coding>
    </category>
    <code>
      <coding>
        <system value="urn:oid:2.16.840.1.113883.6.24"/>
        <code value="131328"/>
        <display value="MDC_ECG_ELEC_POTL"/>
      </coding>
    </code>
    <subject>
      <reference value="Patient/f001"/>
      <display value="P. van de Heuvel"/>
    </subject>
    <effectiveDateTime value="2015-02-19T09:30:35+01:00"/>
    <performer>
      <reference value="Practitioner/f005"/>
      <display value="A. Langeveld"/>
    </performer>
    <device>
      <display value="12 lead EKG Device Metric"/>
    </device>
    <!-- Lead I data chunk 0 -->
    <component>
      <code>
        <coding>
          <system value="urn:oid:2.16.840.1.113883.6.24"/>
          <code value="131329"/>
          <display value="MDC_ECG_ELEC_POTL_I"/>
        </coding>
      </code>
      <valueSampledData>
        <!-- Zero potential -->
        <origin>
          <value value="2048"/>
        </origin>
        <!-- period in ms = 1000/frequency -->
        <period value="10"/>
        <!--  factor.value -->
        <factor value="1.612"/>
        <!--  voltage-min -->
        <lowerLimit value="-3300"/>
        <!--  voltage-max -->
        <upperLimit value="3300"/>
        <dimensions value="1"/>
        <data value="2041 2043 2037 2047 2060 2062 2051 2023 2014 2027 2034 2033 2040 2047 2047 2053 2058 2064 2059
  2063 2061 2052 2053 2038 1966 1885 1884 2009 2129 2166 2137 2102 2086 2077 2067 2067 2060 2059 2062 2062 2060 2057
  2045 2047 2057 2054 2042 2029 2027 2018 2007 1995 2001 2012 2024 2039 2068 2092 2111 2125 2131 2148 2137 2138 2128
  2128 2115 2099 2097 2096 2101 2101 2091 2073 2076 2077 2084 2081 2088 2092 2070 2069 2074 2077 2075 2068 2064 2060
  2062 2074 2075 2074 2075 2063 2058 2058 2064 2064 2070 2074 2067 2060 2062 2063 2061 2059 2048 2052 2049 2048 2051
  2059 2059 2066 2077 2073"/>
      </valueSampledData>
    </component>
    <!-- Lead II data chunk 0 -->
    <component>
```

```
      <code>
        <coding>
          <system value="urn:oid:2.16.840.1.113883.6.24"/>
          <code value="131330"/>
          <display value="MDC_ECG_ELEC_POTL_II"/>
        </coding>
      </code>
      <valueSampledData>
        <!-- Zero potential -->
        <origin>
          <value value="2048"/>
        </origin>
        <!-- period in ms = 1000/frequency -->
        <period value="10"/>
        <!--  factor.value -->
        <factor value="1.612"/>
        <!--  voltage-min -->
        <lowerLimit value="-3300"/>
        <!--  voltage-max -->
        <upperLimit value="3300"/>
        <dimensions value="1"/>
        <data value="2041 2043 2037 2047 2060 2062 2051 2023 2014 2027 2034 2033 2040 2047 2047 2053 2058 2064 2059
  2063 2061 2052 2053 2038 1966 1885 1884 2009 2129 2166 2137 2102 2086 2077 2067 2067 2060 2059 2062 2062 2060 2057
  2045 2047 2057 2054 2042 2029 2027 2018 2007 1995 2001 2012 2024 2039 2068 2092 2111 2125 2131 2148 2137 2138 2128
  2128 2115 2099 2097 2096 2101 2101 2091 2073 2076 2077 2084 2081 2088 2092 2070 2069 2074 2077 2075 2068 2064 2060
  2062 2074 2075 2074 2075 2063 2058 2058 2064 2064 2070 2074 2067 2060 2062 2063 2061 2059 2048 2052 2049 2048 2051
  2059 2059 2066 2077 2073"/>
      </valueSampledData>
    </component>
    <!-- Lead III data chunk 0 -->
    <component>
      <code>
        <coding>
          <system value="urn:oid:2.16.840.1.113883.6.24"/>
          <code value="131389"/>
          <display value="MDC_ECG_ELEC_POTL_III"/>
        </coding>
      </code>
      <valueSampledData>
        <!-- Zero potential -->
        <origin>
          <value value="2048"/>
        </origin>
        <!-- period in ms = 1000/frequency -->
        <period value="10"/>
        <!--  factor.value -->
        <factor value="1.612"/>
        <!--  voltage-min -->
        <lowerLimit value="-3300"/>
        <!--  voltage-max -->
        <upperLimit value="3300"/>
        <dimensions value="1"/>
        <data value="2041 2043 2037 2047 2060 2062 2051 2023 2014 2027 2034 2033 2040 2047 2047 2053 2058 2064 2059
  2063 2061 2052 2053 2038 1966 1885 1884 2009 2129 2166 2137 2102 2086 2077 2067 2067 2060 2059 2062 2062 2060 2057
  2045 2047 2057 2054 2042 2029 2027 2018 2007 1995 2001 2012 2024 2039 2068 2092 2111 2125 2131 2148 2137 2138 2128
  2128 2115 2099 2097 2096 2101 2101 2091 2073 2076 2077 2084 2081 2088 2092 2070 2069 2074 2077 2075 2068 2064 2060
  2062 2074 2075 2074 2075 2063 2058 2058 2064 2064 2070 2074 2067 2060 2062 2063 2061 2059 2048 2052 2049 2048 2051
  2059 2059 2066 2077 2073"/>
      </valueSampledData>
    </component>
   </Observation>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/CarePlan/f003"/>
  <resource>
   <CarePlan xmlns="http://hl7.org/fhir">
     <id value="f003"/>
       <contained>
       <CareTeam>
           <id value="careteam"/>
           <participant>
            <member>
            <reference value="Practitioner/f001"/>
            <display value="E.M. van den broek"/>
            </member>
      </participant>
       </CareTeam>
     </contained>
     <contained>
       <Goal>
         <id value="goal"/>
         <status value="achieved"/>
         <description>
           <text value="Retropharyngeal abscess removal"/>
         </description>
         <subject>
           <reference value="Patient/f001"/>
           <display value="P. van de Heuvel"/>
```

```
        </subject>
        <note>
          <text value="goal accomplished without complications"/>
        </note>
      </Goal>
    </contained>
    <identifier>
      <use value="official"/>
      <!-- urn:oid:2.16.840.1.113883.4.642.1.36 -->
      <system value="http://www.bmc.nl/zorgportal/identifiers/careplans"/>
      <value value="CP3953"/>
    </identifier>
    <status value="completed"/>
    <intent value="plan"/>
    <subject>
      <reference value="Patient/f001"/>
      <display value="P. van de Heuvel"/>
    </subject>
    <period>
      <start value="2013-03-08T09:00:10+01:00"/>
      <end value="2013-03-08T09:30:10+01:00"/>
    </period>
    <careTeam>
      <reference value="#careteam"/>
    </careTeam>
    <addresses>
      <reference value="Condition/f201"/>
      <!--TODO Correcte referentie-->
      <display value="?????"/>
    </addresses>
    <goal>
      <reference value="#goal"/>
    </goal>
    <activity>
      <detail>
        <category><coding><system value="http://hl7.org/fhir/care-plan-activity-category"/><code
  value="procedure"/></coding></category>
        <!-- urn:oid:2.16.840.1.113883.4.642.1.39 -->
        <code>
          <coding>
            <system value="http://snomed.info/sct"/>
            <code value="172960003"/>
            <display value="Incision of retropharyngeal abscess"/>
          </coding>
        </code>
        <status value="completed"/>
        <prohibited value="true"/>
        <scheduledString value="2011-06-27T09:30:10+01:00"/>
        <performer>
          <reference value="Practitioner/f001"/>
          <display value="E.M. van den broek"/>
        </performer>
      </detail>
    </activity>
   </CarePlan>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/Condition/f003"/>
  <resource>
   <Condition xmlns="http://hl7.org/fhir">
    <id value="f003"/>
    <clinicalStatus value="active"/>
    <verificationStatus value="confirmed"/>
    <category>
      <coding>
        <system value="http://snomed.info/sct"/>
        <code value="439401001"/>
        <display value="diagnosis"/>
      </coding>
    </category>
    <severity>
      <coding>
        <system value="http://snomed.info/sct"/>
        <code value="371923003"/>
        <display value="Mild to moderate"/>
      </coding>
    </severity>
    <code>
      <coding>
        <system value="http://snomed.info/sct"/>
        <code value="18099001"/>
        <display value="Retropharyngeal abscess"/>
      </coding>
    </code>

    <bodySite>
      <coding>
        <system value="http://snomed.info/sct"/>
        <code value="280193007"/>
```

```
        <display value="Entire retropharyngeal area"/>
      </coding>
    </bodySite>
    <subject>
      <reference value="Patient/f001"/>
      <display value="P. van de Heuvel"/>
    </subject>
    <context>
      <reference value="Encounter/f003"/>
    </context>
    <onsetDateTime value="2012-02-27"/>
    <assertedDate value="2012-02-20"/>
    <asserter>
      <reference value="Patient/f001"/>
      <display value="P. van de Heuvel"/>
    </asserter>
    <evidence>
      <code>
        <coding>
          <system value="http://snomed.info/sct"/>
          <code value="169068008"/>
          <display value="CT of neck"/>
        </coding>
      </code>
    </evidence>
   </Condition>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/Observation/f004"/>
  <resource>
   <Observation xmlns="http://hl7.org/fhir">
    <id value="f004"/>
   <!--   urn:oid:2.16.840.1.113883.4.642.1.7   -->
    <identifier>
      <use value="official"/>
      <system value="http://www.bmc.nl/zorgportal/identifiers/observations"/>
      <value value="6326"/>
    </identifier>
    <status value="final"/>
    <code>
      <coding>
        <system value="http://loinc.org"/>
        <code value="789-8"/>
        <display value="Erythrocytes [#/volume] in Blood by Automated count"/>
      </coding>
    </code>
    <subject>
      <reference value="Patient/f001"/>
      <display value="P. van de Heuvel"/>
    </subject>
    <effectivePeriod>
      <start value="2013-04-02T10:30:10+01:00"/>
      <end value="2013-04-05T10:30:10+01:00"/>
    </effectivePeriod>
    <issued value="2013-04-03T15:30:10+01:00"/>
    <performer>
      <reference value="Practitioner/f005"/>
      <display value="A. Langeveld"/>
    </performer>
    <valueQuantity>
      <value value="4.12"/>
    <!--  Actually, 4.12E+12  -->
      <unit value="10^12/L"/>
      <system value="http://unitsofmeasure.org"/>
      <code value="10*12/L"/>
    </valueQuantity>
    <interpretation>
      <coding>
        <system value="http://hl7.org/fhir/v2/0078"/>
        <code value="L"/>
        <display value="Low"/>
      </coding>
    </interpretation>
    <!-- this example demonstrates using the reference range text element for stuff that is not easily structured
  !-->
    <referenceRange>
      <text value=" 12-14 y Male: 4.4 - 5.2  x  10^12/L ; 12-14 y Female: 4.2 - 4.8  x  10^12/L ; 15-17 y Male: 4.6 -
   5.4  x  10^12/L ; 15-17 y Female: 4.2 - 4.8  x  10^12/L ; 18-64 y Male: 4.6 - 5.4  x  10^12/L ; 18-64 y Female: 4.0
   - 4.8  x  10^12/L ; 65-74 y Male: 4.3 - 5.3  x  10^12/L ; 65-74 y Female: 4.1 - 4.9  x  10^12/L       "/>
    </referenceRange>
   </Observation>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/Consent/consent-example-notOrg"/>
  <resource>
   <Consent xmlns="http://hl7.org/fhir">
    <id value="consent-example-notOrg"/>
```

```
    <text>
      <status value="generated"/>
      <div xmlns="http://www.w3.org/1999/xhtml">The following scenario is based on existing jurisdictional policy and
   are realized in existing systems in Canada.
    The default policy is one of implied consent for the provision of care, so these scenarios all deal with
  withdrawal or withholding consent for that purpose.
    In other jurisdictions, where an express consent model is used (Opt-In), these would examples would contain the
  phrase "consent to" rather
    than "withhold" or "withdraw" consent for.
      <p>
    specific to use-case 3) Withhold or withdraw consent for disclosure to a specific provider organization
    </p><p>
       Patient "P. van de Heuvel" ex-spouse is a Physician that is employed by Good Health Clinic.  P. decides
      that he would like to withdraw/withhold consent to disclose any instance of his health information
      to Good Health Clinic.
      </p>
      </div>
     </text>

     <status value="active"/>

     <patient>
       <reference value="Patient/f001"/>
       <display value="P. van de Heuvel"/>
     </patient>

     <dateTime value="2015-11-18"/>
      <!-- not bound by a timeframe - Consent.period -->

     <!-- I assume the example given is Canada Infoway wide???  AND I assume it is desired to state that in the
  Consent.authority element -->
     <organization>
       <reference value="Organization/Infoway"/>
      <display value="Canada Infoway"/>
     </organization>

     <!-- the text terms of the consent in lawyer speak -->
     <sourceAttachment>
       <title value="The terms of the consent in lawyer speak."/>
       <!-- likely use url pointer to common text -->
    </sourceAttachment>

     <!-- this is opt-in - e.g. everything approved unless otherwise stated -->
     <policyRule value="http://hl7.org/fhir/ConsentPolicy/opt-in"/>

     <except>
       <!-- no read or write access for this org -->
       <type value="deny"/>
       <actor>
         <role>
           <coding>
             <system value="http://hl7.org/fhir/v3/ParticipationType"/>
             <code value="PRCP"/> <!-- PRCP: recipient of the information -->
           </coding>
         </role>
         <reference>
           <reference value="Organization/2.16.840.1.113883.19.5"/>
           <display value="Good Health Clinic"/>
         </reference>
      </actor>
       <action>
         <coding>
           <system value="http://hl7.org/fhir/consentaction"/>
           <code value="access"/>
         </coding>
       </action>
       <action>
         <coding>
           <system value="http://hl7.org/fhir/consentaction"/>
           <code value="correct"/>
         </coding>
       </action>
     </except>

    </Consent>
   </resource>
 </entry>
 <entry>
   <fullUrl value="http://hl7.org/fhir/Task/example3"/>
   <resource>
    <Task xmlns="http://hl7.org/fhir">
      <id value="example3"/>

     <status value="draft"/>
     <intent value="order"/>
     <code>
       <text value="Refill Request"/>
     </code>
     <focus>
       <reference value="MedicationRequest/medrx002"/>
```

```
     </focus>
     <for>
       <reference value="Patient/f001"/>
     </for>
     <authoredOn value="2016-03-10T22:39:32-04:00"/>
     <lastModified value="2016-03-10T22:39:32-04:00"/>
     <requester>
       <agent>
         <reference value="Patient/example"/>
       </agent>
     </requester>
     <owner>
       <reference value="Practitioner/example"/>
     </owner>
   </Task>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/DiagnosticReport/f001"/>
  <resource>
   <DiagnosticReport xmlns="http://hl7.org/fhir">
    <!--  ISO 8601  -->
    <!--  OID: 2.16.840.1.113883.4.642.1.7  -->
    <id value="f001"/>
    <contained>
      <ProcedureRequest>
        <id value="req"/>
        <extension url="http://example.org/bodysitecode">
          <valueCodeableConcept>
            <coding>
              <system value="http://snomed.info/sct"/>
              <code value="14975008"/>
              <display value="Forearm structure"/>
            </coding>
          </valueCodeableConcept>
        </extension>
        <identifier>
          <system value="http://www.bmc.nl/zorgportal/identifiers/labresults"/>
          <value value="L2381"/>
        </identifier>
        <status value="active"/>
        <intent value="original-order"/>
        <code>
          <coding>
            <system value="http://loinc.org"/>
            <code value="58410-2"/>
            <display value="Complete blood count (hemogram) panel - Blood by Automated count"/>
          </coding>
        </code>
        <subject>
          <reference value="Patient/f001"/>
          <display value="P. van den Heuvel"/>
        </subject>
        <context>
          <reference value="Encounter/f001"/>
        </context>
        <requester>
          <agent>
            <reference value="Practitioner/f001"/>
            <display value="E.van den Broek"/>
          </agent>
        </requester>
        <note>
          <text value="patient almost fainted during procedure"/>
        </note>
      </ProcedureRequest>
    </contained>
    <identifier>
      <use value="official"/>
      <system value="http://www.bmc.nl/zorgportal/identifiers/reports"/>
      <value value="nr1239044"/>
    </identifier>
    <basedOn>
      <reference value="#req"/>
    </basedOn>
    <status value="final"/>
    <category>
      <coding>
        <system value="http://snomed.info/sct"/>
        <code value="252275004"/>
        <display value="Haematology test"/>
      </coding>
      <coding>
        <system value="http://hl7.org/fhir/v2/0074"/>
        <code value="HM"/>
      </coding>
    </category>
    <code>
      <coding>
```

```
        <system value="http://loinc.org"/>
        <code value="58410-2"/>
        <display value="Complete blood count (hemogram) panel - Blood by Automated count"/>
      </coding>
    </code>
    <subject>
      <reference value="Patient/f001"/>
      <display value="P. van den Heuvel"/>
    </subject>
    <issued value="2013-05-15T19:32:52+01:00"/>
    <performer>
      <actor>
        <reference value="Organization/f001"/>
        <display value="Burgers University Medical Centre"/>
      </actor>
    </performer>
    <result>
      <reference value="Observation/f001"/>
    </result>
    <result>
      <reference value="Observation/f002"/>
    </result>
    <result>
      <reference value="Observation/f003"/>
    </result>
    <result>
      <reference value="Observation/f004"/>
    </result>
    <result>
      <reference value="Observation/f005"/>
    </result>
    <conclusion value="Core lab"/>
   </DiagnosticReport>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/Condition/f002"/>
  <resource>
   <Condition xmlns="http://hl7.org/fhir">
     <id value="f002"/>
     <clinicalStatus value="active"/>
     <verificationStatus value="confirmed"/>
     <category>
       <coding>
         <system value="http://snomed.info/sct"/>
         <code value="439401001"/>
         <display value="diagnosis"/>
       </coding>
     </category>
     <severity>
       <coding>
         <system value="http://snomed.info/sct"/>
         <code value="24484000"/>
         <display value="Severe"/>
       </coding>
     </severity>
     <code>
       <coding>
         <system value="http://snomed.info/sct"/>
         <code value="254637007"/>
         <display value="NSCLC - Non-small cell lung cancer"/>
       </coding>
     </code>
     <bodySite>
       <coding>
         <system value="http://snomed.info/sct"/>
         <code value="51185008"/>
         <display value="Thorax"/>
       </coding>
     </bodySite>
     <subject>
       <reference value="Patient/f001"/>
       <display value="P. van de Heuvel"/>
     </subject>
     <context>
       <reference value="Encounter/f002"/>
     </context>
     <onsetDateTime value="2011-05-05"/>
     <assertedDate value="2012-06-03"/>
     <asserter>
       <reference value="Patient/f001"/>
       <display value="P. van de Heuvel"/>
     </asserter>
     <stage>
       <summary>
         <coding>
           <system value="http://snomed.info/sct"/>
           <code value="258219007"/>
           <display value="stage II"/>
```

```
        </coding>
      </summary>
    </stage>
    <evidence>
      <code>
        <coding>
          <system value="http://snomed.info/sct"/>
          <code value="169069000"/>
          <display value="CT of thorax"/>
        </coding>
      </code>
    </evidence>
   </Condition>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/Consent/consent-example-Emergency"/>
  <resource>
   <Consent xmlns="http://hl7.org/fhir">
    <id value="consent-example-Emergency"/>

    <text>
      <status value="generated"/>
      <div xmlns="http://www.w3.org/1999/xhtml">
    <p>
    Withhold Authorization for Treatment except for Emegency Treatment
    </p><p>
       Patient "P. van de Heuvel" wishes to have no data at the Good Health Psychiatric Hospital
      available except for Emergency treatment use.
      An overall Deny consent Directive, with an exception "Permit" of purposeOfUse "ETREAT"
      at "Good Health" hospital, and exception "Deny" all other purposeOfUse from good health hospital.
      </p>
      </div>
     </text>

     <status value="active"/>

     <patient>
       <reference value="Patient/f001"/>
       <display value="P. van de Heuvel"/>
     </patient>

     <dateTime value="2015-11-18"/>
      <!-- not bound by a timeframe - Consent.period -->

     <!-- I assume the example given is Canada Infoway wide???  AND I assume it is desired to state that in the
  Consent.authority element -->
     <organization>
       <reference value="Organization/Infoway"/>
      <display value="Canada Infoway"/>
     </organization>

     <!-- the text terms of the consent in lawyer speak -->
     <sourceAttachment>
       <title value="The terms of the consent in lawyer speak."/>
       <!-- likely use url pointer to common text -->
    </sourceAttachment>

     <!--
       this is opt-out - e.g. nothing approved unless otherwise stated.
     -->
     <policyRule value="http://hl7.org/fhir/ConsentPolicy/opt-out"/>

     <except>
       <type value="permit"/>

       <actor>
         <role>
           <coding>
             <system value="http://hl7.org/fhir/v3/ParticipationType"/>
             <code value="CST"/>
           </coding>
         </role>
         <reference>
            <reference value="Organization/2.16.840.1.113883.19.6"/>
            <display value="Good Health Psychiatric Hospital"/>
         </reference>
       </actor>

       <purpose>
         <system value="http://hl7.org/fhir/v3/ActReason"/>
         <code value="ETREAT"/>
       </purpose>
     </except>
     <!-- this except overrides the other one -->
     <except>
       <type value="deny"/>

       <actor>
```

```
        <role>
          <coding>
            <system value="http://hl7.org/fhir/v3/ParticipationType"/>
            <code value="CST"/>
          </coding>
        </role>
        <reference>
           <reference value="Organization/2.16.840.1.113883.19.6"/>
           <display value="Good Health Psychiatric Hospital"/>
        </reference>
      </actor>
    </except>
   </Consent>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/RelatedPerson/f001"/>
  <resource>
   <RelatedPerson xmlns="http://hl7.org/fhir">
    <id value="f001"/>
     <text>

       <status value="generated"/>
       <div xmlns="http://www.w3.org/1999/xhtml">
        Sarah Abels
       </div>
     </text>

     <identifier>
       <use value="official"/>
       <type><text value="BSN"/></type>
       <system value="urn:oid:2.16.840.1.113883.2.4.6.3"/>
       <!-- Dutch BSN identification system -->
     </identifier>

     <patient>
       <reference value="Patient/f001"/>
     </patient>

     <relationship>
       <coding>
         <system value="http://hl7.org/fhir/v3/RoleCode"/>
         <code value="SIGOTHR"/>
       </coding>
     </relationship>

     <name>
       <use value="usual"/>
       <family value="Abels"/>
       <given value="Sarah"/>
     </name>
     <telecom>
       <system value="phone"/>
       <value value="0690383372"/>
       <use value="mobile"/>
     </telecom>
     <telecom>
       <system value="email"/>
       <value value="s.abels@kpn.nl"/>
       <use value="home"/>
     </telecom>
     <gender value="female"/>

   </RelatedPerson>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/Contract/pcd-example-notOrg"/>
  <resource>
   <Contract xmlns="http://hl7.org/fhir">
     <id value="pcd-example-notOrg"/>

     <text>
       <status value="generated"/>
       <div xmlns="http://www.w3.org/1999/xhtml">The following scenario is based on existing jurisdictional policy and
   are realized in existing systems in Canada.
    The default policy is one of implied consent for the provision of care, so these scenarios all deal with
  withdrawal or withholding consent for that purpose.
    In other jurisdictions, where an express consent model is used (Opt-In), these would examples would contain the
  phrase "consent to" rather
    than "withhold" or "withdraw" consent for.
       <p>
    specific to use-case 3) Withhold or withdraw consent for disclosure to a specific provider organization
    </p><p>
       Patient "P. van de Heuvel" ex-spouse is a Physician that is employed by Good Health Clinic.  P. decides
      that he would like to withdraw/withhold consent to disclose any instance of his health information
      to Good Health Clinic.
      </p>
      </div>
```

```
    </text>

    <issued value="2015-11-18"/>
   <!-- not bound by a timeframe - Contract.applies -->

    <subject>
      <reference value="Patient/f001"/>
      <display value="P. van de Heuvel"/>
    </subject>

    <!-- I assume the example given is Canada Infoway wide???  AND I assume it is desired to state that in the
  Contract.authority element -->
    <authority>
        <reference value="Organization/Infoway"/>
     <display value="Canada Infoway"/>
     </authority>

     <!-- and/or would this [also] go into Contract.domain as a Jurisdiction?? (see the example on UK Pharamacy
  Juristiction) -->
    <domain>
        <reference value="Location/Infoway"/>
     <display value="Canada Infoway"/>
     </domain>

    <type>
       <coding>
      <system value="http://loinc.org"/>
      <code value="57016-8"/>
    </coding>
    </type>

    <!-- made up code-system to represent the set of privacy consent sub-types known and published by Canada Infoway
  -->
    <subType>
       <coding>
      <system value="http://www.infoway-inforoute.ca.org/Consent-subtype-codes"/>
      <code value="Opt-In"/>
      <display value="Default Authorization with exceptions."/>
    </coding>
    </subType>

    <term>
       <type>
         <coding>
        <!-- made up code system -->
        <system value="http://example.org/fhir/consent-term-type-codes"/>
        <code value="withhold-from"/>
      <display value="Withhold all data from specified actor entity."/>
      </coding>
    </type>
    <topic>
          <reference value="Organization/2.16.840.1.113883.19.5"/>
             <display value="Good Health Clinic"/>
    </topic>
       <!-- by not specifying term.actor I assume this means everyone? -->
    <text value="Withhold this order and any results or related objects from any provider."/>
     </term>
     <!-- the terms of the consent in friendly consumer speak -->
     <friendly>
       <contentAttachment>
      <title value="The terms of the consent in friendly consumer speak."/>
      <!-- likely use url pointer to common text -->
    </contentAttachment>
     </friendly>
     <!-- the legal terms of the consent in lawyer speak -->
     <legal>
       <contentAttachment>
      <title value="The terms of the consent in lawyer speak."/>
      <!-- likely use url pointer to common text -->
      <!-- I think some think that this should be the pointer to the law by which this consent is derived under. Such
   as http://lois-laws.justice.gc.ca/eng/acts/P-21/index.html -->
    </contentAttachment>
     </legal>
   </Contract>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/Observation/f005"/>
  <resource>
   <Observation xmlns="http://hl7.org/fhir">
     <id value="f005"/>
   <!--  urn:oid:2.16.840.1.113883.4.642.1.7  -->
     <identifier>
       <use value="official"/>
       <system value="http://www.bmc.nl/zorgportal/identifiers/observations"/>
       <value value="6327"/>
     </identifier>
     <status value="final"/>
     <code>
```

```
      <coding>
        <system value="http://loinc.org"/>
        <code value="718-7"/>
        <display value="Hemoglobin [Mass/volume] in Blood"/>
      </coding>
    </code>
    <subject>
      <reference value="Patient/f001"/>
      <display value="P. van de Heuvel"/>
    </subject>
    <effectivePeriod>
      <start value="2013-04-05T10:30:10+01:00"/>
      <end value="2013-04-05T10:30:10+01:00"/>
    </effectivePeriod>
    <issued value="2013-04-05T15:30:10+01:00"/>
    <performer>
      <reference value="Practitioner/f005"/>
      <display value="A. Langeveld"/>
    </performer>
    <valueQuantity>
      <value value="7.2"/>
      <unit value="g/dl"/>
      <system value="http://unitsofmeasure.org"/>
      <code value="g/dL"/>
    </valueQuantity>
    <interpretation>
      <coding>
        <system value="http://hl7.org/fhir/v2/0078"/>
        <code value="L"/>
        <display value="Low"/>
      </coding>
    </interpretation>
    <referenceRange>
      <low>
        <value value="7.5"/>
        <unit value="g/dl"/>
        <system value="http://unitsofmeasure.org"/>
        <code value="g/dL"/>
      </low>
      <high>
        <value value="10"/>
        <unit value="g/dl"/>
        <system value="http://unitsofmeasure.org"/>
        <code value="g/dL"/>
      </high>
    </referenceRange>
   </Observation>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/CarePlan/f002"/>
  <resource>
   <CarePlan xmlns="http://hl7.org/fhir">
     <id value="f002"/>
     <contained>
       <CareTeam>
           <id value="careteam"/>
           <participant>
            <member>
            <reference value="Practitioner/f003"/>
            <display value="M.I.M. Versteegh"/>
            </member>
      </participant>
       </CareTeam>
     </contained>
     <contained>
       <Goal>
         <id value="goal"/>
         <status value="achieved"/>
         <description>
           <text value="succesful surgery on lung of patient"/>
         </description>
         <subject>
    <reference value="Patient/f001"/>
    <display value="P. van de Heuvel"/>
         </subject>
         <note>
           <text value="goal accomplished with minor complications"/>
         </note>
       </Goal>
     </contained>
     <identifier>
       <use value="official"/>
       <!-- urn:oid:2.16.840.1.113883.4.642.1.36 -->
       <system value="http://www.bmc.nl/zorgportal/identifiers/careplans"/>
       <value value="CP2934"/>
     </identifier>
     <status value="completed"/>
     <intent value="plan"/>
```

```
      <subject>
        <reference value="Patient/f001"/>
        <display value="P. van de Heuvel"/>
      </subject>
      <period>
        <start value="2011-07-06"/>
        <end value="2013-07-07"/>
      </period>
      <careTeam>
        <reference value="#careteam"/>
      </careTeam>
      <addresses>
        <reference value="Condition/f201"/>
        <!--TODO Correcte referentie-->
        <display value="?????"/>
      </addresses>
      <goal>
        <reference value="#goal"/>
      </goal>
      <activity>
        <detail>
          <category><coding><system value="http://hl7.org/fhir/care-plan-activity-category"/><code
  value="procedure"/></coding></category>
          <!-- urn:oid:2.16.840.1.113883.4.642.1.39 -->
          <code>
            <coding>
              <system value="http://snomed.info/sct"/>
              <code value="359615001"/>
              <display value="Partial lobectomy of lung"/>
            </coding>
          </code>
          <status value="completed"/>
          <prohibited value="true"/>
          <scheduledString value="2011-07-07T09:30:10+01:00"/>
          <performer>
            <reference value="Practitioner/f003"/>
            <display value="M.I.M. Versteegh"/>
          </performer>
        </detail>
      </activity>
    </CarePlan>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/Consent/consent-example-Out"/>
  <resource>
   <Consent xmlns="http://hl7.org/fhir">
     <id value="consent-example-Out"/>

     <text>
       <status value="generated"/>
       <div xmlns="http://www.w3.org/1999/xhtml">The following scenario is based on existing jurisdictional policy and
    are realized in existing systems in Canada.
     The default policy is one of implied consent for the provision of care, so these scenarios all deal with
  withdrawal or withholding consent for that purpose.
     In other jurisdictions, where an express consent model is used (Opt-In), these would examples would contain the
  phrase "consent to" rather
     than "withhold" or "withdraw" consent for.
       <p>
     Specific to use-case 0) Withhold Authorization of sharing data for Treatment
     </p><p>
        Patient "P. van de Heuvel" wishes to have no data at the Good Health Psychiatric Hospital
       available for normal treatment use.
       </p>
       </div>
     </text>

      <status value="active"/>

     <patient>
       <reference value="Patient/f001"/>
       <display value="P. van de Heuvel"/>
     </patient>

     <dateTime value="2015-11-18"/>
      <!-- not bound by a timeframe - Consent.period -->

       <actor>
         <role>
           <coding>
             <system value="http://hl7.org/fhir/v3/ParticipationType"/>
             <code value="CST"/>
           </coding>
         </role>
         <reference>
            <reference value="Organization/2.16.840.1.113883.19.6"/>
            <display value="Good Health Psychiatric Hospital"/>
         </reference>
       </actor>
```

```
      <!-- I assume the example given is Canada Infoway wide???  AND I assume it is desired to state that in the
    Consent.authority element -->
       <organization>
         <reference value="Organization/Infoway"/>
        <display value="Canada Infoway"/>
       </organization>

       <!-- the text terms of the consent in lawyer speak -->
       <sourceAttachment>
         <title value="The terms of the consent in lawyer speak."/>
         <!-- likely use url pointer to common text -->
      </sourceAttachment>

       <!-- this is opt-out - e.g. everything approved unless otherwise stated -->
       <policyRule value="http://hl7.org/fhir/ConsentPolicy/opt-out"/>

     </Consent>
    </resource>
  </entry>
  <entry>
    <fullUrl value="http://hl7.org/fhir/Consent/consent-example-basic"/>
    <resource>
     <Consent xmlns="http://hl7.org/fhir">
       <id value="consent-example-basic"/>

       <text>
         <status value="generated"/>
         <div xmlns="http://www.w3.org/1999/xhtml">
         <p>
      Authorize Normal access for Treatment
      </p><p>
         Patient "P. van de Heuvel" wishes to have all of the PHI collected at the Good Health Psychiatric Hospital
        available for normal treatment use.
        </p>
        </div>
       </text>

       <status value="active"/>

       <patient>
         <reference value="Patient/f001"/>
         <display value="P. van de Heuvel"/>
       </patient>

      <!-- bound by a timeframe between 1964 and 2016 - consent.applies -->
         <period>
          <start value="1964-01-01"/>
          <end value="2016-01-01"/>
         </period>

       <dateTime value="2016-05-11"/>

       <organization>
          <reference value="Organization/Infoway"/>
        <display value="Canada Infoway"/>
       </organization>

       <!-- the terms of the consent as signed by the patient in ink -->
       <sourceAttachment>
        <title value="The terms of the consent in lawyer speak."/>
        <!-- likely use url pointer to common text -->
      </sourceAttachment>

       <!-- this is plain out opt-in with no exceptions -->
       <policyRule value="http://goodhealth.org/consent/policy/opt-in"/>

     </Consent>
    </resource>
  </entry>
  <entry>
    <fullUrl value="http://hl7.org/fhir/Contract/pcd-example-notThem"/>
    <resource>
     <Contract xmlns="http://hl7.org/fhir">
       <id value="pcd-example-notThem"/>

       <text>
         <status value="generated"/>
         <div xmlns="http://www.w3.org/1999/xhtml">The following scenario is based on existing
           jurisdictional policy and are realized in existing systems in Canada. The default policy is
           one of implied consent for the provision of care, so these scenarios all deal with withdrawal
           or withholding consent for that purpose. In other jurisdictions, where an express consent
           model is used (Opt-In), these would examples would contain the phrase "consent to" rather than
           "withhold" or "withdraw" consent for. <p> specific to use-case 4) Withhold or withdraw consent
             for disclosure to a specific provider agent (an individual within an organization) </p><p>
             Patient "P. van de Heuvel" ex-spouse, Bill T Lookafter is a Nurse that is employed by Good
             Health Hospital. P. realizes that she may receive treatment at GHH, however she does not
             want her ex-spouse to have any access to her IIHI. She indicates that she would like to
             withdraw/withhold consent to disclose any instance of her health information to her
             ex-spouse as a result of his employment at Good Health Clinic. </p>
```

```
    </div>
  </text>

  <issued value="2015-11-18"/>
  <!-- not bound by a timeframe - Contract.applies -->

  <subject>
    <reference value="Patient/f001"/>
    <display value="P. van de Heuvel"/>
  </subject>

  <!-- I assume the example given is Canada Infoway wide???  AND I assume it is desired to state that in the
Contract.authority element -->
  <authority>
    <reference value="Organization/2"/>
    <display value="Canada Infoway"/>
  </authority>

  <!-- and/or would this [also] go into Contract.domain as a Jurisdiction?? (see the example on UK Pharamacy
Juristiction) -->
  <domain>
    <reference value="Location/ukp"/>
    <display value="Canada Infoway"/>
  </domain>

  <type>
    <coding>
      <system value="http://loinc.org"/>
      <code value="57016-8"/>
    </coding>
  </type>

  <!-- made up code-system to represent the set of privacy consent sub-types known and published by Canada Infoway
-->
  <subType>
    <coding>
      <system value="http://www.infoway-inforoute.ca.org/Consent-subtype-codes"/>
      <code value="Opt-In"/>
      <display value="Default Authorization with exceptions."/>
    </coding>
  </subType>

  <agent>
    <actor>
      <reference value="Patient/f001"/>
    </actor>
  </agent>

  <signer>
    <type>
      <system value="http://www.hl7.org/fhir/contractsignertypecodes"/>
      <code value="COVPTY"/>
    </type>
    <party>
      <reference value="Patient/f001"/>
    </party>
    <signature>
      <type>
        <system value="urn:iso-astm:E1762-95:2013"/>
        <code value="1.2.840.10065.1.12.1.1"/>
      </type>
      <when value="2013-06-08T10:57:34-07:00"/>
      <whoReference>
        <reference value="Patient/f001"/>
      </whoReference>
    </signature>
  </signer>

  <term>
    <type>
      <coding>
        <!-- made up code system -->
        <system value="http://example.org/fhir/consent-term-type-codes"/>
        <code value="withhold-from"/>
        <display value="Withhold all data from specified actor entity."/>
      </coding>
    </type>
    <topic>
      <reference value="Practitioner/f204"/>
      <display value="Fictive Nurse"/>
    </topic>

    <text value="Withhold this order and any results or related objects from specified nurse provider."/>
  </term>
  <!-- the terms of the consent in friendly consumer speak -->
  <friendly>
    <contentAttachment>
      <title value="The terms of the consent in friendly consumer speak."/>
      <!-- likely use url pointer to common text -->
```

```
      </contentAttachment>
    </friendly>
    <!-- the legal terms of the consent in lawyer speak -->
    <legal>
      <contentAttachment>
        <title value="The terms of the consent in lawyer speak."/>
        <!-- likely use url pointer to common text -->
        <!-- I think some think that this should be the pointer to the law by which this consent is derived under.
  Such as http://lois-laws.justice.gc.ca/eng/acts/P-21/index.html -->
      </contentAttachment>
    </legal>
   </Contract>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/Encounter/f002"/>
  <resource>
   <Encounter xmlns="http://hl7.org/fhir">
    <id value="f002"/>
      <identifier>
          <use value="official"/>
          <system value="http://www.bmc.nl/zorgportal/identifiers/encounters"/>
          <value value="v3251"/>
      </identifier>
      <status value="finished"/>
      <class>
          <system value="http://hl7.org/fhir/v3/ActCode"/>
          <code value="AMB"/> <!-- outpatient -->
          <display value="ambulatory"/>
      </class>
      <type>
          <coding>
              <system value="http://snomed.info/sct"/>
              <code value="270427003"/>
              <display value="Patient-initiated encounter"/>
          </coding>
      </type>
      <priority>
          <coding>
              <system value="http://snomed.info/sct"/>
              <code value="103391001"/>
              <display value="Urgent"/>
          </coding>
      </priority>
    <subject>
          <reference value="Patient/f001"/>
          <display value="P. van de Heuvel"/>
      </subject>
      <participant>
          <individual>
              <reference value="Practitioner/f003"/>
              <display value="M.I.M Versteegh"/>
          </individual>
      </participant>
      <length>
          <value value="140"/>
          <unit value="min"/>
          <system value="http://unitsofmeasure.org"/>
          <code value="min"/>
      </length>
      <reason>
          <coding>
              <system value="http://snomed.info/sct"/>
              <code value="34068001"/>
              <display value="Partial lobectomy of lung"/>
          </coding>
      </reason>
      <hospitalization>
          <preAdmissionIdentifier>
              <use value="official"/>
              <system value="http://www.bmc.nl/zorgportal/identifiers/pre-admissions"/>
              <value value="98682"/>
          </preAdmissionIdentifier>
      <!--    <preAdmissionTest>
              <coding>
                  <system value="http://snomed.info/sct"/>
                  <code value="399208008"/>
                  <display value="Chest X-ray"/>
              </coding>
              <coding>
                  <system value="http://snomed.info/sct"/>
                  <code value="396550006"/>
                  <display value="Blood test"/>
              </coding>
          </preAdmissionTest> -->
          <admitSource>
              <coding>
                  <system value="http://snomed.info/sct"/>
                  <code value="305997006"/>
```

```
                    <display value="Referral by radiologist"/>
                </coding>
            </admitSource>
            <dischargeDisposition>
                <coding>
                    <system value="http://snomed.info/sct"/>
                    <code value="306689006"/>
                    <display value="Discharge to home"/>
                </coding>
            </dischargeDisposition>
        </hospitalization>
        <serviceProvider>
            <reference value="Organization/f001"/>
            <display value="BMC"/>
        </serviceProvider>
    </Encounter>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/Consent/consent-example-notThem"/>
  <resource>
   <Consent xmlns="http://hl7.org/fhir">
     <id value="consent-example-notThem"/>

     <text>
       <status value="generated"/>
       <div xmlns="http://www.w3.org/1999/xhtml">The following scenario is based on existing jurisdictional policy and
   are realized in existing systems in Canada.
    The default policy is one of implied consent for the provision of care, so these scenarios all deal with
  withdrawal or withholding consent for that purpose.
    In other jurisdictions, where an express consent model is used (Opt-In), these would examples would contain the
  phrase "consent to" rather
    than "withhold" or "withdraw" consent for.
       <p>
    specific to use-case 4) Withhold or withdraw consent for disclosure to a specific provider agent (an individual
  within an organization)
    </p><p>
       Patient "P. van de Heuvel" ex-spouse, Bill T Lookafter is a Nurse that is employed by Good Health Hospital.
      P. realizes that she may receive treatment at GHH, however she does not want her ex-spouse to have any
      access to her IIHI.  She indicates that she would like to withdraw/withhold consent to disclose any instance
      of her health information to her ex-spouse as a result of his employment at Good Health Clinic.
      </p>
      </div>
     </text>

     <status value="active"/>

     <patient>
       <reference value="Patient/f001"/>
       <display value="P. van de Heuvel"/>
     </patient>

     <dateTime value="2015-11-18"/>
      <!-- not bound by a timeframe - Consent.period -->

     <!-- I assume the example given is Canada Infoway wide???  AND I assume it is desired to state that in the
  Consent.authority element -->
     <organization>
       <reference value="Organization/Infoway"/>
      <display value="Canada Infoway"/>
     </organization>

     <!-- the text terms of the consent in lawyer speak -->
     <sourceAttachment>
       <title value="The terms of the consent in lawyer speak."/>
       <!-- likely use url pointer to common text -->
    </sourceAttachment>

     <!-- this is opt-in - e.g. everything approved unless otherwise stated -->
     <policyRule value="http://hl7.org/fhir/ConsentPolicy/opt-in"/>

     <except>
       <!-- no read or write access for this individual -->
       <type value="deny"/>
       <actor>
         <role>
           <coding>
             <system value="http://hl7.org/fhir/v3/ParticipationType"/>
             <code value="PRCP"/> <!-- PRCP: recipient of the information -->
           </coding>
         </role>
         <reference>
           <reference value="Practitioner/f204"/>
           <display value="Fictive Nurse"/>
         </reference>
      </actor>
       <action>
         <coding>
           <system value="http://hl7.org/fhir/consentaction"/>
```

```
          <code value="access"/>
        </coding>
      </action>
      <action>
        <coding>
          <system value="http://hl7.org/fhir/consentaction"/>
          <code value="correct"/>
        </coding>
      </action>
    </except>
  </Consent>
 </resource>
</entry>
<entry>
 <fullUrl value="http://hl7.org/fhir/Procedure/f001"/>
 <resource>
  <Procedure xmlns="http://hl7.org/fhir">
    <id value="f001"/>
      <status value="completed"/>
      <code>
          <coding>
              <system value="http://snomed.info/sct"/>
              <code value="34068001"/>
              <display value="Heart valve replacement"/>
          </coding>
      </code>
   <subject>
          <reference value="Patient/f001"/>
          <display value="P. van de Heuvel"/>
      </subject>
   <context>
          <reference value="Encounter/f001"/>
      </context>
      <performedPeriod>
          <start value="2011-06-26"/>
          <end value="2011-06-27"/>
      </performedPeriod>
   <performer>
          <role>
              <coding>
                  <system value="urn:oid:2.16.840.1.113883.2.4.15.111"/> <!-- "AssignedRoleType" coding system -->
                  <code value="01.000"/>
                  <display value="Arts"/>
              </coding>
              <text value="Care role"/>
          </role>
     <actor>
              <reference value="Practitioner/f002"/>
              <display value="P. Voigt"/>
          </actor>
      </performer>
   <reasonCode>
        <text value="Heart valve disorder"/>
      </reasonCode>
      <bodySite>
              <coding>
                  <system value="http://snomed.info/sct"/>
                  <code value="17401000"/>
                  <display value="Heart valve structure"/>
              </coding>
      </bodySite>
      <outcome>
        <text value="improved blood circulation"/>
      </outcome>
      <report>
          <reference value="DiagnosticReport/f001"/>
          <display value="Lab results blood test"/>
      </report>
      <followUp>
        <text value="described in care plan"/>
      </followUp>
  </Procedure>
 </resource>
</entry>
<entry>
 <fullUrl value="http://hl7.org/fhir/Consent/consent-example-grantor"/>
 <resource>
  <Consent xmlns="http://hl7.org/fhir">
    <id value="consent-example-grantor"/>

    <text>
      <status value="generated"/>
      <div xmlns="http://www.w3.org/1999/xhtml">The following scenario is based on a question asked on John Moehrke's
  blog in the
   <a href="http://healthcaresecprivacy.blogspot.com/2016/04/consent-given-to-authorized.html">Consent given to
 authorized representative</a>
      <p>
  This is a case where a Patient is giving a limited (read-only) access to an individual.
   </p><p>
```

```
      Patient "P. van de Heuvel" wishes to provide XXX with read-only access to the data at Good Health Psychiatric
  Hospital
      available for patient requested access use.
      </p>
      </div>
     </text>

     <status value="active"/>

     <patient>
       <reference value="Patient/f001"/>
       <display value="P. van de Heuvel"/>
     </patient>

     <dateTime value="2015-11-18"/>
      <!-- not bound by a timeframe - Consent.period -->

     <!-- I assume the example given is Canada Infoway wide???  AND I assume it is desired to state that in the
  Consent.authority element -->
     <organization>
       <reference value="Organization/Infoway"/>
      <display value="Canada Infoway"/>
     </organization>

     <!-- the text terms of the consent in lawyer speak -->
     <sourceAttachment>
       <title value="The terms of the consent in lawyer speak."/>
       <!-- likely use url pointer to common text -->
    </sourceAttachment>

     <!-- this is opt-out - e.g. nothing approved unless otherwise stated -->
     <policyRule value="http://hl7.org/fhir/ConsentPolicy/opt-out"/>

     <except>
       <type value="permit"/>
       <actor>
         <role>
           <coding>
             <system value="http://hl7.org/fhir/v3/ParticipationType"/>
             <code value="CST"/>
           </coding>
         </role>
         <reference>
            <reference value="Organization/2.16.840.1.113883.19.6"/>
            <display value="Good Health Psychiatric Hospital"/>
         </reference>
       </actor>

       <actor>
         <role>
           <coding>
             <system value="http://hl7.org/fhir/v3/ParticipationType"/>
             <code value="PRCP"/>
           </coding>
         </role>
         <reference>
            <reference value="Patient/example"/>
            <display value="Good Health Clinic"/>
         </reference>
       </actor>
       <action>
         <coding>
           <system value="http://hl7.org/fhir/consentaction"/>
           <code value="access"/>
         </coding>
       </action>
     </except>
   </Consent>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/Contract/pcd-example-notThis"/>
  <resource>
   <Contract xmlns="http://hl7.org/fhir">
     <id value="pcd-example-notThis"/>

     <text>
       <status value="generated"/>
       <div xmlns="http://www.w3.org/1999/xhtml">The following scenario is based on existing jurisdictional policy and
   are realized in existing systems in Canada.
    The default policy is one of implied consent for the provision of care, so these scenarios all deal with
  withdrawal or withholding consent for that purpose.
    In other jurisdictions, where an express consent model is used (Opt-In), these would examples would contain the
  phrase "consent to" rather
    than "withhold" or "withdraw" consent for.
       <p>
    specific to use-case 2. Withhold or withdraw consent for disclosure of a specific record (e.g. Lab Order/Result)
    </p><p>
       Patient "P. van de Heuvel" Primary Care Provider, Dr. Philip Primary, has ordered a set of lab test which Adam
```

```
      wishes to keep as private as possible.  At the sample collection facility, he indicates that he would
      like withhold consent to disclose the order and all results associated with that specific order from
      all other providers
      </p>
      </div>
     </text>

     <issued value="2015-11-18"/>
    <!-- not bound by a timeframe - Contract.applies -->

     <subject>
       <reference value="Patient/f001"/>
       <display value="P. van de Heuvel"/>
     </subject>

     <!-- I assume the example given is Canada Infoway wide???  AND I assume it is desired to state that in the
  Contract.authority element -->
     <authority>
         <reference value="Organization/Infoway"/>
      <display value="Canada Infoway"/>
      </authority>

      <!-- and/or would this [also] go into Contract.domain as a Jurisdiction?? (see the example on UK Pharamacy
  Juristiction) -->
     <domain>
         <reference value="Location/Infoway"/>
      <display value="Canada Infoway"/>
      </domain>

     <type>
        <coding>
       <system value="http://loinc.org"/>
       <code value="57016-8"/>
     </coding>
     </type>

     <!-- made up code-system to represent the set of privacy consent sub-types known and published by Canada Infoway
  -->
     <subType>
        <coding>
       <system value="http://www.infoway-inforoute.ca.org/Consent-subtype-codes"/>
       <code value="Opt-In"/>
       <display value="Default Authorization with exceptions."/>
     </coding>
     </subType>

    <!-- I can easily say to exclude a type of FHIR Resource, but lab-orders is not a direct Resource type. Might we
  put in FHIR based query parameters? -->
      <term>
        <type>
          <coding>
         <!-- made up code system -->
         <system value="http://example.org/fhir/consent-term-type-codes"/>
         <code value="withhold-identified-object-and-related"/>
       <display value="Withhold the identified object and any other resources that are related to this object."/>
       </coding>
     </type>
     <topic>
          <reference value="ProcedureRequest/example-lipid"/>
     </topic>
        <!-- by not specifying term.actor I assume this means everyone? -->
     <text value="Withhold this order and any results or related objects from any provider."/>
      </term>
      <!-- the terms of the consent in friendly consumer speak -->
      <friendly>
        <contentAttachment>
       <title value="The terms of the consent in friendly consumer speak."/>
       <!-- likely use url pointer to common text -->
     </contentAttachment>
      </friendly>
      <!-- the legal terms of the consent in lawyer speak -->
      <legal>
        <contentAttachment>
       <title value="The terms of the consent in lawyer speak."/>
       <!-- likely use url pointer to common text -->
       <!-- I think some think that this should be the pointer to the law by which this consent is derived under. Such
   as http://lois-laws.justice.gc.ca/eng/acts/P-21/index.html -->
     </contentAttachment>
      </legal>
   </Contract>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/Encounter/f003"/>
  <resource>
   <Encounter xmlns="http://hl7.org/fhir">
     <id value="f003"/>
       <identifier>
           <use value="official"/>
```

```
            <system value="http://www.bmc.nl/zorgportal/identifiers/encounters"/>
            <value value="v6751"/>
        </identifier>
        <status value="finished"/>
        <class>
            <system value="http://hl7.org/fhir/v3/ActCode"/>
            <code value="AMB"/> <!-- outpatient -->
            <display value="ambulatory"/>
        </class>
        <type>
            <coding>
                <system value="http://snomed.info/sct"/>
                <code value="270427003"/>
                <display value="Patient-initiated encounter"/>
            </coding>
        </type>
        <priority>
            <coding>
                <system value="http://snomed.info/sct"/>
                <code value="103391001"/>
                <display value="Non-urgent ear, nose and throat admission"/>
            </coding>
        </priority>
    <subject>
            <reference value="Patient/f001"/>
            <display value="P. van de Heuvel"/>
        </subject>
        <participant>
            <individual>
                <reference value="Practitioner/f001"/>
                <display value="E.M. van den Broek"/>
            </individual>
        </participant>
        <length>
            <value value="90"/>
            <unit value="min"/>
            <system value="http://unitsofmeasure.org"/>
            <code value="min"/>
        </length>
        <reason>
       <extension url="http://hl7.org/fhir/StructureDefinition/encounter-primaryDiagnosis">
         <valuePositiveInt value="1"/>
       </extension>
            <coding>
                <system value="http://snomed.info/sct"/>
                <code value="18099001"/>
                <display value="Retropharyngeal abscess"/>
            </coding>
        </reason>
        <hospitalization>
            <preAdmissionIdentifier>
                <use value="official"/>
                <system value="http://www.bmc.nl/zorgportal/identifiers/pre-admissions"/>
                <value value="93042"/>
            </preAdmissionIdentifier>
       <!--     <preAdmissionTest>
                <coding>
                    <system value="http://snomed.info/sct"/>
                    <code value="168719007"/>
                    <display value="Neck soft tissue X-ray"/>
                </coding>
                <coding>
                    <system value="http://snomed.info/sct"/>
                    <code value="396550006"/>
                    <display value="Blood test"/>
                </coding>
            </preAdmissionTest> -->
            <admitSource>
                <coding>
                    <system value="http://snomed.info/sct"/>
                    <code value="305956004"/>
                    <display value="Referral by physician"/>
                </coding>
            </admitSource>
            <dischargeDisposition>
                <coding>
                    <system value="http://snomed.info/sct"/>
                    <code value="306689006"/>
                    <display value="Discharge to home"/>
                </coding>
            </dischargeDisposition>
        </hospitalization>
        <serviceProvider>
            <reference value="Organization/f001"/>
        </serviceProvider>
    </Encounter>
   </resource>
</entry>
<entry>
```

```
  <fullUrl value="http://hl7.org/fhir/Procedure/f004"/>
  <resource>
   <Procedure xmlns="http://hl7.org/fhir">
     <id value="f004"/>
       <status value="completed"/>
       <code>
           <coding>
               <system value="http://snomed.info/sct"/>
               <code value="48387007"/>
               <display value="Tracheotomy"/>
           </coding>
       </code>
    <subject>
           <reference value="Patient/f001"/>
           <display value="P. van de Heuvel"/>
       </subject>
    <context>
           <reference value="Encounter/f003"/>
       </context>
       <performedPeriod>
           <start value="2013-03-22T09:30:10+01:00"/>
           <end value="2013-03-22T10:30:10+01:00"/>
       </performedPeriod>
       <performer>
           <role>
               <coding>
                   <system value="urn:oid:2.16.840.1.113883.2.4.15.111"/> <!-- "AssignedRoleType" coding system -->
                   <code value="01.000"/>
                   <display value="Arts"/>
               </coding>
               <text value="Care role"/>
           </role>
      <actor>
                <reference value="Practitioner/f005"/>
                <display value="A. Langeveld"/>
           </actor>
       </performer>
       <reasonCode>
         <text value="ensure breathing during surgery"/>
       </reasonCode>
       <bodySite>
           <coding>
               <system value="http://snomed.info/sct"/>
               <code value="83030008"/>
               <display value="Retropharyngeal area"/>
           </coding>
       </bodySite>
       <outcome>
         <text value="removal of the retropharyngeal abscess"/>
       </outcome>
       <report>
           <reference value="DiagnosticReport/f001"/>
           <display value="???????????"/>
       </report>
       <followUp>
         <text value="described in care plan"/>
       </followUp>
   </Procedure>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/CarePlan/f001"/>
  <resource>
   <CarePlan xmlns="http://hl7.org/fhir">
     <id value="f001"/>
     <contained>
       <CareTeam>
           <id value="careteam"/>
           <participant>
            <member>
            <reference value="Practitioner/f002"/>
            <display value="P. Voigt"/>
               </member>
      </participant>
       </CareTeam>
     </contained>
     <contained>
       <Goal>
         <id value="goal"/>
         <status value="achieved"/>
         <description>
           <text value="recovery surgery on heart of patient"/>
         </description>
         <subject>
    <reference value="Patient/f001"/>
    <display value="P. van de Heuvel"/>
         </subject>
         <note>
           <text value="goal accomplished without complications"/>
```

```
        </note>
      </Goal>
    </contained>
    <identifier>
      <use value="official"/>
      <!-- urn:oid:2.16.840.1.113883.4.642.1.36 -->
      <system value="http://www.bmc.nl/zorgportal/identifiers/careplans"/>
      <value value="CP2903"/>
    </identifier>
    <status value="completed"/>
    <intent value="plan"/>
    <subject>
      <reference value="Patient/f001"/>
      <display value="P. van de Heuvel"/>
    </subject>
    <period>
      <start value="2011-06-26"/>
      <end value="2011-06-27"/>
    </period>
    <careTeam>
      <reference value="#careteam"/>
    </careTeam>
    <addresses>
      <reference value="Condition/f201"/>
      <!--TODO Correcte referentie-->
      <display value="?????"/>
    </addresses>
    <goal>
      <reference value="#goal"/>
    </goal>
    <activity>
      <detail>
        <category><coding><system value="http://hl7.org/fhir/care-plan-activity-category"/><code
  value="procedure"/></coding></category>
        <!-- urn:oid:2.16.840.1.113883.4.642.1.39 -->
        <code>
          <coding>
            <system value="http://snomed.info/sct"/>
            <code value="64915003"/>
            <display value="Operation on heart"/>
          </coding>
        </code>
        <status value="completed"/>
        <prohibited value="true"/>
        <scheduledString value="2011-06-27T09:30:10+01:00"/>
        <performer>
          <reference value="Practitioner/f002"/>
          <display value="P. Voigt"/>
        </performer>
      </detail>
    </activity>
   </CarePlan>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/Observation/f003"/>
  <resource>
   <Observation xmlns="http://hl7.org/fhir">
    <id value="f003"/>
   <!--   urn:oid:2.16.840.1.113883.4.642.1.7   -->
   <!--   2.16.840.1.113883.4.642.1.118   -->
    <identifier>
      <use value="official"/>
      <system value="http://www.bmc.nl/zorgportal/identifiers/observations"/>
      <value value="6325"/>
    </identifier>
    <status value="final"/>
    <code>
      <coding>
        <system value="http://loinc.org"/>
        <code value="11557-6"/>
        <display value="Carbon dioxide in blood"/>
      </coding>
    </code>
    <subject>
      <reference value="Patient/f001"/>
      <display value="P. van de Heuvel"/>
    </subject>
    <effectivePeriod>
      <start value="2013-04-02T10:30:10+01:00"/>
      <end value="2013-04-05T10:30:10+01:00"/>
    </effectivePeriod>
    <issued value="2013-04-03T15:30:10+01:00"/>
    <performer>
      <reference value="Practitioner/f005"/>
      <display value="A. Langeveld"/>
    </performer>
    <valueQuantity>
      <value value="6.2"/>
```

```
      <unit value="kPa"/>
      <system value="http://unitsofmeasure.org"/>
      <code value="kPa"/>
    </valueQuantity>
    <interpretation>
      <coding>
        <system value="http://hl7.org/fhir/v2/0078"/>
        <code value="H"/>
        <display value="High"/>
      </coding>
    </interpretation>
    <referenceRange>
      <low>
        <value value="4.8"/>
        <unit value="kPa"/>
        <system value="http://unitsofmeasure.org"/>
        <code value="kPa"/>
      </low>
      <high>
        <value value="6.0"/>
        <unit value="kPa"/>
        <system value="http://unitsofmeasure.org"/>
        <code value="kPa"/>
      </high>
    </referenceRange>
   </Observation>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/Observation/f001"/>
  <resource>
   <Observation xmlns="http://hl7.org/fhir">
    <id value="f001"/>
   <!--  urn:oid:2.16.840.1.113883.4.642.1.7  -->
   <!--  2.16.840.1.113883.4.642.1.118  -->
    <identifier>
      <use value="official"/>
      <system value="http://www.bmc.nl/zorgportal/identifiers/observations"/>
      <value value="6323"/>
    </identifier>
    <status value="final"/>
    <code>
      <coding>
        <system value="http://loinc.org"/>
        <code value="15074-8"/>
        <display value="Glucose [Moles/volume] in Blood"/>
      </coding>
    </code>
    <subject>
      <reference value="Patient/f001"/>
      <display value="P. van de Heuvel"/>
    </subject>
    <effectivePeriod>
      <start value="2013-04-02T09:30:10+01:00"/>
    </effectivePeriod>
    <issued value="2013-04-03T15:30:10+01:00"/>
    <performer>
      <reference value="Practitioner/f005"/>
      <display value="A. Langeveld"/>
    </performer>
    <valueQuantity>
      <value value="6.3"/>
      <unit value="mmol/l"/>
      <system value="http://unitsofmeasure.org"/>
      <code value="mmol/L"/>
    </valueQuantity>
    <interpretation>
      <coding>
        <system value="http://hl7.org/fhir/v2/0078"/>
        <code value="H"/>
        <display value="High"/>
      </coding>
    </interpretation>
    <referenceRange>
      <low>
        <value value="3.1"/>
        <unit value="mmol/l"/>
        <system value="http://unitsofmeasure.org"/>
        <code value="mmol/L"/>
      </low>
      <high>
        <value value="6.2"/>
        <unit value="mmol/l"/>
        <system value="http://unitsofmeasure.org"/>
        <code value="mmol/L"/>
      </high>
    </referenceRange>
   </Observation>
  </resource>
```

```
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/Procedure/f003"/>
  <resource>
   <Procedure xmlns="http://hl7.org/fhir">
     <id value="f003"/>
       <status value="completed"/>
       <code>
           <coding>
               <system value="http://snomed.info/sct"/>
               <code value="172960003"/>
               <display value="Incision of retropharyngeal abscess"/>
           </coding>
       </code>
    <subject>
           <reference value="Patient/f001"/>
           <display value="P. van de Heuvel"/>
       </subject>
    <context>
           <reference value="Encounter/f003"/>
       </context>
    <performedPeriod>
           <start value="2013-03-24T09:30:10+01:00"/>
           <end value="2013-03-24T10:30:10+01:00"/>
       </performedPeriod>
    <performer>
           <role>
               <coding>
                   <system value="urn:oid:2.16.840.1.113883.2.4.15.111"/> <!-- "AssignedRoleType" coding system -->
                   <code value="01.000"/>
                   <display value="Arts"/>
               </coding>
               <text value="Care role"/>
           </role>
      <actor>
               <reference value="Practitioner/f001"/>
               <display value="E.M.J.M. van den broek"/>
           </actor>
       </performer>
       <reasonCode>
         <text value="abcess in retropharyngeal area"/>
       </reasonCode>
       <bodySite>

               <coding>
                   <system value="http://snomed.info/sct"/>
                   <code value="83030008"/>
                   <display value="Retropharyngeal area"/>
               </coding>

       </bodySite>
       <outcome>
         <text value="removal of the retropharyngeal abscess"/>
       </outcome>
       <report>
           <reference value="DiagnosticReport/f001"/>
           <display value="Lab results blood test"/>
       </report>
       <followUp>
         <text value="described in care plan"/>
       </followUp>
   </Procedure>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/Procedure/f002"/>
  <resource>
   <Procedure xmlns="http://hl7.org/fhir">
     <id value="f002"/>
       <status value="completed"/>
       <code>
           <coding>
               <system value="http://snomed.info/sct"/>
               <code value="359615001"/>
               <display value="Partial lobectomy of lung"/>
           </coding>
       </code>
    <subject>
           <reference value="Patient/f001"/>
           <display value="P. van de Heuvel"/>
       </subject>
    <context>
           <reference value="Encounter/f002"/>
       </context>
       <performedPeriod>
           <start value="2013-03-08T09:00:10+01:00"/>
           <end value="2013-03-08T09:30:10+01:00"/>
       </performedPeriod>
    <performer>
```

```
            <role>
                <coding>
                    <system value="urn:oid:2.16.840.1.113883.2.4.15.111"/> <!-- "AssignedRoleType" coding system -->
                    <code value="01.000"/>
                    <display value="Arts"/>
                </coding>
                <text value="Care role"/>
            </role>
        <actor>
                <reference value="Practitioner/f003"/>
                <display value="M.I.M. Versteegh"/>
            </actor>
         </performer>
      <reasonCode>
           <text value="Malignant tumor of lung"/>
        </reasonCode>
        <bodySite>

                <coding>
                    <system value="http://snomed.info/sct"/>
                    <code value="39607008"/>
                    <display value="Lung structure"/>
                </coding>

        </bodySite>
        <outcome>
          <text value="improved blood circulation"/>
        </outcome>
        <report>
            <reference value="DiagnosticReport/f001"/>
            <display value="Lab results blood test"/>
        </report>
        <followUp>
          <text value="described in care plan"/>
        </followUp>
     </Procedure>
    </resource>
  </entry>
  <entry>
    <fullUrl value="http://hl7.org/fhir/Consent/consent-example-notThis"/>
    <resource>
     <Consent xmlns="http://hl7.org/fhir">
       <id value="consent-example-notThis"/>

       <text>
         <status value="generated"/>
         <div xmlns="http://www.w3.org/1999/xhtml">The following scenario is based on existing jurisdictional policy and
     are realized in existing systems in Canada.
      The default policy is one of implied consent for the provision of care, so these scenarios all deal with
    withdrawal or withholding consent for that purpose.
      In other jurisdictions, where an express consent model is used (Opt-In), these would examples would contain the
    phrase "consent to" rather
      than "withhold" or "withdraw" consent for.
         <p>
      specific to use-case 2. Withhold or withdraw consent for disclosure of a specific record (e.g. Lab Order/Result)
      </p><p>
         Patient "P. van de Heuvel" Primary Care Provider, Dr. Philip Primary, has ordered a set of lab test which Adam
        wishes to keep as private as possible.  At the sample collection facility, he indicates that he would
        like withhold consent to disclose the order and all results associated with that specific order from
        all other providers
        </p>
        </div>
       </text>

       <status value="active"/>

       <patient>
         <reference value="Patient/f001"/>
         <display value="P. van de Heuvel"/>
       </patient>

       <dateTime value="2015-11-18"/>
        <!-- not bound by a timeframe - Consent.period -->

       <!-- I assume the example given is Canada Infoway wide???  AND I assume it is desired to state that in the
    Consent.authority element -->
       <organization>
         <reference value="Organization/Infoway"/>
        <display value="Canada Infoway"/>
       </organization>

       <!-- the text terms of the consent in lawyer speak -->
       <sourceAttachment>
         <title value="The terms of the consent in lawyer speak."/>
         <!-- likely use url pointer to common text -->
      </sourceAttachment>

       <!-- this is opt-in - e.g. everything approved unless otherwise stated -->
       <policyRule value="http://hl7.org/fhir/ConsentPolicy/opt-in"/>
```

```
      <except>
         <type value="deny"/>
         <data>
           <meaning value="related"/>
           <reference>
             <reference value="Task/f201"/>
           </reference>
         </data>
       </except>
    </Consent>
   </resource>
  </entry>
</Bundle>
```

8.1.19 Resource Patient - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/patient-version-maps.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.1.19 Resource Patient - Detailed Descriptions

Patient Administration Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Patient, Practitioner, RelatedPerson

R2 : R3 Converson maps for Patient.

Functional status for this map: 18 tests that all execute ok. All tests pass round-trip testing and 6 r3 resources are invalid (12 errors). (see documentation)

8.1.19.1 R2 to R3

```
map "http://hl7.org/fhir/StructureMap/Patient2to3" = "R2 to R3 Conversions for Patient"

uses "http://hl7.org/fhir/DSTU2/StructureDefinition/Patient" alias PatientR2 as source
uses "http://hl7.org/fhir/StructureDefinition/Patient" alias Patient as target

imports "http://hl7.org/fhir/StructureMap/*2to3"

group for type+types  Patient extends DomainResource
  input src : PatientR2 as source
  input tgt : Patient as target

  "Patient-identifier" : for src.identifier make tgt.identifier
  "Patient-active" : for src.active make tgt.active
  "Patient-name" : for src.name make tgt.name
  "Patient-telecom" : for src.telecom make tgt.telecom
  "Patient-gender" : for src.gender make tgt.gender
  "Patient-birthDate" : for src.birthDate make tgt.birthDate
  "Patient-deceased" : for src.deceased make tgt.deceased
  "Patient-address" : for src.address make tgt.address
  "Patient-maritalStatus" : for src.maritalStatus make tgt.maritalStatus
  "Patient-multipleBirth" : for src.multipleBirth make tgt.multipleBirth
  "Patient-photo" : for src.photo make tgt.photo
  "Patient-contact" : for src.contact as vs0 make tgt.contact as vt0 then patContact(vs0, vt0)
  "Patient-animal" : for src.animal as vs0 make tgt.animal as vt0 then patAnimal(vs0, vt0)
  "Patient-communication" : for src.communication as vs0 make tgt.communication as vt0 then patCommunication(vs0, vt0)
  "Patient-generalPractitioner" : for src.careProvider make tgt.generalPractitioner
  "Patient-managingOrganization" : for src.managingOrganization make tgt.managingOrganization
  "Patient-link" : for src.link as vs0 make tgt.link as vt0 then patLink(vs0, vt0)
endgroup

group patContact extends BackboneElement
  input src as source
  input tgt as target

  "Patient.contact-relationship" : for src.relationship make tgt.relationship
  "Patient.contact-name" : for src.name make tgt.name
  "Patient.contact-telecom" : for src.telecom make tgt.telecom
  "Patient.contact-address" : for src.address make tgt.address
  "Patient.contact-gender" : for src.gender make tgt.gender
  "Patient.contact-organization" : for src.organization make tgt.organization
  "Patient.contact-period" : for src.period make tgt.period
endgroup

group patCommunication extends BackboneElement
  input src as source
  input tgt as target

  "Patient.communication-language" : for src.language make tgt.language
  "Patient.communication-preferred" : for src.preferred make tgt.preferred
endgroup

group patAnimal extends BackboneElement
  input src as source
  input tgt as target

  "Patient.communication-species" : for src.species make tgt.species
  "Patient.communication-breed" : for src.breed make tgt.breed
```

```
  "Patient.communication-genderStatus" : for src.genderStatus make tgt.genderStatus
endgroup

group patLink extends BackboneElement
  input src as source
  input tgt as target

  "Patient.link-other" : for src.other make tgt.other
  "Patient.link-type" : for src.type make tgt.type
endgroup
```

8.1.19.2 R3 to R2

```
map "http://hl7.org/fhir/StructureMap/Patient3to2" = "R3 to R2 Conversion for Patient"

uses "http://hl7.org/fhir/StructureDefinition/Patient" alias Patient as source
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/Patient" alias PatientR2 as target

imports "http://hl7.org/fhir/StructureMap/*3to2"

group for type+types  Patient extends DomainResource
  input src : Patient as source
  input tgt : PatientR2 as target

  "Patient-identifier" : for src.identifier make tgt.identifier
  "Patient-active" : for src.active make tgt.active
  "Patient-name" : for src.name make tgt.name
  "Patient-telecom" : for src.telecom make tgt.telecom
  "Patient-gender" : for src.gender make tgt.gender
  "Patient-birthDate" : for src.birthDate make tgt.birthDate
  "Patient-deceased-boolean" : for src.deceased : boolean as vs make tgt.deceased = create("boolean") as vt then
    boolean(vs, vt)
  "Patient-deceased-dateTime" : for src.deceased : dateTime as vs make tgt.deceased = create("dateTime") as vt then
    dateTime(vs, vt)
  "Patient-address" : for src.address make tgt.address
  "Patient-maritalStatus" : for src.maritalStatus make tgt.maritalStatus
  "Patient-multipleBirth-boolean" : for src.multipleBirth : boolean as vs make tgt.multipleBirth = create("boolean") as
    vt then boolean(vs, vt)
  "Patient-multipleBirth-integer" : for src.multipleBirth : integer as vs make tgt.multipleBirth = create("integer") as
    vt then integer(vs, vt)
  "Patient-photo" : for src.photo make tgt.photo
  "Patient-contact" : for src.contact as vs0 make tgt.contact as vt0 then patContact(vs0, vt0)
  "Patient-animal" : for src.animal as vs0 make tgt.animal as vt0 then patAnimal(vs0, vt0)
  "Patient-communication" : for src.communication as vs0 make tgt.communication as vt0 then patCommunication(vs0, vt0)
  "Patient-careProvider" : for src.generalPractitioner make tgt.careProvider
  "Patient-managingOrganization" : for src.managingOrganization make tgt.managingOrganization
  "Patient-link" : for src.link as vs0 make tgt.link as vt0 then patLink(vs0, vt0)
endgroup

group patContact extends BackboneElement
  input src as source
  input tgt as target

  "Patient.contact-relationship" : for src.relationship make tgt.relationship
  "Patient.contact-name" : for src.name make tgt.name
  "Patient.contact-telecom" : for src.telecom make tgt.telecom
  "Patient.contact-address" : for src.address make tgt.address
  "Patient.contact-gender" : for src.gender make tgt.gender
  "Patient.contact-organization" : for src.organization make tgt.organization
  "Patient.contact-period" : for src.period make tgt.period
endgroup

group patCommunication extends BackboneElement
  input src as source
  input tgt as target

  "Patient.communication-language" : for src.language make tgt.language
  "Patient.communication-preferred" : for src.preferred make tgt.preferred
endgroup

group patAnimal extends BackboneElement
  input src as source
  input tgt as target

  "Patient.communication-species" : for src.species make tgt.species
  "Patient.communication-breed" : for src.breed make tgt.breed
  "Patient.communication-genderStatus" : for src.genderStatus make tgt.genderStatus
endgroup


group patLink extends BackboneElement
  input src as source
  input tgt as target

  "Patient.link-other" : for src.other make tgt.other
  "Patient.link-type" : for src.type make tgt.type
endgroup
```

8.1.19.3 R3 Validation Errors

example | ERROR @ Patient.contact.name.extension: The extension http://hl7.org/fhir/StructureDefinition/HumanName-familyRepeat is unknown, and not allowed here (src = InstanceValidator)

patient-example | ERROR @ Patient.extension[1]: The extension http://hl7.org/fhir/StructureDefinition/us-core-race is unknown, and not allowed here (src = InstanceValidator) ERROR @ Patient.extension[2]: The extension http://hl7.org/fhir/StructureDefinition/us-core-ethnicity is unknown, and not allowed here (src = InstanceValidator)

proband | ERROR @ Patient.extension: The extension http://hl7.org/fhir/StructureDefinition/us-core-race is unknown, and not allowed here (src = InstanceValidator)

us01 | ERROR @ Patient.extension[1]: The extension http://hl7.org/fhir/StructureDefinition/us-core-race is unknown, and not allowed here (src = InstanceValidator) ERROR @ Patient.extension[2]: The extension http://hl7.org/fhir/StructureDefinition/us-core-ethnicity is unknown, and not allowed here (src = InstanceValidator) ERROR @ Patient.telecom[2].extension: The extension http://hl7.org/fhir/StructureDefinition/us-core-direct is unknown, and not allowed here (src = InstanceValidator) ERROR @ Patient.address.extension: The extension http://hl7.org/fhir/StructureDefinition/us-core-county is unknown, and not allowed here (src = InstanceValidator)

patient-uslab-example1 | ERROR @ Patient.extension[1]: The extension http://hl7.org/fhir/StructureDefinition/us-core-race is unknown, and not allowed here (src = InstanceValidator) ERROR @ Patient.extension[2]: The extension http://hl7.org/fhir/StructureDefinition/us-core-ethnicity is unknown, and not allowed here (src = InstanceValidator)

patient-uslab-example2 | ERROR @ Patient.extension[1]: The extension http://hl7.org/fhir/StructureDefinition/us-core-race is unknown, and not allowed here (src = InstanceValidator) ERROR @ Patient.extension[2]: The extension http://hl7.org/fhir/StructureDefinition/us-core-ethnicity is unknown, and not allowed here (src = InstanceValidator)

8.2 Resource RelatedPerson - Content

URL: https://hl7.org/fhir/STU3/relatedperson.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

Patient Administration Work Group | Maturity Level: 2 | Trial Use | Compartments: Patient, RelatedPerson

Information about a person that is involved in the care for a patient, but who is not the target of healthcare, nor has a formal responsibility in the care process.

8.2.1 Scope and Usage

RelatedPersons typically have a personal or non-healthcare-specific professional relationship to the patient. A RelatedPerson resource is primarily used for attribution of information, since RelatedPersons are often a source of information about the patient. For keeping information about people for contact purposes for a patient, use a Patient's Contact element. Some individuals may serve as both a Patient's Contact and a Related Person.

Example RelatedPersons are:

* A patient's wife or husband

* A patient's relatives or friends

* A neighbour bringing a patient to the hospital

* The owner or trainer of a horse

* A patient's attorney or guardian

* A Guide Dog

8.2.2 Boundaries and Relationships

The primary distinction between a Practitioner and a RelatedPerson is based on whether:

* The person/animal operates on behalf of the care delivery organization over multiple patients (Practitioner) or,

* Where the person/animal is not associated with the organization, and instead is allocated tasks specifically for the RelatedPerson's Patient (RelatedPerson).

A standard extension animalSpecies can be used to indicate the species of a service animal.

This resource is referenced by Account, AdverseEvent, AllergyIntolerance, Appointment, AppointmentResponse, AuditEvent, Basic, CarePlan, CareTeam, ChargeItem, Claim, Communication, CommunicationRequest, Composition, Condition, Consent, Contract, Coverage, DeviceRequest, DeviceUseStatement, DocumentManifest, DocumentReference, Encounter, ExplanationOfBenefit, Goal, ImagingManifest, MedicationAdministration, MedicationDispense, MedicationRequest, MedicationStatement, Observation, Patient, Person, Procedure, ProcedureRequest, Provenance, QuestionnaireResponse, ReferralRequest, RequestGroup, Schedule, SupplyRequest and Task

8.2.3 Resource Content

* Structure

* UML

* XML

* JSON

* Turtle

* R2 Diff

* All

Structure

Name | Flags | Card. | Type | Description & Constraints
RelatedPerson | | | DomainResource | An person that is related to a patient, but who is not a direct target of care Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | ? | 0..* | Identifier | A human identifier for this person
active | ?!? | 0..1 | boolean | Whether this related person's record is in active use
patient | ? | 1..1 | Reference(Patient) | The patient this person is related to
relationship | ? | 0..1 | CodeableConcept | The nature of the relationship PatientRelationshipType (Preferred)
name | ? | 0..* | HumanName | A name associated with the person
telecom | ? | 0..* | ContactPoint | A contact detail for the person
gender | ? | 0..1 | code | male | female | other | unknown AdministrativeGender (Required)
birthDate | ? | 0..1 | date | The date on which the related person was born
address | ? | 0..* | Address | Address where the related person can be contacted or visited
photo | | 0..* | Attachment | Image of the person
period | | 0..1 | Period | Period of time that this relationship is considered valid
Documentation for this format

UML Diagram (Legend)

XML Template

```
<RelatedPerson xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier A human identifier for this person --></identifier>
 <active value="[boolean]"/><!-- 0..1 Whether this related person's record is in active use -->
 <patient><!-- 1..1 Reference(Patient) The patient this person is related to --></patient>
 <relationship><!-- 0..1 CodeableConcept The nature of the relationship --></relationship>
 <name><!-- 0..* HumanName A name associated with the person --></name>
 <telecom><!-- 0..* ContactPoint A contact detail for the person --></telecom>
 <gender value="[code]"/><!-- 0..1 male | female | other | unknown -->
 <birthDate value="[date]"/><!-- 0..1 The date on which the related person was born -->
 <address><!-- 0..* Address Address where the related person can be contacted or visited --></address>
 <photo><!-- 0..* Attachment Image of the person --></photo>
 <period><!-- 0..1 Period Period of time that this relationship is considered valid --></period>
</RelatedPerson>
```

JSON Template

```
{
  "resourceType" : "RelatedPerson",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // A human identifier for this person
  "active" : <boolean>, // Whether this related person's record is in active use
  "patient" : { Reference(Patient) }, // R!  The patient this person is related to
  "relationship" : { CodeableConcept }, // The nature of the relationship
  "name" : [{ HumanName }], // A name associated with the person
  "telecom" : [{ ContactPoint }], // A contact detail for the person
  "gender" : "<code>", // male | female | other | unknown
  "birthDate" : "<date>", // The date on which the related person was born
  "address" : [{ Address }], // Address where the related person can be contacted or visited
  "photo" : [{ Attachment }], // Image of the person
  "period" : { Period } // Period of time that this relationship is considered valid
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:RelatedPerson;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:RelatedPerson.identifier [ Identifier ], ... ; # 0..* A human identifier for this person
  fhir:RelatedPerson.active [ boolean ]; # 0..1 Whether this related person's record is in active use
  fhir:RelatedPerson.patient [ Reference(Patient) ]; # 1..1 The patient this person is related to
  fhir:RelatedPerson.relationship [ CodeableConcept ]; # 0..1 The nature of the relationship
  fhir:RelatedPerson.name [ HumanName ], ... ; # 0..* A name associated with the person
  fhir:RelatedPerson.telecom [ ContactPoint ], ... ; # 0..* A contact detail for the person
  fhir:RelatedPerson.gender [ code ]; # 0..1 male | female | other | unknown
  fhir:RelatedPerson.birthDate [ date ]; # 0..1 The date on which the related person was born
  fhir:RelatedPerson.address [ Address ], ... ; # 0..* Address where the related person can be contacted or visited
  fhir:RelatedPerson.photo [ Attachment ], ... ; # 0..* Image of the person
  fhir:RelatedPerson.period [ Period ]; # 0..1 Period of time that this relationship is considered valid
]
```

Changes since DSTU2

RelatedPerson
RelatedPerson.active | Added Element
RelatedPerson.relationship | Remove Binding http://hl7.org/fhir/ValueSet/relatedperson-relationshiptype (required)
RelatedPerson.name | Max Cardinality changed from 1 to *

See the Full Difference for further information

This analysis is available as XML or JSON.

See R2 <--> R3 Conversion Maps (status = 4 tests that all execute ok. All tests pass round-trip testing and 1 r3 resources are invalid (1 errors).).

Structure

Name | Flags | Card. | Type | Description & Constraints

RelatedPerson | | | DomainResource | An person that is related to a patient, but who is not a direct target of care Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension

identifier | ? | 0..* | Identifier | A human identifier for this person

active | ?!? | 0..1 | boolean | Whether this related person's record is in active use

patient | ? | 1..1 | Reference(Patient) | The patient this person is related to

relationship | ? | 0..1 | CodeableConcept | The nature of the relationship PatientRelationshipType (Preferred)

name | ? | 0..* | HumanName | A name associated with the person

telecom | ? | 0..* | ContactPoint | A contact detail for the person

gender | ? | 0..1 | code | male | female | other | unknown AdministrativeGender (Required)

birthDate | ? | 0..1 | date | The date on which the related person was born

address | ? | 0..* | Address | Address where the related person can be contacted or visited

photo | | 0..* | Attachment | Image of the person

period | | 0..1 | Period | Period of time that this relationship is considered valid

Documentation for this format

UML Diagram (Legend)

XML Template

```
<RelatedPerson xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier A human identifier for this person --></identifier>
 <active value="[boolean]"/><!-- 0..1 Whether this related person's record is in active use -->
 <patient><!-- 1..1 Reference(Patient) The patient this person is related to --></patient>
 <relationship><!-- 0..1 CodeableConcept The nature of the relationship --></relationship>
 <name><!-- 0..* HumanName A name associated with the person --></name>
 <telecom><!-- 0..* ContactPoint A contact detail for the person --></telecom>
 <gender value="[code]"/><!-- 0..1 male | female | other | unknown -->
 <birthDate value="[date]"/><!-- 0..1 The date on which the related person was born -->
 <address><!-- 0..* Address Address where the related person can be contacted or visited --></address>
 <photo><!-- 0..* Attachment Image of the person --></photo>
 <period><!-- 0..1 Period Period of time that this relationship is considered valid --></period>
</RelatedPerson>
```

JSON Template

```
{
  "resourceType" : "RelatedPerson",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // A human identifier for this person
  "active" : <boolean>, // Whether this related person's record is in active use
  "patient" : { Reference(Patient) }, // R!  The patient this person is related to
  "relationship" : { CodeableConcept }, // The nature of the relationship
  "name" : [{ HumanName }], // A name associated with the person
  "telecom" : [{ ContactPoint }], // A contact detail for the person
  "gender" : "<code>", // male | female | other | unknown
  "birthDate" : "<date>", // The date on which the related person was born
  "address" : [{ Address }], // Address where the related person can be contacted or visited
  "photo" : [{ Attachment }], // Image of the person
  "period" : { Period } // Period of time that this relationship is considered valid
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:RelatedPerson;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:RelatedPerson.identifier [ Identifier ], ... ; # 0..* A human identifier for this person
  fhir:RelatedPerson.active [ boolean ]; # 0..1 Whether this related person's record is in active use
  fhir:RelatedPerson.patient [ Reference(Patient) ]; # 1..1 The patient this person is related to
  fhir:RelatedPerson.relationship [ CodeableConcept ]; # 0..1 The nature of the relationship
  fhir:RelatedPerson.name [ HumanName ], ... ; # 0..* A name associated with the person
  fhir:RelatedPerson.telecom [ ContactPoint ], ... ; # 0..* A contact detail for the person
  fhir:RelatedPerson.gender [ code ]; # 0..1 male | female | other | unknown
  fhir:RelatedPerson.birthDate [ date ]; # 0..1 The date on which the related person was born
  fhir:RelatedPerson.address [ Address ], ... ; # 0..* Address where the related person can be contacted or visited
  fhir:RelatedPerson.photo [ Attachment ], ... ; # 0..* Image of the person
  fhir:RelatedPerson.period [ Period ]; # 0..1 Period of time that this relationship is considered valid
]
```

Changes since DSTU2

RelatedPerson

RelatedPerson.active | Added Element

RelatedPerson.relationship | Remove Binding http://hl7.org/fhir/ValueSet/relatedperson-relationshiptype (required)
RelatedPerson.name | Max Cardinality changed from 1 to *

See the Full Difference for further information

This analysis is available as XML or JSON.

See R2 <--> R3 Conversion Maps (status = 4 tests that all execute ok. All tests pass round-trip testing and 1 r3 resources are invalid (1 errors).).

Alternate definitions: Master Definition (XML, JSON), XML Schema/Schematron (for) + JSON Schema, ShEx (for Turtle)

8.2.3.1 Terminology Bindings

Path | Definition | Type | Reference
RelatedPerson.relationship | The nature of the relationship between a patient and the related person | Preferred | PatientRelationshipType
RelatedPerson.gender | The gender of a person used for administrative purposes. | Required | AdministrativeGender

8.2.4 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
active | token | Indicates if the related person record is active | RelatedPerson.active
address | string | A server defined search that may match any of the string fields in the Address, including line, city, state, country, postalCode, and/or text | RelatedPerson.address | 3 Resources
address-city | string | A city specified in an address | RelatedPerson.address.city | 3 Resources
address-country | string | A country specified in an address | RelatedPerson.address.country | 3 Resources
address-postalcode | string | A postal code specified in an address | RelatedPerson.address.postalCode | 3 Resources
address-state | string | A state specified in an address | RelatedPerson.address.state | 3 Resources
address-use | token | A use code specified in an address | RelatedPerson.address.use | 3 Resources
birthdate | date | The Related Person's date of birth | RelatedPerson.birthDate | 2 Resources
email | token | A value in an email contact | RelatedPerson.telecom.where(system='email') | 4 Resources
gender | token | Gender of the related person | RelatedPerson.gender | 3 Resources
identifier | token | An Identifier of the RelatedPerson | RelatedPerson.identifier
name | string | A server defined search that may match any of the string fields in the HumanName, including family, give, prefix, suffix, suffix, and/or text | RelatedPerson.name
patient | reference | The patient this related person is related to | RelatedPerson.patient (Patient)
phone | token | A value in a phone contact | RelatedPerson.telecom.where(system='phone') | 4 Resources
phonetic | string | A portion of name using some kind of phonetic matching algorithm | RelatedPerson.name | 3 Resources
telecom | token | The value in any kind of contact | RelatedPerson.telecom | 4 Resources

8.2.5 Resource RelatedPerson - Examples

URL: https://hl7.org/fhir/STU3/relatedperson-examples.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.2.5 Resource RelatedPerson - Examples

Patient Administration Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Patient, RelatedPerson

Example List:

RelatedPerson Benedicte du Marche | benedicte | XML | JSON | Turtle
RelatedPerson Peter Chalmers | peter | XML | JSON | Turtle
Sarah Abels | f001 | XML | JSON | Turtle
Ariadne Bor-Jansma | f002 | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

8.2.6 Resource RelatedPerson - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/relatedperson-definitions.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.2.6 Resource RelatedPerson - Detailed Descriptions

Patient Administration Work Group | Maturity Level: 2 | Trial Use | Compartments: Patient, RelatedPerson

Detailed Descriptions for the elements in the RelatedPerson resource.

RelatedPerson
Definition | Information about a person that is involved in the care for a patient, but who is not the target of healthcare, nor has a formal responsibility in the care process.
Control | 1..1
Requirements | Need to track persons related to the patient or the healthcare process.

RelatedPerson.identifier
Definition | Identifier for a person within a particular scope.
Note | This is a business identifer, not a resource identifier (see discussion)
Control | 0..*
Type | Identifier
Requirements | People are known by a variety of ids. Some institutions maintain several, and most collect identifiers for exchange with other organizations concerning the patient. Examples are national person identifier and local identifier.
Summary | true
RelatedPerson.active
Definition | Whether this related person record is in active use.
Control | 0..1
Type | boolean
Is Modifier | true
Default Value | true
Requirements | Need to be able to mark a related person record as not to be used, such as if it was created in error.
Summary | true
Comments | This element is labeled as a modifier because it may be used to mark that the resource was created in error.
RelatedPerson.patient
Definition | The patient this person is related to.
Control | 1..1
Type | Reference(Patient)
Requirements | We need to know which patient this RelatedPerson is related to.
Summary | true
RelatedPerson.relationship
Definition | The nature of the relationship between a patient and the related person.
Control | 0..1
Terminology Binding | PatientRelationshipType (Preferred)
Type | CodeableConcept
Requirements | We need to know the relationship with the patient since it influences the interpretation of the information attributed to this person.
Summary | true
RelatedPerson.name
Definition | A name associated with the person.
Control | 0..*
Type | HumanName
Requirements | Related persons need to be identified by name, but it is uncommon to need details about multiple other names for that person.
Summary | true
RelatedPerson.telecom
Definition | A contact detail for the person, e.g. a telephone number or an email address.
Control | 0..*
Type | ContactPoint
Requirements | People have (primary) ways to contact them in some way such as phone, email.
Summary | true
Comments | Person may have multiple ways to be contacted with different uses or applicable periods. May need to have options for contacting the person urgently, and also to help with identification.
RelatedPerson.gender
Definition | Administrative Gender - the gender that the person is considered to have for administration and record keeping purposes.
Control | 0..1
Terminology Binding | AdministrativeGender (Required)
Type | code
Requirements | Needed for identification of the person, in combination with (at least) name and birth date.
Summary | true
RelatedPerson.birthDate
Definition | The date on which the related person was born.
Control | 0..1
Type | date
Summary | true
RelatedPerson.address
Definition | Address where the related person can be contacted or visited.
Control | 0..*
Type | Address
Requirements | Need to keep track where the related person can be contacted per postal mail or visited.
Summary | true
RelatedPerson.photo
Definition | Image of the person.
Control | 0..*
Type | Attachment
Requirements | Many EHR systems have the capability to capture an image of persons. Fits with newer social media usage too.
RelatedPerson.period
Definition | The period of time that this relationship is considered to be valid. If there are no dates defined, then the interval is unknown.
Control | 0..1
Type | Period

8.2.7 Resource RelatedPerson - Mappings

URL: https://hl7.org/fhir/STU3/relatedperson-mappings.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.2.7 Resource RelatedPerson - Mappings

Patient Administration Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Patient, RelatedPerson

Mappings:

RIM Mapping

HL7 v2 Mapping

Mappings for the relatedperson resource.

8.2.7.1 Mappings for RIM Mapping (http://hl7.org/v3)

RelatedPerson | role
identifier | .id
active | .statusCode
patient | scoper[classCode=PSN|ANM and determinerCode='INSTANCE']/playedRole[classCode='PAT']/id
relationship | code
name | name
telecom | telecom
gender | administrativeGender
birthDate | player.birthTime
address | addr
photo | player[classCode='PSN' and determinerCode='INSTANCE']/desc
period | .effectiveTime

8.2.7.2 Mappings for HL7 v2 Mapping (http://hl7.org/v2)

RelatedPerson
identifier | NK1-33
active
patient | PID-3
relationship | NK1-3
name | NK1-2
telecom | NK1-5 / NK1-6 / NK1-40
gender | NK1-15
birthDate
address | NK1-4
photo | OBX-5 - needs a profile
period

8.2.8 Resource RelatedPerson - Design Notes

URL: https://hl7.org/fhir/STU3/relatedperson-explanations.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.2.8 Resource RelatedPerson - Design Notes

to do

8.2.9 Resource RelatedPerson - Extensions & Profiles

URL: https://hl7.org/fhir/STU3/relatedperson-profiles.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.2.9 Resource RelatedPerson - Extensions & Profiles

Patient Administration Work Group | Maturity Level: N/A | Ballot Status: Informative

This table lists profiles and extensions for the RelatedPerson resource. For background information, see Profiling Resources and Extensibility. Additional profiles and resources may be found in published Implementation Guides , or in the Conformance resource registry .

8.2.9.1 Profiles

No Profiles defined for this resource

8.2.9.2 Extensions

practitioner-animalSpecies | animalSpecies | for Practitioner HL7 Extensions

Extensions that reference this resource:

consent-Witness | Witness | for Consent HL7 Extensions
cqif-guidanceinitiatingPerson | initiatingPerson | for CQIF Guidance Extensions
cqif-guidancereceivingPerson | receivingPerson | for CQIF Guidance Extensions
goal-acceptance | acceptance | for FHIR Core Goal Profile
questionnaireresponse-author | author | for Core extensions for QuestionnaireResponse
task-candidateList | candidateList | for Task HL7 Extensions

8.2.9.3 Search Extensions

No Search Extensions defined for this resource

8.2.10 Resource RelatedPerson - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/relatedperson-version-maps.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.2.10 Resource RelatedPerson - Detailed Descriptions

Patient Administration Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Patient, RelatedPerson

R2 : R3 Converson maps for RelatedPerson.

Functional status for this map: 4 tests that all execute ok. All tests pass round-trip testing and 1 r3 resources are invalid (1 errors). (see documentation)

8.2.10.1 R2 to R3

```
map "http://hl7.org/fhir/StructureMap/RelatedPerson2to3" = "R2 to R3 Conversions for RelatedPerson"

uses "http://hl7.org/fhir/DSTU2/StructureDefinition/RelatedPerson" alias RelatedPersonR2 as source
uses "http://hl7.org/fhir/StructureDefinition/RelatedPerson" alias RelatedPerson as target

imports "http://hl7.org/fhir/StructureMap/*2to3"

group for type+types  RelatedPerson extends DomainResource
  input src : RelatedPersonR2 as source
  input tgt : RelatedPerson as target

  "RelatedPerson-identifier" : for src.identifier make tgt.identifier
  "RelatedPerson-patient" : for src.patient make tgt.patient
  "RelatedPerson-relationship" : for src.relationship make tgt.relationship
  "RelatedPerson-name" : for src.name make tgt.name
  "RelatedPerson-telecom" : for src.telecom make tgt.telecom
  "RelatedPerson-gender" : for src.gender make tgt.gender
  "RelatedPerson-birthDate" : for src.birthDate make tgt.birthDate
  "RelatedPerson-address" : for src.address make tgt.address
  "RelatedPerson-photo" : for src.photo make tgt.photo
  "RelatedPerson-period" : for src.period make tgt.period

endgroup
```

8.2.10.2 R3 to R2

```
map "http://hl7.org/fhir/StructureMap/RelatedPerson3to2" = "R3 to R2 Conversion for RelatedPerson"

uses "http://hl7.org/fhir/StructureDefinition/RelatedPerson" alias RelatedPerson as source
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/RelatedPerson" alias RelatedPersonR2 as target

imports "http://hl7.org/fhir/StructureMap/*3to2"

group for type+types  RelatedPerson extends DomainResource
  input src : RelatedPerson as source
  input tgt : RelatedPersonR2 as target

  "RelatedPerson-identifier" : for src.identifier make tgt.identifier
  "RelatedPerson-patient" : for src.patient make tgt.patient
  "RelatedPerson-relationship" : for src.relationship make tgt.relationship
  "RelatedPerson-name" : for src.name only_one make tgt.name
  "RelatedPerson-telecom" : for src.telecom make tgt.telecom
  "RelatedPerson-gender" : for src.gender make tgt.gender
  "RelatedPerson-birthDate" : for src.birthDate make tgt.birthDate
  "RelatedPerson-address" : for src.address make tgt.address
  "RelatedPerson-photo" : for src.photo make tgt.photo
  "RelatedPerson-period" : for src.period make tgt.period

endgroup
```

8.2.10.3 R3 Validation Errors

benedicte | ERROR @ RelatedPerson.name.extension: The extension http://hl7.org/fhir/StructureDefinition/HumanName-familyRepeat is unknown, and not allowed here (src = InstanceValidator)

8.3 Resource Group - Content

URL: https://hl7.org/fhir/STU3/group.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

FHIR Infrastructure Work Group | Maturity Level: 1 | Trial Use | Compartments: Device, Patient, Practitioner

Represents a defined collection of entities that may be discussed or acted upon collectively but which are not expected to act collectively and are not formally or legally recognized; i.e. a collection of entities that isn't an Organization.

8.3.1 Scope and Usage

8.3.1.1 Use Cases

The Group resource is used in one of two ways:

* To define a group of specific people, animals, devices, etc. that is being tracked, examined or otherwise referenced as part of healthcare-related activities

* To define a set of possible people, animals, devices, etc. that are of interest for some intended future healthcare-related activities

Examples of the former could include group therapy or treatment sessions, exposed entities tracked as part of public health, etc. The latter might be used to define expected subjects for a clinical study.

Both use cases are handled by a single resource because the data elements captured tend to be similar.

Contained, bundled and remotely referenced resources convey the same meaning.

8.3.2 Boundaries and Relationships

There are five mechanisms in FHIR for communicating collections of resources:

* The List resource - enumerates a flat collection of resources and provides features for managing the collection. While a particular List instance may represent a "snapshot", from a business process perspective the notion of "List" is dynamic – items are added and removed over time. The List resource references other resources. Lists may be curated and have specific business meaning.

* This Group resource - defines a group of specific people, animals, devices etc. by enumerating them, or by describing qualities that group members have. The group resource refers to other resources, possibly implicitly. Groups are intended to be acted upon or observed as a whole; e.g. performing therapy on a group, calculating risk for a group, etc. This resource will commonly be used for public health (e.g. describing an at-risk population), clinical trials (e.g. defining a test subject pool) and similar purposes.

* The Bundle resource - is an infrastructure container for a group of resources. It does not have narrative and is used to group collections of resources for transmission, persistence or processing (e.g. messages, documents, transactions, query responses, etc.) The content of bundles is typically algorithmically determined for a particular exchange or persistence purpose.

* The Composition resource - defines a set of healthcare-related information that is assembled together into a single logical document that provides a single coherent statement of meaning, establishes its own context and that has clinical attestation with regard to who is making the statement. The composition resource provides the basic structure of a FHIR document. The full content of the document is expressed using a bundle. Compositions will often reference Lists as the focus of particular sections.

* The DomainResource.contained element - allows multiple resources to be nested inside any DomainResource. This is a special type of grouping where the grouped resources lose independent existence - they no longer have their own identifiers, can't easily be queried independently, etc. Use of this grouping is a technical mechanism for managing the independence of resources and has no impact on meaning.

This resource is referenced by CarePlan, CareTeam, ChargeItem, ClinicalImpression, Communication, CommunicationRequest, Condition, Consent, Contract, DeviceRequest, DeviceUseStatement, DiagnosticReport, DocumentManifest, DocumentReference, Encounter, Flag, Goal, GuidanceResponse, List, Media, MedicationAdministration, MedicationDispense, MedicationRequest, MedicationStatement, Observation, Procedure, ProcedureRequest, ReferralRequest, RequestGroup, ResearchStudy, RiskAssessment, Specimen and Task

8.3.3 Resource Content

* Structure

* UML

* XML

* JSON

* Turtle

* R2 Diff

* All

Structure

Name | Flags | Card. | Type | Description & Constraints
Group | I | | DomainResource | Group of multiple entities + Can only have members if group is "actual" Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | ? | 0..* | Identifier | Unique id
active | ? | 0..1 | boolean | Whether this group's record is in active use
type | ? | 1..1 | code | person | animal | practitioner | device | medication | substance GroupType (Required)
actual | ?I | 1..1 | boolean | Descriptive or actual
code | ? | 0..1 | CodeableConcept | Kind of Group members
name | ? | 0..1 | string | Label for Group
quantity | ? | 0..1 | unsignedInt | Number of members
characteristic | | 0..* | BackboneElement | Trait of group members
code | | 1..1 | CodeableConcept | Kind of characteristic
value[x] | | 1..1 | | Value held by characteristic
valueCodeableConcept | | | CodeableConcept
valueBoolean | | | boolean
valueQuantity | | | Quantity
valueRange | | | Range
exclude | ?! | 1..1 | boolean | Group includes or excludes
period | | 0..1 | Period | Period over which characteristic is tested
member | I | 0..* | BackboneElement | Who or what is in group
entity | | 1..1 | Reference(Patient | Practitioner | Device | Medication | Substance) | Reference to the group member
period | | 0..1 | Period | Period member belonged to the group
inactive | | 0..1 | boolean | If member is no longer in group
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Group xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Unique id --></identifier>
 <active value="[boolean]"/><!-- 0..1 Whether this group's record is in active use -->
 <type value="[code]"/><!-- 1..1 person | animal | practitioner | device | medication | substance -->
 <actual value="[boolean]"/><!--  1..1 Descriptive or actual -->
 <code><!-- 0..1 CodeableConcept Kind of Group members --></code>
 <name value="[string]"/><!-- 0..1 Label for Group -->
 <quantity value="[unsignedInt]"/><!-- 0..1 Number of members -->
 <characteristic>  <!-- 0..* Trait of group members -->
  <code><!-- 1..1 CodeableConcept Kind of characteristic --></code>
  <value[x]><!-- 1..1 CodeableConcept|boolean|Quantity|Range Value held by characteristic --></value[x]>
  <exclude value="[boolean]"/><!-- 1..1 Group includes or excludes -->
  <period><!-- 0..1 Period Period over which characteristic is tested --></period>
 </characteristic>
 <member>  <!--  0..* Who or what is in group -->
  <entity><!-- 1..1 Reference(Patient|Practitioner|Device|Medication|Substance) Reference to the group member
    --></entity>
  <period><!-- 0..1 Period Period member belonged to the group --></period>
  <inactive value="[boolean]"/><!-- 0..1 If member is no longer in group -->
 </member>
</Group>
```

JSON Template

```
{
  "resourceType" : "Group",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Unique id
  "active" : <boolean>, // Whether this group's record is in active use
  "type" : "<code>", // R!  person | animal | practitioner | device | medication | substance
  "actual" : <boolean>, // C? R!  Descriptive or actual
  "code" : { CodeableConcept }, // Kind of Group members
  "name" : "<string>", // Label for Group
  "quantity" : "<unsignedInt>", // Number of members
  "characteristic" : [{ // Trait of group members
    "code" : { CodeableConcept }, // R!  Kind of characteristic
    // value[x]: Value held by characteristic. One of these 4:
    "valueCodeableConcept" : { CodeableConcept },
    "valueBoolean" : <boolean>,
    "valueQuantity" : { Quantity },
    "valueRange" : { Range },
    "exclude" : <boolean>, // R!  Group includes or excludes
    "period" : { Period } // Period over which characteristic is tested
  }],
  "member" : [{ // C? Who or what is in group
    "entity" : { Reference(Patient|Practitioner|Device|Medication|Substance) }, // R!  Reference to the group member
    "period" : { Period }, // Period member belonged to the group
    "inactive" : <boolean> // If member is no longer in group
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Group;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Group.identifier [ Identifier ], ... ; # 0..* Unique id
  fhir:Group.active [ boolean ]; # 0..1 Whether this group's record is in active use
  fhir:Group.type [ code ]; # 1..1 person | animal | practitioner | device | medication | substance
  fhir:Group.actual [ boolean ]; # 1..1 Descriptive or actual
  fhir:Group.code [ CodeableConcept ]; # 0..1 Kind of Group members
  fhir:Group.name [ string ]; # 0..1 Label for Group
  fhir:Group.quantity [ unsignedInt ]; # 0..1 Number of members
  fhir:Group.characteristic [ # 0..* Trait of group members
    fhir:Group.characteristic.code [ CodeableConcept ]; # 1..1 Kind of characteristic
    # Group.characteristic.value[x] : 1..1 Value held by characteristic. One of these 4
      fhir:Group.characteristic.valueCodeableConcept [ CodeableConcept ]
      fhir:Group.characteristic.valueBoolean [ boolean ]
      fhir:Group.characteristic.valueQuantity [ Quantity ]
      fhir:Group.characteristic.valueRange [ Range ]
    fhir:Group.characteristic.exclude [ boolean ]; # 1..1 Group includes or excludes
    fhir:Group.characteristic.period [ Period ]; # 0..1 Period over which characteristic is tested
  ], ...;
  fhir:Group.member [ # 0..* Who or what is in group
    fhir:Group.member.entity [ Reference(Patient|Practitioner|Device|Medication|Substance) ]; # 1..1 Reference to the
    group member
    fhir:Group.member.period [ Period ]; # 0..1 Period member belonged to the group
    fhir:Group.member.inactive [ boolean ]; # 0..1 If member is no longer in group
  ], ...;
]
```

Changes since DSTU2

Group

Group.active | Added Element

See the Full Difference for further information

This analysis is available as XML or JSON.

See R2 <--> R3 Conversion Maps (status = 2 tests that all execute ok. All tests pass round-trip testing and all r3 resources are valid.).

Structure

Name | Flags | Card. | Type | Description & Constraints

Group | I | | DomainResource | Group of multiple entities + Can only have members if group is "actual" Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension

identifier | ? | 0..* | Identifier | Unique id

active | ? | 0..1 | boolean | Whether this group's record is in active use

type | ? | 1..1 | code | person | animal | practitioner | device | medication | substance GroupType (Required)

actual | ?I | 1..1 | boolean | Descriptive or actual

code | ? | 0..1 | CodeableConcept | Kind of Group members

name | ? | 0..1 | string | Label for Group

quantity | ? | 0..1 | unsignedInt | Number of members

characteristic | | 0..* | BackboneElement | Trait of group members

code | | 1..1 | CodeableConcept | Kind of characteristic

value[x] | | 1..1 | | Value held by characteristic

valueCodeableConcept | | | CodeableConcept

valueBoolean | | | boolean

valueQuantity | | | Quantity

valueRange | | | Range

exclude | ?! | 1..1 | boolean | Group includes or excludes

period | | 0..1 | Period | Period over which characteristic is tested

member | I | 0..* | BackboneElement | Who or what is in group

entity | | 1..1 | Reference(Patient | Practitioner | Device | Medication | Substance) | Reference to the group member

period | | 0..1 | Period | Period member belonged to the group

inactive | | 0..1 | boolean | If member is no longer in group

Documentation for this format

UML Diagram (Legend)

XML Template

```
<Group xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Unique id --></identifier>
 <active value="[boolean]"/><!-- 0..1 Whether this group's record is in active use -->
 <type value="[code]"/><!-- 1..1 person | animal | practitioner | device | medication | substance -->
 <actual value="[boolean]"/><!--  1..1 Descriptive or actual -->
 <code><!-- 0..1 CodeableConcept Kind of Group members --></code>
 <name value="[string]"/><!-- 0..1 Label for Group -->
 <quantity value="[unsignedInt]"/><!-- 0..1 Number of members -->
 <characteristic>  <!-- 0..* Trait of group members -->
```

```
  <code><!-- 1..1 CodeableConcept Kind of characteristic --></code>
  <value[x]><!-- 1..1 CodeableConcept|boolean|Quantity|Range Value held by characteristic --></value[x]>
  <exclude value="[boolean]"/><!-- 1..1 Group includes or excludes -->
  <period><!-- 0..1 Period Period over which characteristic is tested --></period>
 </characteristic>
 <member>  <!--  0..* Who or what is in group -->
  <entity><!-- 1..1 Reference(Patient|Practitioner|Device|Medication|Substance) Reference to the group member
    --></entity>
  <period><!-- 0..1 Period Period member belonged to the group --></period>
  <inactive value="[boolean]"/><!-- 0..1 If member is no longer in group -->
 </member>
</Group>
```

JSON Template

```
{
  "resourceType" : "Group",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Unique id
  "active" : <boolean>, // Whether this group's record is in active use
  "type" : "<code>", // R!  person | animal | practitioner | device | medication | substance
  "actual" : <boolean>, // C? R!  Descriptive or actual
  "code" : { CodeableConcept }, // Kind of Group members
  "name" : "<string>", // Label for Group
  "quantity" : "<unsignedInt>", // Number of members
  "characteristic" : [{ // Trait of group members
    "code" : { CodeableConcept }, // R!  Kind of characteristic
    // value[x]: Value held by characteristic. One of these 4:
    "valueCodeableConcept" : { CodeableConcept },
    "valueBoolean" : <boolean>,
    "valueQuantity" : { Quantity },
    "valueRange" : { Range },
    "exclude" : <boolean>, // R!  Group includes or excludes
    "period" : { Period } // Period over which characteristic is tested
  }],
  "member" : [{ // C? Who or what is in group
    "entity" : { Reference(Patient|Practitioner|Device|Medication|Substance) }, // R!  Reference to the group member
    "period" : { Period }, // Period member belonged to the group
    "inactive" : <boolean> // If member is no longer in group
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Group;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Group.identifier [ Identifier ], ... ; # 0..* Unique id
  fhir:Group.active [ boolean ]; # 0..1 Whether this group's record is in active use
  fhir:Group.type [ code ]; # 1..1 person | animal | practitioner | device | medication | substance
  fhir:Group.actual [ boolean ]; # 1..1 Descriptive or actual
  fhir:Group.code [ CodeableConcept ]; # 0..1 Kind of Group members
  fhir:Group.name [ string ]; # 0..1 Label for Group
  fhir:Group.quantity [ unsignedInt ]; # 0..1 Number of members
  fhir:Group.characteristic [ # 0..* Trait of group members
    fhir:Group.characteristic.code [ CodeableConcept ]; # 1..1 Kind of characteristic
    # Group.characteristic.value[x] : 1..1 Value held by characteristic. One of these 4
      fhir:Group.characteristic.valueCodeableConcept [ CodeableConcept ]
      fhir:Group.characteristic.valueBoolean [ boolean ]
      fhir:Group.characteristic.valueQuantity [ Quantity ]
      fhir:Group.characteristic.valueRange [ Range ]
    fhir:Group.characteristic.exclude [ boolean ]; # 1..1 Group includes or excludes
    fhir:Group.characteristic.period [ Period ]; # 0..1 Period over which characteristic is tested
  ], ...;
  fhir:Group.member [ # 0..* Who or what is in group
    fhir:Group.member.entity [ Reference(Patient|Practitioner|Device|Medication|Substance) ]; # 1..1 Reference to the
    group member
    fhir:Group.member.period [ Period ]; # 0..1 Period member belonged to the group
    fhir:Group.member.inactive [ boolean ]; # 0..1 If member is no longer in group
  ], ...;
]
```

Changes since DSTU2

Group

Group.active | Added Element

See the Full Difference for further information

This analysis is available as XML or JSON.

See R2 <--> R3 Conversion Maps (status = 2 tests that all execute ok. All tests pass round-trip testing and all r3 resources are valid.).

Alternate definitions: Master Definition (XML, JSON), XML Schema/Schematron (for) + JSON Schema, ShEx (for Turtle)

8.3.3.1 Terminology Bindings

Path | Definition | Type | Reference
Group.type | Types of resources that are part of group | Required | GroupType
Group.code | Kind of particular resource; e.g. cow, syringe, lake, etc. | Unknown | No details provided yet
Group.characteristic.code | List of characteristics used to describe group members; e.g. gender, age, owner, location, etc. | Unknown | No details provided yet
Group.characteristic.value[x] | Value of descriptive member characteristic; e.g. red, male, pneumonia, Caucasian, etc. | Unknown | No details provided yet

8.3.3.2 Constraints

* grp-1: Can only have members if group is "actual" (expression : member.empty() or (actual = true))

8.3.4 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
actual | token | Descriptive or actual | Group.actual
characteristic | token | Kind of characteristic | Group.characteristic.code
characteristic-value | composite | A composite of both characteristic and value | Group.characteristic
code | token | The kind of resources contained | Group.code
exclude | token | Group includes or excludes | Group.characteristic.exclude
identifier | token | Unique id | Group.identifier
member | reference | Reference to the group member | Group.member.entity (Practitioner, Device, Medication, Patient, Substance)
type | token | The type of resources the group contains | Group.type
value | token | Value held by characteristic | Group.characteristic.value

8.3.5 Resource Group - Examples

URL: https://hl7.org/fhir/STU3/group-examples.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.3.5 Resource Group - Examples

FHIR Infrastructure Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Device, Patient, Practitioner

Example List:

General Group Example, Characteristic Based | 101 | XML | JSON | Turtle
General Group Example, Member Based | 102 | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

8.3.6 Resource Group - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/group-definitions.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.3.6 Resource Group - Detailed Descriptions

FHIR Infrastructure Work Group | Maturity Level: 1 | Trial Use | Compartments: Device, Patient, Practitioner

Detailed Descriptions for the elements in the Group resource.

Group
Definition | Represents a defined collection of entities that may be discussed or acted upon collectively but which are not expected to act collectively and are not formally or legally recognized; i.e. a collection of entities that isn't an Organization.
Control | 1..1
Invariants | Defined on this element grp-1: Can only have members if group is "actual" (expression : member.empty() or (actual = true), xpath: f:actual/@value=true() or not(exists(f:member)))
Group.identifier
Definition | A unique business identifier for this group.
Note | This is a business identifer, not a resource identifier (see discussion)
Control | 0..*
Type | Identifier
Requirements | Allows the group to be referenced from external specifications.
Summary | true
Group.active
Definition | Indicates whether the record for the group is available for use or is merely being retained for historical purposes.
Control | 0..1
Type | boolean
Default Value | true

Requirements | Need to be able to mark a group record as not to be used because it was created in error or is otherwise no longer available (e.g. a herd that no longer exists).
Summary | true
Group.type
Definition | Identifies the broad classification of the kind of resources the group includes.
Control | 1..1
Terminology Binding | GroupType (Required)
Type | code
Requirements | Identifies what type of resources the group is made up of.
Summary | true
Comments | Group members SHALL be of the appropriate resource type (Patient for person or animal; or Practitioner, Device, Medicaiton or Substance for the other types.).
Group.actual
Definition | If true, indicates that the resource refers to a specific group of real individuals. If false, the group defines a set of intended individuals.
Control | 1..1
Type | boolean
Requirements | There are use-cases for groups that define specific collections of individuals, and other groups that define "types" of intended individuals. The requirements for both kinds of groups are similar, so we use a single resource, distinguished by this flag.
Summary | true
Invariants | Affect this element grp-1: Can only have members if group is "actual" (expression : member.empty() or (actual = true), xpath: f:actual/@value=true() or not(exists(f:member)))
Group.code
Definition | Provides a specific type of resource the group includes; e.g. "cow", "syringe", etc.
Control | 0..1
Terminology Binding | GroupKind:
Type | CodeableConcept
Summary | true
Comments | This would generally be omitted for Person resources.
Group.name
Definition | A label assigned to the group for human identification and communication.
Control | 0..1
Type | string
Requirements | Used to identify the group in human communication.
Summary | true
Group.quantity
Definition | A count of the number of resource instances that are part of the group.
Control | 0..1
Type | unsignedInt
Requirements | Group size is a common defining characteristic.
Summary | true
Comments | Note that the quantity may be less than the number of members if some of the members are not active.
Group.characteristic
Definition | Identifies the traits shared by members of the group.
Control | 0..*
Requirements | Needs to be a generic mechanism for identifying what individuals can be part of a group.
Comments | All the identified characteristics must be true for an entity to a member of the group.
Group.characteristic.code
Definition | A code that identifies the kind of trait being asserted.
Control | 1..1
Terminology Binding | GroupCharacteristicKind:
Type | CodeableConcept
Requirements | Need a formal way of identifying the characteristic being described.
Group.characteristic.value[x]
Definition | The value of the trait that holds (or does not hold - see 'exclude') for members of the group.
Control | 1..1
Terminology Binding | GroupCharacteristicValue:
Type | CodeableConcept|boolean|Quantity|Range
[x] Note | See Choice of Data Types for further information about how to use [x]
Requirements | The value of the characteristic is what determines group membership.
Comments | For Range, it means members of the group have a value that falls somewhere within the specified range.
Group.characteristic.exclude
Definition | If true, indicates the characteristic is one that is NOT held by members of the group.
Control | 1..1
Type | boolean
Is Modifier | true
Requirements | Sometimes group membership is determined by characteristics not possessed.
Comments | This is labeled as "Is Modifier" because applications cannot wrongly include excluded members as included or vice versa.
Group.characteristic.period
Definition | The period over which the characteristic is tested; e.g. the patient had an operation during the month of June.
Control | 0..1
Type | Period
Group.member
Definition | Identifies the resource instances that are members of the group.

Control | 0..*
Requirements | Often the only thing of interest about a group is "who's in it".
Invariants | Affect this element grp-1: Can only have members if group is "actual" (expression : member.empty() or (actual = true), xpath: f:actual/@value=true() or not(exists(f:member)))
Group.member.entity
Definition | A reference to the entity that is a member of the group. Must be consistent with Group.type.
Control | 1..1
Type | Reference(Patient | Practitioner | Device | Medication | Substance)
Group.member.period
Definition | The period that the member was in the group, if known.
Control | 0..1
Type | Period
Meaning if Missing | The member is in the group at this time
Requirements | Need to track who was in a group at a particular time.
Group.member.inactive
Definition | A flag to indicate that the member is no longer in the group, but previously may have been a member.
Control | 0..1
Type | boolean
Default Value | false
Requirements | Sometimes you don't know when someone stopped being in a group, but not when.

8.3.7 Resource Group - Mappings

URL: https://hl7.org/fhir/STU3/group-mappings.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.3.7 Resource Group - Mappings

FHIR Infrastructure Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Device, Patient, Practitioner

Mappings:

RIM Mapping

Mappings for the group resource.

8.3.7.1 Mappings for RIM Mapping (http://hl7.org/v3)

Group | Entity[determinerCode="GRP" or determinerCode="GRP_KIND"]
identifier | unique(./id[displayable="true" and scope="OBJ"])
active | ./statusCode
type | person: ./classCode="PSN" animal: ./classCode="ANM" device: ./classCode="DEV" medication: ./classCode="MMAT" and ./playedRole[isNormalRole() and classCode="THER"] food: ./classCode="FOOD"
actual | true: ./determinerCode="GRP" false: ./determinerCode="GRP_KIND"
code | ./code
name | ./name[type="ST"]
quantity | ./quantity
characteristic | ./playedRole[isNormalRole()]/participation[isNormalParticipation() and typeCode="SBJ"]/act[isNormalAct and classCode="OBS" and moodCode="EVN"]
code | ./code
value[x] | ./value
exclude | ./valueNegationInd
period
member | ./scopedRole[isNormalRole() and classCode="MBR"]/player
entity
period
inactive

8.3.8 Resource Group - Design Notes

URL: https://hl7.org/fhir/STU3/group-explanations.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.3.8 Resource Group - Design Notes

to do

8.3.9 Resource Group - Extensions & Profiles

URL: https://hl7.org/fhir/STU3/group-profiles.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.3.9 Resource Group - Extensions & Profiles

FHIR Infrastructure Work Group | Maturity Level: N/A | Ballot Status: Informative

This table lists profiles and extensions for the Group resource. For background information, see Profiling Resources and Extensibility. Additional profiles and resources may be found in published Implementation Guides , or in the Conformance resource registry .

8.3.9.1 Profiles

No Profiles defined for this resource

8.3.9.2 Extensions

No Extensions defined for this resource

Extensions that reference this resource:

No Extensions refer to this resource

8.3.9.3 Search Extensions

No Search Extensions defined for this resource

8.3.10 Resource Group - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/group-version-maps.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.3.10 Resource Group - Detailed Descriptions

FHIR Infrastructure Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Device, Patient, Practitioner

R2 : R3 Converson maps for Group.

Functional status for this map: 2 tests that all execute ok. All tests pass round-trip testing and all r3 resources are valid. (see documentation)

8.3.10.1 R2 to R3

```
map "http://hl7.org/fhir/StructureMap/Group2to3" = "R2 to R3 Conversions for Group"

uses "http://hl7.org/fhir/DSTU2/StructureDefinition/Group" alias GroupR2 as source
uses "http://hl7.org/fhir/StructureDefinition/Group" alias Group as target

imports "http://hl7.org/fhir/StructureMap/*2to3"

group for type+types  Group extends DomainResource
  input src : GroupR2 as source
  input tgt : Group as target

  "Group-identifier" : for src.identifier make tgt.identifier
  "Group-type" : for src.type make tgt.type
  "Group-actual" : for src.actual make tgt.actual
  "Group-code" : for src.code make tgt.code
  "Group-name" : for src.name make tgt.name
  "Group-quantity" : for src.quantity make tgt.quantity
  "Group-characteristic" : for src.characteristic as vs0 make tgt.characteristic as vt0 then grpCharacteristic(vs0, vt0)
  "Group-member" : for src.member as vs0 make tgt.member as vt0 then grpMember(vs0, vt0)
endgroup

group grpCharacteristic extends BackboneElement
  input src as source
  input tgt as target

  "Group.characteristic-code" : for src.code make tgt.code
  "Group.characteristic-value" : for src.value make tgt.value
  "Group.characteristic-exclude" : for src.exclude make tgt.exclude
  "Group.characteristic-period" : for src.period make tgt.period
endgroup

group grpMember extends BackboneElement
  input src as source
  input tgt as target

  "Group.member-entity" : for src.entity make tgt.entity
  "Group.member-period" : for src.period make tgt.period
  "Group.member-inactive" : for src.inactive make tgt.inactive
endgroup
```

8.3.10.2 R3 to R2

```
map "http://hl7.org/fhir/StructureMap/Group3to2" = "R3 to R2 Conversion for Group"

uses "http://hl7.org/fhir/StructureDefinition/Group" alias Group as source
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/Group" alias GroupR2 as target
```

```
imports "http://hl7.org/fhir/StructureMap/*3to2"

group for type+types  Group extends DomainResource
  input src : Group as source
  input tgt : GroupR2 as target

  "Group-identifier" : for src.identifier make tgt.identifier
  "Group-type" : for src.type make tgt.type
  "Group-actual" : for src.actual make tgt.actual
  "Group-code" : for src.code make tgt.code
  "Group-name" : for src.name make tgt.name
  "Group-quantity" : for src.quantity make tgt.quantity
  "Group-characteristic" : for src.characteristic as vs0 make tgt.characteristic as vt0 then grpCharacteristic(vs0, vt0)
  "Group-member" : for src.member as vs0 make tgt.member as vt0 then grpMember(vs0, vt0)
endgroup

group grpCharacteristic extends BackboneElement
  input src as source
  input tgt as target

  "Group.characteristic-code" : for src.code make tgt.code
  "Group.characteristic-value" : for src.value make tgt.value
  "Group.characteristic-exclude" : for src.exclude make tgt.exclude
  "Group.characteristic-period" : for src.period make tgt.period
endgroup

group grpMember extends BackboneElement
  input src as source
  input tgt as target

  "Group.member-entity" : for src.entity make tgt.entity
  "Group.member-period" : for src.period make tgt.period
  "Group.member-inactive" : for src.inactive make tgt.inactive
endgroup
```

8.3.10.3 R3 Validation Errors

8.4 Resource Practitioner - Content

URL: https://hl7.org/fhir/STU3/practitioner.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

Patient Administration Work Group | Maturity Level: 3 | Trial Use | Compartments: Practitioner

A person who is directly or indirectly involved in the provisioning of healthcare.

8.4.1 Scope and Usage

Practitioner covers all individuals who are engaged in the healthcare process and healthcare-related services as part of their formal responsibilities and this Resource is used for attribution of activities and responsibilities to these individuals. Practitioners include (but are not limited to):

* physicians, dentists, pharmacists

* physician assistants, nurses, scribes

* midwives, dietitians, therapists, optometrists, paramedics

* medical technicians, laboratory scientists, prosthetic technicians, radiographers

* social workers, professional home carers, official volunteers

* receptionists handling patient registration

* IT personnel merging or unmerging patient records

* Service animal (e.g., ward assigned dog capable of detecting cancer in patients)

8.4.2 Boundaries and Relationships

The Resource SHALL NOT be used for persons involved without a formal responsibility like individuals taking care for friends, relatives or neighbors. These can be registered as a Patient's Contact. If performing some action or being referenced by another resource, use the RelatedPerson resource.

The primary distinction between a Practitioner and a RelatedPerson is based on whether:

* The person/animal operates on behalf of the care delivery organization over multiple patients (Practitioner) or,

* Where the person/animal is not associated with the organization, and instead is allocated tasks specifically for the RelatedPerson's Patient (RelatedPerson).

A standard extension animalSpecies can be used to indicate the species of a service animal.

The PractitionerRole resource provides the details of roles that the practitioner is approved to perform for which organizations (an at which locations, and optionally what services too).

Practitioners are also often grouped into CareTeams independently of roles, where the CareTeam defines what specific role that they are fulfilling within the team, and may or may not have actual practitioner role resources created for the practitioner

(and in the care team context, the organization the practitioner is representing)

8.4.3 Background and Context

Practitioner performs different roles within the same or even different organizations. Depending on jurisdiction and custom, it may be necessary to maintain a specific Practitioner Resource for each such role or have a single Practitioner with multiple roles. The role can be limited to a specific period, after which authorization for this role ends. Note that the represented organization need not necessarily be the (direct) employer of a Practitioner.

This resource is referenced by Account, AdverseEvent, AllergyIntolerance, Appointment, AppointmentResponse, AuditEvent, Basic, CarePlan, CareTeam, ChargeItem, Claim, ClaimResponse, ClinicalImpression, Communication, CommunicationRequest, Composition, Condition, Consent, Contract, DetectedIssue, DeviceRequest, DeviceUseStatement, DiagnosticReport, DocumentManifest, DocumentReference, EligibilityRequest, EligibilityResponse, Encounter, EnrollmentRequest, EnrollmentResponse, EpisodeOfCare, ExplanationOfBenefit, Flag, Goal, Group, ImagingManifest, ImagingStudy, Immunization, Linkage, List, Media, MedicationAdministration, MedicationDispense, MedicationRequest, MedicationStatement, MessageHeader, NutritionOrder, Observation, Patient, PaymentNotice, PaymentReconciliation, Person, PractitionerRole, Procedure, ProcedureRequest, ProcessRequest, ProcessResponse, Provenance, QuestionnaireResponse, ReferralRequest, RequestGroup, ResearchStudy, RiskAssessment, Schedule, Specimen, SupplyDelivery, SupplyRequest, Task and VisionPrescription

8.4.4 Resource Content

* Structure

* UML

* XML

* JSON

* Turtle

* R2 Diff

* All

Structure

Name | Flags | Card. | Type | Description & Constraints
Practitioner | | | DomainResource | A person with a formal responsibility in the provisioning of healthcare or related services Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | ? | 0..* | Identifier | A identifier for the person as this agent
active | ? | 0..1 | boolean | Whether this practitioner's record is in active use
name | ? | 0..* | HumanName | The name(s) associated with the practitioner
telecom | ? | 0..* | ContactPoint | A contact detail for the practitioner (that apply to all roles)
address | ? | 0..* | Address | Address(es) of the practitioner that are not role specific (typically home address)
gender | ? | 0..1 | code | male | female | other | unknown AdministrativeGender (Required)
birthDate | ? | 0..1 | date | The date on which the practitioner was born
photo | | 0..* | Attachment | Image of the person
qualification | | 0..* | BackboneElement | Qualifications obtained by training and certification
identifier | | 0..* | Identifier | An identifier for this qualification for the practitioner
code | | 1..1 | CodeableConcept | Coded representation of the qualification v2 table 0360, Version 2.7 (Example)
period | | 0..1 | Period | Period during which the qualification is valid
issuer | | 0..1 | Reference(Organization) | Organization that regulates and issues the qualification
communication | | 0..* | CodeableConcept | A language the practitioner is able to use in patient communication Common Languages (Extensible but limited to All Languages)
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Practitioner xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier A identifier for the person as this agent --></identifier>
 <active value="[boolean]"/><!-- 0..1 Whether this practitioner's record is in active use -->
 <name><!-- 0..* HumanName The name(s) associated with the practitioner --></name>
 <telecom><!-- 0..* ContactPoint A contact detail for the practitioner (that apply to all roles) --></telecom>
 <address><!-- 0..* Address Address(es) of the practitioner that are not role specific (typically home address)
    --></address>
 <gender value="[code]"/><!-- 0..1 male | female | other | unknown -->
 <birthDate value="[date]"/><!-- 0..1 The date  on which the practitioner was born -->
 <photo><!-- 0..* Attachment Image of the person --></photo>
 <qualification>  <!-- 0..* Qualifications obtained by training and certification -->
  <identifier><!-- 0..* Identifier An identifier for this qualification for the practitioner --></identifier>
  <code><!-- 1..1 CodeableConcept Coded representation of the qualification --></code>
  <period><!-- 0..1 Period Period during which the qualification is valid --></period>
  <issuer><!-- 0..1 Reference(Organization) Organization that regulates and issues the qualification --></issuer>
 </qualification>
 <communication><!-- 0..* CodeableConcept A language the practitioner is able to use in patient communication
    --></communication>
</Practitioner>
```

JSON Template

```
{
  "resourceType" : "Practitioner",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // A identifier for the person as this agent
  "active" : <boolean>, // Whether this practitioner's record is in active use
  "name" : [{ HumanName }], // The name(s) associated with the practitioner
  "telecom" : [{ ContactPoint }], // A contact detail for the practitioner (that apply to all roles)
  "address" : [{ Address }], // Address(es) of the practitioner that are not role specific (typically home address)
  "gender" : "<code>", // male | female | other | unknown
  "birthDate" : "<date>", // The date  on which the practitioner was born
  "photo" : [{ Attachment }], // Image of the person
  "qualification" : [{ // Qualifications obtained by training and certification
    "identifier" : [{ Identifier }], // An identifier for this qualification for the practitioner
    "code" : { CodeableConcept }, // R!  Coded representation of the qualification
    "period" : { Period }, // Period during which the qualification is valid
    "issuer" : { Reference(Organization) } // Organization that regulates and issues the qualification
  }],
  "communication" : [{ CodeableConcept }] // A language the practitioner is able to use in patient communication
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Practitioner;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Practitioner.identifier [ Identifier ], ... ; # 0..* A identifier for the person as this agent
  fhir:Practitioner.active [ boolean ]; # 0..1 Whether this practitioner's record is in active use
  fhir:Practitioner.name [ HumanName ], ... ; # 0..* The name(s) associated with the practitioner
  fhir:Practitioner.telecom [ ContactPoint ], ... ; # 0..* A contact detail for the practitioner (that apply to all
    roles)
  fhir:Practitioner.address [ Address ], ... ; # 0..* Address(es) of the practitioner that are not role specific
    (typically home address)
  fhir:Practitioner.gender [ code ]; # 0..1 male | female | other | unknown
  fhir:Practitioner.birthDate [ date ]; # 0..1 The date  on which the practitioner was born
  fhir:Practitioner.photo [ Attachment ], ... ; # 0..* Image of the person
  fhir:Practitioner.qualification [ # 0..* Qualifications obtained by training and certification
    fhir:Practitioner.qualification.identifier [ Identifier ], ... ; # 0..* An identifier for this qualification for the
     practitioner
    fhir:Practitioner.qualification.code [ CodeableConcept ]; # 1..1 Coded representation of the qualification
    fhir:Practitioner.qualification.period [ Period ]; # 0..1 Period during which the qualification is valid
    fhir:Practitioner.qualification.issuer [ Reference(Organization) ]; # 0..1 Organization that regulates and issues
    the qualification
  ], ...;
  fhir:Practitioner.communication [ CodeableConcept ], ... ; # 0..* A language the practitioner is able to use in
    patient communication
]
```

Changes since DSTU2

Practitioner

Practitioner.name | Max Cardinality changed from 1 to *

Practitioner.communication | Change binding strength from required to extensible Change value set from http://tools.ietf.org/html/bcp47 to http://hl7.org/fhir/ValueSet/languages

Practitioner.practitionerRole | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R2 <--> R3 Conversion Maps (status = 18 tests that all execute ok. 15 fail round-trip testing and all r3 resources are valid.).

Structure

Name | Flags | Card. | Type | Description & Constraints

Practitioner | | | DomainResource | A person with a formal responsibility in the provisioning of healthcare or related services Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension

identifier | ? | 0..* | Identifier | A identifier for the person as this agent

active | ? | 0..1 | boolean | Whether this practitioner's record is in active use

name | ? | 0..* | HumanName | The name(s) associated with the practitioner

telecom | ? | 0..* | ContactPoint | A contact detail for the practitioner (that apply to all roles)

address | ? | 0..* | Address | Address(es) of the practitioner that are not role specific (typically home address)

gender | ? | 0..1 | code | male | female | other | unknown AdministrativeGender (Required)

birthDate | ? | 0..1 | date | The date on which the practitioner was born

photo | | 0..* | Attachment | Image of the person

qualification | | 0..* | BackboneElement | Qualifications obtained by training and certification

identifier | | 0..* | Identifier | An identifier for this qualification for the practitioner

code | | 1..1 | CodeableConcept | Coded representation of the qualification v2 table 0360, Version 2.7 (Example)

period | | 0..1 | Period | Period during which the qualification is valid

issuer | | 0..1 | Reference(Organization) | Organization that regulates and issues the qualification

communication | | 0..* | CodeableConcept | A language the practitioner is able to use in patient communication Common Languages (Extensible but limited to All Languages)

Documentation for this format

UML Diagram (Legend)

XML Template

```
<Practitioner xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier A identifier for the person as this agent --></identifier>
 <active value="[boolean]"/><!-- 0..1 Whether this practitioner's record is in active use -->
 <name><!-- 0..* HumanName The name(s) associated with the practitioner --></name>
 <telecom><!-- 0..* ContactPoint A contact detail for the practitioner (that apply to all roles) --></telecom>
 <address><!-- 0..* Address Address(es) of the practitioner that are not role specific (typically home address)
    --></address>
 <gender value="[code]"/><!-- 0..1 male | female | other | unknown -->
 <birthDate value="[date]"/><!-- 0..1 The date  on which the practitioner was born -->
 <photo><!-- 0..* Attachment Image of the person --></photo>
 <qualification>  <!-- 0..* Qualifications obtained by training and certification -->
  <identifier><!-- 0..* Identifier An identifier for this qualification for the practitioner --></identifier>
  <code><!-- 1..1 CodeableConcept Coded representation of the qualification --></code>
  <period><!-- 0..1 Period Period during which the qualification is valid --></period>
  <issuer><!-- 0..1 Reference(Organization) Organization that regulates and issues the qualification --></issuer>
 </qualification>
 <communication><!-- 0..* CodeableConcept A language the practitioner is able to use in patient communication
    --></communication>
</Practitioner>
```

JSON Template

```
{
  "resourceType" : "Practitioner",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // A identifier for the person as this agent
  "active" : <boolean>, // Whether this practitioner's record is in active use
  "name" : [{ HumanName }], // The name(s) associated with the practitioner
  "telecom" : [{ ContactPoint }], // A contact detail for the practitioner (that apply to all roles)
  "address" : [{ Address }], // Address(es) of the practitioner that are not role specific (typically home address)
  "gender" : "<code>", // male | female | other | unknown
  "birthDate" : "<date>", // The date  on which the practitioner was born
  "photo" : [{ Attachment }], // Image of the person
  "qualification" : [{ // Qualifications obtained by training and certification
    "identifier" : [{ Identifier }], // An identifier for this qualification for the practitioner
    "code" : { CodeableConcept }, // R!  Coded representation of the qualification
    "period" : { Period }, // Period during which the qualification is valid
    "issuer" : { Reference(Organization) } // Organization that regulates and issues the qualification
  }],
  "communication" : [{ CodeableConcept }] // A language the practitioner is able to use in patient communication
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Practitioner;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Practitioner.identifier [ Identifier ], ... ; # 0..* A identifier for the person as this agent
  fhir:Practitioner.active [ boolean ]; # 0..1 Whether this practitioner's record is in active use
  fhir:Practitioner.name [ HumanName ], ... ; # 0..* The name(s) associated with the practitioner
  fhir:Practitioner.telecom [ ContactPoint ], ... ; # 0..* A contact detail for the practitioner (that apply to all
    roles)
  fhir:Practitioner.address [ Address ], ... ; # 0..* Address(es) of the practitioner that are not role specific
    (typically home address)
  fhir:Practitioner.gender [ code ]; # 0..1 male | female | other | unknown
  fhir:Practitioner.birthDate [ date ]; # 0..1 The date  on which the practitioner was born
  fhir:Practitioner.photo [ Attachment ], ... ; # 0..* Image of the person
  fhir:Practitioner.qualification [ # 0..* Qualifications obtained by training and certification
    fhir:Practitioner.qualification.identifier [ Identifier ], ... ; # 0..* An identifier for this qualification for the
     practitioner
    fhir:Practitioner.qualification.code [ CodeableConcept ]; # 1..1 Coded representation of the qualification
    fhir:Practitioner.qualification.period [ Period ]; # 0..1 Period during which the qualification is valid
    fhir:Practitioner.qualification.issuer [ Reference(Organization) ]; # 0..1 Organization that regulates and issues
    the qualification
  ], ...;
  fhir:Practitioner.communication [ CodeableConcept ], ... ; # 0..* A language the practitioner is able to use in
    patient communication
]
```

Changes since DSTU2

Practitioner

Practitioner.name | Max Cardinality changed from 1 to *

Practitioner.communication | Change binding strength from required to extensible Change value set from http://tools.ietf.org/html/bcp47 to http://hl7.org/fhir/ValueSet/languages

Practitioner.practitionerRole | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R2 <--> R3 Conversion Maps (status = 18 tests that all execute ok. 15 fail round-trip testing and all r3 resources are valid.).

Alternate definitions: Master Definition (XML, JSON), XML Schema/Schematron (for) + JSON Schema, ShEx (for Turtle)

8.4.4.1 Terminology Bindings

Path | Definition | Type | Reference
Practitioner.gender | The gender of a person used for administrative purposes. | Required | AdministrativeGender
Practitioner.qualification.code | Specific qualification the practitioner has to provide a service | Example | v2 table 0360, Version 2.7
Practitioner.communication | A human language. | Extensible, but limited to All Languages | Common Languages

8.4.5 Notes:

* The practitioner's Qualifications are aquired by the practitioner independant of any organization or role, and do not imply that they are allowed/authorzied to perform roles relevant to the qualification at any specific Organization/Location.

8.4.6 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
active | token | Whether the practitioner record is active | Practitioner.active
address | string | A server defined search that may match any of the string fields in the Address, including line, city, state, country, postalCode, and/or text | Practitioner.address | 3 Resources
address-city | string | A city specified in an address | Practitioner.address.city | 3 Resources
address-country | string | A country specified in an address | Practitioner.address.country | 3 Resources
address-postalcode | string | A postalCode specified in an address | Practitioner.address.postalCode | 3 Resources
address-state | string | A state specified in an address | Practitioner.address.state | 3 Resources
address-use | token | A use code specified in an address | Practitioner.address.use | 3 Resources
communication | token | One of the languages that the practitioner can communicate with | Practitioner.communication
email | token | A value in an email contact | Practitioner.telecom.where(system='email') | 4 Resources
family | string | A portion of the family name | Practitioner.name.family | 1 Resources
gender | token | Gender of the practitioner | Practitioner.gender | 3 Resources
given | string | A portion of the given name | Practitioner.name.given | 1 Resources
identifier | token | A practitioner's Identifier | Practitioner.identifier
name | string | A server defined search that may match any of the string fields in the HumanName, including family, give, prefix, suffix, suffix, and/or text | Practitioner.name
phone | token | A value in a phone contact | Practitioner.telecom.where(system='phone') | 4 Resources
phonetic | string | A portion of either family or given name using some kind of phonetic matching algorithm | Practitioner.name | 3 Resources
telecom | token | The value in any kind of contact | Practitioner.telecom | 4 Resources

8.4.7 Resource Practitioner - Examples

URL: https://hl7.org/fhir/STU3/practitioner-examples.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.4.7 Resource Practitioner - Examples

Patient Administration Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Practitioner

Example List:

General Person Example | example | XML | JSON | Turtle
CDA Example Author | xcda-author | XML | JSON | Turtle
HL7 Defined Practitioner Examples | 3ad0687e-f477-468c-afd5-fcc2bf897809 | XML | JSON | Turtle
Fictive KNO-physician | f001 | XML | JSON | Turtle
Fictive Cardiothoracal surgeon | f002 | XML | JSON | Turtle
Fictive Cardiothoracal surgeon | f003 | XML | JSON | Turtle
Fictive KNO-physician | f004 | XML | JSON | Turtle
Fictive KNO-physician | f005 | XML | JSON | Turtle
Fictive Oncologist/Pulmonologist | f201 | XML | JSON | Turtle
Fictive Lab worker | f202 | XML | JSON | Turtle
Fictive Physiotherapist | f203 | XML | JSON | Turtle
Fictive Nurse | f204 | XML | JSON | Turtle
Fictive Pharmacist | f006 | XML | JSON | Turtle
Fictive physician | f007 | XML | JSON | Turtle
2nd CDA Example Author | xcda1 | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

8.4.8 Resource Practitioner - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/practitioner-definitions.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.4.8 Resource Practitioner - Detailed Descriptions

Patient Administration Work Group | Maturity Level: 3 | Trial Use | Compartments: Practitioner

Detailed Descriptions for the elements in the Practitioner resource.

Practitioner
Definition | A person who is directly or indirectly involved in the provisioning of healthcare.
Control | 1..1
Requirements | Need to track doctors, staff, locums etc. for both healthcare practitioners, funders, etc.
Comments | Note that a cab driver no longer fits the bill. You probably would be interested in the organization rather than the individual?
Practitioner.identifier
Definition | An identifier that applies to this person in this role.
Note | This is a business identifer, not a resource identifier (see discussion)
Control | 0..*
Type | Identifier
Requirements | Often, specific identities are assigned for the agent.
Summary | true
Practitioner.active
Definition | Whether this practitioner's record is in active use.
Control | 0..1
Type | boolean
Default Value | true
Requirements | Need to be able to mark a practitioner record as not to be used because it was created in error.
Summary | true
Comments | Default is true. If the practitioner is not in use by one organization, then it should mark the period on the PractitonerRole with an end date (even if they are active) as they may be active in another role.
Practitioner.name
Definition | The name(s) associated with the practitioner.
Control | 0..*
Type | HumanName
Requirements | The name(s) that a Practitioner is known by. Where there are multiple, the name that the practitioner is usually known as should be used in the display.
Summary | true
Comments | The selection of the use property should ensure that there is a single usual name specified, and others use the nickname (alias), old, or other values as appropriate. In general select the value to be used in the ResourceReference.display based on this: There is more than 1 name Use = usual Period is current to the date of the usage Use = official Other order as decided by internal business rules.
Practitioner.telecom
Definition | A contact detail for the practitioner, e.g. a telephone number or an email address.
Control | 0..*
Type | ContactPoint
Requirements | Need to know how to reach a practitioner independent to any roles the practitioner may have.
Summary | true
Comments | Person may have multiple ways to be contacted with different uses or applicable periods. May need to have options for contacting the person urgently, and also to help with identification These typically will have home numbers, or mobile numbers that are not role specific.
Practitioner.address
Definition | Address(es) of the practitioner that are not role specific (typically home address). Work addresses are not typically entered in this property as they are usually role dependent.
Control | 0..*
Type | Address
Requirements | The home/mailing address of the practitioner is often required for employee administration purposes, and also for some rostering services where the start point (practitioners home) can be used in calculations.
Summary | true
Comments | The PractitionerRole does not have an address value on it, as it is expected that the location property be used for this purpose (which has an address).
Practitioner.gender
Definition | Administrative Gender - the gender that the person is considered to have for administration and record keeping purposes.
Control | 0..1
Terminology Binding | AdministrativeGender (Required)
Type | code
Requirements | Needed to address the person correctly.
Summary | true
Practitioner.birthDate
Definition | The date of birth for the practitioner.
Control | 0..1

Type | date
Requirements | Needed for identification.
Summary | true
Practitioner.photo
Definition | Image of the person.
Control | 0..*
Type | Attachment
Requirements | Many EHR systems have the capability to capture an image of patients and personnel. Fits with newer social media usage too.
Practitioner.qualification
Definition | Qualifications obtained by training and certification.
Control | 0..*
Practitioner.qualification.identifier
Definition | An identifier that applies to this person's qualification in this role.
Note | This is a business identifer, not a resource identifier (see discussion)
Control | 0..*
Type | Identifier
Requirements | Often, specific identities are assigned for the qualification.
Practitioner.qualification.code
Definition | Coded representation of the qualification.
Control | 1..1
Terminology Binding | v2 table 0360, Version 2.7 (Example)
Type | CodeableConcept
Practitioner.qualification.period
Definition | Period during which the qualification is valid.
Control | 0..1
Type | Period
Requirements | Qualifications are often for a limited period of time, and can be revoked.
Practitioner.qualification.issuer
Definition | Organization that regulates and issues the qualification.
Control | 0..1
Type | Reference(Organization)
Practitioner.communication
Definition | A language the practitioner is able to use in patient communication.
Control | 0..*
Terminology Binding | Common Languages (Extensible but limited to ??)
Type | CodeableConcept
Requirements | Knowing which language a practitioner speaks can help in facilitating communication with patients.
Comments | The structure aa-BB with this exact casing is one the most widely used notations for locale. However not all systems actually code this but instead have it as free text. Hence CodeableConcept instead of code as the data type.

8.4.9 Resource Practitioner - Mappings

URL: https://hl7.org/fhir/STU3/practitioner-mappings.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.4.9 Resource Practitioner - Mappings

Patient Administration Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Practitioner

Mappings:

RIM Mapping

HL7 v2 Mapping

ServD

Mappings for the practitioner resource.

8.4.9.1 Mappings for RIM Mapping (http://hl7.org/v3)

Practitioner | Role
identifier | ./id
active | ./statusCode
name | ./name
telecom | ./telecom
address | ./addr
gender | ./administrativeGender
birthDate | ./birthTime
photo | ./subjectOf/ObservationEvent[code="photo"]/value
qualification | .playingEntity.playingRole[classCode=QUAL].code
identifier | .playingEntity.playingRole[classCode=QUAL].id
code | .playingEntity.playingRole[classCode=QUAL].code
period | .playingEntity.playingRole[classCode=QUAL].effectiveTime
issuer | .playingEntity.playingRole[classCode=QUAL].scoper

communication | ./languageCommunication

8.4.9.2 Mappings for HL7 v2 Mapping (http://hl7.org/v2)

Practitioner | PRD (as one example)
identifier | PRD-7 (or XCN.1)
active
name | XCN Components
telecom | PRT-15, STF-10, ROL-12
address | ORC-24, STF-11, ROL-11, PRT-14
gender | STF-5
birthDate | STF-6
photo
qualification | CER?
identifier
code
period
issuer
communication | PID-15, NK1-20, LAN-2

8.4.9.3 Mappings for ServD (http://www.omg.org/spec/ServD/1.0/)

Practitioner | Provider
identifier | ./Identifiers
active
name | ./PreferredName (GivenNames, FamilyName, TitleCode)
telecom | ./ContactPoints
address | ./Addresses
gender | ./GenderCode
birthDate | (not represented in ServD)
photo | ./ImageURI (only supports the URI reference)
qualification | ./Qualifications
identifier
code | ./Qualifications.Value
period | ./Qualifications.StartDate and ./Qualifications.EndDate
issuer
communication | ./Languages.LanguageSpokenCode

8.4.10 Resource Practitioner - Design Notes

URL: https://hl7.org/fhir/STU3/practitioner-explanations.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.4.10 Resource Practitioner - Design Notes

to do

8.4.11 Resource Practitioner - Extensions & Profiles

URL: https://hl7.org/fhir/STU3/practitioner-profiles.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.4.11 Resource Practitioner - Extensions & Profiles

Patient Administration Work Group | Maturity Level: N/A | Ballot Status: Informative

This table lists profiles and extensions for the Practitioner resource. For background information, see Profiling Resources and Extensibility. Additional profiles and resources may be found in published Implementation Guides , or in the Conformance resource registry .

8.4.11.1 Profiles

No Profiles defined for this resource

8.4.11.2 Extensions

practitioner-animalSpecies | animalSpecies | for Practitioner HL7 Extensions
practitioner-classification | classification | for Practitioner HL7 Extensions

Extensions that reference this resource:

communicationrequest-orderedBy | orderedBy | for CommunicationRequest HL7 Extensions
consent-Witness | Witness | for Consent HL7 Extensions
cqif-guidanceinitiatingPerson | initiatingPerson | for CQIF Guidance Extensions
cqif-guidancereceivingPerson | receivingPerson | for CQIF Guidance Extensions

goal-acceptance | acceptance | for FHIR Core Goal Profile
medicationstatement-Prescriber | Prescriber | for MedicationStatement HL7 Extensions
procedurerequest-authorizedBy | authorizedBy | for Procedure Request HL7 Extensions
questionnaireresponse-author | author | for Core extensions for QuestionnaireResponse
questionnaireresponse-reviewer | reviewer | for Core extensions for QuestionnaireResponse
task-candidateList | candidateList | for Task HL7 Extensions

8.4.11.3 Search Extensions

No Search Extensions defined for this resource

8.4.12 Practitioner HL7 Extensions

URL: https://hl7.org/fhir/STU3/practitioner-extensions.html

Defines common extensions used with or related to the Practitioner resource

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.4.12 Practitioner HL7 Extensions

(No assigned work group) | Maturity Level: N/A | Ballot Status: Informative

8.4.12.1 Content

Extensions:
practitioner-classification | classification : The more specific service or occupation within the health care provider type of the practitioner at a specific organization unit, for example, counselor.
practitioner-animalSpecies | animalSpecies : This extension should be used to specifiy that a practioner or RelatedPerson resource is a service animal.

8.4.13 Resource Practitioner - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/practitioner-version-maps.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.4.13 Resource Practitioner - Detailed Descriptions

Patient Administration Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Practitioner

R2 : R3 Converson maps for Practitioner.

Functional status for this map: 18 tests that all execute ok. 15 fail round-trip testing and all r3 resources are valid. (see documentation)

8.4.13.1 R2 to R3

```
map "http://hl7.org/fhir/StructureMap/Practitioner2to3" = "R2 to R3 Conversions for Practitioner"

uses "http://hl7.org/fhir/DSTU2/StructureDefinition/Practitioner" alias PractitionerR2 as source
uses "http://hl7.org/fhir/StructureDefinition/Practitioner" alias Practitioner as target

imports "http://hl7.org/fhir/StructureMap/*2to3"

group for type+types  Practitioner extends DomainResource
  input src : PractitionerR2 as source
  input tgt : Practitioner as target

  "Practitioner-identifier" : for src.identifier make tgt.identifier
  "Practitioner-active" : for src.active make tgt.active
  "Practitioner-name" : for src.name make tgt.name
  "Practitioner-telecom" : for src.telecom make tgt.telecom
  "Practitioner-address" : for src.address make tgt.address
  "Practitioner-gender" : for src.gender make tgt.gender
  "Practitioner-birthDate" : for src.birthDate make tgt.birthDate
  "Practitioner-photo" : for src.photo make tgt.photo
  "Practitioner-role" : for src.practitionerRole as vs0 make create("PractitionerRole") as vt0 then prcRole(src, tgt,
    vs0, vt0)
  "Practitioner-qualification" : for src.qualification as vs0 make tgt.qualification as vt0 then prcQualification(vs0,
    vt0)
  "Practitioner-communication" : for src.communication make tgt.communication
endgroup

group prcRole extends BackboneElement
  input pr as source // the source r2 practitioner resource
  input prtgt as target // the target r3 practitioner resource
  input src as source // the source practitioner role
  input tgt as target // the target PractitionerRole resource

  "Practitioner.role-id" : for src make tgt.practitioner = create("Reference") as p, p.reference =
    ('Practitioner/'+pr.id)
  "Practitioner.role-organization" : for src.managingOrganization make tgt.organization
```

```
  "Practitioner.role-code" : for src.role make tgt.code
  "Practitioner.role-specialty" : for src.specialty make tgt.specialty
  "Practitioner.role-period" : for src.period make tgt.period
  "Practitioner.role-location" : for src.location make tgt.location
  "Practitioner.role-healthcareService" : for src.healthcareService make tgt.healthcareService
endgroup

group prcQualification extends BackboneElement
  input src as source
  input tgt as target

  "Practitioner.qualification-identifier" : for src.identifier make tgt.identifier
  "Practitioner.qualification-code" : for src.code make tgt.code
  "Practitioner.qualification-period" : for src.period make tgt.period
  "Practitioner.qualification-issuer" : for src.issuer make tgt.issuer
endgroup
```

8.4.13.2 R3 to R2

```
map "http://hl7.org/fhir/StructureMap/Practitioner3to2" = "R3 to R2 Conversion for Practitioner"

uses "http://hl7.org/fhir/StructureDefinition/Practitioner" alias Practitioner as source
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/Practitioner" alias PractitionerR2 as target

imports "http://hl7.org/fhir/StructureMap/*3to2"

group for type+types  Practitioner extends DomainResource
  input src : Practitioner as source
  input tgt : PractitionerR2 as target

  "Practitioner-identifier" : for src.identifier make tgt.identifier
  "Practitioner-active" : for src.active make tgt.active
  "Practitioner-name" : for src.name make tgt.name
  "Practitioner-telecom" : for src.telecom make tgt.telecom
  "Practitioner-address" : for src.address make tgt.address
  "Practitioner-gender" : for src.gender make tgt.gender
  "Practitioner-birthDate" : for src.birthDate make tgt.birthDate
  "Practitioner-photo" : for src.photo make tgt.photo
  "Practitioner-role" : for search('/PractitionerRole?practitioner='+src.id) : PractitionerRole as vs0 make
    tgt.practitionerRole as vt0 then prcRole(vs0, vt0)
  "Practitioner-qualification" : for src.qualification as vs0 make tgt.qualification as vt0 then prcQualification(vs0,
    vt0)
  "Practitioner-communication" : for src.communication make tgt.communication
endgroup

group prcRole extends BackboneElement
  input src as source
  input tgt as target

  "Practitioner.role-organization" : for src.organization make tgt.managingOrganization
  "Practitioner.role-code" : for src.code make tgt.role
  "Practitioner.role-specialty" : for src.specialty make tgt.specialty
  "Practitioner.role-period" : for src.period make tgt.period
  "Practitioner.role-location" : for src.location make tgt.location
  "Practitioner.role-healthcareService" : for src.healthcareService make tgt.healthcareService
endgroup

group prcQualification extends BackboneElement
  input src as source
  input tgt as target

  "Practitioner.qualification-identifier" : for src.identifier make tgt.identifier
  "Practitioner.qualification-code" : for src.code make tgt.code
  "Practitioner.qualification-period" : for src.period make tgt.period
  "Practitioner.qualification-issuer" : for src.issuer make tgt.issuer
endgroup
```

8.4.13.3 R3 Validation Errors

8.5 Resource PractitionerRole - Content

URL: https://hl7.org/fhir/STU3/practitionerrole.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

Patient Administration Work Group | Maturity Level: 2 | Trial Use | Compartments: Practitioner

A specific set of Roles/Locations/specialties/services that a practitioner may perform at an organization for a period of time.

8.5.1 Scope and Usage

PractitionerRole covers the recording of the location and types of services that Practitioners are able to provide for an organization.

The role, specialty, Location telecom and HealthcareService properties can be repeated if required in other instances of the PractitionerRole. Some systems record a collection of service values for a single location, others record the single service and the list of locations it is available. Both are acceptable options for representing this data.

8.5.2 Boundaries and Relationships

Qualifications (from the Practitioner resource) do not imply a Role, but might be considered when an Organization allocates practitioners to roles within their organization, and could provide useful information (such as expiry information) which could need to be tracked in some situations to ensure they continue to be eligible for a specific role.

The CareTeam resource is also often used to provide details of a role that a practitioner is allocated to perform, but is usually scoped to a much finer granularity of care, and often within the specific context of a Patient, or functional role (e.g. Crisis planning team). In contrast the PractitionerRole is used in a more general sense to cover all the places that the practitioner is allocated to work (and specific details relevant to that role - such as a specific contact number, or electronic services endpoint).

8.5.3 Background and Context

Practitioner performs different roles within the same or even different organizations. Depending on jurisdiction and custom, it may be necessary to maintain a specific Practitioner Resource for each such role or have a single Practitioner with multiple roles. The role can be limited to a specific period, after which authorization for this role ends. Note that the represented organization need not necessarily be the (direct) employer of a Practitioner.

This resource is referenced by AdverseEvent and Schedule

8.5.4 Resource Content

* Structure

* UML

* XML

* JSON

* Turtle

* R2 Diff

* All

Structure

Name | Flags | Card. | Type | Description & Constraints
PractitionerRole | | | DomainResource | Roles/organizations the practitioner is associated with Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | ? | 0..* | Identifier | Business Identifiers that are specific to a role/location
active | ? | 0..1 | boolean | Whether this practitioner's record is in active use
period | ? | 0..1 | Period | The period during which the practitioner is authorized to perform in these role(s)
practitioner | ? | 0..1 | Reference(Practitioner) | Practitioner that is able to provide the defined services for the organation
organization | ? | 0..1 | Reference(Organization) | Organization where the roles are available
code | ? | 0..* | CodeableConcept | Roles which this practitioner may perform PractitionerRole (Example)
specialty | ? | 0..* | CodeableConcept | Specific specialty of the practitioner Practice Setting Code Value Set (Preferred)
location | ? | 0..* | Reference(Location) | The location(s) at which this practitioner provides care
healthcareService | | 0..* | Reference(HealthcareService) | The list of healthcare services that this worker provides for this role's Organization/Location(s)
telecom | ? | 0..* | ContactPoint | Contact details that are specific to the role/location/service
availableTime | | 0..* | BackboneElement | Times the Service Site is available
daysOfWeek | | 0..* | code | mon | tue | wed | thu | fri | sat | sun DaysOfWeek (Required)
allDay | | 0..1 | boolean | Always available? e.g. 24 hour service
availableStartTime | | 0..1 | time | Opening time of day (ignored if allDay = true)
availableEndTime | | 0..1 | time | Closing time of day (ignored if allDay = true)
notAvailable | | 0..* | BackboneElement | Not available during this time due to provided reason
description | | 1..1 | string | Reason presented to the user explaining why time not available
during | | 0..1 | Period | Service not availablefrom this date
availabilityExceptions | | 0..1 | string | Description of availability exceptions
endpoint | | 0..* | Reference(Endpoint) | Technical endpoints providing access to services operated for the practitioner with this role
Documentation for this format

UML Diagram (Legend)

XML Template

```
<PractitionerRole xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Business Identifiers that are specific to a role/location --></identifier>
 <active value="[boolean]"/><!-- 0..1 Whether this practitioner's record is in active use -->
 <period><!-- 0..1 Period The period during which the practitioner is authorized to perform in these role(s)
    --></period>
 <practitioner><!-- 0..1 Reference(Practitioner) Practitioner that is able to provide the defined services for the
    organation --></practitioner>
 <organization><!-- 0..1 Reference(Organization) Organization where the roles are available --></organization>
 <code><!-- 0..* CodeableConcept Roles which this practitioner may perform --></code>
 <specialty><!-- 0..* CodeableConcept Specific specialty of the practitioner --></specialty>
 <location><!-- 0..* Reference(Location) The location(s) at which this practitioner provides care --></location>
 <healthcareService><!-- 0..* Reference(HealthcareService) The list of healthcare services that this worker provides for
     this role's Organization/Location(s) --></healthcareService>
```

```
 <telecom><!-- 0..* ContactPoint Contact details that are specific to the role/location/service --></telecom>
 <availableTime>  <!-- 0..* Times the Service Site is available -->
  <daysOfWeek value="[code]"/><!-- 0..* mon | tue | wed | thu | fri | sat | sun -->
  <allDay value="[boolean]"/><!-- 0..1 Always available? e.g. 24 hour service -->
  <availableStartTime value="[time]"/><!-- 0..1 Opening time of day (ignored if allDay = true) -->
  <availableEndTime value="[time]"/><!-- 0..1 Closing time of day (ignored if allDay = true) -->
 </availableTime>
 <notAvailable>  <!-- 0..* Not available during this time due to provided reason -->
  <description value="[string]"/><!-- 1..1 Reason presented to the user explaining why time not available -->
  <during><!-- 0..1 Period Service not availablefrom this date --></during>
 </notAvailable>
 <availabilityExceptions value="[string]"/><!-- 0..1 Description of availability exceptions -->
 <endpoint><!-- 0..* Reference(Endpoint) Technical endpoints providing access to services operated for the practitioner
    with this role --></endpoint>
</PractitionerRole>
```

JSON Template

```
{
  "resourceType" : "PractitionerRole",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Business Identifiers that are specific to a role/location
  "active" : <boolean>, // Whether this practitioner's record is in active use
  "period" : { Period }, // The period during which the practitioner is authorized to perform in these role(s)
  "practitioner" : { Reference(Practitioner) }, // Practitioner that is able to provide the defined services for the
    organation
  "organization" : { Reference(Organization) }, // Organization where the roles are available
  "code" : [{ CodeableConcept }], // Roles which this practitioner may perform
  "specialty" : [{ CodeableConcept }], // Specific specialty of the practitioner
  "location" : [{ Reference(Location) }], // The location(s) at which this practitioner provides care
  "healthcareService" : [{ Reference(HealthcareService) }], // The list of healthcare services that this worker provides
     for this role's Organization/Location(s)
  "telecom" : [{ ContactPoint }], // Contact details that are specific to the role/location/service
  "availableTime" : [{ // Times the Service Site is available
    "daysOfWeek" : ["<code>"], // mon | tue | wed | thu | fri | sat | sun
    "allDay" : <boolean>, // Always available? e.g. 24 hour service
    "availableStartTime" : "<time>", // Opening time of day (ignored if allDay = true)
    "availableEndTime" : "<time>" // Closing time of day (ignored if allDay = true)
  }],
  "notAvailable" : [{ // Not available during this time due to provided reason
    "description" : "<string>", // R!  Reason presented to the user explaining why time not available
    "during" : { Period } // Service not availablefrom this date
  }],
  "availabilityExceptions" : "<string>", // Description of availability exceptions
  "endpoint" : [{ Reference(Endpoint) }] // Technical endpoints providing access to services operated for the
    practitioner with this role
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:PractitionerRole;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:PractitionerRole.identifier [ Identifier ], ... ; # 0..* Business Identifiers that are specific to a
    role/location
  fhir:PractitionerRole.active [ boolean ]; # 0..1 Whether this practitioner's record is in active use
  fhir:PractitionerRole.period [ Period ]; # 0..1 The period during which the practitioner is authorized to perform in
    these role(s)
  fhir:PractitionerRole.practitioner [ Reference(Practitioner) ]; # 0..1 Practitioner that is able to provide the
    defined services for the organation
  fhir:PractitionerRole.organization [ Reference(Organization) ]; # 0..1 Organization where the roles are available
  fhir:PractitionerRole.code [ CodeableConcept ], ... ; # 0..* Roles which this practitioner may perform
  fhir:PractitionerRole.specialty [ CodeableConcept ], ... ; # 0..* Specific specialty of the practitioner
  fhir:PractitionerRole.location [ Reference(Location) ], ... ; # 0..* The location(s) at which this practitioner
    provides care
  fhir:PractitionerRole.healthcareService [ Reference(HealthcareService) ], ... ; # 0..* The list of healthcare services
     that this worker provides for this role's Organization/Location(s)
  fhir:PractitionerRole.telecom [ ContactPoint ], ... ; # 0..* Contact details that are specific to the
    role/location/service
  fhir:PractitionerRole.availableTime [ # 0..* Times the Service Site is available
    fhir:PractitionerRole.availableTime.daysOfWeek [ code ], ... ; # 0..* mon | tue | wed | thu | fri | sat | sun
    fhir:PractitionerRole.availableTime.allDay [ boolean ]; # 0..1 Always available? e.g. 24 hour service
    fhir:PractitionerRole.availableTime.availableStartTime [ time ]; # 0..1 Opening time of day (ignored if allDay =
    true)
    fhir:PractitionerRole.availableTime.availableEndTime [ time ]; # 0..1 Closing time of day (ignored if allDay = true)
  ], ...;
  fhir:PractitionerRole.notAvailable [ # 0..* Not available during this time due to provided reason
    fhir:PractitionerRole.notAvailable.description [ string ]; # 1..1 Reason presented to the user explaining why time
    not available
    fhir:PractitionerRole.notAvailable.during [ Period ]; # 0..1 Service not availablefrom this date
  ], ...;
  fhir:PractitionerRole.availabilityExceptions [ string ]; # 0..1 Description of availability exceptions
  fhir:PractitionerRole.endpoint [ Reference(Endpoint) ], ... ; # 0..* Technical endpoints providing access to services
    operated for the practitioner with this role
]
```

Changes since DSTU2

This resource did not exist in Release 2

This analysis is available as XML or JSON.

Structure

Name | Flags | Card. | Type | Description & Constraints

PractitionerRole | | | DomainResource | Roles/organizations the practitioner is associated with Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension

identifier | ? | 0..* | Identifier | Business Identifiers that are specific to a role/location

active | ? | 0..1 | boolean | Whether this practitioner's record is in active use

period | ? | 0..1 | Period | The period during which the practitioner is authorized to perform in these role(s)

practitioner | ? | 0..1 | Reference(Practitioner) | Practitioner that is able to provide the defined services for the organation

organization | ? | 0..1 | Reference(Organization) | Organization where the roles are available

code | ? | 0..* | CodeableConcept | Roles which this practitioner may perform PractitionerRole (Example)

specialty | ? | 0..* | CodeableConcept | Specific specialty of the practitioner Practice Setting Code Value Set (Preferred)

location | ? | 0..* | Reference(Location) | The location(s) at which this practitioner provides care

healthcareService | | 0..* | Reference(HealthcareService) | The list of healthcare services that this worker provides for this role's Organization/Location(s)

telecom | ? | 0..* | ContactPoint | Contact details that are specific to the role/location/service

availableTime | | 0..* | BackboneElement | Times the Service Site is available

daysOfWeek | | 0..* | code | mon | tue | wed | thu | fri | sat | sun DaysOfWeek (Required)

allDay | | 0..1 | boolean | Always available? e.g. 24 hour service

availableStartTime | | 0..1 | time | Opening time of day (ignored if allDay = true)

availableEndTime | | 0..1 | time | Closing time of day (ignored if allDay = true)

notAvailable | | 0..* | BackboneElement | Not available during this time due to provided reason

description | | 1..1 | string | Reason presented to the user explaining why time not available

during | | 0..1 | Period | Service not availablefrom this date

availabilityExceptions | | 0..1 | string | Description of availability exceptions

endpoint | | 0..* | Reference(Endpoint) | Technical endpoints providing access to services operated for the practitioner with this role

Documentation for this format

UML Diagram (Legend)

XML Template

```
<PractitionerRole xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Business Identifiers that are specific to a role/location --></identifier>
 <active value="[boolean]"/><!-- 0..1 Whether this practitioner's record is in active use -->
 <period><!-- 0..1 Period The period during which the practitioner is authorized to perform in these role(s)
    --></period>
 <practitioner><!-- 0..1 Reference(Practitioner) Practitioner that is able to provide the defined services for the
    organation --></practitioner>
 <organization><!-- 0..1 Reference(Organization) Organization where the roles are available --></organization>
 <code><!-- 0..* CodeableConcept Roles which this practitioner may perform --></code>
 <specialty><!-- 0..* CodeableConcept Specific specialty of the practitioner --></specialty>
 <location><!-- 0..* Reference(Location) The location(s) at which this practitioner provides care --></location>
 <healthcareService><!-- 0..* Reference(HealthcareService) The list of healthcare services that this worker provides for
     this role's Organization/Location(s) --></healthcareService>
 <telecom><!-- 0..* ContactPoint Contact details that are specific to the role/location/service --></telecom>
 <availableTime>  <!-- 0..* Times the Service Site is available -->
  <daysOfWeek value="[code]"/><!-- 0..* mon | tue | wed | thu | fri | sat | sun -->
  <allDay value="[boolean]"/><!-- 0..1 Always available? e.g. 24 hour service -->
  <availableStartTime value="[time]"/><!-- 0..1 Opening time of day (ignored if allDay = true) -->
  <availableEndTime value="[time]"/><!-- 0..1 Closing time of day (ignored if allDay = true) -->
 </availableTime>
 <notAvailable>  <!-- 0..* Not available during this time due to provided reason -->
  <description value="[string]"/><!-- 1..1 Reason presented to the user explaining why time not available -->
  <during><!-- 0..1 Period Service not availablefrom this date --></during>
 </notAvailable>
 <availabilityExceptions value="[string]"/><!-- 0..1 Description of availability exceptions -->
 <endpoint><!-- 0..* Reference(Endpoint) Technical endpoints providing access to services operated for the practitioner
    with this role --></endpoint>
</PractitionerRole>
```

JSON Template

```
{
  "resourceType" : "PractitionerRole",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Business Identifiers that are specific to a role/location
  "active" : <boolean>, // Whether this practitioner's record is in active use
  "period" : { Period }, // The period during which the practitioner is authorized to perform in these role(s)
  "practitioner" : { Reference(Practitioner) }, // Practitioner that is able to provide the defined services for the
    organation
  "organization" : { Reference(Organization) }, // Organization where the roles are available
  "code" : [{ CodeableConcept }], // Roles which this practitioner may perform
  "specialty" : [{ CodeableConcept }], // Specific specialty of the practitioner
  "location" : [{ Reference(Location) }], // The location(s) at which this practitioner provides care
  "healthcareService" : [{ Reference(HealthcareService) }], // The list of healthcare services that this worker provides
     for this role's Organization/Location(s)
  "telecom" : [{ ContactPoint }], // Contact details that are specific to the role/location/service
```

```
  "availableTime" : [{ // Times the Service Site is available
    "daysOfWeek" : ["<code>"], // mon | tue | wed | thu | fri | sat | sun
    "allDay" : <boolean>, // Always available? e.g. 24 hour service
    "availableStartTime" : "<time>", // Opening time of day (ignored if allDay = true)
    "availableEndTime" : "<time>" // Closing time of day (ignored if allDay = true)
  }],
  "notAvailable" : [{ // Not available during this time due to provided reason
    "description" : "<string>", // R!  Reason presented to the user explaining why time not available
    "during" : { Period } // Service not availablefrom this date
  }],
  "availabilityExceptions" : "<string>", // Description of availability exceptions
  "endpoint" : [{ Reference(Endpoint) }] // Technical endpoints providing access to services operated for the
    practitioner with this role
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:PractitionerRole;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:PractitionerRole.identifier [ Identifier ], ... ; # 0..* Business Identifiers that are specific to a
    role/location
  fhir:PractitionerRole.active [ boolean ]; # 0..1 Whether this practitioner's record is in active use
  fhir:PractitionerRole.period [ Period ]; # 0..1 The period during which the practitioner is authorized to perform in
    these role(s)
  fhir:PractitionerRole.practitioner [ Reference(Practitioner) ]; # 0..1 Practitioner that is able to provide the
    defined services for the organation
  fhir:PractitionerRole.organization [ Reference(Organization) ]; # 0..1 Organization where the roles are available
  fhir:PractitionerRole.code [ CodeableConcept ], ... ; # 0..* Roles which this practitioner may perform
  fhir:PractitionerRole.specialty [ CodeableConcept ], ... ; # 0..* Specific specialty of the practitioner
  fhir:PractitionerRole.location [ Reference(Location) ], ... ; # 0..* The location(s) at which this practitioner
    provides care
  fhir:PractitionerRole.healthcareService [ Reference(HealthcareService) ], ... ; # 0..* The list of healthcare services
     that this worker provides for this role's Organization/Location(s)
  fhir:PractitionerRole.telecom [ ContactPoint ], ... ; # 0..* Contact details that are specific to the
    role/location/service
  fhir:PractitionerRole.availableTime [ # 0..* Times the Service Site is available
    fhir:PractitionerRole.availableTime.daysOfWeek [ code ], ... ; # 0..* mon | tue | wed | thu | fri | sat | sun
    fhir:PractitionerRole.availableTime.allDay [ boolean ]; # 0..1 Always available? e.g. 24 hour service
    fhir:PractitionerRole.availableTime.availableStartTime [ time ]; # 0..1 Opening time of day (ignored if allDay =
    true)
    fhir:PractitionerRole.availableTime.availableEndTime [ time ]; # 0..1 Closing time of day (ignored if allDay = true)
  ], ...;
  fhir:PractitionerRole.notAvailable [ # 0..* Not available during this time due to provided reason
    fhir:PractitionerRole.notAvailable.description [ string ]; # 1..1 Reason presented to the user explaining why time
    not available
    fhir:PractitionerRole.notAvailable.during [ Period ]; # 0..1 Service not availablefrom this date
  ], ...;
  fhir:PractitionerRole.availabilityExceptions [ string ]; # 0..1 Description of availability exceptions
  fhir:PractitionerRole.endpoint [ Reference(Endpoint) ], ... ; # 0..* Technical endpoints providing access to services
    operated for the practitioner with this role
]
```

Changes since DSTU2

This resource did not exist in Release 2

This analysis is available as XML or JSON.

Alternate definitions: Master Definition (XML, JSON), XML Schema/Schematron (for) + JSON Schema, ShEx (for Turtle)

8.5.4.1 Terminology Bindings

Path	Definition	Type	Reference
PractitionerRole.code	The role a person plays representing an organization	Example	PractitionerRole
PractitionerRole.specialty	Specific specialty associated with the agency	Preferred	Practice Setting Code Value Set
PractitionerRole.availableTime.daysOfWeek	The days of the week.	Required	DaysOfWeek

8.5.5 Notes:

* There is no address on the PractitionerRole as the location that is defined here contains the address. This prevents having to duplicate the address values across multiple resources.

8.5.6 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name	Type	Description	Expression	In Common
active	token	Whether this practitioner's record is in active use	PractitionerRole.active	
date	date	The period during which the practitioner is authorized to perform in these role(s)	PractitionerRole.period	
email	token	A value in an email contact	PractitionerRole.telecom.where(system='email')	4 Resources

endpoint | reference | Technical endpoints providing access to services operated for the practitioner with this role | PractitionerRole.endpoint (Endpoint)
identifier | token | A practitioner's Identifier | PractitionerRole.identifier
location | reference | One of the locations at which this practitioner provides care | PractitionerRole.location (Location)
organization | reference | The identity of the organization the practitioner represents / acts on behalf of | PractitionerRole.organization (Organization)
phone | token | A value in a phone contact | PractitionerRole.telecom.where(system='phone') | 4 Resources
practitioner | reference | Practitioner that is able to provide the defined services for the organation | PractitionerRole.practitioner (Practitioner)
role | token | The practitioner can perform this role at for the organization | PractitionerRole.code
service | reference | The list of healthcare services that this worker provides for this role's Organization/Location(s) | PractitionerRole.healthcareService (HealthcareService)
specialty | token | The practitioner has this specialty at an organization | PractitionerRole.specialty
telecom | token | The value in any kind of contact | PractitionerRole.telecom | 4 Resources

8.5.7 Resource PractitionerRole - Examples

URL: https://hl7.org/fhir/STU3/practitionerrole-examples.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.5.7 Resource PractitionerRole - Examples

Patient Administration Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Practitioner

Example List:

Dr Adam Careful's role at the Acme Hospital | example | XML | JSON | Turtle
HL7 Defined Practitioner Role examples | 3ad0687e-f477-468c-afd5-fcc2bf897808 | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

8.5.8 Resource PractitionerRole - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/practitionerrole-definitions.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.5.8 Resource PractitionerRole - Detailed Descriptions

Patient Administration Work Group | Maturity Level: 2 | Trial Use | Compartments: Practitioner

Detailed Descriptions for the elements in the PractitionerRole resource.

PractitionerRole
Definition | A specific set of Roles/Locations/specialties/services that a practitioner may perform at an organization for a period of time.
Control | 1..1
Requirements | Need to track services that a healthcare provider is able to provide at an organization's location, and the services that they can perform there.
PractitionerRole.identifier
Definition | Business Identifiers that are specific to a role/location.
Note | This is a business identifer, not a resource identifier (see discussion)
Control | 0..*
Type | Identifier
Requirements | Often, specific identities are assigned for the agent.
Summary | true
PractitionerRole.active
Definition | Whether this practitioner's record is in active use.
Control | 0..1
Type | boolean
Default Value | true
Requirements | Need to be able to mark a practitioner record as not to be used because it was created in error.
Summary | true
Comments | Default is true. If the practitioner is not in use by one organization, then it should mark the period on the PractitonerRole with an end date (even if they are active) as they may be active in another role.
PractitionerRole.period
Definition | The period during which the person is authorized to act as a practitioner in these role(s) for the organization.
Control | 0..1
Type | Period
Requirements | Even after the agencies is revoked, the fact that it existed must still be recorded.
Summary | true
PractitionerRole.practitioner
Definition | Practitioner that is able to provide the defined services for the organation.
Control | 0..1
Type | Reference(Practitioner)
Summary | true
PractitionerRole.organization

Definition | The organization where the Practitioner performs the roles associated.
Control | 0..1
Type | Reference(Organization)
Summary | true
PractitionerRole.code
Definition | Roles which this practitioner is authorized to perform for the organization.
Control | 0..*
Terminology Binding | PractitionerRole (Example)
Type | CodeableConcept
Requirements | Need to know what authority the practitioner has - what can they do?
Summary | true
Comments | A person may have more than one role. At least one role is required.
PractitionerRole.specialty
Definition | Specific specialty of the practitioner.
Control | 0..*
Terminology Binding | Practice Setting Code Value Set (Preferred)
Type | CodeableConcept
Summary | true
PractitionerRole.location
Definition | The location(s) at which this practitioner provides care.
Control | 0..*
Type | Reference(Location)
Summary | true
PractitionerRole.healthcareService
Definition | The list of healthcare services that this worker provides for this role's Organization/Location(s).
Control | 0..*
Type | Reference(HealthcareService)
PractitionerRole.telecom
Definition | Contact details that are specific to the role/location/service.
Control | 0..*
Type | ContactPoint
Requirements | Often practitioners have a dedicated line for each location (or service) that they work at, and need to be able to define separate contact details for each of these.
Summary | true
PractitionerRole.availableTime
Definition | A collection of times that the Service Site is available.
Control | 0..*
Comments | More detailed availability information may be provided in associated Schedule/Slot resources.
PractitionerRole.availableTime.daysOfWeek
Definition | Indicates which days of the week are available between the start and end Times.
Control | 0..*
Terminology Binding | DaysOfWeek (Required)
Type | code
PractitionerRole.availableTime.allDay
Definition | Is this always available? (hence times are irrelevant) e.g. 24 hour service.
Control | 0..1
Type | boolean
PractitionerRole.availableTime.availableStartTime
Definition | The opening time of day. Note: If the AllDay flag is set, then this time is ignored.
Control | 0..1
Type | time
Comments | The timezone is expected to be for where this HealthcareService is provided at.
PractitionerRole.availableTime.availableEndTime
Definition | The closing time of day. Note: If the AllDay flag is set, then this time is ignored.
Control | 0..1
Type | time
Comments | The timezone is expected to be for where this HealthcareService is provided at.
PractitionerRole.notAvailable
Definition | The HealthcareService is not available during this period of time due to the provided reason.
Control | 0..*
PractitionerRole.notAvailable.description
Definition | The reason that can be presented to the user as to why this time is not available.
Control | 1..1
Type | string
PractitionerRole.notAvailable.during
Definition | Service is not available (seasonally or for a public holiday) from this date.
Control | 0..1
Type | Period
PractitionerRole.availabilityExceptions
Definition | A description of site availability exceptions, e.g. public holiday availability. Succinctly describing all possible exceptions to normal site availability as details in the available Times and not available Times.
Control | 0..1

Type | string
PractitionerRole.endpoint
Definition | Technical endpoints providing access to services operated for the practitioner with this role.
Control | 0..*
Type | Reference(Endpoint)
Requirements | Organizations have multiple systems that provide various services and ,ay also be different for practitioners too. So the endpoint satisfies the need to be able to define the technical connection details for how to connect to them, and for what purpose.

8.5.9 Resource PractitionerRole - Mappings

URL: https://hl7.org/fhir/STU3/practitionerrole-mappings.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.5.9 Resource PractitionerRole - Mappings

Patient Administration Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Practitioner

Mappings:

RIM Mapping

HL7 v2 Mapping

ServD

Mappings for the practitionerrole resource.

8.5.9.1 Mappings for RIM Mapping (http://hl7.org/v3)

PractitionerRole | Role
identifier | .id
active | .statusCode
period | .performance[@typeCode <= 'PPRF'].ActDefinitionOrEvent.effectiveTime
practitioner | .player
organization | .scoper
code | .code
specialty | .player.HealthCareProvider[@classCode = 'PROV'].code
location | .performance.ActDefinitionOrEvent.ServiceDeliveryLocation[@classCode = 'SDLOC']
healthcareService | .player.QualifiedEntity[@classCode = 'QUAL'].code
telecom | .telecom
availableTime | .effectiveTime
daysOfWeek | .effectiveTime
allDay | .effectiveTime
availableStartTime | .effectiveTime
availableEndTime | .effectiveTime
notAvailable | .effectiveTime
description | n/a
during | .effectiveTime
availabilityExceptions | .effectiveTime
endpoint | n/a

8.5.9.2 Mappings for HL7 v2 Mapping (http://hl7.org/v2)

PractitionerRole | PRD (as one example)
identifier | PRD-7 (or XCN.1)
active | STF-7
period | PRD-8/9 / PRA-5.4
practitioner
organization
code | PRD-1 / STF-18 / PRA-3 / PRT-4 / ROL-3 / ORC-12 / OBR-16 / PV1-7 / PV1-8 / PV1-9 / PV1-17
specialty | PRA-5
location
healthcareService | EDU-2 / AFF-3
telecom
availableTime
daysOfWeek
allDay
availableStartTime
availableEndTime
notAvailable
description
during
availabilityExceptions
endpoint

8.5.9.3 Mappings for ServD (http://www.omg.org/spec/ServD/1.0/)

PractitionerRole | ServiceSiteProvider
identifier | ./Identifiers
active
period | (ServD maps Practitioners and Organizations via another entity, so this concept is not available)
practitioner
organization
code | (ServD maps Practitioners and Organizations via another entity, so this concept is not available)
specialty | ./Specialty
location | (ServD maps Practitioners and Organizations via another entity, so this concept is not available)
 However these are accessed via the Site.ServiceSite.ServiceSiteProvider record. (The Site has the location)
healthcareService
telecom
availableTime
daysOfWeek
allDay
availableStartTime
availableEndTime
notAvailable
description
during
availabilityExceptions
endpoint

8.5.10 Resource PractitionerRole - Design Notes

URL: https://hl7.org/fhir/STU3/practitionerrole-explanations.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.5.10 Resource PractitionerRole - Design Notes

to do

8.5.11 Resource PractitionerRole - Extensions & Profiles

URL: https://hl7.org/fhir/STU3/practitionerrole-profiles.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.5.11 Resource PractitionerRole - Extensions & Profiles

Patient Administration Work Group | Maturity Level: N/A | Ballot Status: Informative

This table lists profiles and extensions for the PractitionerRole resource. For background information, see Profiling Resources and Extensibility. Additional profiles and resources may be found in published Implementation Guides , or in the Conformance resource registry .

8.5.11.1 Profiles

No Profiles defined for this resource

8.5.11.2 Extensions

practitionerrole-primaryInd | primaryInd | for PractitionerRole HL7 Extensions

Extensions that reference this resource:

No Extensions refer to this resource

8.5.11.3 Search Extensions

No Search Extensions defined for this resource

8.5.12 PractitionerRole HL7 Extensions

URL: https://hl7.org/fhir/STU3/practitionerrole-extensions.html

Defines common extensions used with or related to the PractitionerRole resource

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.5.12 PractitionerRole HL7 Extensions

(No assigned work group) | Maturity Level: N/A | Ballot Status: Informative

8.5.12.1 Content

Extensions:
practitionerrole-primaryInd | primaryInd : Flag indicating if the specialty is the primary specialty of the provider. Normally, a practitioner will have one primary specialty, but in some cases more than one can be primary.

8.6 Resource Organization - Content

URL: https://hl7.org/fhir/STU3/organization.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

Patient Administration Work Group | Maturity Level: 3 | Trial Use | Compartments: Not linked to any defined compartments

A formally or informally recognized grouping of people or organizations formed for the purpose of achieving some form of collective action. Includes companies, institutions, corporations, departments, community groups, healthcare practice groups, etc.

8.6.1 Scope and Usage

This resource may be used in a shared registry of contact and other information for various organizations or it can be used merely as a support for other resources that need to reference organizations, perhaps as a document, message or as a contained resource. If using a registry approach, it's entirely possible for multiple registries to exist, each dealing with different types or levels of organization.

8.6.2 Boundaries and Relationships

The Organization resource is used for collections of people that have come together to achieve an objective. The Group resource is used to identify a collection of people (or animals, devices, etc.) that are gathered for the purpose of analysis or acting upon, but are not expected to act themselves.

The Organization resource often exists as a hierarchy of organization resources, using the part-of property to provide the association of the child to its parent organization. This organizational hierarchy helps communicate the conceptual structure, whereas the Location resource provides the physical representation of the hierarchy. The linkage between Organization and Location is from each point in the location hierarchy to the appropriate level in the Organization hierarchy. These links don't all have to be to the top level Organization. When populating the organization and location hierarchies there is often not a clear distinction between these 2, however to assist in making the decision, Locations are always used for recording where a service occurs, and hence where encounters and observations are associated. The Organization property on these resources may not be the location where the service took place.

This resource is referenced by Account, AuditEvent, CarePlan, CareTeam, ChargeItem, Claim, ClaimResponse, Communication, CommunicationRequest, Composition, Consent, Contract, Coverage, Device, DeviceRequest, DiagnosticReport, DocumentManifest, DocumentReference, EligibilityRequest, EligibilityResponse, Encounter, Endpoint, EnrollmentRequest, EnrollmentResponse, EpisodeOfCare, ExplanationOfBenefit, Flag, Goal, HealthcareService, ImagingManifest, Immunization, ImmunizationRecommendation, Linkage, Location, MeasureReport, Medication, MedicationAdministration, MedicationDispense, MedicationRequest, MedicationStatement, MessageHeader, Observation, Patient, PaymentNotice, PaymentReconciliation, Person, Practitioner, PractitionerRole, Procedure, ProcedureRequest, ProcessRequest, ProcessResponse, Provenance, ReferralRequest, ResearchStudy, Sequence, SupplyDelivery, SupplyRequest and Task

8.6.3 Resource Content

* Structure
* UML
* XML
* JSON
* Turtle
* R2 Diff
* All

Structure

Name | Flags | Card. | Type | Description & Constraints
Organization | I | | DomainResource | A grouping of people or organizations with a common purpose + The organization SHALL at least have a name or an id, and possibly more than one Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | ?I | 0..* | Identifier | Identifies this organization across multiple systems
active | ?!? | 0..1 | boolean | Whether the organization's record is still in active use
type | ? | 0..* | CodeableConcept | Kind of organization OrganizationType (Example)
name | ?I | 0..1 | string | Name used for the organization
alias | | 0..* | string | A list of alternate names that the organization is known as, or was known as in the past
telecom | I | 0..* | ContactPoint | A contact detail for the organization + The telecom of an organization can never be of use 'home'
address | I | 0..* | Address | An address for the organization + An address of an organization can never be of use 'home'
partOf | ? | 0..1 | Reference(Organization) | The organization of which this organization forms a part
contact | | 0..* | BackboneElement | Contact for the organization for a certain purpose
purpose | | 0..1 | CodeableConcept | The type of contact ContactEntityType (Extensible)

name | | 0..1 | HumanName | A name associated with the contact
telecom | | 0..* | ContactPoint | Contact details (telephone, email, etc.) for a contact
address | | 0..1 | Address | Visiting or postal addresses for the contact
endpoint | | 0..* | Reference(Endpoint) | Technical endpoints providing access to services operated for the organization
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Organization xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!--  0..* Identifier Identifies this organization  across multiple systems --></identifier>
 <active value="[boolean]"/><!-- 0..1 Whether the organization's record is still in active use -->
 <type><!-- 0..* CodeableConcept Kind of organization --></type>
 <name value="[string]"/><!--  0..1 Name used for the organization -->
 <alias value="[string]"/><!-- 0..* A list of alternate names that the organization is known as, or was known as in the
    past -->
 <telecom><!--  0..* ContactPoint A contact detail for the organization --></telecom>
 <address><!--  0..* Address An address for the organization --></address>
 <partOf><!-- 0..1 Reference(Organization) The organization of which this organization forms a part --></partOf>
 <contact>  <!-- 0..* Contact for the organization for a certain purpose -->
  <purpose><!-- 0..1 CodeableConcept The type of contact --></purpose>
  <name><!-- 0..1 HumanName A name associated with the contact --></name>
  <telecom><!-- 0..* ContactPoint Contact details (telephone, email, etc.)  for a contact --></telecom>
  <address><!-- 0..1 Address Visiting or postal addresses for the contact --></address>
 </contact>
 <endpoint><!-- 0..* Reference(Endpoint) Technical endpoints providing access to services operated for the organization
    --></endpoint>
</Organization>
```

JSON Template

```
{
  "resourceType" : "Organization",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // C? Identifies this organization  across multiple systems
  "active" : <boolean>, // Whether the organization's record is still in active use
  "type" : [{ CodeableConcept }], // Kind of organization
  "name" : "<string>", // C? Name used for the organization
  "alias" : ["<string>"], // A list of alternate names that the organization is known as, or was known as in the past
  "telecom" : [{ ContactPoint }], // C? A contact detail for the organization
  "address" : [{ Address }], // C? An address for the organization
  "partOf" : { Reference(Organization) }, // The organization of which this organization forms a part
  "contact" : [{ // Contact for the organization for a certain purpose
    "purpose" : { CodeableConcept }, // The type of contact
    "name" : { HumanName }, // A name associated with the contact
    "telecom" : [{ ContactPoint }], // Contact details (telephone, email, etc.)  for a contact
    "address" : { Address } // Visiting or postal addresses for the contact
  }],
  "endpoint" : [{ Reference(Endpoint) }] // Technical endpoints providing access to services operated for the
    organization
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Organization;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Organization.identifier [ Identifier ], ... ; # 0..* Identifies this organization  across multiple systems
  fhir:Organization.active [ boolean ]; # 0..1 Whether the organization's record is still in active use
  fhir:Organization.type [ CodeableConcept ], ... ; # 0..* Kind of organization
  fhir:Organization.name [ string ]; # 0..1 Name used for the organization
  fhir:Organization.alias [ string ], ... ; # 0..* A list of alternate names that the organization is known as, or was
    known as in the past
  fhir:Organization.telecom [ ContactPoint ], ... ; # 0..* A contact detail for the organization
  fhir:Organization.address [ Address ], ... ; # 0..* An address for the organization
  fhir:Organization.partOf [ Reference(Organization) ]; # 0..1 The organization of which this organization forms a part
  fhir:Organization.contact [ # 0..* Contact for the organization for a certain purpose
    fhir:Organization.contact.purpose [ CodeableConcept ]; # 0..1 The type of contact
    fhir:Organization.contact.name [ HumanName ]; # 0..1 A name associated with the contact
    fhir:Organization.contact.telecom [ ContactPoint ], ... ; # 0..* Contact details (telephone, email, etc.)  for a
    contact
    fhir:Organization.contact.address [ Address ]; # 0..1 Visiting or postal addresses for the contact
  ], ...;
  fhir:Organization.endpoint [ Reference(Endpoint) ], ... ; # 0..* Technical endpoints providing access to services
    operated for the organization
]
```

Changes since DSTU2

Organization
Organization.type | Max Cardinality changed from 1 to *
Organization.alias | Added Element

Organization.endpoint | Added Element

See the Full Difference for further information

This analysis is available as XML or JSON.

See R2 <--> R3 Conversion Maps (status = 14 tests that all execute ok. All tests pass round-trip testing and 2 r3 resources are invalid (2 errors).).

Structure

Name | Flags | Card. | Type | Description & Constraints
Organization | I | | DomainResource | A grouping of people or organizations with a common purpose + The organization SHALL at least have a name or an id, and possibly more than one Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | ?I | 0..* | Identifier | Identifies this organization across multiple systems
active | ?!? | 0..1 | boolean | Whether the organization's record is still in active use
type | ? | 0..* | CodeableConcept | Kind of organization OrganizationType (Example)
name | ?I | 0..1 | string | Name used for the organization
alias | | 0..* | string | A list of alternate names that the organization is known as, or was known as in the past
telecom | I | 0..* | ContactPoint | A contact detail for the organization + The telecom of an organization can never be of use 'home'
address | I | 0..* | Address | An address for the organization + An address of an organization can never be of use 'home'
partOf | ? | 0..1 | Reference(Organization) | The organization of which this organization forms a part
contact | | 0..* | BackboneElement | Contact for the organization for a certain purpose
purpose | | 0..1 | CodeableConcept | The type of contact ContactEntityType (Extensible)
name | | 0..1 | HumanName | A name associated with the contact
telecom | | 0..* | ContactPoint | Contact details (telephone, email, etc.) for a contact
address | | 0..1 | Address | Visiting or postal addresses for the contact
endpoint | | 0..* | Reference(Endpoint) | Technical endpoints providing access to services operated for the organization
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Organization xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!--  0..* Identifier Identifies this organization  across multiple systems --></identifier>
 <active value="[boolean]"/><!-- 0..1 Whether the organization's record is still in active use -->
 <type><!-- 0..* CodeableConcept Kind of organization --></type>
 <name value="[string]"/><!--  0..1 Name used for the organization -->
 <alias value="[string]"/><!-- 0..* A list of alternate names that the organization is known as, or was known as in the
    past -->
 <telecom><!--  0..* ContactPoint A contact detail for the organization --></telecom>
 <address><!--  0..* Address An address for the organization --></address>
 <partOf><!-- 0..1 Reference(Organization) The organization of which this organization forms a part --></partOf>
 <contact>  <!-- 0..* Contact for the organization for a certain purpose -->
  <purpose><!-- 0..1 CodeableConcept The type of contact --></purpose>
  <name><!-- 0..1 HumanName A name associated with the contact --></name>
  <telecom><!-- 0..* ContactPoint Contact details (telephone, email, etc.)  for a contact --></telecom>
  <address><!-- 0..1 Address Visiting or postal addresses for the contact --></address>
 </contact>
 <endpoint><!-- 0..* Reference(Endpoint) Technical endpoints providing access to services operated for the organization
    --></endpoint>
</Organization>
```

JSON Template

```
{
  "resourceType" : "Organization",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // C? Identifies this organization  across multiple systems
  "active" : <boolean>, // Whether the organization's record is still in active use
  "type" : [{ CodeableConcept }], // Kind of organization
  "name" : "<string>", // C? Name used for the organization
  "alias" : ["<string>"], // A list of alternate names that the organization is known as, or was known as in the past
  "telecom" : [{ ContactPoint }], // C? A contact detail for the organization
  "address" : [{ Address }], // C? An address for the organization
  "partOf" : { Reference(Organization) }, // The organization of which this organization forms a part
  "contact" : [{ // Contact for the organization for a certain purpose
    "purpose" : { CodeableConcept }, // The type of contact
    "name" : { HumanName }, // A name associated with the contact
    "telecom" : [{ ContactPoint }], // Contact details (telephone, email, etc.)  for a contact
    "address" : { Address } // Visiting or postal addresses for the contact
  }],
  "endpoint" : [{ Reference(Endpoint) }] // Technical endpoints providing access to services operated for the
    organization
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Organization;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
```

```
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Organization.identifier [ Identifier ], ... ; # 0..* Identifies this organization  across multiple systems
  fhir:Organization.active [ boolean ]; # 0..1 Whether the organization's record is still in active use
  fhir:Organization.type [ CodeableConcept ], ... ; # 0..* Kind of organization
  fhir:Organization.name [ string ]; # 0..1 Name used for the organization
  fhir:Organization.alias [ string ], ... ; # 0..* A list of alternate names that the organization is known as, or was
    known as in the past
  fhir:Organization.telecom [ ContactPoint ], ... ; # 0..* A contact detail for the organization
  fhir:Organization.address [ Address ], ... ; # 0..* An address for the organization
  fhir:Organization.partOf [ Reference(Organization) ]; # 0..1 The organization of which this organization forms a part
  fhir:Organization.contact [ # 0..* Contact for the organization for a certain purpose
    fhir:Organization.contact.purpose [ CodeableConcept ]; # 0..1 The type of contact
    fhir:Organization.contact.name [ HumanName ]; # 0..1 A name associated with the contact
    fhir:Organization.contact.telecom [ ContactPoint ], ... ; # 0..* Contact details (telephone, email, etc.)  for a
    contact
    fhir:Organization.contact.address [ Address ]; # 0..1 Visiting or postal addresses for the contact
  ], ...;
  fhir:Organization.endpoint [ Reference(Endpoint) ], ... ; # 0..* Technical endpoints providing access to services
    operated for the organization
]
```

Changes since DSTU2

Organization

Organization.type | Max Cardinality changed from 1 to *

Organization.alias | Added Element

Organization.endpoint | Added Element

See the Full Difference for further information

This analysis is available as XML or JSON.

See R2 <--> R3 Conversion Maps (status = 14 tests that all execute ok. All tests pass round-trip testing and 2 r3 resources are invalid (2 errors).).

Alternate definitions: Master Definition (XML, JSON), XML Schema/Schematron (for) + JSON Schema, ShEx (for Turtle)

8.6.3.1 Terminology Bindings

Path | Definition | Type | Reference

Organization.type | Used to categorize the organization | Example | OrganizationType

Organization.contact.purpose | The purpose for which you would contact a contact party | Extensible | ContactEntityType

8.6.3.2 Constraints

* org-1: The organization SHALL at least have a name or an id, and possibly more than one (expression : (identifier.count() + name.count()) > 0)

* org-2: On Organization.address: An address of an organization can never be of use 'home' (expression on Organization.address: where(use = 'home').empty())

* org-3: On Organization.telecom: The telecom of an organization can never be of use 'home' (expression on Organization.telecom: where(use = 'home').empty())

8.6.4 Notes:

* There are two places for contact information: one on Organization itself and zero or more using the ContactEntity construct. The first one is to be used for the generic, public organization point of contact. The ContactEntity is to be used for reaching a person or party that has been designated by the organization to be contacted for a specific purpose or goal.

8.6.5 Example Organization Hierarchy:

An example organization hierarchy should help give some guidance as to one example of how a location hierarchy could look within a fictitious Medical Organization. (The nesting here would be the "part-of" structure of the Organization resource)

```
Burgers University Medical Center
    Eastern Services (prov)
        Emergency Dept
        Oncology Dept
            Neuclear Medicine Research Trials (edu)
        Maternity Ward
        Childrens Ward
        Day Procedures Unit
    Mobile Services (Ambulance)
    Research Center (edu)
        Neuclear Medicine  (edu)
    Burgers University (edu)
        Neuclear Medicine Faculty (edu)
        Undergraduate Medicine (edu)
        ...
```

Note that physical structures of this hierarchy are not present - these are defined by a Location hierarchy.

8.6.6 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
active | token | A server defined search that may match any of the string fields in the Address, including line, city, state, country, postalCode, and/or text | Organization.active
address | string | A (part of the) address of the organization | Organization.address
address-city | string | A city specified in an address | Organization.address.city
address-country | string | A country specified in an address | Organization.address.country
address-postalcode | string | A postal code specified in an address | Organization.address.postalCode
address-state | string | A state specified in an address | Organization.address.state
address-use | token | A use code specified in an address | Organization.address.use
endpoint | reference | Technical endpoints providing access to services operated for the organization | Organization.endpoint (Endpoint)
identifier | token | Any identifier for the organization (not the accreditation issuer's identifier) | Organization.identifier
name | string | A portion of the organization's name or alias | Organization.name | Organization.alias
partof | reference | An organization of which this organization forms a part | Organization.partOf (Organization)
phonetic | string | A portion of the organization's name using some kind of phonetic matching algorithm | Organization.name
type | token | A code for the type of organization | Organization.type

8.6.7 Resource Organization - Examples

URL: https://hl7.org/fhir/STU3/organization-examples.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.6.7 Resource Organization - Examples

Patient Administration Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Not linked to any defined compartments

Example List:

HL7 itself | hl7 | XML | JSON | Turtle
Gastroenterology team at ACME Healthcare | 1 | XML | JSON | Turtle
Clinical Laboratory at ACME Healthcare | 1832473e-2fe0-452d-abe9-3cdb9879522f | XML | JSON | Turtle
XYZ Insurance | 2 | XML | JSON | Turtle
Good Health Clinic | 2.16.840.1.113883.19.5 | XML | JSON | Turtle
Real-world organization example (Burgers MC) | f001 | XML | JSON | Turtle
Real-world organization example (Burgers MC Cardiology) | f002 | XML | JSON | Turtle
Real-world organization example (Artis UMC) | f201 | XML | JSON | Turtle
Real-world organization example (Blijdorp MC) | f203 | XML | JSON | Turtle
Real-world organization example (ENT) | f003 | XML | JSON | Turtle
Medication Manufacturer | mmanu | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

8.6.8 Resource Organization - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/organization-definitions.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.6.8 Resource Organization - Detailed Descriptions

Patient Administration Work Group | Maturity Level: 3 | Trial Use | Compartments: Not linked to any defined compartments

Detailed Descriptions for the elements in the Organization resource.

Organization
Definition | A formally or informally recognized grouping of people or organizations formed for the purpose of achieving some form of collective action. Includes companies, institutions, corporations, departments, community groups, healthcare practice groups, etc.
Control | 1..1
Invariants | Defined on this element org-1: The organization SHALL at least have a name or an id, and possibly more than one (expression : (identifier.count() + name.count()) > 0, xpath: count(f:identifier | f:name) > 0)
Organization.identifier
Definition | Identifier for the organization that is used to identify the organization across multiple disparate systems.
Note | This is a business identifer, not a resource identifier (see discussion)
Control | 0..*
Type | Identifier
Requirements | Organizations are known by a variety of ids. Some institutions maintain several, and most collect identifiers for exchange with other organizations concerning the organization.
Summary | true
Invariants | Affect this element org-1: The organization SHALL at least have a name or an id, and possibly more than one (expression : (identifier.count() + name.count()) > 0, xpath: count(f:identifier | f:name) > 0)
Organization.active
Definition | Whether the organization's record is still in active use.
Control | 0..1
Type | boolean
Is Modifier | true

Default Value | true
Requirements | Need a flag to indicate a record is no longer to be used and should generally be hidden for the user in the UI.
Summary | true
Comments | Default is true. This active flag is not intended to be used to mark an organizations as temporarily closed or under construction. Instead the Location(s) within the Organization should have the suspended status. If further details of the reason for the suspension are required, then an extension on this element should be used. This element is labeled as a modifier because it may be used to mark that the resource was created in error.
Organization.type
Definition | The kind(s) of organization that this is.
Control | 0..*
Terminology Binding | OrganizationType (Example)
Type | CodeableConcept
Requirements | Need to be able to track the kind of organization that this is - different organization types have different uses.
Summary | true
Comments | Organizations can be corporations, wards, sections, clinical teams, government departments, etc. Note that code is generally a classifier of the type of organization; in many applications, codes are used to identity a particular organization (say, ward) as opposed to another of the same type - these are identifiers, not codes When considering if multiple types are appropriate, you should evaluate if child organizations would be a more appropriate use of the concept, as different types likely are in different sub-areas of the organization. This is most likely to be used where type values have orthogonal values, such as a religious, academic and medical center. We expect that some jurisdictions will profile this optionality to be a single cardinality.
Organization.name
Definition | A name associated with the organization.
Control | 0..1
Type | string
Requirements | Need to use the name as the label of the organization.
Summary | true
Comments | If the name of an organization changes, consider putting the old name in the alias column so that it can still be located through searches.
Invariants | Affect this element org-1: The organization SHALL at least have a name or an id, and possibly more than one (expression : (identifier.count() + name.count()) > 0, xpath: count(f:identifier | f:name) > 0)
Organization.alias
Definition | A list of alternate names that the organization is known as, or was known as in the past.
Control | 0..*
Type | string
Requirements | Over time locations and organizations go through many changes and can be known by different names. For searching knowing previous names that the organization was known by can be very useful.
Comments | There are no dates associated with the alias/historic names, as this is not intended to track when names were used, but to assist in searching so that older names can still result in identifying the organization.
Organization.telecom
Definition | A contact detail for the organization.
Control | 0..*
Type | ContactPoint
Requirements | Human contact for the organization.
Comments | The use code home is not to be used. Note that these contacts are not the contact details of people who are employed by or represent the organization, but official contacts for the organization itself.
Invariants | Defined on this element org-3: The telecom of an organization can never be of use 'home' (expression : where(use = 'home').empty(), xpath: count(f:use[@value='home']) = 0) Affect this element
Organization.address
Definition | An address for the organization.
Control | 0..*
Type | Address
Requirements | May need to keep track of the organization's addresses for contacting, billing or reporting requirements.
Comments | Organization may have multiple addresses with different uses or applicable periods. The use code home is not to be used.
Invariants | Defined on this element org-2: An address of an organization can never be of use 'home' (expression : where(use = 'home').empty(), xpath: count(f:use[@value='home']) = 0) Affect this element
Organization.partOf
Definition | The organization of which this organization forms a part.
Control | 0..1
Type | Reference(Organization)
Requirements | Need to be able to track the hierarchy of organizations within an organization.
Summary | true
Organization.contact
Definition | Contact for the organization for a certain purpose.
Control | 0..*
Requirements | Need to keep track of assigned contact points within bigger organization.
Comments | Where multiple contacts for the same purpose are provided there is a standard extension that can be used to determine which one is the preferred contact to use.
Organization.contact.purpose
Definition | Indicates a purpose for which the contact can be reached.
Control | 0..1
Terminology Binding | ContactEntityType (Extensible)
Type | CodeableConcept
Requirements | Need to distinguish between multiple contact persons.
Organization.contact.name

Definition | A name associated with the contact.
Control | 0..1
Type | HumanName
Requirements | Need to be able to track the person by name.
Organization.contact.telecom
Definition | A contact detail (e.g. a telephone number or an email address) by which the party may be contacted.
Control | 0..*
Type | ContactPoint
Requirements | People have (primary) ways to contact them in some way such as phone, email.
Organization.contact.address
Definition | Visiting or postal addresses for the contact.
Control | 0..1
Type | Address
Requirements | May need to keep track of a contact party's address for contacting, billing or reporting requirements.
Organization.endpoint
Definition | Technical endpoints providing access to services operated for the organization.
Control | 0..*
Type | Reference(Endpoint)
Requirements | Organizations have multiple systems that provide various services and need to be able to define the technical connection details for how to connect to them, and for what purpose.

8.6.9 Resource Organization - Mappings

URL: https://hl7.org/fhir/STU3/organization-mappings.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.6.9 Resource Organization - Mappings

Patient Administration Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Not linked to any defined compartments

Mappings:

RIM Mapping

HL7 v2 Mapping

ServD

Mappings for the organization resource.

8.6.9.1 Mappings for RIM Mapping (http://hl7.org/v3)

Organization | Organization(classCode=ORG, determinerCode=INST)
identifier | .scopes[Role](classCode=IDENT)
active | .status
type | .code
name | .name
alias | .name
telecom | .telecom
address | .address
partOf | .playedBy[classCode=Part].scoper
contact | .contactParty
purpose | ./type
name | ./name
telecom | ./telecom
address | ./addr
endpoint | n/a

8.6.9.2 Mappings for HL7 v2 Mapping (http://hl7.org/v2)

Organization | (also see master files messages)
identifier | XON.10 / XON.3
active | No equivalent in HL7 v2
type | No equivalent in v2
name | XON.1
alias
telecom | ORC-22?
address | ORC-23?
partOf | No equivalent in HL7 v2
contact
purpose
name | PID-5, PID-9
telecom | PID-13, PID-14
address | PID-11
endpoint

8.6.9.3 Mappings for ServD (http://www.omg.org/spec/ServD/1.0/)

Organization | Organization
identifier | ./Identifiers
active | ./Status (however this concept in ServD more covers why the organization is active or not, could be delisted, deregistered, not operational yet) this could alternatively be derived from ./StartDate and ./EndDate and given a context date.
type | n/a
name | .PreferredName/Name
alias
telecom | ./ContactPoints
address | ./PrimaryAddress and ./OtherAddresses
partOf | n/a
contact
purpose
name
telecom
address
endpoint

8.6.10 Resource Organization - Design Notes

URL: https://hl7.org/fhir/STU3/organization-explanations.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.6.10 Resource Organization - Design Notes

to do

8.6.11 Resource Organization - Extensions & Profiles

URL: https://hl7.org/fhir/STU3/organization-profiles.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.6.11 Resource Organization - Extensions & Profiles

Patient Administration Work Group | Maturity Level: N/A | Ballot Status: Informative

This table lists profiles and extensions for the Organization resource. For background information, see Profiling Resources and Extensibility. Additional profiles and resources may be found in published Implementation Guides , or in the Conformance resource registry .

8.6.11.1 Profiles

No Profiles defined for this resource

8.6.11.2 Extensions

organization-alias | alias | for Organization HL7 Extensions
organization-period | period | for Organization HL7 Extensions
organization-preferredContact | preferredContact | for Organization HL7 Extensions

Extensions that reference this resource:

11179-de-registry-org | registry-org | for ISO 11179 Data Element Profile
11179-de-submitter-org | submitter-org | for ISO 11179 Data Element Profile
communicationrequest-orderedBy | orderedBy | for CommunicationRequest HL7 Extensions
consent-Witness | Witness | for Consent HL7 Extensions
cqif-guidanceinitiatingOrganization | initiatingOrganization | for CQIF Guidance Extensions
cqif-guidancereceivingOrganization | receivingOrganization | for CQIF Guidance Extensions
event-OnBehalfOf | OnBehalfOf | for Event Pattern HL7 Extensions
task-candidateList | candidateList | for Task HL7 Extensions

8.6.11.3 Search Extensions

No Search Extensions defined for this resource

8.6.12 Organization HL7 Extensions

URL: https://hl7.org/fhir/STU3/organization-extensions.html

Defines common extensions used with or related to the Organization resource

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.6.12 Organization HL7 Extensions

(No assigned work group) | Maturity Level: N/A | Ballot Status: Informative

8.6.12.1 Content

Extensions:
organization-alias | alias : Other names that this Organization is known as.
organization-period | period : The date range that this organization should be considered available.
organization-preferredContact | preferredContact : This Contact is the preferred contact at this organization for the purpose of the contact There can be multiple contacts on an Organizations record with this value set to true, but these should all have different purpose values.

8.6.12.2 Search Parameters

Search parameters defined by this package. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Paths | Source
alias | string | Include Organizations that contain the provided alias |
f:Organization/f:extension[@url='http://hl7.org/fhir/StructureDefinition/organization-alias'] | /f:#alias | XML / JSON

8.6.12.3 Organization-extensions-Organization-alias

URL: https://hl7.org/fhir/STU3/organization-extensions-Organization-alias.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.6.12.3 Organization-extensions-Organization-alias

Search Parameter alias (type: string) defined as part of the organization-extensions Profile.

Include Organizations that contain the provided alias

Other Details:

* Full URL : http://hl7.org/fhir/SearchParameter/organization-extensions-Organization-alias

* Paths : f:Organization/f:extension[@url='http://hl7.org/fhir/StructureDefinition/organization-alias'] | /f:#alias

* Targets :

8.6.13 Resource Organization - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/organization-version-maps.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.6.13 Resource Organization - Detailed Descriptions

Patient Administration Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Not linked to any defined compartments

R2 : R3 Converson maps for Organization.

Functional status for this map: 14 tests that all execute ok. All tests pass round-trip testing and 2 r3 resources are invalid (2 errors). (see documentation)

8.6.13.1 R2 to R3

```
map "http://hl7.org/fhir/StructureMap/Organization2to3" = "R2 to R3 Conversions for Organization"

uses "http://hl7.org/fhir/DSTU2/StructureDefinition/Organization" alias OrganizationR2 as source
uses "http://hl7.org/fhir/StructureDefinition/Organization" alias Organization as target

imports "http://hl7.org/fhir/StructureMap/*2to3"

group for type+types  Organization extends DomainResource
  input src : OrganizationR2 as source
  input tgt : Organization as target

  "Organization-identifier" : for src.identifier make tgt.identifier
  "Organization-active" : for src.active make tgt.active
  "Organization-type" : for src.type make tgt.type
  "Organization-name" : for src.name make tgt.name
  "Organization-telecom" : for src.telecom make tgt.telecom
  "Organization-address" : for src.address make tgt.address
  "Organization-partOf" : for src.partOf make tgt.partOf
  "Organization-contact" : for src.contact as vs0 make tgt.contact as vt0 then orgContact(vs0, vt0)
endgroup

group orgContact extends BackboneElement
  input src as source
  input tgt as target

  "Organization.contact-purpose" : for src.purpose make tgt.purpose
  "Organization.contact-name" : for src.name make tgt.name
  "Organization.contact-telecom" : for src.telecom make tgt.telecom
  "Organization.contact-address" : for src.address make tgt.address
```

```
endgroup
```

8.6.13.2 R3 to R2

```
map "http://hl7.org/fhir/StructureMap/Organization3to2" = "R3 to R2 Conversion for Organization"

uses "http://hl7.org/fhir/StructureDefinition/Organization" alias Organization as source
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/Organization" alias OrganizationR2 as target

imports "http://hl7.org/fhir/StructureMap/*3to2"

group for type+types  Organization extends DomainResource
  input src : Organization as source
  input tgt : OrganizationR2 as target

  "Organization-identifier" : for src.identifier make tgt.identifier
  "Organization-active" : for src.active make tgt.active
  "Organization-type" : for src.type make tgt.type
  "Organization-name" : for src.name make tgt.name
  "Organization-telecom" : for src.telecom make tgt.telecom
  "Organization-address" : for src.address make tgt.address
  "Organization-partOf" : for src.partOf make tgt.partOf
  "Organization-contact" : for src.contact as vs0 make tgt.contact as vt0 then orgContact(vs0, vt0)
endgroup

group orgContact extends BackboneElement
  input src as source
  input tgt as target

  "Organization.contact-purpose" : for src.purpose make tgt.purpose
  "Organization.contact-name" : for src.name make tgt.name
  "Organization.contact-telecom" : for src.telecom make tgt.telecom
  "Organization.contact-address" : for src.address make tgt.address
endgroup
```

8.6.13.3 R3 Validation Errors

org-uslab-example1 | ERROR @ Organization.address.extension: The extension http://hl7.org/fhir/StructureDefinition/us-core-county is unknown, and not allowed here (src = InstanceValidator)

org-uslab-example3 | ERROR @ Organization.address.extension: The extension http://hl7.org/fhir/StructureDefinition/us-core-county is unknown, and not allowed here (src = InstanceValidator)

8.7 Resource Location - Content

URL: https://hl7.org/fhir/STU3/location.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

Patient Administration Work Group | Maturity Level: 3 | Trial Use | Compartments: Not linked to any defined compartments

Details and position information for a physical place where services are provided and resources and participants may be stored, found, contained or accommodated.

8.7.1 Scope and Usage

A Location includes both incidental locations (a place which is used for healthcare without prior designation or authorization) and dedicated, formally appointed locations. Locations may be private, public, mobile or fixed and scale from small freezers to full hospital buildings or parking garages.

Examples of Locations are:

* Building, ward, corridor, room or bed
* Mobile Clinic
* Freezer, incubator
* Vehicle or lift
* Home, shed, or a garage
* Road, parking place, a park
* Ambulance (generic)
* Ambulance (specific)
* Patient's Home (generic)
* Jurisdiction

These locations are not intended to cover locations on a patient where something occurred (i.e. a patient's broken leg), but can hapily cover the location where the patient broke the leg (the playground)

8.7.2 Boundaries and Relationships

Locations and Organizations are very closely related resources and can often be mixed/matched/confused. The Location is intended to describe the more physical structures managed/operated by an organization, whereas the Organization is

intended to represent the more conceptual hierarchies, such as a ward.

A Location is valid without an address in cases where it could be purely described by a geo-coded location in remote areas, or when recorded by a device. Locations with a mode = "kind" would also likely not have an address, as they are just a type of location, but could also have an address where they can be found at the address.

Another use of location could be for describing a Jurisdiction. This jurisdiction may be considered a classified boundary which could be a combination of a physical boundary, and some other discriminator(s):

* Nation - Country wide community or Federal Government (Ministry of Health)

* Province or State (community or Government)

* Business (throughout an enterprise)

* Business scope (CDC/FDA)

* Business segment (UK Pharmacy)

This resource is referenced by Account, ActivityDefinition, AdverseEvent, Appointment, AppointmentResponse, AuditEvent, CarePlan, Claim, Contract, Device, DeviceRequest, DiagnosticReport, EligibilityRequest, Encounter, ExplanationOfBenefit, Flag, HealthcareService, Immunization, List, MedicationDispense, Observation, PractitionerRole, Procedure, ProcedureRequest, Provenance, ResearchStudy, Schedule, SupplyDelivery and SupplyRequest

8.7.3 Resource Content

* Structure

* UML

* XML

* JSON

* Turtle

* R2 Diff

* All

Structure

Name | Flags | Card. | Type | Description & Constraints
Location | | | DomainResource | Details and position information for a physical place Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | ? | 0..* | Identifier | Unique code or number identifying the location to its users
status | ?!? | 0..1 | code | active | suspended | inactive LocationStatus (Required)
operationalStatus | ? | 0..1 | Coding | The Operational status of the location (typically only for a bed/room) v2 Bed Status (Preferred)
name | ? | 0..1 | string | Name of the location as used by humans
alias | | 0..* | string | A list of alternate names that the location is known as, or was known as in the past
description | ? | 0..1 | string | Additional details about the location that could be displayed as further information to identify the location beyond its name
mode | ?!? | 0..1 | code | instance | kind LocationMode (Required)
type | ? | 0..1 | CodeableConcept | Type of function performed ServiceDeliveryLocationRoleType (Extensible)
telecom | | 0..* | ContactPoint | Contact details of the location
address | | 0..1 | Address | Physical location
physicalType | ? | 0..1 | CodeableConcept | Physical form of the location LocationType (Example)
position | | 0..1 | BackboneElement | The absolute geographic location
longitude | | 1..1 | decimal | Longitude with WGS84 datum
latitude | | 1..1 | decimal | Latitude with WGS84 datum
altitude | | 0..1 | decimal | Altitude with WGS84 datum
managingOrganization | ? | 0..1 | Reference(Organization) | Organization responsible for provisioning and upkeep
partOf | | 0..1 | Reference(Location) | Another Location this one is physically part of
endpoint | | 0..* | Reference(Endpoint) | Technical endpoints providing access to services operated for the location
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Location xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Unique code or number identifying the location to its users --></identifier>
 <status value="[code]"/><!-- 0..1 active | suspended | inactive -->
 <operationalStatus><!-- 0..1 Coding The Operational status of the location (typically only for a bed/room)
    --></operationalStatus>
 <name value="[string]"/><!-- 0..1 Name of the location as used by humans -->
 <alias value="[string]"/><!-- 0..* A list of alternate names that the location is known as, or was known as in the past
     -->
 <description value="[string]"/><!-- 0..1 Additional details about the location that could be displayed as further
    information to identify the location beyond its name -->
 <mode value="[code]"/><!-- 0..1 instance | kind -->
 <type><!-- 0..1 CodeableConcept Type of function performed --></type>
 <telecom><!-- 0..* ContactPoint Contact details of the location --></telecom>
 <address><!-- 0..1 Address Physical location --></address>
 <physicalType><!-- 0..1 CodeableConcept Physical form of the location --></physicalType>
```

```
 <position>  <!-- 0..1 The absolute geographic location -->
  <longitude value="[decimal]"/><!-- 1..1 Longitude with WGS84 datum -->
  <latitude value="[decimal]"/><!-- 1..1 Latitude with WGS84 datum -->
  <altitude value="[decimal]"/><!-- 0..1 Altitude with WGS84 datum -->
 </position>
 <managingOrganization><!-- 0..1 Reference(Organization) Organization responsible for provisioning and upkeep
    --></managingOrganization>
 <partOf><!-- 0..1 Reference(Location) Another Location this one is physically part of --></partOf>
 <endpoint><!-- 0..* Reference(Endpoint) Technical endpoints providing access to services operated for the location
    --></endpoint>
</Location>
```

JSON Template

```
{
  "resourceType" : "Location",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Unique code or number identifying the location to its users
  "status" : "<code>", // active | suspended | inactive
  "operationalStatus" : { Coding }, // The Operational status of the location (typically only for a bed/room)
  "name" : "<string>", // Name of the location as used by humans
  "alias" : ["<string>"], // A list of alternate names that the location is known as, or was known as in the past
  "description" : "<string>", // Additional details about the location that could be displayed as further information to
     identify the location beyond its name
  "mode" : "<code>", // instance | kind
  "type" : { CodeableConcept }, // Type of function performed
  "telecom" : [{ ContactPoint }], // Contact details of the location
  "address" : { Address }, // Physical location
  "physicalType" : { CodeableConcept }, // Physical form of the location
  "position" : { // The absolute geographic location
    "longitude" : <decimal>, // R!  Longitude with WGS84 datum
    "latitude" : <decimal>, // R!  Latitude with WGS84 datum
    "altitude" : <decimal> // Altitude with WGS84 datum
  },
  "managingOrganization" : { Reference(Organization) }, // Organization responsible for provisioning and upkeep
  "partOf" : { Reference(Location) }, // Another Location this one is physically part of
  "endpoint" : [{ Reference(Endpoint) }] // Technical endpoints providing access to services operated for the location
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Location;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Location.identifier [ Identifier ], ... ; # 0..* Unique code or number identifying the location to its users
  fhir:Location.status [ code ]; # 0..1 active | suspended | inactive
  fhir:Location.operationalStatus [ Coding ]; # 0..1 The Operational status of the location (typically only for a
    bed/room)
  fhir:Location.name [ string ]; # 0..1 Name of the location as used by humans
  fhir:Location.alias [ string ], ... ; # 0..* A list of alternate names that the location is known as, or was known as
    in the past
  fhir:Location.description [ string ]; # 0..1 Additional details about the location that could be displayed as further
    information to identify the location beyond its name
  fhir:Location.mode [ code ]; # 0..1 instance | kind
  fhir:Location.type [ CodeableConcept ]; # 0..1 Type of function performed
  fhir:Location.telecom [ ContactPoint ], ... ; # 0..* Contact details of the location
  fhir:Location.address [ Address ]; # 0..1 Physical location
  fhir:Location.physicalType [ CodeableConcept ]; # 0..1 Physical form of the location
  fhir:Location.position [ # 0..1 The absolute geographic location
    fhir:Location.position.longitude [ decimal ]; # 1..1 Longitude with WGS84 datum
    fhir:Location.position.latitude [ decimal ]; # 1..1 Latitude with WGS84 datum
    fhir:Location.position.altitude [ decimal ]; # 0..1 Altitude with WGS84 datum
  ];
  fhir:Location.managingOrganization [ Reference(Organization) ]; # 0..1 Organization responsible for provisioning and
    upkeep
  fhir:Location.partOf [ Reference(Location) ]; # 0..1 Another Location this one is physically part of
  fhir:Location.endpoint [ Reference(Endpoint) ], ... ; # 0..* Technical endpoints providing access to services operated
     for the location
]
```

Changes since DSTU2

Location

Location.operationalStatus | Added Element

Location.alias | Added Element

Location.endpoint | Added Element

See the Full Difference for further information

This analysis is available as XML or JSON.

See R2 <--> R3 Conversion Maps (status = 6 tests that all execute ok. All tests pass round-trip testing and all r3 resources are valid.).

Structure

Name | Flags | Card. | Type | Description & Constraints
Location | | | DomainResource | Details and position information for a physical place Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | ? | 0..* | Identifier | Unique code or number identifying the location to its users
status | ?!? | 0..1 | code | active | suspended | inactive LocationStatus (Required)
operationalStatus | ? | 0..1 | Coding | The Operational status of the location (typically only for a bed/room) v2 Bed Status (Preferred)
name | ? | 0..1 | string | Name of the location as used by humans
alias | | 0..* | string | A list of alternate names that the location is known as, or was known as in the past
description | ? | 0..1 | string | Additional details about the location that could be displayed as further information to identify the location beyond its name
mode | ?!? | 0..1 | code | instance | kind LocationMode (Required)
type | ? | 0..1 | CodeableConcept | Type of function performed ServiceDeliveryLocationRoleType (Extensible)
telecom | | 0..* | ContactPoint | Contact details of the location
address | | 0..1 | Address | Physical location
physicalType | ? | 0..1 | CodeableConcept | Physical form of the location LocationType (Example)
position | | 0..1 | BackboneElement | The absolute geographic location
longitude | | 1..1 | decimal | Longitude with WGS84 datum
latitude | | 1..1 | decimal | Latitude with WGS84 datum
altitude | | 0..1 | decimal | Altitude with WGS84 datum
managingOrganization | ? | 0..1 | Reference(Organization) | Organization responsible for provisioning and upkeep
partOf | | 0..1 | Reference(Location) | Another Location this one is physically part of
endpoint | | 0..* | Reference(Endpoint) | Technical endpoints providing access to services operated for the location
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Location xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Unique code or number identifying the location to its users --></identifier>
 <status value="[code]"/><!-- 0..1 active | suspended | inactive -->
 <operationalStatus><!-- 0..1 Coding The Operational status of the location (typically only for a bed/room)
    --></operationalStatus>
 <name value="[string]"/><!-- 0..1 Name of the location as used by humans -->
 <alias value="[string]"/><!-- 0..* A list of alternate names that the location is known as, or was known as in the past
     -->
 <description value="[string]"/><!-- 0..1 Additional details about the location that could be displayed as further
    information to identify the location beyond its name -->
 <mode value="[code]"/><!-- 0..1 instance | kind -->
 <type><!-- 0..1 CodeableConcept Type of function performed --></type>
 <telecom><!-- 0..* ContactPoint Contact details of the location --></telecom>
 <address><!-- 0..1 Address Physical location --></address>
 <physicalType><!-- 0..1 CodeableConcept Physical form of the location --></physicalType>
 <position>  <!-- 0..1 The absolute geographic location -->
  <longitude value="[decimal]"/><!-- 1..1 Longitude with WGS84 datum -->
  <latitude value="[decimal]"/><!-- 1..1 Latitude with WGS84 datum -->
  <altitude value="[decimal]"/><!-- 0..1 Altitude with WGS84 datum -->
 </position>
 <managingOrganization><!-- 0..1 Reference(Organization) Organization responsible for provisioning and upkeep
    --></managingOrganization>
 <partOf><!-- 0..1 Reference(Location) Another Location this one is physically part of --></partOf>
 <endpoint><!-- 0..* Reference(Endpoint) Technical endpoints providing access to services operated for the location
    --></endpoint>
</Location>
```

JSON Template

```
{
  "resourceType" : "Location",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Unique code or number identifying the location to its users
  "status" : "<code>", // active | suspended | inactive
  "operationalStatus" : { Coding }, // The Operational status of the location (typically only for a bed/room)
  "name" : "<string>", // Name of the location as used by humans
  "alias" : ["<string>"], // A list of alternate names that the location is known as, or was known as in the past
  "description" : "<string>", // Additional details about the location that could be displayed as further information to
     identify the location beyond its name
  "mode" : "<code>", // instance | kind
  "type" : { CodeableConcept }, // Type of function performed
  "telecom" : [{ ContactPoint }], // Contact details of the location
  "address" : { Address }, // Physical location
  "physicalType" : { CodeableConcept }, // Physical form of the location
  "position" : { // The absolute geographic location
    "longitude" : <decimal>, // R!  Longitude with WGS84 datum
    "latitude" : <decimal>, // R!  Latitude with WGS84 datum
    "altitude" : <decimal> // Altitude with WGS84 datum
  },
  "managingOrganization" : { Reference(Organization) }, // Organization responsible for provisioning and upkeep
  "partOf" : { Reference(Location) }, // Another Location this one is physically part of
  "endpoint" : [{ Reference(Endpoint) }] // Technical endpoints providing access to services operated for the location
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Location;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Location.identifier [ Identifier ], ... ; # 0..* Unique code or number identifying the location to its users
  fhir:Location.status [ code ]; # 0..1 active | suspended | inactive
  fhir:Location.operationalStatus [ Coding ]; # 0..1 The Operational status of the location (typically only for a
    bed/room)
  fhir:Location.name [ string ]; # 0..1 Name of the location as used by humans
  fhir:Location.alias [ string ], ... ; # 0..* A list of alternate names that the location is known as, or was known as
    in the past
  fhir:Location.description [ string ]; # 0..1 Additional details about the location that could be displayed as further
    information to identify the location beyond its name
  fhir:Location.mode [ code ]; # 0..1 instance | kind
  fhir:Location.type [ CodeableConcept ]; # 0..1 Type of function performed
  fhir:Location.telecom [ ContactPoint ], ... ; # 0..* Contact details of the location
  fhir:Location.address [ Address ]; # 0..1 Physical location
  fhir:Location.physicalType [ CodeableConcept ]; # 0..1 Physical form of the location
  fhir:Location.position [ # 0..1 The absolute geographic location
    fhir:Location.position.longitude [ decimal ]; # 1..1 Longitude with WGS84 datum
    fhir:Location.position.latitude [ decimal ]; # 1..1 Latitude with WGS84 datum
    fhir:Location.position.altitude [ decimal ]; # 0..1 Altitude with WGS84 datum
  ];
  fhir:Location.managingOrganization [ Reference(Organization) ]; # 0..1 Organization responsible for provisioning and
    upkeep
  fhir:Location.partOf [ Reference(Location) ]; # 0..1 Another Location this one is physically part of
  fhir:Location.endpoint [ Reference(Endpoint) ], ... ; # 0..* Technical endpoints providing access to services operated
     for the location
]
```

Changes since DSTU2

Location
Location.operationalStatus | Added Element
Location.alias | Added Element
Location.endpoint | Added Element

See the Full Difference for further information

This analysis is available as XML or JSON.

See R2 <--> R3 Conversion Maps (status = 6 tests that all execute ok. All tests pass round-trip testing and all r3 resources are valid.).

Alternate definitions: Master Definition (XML, JSON), XML Schema/Schematron (for) + JSON Schema, ShEx (for Turtle)

8.7.3.1 Terminology Bindings

Path | Definition | Type | Reference
Location.status | Indicates whether the location is still in use. | Required | LocationStatus
Location.operationalStatus | The operational status if the location (where typically a bed/room) | Preferred | v2 Bed Status
Location.mode | Indicates whether a resource instance represents a specific location or a class of locations. | Required | LocationMode
Location.type | Indicates the type of function performed at the location. | Extensible | ServiceDeliveryLocationRoleType
Location.physicalType | Physical form of the location | Example | LocationType

8.7.4 Notes

* Multiple Organizations or Practitioners may provide services at a Location. These references are not kept in Location, but can be found in the models for Organization and Practitioner instead.

* Locations may range from whole buildings to cabinets; it is possible to relate smaller Locations to their containing bigger Location using the Location.partOf element.

* Location.position is expressed using the same syntax, datum and reference system as used in Google Earth's KML files, see Google/OGS's KML .

8.7.4.1 Location Mode

The Location.mode element can be used to indicate whether a Location resource represents a specific (potentially identifiable) Location ('instance'), or a class of Locations ('kind'). Especially Resources capturing orders, resource scheduling, plans and definitions may refer to Locations in 'kind' mode. For these domains, it is often not necessary to refer to a specific Location, but rather to a class of Locations. An example of this is found in planning, where we need to allocate an "isolation room" for a patient, or need to dispatch "an ambulance" at a certain time. In these cases it is not important exactly which isolation room or ambulance is allocated, and it is sufficient to just indicate a 'kind' of Location.

Note that 'kind' should not be used to represent Locations where an actual instance of a Location was involved, but identifying information is missing. E.g. when a patient arrived 'by ambulance', but it is not known by which ambulance, this should be represented using a Location in 'instance' mode with a missing identifier, not a Location of 'kind' ambulance.

Some of Location's data elements are only relevant when mode is 'instance' and should not be used when mode is 'kind': (however this information could still be included if was relevant, such as when it is a generic item, but not globally generic,

e.g. a Burgers MU ambulance)

* Location.identifier

* Location.telecom

* Location.address

* Location.position

* Location.status

* Location.managingOrganization

8.7.5 Example Location Hierarchy

An example location hierarchy should help give some guidance as to one example of how a location hierarchy could look within a fictitious Hospital. (The nesting here would be the "part-of" structure of the location)

```
Hospital A Building C (instance)
    East Wing (instance)
        Level 1 (instance)
            Reception (instance)
            Nurses Station EM-ns1 (instance)
                Medication Cupboard A (instance)
            Room 1 (instance)
                Room 1a (instance) - space in room separatable via a curtain
                    Bed 1a (instance) - always in this room
                Room 1b (instance)
                    Trolley 43 (instance) - moves about
                Room 1d (instance)
                    Trolley 19 (instance) - moves about
                Room 2 (instance)
                    ...
            Theatre EM-TA (instance)
            Coridor (generic)
        Level 2 (instance)
            Reception (instance)
                ...
            Nurses Station EM-ns1 (instance)
                Medication Cupboard A (instance)
            Coridor (generic)
Mobile Services (kind)
    Ambulance (kind)
        Ambulance AMB1 (instance)
        Ambulance AMB2 (instance)
```

Note: Wards/departments are not part of this structure - these would form part of the Organizational Hierarchy.

8.7.5.1 Positional Searching

Searching for locations often require that a facility is within a specified distance of a particular point. For example, to locate healthcare facilities within 2kms of a clients home, or the current geo-coded position of a practitioner travelling between patients (read from a mobile phone or device).

```
 GET [base]/Location?near=-83.694810:42.256500&near-distance=11.20||km...
```

Note: The near search parameters are not able to be used with multiples thus the multiple seperator , should not be used with near or near-distance.

The distance between the location and the provided point is often used as one of the determining factors for selection of the location. So this value is included in the results. However the value cannot be inside the Location resource as it is different depending on the point of reference in the search. So the distance between is included in the search section of the bundle entry.

```
<entry>
    <resource>
        <Location>
            <!-- location details -->
        </Location>
    </resource>
    <search>
        <extension url="http://hl7.org/fhir/StructureDefinition/location-distance">
            <valueDistance >
                <!-- The distance that this location resource is from the provided point in the query --<
                </value value="10.5"/>
                </unit value="km"/>
            </valueDistance>
        </extension>
    </search>
</entry>
```

8.7.6 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
address | string | A (part of the) address of the location | Location.address
address-city | string | A city specified in an address | Location.address.city

address-country | string | A country specified in an address | Location.address.country
address-postalcode | string | A postal code specified in an address | Location.address.postalCode
address-state | string | A state specified in an address | Location.address.state
address-use | token | A use code specified in an address | Location.address.use
endpoint | reference | Technical endpoints providing access to services operated for the location | Location.endpoint (Endpoint)
identifier | token | An identifier for the location | Location.identifier
name | string | A portion of the location's name or alias | Location.name | Location.alias
near | token | The coordinates expressed as [latitude]:[longitude] (using the WGS84 datum, see notes) to find locations near to (servers may search using a square rather than a circle for efficiency) Requires the near-distance parameter to be provided also | Location.position
near-distance | quantity | A distance quantity to limit the near search to locations within a specific distance Requires the near parameter to also be included | Location.position
operational-status | token | Searches for locations (typically bed/room) that have an operational status (e.g. contaminated, housekeeping) | Location.operationalStatus
organization | reference | Searches for locations that are managed by the provided organization | Location.managingOrganization (Organization)
partof | reference | A location of which this location is a part | Location.partOf (Location)
status | token | Searches for locations with a specific kind of status | Location.status
type | token | A code for the type of location | Location.type

8.7.7 Resource Location - Examples

URL: https://hl7.org/fhir/STU3/location-examples.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.7.7 Resource Location - Examples

Patient Administration Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Not linked to any defined compartments

Example List:

Wing within a hospital | 1 | XML | JSON | Turtle
Operation room within hospital | 2 | XML | JSON | Turtle
Ambulance provided by Burgers Healthcare | amb | XML | JSON | Turtle
Patient's Home which can be used for remote encounters | ph | XML | JSON | Turtle
UK Pharmacy Jurisdiction | ukp | XML | JSON | Turtle
HL7 HQ Location | hl7 | XML | JSON | Turtle
General location examples | 3ad0687e-f477-468c-afd5-fcc2bf897819 | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

8.7.8 Resource Location - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/location-definitions.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.7.8 Resource Location - Detailed Descriptions

Patient Administration Work Group | Maturity Level: 3 | Trial Use | Compartments: Not linked to any defined compartments

Detailed Descriptions for the elements in the Location resource.

Location
Definition | Details and position information for a physical place where services are provided and resources and participants may be stored, found, contained or accommodated.
Control | 1..1
Location.identifier
Definition | Unique code or number identifying the location to its users.
Note | This is a business identifer, not a resource identifier (see discussion)
Control | 0..*
Type | Identifier
Requirements | Organization label locations in registries, need to keep track of those.
Summary | true
Location.status
Definition | The status property covers the general availability of the resource, not the current value which may be covered by the operationStatus, or by a schedule/slots if they are configured for the location.
Control | 0..1
Terminology Binding | LocationStatus (Required)
Type | code
Is Modifier | true
Summary | true
Comments | This element is labeled as a modifier because the status contains codes that mark the resource as not currently valid.
Location.operationalStatus
Definition | The Operational status covers operation values most relevant to beds (but can also apply to rooms/units/chair/etc such as an isolation unit/dialisys chair). This typically covers concepts such as contamination, housekeeping and other activities like maintenance.

Control | 0..1
Terminology Binding | v2 Bed Status (Preferred)
Type | Coding
Summary | true
Location.name
Definition | Name of the location as used by humans. Does not need to be unique.
Control | 0..1
Type | string
Summary | true
Comments | If the name of an location changes, consider putting the old name in the alias column so that it can still be located through searches.
Location.alias
Definition | A list of alternate names that the location is known as, or was known as in the past.
Control | 0..*
Type | string
Requirements | Over time locations and organizations go through many changes and can be known by different names. For searching knowing previous names that the location was known by can be very useful.
Comments | There are no dates associated with the alias/historic names, as this is not intended to track when names were used, but to assist in searching so that older names can still result in identifying the location.
Location.description
Definition | Description of the Location, which helps in finding or referencing the place.
Control | 0..1
Type | string
Requirements | Humans need additional information to verify a correct location has been identified.
Summary | true
Location.mode
Definition | Indicates whether a resource instance represents a specific location or a class of locations.
Control | 0..1
Terminology Binding | LocationMode (Required)
Type | code
Is Modifier | true
Requirements | When using a Location resource for scheduling or orders, we need to be able to refer to a class of Locations instead of a specific Location.
Summary | true
Comments | This is labeled as a modifer because whether or not the location is a class of locations changes how it can be used and understood.
Location.type
Definition | Indicates the type of function performed at the location.
Control | 0..1
Terminology Binding | ServiceDeliveryLocationRoleType (Extensible)
Type | CodeableConcept
Summary | true
Location.telecom
Definition | The contact details of communication devices available at the location. This can include phone numbers, fax numbers, mobile numbers, email addresses and web sites.
Control | 0..*
Type | ContactPoint
Location.address
Definition | Physical location.
Control | 0..1
Type | Address
Requirements | If locations can be visited, we need to keep track of their address.
Comments | This was kept as 0..1 as there is no use property on the address, so wouldn't be able to identify different address types.
Location.physicalType
Definition | Physical form of the location, e.g. building, room, vehicle, road.
Control | 0..1
Terminology Binding | LocationType (Example)
Type | CodeableConcept
Requirements | For purposes of showing relevant locations in queries, we need to categorize locations.
Summary | true
Location.position
Definition | The absolute geographic location of the Location, expressed using the WGS84 datum (This is the same co-ordinate system used in KML).
Control | 0..1
Requirements | For mobile applications and automated route-finding knowing the exact location of the Location is required.
Location.position.longitude
Definition | Longitude. The value domain and the interpretation are the same as for the text of the longitude element in KML (see notes below).
Control | 1..1
Type | decimal
Location.position.latitude
Definition | Latitude. The value domain and the interpretation are the same as for the text of the latitude element in KML (see notes below).
Control | 1..1
Type | decimal
Location.position.altitude
Definition | Altitude. The value domain and the interpretation are the same as for the text of the altitude element in KML (see notes below).

Control | 0..1
Type | decimal
Location.managingOrganization
Definition | The organization responsible for the provisioning and upkeep of the location.
Control | 0..1
Type | Reference(Organization)
Requirements | Need to know who manages the location.
Summary | true
Comments | This can also be used as the part of the organization hierarchy where this location provides services. These services can be defined through the HealthcareService resource.
Location.partOf
Definition | Another Location which this Location is physically part of.
Control | 0..1
Type | Reference(Location)
Requirements | For purposes of location, display and identification, knowing which locations are located within other locations is important.
Location.endpoint
Definition | Technical endpoints providing access to services operated for the location.
Control | 0..*
Type | Reference(Endpoint)
Requirements | Organizations may have different systems at different locations that provide various services and need to be able to define the technical connection details for how to connect to them, and for what purpose.

8.7.9 Resource Location - Mappings

URL: https://hl7.org/fhir/STU3/location-mappings.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.7.9 Resource Location - Mappings

Patient Administration Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Not linked to any defined compartments

Mappings:

RIM Mapping

Mappings for the location resource.

8.7.9.1 Mappings for RIM Mapping (http://hl7.org/v3)

Location | .Role[classCode=SDLC]
identifier | .id
status | .statusCode
operationalStatus | n/a
name | .name
alias | .name
description | .playingEntity[classCode=PLC determinerCode=INSTANCE].desc
mode | .playingEntity[classCode=PLC].determinerCode
type | .code
telecom | .telecom
address | .addr
physicalType | .playingEntity [classCode=PLC].code
position | .playingEntity [classCode=PLC determinerCode=INSTANCE].positionText
longitude | (RIM Opted not to map the sub-elements of GPS location, is now an OBS)
latitude | (RIM Opted not to map the sub-elements of GPS location, is now an OBS)
altitude | (RIM Opted not to map the sub-elements of GPS location, is now an OBS)
managingOrganization | .scopingEntity[classCode=ORG determinerKind=INSTANCE]
partOf | .inboundLink[typeCode=PART].source[classCode=SDLC]
endpoint | n/a

8.7.10 Resource Location - Design Notes

URL: https://hl7.org/fhir/STU3/location-explanations.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.7.10 Resource Location - Design Notes

to do

8.7.11 Resource Location - Extensions & Profiles

URL: https://hl7.org/fhir/STU3/location-profiles.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.7.11 Resource Location - Extensions & Profiles

Patient Administration Work Group | Maturity Level: N/A | Ballot Status: Informative

This table lists profiles and extensions for the Location resource. For background information, see Profiling Resources and Extensibility. Additional profiles and resources may be found in published Implementation Guides , or in the Conformance resource registry .

8.7.11.1 Profiles

No Profiles defined for this resource

8.7.11.2 Extensions

location-alias | alias | for Location HL7 Extensions

Extensions that reference this resource:

consent-location | location | for Consent HL7 Extensions
diagnosticReport-locationPerformed | locationPerformed | for DiagnosticReport HL7 Extensions

8.7.11.3 Search Extensions

No Search Extensions defined for this resource

8.7.12 Location HL7 Extensions

URL: https://hl7.org/fhir/STU3/location-extensions.html

Defines common extensions used with or related to the Location resource

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.7.12 Location HL7 Extensions

(No assigned work group) | Maturity Level: N/A | Ballot Status: Informative

8.7.12.1 Content

Extensions:
location-alias | alias : Other names that this location is known as. (for Example, "Pembo HC" and "Pembo Health Clinic").

8.7.12.2 Search Parameters

Search parameters defined by this package. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Paths | Source
alias | string | Include Locations that contain the provided alias | f:Location/f:extension[@url='http://hl7.org/fhir/StructureDefinition/location-alias'] | /f:#alias | XML / JSON

8.7.12.3 Location-extensions-Location-alias

URL: https://hl7.org/fhir/STU3/location-extensions-Location-alias.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.7.12.3 Location-extensions-Location-alias

Search Parameter alias (type: string) defined as part of the location-extensions Profile.

Include Locations that contain the provided alias

Other Details:

* Full URL : http://hl7.org/fhir/SearchParameter/location-extensions-Location-alias

* Paths : f:Location/f:extension[@url='http://hl7.org/fhir/StructureDefinition/location-alias'] | /f:#alias

* Targets :

8.7.13 Resource Location - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/location-version-maps.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.7.13 Resource Location - Detailed Descriptions

Patient Administration Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Not linked to any defined compartments

R2 : R3 Converson maps for Location.

Functional status for this map: 6 tests that all execute ok. All tests pass round-trip testing and all r3 resources are valid. (see documentation)

8.7.13.1 R2 to R3

```
map "http://hl7.org/fhir/StructureMap/Location2to3" = "R2 to R3 Conversions for Location"

uses "http://hl7.org/fhir/DSTU2/StructureDefinition/Location" alias LocationR2 as source
uses "http://hl7.org/fhir/StructureDefinition/Location" alias Location as target

imports "http://hl7.org/fhir/StructureMap/*2to3"

group for type+types  Location extends DomainResource
  input src : LocationR2 as source
  input tgt : Location as target

  "Location-identifier" : for src.identifier make tgt.identifier
  "Location-status" : for src.status make tgt.status
  "Location-name" : for src.name make tgt.name
  "Location-description" : for src.description make tgt.description
  "Location-mode" : for src.mode make tgt.mode
  "Location-type" : for src.type make tgt.type
  "Location-telecom" : for src.telecom make tgt.telecom
  "Location-address" : for src.address make tgt.address
  "Location-physicalType" : for src.physicalType make tgt.physicalType
  "Location-position" : for src.position as vs0 make tgt.position as vt0 then locPosition(vs0, vt0)
  "Location-managingOrganization" : for src.managingOrganization make tgt.managingOrganization
  "Location-partOf" : for src.partOf make tgt.partOf
endgroup

group locPosition extends BackboneElement
  input src : Location as source
  input tgt : LocationR2 as target

  "Location.position-longitude" : for src.longitude make tgt.longitude
  "Location.position-latitude" : for src.latitude make tgt.latitude
  "Location.position-altitude" : for src.altitude make tgt.altitude
endgroup
```

8.7.13.2 R3 to R2

```
map "http://hl7.org/fhir/StructureMap/Location3to2" = "R3 to R2 Conversion for Location"

uses "http://hl7.org/fhir/StructureDefinition/Location" alias Location as source
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/Location" alias LocationR2 as target

imports "http://hl7.org/fhir/StructureMap/*3to2"

group for type+types  Location extends DomainResource
  input src : Location as source
  input tgt : LocationR2 as target

  "Location-identifier" : for src.identifier make tgt.identifier
  "Location-status" : for src.status make tgt.status
  "Location-name" : for src.name make tgt.name
  "Location-description" : for src.description make tgt.description
  "Location-mode" : for src.mode make tgt.mode
  "Location-type" : for src.type make tgt.type
  "Location-telecom" : for src.telecom make tgt.telecom
  "Location-address" : for src.address make tgt.address
  "Location-physicalType" : for src.physicalType make tgt.physicalType
  "Location-position" : for src.position as vs0 make tgt.position as vt0 then locPosition(vs0, vt0)
  "Location-managingOrganization" : for src.managingOrganization make tgt.managingOrganization
  "Location-partOf" : for src.partOf make tgt.partOf
endgroup

group locPosition extends BackboneElement
  input src : Location as source
  input tgt : LocationR2 as target

  "Location.position-longitude" : for src.longitude make tgt.longitude
  "Location.position-latitude" : for src.latitude make tgt.latitude
  "Location.position-altitude" : for src.altitude make tgt.altitude
endgroup
```

8.7.13.3 R3 Validation Errors

8.8 Resource HealthcareService - Content

URL: https://hl7.org/fhir/STU3/healthcareservice.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

Patient Administration Work Group | Maturity Level: 2 | Trial Use | Compartments: Not linked to any defined compartments

The details of a healthcare service available at a location.

8.8.1 Scope and Usage

The HealthcareService resource is used to describe a single healthcare service or category of services that are provided by an organization at a location. The location of the services could be virtual, as with TeleMedicine Services.

Common examples of HealthcareServices resources are:

* Allied Health
* Clinical Neuropsychologist
* Podiatry Service
* Smallville Hospital Emergency Services
* Respite care provided at a nursing home or hostel
* 24hr crisis telephone counseling service
* Information, advice and/or referral services; Disability, Telecommunications
* Rural TeleHealth Services
* Hospital in the home
* Yellow Cabs
* Pharmacy
* Active Rehab
* Social Support
* Drug and/or alcohol counseling
* Day Programs, Adult Training & Support Services
* Consulting psychologists and/or psychology services
* Group Hydrotherapy
* Little River Home Maintenance

HealthcareService resources do not represent Computer related Services (not SOA)

Example uses of HealthcareService resources are:

* National Services Directory - Consumer Focus
* National Services Directory - Practitioner Referrals Searching
* Organization's Client Portal - to locate services / book appointments
* Address book of services for Referrals including references to Questionnaires for assessments that are required as part of the referral
* Health Network internal directory Used for tracking available services offered internally, and also those offered by business partners. This information may also include costing information.

8.8.2 Boundaries and Relationships

The HealthcareService resource can be used with the Schedule resource to define actual availability of the service. This would be done by using the Schedule's Actor property.

When creating an Appointment, the HealthcareService is to be assigned to one of the participants. It is up to the scheduling system to determine if the service is available, and can be accepted.

This resource is referenced by Account, Appointment, AppointmentResponse, DeviceRequest, PractitionerRole, Procedure, ProcedureRequest, ReferralRequest and Schedule

8.8.3 Resource Content

* Structure
* UML
* XML
* JSON
* Turtle
* R2 Diff
* All

Structure

Name | Flags | Card. | Type | Description & Constraints
HealthcareService | | | DomainResource | The details of a healthcare service available at a location Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | ? | 0..* | Identifier | External identifiers for this item
active | ?!? | 0..1 | boolean | Whether this healthcareservice is in active use
providedBy | ? | 0..1 | Reference(Organization) | Organization that provides this service
category | ? | 0..1 | CodeableConcept | Broad category of service being performed or delivered ServiceCategory (Example)

type | ? | 0..* | CodeableConcept | Type of service that may be delivered or performed ServiceType (Example)
specialty | ? | 0..* | CodeableConcept | Specialties handled by the HealthcareService Practice Setting Code Value Set (Preferred)
location | ? | 0..* | Reference(Location) | Location(s) where service may be provided
name | ? | 0..1 | string | Description of service as presented to a consumer while searching
comment | ? | 0..1 | string | Additional description and/or any specific issues not covered elsewhere
extraDetails | | 0..1 | string | Extra details about the service that can't be placed in the other fields
photo | ? | 0..1 | Attachment | Facilitates quick identification of the service
telecom | | 0..* | ContactPoint | Contacts related to the healthcare service
coverageArea | | 0..* | Reference(Location) | Location(s) service is inteded for/available to
serviceProvisionCode | | 0..* | CodeableConcept | Conditions under which service is available/offered ServiceProvisionConditions (Example)
eligibility | | 0..1 | CodeableConcept | Specific eligibility requirements required to use the service
eligibilityNote | | 0..1 | string | Describes the eligibility conditions for the service
programName | | 0..* | string | Program Names that categorize the service
characteristic | | 0..* | CodeableConcept | Collection of characteristics (attributes)
referralMethod | | 0..* | CodeableConcept | Ways that the service accepts referrals ReferralMethod (Example)
appointmentRequired | | 0..1 | boolean | If an appointment is required for access to this service
availableTime | | 0..* | BackboneElement | Times the Service Site is available
daysOfWeek | | 0..* | code | mon | tue | wed | thu | fri | sat | sun DaysOfWeek (Required)
allDay | | 0..1 | boolean | Always available? e.g. 24 hour service
availableStartTime | | 0..1 | time | Opening time of day (ignored if allDay = true)
availableEndTime | | 0..1 | time | Closing time of day (ignored if allDay = true)
notAvailable | | 0..* | BackboneElement | Not available during this time due to provided reason
description | | 1..1 | string | Reason presented to the user explaining why time not available
during | | 0..1 | Period | Service not availablefrom this date
availabilityExceptions | | 0..1 | string | Description of availability exceptions
endpoint | | 0..* | Reference(Endpoint) | Technical endpoints providing access to services operated for the location
Documentation for this format

UML Diagram (Legend)

XML Template

```
<HealthcareService xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier External identifiers for this item --></identifier>
 <active value="[boolean]"/><!-- 0..1 Whether this healthcareservice is in active use -->
 <providedBy><!-- 0..1 Reference(Organization) Organization that provides this service --></providedBy>
 <category><!-- 0..1 CodeableConcept Broad category of service being performed or delivered --></category>
 <type><!-- 0..* CodeableConcept Type of service that may be delivered or performed --></type>
 <specialty><!-- 0..* CodeableConcept Specialties handled by the HealthcareService --></specialty>
 <location><!-- 0..* Reference(Location) Location(s) where service may be provided --></location>
 <name value="[string]"/><!-- 0..1 Description of service as presented to a consumer while searching -->
 <comment value="[string]"/><!-- 0..1 Additional description and/or any specific issues not covered elsewhere -->
 <extraDetails value="[string]"/><!-- 0..1 Extra details about the service that can't be placed in the other fields -->
 <photo><!-- 0..1 Attachment Facilitates quick identification of the service --></photo>
 <telecom><!-- 0..* ContactPoint Contacts related to the healthcare service --></telecom>
 <coverageArea><!-- 0..* Reference(Location) Location(s) service is inteded for/available to --></coverageArea>
 <serviceProvisionCode><!-- 0..* CodeableConcept Conditions under which service is available/offered
    --></serviceProvisionCode>
 <eligibility><!-- 0..1 CodeableConcept Specific eligibility requirements required to use the service --></eligibility>
 <eligibilityNote value="[string]"/><!-- 0..1 Describes the eligibility conditions for the service -->
 <programName value="[string]"/><!-- 0..* Program Names that categorize the service -->
 <characteristic><!-- 0..* CodeableConcept Collection of characteristics (attributes) --></characteristic>
 <referralMethod><!-- 0..* CodeableConcept Ways that the service accepts referrals --></referralMethod>
 <appointmentRequired value="[boolean]"/><!-- 0..1 If an appointment is required for access to this service -->
 <availableTime>  <!-- 0..* Times the Service Site is available -->
  <daysOfWeek value="[code]"/><!-- 0..* mon | tue | wed | thu | fri | sat | sun -->
  <allDay value="[boolean]"/><!-- 0..1 Always available? e.g. 24 hour service -->
  <availableStartTime value="[time]"/><!-- 0..1 Opening time of day (ignored if allDay = true) -->
  <availableEndTime value="[time]"/><!-- 0..1 Closing time of day (ignored if allDay = true) -->
 </availableTime>
 <notAvailable>  <!-- 0..* Not available during this time due to provided reason -->
  <description value="[string]"/><!-- 1..1 Reason presented to the user explaining why time not available -->
  <during><!-- 0..1 Period Service not availablefrom this date --></during>
 </notAvailable>
 <availabilityExceptions value="[string]"/><!-- 0..1 Description of availability exceptions -->
 <endpoint><!-- 0..* Reference(Endpoint) Technical endpoints providing access to services operated for the location
    --></endpoint>
</HealthcareService>
```

JSON Template

```
{
  "resourceType" : "HealthcareService",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // External identifiers for this item
  "active" : <boolean>, // Whether this healthcareservice is in active use
  "providedBy" : { Reference(Organization) }, // Organization that provides this service
  "category" : { CodeableConcept }, // Broad category of service being performed or delivered
  "type" : [{ CodeableConcept }], // Type of service that may be delivered or performed
  "specialty" : [{ CodeableConcept }], // Specialties handled by the HealthcareService
  "location" : [{ Reference(Location) }], // Location(s) where service may be provided
  "name" : "<string>", // Description of service as presented to a consumer while searching
```

```
  "comment" : "<string>", // Additional description and/or any specific issues not covered elsewhere
  "extraDetails" : "<string>", // Extra details about the service that can't be placed in the other fields
  "photo" : { Attachment }, // Facilitates quick identification of the service
  "telecom" : [{ ContactPoint }], // Contacts related to the healthcare service
  "coverageArea" : [{ Reference(Location) }], // Location(s) service is inteded for/available to
  "serviceProvisionCode" : [{ CodeableConcept }], // Conditions under which service is available/offered
  "eligibility" : { CodeableConcept }, // Specific eligibility requirements required to use the service
  "eligibilityNote" : "<string>", // Describes the eligibility conditions for the service
  "programName" : ["<string>"], // Program Names that categorize the service
  "characteristic" : [{ CodeableConcept }], // Collection of characteristics (attributes)
  "referralMethod" : [{ CodeableConcept }], // Ways that the service accepts referrals
  "appointmentRequired" : <boolean>, // If an appointment is required for access to this service
  "availableTime" : [{ // Times the Service Site is available
    "daysOfWeek" : ["<code>"], // mon | tue | wed | thu | fri | sat | sun
    "allDay" : <boolean>, // Always available? e.g. 24 hour service
    "availableStartTime" : "<time>", // Opening time of day (ignored if allDay = true)
    "availableEndTime" : "<time>" // Closing time of day (ignored if allDay = true)
  }],
  "notAvailable" : [{ // Not available during this time due to provided reason
    "description" : "<string>", // R!  Reason presented to the user explaining why time not available
    "during" : { Period } // Service not availablefrom this date
  }],
  "availabilityExceptions" : "<string>", // Description of availability exceptions
  "endpoint" : [{ Reference(Endpoint) }] // Technical endpoints providing access to services operated for the location
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:HealthcareService;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:HealthcareService.identifier [ Identifier ], ... ; # 0..* External identifiers for this item
  fhir:HealthcareService.active [ boolean ]; # 0..1 Whether this healthcareservice is in active use
  fhir:HealthcareService.providedBy [ Reference(Organization) ]; # 0..1 Organization that provides this service
  fhir:HealthcareService.category [ CodeableConcept ]; # 0..1 Broad category of service being performed or delivered
  fhir:HealthcareService.type [ CodeableConcept ], ... ; # 0..* Type of service that may be delivered or performed
  fhir:HealthcareService.specialty [ CodeableConcept ], ... ; # 0..* Specialties handled by the HealthcareService
  fhir:HealthcareService.location [ Reference(Location) ], ... ; # 0..* Location(s) where service may be provided
  fhir:HealthcareService.name [ string ]; # 0..1 Description of service as presented to a consumer while searching
  fhir:HealthcareService.comment [ string ]; # 0..1 Additional description and/or any specific issues not covered
    elsewhere
  fhir:HealthcareService.extraDetails [ string ]; # 0..1 Extra details about the service that can't be placed in the
    other fields
  fhir:HealthcareService.photo [ Attachment ]; # 0..1 Facilitates quick identification of the service
  fhir:HealthcareService.telecom [ ContactPoint ], ... ; # 0..* Contacts related to the healthcare service
  fhir:HealthcareService.coverageArea [ Reference(Location) ], ... ; # 0..* Location(s) service is inteded for/available
     to
  fhir:HealthcareService.serviceProvisionCode [ CodeableConcept ], ... ; # 0..* Conditions under which service is
    available/offered
  fhir:HealthcareService.eligibility [ CodeableConcept ]; # 0..1 Specific eligibility requirements required to use the
    service
  fhir:HealthcareService.eligibilityNote [ string ]; # 0..1 Describes the eligibility conditions for the service
  fhir:HealthcareService.programName [ string ], ... ; # 0..* Program Names that categorize the service
  fhir:HealthcareService.characteristic [ CodeableConcept ], ... ; # 0..* Collection of characteristics (attributes)
  fhir:HealthcareService.referralMethod [ CodeableConcept ], ... ; # 0..* Ways that the service accepts referrals
  fhir:HealthcareService.appointmentRequired [ boolean ]; # 0..1 If an appointment is required for access to this
    service
  fhir:HealthcareService.availableTime [ # 0..* Times the Service Site is available
    fhir:HealthcareService.availableTime.daysOfWeek [ code ], ... ; # 0..* mon | tue | wed | thu | fri | sat | sun
    fhir:HealthcareService.availableTime.allDay [ boolean ]; # 0..1 Always available? e.g. 24 hour service
    fhir:HealthcareService.availableTime.availableStartTime [ time ]; # 0..1 Opening time of day (ignored if allDay =
    true)
    fhir:HealthcareService.availableTime.availableEndTime [ time ]; # 0..1 Closing time of day (ignored if allDay =
    true)
  ], ...;
  fhir:HealthcareService.notAvailable [ # 0..* Not available during this time due to provided reason
    fhir:HealthcareService.notAvailable.description [ string ]; # 1..1 Reason presented to the user explaining why time
    not available
    fhir:HealthcareService.notAvailable.during [ Period ]; # 0..1 Service not availablefrom this date
  ], ...;
  fhir:HealthcareService.availabilityExceptions [ string ]; # 0..1 Description of availability exceptions
  fhir:HealthcareService.endpoint [ Reference(Endpoint) ], ... ; # 0..* Technical endpoints providing access to services
     operated for the location
]
```

Changes since DSTU2

HealthcareService

HealthcareService.active | Added Element

HealthcareService.category | Renamed from serviceCategory to category

HealthcareService.type | Added Element

HealthcareService.specialty | Added Element

HealthcareService.location | Min Cardinality changed from 1 to 0 Max Cardinality changed from 1 to *

HealthcareService.name | Renamed from serviceName to name

HealthcareService.endpoint | Added Element
HealthcareService.serviceType | deleted
HealthcareService.publicKey | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R2 <--> R3 Conversion Maps (status = 1 test that all execute ok. 1 fail round-trip testing and all r3 resources are valid.).

Structure

Name | Flags | Card. | Type | Description & Constraints
HealthcareService | | | DomainResource | The details of a healthcare service available at a location Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | ? | 0..* | Identifier | External identifiers for this item
active | ?!? | 0..1 | boolean | Whether this healthcareservice is in active use
providedBy | ? | 0..1 | Reference(Organization) | Organization that provides this service
category | ? | 0..1 | CodeableConcept | Broad category of service being performed or delivered ServiceCategory (Example)
type | ? | 0..* | CodeableConcept | Type of service that may be delivered or performed ServiceType (Example)
specialty | ? | 0..* | CodeableConcept | Specialties handled by the HealthcareService Practice Setting Code Value Set (Preferred)
location | ? | 0..* | Reference(Location) | Location(s) where service may be provided
name | ? | 0..1 | string | Description of service as presented to a consumer while searching
comment | ? | 0..1 | string | Additional description and/or any specific issues not covered elsewhere
extraDetails | | 0..1 | string | Extra details about the service that can't be placed in the other fields
photo | ? | 0..1 | Attachment | Facilitates quick identification of the service
telecom | | 0..* | ContactPoint | Contacts related to the healthcare service
coverageArea | | 0..* | Reference(Location) | Location(s) service is inteded for/available to
serviceProvisionCode | | 0..* | CodeableConcept | Conditions under which service is available/offered ServiceProvisionConditions (Example)
eligibility | | 0..1 | CodeableConcept | Specific eligibility requirements required to use the service
eligibilityNote | | 0..1 | string | Describes the eligibility conditions for the service
programName | | 0..* | string | Program Names that categorize the service
characteristic | | 0..* | CodeableConcept | Collection of characteristics (attributes)
referralMethod | | 0..* | CodeableConcept | Ways that the service accepts referrals ReferralMethod (Example)
appointmentRequired | | 0..1 | boolean | If an appointment is required for access to this service
availableTime | | 0..* | BackboneElement | Times the Service Site is available
daysOfWeek | | 0..* | code | mon | tue | wed | thu | fri | sat | sun DaysOfWeek (Required)
allDay | | 0..1 | boolean | Always available? e.g. 24 hour service
availableStartTime | | 0..1 | time | Opening time of day (ignored if allDay = true)
availableEndTime | | 0..1 | time | Closing time of day (ignored if allDay = true)
notAvailable | | 0..* | BackboneElement | Not available during this time due to provided reason
description | | 1..1 | string | Reason presented to the user explaining why time not available
during | | 0..1 | Period | Service not availablefrom this date
availabilityExceptions | | 0..1 | string | Description of availability exceptions
endpoint | | 0..* | Reference(Endpoint) | Technical endpoints providing access to services operated for the location
Documentation for this format

UML Diagram (Legend)

XML Template

```
<HealthcareService xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier External identifiers for this item --></identifier>
 <active value="[boolean]"/><!-- 0..1 Whether this healthcareservice is in active use -->
 <providedBy><!-- 0..1 Reference(Organization) Organization that provides this service --></providedBy>
 <category><!-- 0..1 CodeableConcept Broad category of service being performed or delivered --></category>
 <type><!-- 0..* CodeableConcept Type of service that may be delivered or performed --></type>
 <specialty><!-- 0..* CodeableConcept Specialties handled by the HealthcareService --></specialty>
 <location><!-- 0..* Reference(Location) Location(s) where service may be provided --></location>
 <name value="[string]"/><!-- 0..1 Description of service as presented to a consumer while searching -->
 <comment value="[string]"/><!-- 0..1 Additional description and/or any specific issues not covered elsewhere -->
 <extraDetails value="[string]"/><!-- 0..1 Extra details about the service that can't be placed in the other fields -->
 <photo><!-- 0..1 Attachment Facilitates quick identification of the service --></photo>
 <telecom><!-- 0..* ContactPoint Contacts related to the healthcare service --></telecom>
 <coverageArea><!-- 0..* Reference(Location) Location(s) service is inteded for/available to --></coverageArea>
 <serviceProvisionCode><!-- 0..* CodeableConcept Conditions under which service is available/offered
    --></serviceProvisionCode>
 <eligibility><!-- 0..1 CodeableConcept Specific eligibility requirements required to use the service --></eligibility>
 <eligibilityNote value="[string]"/><!-- 0..1 Describes the eligibility conditions for the service -->
 <programName value="[string]"/><!-- 0..* Program Names that categorize the service -->
 <characteristic><!-- 0..* CodeableConcept Collection of characteristics (attributes) --></characteristic>
 <referralMethod><!-- 0..* CodeableConcept Ways that the service accepts referrals --></referralMethod>
 <appointmentRequired value="[boolean]"/><!-- 0..1 If an appointment is required for access to this service -->
 <availableTime>  <!-- 0..* Times the Service Site is available -->
  <daysOfWeek value="[code]"/><!-- 0..* mon | tue | wed | thu | fri | sat | sun -->
  <allDay value="[boolean]"/><!-- 0..1 Always available? e.g. 24 hour service -->
  <availableStartTime value="[time]"/><!-- 0..1 Opening time of day (ignored if allDay = true) -->
  <availableEndTime value="[time]"/><!-- 0..1 Closing time of day (ignored if allDay = true) -->
 </availableTime>
 <notAvailable>  <!-- 0..* Not available during this time due to provided reason -->
  <description value="[string]"/><!-- 1..1 Reason presented to the user explaining why time not available -->
  <during><!-- 0..1 Period Service not availablefrom this date --></during>
```

```
 </notAvailable>
 <availabilityExceptions value="[string]"/><!-- 0..1 Description of availability exceptions -->
 <endpoint><!-- 0..* Reference(Endpoint) Technical endpoints providing access to services operated for the location
    --></endpoint>
</HealthcareService>
```

JSON Template

```
{
  "resourceType" : "HealthcareService",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // External identifiers for this item
  "active" : <boolean>, // Whether this healthcareservice is in active use
  "providedBy" : { Reference(Organization) }, // Organization that provides this service
  "category" : { CodeableConcept }, // Broad category of service being performed or delivered
  "type" : [{ CodeableConcept }], // Type of service that may be delivered or performed
  "specialty" : [{ CodeableConcept }], // Specialties handled by the HealthcareService
  "location" : [{ Reference(Location) }], // Location(s) where service may be provided
  "name" : "<string>", // Description of service as presented to a consumer while searching
  "comment" : "<string>", // Additional description and/or any specific issues not covered elsewhere
  "extraDetails" : "<string>", // Extra details about the service that can't be placed in the other fields
  "photo" : { Attachment }, // Facilitates quick identification of the service
  "telecom" : [{ ContactPoint }], // Contacts related to the healthcare service
  "coverageArea" : [{ Reference(Location) }], // Location(s) service is inteded for/available to
  "serviceProvisionCode" : [{ CodeableConcept }], // Conditions under which service is available/offered
  "eligibility" : { CodeableConcept }, // Specific eligibility requirements required to use the service
  "eligibilityNote" : "<string>", // Describes the eligibility conditions for the service
  "programName" : ["<string>"], // Program Names that categorize the service
  "characteristic" : [{ CodeableConcept }], // Collection of characteristics (attributes)
  "referralMethod" : [{ CodeableConcept }], // Ways that the service accepts referrals
  "appointmentRequired" : <boolean>, // If an appointment is required for access to this service
  "availableTime" : [{ // Times the Service Site is available
    "daysOfWeek" : ["<code>"], // mon | tue | wed | thu | fri | sat | sun
    "allDay" : <boolean>, // Always available? e.g. 24 hour service
    "availableStartTime" : "<time>", // Opening time of day (ignored if allDay = true)
    "availableEndTime" : "<time>" // Closing time of day (ignored if allDay = true)
  }],
  "notAvailable" : [{ // Not available during this time due to provided reason
    "description" : "<string>", // R!  Reason presented to the user explaining why time not available
    "during" : { Period } // Service not availablefrom this date
  }],
  "availabilityExceptions" : "<string>", // Description of availability exceptions
  "endpoint" : [{ Reference(Endpoint) }] // Technical endpoints providing access to services operated for the location
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:HealthcareService;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:HealthcareService.identifier [ Identifier ], ... ; # 0..* External identifiers for this item
  fhir:HealthcareService.active [ boolean ]; # 0..1 Whether this healthcareservice is in active use
  fhir:HealthcareService.providedBy [ Reference(Organization) ]; # 0..1 Organization that provides this service
  fhir:HealthcareService.category [ CodeableConcept ]; # 0..1 Broad category of service being performed or delivered
  fhir:HealthcareService.type [ CodeableConcept ], ... ; # 0..* Type of service that may be delivered or performed
  fhir:HealthcareService.specialty [ CodeableConcept ], ... ; # 0..* Specialties handled by the HealthcareService
  fhir:HealthcareService.location [ Reference(Location) ], ... ; # 0..* Location(s) where service may be provided
  fhir:HealthcareService.name [ string ]; # 0..1 Description of service as presented to a consumer while searching
  fhir:HealthcareService.comment [ string ]; # 0..1 Additional description and/or any specific issues not covered
    elsewhere
  fhir:HealthcareService.extraDetails [ string ]; # 0..1 Extra details about the service that can't be placed in the
    other fields
  fhir:HealthcareService.photo [ Attachment ]; # 0..1 Facilitates quick identification of the service
  fhir:HealthcareService.telecom [ ContactPoint ], ... ; # 0..* Contacts related to the healthcare service
  fhir:HealthcareService.coverageArea [ Reference(Location) ], ... ; # 0..* Location(s) service is inteded for/available
     to
  fhir:HealthcareService.serviceProvisionCode [ CodeableConcept ], ... ; # 0..* Conditions under which service is
    available/offered
  fhir:HealthcareService.eligibility [ CodeableConcept ]; # 0..1 Specific eligibility requirements required to use the
    service
  fhir:HealthcareService.eligibilityNote [ string ]; # 0..1 Describes the eligibility conditions for the service
  fhir:HealthcareService.programName [ string ], ... ; # 0..* Program Names that categorize the service
  fhir:HealthcareService.characteristic [ CodeableConcept ], ... ; # 0..* Collection of characteristics (attributes)
  fhir:HealthcareService.referralMethod [ CodeableConcept ], ... ; # 0..* Ways that the service accepts referrals
  fhir:HealthcareService.appointmentRequired [ boolean ]; # 0..1 If an appointment is required for access to this
    service
  fhir:HealthcareService.availableTime [ # 0..* Times the Service Site is available
    fhir:HealthcareService.availableTime.daysOfWeek [ code ], ... ; # 0..* mon | tue | wed | thu | fri | sat | sun
    fhir:HealthcareService.availableTime.allDay [ boolean ]; # 0..1 Always available? e.g. 24 hour service
    fhir:HealthcareService.availableTime.availableStartTime [ time ]; # 0..1 Opening time of day (ignored if allDay =
    true)
    fhir:HealthcareService.availableTime.availableEndTime [ time ]; # 0..1 Closing time of day (ignored if allDay =
    true)
  ], ...;
  fhir:HealthcareService.notAvailable [ # 0..* Not available during this time due to provided reason
```

```
    fhir:HealthcareService.notAvailable.description [ string ]; # 1..1 Reason presented to the user explaining why time
    not available
    fhir:HealthcareService.notAvailable.during [ Period ]; # 0..1 Service not availablefrom this date
  ], ...;
  fhir:HealthcareService.availabilityExceptions [ string ]; # 0..1 Description of availability exceptions
  fhir:HealthcareService.endpoint [ Reference(Endpoint) ], ... ; # 0..* Technical endpoints providing access to services
     operated for the location
]
```

Changes since DSTU2

HealthcareService
HealthcareService.active | Added Element
HealthcareService.category | Renamed from serviceCategory to category
HealthcareService.type | Added Element
HealthcareService.specialty | Added Element
HealthcareService.location | Min Cardinality changed from 1 to 0 Max Cardinality changed from 1 to *
HealthcareService.name | Renamed from serviceName to name
HealthcareService.endpoint | Added Element
HealthcareService.serviceType | deleted
HealthcareService.publicKey | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R2 <--> R3 Conversion Maps (status = 1 test that all execute ok. 1 fail round-trip testing and all r3 resources are valid.).

Alternate definitions: Master Definition (XML, JSON), XML Schema/Schematron (for) + JSON Schema, ShEx (for Turtle)

8.8.3.1 Terminology Bindings

Path | Definition | Type | Reference
HealthcareService.category | A category of the service(s) that could be provided. | Example | ServiceCategory
HealthcareService.type | Additional details about where the content was created (e.g. clinical specialty). | Example | ServiceType
HealthcareService.specialty | A specialty that a healthcare service may provide. | Preferred | Practice Setting Code Value Set
HealthcareService.serviceProvisionCode | The code(s) that detail the conditions under which the healthcare service is available/offered. | Example | ServiceProvisionConditions
HealthcareService.eligibility | Coded values underwhich a specific service is made available. | Unknown | No details provided yet
HealthcareService.characteristic | A custom attribute that could be provided at a service (e.g. Wheelchair accessibiliy). | Unknown | No details provided yet
HealthcareService.referralMethod | The methods of referral can be used when referring to a specific HealthCareService resource. | Example | ReferralMethod
HealthcareService.availableTime.daysOfWeek | The days of the week. | Required | DaysOfWeek

8.8.4 Notes:

* The HealthcareService could be mapped to components of the IHE Care Services Directory, and/or the OMG ServD standards

8.8.5 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
active | token | The Healthcare Service is currently marked as active | HealthcareService.active
category | token | Service Category of the Healthcare Service | HealthcareService.category
characteristic | token | One of the HealthcareService's characteristics | HealthcareService.characteristic
endpoint | reference | Technical endpoints providing access to services operated for the location | HealthcareService.endpoint (Endpoint)
identifier | token | External identifiers for this item | HealthcareService.identifier
location | reference | The location of the Healthcare Service | HealthcareService.location (Location)
name | string | A portion of the Healthcare service name | HealthcareService.name
organization | reference | The organization that provides this Healthcare Service | HealthcareService.providedBy (Organization)
programname | string | One of the Program Names serviced by this HealthcareService | HealthcareService.programName
type | token | The type of service provided by this healthcare service | HealthcareService.type

8.8.6 Resource HealthcareService - Examples

URL: https://hl7.org/fhir/STU3/healthcareservice-examples.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.8.6 Resource HealthcareService - Examples

Patient Administration Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Not linked to any defined compartments

Example List:

Example of healthcareservice | example | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

8.8.7 Resource HealthcareService - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/healthcareservice-definitions.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.8.7 Resource HealthcareService - Detailed Descriptions

Patient Administration Work Group | Maturity Level: 2 | Trial Use | Compartments: Not linked to any defined compartments

Detailed Descriptions for the elements in the HealthcareService resource.

HealthcareService
Definition | The details of a healthcare service available at a location.
Control | 1..1
HealthcareService.identifier
Definition | External identifiers for this item.
Note | This is a business identifer, not a resource identifier (see discussion)
Control | 0..*
Type | Identifier
Summary | true
HealthcareService.active
Definition | Whether this healthcareservice record is in active use.
Control | 0..1
Type | boolean
Is Modifier | true
Default Value | true
Summary | true
Comments | This element is labeled as a modifier because it may be used to mark that the resource was created in error.
HealthcareService.providedBy
Definition | The organization that provides this healthcare service.
Control | 0..1
Type | Reference(Organization)
Summary | true
Comments | This property is recommended to be the same as the Location's managingOrganization, and if not provided should be interpreted as such. If the Location does not have a managing Organization, then this property should be populated.
HealthcareService.category
Definition | Identifies the broad category of service being performed or delivered.
Control | 0..1
Terminology Binding | ServiceCategory (Example)
Type | CodeableConcept
Alternate Names | service category
Summary | true
Comments | Selecting a Service Category then determines the list of relevant service types that can be selected in the primary service type.
HealthcareService.type
Definition | The specific type of service that may be delivered or performed.
Control | 0..*
Terminology Binding | ServiceType (Example)
Type | CodeableConcept
Alternate Names | service type
Summary | true
HealthcareService.specialty
Definition | Collection of specialties handled by the service site. This is more of a medical term.
Control | 0..*
Terminology Binding | Practice Setting Code Value Set (Preferred)
Type | CodeableConcept
Summary | true
HealthcareService.location
Definition | The location(s) where this healthcare service may be provided.
Control | 0..*
Type | Reference(Location)
Summary | true
HealthcareService.name
Definition | Further description of the service as it would be presented to a consumer while searching.
Control | 0..1
Type | string
Summary | true
HealthcareService.comment
Definition | Any additional description of the service and/or any specific issues not covered by the other attributes, which can be displayed as further detail under the serviceName.
Control | 0..1

Type | string
Summary | true
Comments | Would expect that a user would not see this information on a search results, and it would only be available when viewing the complete details of the service.
HealthcareService.extraDetails
Definition | Extra details about the service that can't be placed in the other fields.
Control | 0..1
Type | string
HealthcareService.photo
Definition | If there is a photo/symbol associated with this HealthcareService, it may be included here to facilitate quick identification of the service in a list.
Control | 0..1
Type | Attachment
Summary | true
HealthcareService.telecom
Definition | List of contacts related to this specific healthcare service.
Control | 0..*
Type | ContactPoint
Comments | If this is empty, then refer to the location's contacts.
HealthcareService.coverageArea
Definition | The location(s) that this service is available to (not where the service is provided).
Control | 0..*
Type | Reference(Location)
Comments | The locations referenced by the coverage area can include both specific locations, including areas, and also conceptual domains too (mode = kind), such as a physical area (tri-state area) and some other attribute (covered by Example Care Organization). These types of Locations are often not managed by any specific organization. This could also include generic locations such as "in-home".
HealthcareService.serviceProvisionCode
Definition | The code(s) that detail the conditions under which the healthcare service is available/offered.
Control | 0..*
Terminology Binding | ServiceProvisionConditions (Example)
Type | CodeableConcept
Comments | The provision means being commissioned by, contractually obliged or financially sourced. Types of costings that may apply to this healthcare service, such if the service may be available for free, some discounts available, or fees apply.
HealthcareService.eligibility
Definition | Does this service have specific eligibility requirements that need to be met in order to use the service?
Control | 0..1
Terminology Binding | ServiceEligibility:
Type | CodeableConcept
HealthcareService.eligibilityNote
Definition | Describes the eligibility conditions for the service.
Control | 0..1
Type | string
Comments | The description of service eligibility should, in general, not exceed one or two paragraphs. It should be sufficient for a prospective consumer to determine if they are likely to be eligible or not. Where eligibility requirements and conditions are complex, it may simply be noted that an eligibility assessment is required. Where eligibility is determined by an outside source, such as an Act of Parliament, this should be noted, preferably with a reference to a commonly available copy of the source document such as a web page.
HealthcareService.programName
Definition | Program Names that can be used to categorize the service.
Control | 0..*
Type | string
Comments | Programs are often defined externally to an Organization, commonly by governments; e.g. Home and Community Care Programs, Homeless Program, ….
HealthcareService.characteristic
Definition | Collection of characteristics (attributes).
Control | 0..*
Terminology Binding | ServiceCharacteristic:
Type | CodeableConcept
Comments | These could be such things as is wheelchair accessible.
HealthcareService.referralMethod
Definition | Ways that the service accepts referrals, if this is not provided then it is implied that no referral is required.
Control | 0..*
Terminology Binding | ReferralMethod (Example)
Type | CodeableConcept
HealthcareService.appointmentRequired
Definition | Indicates whether or not a prospective consumer will require an appointment for a particular service at a site to be provided by the Organization. Indicates if an appointment is required for access to this service.
Control | 0..1
Type | boolean
HealthcareService.availableTime
Definition | A collection of times that the Service Site is available.
Control | 0..*
Comments | More detailed availability information may be provided in associated Schedule/Slot resources.
HealthcareService.availableTime.daysOfWeek

Definition | Indicates which days of the week are available between the start and end Times.
Control | 0..*
Terminology Binding | DaysOfWeek (Required)
Type | code
HealthcareService.availableTime.allDay
Definition | Is this always available? (hence times are irrelevant) e.g. 24 hour service.
Control | 0..1
Type | boolean
HealthcareService.availableTime.availableStartTime
Definition | The opening time of day. Note: If the AllDay flag is set, then this time is ignored.
Control | 0..1
Type | time
Comments | The timezone is expected to be for where this HealthcareService is provided at.
HealthcareService.availableTime.availableEndTime
Definition | The closing time of day. Note: If the AllDay flag is set, then this time is ignored.
Control | 0..1
Type | time
Comments | The timezone is expected to be for where this HealthcareService is provided at.
HealthcareService.notAvailable
Definition | The HealthcareService is not available during this period of time due to the provided reason.
Control | 0..*
HealthcareService.notAvailable.description
Definition | The reason that can be presented to the user as to why this time is not available.
Control | 1..1
Type | string
HealthcareService.notAvailable.during
Definition | Service is not available (seasonally or for a public holiday) from this date.
Control | 0..1
Type | Period
HealthcareService.availabilityExceptions
Definition | A description of site availability exceptions, e.g. public holiday availability. Succinctly describing all possible exceptions to normal site availability as details in the available Times and not available Times.
Control | 0..1
Type | string
HealthcareService.endpoint
Definition | Technical endpoints providing access to services operated for the specific healthcare services defined at this resource.
Control | 0..*
Type | Reference(Endpoint)
Requirements | Organizations may have different systems covering different services at the different locations and need to be able to define the technical connection details for how to connect to them, and for what purpose.

8.8.8 Resource HealthcareService - Mappings

URL: https://hl7.org/fhir/STU3/healthcareservice-mappings.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.8.8 Resource HealthcareService - Mappings

Patient Administration Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Not linked to any defined compartments

Mappings:

RIM Mapping

Mappings for the healthcareservice resource.

8.8.8.1 Mappings for RIM Mapping (http://hl7.org/v3)

HealthcareService | act[classCode=ACT][moodCode=DEF]
identifier | .id
active | .statusCode
providedBy | .scopingRole.Organization
category | .code
type | .actrelationship[typeCode=COMP.act[classCode=ACT][moodCode=DEF].code
specialty | .actrelationship[typeCode=COMP.act[classCode=ACT][moodCode=DEF].code
location | .location.role[classCode=SDLOC]
name | .name
comment | .location.role[classCode=SDLOC].desc
extraDetails | .actrelationship[typeCode=COMP.act[classCode=ACT][moodCode=DEF].text
photo | .actrelationship[typeCode=SBJ].observation.value
telecom | .telecom
coverageArea | .location.role[classCode=SDLOC].subjectOf.A_SpatialCoordinate
serviceProvisionCode | .actrelationship[typeCode=PRCN].observation[moodCode=EVN.CRT]
eligibility | .actrelationship[typeCode=PRCN].observation[moodCode=EVN.CRT]

eligibilityNote | .actrelationship[typeCode=PRCN].observation[moodCode=EVN.CRT].text
programName | .actrelationship[typeCode=PERT].observation
characteristic | .actrelationship[typeCode=PERT].observation
referralMethod | .actrelationship[typeCode=PERT].observation
appointmentRequired | .actrelationship[typeCode=PERT].observation
availableTime | .effectiveTime
daysOfWeek | .effectiveTime
allDay | .effectiveTime
availableStartTime | .effectiveTime
availableEndTime | .effectiveTime
notAvailable | .effectiveTime
description | n/a
during | .effectiveTime
availabilityExceptions | .effectiveTime
endpoint | n/a

8.8.9 Resource HealthcareService - Design Notes

URL: https://hl7.org/fhir/STU3/healthcareservice-explanations.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.8.9 Resource HealthcareService - Design Notes

to do

8.8.10 Resource HealthcareService - Extensions & Profiles

URL: https://hl7.org/fhir/STU3/healthcareservice-profiles.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.8.10 Resource HealthcareService - Extensions & Profiles

Patient Administration Work Group | Maturity Level: N/A | Ballot Status: Informative

This table lists profiles and extensions for the HealthcareService resource. For background information, see Profiling Resources and Extensibility. Additional profiles and resources may be found in published Implementation Guides , or in the Conformance resource registry .

8.8.10.1 Profiles

No Profiles defined for this resource

8.8.10.2 Extensions

No Extensions defined for this resource

Extensions that reference this resource:

No Extensions refer to this resource

8.8.10.3 Search Extensions

No Search Extensions defined for this resource

8.8.11 Resource HealthcareService - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/healthcareservice-version-maps.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.8.11 Resource HealthcareService - Detailed Descriptions

Patient Administration Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Not linked to any defined compartments

R2 : R3 Converson maps for HealthcareService.

Functional status for this map: 1 test that all execute ok. 1 fail round-trip testing and all r3 resources are valid. (see documentation)

8.8.11.1 R2 to R3

```
map "http://hl7.org/fhir/StructureMap/HealthcareService2to3" = "R2 to R3 Conversions for HealthcareService"

uses "http://hl7.org/fhir/DSTU2/StructureDefinition/HealthcareService" alias HealthcareServiceR2 as source
uses "http://hl7.org/fhir/StructureDefinition/HealthcareService" alias HealthcareService as target

imports "http://hl7.org/fhir/StructureMap/*2to3"
```

```
group for type+types HealthcareService extends DomainResource
  input src : HealthcareServiceR2 as source
  input tgt : HealthcareService as target

  "HealthcareService-identifier" : for src.identifier make tgt.identifier
  "HealthcareService-providedBy" : for src.providedBy make tgt.providedBy
  "HealthcareService-serviceCategory" : for src.serviceCategory make tgt.category
  "HealthcareService-serviceType" : for src.serviceType as v then hsvcType(v, tgt)
  "HealthcareService-location" : for src.location make tgt.location
  "HealthcareService-serviceName" : for src.serviceName make tgt.name
  "HealthcareService-comment" : for src.comment make tgt.comment
  "HealthcareService-extraDetails" : for src.extraDetails make tgt.extraDetails
  "HealthcareService-photo" : for src.photo make tgt.photo
  "HealthcareService-telecom" : for src.telecom make tgt.telecom
  "HealthcareService-coverageArea" : for src.coverageArea make tgt.coverageArea
  "HealthcareService-serviceProvisionCode" : for src.serviceProvisionCode make tgt.serviceProvisionCode
  "HealthcareService-eligibility" : for src.eligibility make tgt.eligibility
  "HealthcareService-eligibilityNote" : for src.eligibilityNote make tgt.eligibilityNote
  "HealthcareService-programName" : for src.programName make tgt.programName
  "HealthcareService-characteristic" : for src.characteristic make tgt.characteristic
  "HealthcareService-referralMethod" : for src.referralMethod make tgt.referralMethod
  "HealthcareService-appointmentRequired" : for src.appointmentRequired make tgt.appointmentRequired
  "HealthcareService-availableTime" : for src.availableTime as vs0 make tgt.availableTime as vt0 then hsvcAvailable(vs0,
     vt0)
  "HealthcareService-notAvailable" : for src.notAvailable as vs0 make tgt.notAvailable as vt0 then hsvcNotAvailable(vs0,
     vt0)
  "HealthcareService-availabilityExceptions" : for src.availabilityExceptions make tgt.availabilityExceptions
endgroup

group hsvcAvailable extends BackboneElement
  input src as source
  input tgt as target

  "HealthcareService.availableTime-daysOfWeek" : for src.daysOfWeek make tgt.daysOfWeek
  "HealthcareService.availableTime-allDay" : for src.allDay make tgt.allDay
  "HealthcareService.availableTime-availableStartTime" : for src.availableStartTime make tgt.availableStartTime
  "HealthcareService.availableTime-availableEndTime" : for src.availableEndTime make tgt.availableEndTime
endgroup

group hsvcNotAvailable extends BackboneElement
  input src as source
  input tgt as target

  "HealthcareService.notAvailable-description" : for src.description make tgt.description
  "HealthcareService.notAvailable-during" : for src.during make tgt.during
endgroup

group hsvcType extends BackboneElement
  input src as source
  input tgt as target

  "HealthcareService.serviceType-type" : for src.type make tgt.type
  "HealthcareService.serviceType-speciality" : for src.specialty make tgt.specialty
endgroup
```

8.8.11.2 R3 to R2

```
map "http://hl7.org/fhir/StructureMap/HealthcareService3to2" = "R3 to R2 Conversion for HealthcareService"

uses "http://hl7.org/fhir/StructureDefinition/HealthcareService" alias HealthcareService as source
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/HealthcareService" alias HealthcareServiceR2 as target

imports "http://hl7.org/fhir/StructureMap/*3to2"

group for type+types  HealthcareService extends DomainResource
  input src : HealthcareService as source
  input tgt : HealthcareServiceR2 as target

  "HealthcareService-identifier" : for src.identifier make tgt.identifier
  "HealthcareService-providedBy" : for src.providedBy make tgt.providedBy
  "HealthcareService-serviceCategory" : for src.category make tgt.serviceCategory
  "HealthcareService-serviceType" : for src.type as vs make tgt.serviceType as vt, vt.type as vt then
    CodeableConcept(vs, vt)
  "HealthcareService-speciality" : for src.specialty as vs make tgt.serviceType as vt first, vt.specialty as vt then
    CodeableConcept(vs, vt) // put all the specialties on the first type
  "HealthcareService-location" : for src.location make tgt.location
  "HealthcareService-serviceName" : for src.name make tgt.serviceName
  "HealthcareService-comment" : for src.comment make tgt.comment
  "HealthcareService-extraDetails" : for src.extraDetails make tgt.extraDetails
  "HealthcareService-photo" : for src.photo make tgt.photo
  "HealthcareService-telecom" : for src.telecom make tgt.telecom
  "HealthcareService-coverageArea" : for src.coverageArea make tgt.coverageArea
  "HealthcareService-serviceProvisionCode" : for src.serviceProvisionCode make tgt.serviceProvisionCode
  "HealthcareService-eligibility" : for src.eligibility make tgt.eligibility
  "HealthcareService-eligibilityNote" : for src.eligibilityNote make tgt.eligibilityNote
  "HealthcareService-programName" : for src.programName make tgt.programName
  "HealthcareService-characteristic" : for src.characteristic make tgt.characteristic
  "HealthcareService-referralMethod" : for src.referralMethod make tgt.referralMethod
  "HealthcareService-appointmentRequired" : for src.appointmentRequired make tgt.appointmentRequired
  "HealthcareService-availableTime" : for src.availableTime as vs0 make tgt.availableTime as vt0 then hsvcAvailable(vs0,
```

```
    vt0)
  "HealthcareService-notAvailable" : for src.notAvailable as vs0 make tgt.notAvailable as vt0 then hsvcNotAvailable(vs0,
    vt0)
  "HealthcareService-availabilityExceptions" : for src.availabilityExceptions make tgt.availabilityExceptions
endgroup

group hsvcAvailable extends BackboneElement
  input src as source
  input tgt as target

  "HealthcareService.availableTime-daysOfWeek" : for src.daysOfWeek make tgt.daysOfWeek
  "HealthcareService.availableTime-allDay" : for src.allDay make tgt.allDay
  "HealthcareService.availableTime-availableStartTime" : for src.availableStartTime make tgt.availableStartTime
  "HealthcareService.availableTime-availableEndTime" : for src.availableEndTime make tgt.availableEndTime
endgroup

group hsvcNotAvailable extends BackboneElement
  input src as source
  input tgt as target

  "HealthcareService.notAvailable-description" : for src.description make tgt.description
  "HealthcareService.notAvailable-during" : for src.during make tgt.during
endgroup
```

8.8.11.3 R3 Validation Errors

8.9 Resource Endpoint - Content

URL: https://hl7.org/fhir/STU3/endpoint.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

Patient Administration Work Group | Maturity Level: 2 | Trial Use | Compartments: Not linked to any defined compartments

The technical details of an endpoint that can be used for electronic services, such as for web services providing XDS.b or a REST endpoint for another FHIR server. This may include any security context information.

8.9.1 Scope and Usage

An endpoint describes the technical details of a location that can be connected to for the delivery/retrieval of information. Sufficient information is required to ensure that a connection can be made securely, and appropriate data transmitted as defined by the endpoint owner. This is not a description of details of the current system, as found in CapabilityStatement, but of another (potentially external) system. These may be locally hosted services, regional services, or national service.

These resources are typically used to identify where to locate endpoint details for:

* Questionnaires: Where to send information (currently an SDC extension with just the address)

* ValueSet: Where related Terminology Services can be found (where not local)

* Subscription: The destination to send the subscribed data (or to pull)

* Messaging: (currently defined in the Message Header, but only as the address)

* Referrals: Where to send referral requests (linked to the services directory resources - Organization/Location/Practitioner/HealthcareService)

* Referrals - Templates: Where to locate referral templates (Questionnaires) (linked to the services directory resources - Organization/Location/Practitioner/HealthcareService)

* CarePlans: Where a shared CarePlan can be found

* Scheduling: Where to lookup to discover schedules/availability information (linked to the services directory resources - Organization/Location/Practitioner/HealthcareService)

* Scheduling: Where to lookup to send appointment requests (linked to the services directory resources - Organization/Location/Practitioner/HealthcareService)

* Patient/Person: Location of Master Patient/Person Indexes

* Service Provider Directories: Location of related directories (parent/child/federated)

8.9.2 Boundaries and Relationships

The endpoint is distinct from a capability statement in that the CapabilityStatement statement describes the entire capability of a server (and in the metadata case, just this server) Where the endpoint resource describes the technical details for how to connect, and for what purposes (which could be a small sub-set of the server's capabilities, and may not be a FHIR endpoint).

8.9.3 Background and Context

8.9.3.1 Expected Implementations

* Any solution where there are distributed FHIR servers deployed and need discovery/configuration

* (Refer to the scope and usage section for common uses)

This resource is referenced by HealthcareService, ImagingManifest, ImagingStudy, Location, Organization and PractitionerRole

8.9.4 Resource Content

* Structure

* UML

* XML

* JSON

* Turtle

* R2 Diff

* All

Structure

Name | Flags | Card. | Type | Description & Constraints

Endpoint | | | DomainResource | The technical details of an endpoint that can be used for electronic services Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension

identifier | ? | 0..* | Identifier | Identifies this endpoint across multiple systems

status | ?!? | 1..1 | code | active | suspended | error | off | entered-in-error | test EndpointStatus (Required)

connectionType | ? | 1..1 | Coding | Protocol/Profile/Standard to be used with this endpoint connection Endpoint Connection Type (Extensible)

name | ? | 0..1 | string | A name that this endpoint can be identified by

managingOrganization | ? | 0..1 | Reference(Organization) | Organization that manages this endpoint (may not be the organization that exposes the endpoint)

contact | | 0..* | ContactPoint | Contact details for source (e.g. troubleshooting)

period | ? | 0..1 | Period | Interval the endpoint is expected to be operational

payloadType | ? | 1..* | CodeableConcept | The type of content that may be used at this endpoint (e.g. XDS Discharge summaries) Endpoint Payload Type (Example)

payloadMimeType | ? | 0..* | code | Mimetype to send. If not specified, the content could be anything (including no payload, if the connectionType defined this) MimeType (Required)

address | ? | 1..1 | uri | The technical base address for connecting to this endpoint

header | | 0..* | string | Usage depends on the channel type

Documentation for this format

UML Diagram (Legend)

XML Template

```
<Endpoint xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Identifies this endpoint across multiple systems --></identifier>
 <status value="[code]"/><!-- 1..1 active | suspended | error | off | entered-in-error | test -->
 <connectionType><!-- 1..1 Coding Protocol/Profile/Standard to be used with this endpoint connection
    --></connectionType>
 <name value="[string]"/><!-- 0..1 A name that this endpoint can be identified by -->
 <managingOrganization><!-- 0..1 Reference(Organization) Organization that manages this endpoint (may not be the
    organization that exposes the endpoint) --></managingOrganization>
 <contact><!-- 0..* ContactPoint Contact details for source (e.g. troubleshooting) --></contact>
 <period><!-- 0..1 Period Interval the endpoint is expected to be operational --></period>
 <payloadType><!-- 1..* CodeableConcept The type of content that may be used at this endpoint (e.g. XDS Discharge
    summaries) --></payloadType>
 <payloadMimeType value="[code]"/><!-- 0..* Mimetype to send. If not specified, the content could be anything (including
     no payload, if the connectionType defined this)  -->
 <address value="[uri]"/><!-- 1..1 The technical base address for connecting to this endpoint -->
 <header value="[string]"/><!-- 0..* Usage depends on the channel type -->
</Endpoint>
```

JSON Template

```
{
  "resourceType" : "Endpoint",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Identifies this endpoint across multiple systems
  "status" : "<code>", // R!  active | suspended | error | off | entered-in-error | test
  "connectionType" : { Coding }, // R!  Protocol/Profile/Standard to be used with this endpoint connection
  "name" : "<string>", // A name that this endpoint can be identified by
  "managingOrganization" : { Reference(Organization) }, // Organization that manages this endpoint (may not be the
    organization that exposes the endpoint)
  "contact" : [{ ContactPoint }], // Contact details for source (e.g. troubleshooting)
  "period" : { Period }, // Interval the endpoint is expected to be operational
  "payloadType" : [{ CodeableConcept }], // R!  The type of content that may be used at this endpoint (e.g. XDS
    Discharge summaries)
  "payloadMimeType" : ["<code>"], // Mimetype to send. If not specified, the content could be anything (including no
    payload, if the connectionType defined this)
  "address" : "<uri>", // R!  The technical base address for connecting to this endpoint
  "header" : ["<string>"] // Usage depends on the channel type
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .
```

```
[ a fhir:Endpoint;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Endpoint.identifier [ Identifier ], ... ; # 0..* Identifies this endpoint across multiple systems
  fhir:Endpoint.status [ code ]; # 1..1 active | suspended | error | off | entered-in-error | test
  fhir:Endpoint.connectionType [ Coding ]; # 1..1 Protocol/Profile/Standard to be used with this endpoint connection
  fhir:Endpoint.name [ string ]; # 0..1 A name that this endpoint can be identified by
  fhir:Endpoint.managingOrganization [ Reference(Organization) ]; # 0..1 Organization that manages this endpoint (may
    not be the organization that exposes the endpoint)
  fhir:Endpoint.contact [ ContactPoint ], ... ; # 0..* Contact details for source (e.g. troubleshooting)
  fhir:Endpoint.period [ Period ]; # 0..1 Interval the endpoint is expected to be operational
  fhir:Endpoint.payloadType [ CodeableConcept ], ... ; # 1..* The type of content that may be used at this endpoint
    (e.g. XDS Discharge summaries)
  fhir:Endpoint.payloadMimeType [ code ], ... ; # 0..* Mimetype to send. If not specified, the content could be anything
     (including no payload, if the connectionType defined this)
  fhir:Endpoint.address [ uri ]; # 1..1 The technical base address for connecting to this endpoint
  fhir:Endpoint.header [ string ], ... ; # 0..* Usage depends on the channel type
]
```

Changes since DSTU2

This resource did not exist in Release 2

This analysis is available as XML or JSON.

Structure

Name | Flags | Card. | Type | Description & Constraints

Endpoint | | | DomainResource | The technical details of an endpoint that can be used for electronic services Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension

identifier | ? | 0..* | Identifier | Identifies this endpoint across multiple systems

status | ?!? | 1..1 | code | active | suspended | error | off | entered-in-error | test EndpointStatus (Required)

connectionType | ? | 1..1 | Coding | Protocol/Profile/Standard to be used with this endpoint connection Endpoint Connection Type (Extensible)

name | ? | 0..1 | string | A name that this endpoint can be identified by

managingOrganization | ? | 0..1 | Reference(Organization) | Organization that manages this endpoint (may not be the organization that exposes the endpoint)

contact | | 0..* | ContactPoint | Contact details for source (e.g. troubleshooting)

period | ? | 0..1 | Period | Interval the endpoint is expected to be operational

payloadType | ? | 1..* | CodeableConcept | The type of content that may be used at this endpoint (e.g. XDS Discharge summaries) Endpoint Payload Type (Example)

payloadMimeType | ? | 0..* | code | Mimetype to send. If not specified, the content could be anything (including no payload, if the connectionType defined this) MimeType (Required)

address | ? | 1..1 | uri | The technical base address for connecting to this endpoint

header | | 0..* | string | Usage depends on the channel type

Documentation for this format

UML Diagram (Legend)

XML Template

```
<Endpoint xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Identifies this endpoint across multiple systems --></identifier>
 <status value="[code]"/><!-- 1..1 active | suspended | error | off | entered-in-error | test -->
 <connectionType><!-- 1..1 Coding Protocol/Profile/Standard to be used with this endpoint connection
    --></connectionType>
 <name value="[string]"/><!-- 0..1 A name that this endpoint can be identified by -->
 <managingOrganization><!-- 0..1 Reference(Organization) Organization that manages this endpoint (may not be the
    organization that exposes the endpoint) --></managingOrganization>
 <contact><!-- 0..* ContactPoint Contact details for source (e.g. troubleshooting) --></contact>
 <period><!-- 0..1 Period Interval the endpoint is expected to be operational --></period>
 <payloadType><!-- 1..* CodeableConcept The type of content that may be used at this endpoint (e.g. XDS Discharge
    summaries) --></payloadType>
 <payloadMimeType value="[code]"/><!-- 0..* Mimetype to send. If not specified, the content could be anything (including
     no payload, if the connectionType defined this)  -->
 <address value="[uri]"/><!-- 1..1 The technical base address for connecting to this endpoint -->
 <header value="[string]"/><!-- 0..* Usage depends on the channel type -->
</Endpoint>
```

JSON Template

```
{
  "resourceType" : "Endpoint",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Identifies this endpoint across multiple systems
  "status" : "<code>", // R!  active | suspended | error | off | entered-in-error | test
  "connectionType" : { Coding }, // R!  Protocol/Profile/Standard to be used with this endpoint connection
  "name" : "<string>", // A name that this endpoint can be identified by
  "managingOrganization" : { Reference(Organization) }, // Organization that manages this endpoint (may not be the
    organization that exposes the endpoint)
  "contact" : [{ ContactPoint }], // Contact details for source (e.g. troubleshooting)
  "period" : { Period }, // Interval the endpoint is expected to be operational
  "payloadType" : [{ CodeableConcept }], // R!  The type of content that may be used at this endpoint (e.g. XDS
    Discharge summaries)
  "payloadMimeType" : ["<code>"], // Mimetype to send. If not specified, the content could be anything (including no
```

```
    payload, if the connectionType defined this)
  "address" : "<uri>", // R!  The technical base address for connecting to this endpoint
  "header" : ["<string>"] // Usage depends on the channel type
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Endpoint;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Endpoint.identifier [ Identifier ], ... ; # 0..* Identifies this endpoint across multiple systems
  fhir:Endpoint.status [ code ]; # 1..1 active | suspended | error | off | entered-in-error | test
  fhir:Endpoint.connectionType [ Coding ]; # 1..1 Protocol/Profile/Standard to be used with this endpoint connection
  fhir:Endpoint.name [ string ]; # 0..1 A name that this endpoint can be identified by
  fhir:Endpoint.managingOrganization [ Reference(Organization) ]; # 0..1 Organization that manages this endpoint (may
    not be the organization that exposes the endpoint)
  fhir:Endpoint.contact [ ContactPoint ], ... ; # 0..* Contact details for source (e.g. troubleshooting)
  fhir:Endpoint.period [ Period ]; # 0..1 Interval the endpoint is expected to be operational
  fhir:Endpoint.payloadType [ CodeableConcept ], ... ; # 1..* The type of content that may be used at this endpoint
    (e.g. XDS Discharge summaries)
  fhir:Endpoint.payloadMimeType [ code ], ... ; # 0..* Mimetype to send. If not specified, the content could be anything
     (including no payload, if the connectionType defined this)
  fhir:Endpoint.address [ uri ]; # 1..1 The technical base address for connecting to this endpoint
  fhir:Endpoint.header [ string ], ... ; # 0..* Usage depends on the channel type
]
```

Changes since DSTU2

This resource did not exist in Release 2

This analysis is available as XML or JSON.

Alternate definitions: Master Definition (XML, JSON), XML Schema/Schematron (for) + JSON Schema, ShEx (for Turtle)

8.9.4.1 Terminology Bindings

Path | Definition | Type | Reference
Endpoint.status | The status of the endpoint | Required | EndpointStatus
Endpoint.connectionType | | Extensible | Endpoint Connection Type
Endpoint.payloadType | | Example | Endpoint Payload Type
Endpoint.payloadMimeType | The mime type of an attachment. Any valid mime type is allowed. | Required | BCP 13 (RFCs 2045, 2046, 2047, 4288, 4289 and 2049)

8.9.5 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
connection-type | token | Protocol/Profile/Standard to be used with this endpoint connection | Endpoint.connectionType
identifier | token | Identifies this endpoint across multiple systems | Endpoint.identifier
name | string | A name that this endpoint can be identified by | Endpoint.name
organization | reference | The organization that is managing the endpoint | Endpoint.managingOrganization (Organization)
payload-type | token | The type of content that may be used at this endpoint (e.g. XDS Discharge summaries) | Endpoint.payloadType
status | token | The current status of the Endpoint (usually expected to be active) | Endpoint.status

8.9.6 Resource Endpoint - Examples

URL: https://hl7.org/fhir/STU3/endpoint-examples.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.9.6 Resource Endpoint - Examples

Patient Administration Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Not linked to any defined compartments

Example List:

Example Endpoint | example | XML | JSON | Turtle
Collection of endpoint examples | b0a5e4277-83c4-4adb-87e2-e3efe3369b6f | XML | JSON | Turtle
Example of IHE-IID endpoint | example-iid | XML | JSON | Turtle
Example of DICOM WADO-RS endpoint | example-wadors | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

8.9.7 Resource Endpoint - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/endpoint-definitions.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.9.7 Resource Endpoint - Detailed Descriptions

Patient Administration Work Group | Maturity Level: 2 | Trial Use | Compartments: Not linked to any defined compartments

Detailed Descriptions for the elements in the Endpoint resource.

Endpoint
Definition | The technical details of an endpoint that can be used for electronic services, such as for web services providing XDS.b or a REST endpoint for another FHIR server. This may include any security context information.
Control | 1..1
Endpoint.identifier
Definition | Identifier for the organization that is used to identify the endpoint across multiple disparate systems.
Note | This is a business identifer, not a resource identifier (see discussion)
Control | 0..*
Type | Identifier
Summary | true
Endpoint.status
Definition | active | suspended | error | off | test.
Control | 1..1
Terminology Binding | EndpointStatus (Required)
Type | code
Is Modifier | true
Summary | true
Comments | This element is labeled as a modifier because the status contains codes that mark the endpoint as not currently valid.
Endpoint.connectionType
Definition | A coded value that represents the technical details of the usage of this endpoint, such as what WSDLs should be used in what way. (e.g. XDS.b/DICOM/cds-hook).
Control | 1..1
Terminology Binding | Endpoint Connection Type (Extensible)
Type | Coding
Summary | true
Comments | For additional connectivity details for the protocol, extensions will be used at this point, as in the XDS example.
Endpoint.name
Definition | A friendly name that this endpoint can be referred to with.
Control | 0..1
Type | string
Summary | true
Endpoint.managingOrganization
Definition | The organization that manages this endpoint (even if technically another organisation is hosting this in the cloud, it is the organisation associated with the data).
Control | 0..1
Type | Reference(Organization)
Summary | true
Comments | This property is not typically used when searching for Endpoint resources for usage. The typical usage is via the reference from an aplicable Organization/Location/Practitioner resource, which is where the context is provided. Multiple Locations may reference a single endpoint, and don't have to be within the same organization resource, but most likely within the same organizational hierarchy.
Endpoint.contact
Definition | Contact details for a human to contact about the subscription. The primary use of this for system administrator troubleshooting.
Control | 0..*
Type | ContactPoint
Endpoint.period
Definition | The interval during which the endpoint is expected to be operational.
Control | 0..1
Type | Period
Summary | true
Endpoint.payloadType
Definition | The payload type describes the acceptable content that can be communicated on the endpoint.
Control | 1..*
Terminology Binding | Endpoint Payload Type (Example)
Type | CodeableConcept
Summary | true
Comments | The payloadFormat describes the serialization format of the data, where the payloadType indicates the specific document/schema that is being transferred; e.g. DischargeSummary or CarePlan.
Endpoint.payloadMimeType
Definition | The mime type to send the payload in - e.g. application/fhir+xml, application/fhir+json. If the mime type is not specified, then the sender could send any content (including no content depending on the connectionType).
Control | 0..*
Terminology Binding | MimeType :
Type | code
Summary | true
Comments | Sending the payload has obvious security consequences. The server is responsible for ensuring that the content is appropriately secured.

Endpoint.address
Definition | The uri that describes the actual end-point to connect to.
Control | 1..1
Type | uri
Summary | true
Comments | For rest-hook, and websocket, the end-point must be an http: or https: URL; for email, a mailto: url, for sms, a tel: url, and for message the endpoint can be in any form of url the server understands (usually, http: or mllp:). The URI is allowed to be relative; in which case, it is relative to the server end-point (since their may be more than one, clients should avoid using relative URIs) This address will be to the service base, without any parameters, or sub-services or resources tacked on. E.g. for a WADO-RS endpoint, the url should be "https://pacs.hospital.org/wado-rs" and not "https://pacs.hospital.org/wado-rs/studies/1.2.250.1.59.40211.12345678.678910/series/1.2.250.1.59.40211.789001276.14556172.67789/instances/...".
Endpoint.header
Definition | Additional headers / information to send as part of the notification.
Control | 0..*
Type | string
Summary | false
Comments | Exactly what these mean depend on the channel type. The can convey additional information to the recipient and/or meet security requirements.

8.9.8 Resource Endpoint - Mappings

URL: https://hl7.org/fhir/STU3/endpoint-mappings.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.9.8 Resource Endpoint - Mappings

Patient Administration Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Not linked to any defined compartments

Mappings:

RIM Mapping

Mappings for the endpoint resource.

8.9.8.1 Mappings for RIM Mapping (http://hl7.org/v3)

Endpoint | n/a

8.9.9 Resource Endpoint - Design Notes

URL: https://hl7.org/fhir/STU3/endpoint-explanations.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.9.9 Resource Endpoint - Design Notes

to do

8.9.10 Resource Endpoint - Extensions & Profiles

URL: https://hl7.org/fhir/STU3/endpoint-profiles.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.9.10 Resource Endpoint - Extensions & Profiles

Patient Administration Work Group | Maturity Level: N/A | Ballot Status: Informative

This table lists profiles and extensions for the Endpoint resource. For background information, see Profiling Resources and Extensibility. Additional profiles and resources may be found in published Implementation Guides , or in the Conformance resource registry .

8.9.10.1 Profiles

No Profiles defined for this resource

8.9.10.2 Extensions

No Extensions defined for this resource

Extensions that reference this resource:

No Extensions refer to this resource

8.9.10.3 Search Extensions

No Search Extensions defined for this resource

8.10 Resource EpisodeOfCare - Content

URL: https://hl7.org/fhir/STU3/episodeofcare.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

Patient Administration Work Group | Maturity Level: 2 | Trial Use | Compartments: Patient, Practitioner

An association between a patient and an organization / healthcare provider(s) during which time encounters may occur. The managing organization assumes a level of responsibility for the patient during this time.

8.10.1 Scope and Usage

The EpisodeOfCare Resource contains information about an association of a Patient with a Healthcare Provider for a period of time under which related healthcare activities may occur.

In many cases, this represents a period of time where the Healthcare Provider has some level of responsibility for the care of the patient regarding a specific condition or problem, even if not currently participating in an encounter.

These resources are typically known in existing systems as:

* EpisodeOfCare: Case, Program, Problem, Episode

* Encounter: Visit, Contact

8.10.1.1 Multiple Organizations and Transfer of Care

Many organizations can be involved in an EpisodeOfCare, however each organization will have their own EpisodeOfCare which tracks their responsibility with the patient.

When an Organization completes their involvement with the patient and transfers care to another Organization. This is often in the form of a referral to another Organization (or Organizations).

When an incoming referral is received a new EpisodeOfCare may be created for this organization. The initial step(s) in the intake workflow for the referral often involve some form of assessment(s), eligibility, capacity, care levels, which could take some time. Once the intake process is completed and the patient is accepted, a CarePlan is often created.

8.10.2 Boundaries and Relationships

The primary difference between the EpisodeOfCare and the Encounter is that the Encounter records the details of an activity directly relating to the patient, while the EpisodeOfCare is the container that can link a series of Encounters together for problems/issues. The Example scenarios below give some good examples as to when you might want to be using an EpisodeOfCare.

This difference is a similar difference between the EpisodeOfCare and a CarePlan. The EpisodeOfCare is a tracking resource, rather than a planning resource. The EpisodeOfCare usually exists before the CarePlan. You don't need a CarePlan to use an EpisodeOfCare.

8.10.3 Background and Context

Systems collect a coherent group of activities (such as encounters) related to a patient's health condition or problem often referred to as a Care Episode. Information about an episode is often shared across systems, and in some cases organizational and disciplinary boundaries. An Episode Of Care contains details about the purpose of the care and can exist without any activities. The minimal information that would be required in an episode of care would be a patient, organization and a reason for the ongoing association. Other reasons for creating an EpisodeOfCare could be for tracking the details required for government reporting or billing.

8.10.3.1 Expected Implementations

* Chronic Disease Management Systems

* Community Care Systems

* Tracking progress of a specific condition

* Tracking government funding

* Problem based General Practice systems

* Disability Support Systems

* Aged Care Systems (Community and Residential)

This resource is referenced by CarePlan, CareTeam, ChargeItem, ClinicalImpression, Communication, CommunicationRequest, Condition, DeviceRequest, DiagnosticReport, Encounter, GuidanceResponse, ImagingStudy, Media, MedicationAdministration, MedicationDispense, MedicationRequest, MedicationStatement, Observation, Procedure, ProcedureRequest, QuestionnaireResponse, ReferralRequest, RequestGroup, RiskAssessment and Task

8.10.4 Resource Content

* Structure

* UML

* XML
* JSON
* Turtle
* R2 Diff
* All

Structure

Name | Flags | Card. | Type | Description & Constraints
EpisodeOfCare | | | DomainResource | An association of a Patient with an Organization and Healthcare Provider(s) for a period of time that the Organization assumes some level of responsibility Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | | 0..* | Identifier | Business Identifier(s) relevant for this EpisodeOfCare
status | ?!? | 1..1 | code | planned | waitlist | active | onhold | finished | cancelled | entered-in-error EpisodeOfCareStatus (Required)
statusHistory | | 0..* | BackboneElement | Past list of status codes (the current status may be included to cover the start date of the status)
status | | 1..1 | code | planned | waitlist | active | onhold | finished | cancelled | entered-in-error EpisodeOfCareStatus (Required)
period | | 1..1 | Period | Duration the EpisodeOfCare was in the specified status
type | ? | 0..* | CodeableConcept | Type/class - e.g. specialist referral, disease management EpisodeOfCareType (Example)
diagnosis | ? | 0..* | BackboneElement | The list of diagnosis relevant to this episode of care
condition | ? | 1..1 | Reference(Condition) | Conditions/problems/diagnoses this episode of care is for
role | ? | 0..1 | CodeableConcept | Role that this diagnosis has within the episode of care (e.g. admission, billing, discharge …) DiagnosisRole (Preferred)
rank | ? | 0..1 | positiveInt | Ranking of the diagnosis (for each role type)
patient | ? | 1..1 | Reference(Patient) | The patient who is the focus of this episode of care
managingOrganization | ? | 0..1 | Reference(Organization) | Organization that assumes care
period | ? | 0..1 | Period | Interval during responsibility is assumed
referralRequest | | 0..* | Reference(ReferralRequest) | Originating Referral Request(s)
careManager | | 0..1 | Reference(Practitioner) | Care manager/care co-ordinator for the patient
team | | 0..* | Reference(CareTeam) | Other practitioners facilitating this episode of care
account | | 0..* | Reference(Account) | The set of accounts that may be used for billing for this EpisodeOfCare
Documentation for this format

UML Diagram (Legend)

XML Template

```
<EpisodeOfCare xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Business Identifier(s) relevant for this EpisodeOfCare --></identifier>
 <status value="[code]"/><!-- 1..1 planned | waitlist | active | onhold | finished | cancelled | entered-in-error -->
 <statusHistory>  <!-- 0..* Past list of status codes (the current status may be included to cover the start date of the
     status) -->
  <status value="[code]"/><!-- 1..1 planned | waitlist | active | onhold | finished | cancelled | entered-in-error -->
  <period><!-- 1..1 Period Duration the EpisodeOfCare was in the specified status --></period>
 </statusHistory>
 <type><!-- 0..* CodeableConcept Type/class  - e.g. specialist referral, disease management --></type>
 <diagnosis>  <!-- 0..* The list of diagnosis relevant to this episode of care -->
  <condition><!-- 1..1 Reference(Condition) Conditions/problems/diagnoses this episode of care is for --></condition>
  <role><!-- 0..1 CodeableConcept Role that this diagnosis has within the episode of care (e.g. admission, billing,
    discharge …) --></role>
  <rank value="[positiveInt]"/><!-- 0..1 Ranking of the diagnosis (for each role type) -->
 </diagnosis>
 <patient><!-- 1..1 Reference(Patient) The patient who is the focus of this episode of care --></patient>
 <managingOrganization><!-- 0..1 Reference(Organization) Organization that assumes care --></managingOrganization>
 <period><!-- 0..1 Period Interval during responsibility is assumed --></period>
 <referralRequest><!-- 0..* Reference(ReferralRequest) Originating Referral Request(s) --></referralRequest>
 <careManager><!-- 0..1 Reference(Practitioner) Care manager/care co-ordinator for the patient --></careManager>
 <team><!-- 0..* Reference(CareTeam) Other practitioners facilitating this episode of care --></team>
 <account><!-- 0..* Reference(Account) The set of accounts that may be used for billing for this EpisodeOfCare
    --></account>
</EpisodeOfCare>
```

JSON Template

```
{
  "resourceType" : "EpisodeOfCare",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Business Identifier(s) relevant for this EpisodeOfCare
  "status" : "<code>", // R!  planned | waitlist | active | onhold | finished | cancelled | entered-in-error
  "statusHistory" : [{ // Past list of status codes (the current status may be included to cover the start date of the
    status)
    "status" : "<code>", // R!  planned | waitlist | active | onhold | finished | cancelled | entered-in-error
    "period" : { Period } // R!  Duration the EpisodeOfCare was in the specified status
  }],
  "type" : [{ CodeableConcept }], // Type/class  - e.g. specialist referral, disease management
  "diagnosis" : [{ // The list of diagnosis relevant to this episode of care
    "condition" : { Reference(Condition) }, // R!  Conditions/problems/diagnoses this episode of care is for
    "role" : { CodeableConcept }, // Role that this diagnosis has within the episode of care (e.g. admission, billing,
    discharge …)
    "rank" : "<positiveInt>" // Ranking of the diagnosis (for each role type)
  }],
  "patient" : { Reference(Patient) }, // R!  The patient who is the focus of this episode of care
```

```
  "managingOrganization" : { Reference(Organization) }, // Organization that assumes care
  "period" : { Period }, // Interval during responsibility is assumed
  "referralRequest" : [{ Reference(ReferralRequest) }], // Originating Referral Request(s)
  "careManager" : { Reference(Practitioner) }, // Care manager/care co-ordinator for the patient
  "team" : [{ Reference(CareTeam) }], // Other practitioners facilitating this episode of care
  "account" : [{ Reference(Account) }] // The set of accounts that may be used for billing for this EpisodeOfCare
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:EpisodeOfCare;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:EpisodeOfCare.identifier [ Identifier ], ... ; # 0..* Business Identifier(s) relevant for this EpisodeOfCare
  fhir:EpisodeOfCare.status [ code ]; # 1..1 planned | waitlist | active | onhold | finished | cancelled | entered-in-
    error
  fhir:EpisodeOfCare.statusHistory [ # 0..* Past list of status codes (the current status may be included to cover the
    start date of the status)
    fhir:EpisodeOfCare.statusHistory.status [ code ]; # 1..1 planned | waitlist | active | onhold | finished | cancelled
     | entered-in-error
    fhir:EpisodeOfCare.statusHistory.period [ Period ]; # 1..1 Duration the EpisodeOfCare was in the specified status
  ], ...;
  fhir:EpisodeOfCare.type [ CodeableConcept ], ... ; # 0..* Type/class  - e.g. specialist referral, disease management
  fhir:EpisodeOfCare.diagnosis [ # 0..* The list of diagnosis relevant to this episode of care
    fhir:EpisodeOfCare.diagnosis.condition [ Reference(Condition) ]; # 1..1 Conditions/problems/diagnoses this episode
    of care is for
    fhir:EpisodeOfCare.diagnosis.role [ CodeableConcept ]; # 0..1 Role that this diagnosis has within the episode of
    care (e.g. admission, billing, discharge …)
    fhir:EpisodeOfCare.diagnosis.rank [ positiveInt ]; # 0..1 Ranking of the diagnosis (for each role type)
  ], ...;
  fhir:EpisodeOfCare.patient [ Reference(Patient) ]; # 1..1 The patient who is the focus of this episode of care
  fhir:EpisodeOfCare.managingOrganization [ Reference(Organization) ]; # 0..1 Organization that assumes care
  fhir:EpisodeOfCare.period [ Period ]; # 0..1 Interval during responsibility is assumed
  fhir:EpisodeOfCare.referralRequest [ Reference(ReferralRequest) ], ... ; # 0..* Originating Referral Request(s)
  fhir:EpisodeOfCare.careManager [ Reference(Practitioner) ]; # 0..1 Care manager/care co-ordinator for the patient
  fhir:EpisodeOfCare.team [ Reference(CareTeam) ], ... ; # 0..* Other practitioners facilitating this episode of care
  fhir:EpisodeOfCare.account [ Reference(Account) ], ... ; # 0..* The set of accounts that may be used for billing for
    this EpisodeOfCare
]
```

Changes since DSTU2

EpisodeOfCare
EpisodeOfCare.diagnosis | Added Element
EpisodeOfCare.diagnosis.condition | Added Element
EpisodeOfCare.diagnosis.role | Added Element
EpisodeOfCare.diagnosis.rank | Added Element
EpisodeOfCare.team | Renamed from careTeam to team Type changed from BackboneElement to Reference(CareTeam)
EpisodeOfCare.account | Added Element
EpisodeOfCare.condition | deleted
EpisodeOfCare.careTeam.role | deleted
EpisodeOfCare.careTeam.period | deleted
EpisodeOfCare.careTeam.member | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R2 <--> R3 Conversion Maps (status = 1 test that all execute ok. All tests pass round-trip testing and all r3 resources are valid.).

Structure

Name | Flags | Card. | Type | Description & Constraints
EpisodeOfCare | | | DomainResource | An association of a Patient with an Organization and Healthcare Provider(s) for a period of time that the Organization assumes some level of responsibility Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | | 0..* | Identifier | Business Identifier(s) relevant for this EpisodeOfCare
status | ?!? | 1..1 | code | planned | waitlist | active | onhold | finished | cancelled | entered-in-error EpisodeOfCareStatus (Required)
statusHistory | | 0..* | BackboneElement | Past list of status codes (the current status may be included to cover the start date of the status)
status | | 1..1 | code | planned | waitlist | active | onhold | finished | cancelled | entered-in-error EpisodeOfCareStatus (Required)
period | | 1..1 | Period | Duration the EpisodeOfCare was in the specified status
type | ? | 0..* | CodeableConcept | Type/class - e.g. specialist referral, disease management EpisodeOfCareType (Example)
diagnosis | ? | 0..* | BackboneElement | The list of diagnosis relevant to this episode of care
condition | ? | 1..1 | Reference(Condition) | Conditions/problems/diagnoses this episode of care is for
role | ? | 0..1 | CodeableConcept | Role that this diagnosis has within the episode of care (e.g. admission, billing, discharge …) DiagnosisRole (Preferred)
rank | ? | 0..1 | positiveInt | Ranking of the diagnosis (for each role type)
patient | ? | 1..1 | Reference(Patient) | The patient who is the focus of this episode of care
managingOrganization | ? | 0..1 | Reference(Organization) | Organization that assumes care
period | ? | 0..1 | Period | Interval during responsibility is assumed

referralRequest | | 0..* | Reference(ReferralRequest) | Originating Referral Request(s)
careManager | | 0..1 | Reference(Practitioner) | Care manager/care co-ordinator for the patient
team | | 0..* | Reference(CareTeam) | Other practitioners facilitating this episode of care
account | | 0..* | Reference(Account) | The set of accounts that may be used for billing for this EpisodeOfCare
Documentation for this format

UML Diagram (Legend)

XML Template

```
<EpisodeOfCare xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Business Identifier(s) relevant for this EpisodeOfCare --></identifier>
 <status value="[code]"/><!-- 1..1 planned | waitlist | active | onhold | finished | cancelled | entered-in-error -->
 <statusHistory>  <!-- 0..* Past list of status codes (the current status may be included to cover the start date of the
     status) -->
  <status value="[code]"/><!-- 1..1 planned | waitlist | active | onhold | finished | cancelled | entered-in-error -->
  <period><!-- 1..1 Period Duration the EpisodeOfCare was in the specified status --></period>
 </statusHistory>
 <type><!-- 0..* CodeableConcept Type/class  - e.g. specialist referral, disease management --></type>
 <diagnosis>  <!-- 0..* The list of diagnosis relevant to this episode of care -->
  <condition><!-- 1..1 Reference(Condition) Conditions/problems/diagnoses this episode of care is for --></condition>
  <role><!-- 0..1 CodeableConcept Role that this diagnosis has within the episode of care (e.g. admission, billing,
    discharge …) --></role>
  <rank value="[positiveInt]"/><!-- 0..1 Ranking of the diagnosis (for each role type) -->
 </diagnosis>
 <patient><!-- 1..1 Reference(Patient) The patient who is the focus of this episode of care --></patient>
 <managingOrganization><!-- 0..1 Reference(Organization) Organization that assumes care --></managingOrganization>
 <period><!-- 0..1 Period Interval during responsibility is assumed --></period>
 <referralRequest><!-- 0..* Reference(ReferralRequest) Originating Referral Request(s) --></referralRequest>
 <careManager><!-- 0..1 Reference(Practitioner) Care manager/care co-ordinator for the patient --></careManager>
 <team><!-- 0..* Reference(CareTeam) Other practitioners facilitating this episode of care --></team>
 <account><!-- 0..* Reference(Account) The set of accounts that may be used for billing for this EpisodeOfCare
    --></account>
</EpisodeOfCare>
```

JSON Template

```
{
  "resourceType" : "EpisodeOfCare",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Business Identifier(s) relevant for this EpisodeOfCare
  "status" : "<code>", // R!  planned | waitlist | active | onhold | finished | cancelled | entered-in-error
  "statusHistory" : [{ // Past list of status codes (the current status may be included to cover the start date of the
    status)
    "status" : "<code>", // R!  planned | waitlist | active | onhold | finished | cancelled | entered-in-error
    "period" : { Period } // R!  Duration the EpisodeOfCare was in the specified status
  }],
  "type" : [{ CodeableConcept }], // Type/class  - e.g. specialist referral, disease management
  "diagnosis" : [{ // The list of diagnosis relevant to this episode of care
    "condition" : { Reference(Condition) }, // R!  Conditions/problems/diagnoses this episode of care is for
    "role" : { CodeableConcept }, // Role that this diagnosis has within the episode of care (e.g. admission, billing,
    discharge …)
    "rank" : "<positiveInt>" // Ranking of the diagnosis (for each role type)
  }],
  "patient" : { Reference(Patient) }, // R!  The patient who is the focus of this episode of care
  "managingOrganization" : { Reference(Organization) }, // Organization that assumes care
  "period" : { Period }, // Interval during responsibility is assumed
  "referralRequest" : [{ Reference(ReferralRequest) }], // Originating Referral Request(s)
  "careManager" : { Reference(Practitioner) }, // Care manager/care co-ordinator for the patient
  "team" : [{ Reference(CareTeam) }], // Other practitioners facilitating this episode of care
  "account" : [{ Reference(Account) }] // The set of accounts that may be used for billing for this EpisodeOfCare
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:EpisodeOfCare;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:EpisodeOfCare.identifier [ Identifier ], ... ; # 0..* Business Identifier(s) relevant for this EpisodeOfCare
  fhir:EpisodeOfCare.status [ code ]; # 1..1 planned | waitlist | active | onhold | finished | cancelled | entered-in-
    error
  fhir:EpisodeOfCare.statusHistory [ # 0..* Past list of status codes (the current status may be included to cover the
    start date of the status)
    fhir:EpisodeOfCare.statusHistory.status [ code ]; # 1..1 planned | waitlist | active | onhold | finished | cancelled
     | entered-in-error
    fhir:EpisodeOfCare.statusHistory.period [ Period ]; # 1..1 Duration the EpisodeOfCare was in the specified status
  ], ...;
  fhir:EpisodeOfCare.type [ CodeableConcept ], ... ; # 0..* Type/class  - e.g. specialist referral, disease management
  fhir:EpisodeOfCare.diagnosis [ # 0..* The list of diagnosis relevant to this episode of care
    fhir:EpisodeOfCare.diagnosis.condition [ Reference(Condition) ]; # 1..1 Conditions/problems/diagnoses this episode
    of care is for
    fhir:EpisodeOfCare.diagnosis.role [ CodeableConcept ]; # 0..1 Role that this diagnosis has within the episode of
    care (e.g. admission, billing, discharge …)
```

```
    fhir:EpisodeOfCare.diagnosis.rank [ positiveInt ]; # 0..1 Ranking of the diagnosis (for each role type)
  ], ...;
  fhir:EpisodeOfCare.patient [ Reference(Patient) ]; # 1..1 The patient who is the focus of this episode of care
  fhir:EpisodeOfCare.managingOrganization [ Reference(Organization) ]; # 0..1 Organization that assumes care
  fhir:EpisodeOfCare.period [ Period ]; # 0..1 Interval during responsibility is assumed
  fhir:EpisodeOfCare.referralRequest [ Reference(ReferralRequest) ], ... ; # 0..* Originating Referral Request(s)
  fhir:EpisodeOfCare.careManager [ Reference(Practitioner) ]; # 0..1 Care manager/care co-ordinator for the patient
  fhir:EpisodeOfCare.team [ Reference(CareTeam) ], ... ; # 0..* Other practitioners facilitating this episode of care
  fhir:EpisodeOfCare.account [ Reference(Account) ], ... ; # 0..* The set of accounts that may be used for billing for
    this EpisodeOfCare
]
```

Changes since DSTU2

EpisodeOfCare
EpisodeOfCare.diagnosis | Added Element
EpisodeOfCare.diagnosis.condition | Added Element
EpisodeOfCare.diagnosis.role | Added Element
EpisodeOfCare.diagnosis.rank | Added Element
EpisodeOfCare.team | Renamed from careTeam to team Type changed from BackboneElement to Reference(CareTeam)
EpisodeOfCare.account | Added Element
EpisodeOfCare.condition | deleted
EpisodeOfCare.careTeam.role | deleted
EpisodeOfCare.careTeam.period | deleted
EpisodeOfCare.careTeam.member | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R2 <--> R3 Conversion Maps (status = 1 test that all execute ok. All tests pass round-trip testing and all r3 resources are valid.).

Alternate definitions: Master Definition (XML, JSON), XML Schema/Schematron (for) + JSON Schema, ShEx (for Turtle)

8.10.4.1 Terminology Bindings

Path | Definition | Type | Reference
EpisodeOfCare.status EpisodeOfCare.statusHistory.status | The status of the episode of care. | Required | EpisodeOfCareStatus
EpisodeOfCare.type | The type of the episode of care | Example | EpisodeOfCareType
EpisodeOfCare.diagnosis.role | The type of diagnosis this condition represents | Preferred | DiagnosisRole

8.10.5 Status Management

8.10.5.1 History and Period

When an organization assumes responsibility for a patient, then the EpisodeOfCare is created and a start date entered to show when it has begun. As the organization's responsibility changes, so does the status of the EpisodeOfCare. This is described via an example below for an intake workflow.

8.10.5.2 Leave Handling

With long term care there is often a concept of the provision of care being suspended for various reasons. Many systems have extensive Leave Management/Tracking solutions which consider the complexities of this space, however this EpisodeOfCare resource is NOT intended to provide this level of tracking. Extension(s) may be used on the status/status history to track the on hold reason, which can facilitate the processing.

A more complete Leave Management solution may have to deal with:

* Leave Types

* Leave Entitlements

* Billing/Funding implications while on different types of leave

8.10.5.3 Example Intake Workflow

This example sequence demonstrates some status transitions and how other resources interact. The context could be in a Community/Aged Care/Disability/Mental Health setting.

* ReferralRequest received

* intake clerk processes referral and decides that the first level eligibility has been met (e.g. Have capacity in the facility for the patient, the patient is covered by VA)

* EpisodeOfCare created with status of planned which is allocated as fulfilling the ReferralRequest

* Further assessment of needs is scheduled to be taken, a care manager is probably allocated at this point

* Assessment Practitioner sees the Patient and completes a series of relevant Questionnaires to rank the patient

* The assessments are reviewed and a formal CarePlan is created

* The EpisodeOfCare is updated to be marked as active, and the CareTeam is likely filled in

* The provision of care is then managed through the care plan, with all activities will also being linked to the EpisodeOfCare

* The patient is admitted to hospital for some procedures, and the EpisodeOfCare is marked as on hold Some of the services on the CarePlan (or scheduled appointments) would be reviewed to determine if they can be performed without the patient (e.g., home maintenance), or if they should be suspended while the patient is on hold.

* The patient returns from the hospital and the EpisodeOfCare is marked as active again (and services reviewed again)

* Patient wished to move to another area to be closer to family

* Organization creates an outgoing ReferralRequest to a new Organization to continue the care

* The EpisodeOfCare is closed

In some jurisdictions an Organization may be funded by a government body for the days that a patient is under their care. These are known as "active days". This does not mean that they are actively receiving a service (an encounter), but that the organization is responsible for managing their care. This monthly reporting value can be easily extracted from the status history as described above. The actual provision of services may also be funded separately, and this would be via the Encounters.

8.10.5.4 EpisodeOfCare Outcomes Review

An Organization may perform analytics on their Episodes Of Care to have an understanding of how their business is performing. Observing that there was a 60/40 split of episodes being finished/cancelled is not very informative. The organization would prefer to know the reason why the episodes are completing so that they can plan their business effectively. They'd be more interested in knowing whether it was due to services hitting their mandatory end date, client passing away, client transitioning to a higher level of services provided by them or to another provider etc.

Currently there are no attributes on this resource to provide this information. This would be very specific to each implementation and usage, so it would be recommended to use extensions to achieve this functionality.

8.10.6 Example Scenarios

A General Practitioner wants to review how well his patient is managing his diabetes over time from information within his clinic and also the regional community care organization's system(s).

The EpisodeOfCare enables the practitioner to easily separate the diabetes activities from the mental health problem's activities.

A Community Care organization wants to track all activities that occur with a patient relating to their disability to simplify the reporting to the government to receive funding to care for the patient

8.10.7 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
care-manager | reference | Care manager/care co-ordinator for the patient | EpisodeOfCare.careManager (Practitioner)
condition | reference | Conditions/problems/diagnoses this episode of care is for | EpisodeOfCare.diagnosis.condition (Condition)
date | date | The provided date search value falls within the episode of care's period | EpisodeOfCare.period | 18 Resources
identifier | token | Business Identifier(s) relevant for this EpisodeOfCare | EpisodeOfCare.identifier | 26 Resources
incomingreferral | reference | Incoming Referral Request | EpisodeOfCare.referralRequest (ReferralRequest)
organization | reference | The organization that has assumed the specific responsibilities of this EpisodeOfCare | EpisodeOfCare.managingOrganization (Organization)
patient | reference | The patient who is the focus of this episode of care | EpisodeOfCare.patient (Patient) | 31 Resources
status | token | The current status of the Episode of Care as provided (does not check the status history collection) | EpisodeOfCare.status
type | token | Type/class - e.g. specialist referral, disease management | EpisodeOfCare.type | 6 Resources

8.10.8 Resource EpisodeOfCare - Examples

URL: https://hl7.org/fhir/STU3/episodeofcare-examples.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.10.8 Resource EpisodeOfCare - Examples

Patient Administration Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Patient, Practitioner

Example List:

Episode of Care example | example | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

8.10.9 Resource EpisodeOfCare - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/episodeofcare-definitions.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.10.9 Resource EpisodeOfCare - Detailed Descriptions

Patient Administration Work Group | Maturity Level: 2 | Trial Use | Compartments: Patient, Practitioner

Detailed Descriptions for the elements in the EpisodeOfCare resource.

EpisodeOfCare
Definition | An association between a patient and an organization / healthcare provider(s) during which time encounters may occur. The managing organization assumes a level of responsibility for the patient during this time.
Control | 1..1
Alternate Names | Case Program Problem
EpisodeOfCare.identifier
Definition | The EpisodeOfCare may be known by different identifiers for different contexts of use, such as when an external agency is tracking the Episode for funding purposes.
Note | This is a business identifer, not a resource identifier (see discussion)
Control | 0..*
Type | Identifier
EpisodeOfCare.status
Definition | planned | waitlist | active | onhold | finished | cancelled.
Control | 1..1
Terminology Binding | EpisodeOfCareStatus (Required)
Type | code
Is Modifier | true
Summary | true
Comments | This element is labeled as a modifier because the status contains codes that mark the episode as not currently valid.
EpisodeOfCare.statusHistory
Definition | The history of statuses that the EpisodeOfCare has been through (without requiring processing the history of the resource).
Control | 0..*
EpisodeOfCare.statusHistory.status
Definition | planned | waitlist | active | onhold | finished | cancelled.
Control | 1..1
Terminology Binding | EpisodeOfCareStatus (Required)
Type | code
EpisodeOfCare.statusHistory.period
Definition | The period during this EpisodeOfCare that the specific status applied.
Control | 1..1
Type | Period
EpisodeOfCare.type
Definition | A classification of the type of episode of care; e.g. specialist referral, disease management, type of funded care.
Control | 0..*
Terminology Binding | EpisodeOfCareType (Example)
Type | CodeableConcept
Summary | true
Comments | The type can be very important in processing as this could be used in determining if the EpisodeOfCare is relevant to specific government reporting, or other types of classifications.
EpisodeOfCare.diagnosis
Definition | The list of diagnosis relevant to this episode of care.
Control | 0..*
Summary | true
EpisodeOfCare.diagnosis.condition
Definition | A list of conditions/problems/diagnoses that this episode of care is intended to be providing care for.
Control | 1..1
Type | Reference(Condition)
Summary | true
EpisodeOfCare.diagnosis.role
Definition | Role that this diagnosis has within the episode of care (e.g. admission, billing, discharge …).
Control | 0..1
Terminology Binding | DiagnosisRole (Preferred)
Type | CodeableConcept
Summary | true
EpisodeOfCare.diagnosis.rank
Definition | Ranking of the diagnosis (for each role type).
Control | 0..1
Type | positiveInt
Summary | true
EpisodeOfCare.patient
Definition | The patient who is the focus of this episode of care.
Control | 1..1
Type | Reference(Patient)
Summary | true
EpisodeOfCare.managingOrganization
Definition | The organization that has assumed the specific responsibilities for the specified duration.
Control | 0..1
Type | Reference(Organization)

Summary | true
EpisodeOfCare.period
Definition | The interval during which the managing organization assumes the defined responsibility.
Control | 0..1
Type | Period
Summary | true
EpisodeOfCare.referralRequest
Definition | Referral Request(s) that are fulfilled by this EpisodeOfCare, incoming referrals.
Control | 0..*
Type | Reference(ReferralRequest)
EpisodeOfCare.careManager
Definition | The practitioner that is the care manager/care co-ordinator for this patient.
Control | 0..1
Type | Reference(Practitioner)
EpisodeOfCare.team
Definition | The list of practitioners that may be facilitating this episode of care for specific purposes.
Control | 0..*
Type | Reference(CareTeam)
Alternate Names | CareTeam
EpisodeOfCare.account
Definition | The set of accounts that may be used for billing for this EpisodeOfCare.
Control | 0..*
Type | Reference(Account)
Comments | The billing system may choose to allocate billable items associated with the EpisodeOfCare to different referenced Accounts based on internal business rules.

8.10.10 Resource EpisodeOfCare - Mappings

URL: https://hl7.org/fhir/STU3/episodeofcare-mappings.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.10.10 Resource EpisodeOfCare - Mappings

Patient Administration Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Patient, Practitioner

Mappings:

RIM Mapping

Mappings for the episodeofcare resource.

8.10.10.1 Mappings for RIM Mapping (http://hl7.org/v3)

EpisodeOfCare | n/a

8.10.11 Resource EpisodeOfCare - Design Notes

URL: https://hl7.org/fhir/STU3/episodeofcare-explanations.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.10.11 Resource EpisodeOfCare - Design Notes

to do

8.10.12 Resource EpisodeOfCare - Extensions & Profiles

URL: https://hl7.org/fhir/STU3/episodeofcare-profiles.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.10.12 Resource EpisodeOfCare - Extensions & Profiles

Patient Administration Work Group | Maturity Level: N/A | Ballot Status: Informative

This table lists profiles and extensions for the EpisodeOfCare resource. For background information, see Profiling Resources and Extensibility. Additional profiles and resources may be found in published Implementation Guides , or in the Conformance resource registry .

8.10.12.1 Profiles

No Profiles defined for this resource

8.10.12.2 Extensions

No Extensions defined for this resource

Extensions that reference this resource:

No Extensions refer to this resource

8.10.12.3 Search Extensions

No Search Extensions defined for this resource

8.10.13 Resource EpisodeOfCare - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/episodeofcare-version-maps.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.10.13 Resource EpisodeOfCare - Detailed Descriptions

Patient Administration Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Patient, Practitioner

R2 : R3 Converson maps for EpisodeOfCare.

Functional status for this map: 1 test that all execute ok. All tests pass round-trip testing and all r3 resources are valid. (see documentation)

8.10.13.1 R2 to R3

```
map "http://hl7.org/fhir/StructureMap/EpisodeOfCare2to3" = "R2 to R3 Conversions for EpisodeOfCare"

uses "http://hl7.org/fhir/DSTU2/StructureDefinition/EpisodeOfCare" alias EpisodeOfCareR2 as source
uses "http://hl7.org/fhir/StructureDefinition/EpisodeOfCare" alias EpisodeOfCare as target

imports "http://hl7.org/fhir/StructureMap/*2to3"

group for type+types  EpisodeOfCare extends DomainResource
  input src : EpisodeOfCareR2 as source
  input tgt : EpisodeOfCare as target

  "EpisodeOfCare-identifier" : for src.identifier make tgt.identifier
  "EpisodeOfCare-status" : for src.status make tgt.status
  "EpisodeOfCare-statusHistory" : for src.statusHistory as vs0 make tgt.statusHistory as vt0 then epcStatusHistory(vs0,
    vt0)
  "EpisodeOfCare-type" : for src.type make tgt.type
  "EpisodeOfCare-condition" : for src.condition as v make tgt.diagnosis as d, d.condition = v
  "EpisodeOfCare-patient" : for src.patient make tgt.patient
  "EpisodeOfCare-managingOrganization" : for src.managingOrganization make tgt.managingOrganization
  "EpisodeOfCare-period" : for src.period make tgt.period
  "EpisodeOfCare-referralRequest" : for src.referralRequest make tgt.referralRequest
  "EpisodeOfCare-careManager" : for src.careManager make tgt.careManager

  "EpisodeOfCare-careteam" : for src where src.careTeam.exists() make create("CareTeam") as vt0 then epiCareTeam(src,
    vt0, tgt)
endgroup

group epcStatusHistory extends BackboneElement
  input src as source
  input tgt as target

  "EpisodeOfCare.statusHistory-status" : for src.status make tgt.status
  "EpisodeOfCare.statusHistory-period" : for src.period make tgt.period
endgroup

group epiCareTeam
  input src as source
  input tgt as target
  input tgtE as target

  "EpisodeOfCare.careteam-link" : for src make tgtE.team = reference(tgt)
  "EpisodeOfCare.careteam-member" : for src.careTeam as vs make tgt.participant as vt then epiCareTeamMember(vs, vt)
endgroup

group epiCareTeamMember extends BackboneElement
  input src as source
  input tgt as target

  "EpisodeOfCare.careteam-role" : for src.role make tgt.role
  "EpisodeOfCare.careteam-period" : for src.period make tgt.period
  "EpisodeOfCare.careteam-member" : for src.member make tgt.member
endgroup
```

8.10.13.2 R3 to R2

```
map "http://hl7.org/fhir/StructureMap/EpisodeOfCare3to2" = "R3 to R2 Conversion for EpisodeOfCare"

uses "http://hl7.org/fhir/StructureDefinition/EpisodeOfCare" alias EpisodeOfCare as source
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/EpisodeOfCare" alias EpisodeOfCareR2 as target

imports "http://hl7.org/fhir/StructureMap/*3to2"

group for type+types  EpisodeOfCare extends DomainResource
```

```
  input src : EpisodeOfCare as source
  input tgt : EpisodeOfCareR2 as target

  "EpisodeOfCare-identifier" : for src.identifier make tgt.identifier
  "EpisodeOfCare-status" : for src.status make tgt.status
  "EpisodeOfCare-statusHistory" : for src.statusHistory as vs0 make tgt.statusHistory as vt0 then epcStatusHistory(vs0,
    vt0)
  "EpisodeOfCare-type" : for src.type make tgt.type
  "EpisodeOfCare-condition" : for src.diagnosis as d then epcDiagnosis(d, tgt)
  "EpisodeOfCare-patient" : for src.patient make tgt.patient
  "EpisodeOfCare-managingOrganization" : for src.managingOrganization make tgt.managingOrganization
  "EpisodeOfCare-period" : for src.period make tgt.period
  "EpisodeOfCare-referralRequest" : for src.referralRequest make tgt.referralRequest
  "EpisodeOfCare-careManager" : for src.careManager make tgt.careManager
  "EpisodeOfCare-careTeam" : for src.team as vs where reference.resolve().exists() make (src.team.reference.resolve())
    as ct then epiCareTeam(ct, tgt)

endgroup

group epcDiagnosis extends BackboneElement
  input src as source
  input tgt as target

  "EpisodeOfCare.indication-condition1" : for src.condition as v where src.role.coding.code.contains("AD").not() and
    role.coding.code.contains("DD").not() make tgt.condition = v
endgroup


group epcStatusHistory extends BackboneElement
  input src as source
  input tgt as target

  "EpisodeOfCare.statusHistory-status" : for src.status make tgt.status
  "EpisodeOfCare.statusHistory-period" : for src.period make tgt.period
endgroup


group epiCareTeam
  input src as source
  input tgt as target

  "EpisodeOfCare.careteam-member" : for src.participant as vs make tgt.careTeam as vt then epiCareTeamMember(vs, vt)
endgroup

group epiCareTeamMember extends BackboneElement
  input src as source
  input tgt as target

  "EpisodeOfCare.careteam-role" : for src.role make tgt.role
  "EpisodeOfCare.careteam-period" : for src.period make tgt.period
  "EpisodeOfCare.careteam-member" : for src.member make tgt.member
endgroup
```

8.10.13.3 R3 Validation Errors

8.11 Resource Encounter - Content

URL: https://hl7.org/fhir/STU3/encounter.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

Patient Administration Work Group | Maturity Level: 2 | Trial Use | Compartments: Encounter, Patient, Practitioner, RelatedPerson

An interaction between a patient and healthcare provider(s) for the purpose of providing healthcare service(s) or assessing the health status of a patient.

8.11.1 Scope and Usage

A patient encounter is further characterized by the setting in which it takes place. Amongst them are ambulatory, emergency, home health, inpatient and virtual encounters. An Encounter encompasses the lifecycle from pre-admission, the actual encounter (for ambulatory encounters), and admission, stay and discharge (for inpatient encounters). During the encounter the patient may move from practitioner to practitioner and location to location.

Because of the broad scope of Encounter, not all elements will be relevant in all settings. For this reason, admission/discharge related information is kept in a separate Hospitalization component within Encounter. The class element is used to distinguish between these settings, which will guide further validation and application of business rules.

There is also substantial variance from organization to organization (and between jurisdictions and countries) on which business events translate to the start of a new Encounter, or what level of aggregation is used for Encounter. For example, each single visit of a practitioner during a hospitalization may lead to a new instance of Encounter, but depending on local practice and the systems involved, it may well be that this is aggregated to a single instance for a whole hospitalization. Even more aggregation may occur where jurisdictions introduce groups of Encounters for financial or other reasons. Encounters can be aggregated or grouped under other Encounters using the partOf element. See below for examples.

Encounter instances may exist before the actual encounter takes place to convey pre-admission information, including using Encounters elements to reflect the planned start date or planned encounter locations. In this case the status element is set to 'planned'.

The Hospitalization component is intended to store the extended information relating to a hospitalization event. It is always expected to be the same period as the encounter itself. Where the period is different, another encounter instance should be used to capture this information as a partOf this encounter instance.

The Procedure and encounter have references to each other, and these should be to different procedures; one for the procedure that was performed during the encounter (stored in Procedure.encounter), and another for cases where an encounter is a result of another procedure (stored in Encounter.indication) such as a followup encounter to resolve complications from an earlier procedure.

8.11.1.1 Status Management

During the life-cycle of an encounter it will pass through many statuses. Typically these are in order or the organization's workflow: planned, in-progress, finished/cancelled. This status information is often used for other things, and often an analysis of the status history is required. This could be done by scanning through all the versions of the encounter and then checking the period of each, and doing some form of post processing. To ease the burden of this (or where a system doesn't support resource histories) a status history component is included.

There is no direct indication purely by the status field as to whether an encounter is considered "admitted". The context of the encounter and business practices/policies/workflows/types can influence this definition. (e.g., acute care facility, aged care center, outpatient clinic, emergency department, community based clinic). Statuses of "arrived", "triaged" or "in progress" could be considered the start of the admission, and also have the presence of the hospitalization sub-component entered.

The "on leave" status may or may not be a part of the admission, for example if the patient was permitted to go home for a weekend or some other form of external event. The location is also likely to be filled in with a location status of "present". For other examples such as an outpatient visit (day procedure - colonoscopy), the patient could also be considered to be admitted, hence the encounter doesn't have a fixed definition of admitted. At a minimum, we do believe that a patient IS admitted when the status is in-progress.

8.11.2 Boundaries and Relationships

The Encounter resource is not to be used to store appointment information, the Appointment resource is intended to be used for that. Note that in many systems outpatient encounters (which are in scope for Encounter) and Appointment are used concurrently. In FHIR, Appointment is used for establishing a date for the encounter, while Encounter is applicable to information about the actual Encounter, i.e., the patient showing up. As such, an encounter in the "planned" status is not identical to the appointment that scheduled it, but it is the encounter prior to its actual occurrence, with the expectation that encounter will be updated as it progresses to completion. Patient arrival at a location does not necessarily mean the start of the encounter (e.g. a patient arrives an hour earlier than he is actually seen by a practitioner).

An appointment is normally used for the planning stage of an appointment, searching, locating an available time, then making the appointment. Once this process is completed and the appointment is about to start, then the appointment will be marked as fulfilled, and linked to the newly created encounter. This new encounter may start in an "arrived" status when they are admitted at a location of the facility, and then will move to the ward where another part-of encounter may begin.

Communication resources are used for a simultaneous interaction between a practitioner and a patient where there is no direct contact. Examples include a phone message, or transmission of some correspondence documentation. There is no duration recorded for a communication resource, but it could contain sent and received times.

Standard Extension: Associated Encounter This extension should be used to reference an encounter where there is no property that already defines this association on the resource.

This resource is referenced by CarePlan, CareTeam, ChargeItem, Claim, ClinicalImpression, Communication, CommunicationRequest, Composition, Condition, DeviceRequest, DiagnosticReport, DocumentReference, ExplanationOfBenefit, Flag, GuidanceResponse, ImagingStudy, Immunization, List, Media, MedicationAdministration, MedicationDispense, MedicationRequest, MedicationStatement, NutritionOrder, Observation, Procedure, ProcedureRequest, QuestionnaireResponse, ReferralRequest, RequestGroup, RiskAssessment, Task and VisionPrescription

8.11.3 Resource Content

* Structure

* UML

* XML

* JSON

* Turtle

* R2 Diff

* All

Structure

Name | Flags | Card. | Type | Description & Constraints

Encounter | | | DomainResource | An interaction during which services are provided to the patient Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension

identifier | ? | 0..* | Identifier | Identifier(s) by which this encounter is known
status | ?!? | 1..1 | code | planned | arrived | triaged | in-progress | onleave | finished | cancelled + EncounterStatus (Required)
statusHistory | | 0..* | BackboneElement | List of past encounter statuses
status | | 1..1 | code | planned | arrived | triaged | in-progress | onleave | finished | cancelled + EncounterStatus (Required)
period | | 1..1 | Period | The time that the episode was in the specified status
class | ? | 0..1 | Coding | inpatient | outpatient | ambulatory | emergency + ActEncounterCode (Extensible)
classHistory | | 0..* | BackboneElement | List of past encounter classes
class | | 1..1 | Coding | inpatient | outpatient | ambulatory | emergency + ActEncounterCode (Extensible)
period | | 1..1 | Period | The time that the episode was in the specified class
type | ? | 0..* | CodeableConcept | Specific type of encounter EncounterType (Example)
priority | | 0..1 | CodeableConcept | Indicates the urgency of the encounter v3 Code System ActPriority (Example)
subject | ? | 0..1 | Reference(Patient | Group) | The patient ro group present at the encounter
episodeOfCare | ? | 0..* | Reference(EpisodeOfCare) | Episode(s) of care that this encounter should be recorded against
incomingReferral | | 0..* | Reference(ReferralRequest) | The ReferralRequest that initiated this encounter
participant | ? | 0..* | BackboneElement | List of participants involved in the encounter
type | ? | 0..* | CodeableConcept | Role of participant in encounter ParticipantType (Extensible)
period | | 0..1 | Period | Period of time during the encounter that the participant participated
individual | ? | 0..1 | Reference(Practitioner | RelatedPerson) | Persons involved in the encounter other than the patient
appointment | ? | 0..1 | Reference(Appointment) | The appointment that scheduled this encounter
period | | 0..1 | Period | The start and end time of the encounter
length | | 0..1 | Duration | Quantity of time the encounter lasted (less time absent)
reason | ? | 0..* | CodeableConcept | Reason the encounter takes place (code) Encounter Reason Codes (Preferred)
diagnosis | ? | 0..* | BackboneElement | The list of diagnosis relevant to this encounter
condition | | 1..1 | Reference(Condition | Procedure) | Reason the encounter takes place (resource)
role | | 0..1 | CodeableConcept | Role that this diagnosis has within the encounter (e.g. admission, billing, discharge …) DiagnosisRole (Preferred)
rank | | 0..1 | positiveInt | Ranking of the diagnosis (for each role type)
account | | 0..* | Reference(Account) | The set of accounts that may be used for billing for this Encounter
hospitalization | | 0..1 | BackboneElement | Details about the admission to a healthcare service
preAdmissionIdentifier | | 0..1 | Identifier | Pre-admission identifier
origin | | 0..1 | Reference(Location) | The location from which the patient came before admission
admitSource | | 0..1 | CodeableConcept | From where patient was admitted (physician referral, transfer) AdmitSource (Preferred)
reAdmission | | 0..1 | CodeableConcept | The type of hospital re-admission that has occurred (if any). If the value is absent, then this is not identified as a readmission v2 Re-Admission Indicator (Example)
dietPreference | | 0..* | CodeableConcept | Diet preferences reported by the patient Diet (Example)
specialCourtesy | | 0..* | CodeableConcept | Special courtesies (VIP, board member) SpecialCourtesy (Preferred)
specialArrangement | | 0..* | CodeableConcept | Wheelchair, translator, stretcher, etc. SpecialArrangements (Preferred)
destination | | 0..1 | Reference(Location) | Location to which the patient is discharged
dischargeDisposition | | 0..1 | CodeableConcept | Category or kind of location after discharge DischargeDisposition (Example)
location | | 0..* | BackboneElement | List of locations where the patient has been
location | | 1..1 | Reference(Location) | Location the encounter takes place
status | | 0..1 | code | planned | active | reserved | completed EncounterLocationStatus (Required)
period | | 0..1 | Period | Time period during which the patient was present at the location
serviceProvider | | 0..1 | Reference(Organization) | The custodian organization of this Encounter record
partOf | | 0..1 | Reference(Encounter) | Another Encounter this encounter is part of
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Encounter xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Identifier(s) by which this encounter is known --></identifier>
 <status value="[code]"/><!-- 1..1 planned | arrived | triaged | in-progress | onleave | finished | cancelled + -->
 <statusHistory>  <!-- 0..* List of past encounter statuses -->
  <status value="[code]"/><!-- 1..1 planned | arrived | triaged | in-progress | onleave | finished | cancelled + -->
  <period><!-- 1..1 Period The time that the episode was in the specified status --></period>
 </statusHistory>
 <class><!-- 0..1 Coding inpatient | outpatient | ambulatory | emergency + --></class>
 <classHistory>  <!-- 0..* List of past encounter classes -->
  <class><!-- 1..1 Coding inpatient | outpatient | ambulatory | emergency + --></class>
  <period><!-- 1..1 Period The time that the episode was in the specified class --></period>
 </classHistory>
 <type><!-- 0..* CodeableConcept Specific type of encounter --></type>
 <priority><!-- 0..1 CodeableConcept Indicates the urgency of the encounter --></priority>
 <subject><!-- 0..1 Reference(Patient|Group) The patient ro group present at the encounter --></subject>
 <episodeOfCare><!-- 0..* Reference(EpisodeOfCare) Episode(s) of care that this encounter should be recorded against
    --></episodeOfCare>
 <incomingReferral><!-- 0..* Reference(ReferralRequest) The ReferralRequest that initiated this encounter
    --></incomingReferral>
 <participant>  <!-- 0..* List of participants involved in the encounter -->
  <type><!-- 0..* CodeableConcept Role of participant in encounter --></type>
  <period><!-- 0..1 Period Period of time during the encounter that the participant participated --></period>
  <individual><!-- 0..1 Reference(Practitioner|RelatedPerson) Persons involved in the encounter other than the patient
    --></individual>
 </participant>
 <appointment><!-- 0..1 Reference(Appointment) The appointment that scheduled this encounter --></appointment>
 <period><!-- 0..1 Period The start and end time of the encounter --></period>
 <length><!-- 0..1 Duration Quantity of time the encounter lasted (less time absent) --></length>
```

```
 <reason><!-- 0..* CodeableConcept Reason the encounter takes place (code) --></reason>
 <diagnosis>  <!-- 0..* The list of diagnosis relevant to this encounter -->
  <condition><!-- 1..1 Reference(Condition|Procedure) Reason the encounter takes place (resource) --></condition>
  <role><!-- 0..1 CodeableConcept Role that this diagnosis has within the encounter (e.g. admission, billing, discharge
    …) --></role>
  <rank value="[positiveInt]"/><!-- 0..1 Ranking of the diagnosis (for each role type) -->
 </diagnosis>
 <account><!-- 0..* Reference(Account) The set of accounts that may be used for billing for this Encounter --></account>
 <hospitalization>  <!-- 0..1 Details about the admission to a healthcare service -->
  <preAdmissionIdentifier><!-- 0..1 Identifier Pre-admission identifier --></preAdmissionIdentifier>
  <origin><!-- 0..1 Reference(Location) The location from which the patient came before admission --></origin>
  <admitSource><!-- 0..1 CodeableConcept From where patient was admitted (physician referral, transfer)
    --></admitSource>
  <reAdmission><!-- 0..1 CodeableConcept The type of hospital re-admission that has occurred (if any). If the value is
    absent, then this is not identified as a readmission --></reAdmission>
  <dietPreference><!-- 0..* CodeableConcept Diet preferences reported by the patient --></dietPreference>
  <specialCourtesy><!-- 0..* CodeableConcept Special courtesies (VIP, board member) --></specialCourtesy>
  <specialArrangement><!-- 0..* CodeableConcept Wheelchair, translator, stretcher, etc. --></specialArrangement>
  <destination><!-- 0..1 Reference(Location) Location to which the patient is discharged --></destination>
  <dischargeDisposition><!-- 0..1 CodeableConcept Category or kind of location after discharge
    --></dischargeDisposition>
 </hospitalization>
 <location>  <!-- 0..* List of locations where the patient has been -->
  <location><!-- 1..1 Reference(Location) Location the encounter takes place --></location>
  <status value="[code]"/><!-- 0..1 planned | active | reserved | completed -->
  <period><!-- 0..1 Period Time period during which the patient was present at the location --></period>
 </location>
 <serviceProvider><!-- 0..1 Reference(Organization) The custodian organization of this Encounter record
    --></serviceProvider>
 <partOf><!-- 0..1 Reference(Encounter) Another Encounter this encounter is part of --></partOf>
</Encounter>
```

JSON Template

```
{
  "resourceType" : "Encounter",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Identifier(s) by which this encounter is known
  "status" : "<code>", // R!  planned | arrived | triaged | in-progress | onleave | finished | cancelled +
  "statusHistory" : [{ // List of past encounter statuses
    "status" : "<code>", // R!  planned | arrived | triaged | in-progress | onleave | finished | cancelled +
    "period" : { Period } // R!  The time that the episode was in the specified status
  }],
  "class" : { Coding }, // inpatient | outpatient | ambulatory | emergency +
  "classHistory" : [{ // List of past encounter classes
    "class" : { Coding }, // R!  inpatient | outpatient | ambulatory | emergency +
    "period" : { Period } // R!  The time that the episode was in the specified class
  }],
  "type" : [{ CodeableConcept }], // Specific type of encounter
  "priority" : { CodeableConcept }, // Indicates the urgency of the encounter
  "subject" : { Reference(Patient|Group) }, // The patient ro group present at the encounter
  "episodeOfCare" : [{ Reference(EpisodeOfCare) }], // Episode(s) of care that this encounter should be recorded against
  "incomingReferral" : [{ Reference(ReferralRequest) }], // The ReferralRequest that initiated this encounter
  "participant" : [{ // List of participants involved in the encounter
    "type" : [{ CodeableConcept }], // Role of participant in encounter
    "period" : { Period }, // Period of time during the encounter that the participant participated
    "individual" : { Reference(Practitioner|RelatedPerson) } // Persons involved in the encounter other than the patient
  }],
  "appointment" : { Reference(Appointment) }, // The appointment that scheduled this encounter
  "period" : { Period }, // The start and end time of the encounter
  "length" : { Duration }, // Quantity of time the encounter lasted (less time absent)
  "reason" : [{ CodeableConcept }], // Reason the encounter takes place (code)
  "diagnosis" : [{ // The list of diagnosis relevant to this encounter
    "condition" : { Reference(Condition|Procedure) }, // R!  Reason the encounter takes place (resource)
    "role" : { CodeableConcept }, // Role that this diagnosis has within the encounter (e.g. admission, billing,
    discharge …)
    "rank" : "<positiveInt>" // Ranking of the diagnosis (for each role type)
  }],
  "account" : [{ Reference(Account) }], // The set of accounts that may be used for billing for this Encounter
  "hospitalization" : { // Details about the admission to a healthcare service
    "preAdmissionIdentifier" : { Identifier }, // Pre-admission identifier
    "origin" : { Reference(Location) }, // The location from which the patient came before admission
    "admitSource" : { CodeableConcept }, // From where patient was admitted (physician referral, transfer)
    "reAdmission" : { CodeableConcept }, // The type of hospital re-admission that has occurred (if any). If the value
    is absent, then this is not identified as a readmission
    "dietPreference" : [{ CodeableConcept }], // Diet preferences reported by the patient
    "specialCourtesy" : [{ CodeableConcept }], // Special courtesies (VIP, board member)
    "specialArrangement" : [{ CodeableConcept }], // Wheelchair, translator, stretcher, etc.
    "destination" : { Reference(Location) }, // Location to which the patient is discharged
    "dischargeDisposition" : { CodeableConcept } // Category or kind of location after discharge
  },
  "location" : [{ // List of locations where the patient has been
    "location" : { Reference(Location) }, // R!  Location the encounter takes place
    "status" : "<code>", // planned | active | reserved | completed
    "period" : { Period } // Time period during which the patient was present at the location
  }],
  "serviceProvider" : { Reference(Organization) }, // The custodian organization of this Encounter record
  "partOf" : { Reference(Encounter) } // Another Encounter this encounter is part of
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Encounter;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Encounter.identifier [ Identifier ], ... ; # 0..* Identifier(s) by which this encounter is known
  fhir:Encounter.status [ code ]; # 1..1 planned | arrived | triaged | in-progress | onleave | finished | cancelled +
  fhir:Encounter.statusHistory [ # 0..* List of past encounter statuses
    fhir:Encounter.statusHistory.status [ code ]; # 1..1 planned | arrived | triaged | in-progress | onleave | finished
    | cancelled +
    fhir:Encounter.statusHistory.period [ Period ]; # 1..1 The time that the episode was in the specified status
  ], ...;
  fhir:Encounter.class [ Coding ]; # 0..1 inpatient | outpatient | ambulatory | emergency +
  fhir:Encounter.classHistory [ # 0..* List of past encounter classes
    fhir:Encounter.classHistory.class [ Coding ]; # 1..1 inpatient | outpatient | ambulatory | emergency +
    fhir:Encounter.classHistory.period [ Period ]; # 1..1 The time that the episode was in the specified class
  ], ...;
  fhir:Encounter.type [ CodeableConcept ], ... ; # 0..* Specific type of encounter
  fhir:Encounter.priority [ CodeableConcept ]; # 0..1 Indicates the urgency of the encounter
  fhir:Encounter.subject [ Reference(Patient|Group) ]; # 0..1 The patient ro group present at the encounter
  fhir:Encounter.episodeOfCare [ Reference(EpisodeOfCare) ], ... ; # 0..* Episode(s) of care that this encounter should
    be recorded against
  fhir:Encounter.incomingReferral [ Reference(ReferralRequest) ], ... ; # 0..* The ReferralRequest that initiated this
    encounter
  fhir:Encounter.participant [ # 0..* List of participants involved in the encounter
    fhir:Encounter.participant.type [ CodeableConcept ], ... ; # 0..* Role of participant in encounter
    fhir:Encounter.participant.period [ Period ]; # 0..1 Period of time during the encounter that the participant
    participated
    fhir:Encounter.participant.individual [ Reference(Practitioner|RelatedPerson) ]; # 0..1 Persons involved in the
    encounter other than the patient
  ], ...;
  fhir:Encounter.appointment [ Reference(Appointment) ]; # 0..1 The appointment that scheduled this encounter
  fhir:Encounter.period [ Period ]; # 0..1 The start and end time of the encounter
  fhir:Encounter.length [ Duration ]; # 0..1 Quantity of time the encounter lasted (less time absent)
  fhir:Encounter.reason [ CodeableConcept ], ... ; # 0..* Reason the encounter takes place (code)
  fhir:Encounter.diagnosis [ # 0..* The list of diagnosis relevant to this encounter
    fhir:Encounter.diagnosis.condition [ Reference(Condition|Procedure) ]; # 1..1 Reason the encounter takes place
    (resource)
    fhir:Encounter.diagnosis.role [ CodeableConcept ]; # 0..1 Role that this diagnosis has within the encounter (e.g.
    admission, billing, discharge …)
    fhir:Encounter.diagnosis.rank [ positiveInt ]; # 0..1 Ranking of the diagnosis (for each role type)
  ], ...;
  fhir:Encounter.account [ Reference(Account) ], ... ; # 0..* The set of accounts that may be used for billing for this
    Encounter
  fhir:Encounter.hospitalization [ # 0..1 Details about the admission to a healthcare service
    fhir:Encounter.hospitalization.preAdmissionIdentifier [ Identifier ]; # 0..1 Pre-admission identifier
    fhir:Encounter.hospitalization.origin [ Reference(Location) ]; # 0..1 The location from which the patient came
    before admission
    fhir:Encounter.hospitalization.admitSource [ CodeableConcept ]; # 0..1 From where patient was admitted (physician
    referral, transfer)
    fhir:Encounter.hospitalization.reAdmission [ CodeableConcept ]; # 0..1 The type of hospital re-admission that has
    occurred (if any). If the value is absent, then this is not identified as a readmission
    fhir:Encounter.hospitalization.dietPreference [ CodeableConcept ], ... ; # 0..* Diet preferences reported by the
    patient
    fhir:Encounter.hospitalization.specialCourtesy [ CodeableConcept ], ... ; # 0..* Special courtesies (VIP, board
    member)
    fhir:Encounter.hospitalization.specialArrangement [ CodeableConcept ], ... ; # 0..* Wheelchair, translator,
    stretcher, etc.
    fhir:Encounter.hospitalization.destination [ Reference(Location) ]; # 0..1 Location to which the patient is
    discharged
    fhir:Encounter.hospitalization.dischargeDisposition [ CodeableConcept ]; # 0..1 Category or kind of location after
    discharge
  ];
  fhir:Encounter.location [ # 0..* List of locations where the patient has been
    fhir:Encounter.location.location [ Reference(Location) ]; # 1..1 Location the encounter takes place
    fhir:Encounter.location.status [ code ]; # 0..1 planned | active | reserved | completed
    fhir:Encounter.location.period [ Period ]; # 0..1 Time period during which the patient was present at the location
  ], ...;
  fhir:Encounter.serviceProvider [ Reference(Organization) ]; # 0..1 The custodian organization of this Encounter record
  fhir:Encounter.partOf [ Reference(Encounter) ]; # 0..1 Another Encounter this encounter is part of
]
```

Changes since DSTU2

Encounter

Encounter.status | Change value set from http://hl7.org/fhir/ValueSet/encounter-state to http://hl7.org/fhir/ValueSet/encounter-status

Encounter.statusHistory.status | Change value set from http://hl7.org/fhir/ValueSet/encounter-state to http://hl7.org/fhir/ValueSet/encounter-status

Encounter.class | Type changed from code to Coding Change binding strength from required to extensible Change value set from http://hl7.org/fhir/ValueSet/encounter-class to http://hl7.org/fhir/ValueSet/v3-ActEncounterCode

Encounter.classHistory | Added Element

Encounter.classHistory.class | Added Element

Encounter.classHistory.period | Added Element

Encounter.subject | Renamed from patient to subject Add Reference(Group)

Encounter.length | Type changed from Quantity{http://hl7.org/fhir/StructureDefinition/Duration} to Duration
Encounter.diagnosis | Added Element
Encounter.diagnosis.condition | Added Element
Encounter.diagnosis.role | Added Element
Encounter.diagnosis.rank | Added Element
Encounter.account | Added Element
Encounter.indication | deleted
Encounter.hospitalization.admittingDiagnosis | deleted
Encounter.hospitalization.dischargeDiagnosis | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R2 <--> R3 Conversion Maps (status = 10 tests that all execute ok. 1 fail round-trip testing and 2 r3 resources are invalid (2 errors).).

Structure

Name | Flags | Card. | Type | Description & Constraints
Encounter | | | DomainResource | An interaction during which services are provided to the patient Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | ? | 0..* | Identifier | Identifier(s) by which this encounter is known
status | ?!? | 1..1 | code | planned | arrived | triaged | in-progress | onleave | finished | cancelled + EncounterStatus (Required)
statusHistory | | 0..* | BackboneElement | List of past encounter statuses
status | | 1..1 | code | planned | arrived | triaged | in-progress | onleave | finished | cancelled + EncounterStatus (Required)
period | | 1..1 | Period | The time that the episode was in the specified status
class | ? | 0..1 | Coding | inpatient | outpatient | ambulatory | emergency + ActEncounterCode (Extensible)
classHistory | | 0..* | BackboneElement | List of past encounter classes
class | | 1..1 | Coding | inpatient | outpatient | ambulatory | emergency + ActEncounterCode (Extensible)
period | | 1..1 | Period | The time that the episode was in the specified class
type | ? | 0..* | CodeableConcept | Specific type of encounter EncounterType (Example)
priority | | 0..1 | CodeableConcept | Indicates the urgency of the encounter v3 Code System ActPriority (Example)
subject | ? | 0..1 | Reference(Patient | Group) | The patient ro group present at the encounter
episodeOfCare | ? | 0..* | Reference(EpisodeOfCare) | Episode(s) of care that this encounter should be recorded against
incomingReferral | | 0..* | Reference(ReferralRequest) | The ReferralRequest that initiated this encounter
participant | ? | 0..* | BackboneElement | List of participants involved in the encounter
type | ? | 0..* | CodeableConcept | Role of participant in encounter ParticipantType (Extensible)
period | | 0..1 | Period | Period of time during the encounter that the participant participated
individual | ? | 0..1 | Reference(Practitioner | RelatedPerson) | Persons involved in the encounter other than the patient
appointment | ? | 0..1 | Reference(Appointment) | The appointment that scheduled this encounter
period | | 0..1 | Period | The start and end time of the encounter
length | | 0..1 | Duration | Quantity of time the encounter lasted (less time absent)
reason | ? | 0..* | CodeableConcept | Reason the encounter takes place (code) Encounter Reason Codes (Preferred)
diagnosis | ? | 0..* | BackboneElement | The list of diagnosis relevant to this encounter
condition | | 1..1 | Reference(Condition | Procedure) | Reason the encounter takes place (resource)
role | | 0..1 | CodeableConcept | Role that this diagnosis has within the encounter (e.g. admission, billing, discharge …) DiagnosisRole (Preferred)
rank | | 0..1 | positiveInt | Ranking of the diagnosis (for each role type)
account | | 0..* | Reference(Account) | The set of accounts that may be used for billing for this Encounter
hospitalization | | 0..1 | BackboneElement | Details about the admission to a healthcare service
preAdmissionIdentifier | | 0..1 | Identifier | Pre-admission identifier
origin | | 0..1 | Reference(Location) | The location from which the patient came before admission
admitSource | | 0..1 | CodeableConcept | From where patient was admitted (physician referral, transfer) AdmitSource (Preferred)
reAdmission | | 0..1 | CodeableConcept | The type of hospital re-admission that has occurred (if any). If the value is absent, then this is not identified as a readmission v2 Re-Admission Indicator (Example)
dietPreference | | 0..* | CodeableConcept | Diet preferences reported by the patient Diet (Example)
specialCourtesy | | 0..* | CodeableConcept | Special courtesies (VIP, board member) SpecialCourtesy (Preferred)
specialArrangement | | 0..* | CodeableConcept | Wheelchair, translator, stretcher, etc. SpecialArrangements (Preferred)
destination | | 0..1 | Reference(Location) | Location to which the patient is discharged
dischargeDisposition | | 0..1 | CodeableConcept | Category or kind of location after discharge DischargeDisposition (Example)
location | | 0..* | BackboneElement | List of locations where the patient has been
location | | 1..1 | Reference(Location) | Location the encounter takes place
status | | 0..1 | code | planned | active | reserved | completed EncounterLocationStatus (Required)
period | | 0..1 | Period | Time period during which the patient was present at the location
serviceProvider | | 0..1 | Reference(Organization) | The custodian organization of this Encounter record
partOf | | 0..1 | Reference(Encounter) | Another Encounter this encounter is part of
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Encounter xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Identifier(s) by which this encounter is known --></identifier>
 <status value="[code]"/><!-- 1..1 planned | arrived | triaged | in-progress | onleave | finished | cancelled + -->
 <statusHistory>  <!-- 0..* List of past encounter statuses -->
  <status value="[code]"/><!-- 1..1 planned | arrived | triaged | in-progress | onleave | finished | cancelled + -->
```

```
 <period><!-- 1..1 Period The time that the episode was in the specified status --></period>
</statusHistory>
<class><!-- 0..1 Coding inpatient | outpatient | ambulatory | emergency + --></class>
<classHistory>  <!-- 0..* List of past encounter classes -->
 <class><!-- 1..1 Coding inpatient | outpatient | ambulatory | emergency + --></class>
 <period><!-- 1..1 Period The time that the episode was in the specified class --></period>
</classHistory>
<type><!-- 0..* CodeableConcept Specific type of encounter --></type>
<priority><!-- 0..1 CodeableConcept Indicates the urgency of the encounter --></priority>
<subject><!-- 0..1 Reference(Patient|Group) The patient ro group present at the encounter --></subject>
<episodeOfCare><!-- 0..* Reference(EpisodeOfCare) Episode(s) of care that this encounter should be recorded against
   --></episodeOfCare>
<incomingReferral><!-- 0..* Reference(ReferralRequest) The ReferralRequest that initiated this encounter
   --></incomingReferral>
<participant>  <!-- 0..* List of participants involved in the encounter -->
 <type><!-- 0..* CodeableConcept Role of participant in encounter --></type>
 <period><!-- 0..1 Period Period of time during the encounter that the participant participated --></period>
 <individual><!-- 0..1 Reference(Practitioner|RelatedPerson) Persons involved in the encounter other than the patient
   --></individual>
</participant>
<appointment><!-- 0..1 Reference(Appointment) The appointment that scheduled this encounter --></appointment>
<period><!-- 0..1 Period The start and end time of the encounter --></period>
<length><!-- 0..1 Duration Quantity of time the encounter lasted (less time absent) --></length>
<reason><!-- 0..* CodeableConcept Reason the encounter takes place (code) --></reason>
<diagnosis>  <!-- 0..* The list of diagnosis relevant to this encounter -->
 <condition><!-- 1..1 Reference(Condition|Procedure) Reason the encounter takes place (resource) --></condition>
 <role><!-- 0..1 CodeableConcept Role that this diagnosis has within the encounter (e.g. admission, billing, discharge
   …) --></role>
 <rank value="[positiveInt]"/><!-- 0..1 Ranking of the diagnosis (for each role type) -->
</diagnosis>
<account><!-- 0..* Reference(Account) The set of accounts that may be used for billing for this Encounter --></account>
<hospitalization>  <!-- 0..1 Details about the admission to a healthcare service -->
 <preAdmissionIdentifier><!-- 0..1 Identifier Pre-admission identifier --></preAdmissionIdentifier>
 <origin><!-- 0..1 Reference(Location) The location from which the patient came before admission --></origin>
 <admitSource><!-- 0..1 CodeableConcept From where patient was admitted (physician referral, transfer)
   --></admitSource>
 <reAdmission><!-- 0..1 CodeableConcept The type of hospital re-admission that has occurred (if any). If the value is
   absent, then this is not identified as a readmission --></reAdmission>
 <dietPreference><!-- 0..* CodeableConcept Diet preferences reported by the patient --></dietPreference>
 <specialCourtesy><!-- 0..* CodeableConcept Special courtesies (VIP, board member) --></specialCourtesy>
 <specialArrangement><!-- 0..* CodeableConcept Wheelchair, translator, stretcher, etc. --></specialArrangement>
 <destination><!-- 0..1 Reference(Location) Location to which the patient is discharged --></destination>
 <dischargeDisposition><!-- 0..1 CodeableConcept Category or kind of location after discharge
   --></dischargeDisposition>
</hospitalization>
<location>  <!-- 0..* List of locations where the patient has been -->
 <location><!-- 1..1 Reference(Location) Location the encounter takes place --></location>
 <status value="[code]"/><!-- 0..1 planned | active | reserved | completed -->
 <period><!-- 0..1 Period Time period during which the patient was present at the location --></period>
</location>
<serviceProvider><!-- 0..1 Reference(Organization) The custodian organization of this Encounter record
   --></serviceProvider>
<partOf><!-- 0..1 Reference(Encounter) Another Encounter this encounter is part of --></partOf>
</Encounter>
```

JSON Template

```
{
  "resourceType" : "Encounter",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Identifier(s) by which this encounter is known
  "status" : "<code>", // R!  planned | arrived | triaged | in-progress | onleave | finished | cancelled +
  "statusHistory" : [{ // List of past encounter statuses
    "status" : "<code>", // R!  planned | arrived | triaged | in-progress | onleave | finished | cancelled +
    "period" : { Period } // R!  The time that the episode was in the specified status
  }],
  "class" : { Coding }, // inpatient | outpatient | ambulatory | emergency +
  "classHistory" : [{ // List of past encounter classes
    "class" : { Coding }, // R!  inpatient | outpatient | ambulatory | emergency +
    "period" : { Period } // R!  The time that the episode was in the specified class
  }],
  "type" : [{ CodeableConcept }], // Specific type of encounter
  "priority" : { CodeableConcept }, // Indicates the urgency of the encounter
  "subject" : { Reference(Patient|Group) }, // The patient ro group present at the encounter
  "episodeOfCare" : [{ Reference(EpisodeOfCare) }], // Episode(s) of care that this encounter should be recorded against
  "incomingReferral" : [{ Reference(ReferralRequest) }], // The ReferralRequest that initiated this encounter
  "participant" : [{ // List of participants involved in the encounter
    "type" : [{ CodeableConcept }], // Role of participant in encounter
    "period" : { Period }, // Period of time during the encounter that the participant participated
    "individual" : { Reference(Practitioner|RelatedPerson) } // Persons involved in the encounter other than the patient
  }],
  "appointment" : { Reference(Appointment) }, // The appointment that scheduled this encounter
  "period" : { Period }, // The start and end time of the encounter
  "length" : { Duration }, // Quantity of time the encounter lasted (less time absent)
  "reason" : [{ CodeableConcept }], // Reason the encounter takes place (code)
  "diagnosis" : [{ // The list of diagnosis relevant to this encounter
    "condition" : { Reference(Condition|Procedure) }, // R!  Reason the encounter takes place (resource)
    "role" : { CodeableConcept }, // Role that this diagnosis has within the encounter (e.g. admission, billing,
    discharge …)
```

```
    "rank" : "<positiveInt>" // Ranking of the diagnosis (for each role type)
  }],
  "account" : [{ Reference(Account) }], // The set of accounts that may be used for billing for this Encounter
  "hospitalization" : { // Details about the admission to a healthcare service
    "preAdmissionIdentifier" : { Identifier }, // Pre-admission identifier
    "origin" : { Reference(Location) }, // The location from which the patient came before admission
    "admitSource" : { CodeableConcept }, // From where patient was admitted (physician referral, transfer)
    "reAdmission" : { CodeableConcept }, // The type of hospital re-admission that has occurred (if any). If the value
    is absent, then this is not identified as a readmission
    "dietPreference" : [{ CodeableConcept }], // Diet preferences reported by the patient
    "specialCourtesy" : [{ CodeableConcept }], // Special courtesies (VIP, board member)
    "specialArrangement" : [{ CodeableConcept }], // Wheelchair, translator, stretcher, etc.
    "destination" : { Reference(Location) }, // Location to which the patient is discharged
    "dischargeDisposition" : { CodeableConcept } // Category or kind of location after discharge
  },
  "location" : [{ // List of locations where the patient has been
    "location" : { Reference(Location) }, // R!  Location the encounter takes place
    "status" : "<code>", // planned | active | reserved | completed
    "period" : { Period } // Time period during which the patient was present at the location
  }],
  "serviceProvider" : { Reference(Organization) }, // The custodian organization of this Encounter record
  "partOf" : { Reference(Encounter) } // Another Encounter this encounter is part of
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Encounter;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Encounter.identifier [ Identifier ], ... ; # 0..* Identifier(s) by which this encounter is known
  fhir:Encounter.status [ code ]; # 1..1 planned | arrived | triaged | in-progress | onleave | finished | cancelled +
  fhir:Encounter.statusHistory [ # 0..* List of past encounter statuses
    fhir:Encounter.statusHistory.status [ code ]; # 1..1 planned | arrived | triaged | in-progress | onleave | finished
    | cancelled +
    fhir:Encounter.statusHistory.period [ Period ]; # 1..1 The time that the episode was in the specified status
  ], ...;
  fhir:Encounter.class [ Coding ]; # 0..1 inpatient | outpatient | ambulatory | emergency +
  fhir:Encounter.classHistory [ # 0..* List of past encounter classes
    fhir:Encounter.classHistory.class [ Coding ]; # 1..1 inpatient | outpatient | ambulatory | emergency +
    fhir:Encounter.classHistory.period [ Period ]; # 1..1 The time that the episode was in the specified class
  ], ...;
  fhir:Encounter.type [ CodeableConcept ], ... ; # 0..* Specific type of encounter
  fhir:Encounter.priority [ CodeableConcept ]; # 0..1 Indicates the urgency of the encounter
  fhir:Encounter.subject [ Reference(Patient|Group) ]; # 0..1 The patient ro group present at the encounter
  fhir:Encounter.episodeOfCare [ Reference(EpisodeOfCare) ], ... ; # 0..* Episode(s) of care that this encounter should
    be recorded against
  fhir:Encounter.incomingReferral [ Reference(ReferralRequest) ], ... ; # 0..* The ReferralRequest that initiated this
    encounter
  fhir:Encounter.participant [ # 0..* List of participants involved in the encounter
    fhir:Encounter.participant.type [ CodeableConcept ], ... ; # 0..* Role of participant in encounter
    fhir:Encounter.participant.period [ Period ]; # 0..1 Period of time during the encounter that the participant
    participated
    fhir:Encounter.participant.individual [ Reference(Practitioner|RelatedPerson) ]; # 0..1 Persons involved in the
    encounter other than the patient
  ], ...;
  fhir:Encounter.appointment [ Reference(Appointment) ]; # 0..1 The appointment that scheduled this encounter
  fhir:Encounter.period [ Period ]; # 0..1 The start and end time of the encounter
  fhir:Encounter.length [ Duration ]; # 0..1 Quantity of time the encounter lasted (less time absent)
  fhir:Encounter.reason [ CodeableConcept ], ... ; # 0..* Reason the encounter takes place (code)
  fhir:Encounter.diagnosis [ # 0..* The list of diagnosis relevant to this encounter
    fhir:Encounter.diagnosis.condition [ Reference(Condition|Procedure) ]; # 1..1 Reason the encounter takes place
    (resource)
    fhir:Encounter.diagnosis.role [ CodeableConcept ]; # 0..1 Role that this diagnosis has within the encounter (e.g.
    admission, billing, discharge …)
    fhir:Encounter.diagnosis.rank [ positiveInt ]; # 0..1 Ranking of the diagnosis (for each role type)
  ], ...;
  fhir:Encounter.account [ Reference(Account) ], ... ; # 0..* The set of accounts that may be used for billing for this
    Encounter
  fhir:Encounter.hospitalization [ # 0..1 Details about the admission to a healthcare service
    fhir:Encounter.hospitalization.preAdmissionIdentifier [ Identifier ]; # 0..1 Pre-admission identifier
    fhir:Encounter.hospitalization.origin [ Reference(Location) ]; # 0..1 The location from which the patient came
    before admission
    fhir:Encounter.hospitalization.admitSource [ CodeableConcept ]; # 0..1 From where patient was admitted (physician
    referral, transfer)
    fhir:Encounter.hospitalization.reAdmission [ CodeableConcept ]; # 0..1 The type of hospital re-admission that has
    occurred (if any). If the value is absent, then this is not identified as a readmission
    fhir:Encounter.hospitalization.dietPreference [ CodeableConcept ], ... ; # 0..* Diet preferences reported by the
    patient
    fhir:Encounter.hospitalization.specialCourtesy [ CodeableConcept ], ... ; # 0..* Special courtesies (VIP, board
    member)
    fhir:Encounter.hospitalization.specialArrangement [ CodeableConcept ], ... ; # 0..* Wheelchair, translator,
    stretcher, etc.
    fhir:Encounter.hospitalization.destination [ Reference(Location) ]; # 0..1 Location to which the patient is
    discharged
    fhir:Encounter.hospitalization.dischargeDisposition [ CodeableConcept ]; # 0..1 Category or kind of location after
```

```
    discharge
  ];
  fhir:Encounter.location [ # 0..* List of locations where the patient has been
    fhir:Encounter.location.location [ Reference(Location) ]; # 1..1 Location the encounter takes place
    fhir:Encounter.location.status [ code ]; # 0..1 planned | active | reserved | completed
    fhir:Encounter.location.period [ Period ]; # 0..1 Time period during which the patient was present at the location
  ], ...;
  fhir:Encounter.serviceProvider [ Reference(Organization) ]; # 0..1 The custodian organization of this Encounter record
  fhir:Encounter.partOf [ Reference(Encounter) ]; # 0..1 Another Encounter this encounter is part of
]
```

Changes since DSTU2

Encounter

Encounter.status | Change value set from http://hl7.org/fhir/ValueSet/encounter-state to http://hl7.org/fhir/ValueSet/encounter-status

Encounter.statusHistory.status | Change value set from http://hl7.org/fhir/ValueSet/encounter-state to http://hl7.org/fhir/ValueSet/encounter-status

Encounter.class | Type changed from code to Coding Change binding strength from required to extensible Change value set from http://hl7.org/fhir/ValueSet/encounter-class to http://hl7.org/fhir/ValueSet/v3-ActEncounterCode

Encounter.classHistory | Added Element

Encounter.classHistory.class | Added Element

Encounter.classHistory.period | Added Element

Encounter.subject | Renamed from patient to subject Add Reference(Group)

Encounter.length | Type changed from Quantity{http://hl7.org/fhir/StructureDefinition/Duration} to Duration

Encounter.diagnosis | Added Element

Encounter.diagnosis.condition | Added Element

Encounter.diagnosis.role | Added Element

Encounter.diagnosis.rank | Added Element

Encounter.account | Added Element

Encounter.indication | deleted

Encounter.hospitalization.admittingDiagnosis | deleted

Encounter.hospitalization.dischargeDiagnosis | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R2 <--> R3 Conversion Maps (status = 10 tests that all execute ok. 1 fail round-trip testing and 2 r3 resources are invalid (2 errors).).

Alternate definitions: Master Definition (XML, JSON), XML Schema/Schematron (for) + JSON Schema, ShEx (for Turtle)

8.11.3.1 Terminology Bindings

Path | Definition | Type | Reference

Encounter.status Encounter.statusHistory.status | Current state of the encounter | Required | EncounterStatus

Encounter.class Encounter.classHistory.class | Classification of the encounter | Extensible | ActEncounterCode

Encounter.type | The type of encounter | Example | EncounterType

Encounter.priority | Indicates the urgency of the encounter. | Example | v3 Code System ActPriority

Encounter.participant.type | Role of participant in encounter | Extensible | ParticipantType

Encounter.reason | Reason why the encounter takes place. | Preferred | Encounter Reason Codes

Encounter.diagnosis.role | The type of diagnosis this condition represents | Preferred | DiagnosisRole

Encounter.hospitalization.admitSource | From where the patient was admitted. | Preferred | AdmitSource

Encounter.hospitalization.reAdmission | The reason for re-admission of this hospitalization encounter. | Example | v2 Re-Admission Indicator

Encounter.hospitalization.dietPreference | Medical, cultural or ethical food preferences to help with catering requirements. | Example | Diet

Encounter.hospitalization.specialCourtesy | Special courtesies | Preferred | SpecialCourtesy

Encounter.hospitalization.specialArrangement | Special arrangements | Preferred | SpecialArrangements

Encounter.hospitalization.dischargeDisposition | Discharge Disposition | Example | DischargeDisposition

Encounter.location.status | The status of the location. | Required | EncounterLocationStatus

8.11.4 Notes

* The class element describes the setting (in/outpatient etc.) in which the Encounter took place. Since this is important for interpreting the context of the encounter, choosing the appropriate business rules to enforce and for the management of the process, this element is required.

* In future versions of FHIR, some kind of charge posting vehicle (e.g. Account) will be added.

8.11.5 Example usage

As stated, Encounter allows a flexible nesting of Encounters using the partOf element. For example:

* A patient is admitted for two weeks - This could be modeled using a single Encounter instance, in which the start and length are given for the duration of the whole stay. The admitting doctor and the responsible doctor during the stay are specified using the Participant component.

* During the encounter, the patient moves from the admitting department to the Intensive Care unit and back - Three more detailed additional Encounters can be created, one for each location in which the patient stayed. Each of these Encounters has a single location (twice the admitting department and once the Intensive Care unit) and one or more participants at that location. These Encounters may use the partOf relationship to indicate these movements occurred during the longer overarching Encounter.

* During the last part of the stay, the patient is visited by the members of the multi-disciplinary team that treated him for final evaluation - If relevant, for each of these short visits, an Encounter may be created with a single participant. Since these took place during the last part of the stay, the partOf element can be used to associate these short visits with either the third patient movement or the bigger overall encounter.

Exactly how the Encounter is used depends on information available in the source system, the relevance of exchange of each level of Encounter and demands specific to the communicating partners. The expectation is that for each domain of exchange, profiles are used to limit the flexibility of Encounter to meet the demands of the use case.

8.11.6 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
appointment | reference | The appointment that scheduled this encounter | Encounter.appointment (Appointment)
class | token | inpatient | outpatient | ambulatory | emergency + | Encounter.class
date | date | A date within the period the Encounter lasted | Encounter.period | 18 Resources
diagnosis | reference | Reason the encounter takes place (resource) | Encounter.diagnosis.condition (Condition, Procedure)
episodeofcare | reference | Episode(s) of care that this encounter should be recorded against | Encounter.episodeOfCare (EpisodeOfCare)
identifier | token | Identifier(s) by which this encounter is known | Encounter.identifier | 26 Resources
incomingreferral | reference | The ReferralRequest that initiated this encounter | Encounter.incomingReferral (ReferralRequest)
length | number | Length of encounter in days | Encounter.length
location | reference | Location the encounter takes place | Encounter.location.location (Location)
location-period | date | Time period during which the patient was present at the location | Encounter.location.period
part-of | reference | Another Encounter this encounter is part of | Encounter.partOf (Encounter)
participant | reference | Persons involved in the encounter other than the patient | Encounter.participant.individual (Practitioner, RelatedPerson)
participant-type | token | Role of participant in encounter | Encounter.participant.type
patient | reference | The patient ro group present at the encounter | Encounter.subject (Patient) | 31 Resources
practitioner | reference | Persons involved in the encounter other than the patient | Encounter.participant.individual (Practitioner)
reason | token | Reason the encounter takes place (code) | Encounter.reason
service-provider | reference | The custodian organization of this Encounter record | Encounter.serviceProvider (Organization)
special-arrangement | token | Wheelchair, translator, stretcher, etc. | Encounter.hospitalization.specialArrangement
status | token | planned | arrived | triaged | in-progress | onleave | finished | cancelled + | Encounter.status
subject | reference | The patient ro group present at the encounter | Encounter.subject (Group, Patient)
type | token | Specific type of encounter | Encounter.type | 6 Resources

8.11.7 Resource Encounter - Examples

URL: https://hl7.org/fhir/STU3/encounter-examples.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.11.7 Resource Encounter - Examples

Patient Administration Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Encounter, Patient, Practitioner, RelatedPerson

Example List:

Encounter example | example | XML | JSON | Turtle
Encounter example - virtual encounter | home | XML | JSON | Turtle
Real-world encounter example | f201 | XML | JSON | Turtle
Real-world encounter example (with primaryDiagnosis extension added) | f202 | XML | JSON | Turtle
Real-world encounter example | f203 | XML | JSON | Turtle
Real-world encounter example | f001 | XML | JSON | Turtle
Real-world encounter example | f002 | XML | JSON | Turtle
Real-world encounter example | f003 | XML | JSON | Turtle
for Clinical Document example patient | xcda | XML | JSON | Turtle
Emergency transitioning into inpatient example | emerg | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

8.11.8 Resource Encounter - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/encounter-definitions.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.11.8 Resource Encounter - Detailed Descriptions

Patient Administration Work Group | Maturity Level: 2 | Trial Use | Compartments: Encounter, Patient, Practitioner, RelatedPerson

Detailed Descriptions for the elements in the Encounter resource.

Encounter

Definition | An interaction between a patient and healthcare provider(s) for the purpose of providing healthcare service(s) or assessing the health status of a patient.
Control | 1..1
Alternate Names | Visit
Encounter.identifier
Definition | Identifier(s) by which this encounter is known.
Note | This is a business identifer, not a resource identifier (see discussion)
Control | 0..*
Type | Identifier
Summary | true
Encounter.status
Definition | planned | arrived | triaged | in-progress | onleave | finished | cancelled +.
Control | 1..1
Terminology Binding | EncounterStatus (Required)
Type | code
Is Modifier | true
Summary | true
Comments | Note that internal business rules will detemine the appropraite transitions that may occur between statuses (and also classes). This element is labeled as a modifier because the status contains codes that mark the encounter as not currently valid.
Encounter.statusHistory
Definition | The status history permits the encounter resource to contain the status history without needing to read through the historical versions of the resource, or even have the server store them.
Control | 0..*
Comments | The current status is always found in the current version of the resource, not the status history.
Encounter.statusHistory.status
Definition | planned | arrived | triaged | in-progress | onleave | finished | cancelled +.
Control | 1..1
Terminology Binding | EncounterStatus (Required)
Type | code
Encounter.statusHistory.period
Definition | The time that the episode was in the specified status.
Control | 1..1
Type | Period
Encounter.class
Definition | inpatient | outpatient | ambulatory | emergency +.
Control | 0..1
Terminology Binding | ActEncounterCode (Extensible)
Type | Coding
Summary | true
Encounter.classHistory
Definition | The class history permits the tracking of the encounters transitions without needing to go through the resource history. This would be used for a case where an admission starts of as an emergency encounter, then transisions into an inpatient scenario. Doing this and not restarting a new encounter ensures that any lab/diagnostic results can more easily follow the patient and not require re-processing and not get lost or cancelled during a kindof discharge from emergency to inpatient.
Control | 0..*
Encounter.classHistory.class
Definition | inpatient | outpatient | ambulatory | emergency +.
Control | 1..1
Terminology Binding | ActEncounterCode (Extensible)
Type | Coding
Encounter.classHistory.period
Definition | The time that the episode was in the specified class.
Control | 1..1
Type | Period
Encounter.type
Definition | Specific type of encounter (e.g. e-mail consultation, surgical day-care, skilled nursing, rehabilitation).
Control | 0..*
Terminology Binding | EncounterType (Example)
Type | CodeableConcept
Summary | true
Comments | Since there are many ways to further classify encounters, this element is 0..*.
Encounter.priority
Definition | Indicates the urgency of the encounter.
Control | 0..1
Terminology Binding | v3 Code System ActPriority (Example)
Type | CodeableConcept
Encounter.subject
Definition | The patient ro group present at the encounter.
Control | 0..1
Type | Reference(Patient | Group)
Alternate Names | patient
Summary | true

Comments | While the encounter is always about the patient, the patient may not actually be known in all contexts of use, and there may be a group of patients that could be anonymous (such as in a group therapy for Alcoholics Anonymous - where the recording of the encounter could be used for billing on the number of people/staff and not important to the context of the specific patients) or alternately in veterinary care a herd of sheep receiving treatment (where the animals are not individually tracked).
Encounter.episodeOfCare
Definition | Where a specific encounter should be classified as a part of a specific episode(s) of care this field should be used. This association can facilitate grouping of related encounters together for a specific purpose, such as government reporting, issue tracking, association via a common problem. The association is recorded on the encounter as these are typically created after the episode of care, and grouped on entry rather than editing the episode of care to append another encounter to it (the episode of care could span years).
Control | 0..*
Type | Reference(EpisodeOfCare)
Summary | true
Encounter.incomingReferral
Definition | The referral request this encounter satisfies (incoming referral).
Control | 0..*
Type | Reference(ReferralRequest)
Encounter.participant
Definition | The list of people responsible for providing the service.
Control | 0..*
Summary | true
Encounter.participant.type
Definition | Role of participant in encounter.
Control | 0..*
Terminology Binding | ParticipantType (Extensible)
Type | CodeableConcept
Summary | true
Comments | The participant type indicates how an individual partitipates in an encounter. It includes non-practitioner participants, and for practitioners this is to describe the action type in the context of this encounter (e.g. Admitting Dr, Attending Dr, Translator, Consulting Dr). This is different to the practitioner roles which are functional roles, derived from terms of employment, education, licensing, etc.
Encounter.participant.period
Definition | The period of time that the specified participant participated in the encounter. These can overlap or be sub-sets of the overall encounter's period.
Control | 0..1
Type | Period
Encounter.participant.individual
Definition | Persons involved in the encounter other than the patient.
Control | 0..1
Type | Reference(Practitioner | RelatedPerson)
Summary | true
Encounter.appointment
Definition | The appointment that scheduled this encounter.
Control | 0..1
Type | Reference(Appointment)
Summary | true
Encounter.period
Definition | The start and end time of the encounter.
Control | 0..1
Type | Period
Comments | If not (yet) known, the end of the Period may be omitted.
Encounter.length
Definition | Quantity of time the encounter lasted. This excludes the time during leaves of absence.
Control | 0..1
Type | Duration
Comments | May differ from the time the Encounter.period lasted because of leave of absence.
Encounter.reason
Definition | Reason the encounter takes place, expressed as a code. For admissions, this can be used for a coded admission diagnosis.
Control | 0..*
Terminology Binding | Encounter Reason Codes (Preferred)
Type | CodeableConcept
Alternate Names | Indication; Admission diagnosis
Summary | true
Comments | For systems that need to know which was the primary diagnosis, these will be marked with the standard extension primaryDiagnosis (which is a sequence value rather than a flag, 1 = primary diagnosis).
Encounter.diagnosis
Definition | The list of diagnosis relevant to this encounter.
Control | 0..*
Summary | true
Encounter.diagnosis.condition
Definition | Reason the encounter takes place, as specified using information from another resource. For admissions, this is the admission diagnosis. The indication will typically be a Condition (with other resources referenced in the evidence.detail), or a Procedure.
Control | 1..1
Type | Reference(Condition | Procedure)
Alternate Names | Admission diagnosis; discharge diagnosis; indication

Comments | For systems that need to know which was the primary diagnosis, these will be marked with the standard extension primaryDiagnosis (which is a sequence value rather than a flag, 1 = primary diagnosis).
Encounter.diagnosis.role
Definition | Role that this diagnosis has within the encounter (e.g. admission, billing, discharge …).
Control | 0..1
Terminology Binding | DiagnosisRole (Preferred)
Type | CodeableConcept
Encounter.diagnosis.rank
Definition | Ranking of the diagnosis (for each role type).
Control | 0..1
Type | positiveInt
Encounter.account
Definition | The set of accounts that may be used for billing for this Encounter.
Control | 0..*
Type | Reference(Account)
Comments | The billing system may choose to allocate billable items associated with the Encounter to different referenced Accounts based on internal business rules.
Encounter.hospitalization
Definition | Details about the admission to a healthcare service.
Control | 0..1
Comments | An Encounter may cover more than just the inpatient stay. Contexts such as outpatients, community clinics, and aged care facilities are also included. The duration recorded in the period of this encounter covers the entire scope of this hospitalization record.
Encounter.hospitalization.preAdmissionIdentifier
Definition | Pre-admission identifier.
Control | 0..1
Type | Identifier
Encounter.hospitalization.origin
Definition | The location from which the patient came before admission.
Control | 0..1
Type | Reference(Location)
Encounter.hospitalization.admitSource
Definition | From where patient was admitted (physician referral, transfer).
Control | 0..1
Terminology Binding | AdmitSource (Preferred)
Type | CodeableConcept
Encounter.hospitalization.reAdmission
Definition | Whether this hospitalization is a readmission and why if known.
Control | 0..1
Terminology Binding | v2 Re-Admission Indicator (Example)
Type | CodeableConcept
Encounter.hospitalization.dietPreference
Definition | Diet preferences reported by the patient.
Control | 0..*
Terminology Binding | Diet (Example)
Type | CodeableConcept
Requirements | Used to track patient's diet restrictions and/or preference. For a complete description of the nutrition needs of a patient during their stay, one should use the nutritionOrder resource which links to Encounter.
Comments | For example a patient may request both a dairy-free and nut-free diet preference (not mutually exclusive).
Encounter.hospitalization.specialCourtesy
Definition | Special courtesies (VIP, board member).
Control | 0..*
Terminology Binding | SpecialCourtesy (Preferred)
Type | CodeableConcept
Encounter.hospitalization.specialArrangement
Definition | Any special requests that have been made for this hospitalization encounter, such as the provision of specific equipment or other things.
Control | 0..*
Terminology Binding | SpecialArrangements (Preferred)
Type | CodeableConcept
Encounter.hospitalization.destination
Definition | Location to which the patient is discharged.
Control | 0..1
Type | Reference(Location)
Encounter.hospitalization.dischargeDisposition
Definition | Category or kind of location after discharge.
Control | 0..1
Terminology Binding | DischargeDisposition (Example)
Type | CodeableConcept
Encounter.location
Definition | List of locations where the patient has been during this encounter.
Control | 0..*
Comments | Virtual encounters can be recorded in the Encounter by specifying a location reference to a location of type "kind" such as "client's home" and an encounter.class = "virtual".
Encounter.location.location

Definition | The location where the encounter takes place.
Control | 1..1
Type | Reference(Location)
Encounter.location.status
Definition | The status of the participants' presence at the specified location during the period specified. If the participant is is no longer at the location, then the period will have an end date/time.
Control | 0..1
Terminology Binding | EncounterLocationStatus (Required)
Type | code
Comments | When the patient is no longer active at a location, then the period end date is entered, and the status may be changed to completed.
Encounter.location.period
Definition | Time period during which the patient was present at the location.
Control | 0..1
Type | Period
Encounter.serviceProvider
Definition | An organization that is in charge of maintaining the information of this Encounter (e.g. who maintains the report or the master service catalog item, etc.). This MAY be the same as the organization on the Patient record, however it could be different. This MAY not be not the Service Delivery Location's Organization.
Control | 0..1
Type | Reference(Organization)
Encounter.partOf
Definition | Another Encounter of which this encounter is a part of (administratively or in time).
Control | 0..1
Type | Reference(Encounter)
Comments | This is also used for associating a child's encounter back to the mother's encounter. Refer to the Notes section in the Patient resource for further details.

8.11.9 Resource Encounter - Mappings

URL: https://hl7.org/fhir/STU3/encounter-mappings.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.11.9 Resource Encounter - Mappings

Patient Administration Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Encounter, Patient, Practitioner, RelatedPerson

Mappings:

RIM Mapping

HL7 v2 Mapping

Mappings for the encounter resource.

8.11.9.1 Mappings for RIM Mapping (http://hl7.org/v3)

Encounter | Encounter[@moodCode='EVN']
identifier | .id
status | .statusCode
statusHistory | n/a
status | n/a
period | n/a
class | .inboundRelationship[typeCode=SUBJ].source[classCode=LIST].code
classHistory | n/a
class | n/a
period | n/a
type | .code
priority | .priorityCode
subject | .participation[typeCode=SBJ]/role[classCode=PAT]
episodeOfCare | n/a
incomingReferral | .reason.ClinicalDocument
participant | .participation[typeCode=PFM]
type | .functionCode
period | .time
individual | .role
appointment | .outboundRelationship[typeCode=FLFS].target[classCode=ENC, moodCode=APT]
period | .effectiveTime (low & high)
length | .lengthOfStayQuantity
reason | .reasonCode
diagnosis | .outboundRelationship[typeCode=RSON]
condition | .outboundRelationship[typeCode=RSON].target
role | n/a
rank | .outboundRelationship[typeCode=RSON].priority
account | .pertains.A_Account
hospitalization | .outboundRelationship[typeCode=COMP].target[classCode=ENC, moodCode=EVN]

preAdmissionIdentifier | .id
origin | .participation[typeCode=ORG].role
admitSource | .admissionReferralSourceCode
reAdmission | n/a
dietPreference | .outboundRelationship[typeCode=COMP].target[classCode=SBADM, moodCode=EVN, code="diet"]
specialCourtesy | .specialCourtesiesCode
specialArrangement | .specialArrangementCode
destination | .participation[typeCode=DST]
dischargeDisposition | .dischargeDispositionCode
location | .participation[typeCode=LOC]
location | .role
status | .role.statusCode
period | .time
serviceProvider | .particiaption[typeCode=PFM].role
partOf | .inboundRelationship[typeCode=COMP].source[classCode=COMP, moodCode=EVN]

8.11.9.2 Mappings for HL7 v2 Mapping (http://hl7.org/v2)

Encounter
identifier | PV1-19
status | No clear equivalent in HL7 v2; active/finished could be inferred from PV1-44, PV1-45, PV2-24; inactive could be inferred from PV2-16
statusHistory
status
period
class | PV1-2
classHistory
class
period
type | PV1-4 / PV1-18
priority | PV2-25
subject | PID-3
episodeOfCare | PV1-54, PV1-53
incomingReferral
participant | ROL
type | ROL-3 (or maybe PRT-4)
period | ROL-5, ROL-6 (or maybe PRT-5)
individual | ROL-4
appointment | SCH-1 / SCH-2
period | PV1-44, PV1-45
length | (PV1-45 less PV1-44) iff ((PV1-44 not empty) and (PV1-45 not empty)); units in minutes
reason | EVN-4 / PV2-3 (note: PV2-3 is nominally constrained to inpatient admissions; HL7 v2 makes no vocabulary suggestions for PV2-3; would not expect PV2 segment or PV2-3 to be in use in all implementations)
diagnosis
condition | Resources that would commonly referenced at Encounter.indication would be Condition and/or Procedure. These most closely align with DG1/PRB and PR1 respectively.
role
rank
account
hospitalization
preAdmissionIdentifier | PV1-5
origin
admitSource | PV1-14
reAdmission | PV1-13
dietPreference | PV1-38
specialCourtesy | PV1-16
specialArrangement | PV1-15 / OBR-30 / OBR-43
destination | PV1-37
dischargeDisposition | PV1-36
location
location | PV1-3 / PV1-6 / PV1-11 / PV1-42 / PV1-43
status
period
serviceProvider | PV1-10 / PL.6 & PL.1 (note: HL7 v2 definition is "the treatment or type of surgery that the patient is scheduled to receive"; seems slightly out of alignment with the concept name 'hospital service'. Would not trust that implementations apply this semantic by default)
partOf

8.11.10 Resource Encounter - Design Notes

URL: https://hl7.org/fhir/STU3/encounter-explanations.html

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8.11.10 Resource Encounter - Design Notes

to do

8.11.11 Resource Encounter - Extensions & Profiles

URL: https://hl7.org/fhir/STU3/encounter-profiles.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.11.11 Resource Encounter - Extensions & Profiles

Patient Administration Work Group | Maturity Level: N/A | Ballot Status: Informative

This table lists profiles and extensions for the Encounter resource. For background information, see Profiling Resources and Extensibility. Additional profiles and resources may be found in published Implementation Guides , or in the Conformance resource registry .

8.11.11.1 Profiles

No Profiles defined for this resource

8.11.11.2 Extensions

encounter-associatedEncounter | associatedEncounter | for Encounter HL7 Extensions
encounter-modeOfArrival | modeOfArrival | for Encounter HL7 Extensions
encounter-primaryDiagnosis | primaryDiagnosis | for Encounter HL7 Extensions
encounter-reasonCancelled | reasonCancelled | for Encounter HL7 Extensions

Extensions that reference this resource:

encounter-associatedEncounter | associatedEncounter | for Encounter HL7 Extensions

8.11.11.3 Search Extensions

No Search Extensions defined for this resource

8.11.12 Encounter HL7 Extensions

URL: https://hl7.org/fhir/STU3/encounter-extensions.html

Defines common extensions used with or related to the Encounter resource

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.11.12 Encounter HL7 Extensions

(No assigned work group) | Maturity Level: N/A | Ballot Status: Informative

8.11.12.1 Content

Extensions:
encounter-reasonCancelled | reasonCancelled : If the encountered was cancelled after it was planned, why? Applies only if the status is cancelled.
encounter-primaryDiagnosis | primaryDiagnosis : The order of diagnosis importance (1 = highest in importance), from the clinical perspective, may be used in billing.
encounter-associatedEncounter | associatedEncounter : This encounter occurs within the scope of the referenced encounter.
encounter-modeOfArrival | modeOfArrival : Identifies whether a patient arrives at the reporting facility via ambulance and the type of ambulance that was used.

8.11.13 Resource Encounter - Operations

URL: https://hl7.org/fhir/STU3/encounter-operations.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.11.13 Resource Encounter - Operations

Patient Administration Work Group | Maturity Level: 2 | Trial Use | Compartments: Encounter, Patient, Practitioner, RelatedPerson

This resource has 1 operation associated with it:

$everything | Fetch Encounter Record

For more information about operations, including how they are invoked, see Operations.

8.11.13.1 Fetch Encounter Record

This operation is used to return all the information related to an encounter described in the resource on which this operation is invoked. The response is a bundle of type "searchset". At a minimum, the encounter resource itself is returned, along with any other resources that the server has available for the given encounter for the user. The server also returns whatever resources

are needed to support the records - e.g. linked practitioners, locations, organizations etc. The principle intended use for this operation is to provide a patient with access to their record, or to allow a client to retrieve everything for an encounter for efficient display).

The server SHOULD return all resources it has that:

* are included in the encounter compartment for the identified encounter (have a reference to the encounter)

* are referenced by the standard extenstion for associating an encounter (where no reference element exists) http://hl7.org/fhir/StructureDefinition/encounter-associatedEncounter

* the server believes are relevant to the context of the encounter for any other reason (internally defined/decided)

* any resource referenced by the above, including binaries and attachments (to make a more complete package)

In the US Realm, at a mimimum, the resources returned SHALL include all the data covered by the meaningful use common data elements (see DAF for further guidance). Other applicable implementation guides may make additional rules about the information that is returned. Note that for many resources, the exact nature of the link to encounter can be ambiguous (e.g. for a DiagnosticReport, is it the encounter when it was initiated, or when it was reported?)

The official URL for this operation definition is

```
http://hl7.org/fhir/OperationDefinition/Encounter-everything
```

Formal Definition (as a OperationDefinition).

URL: [base]/Encounter/[id]/$everything

This is an idempotent operation

Out Parameters:

Name | Cardinality | Type | Binding | Profile | Documentation

return | 1..1 | Bundle | | | The bundle type is "searchset" Note: as this the only out parameter, it is a resource, and it has the name 'return', the result of this operation is returned directly as a resource

The key differences between this operation and simply searching the encounter compartment are: * unless the client requests otherwise, the server returns the entire result set in a single bundle (rather than using paging) * the server is responsible for determining what resources to return as included resources (rather than the client specifying which ones)

8.11.13.1.1 Examples

Get all resources linked to a single patient (Request):

```
GET /open/Encounter/example/$everything
[some headers]
```

Everything linked to the nominated patient (in this case, built automatically by the publication tooling) (Response):

```
HTTP/1.1 200 OK
[other headers]

<Bundle xml="http://hl7.org/fhir">
  <id value="abe363f2-c635-45e0-b9e2-f95dd12063de"/>
  <type value="searchset"/>
  <entry>
    <fullUrl value="http://hl7.org/fhir/Encounter/example"/>
    <resource>
     <Encounter xmlns="http://hl7.org/fhir">
       <id value="example"/>
       <text>
         <status value="generated"/>
         <div xmlns="http://www.w3.org/1999/xhtml">Encounter with patient @example</div>
       </text>
       <status value="in-progress"/>
         <class>
             <system value="http://hl7.org/fhir/v3/ActCode"/>
             <code value="IMP"/>
             <display value="inpatient encounter"/>
         </class>
      <subject>
         <reference value="Patient/example"/>
       </subject>
     </Encounter>
    </resource>
  </entry>
  <entry>
    <fullUrl value="http://hl7.org/fhir/Flag/example-encounter"/>
    <resource>
     <Flag xmlns="http://hl7.org/fhir">
       <id value="example-encounter"/>
       <text>
         <status value="generated"/>
         <div xmlns="http://www.w3.org/1999/xhtml">Follow Infection Control Level 3 Protocol</div>
       </text>
       <status value="active"/>
       <category>
         <coding>
           <system value="http://example.org/local"/>
           <code value="infection"/>
           <display value="Infection Control Level"/>
         </coding>
```

```
     </category>
     <code>
       <coding>
         <system value="http://example.org/local/if1"/>
         <code value="l3"/>
         <display value="Follow Level 3 Protocol"/>
       </coding>
     </code>

     <subject>
       <reference value="Patient/example"/>
       <display value="Peter Patient"/>
     </subject>

     <encounter>
       <reference value="Encounter/example"/>
     </encounter>


   </Flag>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/Task/example4"/>
  <resource>
   <Task xmlns="http://hl7.org/fhir">
    <!--This example show an updated status and output - a generated task to collect a specimen
   example1 action lipid panel -in progress =>. example2 specimen collection subtask -accepted => example4  specimen
  collection subtask - completed  with output Specimen=>example 5 action lipid panel -in progress specimen collected
  with output Specimen => example 6 action lipid panel - completed with output Specimen and DiagnosticReport -->
    <id value="example4"/>
    <identifier>
      <use value="official"/>
      <system value="http:/goodhealth.org/identifiers"/>
      <value value="20170201-002"/>
    </identifier>
    <groupIdentifier>
      <use value="official"/>
      <system value="http:/goodhealth.org/accession/identifiers"/>
      <value value="G20170201-001"/>
    </groupIdentifier>
    <!--  This Task is a child task of the submitted lipid panel requet
   -->
    <partOf>
      <reference value="Task/example1"/>
      <display value="Lipid Panel"/>
    </partOf>
    <status value="completed"/>
    <!--fictive business status populated for this example-->
    <intent value="filler-order"/>
    <priority value="routine"/>
    <!--fictive code and description elements populated for this example -->
    <code>
      <text value="Specimen Collection"/>
    </code>
    <!--  Filler generated order using a fictive  procedurerequest to complete the loop-->
    <focus>
      <display value="BloodDraw ProcedureRequest"/>
    </focus>
    <!--  who the task is for
   -->
    <for>
      <reference value="Patient/example"/>
      <display value="Peter James Chalmers"/>
    </for>
    <context>
      <reference value="Encounter/example"/>
      <display value="Example In-Patient Encounter"/>
    </context>
    <!--  Lots of dates -->
    <executionPeriod>
      <start value="2016-10-31T08:45:05+10:00"/>
      <end value="2016-10-31T14:45:05+10:00"/>
    </executionPeriod>
    <authoredOn value="2016-10-31T08:45:05+10:00"/>
    <lastModified value="2016-10-31T09:45:05+10:00"/>
    <!--requester is filler-->
    <requester>
      <agent>
        <reference value="Organization/1832473e-2fe0-452d-abe9-3cdb9879522f"/>
        <display value="Clinical Laboratory @ Acme Hospital"/>
      </agent>
    </requester>
    <performerType>
      <coding>
        <system value="http://hl7.org/fhir/task-performer-type"/>
        <code value="performer"/>
        <display value="Performer"/>
      </coding>
```

```
      <text value="Performer"/>
    </performerType>
    <!--assume at draw station-->
    <owner>
      <reference value="Practitioner/f202"/>
      <display value="Luigi Maas"/>
    </owner>
    <restriction>
      <repetitions value="1"/>
      <period>
        <end value="2016-11-01T09:45:05+10:00"/>
      </period>
    </restriction>
    <!--The expected output of this task is a Specimen and will generated once the task is completed -->
    <output>
      <type>
        <text value="collected specimen"/>
      </type>
      <valueReference>
        <reference value="Specimen/101"/>
      </valueReference>
    </output>
   </Task>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/Task/example2"/>
  <resource>
   <Task xmlns="http://hl7.org/fhir">
    <!--This is an example to demonstrate using task for actioning a filler generated procedurerequest and to
  illustrate how to populate many of the task elements - this is the child task from task example1 to grab the
  specimen
   example1 action lipid panel -in progress =>. example2 specimen collection subtask -accepted => example4  specimen
  collection subtask - completed  with output Specimen=>example 5 action lipid panel -in progress specimen collected
  with output Specimen => example 6 action lipid panel - completed with output Specimen and DiagnosticReport -->
    <id value="example2"/>
    <identifier>
      <use value="official"/>
      <system value="http:/goodhealth.org/identifiers"/>
      <value value="20170201-002"/>
    </identifier>
    <groupIdentifier>
      <use value="official"/>
      <system value="http:/goodhealth.org/accession/identifiers"/>
      <value value="G20170201-001"/>
    </groupIdentifier>
    <!--  This Task is a child task of the submitted lipid panel requet
   -->
    <partOf>
      <reference value="Task/example1"/>
      <display value="Lipid Panel"/>
    </partOf>
    <status value="accepted"/>
    <!--fictive business status populated for this example-->
    <businessStatus>
      <text value="waiting for patient"/>
    </businessStatus>
    <intent value="filler-order"/>
    <priority value="routine"/>
    <!--fictive code and description elements populated for this example -->
    <code>
      <text value="Specimen Collection"/>
    </code>
    <!--  Filler generated order using a fictive  procedurerequest to complete the loop-->
    <focus>
      <display value="BloodDraw ProcedureRequest"/>
    </focus>
    <!--  who the task is for
   -->
    <for>
      <reference value="Patient/example"/>
      <display value="Peter James Chalmers"/>
    </for>
    <context>
      <reference value="Encounter/example"/>
      <display value="Example In-Patient Encounter"/>
    </context>
    <!--  Lots of dates -->
    <executionPeriod>
      <start value="2016-10-31T08:45:05+10:00"/>
    </executionPeriod>
    <authoredOn value="2016-10-31T08:45:05+10:00"/>
    <lastModified value="2016-10-31T09:45:05+10:00"/>
    <!--requester is filler-->
    <requester>
      <agent>
        <reference value="Organization/1832473e-2fe0-452d-abe9-3cdb9879522f"/>
        <display value="Clinical Laboratory @ Acme Hospital"/>
      </agent>
```

```
    </requester>
    <performerType>
      <coding>
        <system value="http://hl7.org/fhir/task-performer-type"/>
        <code value="performer"/>
        <display value="Performer"/>
      </coding>
      <text value="Performer"/>
    </performerType>
    <!--assume at draw station-->
    <owner>
      <reference value="Organization/1832473e-2fe0-452d-abe9-3cdb9879522f"/>
      <display value="Clinical Laboratory @ Acme Hospital"/>
    </owner>
    <restriction>
      <repetitions value="1"/>
      <period>
        <end value="2016-11-01T09:45:05+10:00"/>
      </period>
    </restriction>
    <!--The expected output of this task is a Specimen and will generated once the task is completed -->
   </Task>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/NutritionOrder/diabeticdiet"/>
  <resource>
   <NutritionOrder xmlns="http://hl7.org/fhir">
   edited by Eric Haas Health eData Inc-->
   <!--EH  Show all the elements in this examples for preference, and intolerance --><!-- EH: use oral Diet element
  --><id value="diabeticdiet"/>
    <identifier>
      <system value="http://goodhealthhospital.org/nutrition-requests"/>
      <value value="123"/>
    </identifier>
    <status value="active"/>
    <patient>
      <reference value="Patient/example"/>
      <display value="Peter Chalmers"/>
    </patient>
    <encounter>
      <reference value="Encounter/example"/>
      <display value="Inpatient"/>
    </encounter>
    <dateTime value="2014-09-17"/>
    <orderer>
      <reference value="Practitioner/example"/>
      <display value="Dr Adam Careful"/>
    </orderer>
    <allergyIntolerance>
      <reference value="AllergyIntolerance/example"/>
      <display value="Cashew Nuts"/>
    </allergyIntolerance>
    <foodPreferenceModifier>
      <coding>
        <system value="http://hl7.org/fhir/diet"/>
        <code value="dairy-free"/>
      </coding>
    </foodPreferenceModifier>
    <excludeFoodModifier>
      <coding>
        <system value="http://snomed.info/sct"/>
        <version value="20140730"/>
        <code value="227493005"/>
        <display value="Cashew Nut"/>
      </coding>
    </excludeFoodModifier>
    <oralDiet>
      <!-- EH:  list type of diet showing standard code and local codes -->
      <type>
        <coding>
          <system value="http://snomed.info/sct"/>
          <code value="160670007"/>
          <display value="Diabetic diet"/>
        </coding>
        <coding>
          <system value="http://goodhealthhospital.org/diet-type-codes"/>
          <code value="1030"/>
          <display value="DD - Diabetic diet"/>
        </coding>
        <text value="DD - Diabetic diet"/>
      </type>
      <!-- EH example of three times a day starting on 2/10 using schedule -->
      <schedule>
        <repeat>
           <boundsPeriod>
             <start value="2015-02-10"/>
           </boundsPeriod>
          <frequency value="3"/>
```

```
          <period value="1"/>
          <periodUnit value="d"/>
        </repeat>
      </schedule>
      <!-- list nutrients and amounts here -->
      <nutrient>
        <modifier>
          <coding>
            <system value="http://snomed.info/sct"/>
            <code value="2331003"/>
            <display value="Carbohydrate"/>
          </coding>
        </modifier>
        <amount>
          <value value="75"/>
          <unit value="grams"/>
          <system value="http://unitsofmeasure.org"/>
          <code value="g"/>
        </amount>
      </nutrient>
    </oralDiet>
   </NutritionOrder>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/NutritionOrder/pureeddiet"/>
  <resource>
   <NutritionOrder xmlns="http://hl7.org/fhir">
   edited by Eric Haas Health eData Inc
   --><!--EH  Show all the elements in this examples for preference, and intolerance --><!-- EH: use oral Diet element
   --><id value="pureeddiet"/>
    <identifier>
      <system value="http://goodhealthhospital.org/nutrition-requests"/>
      <value value="123"/>
    </identifier>
    <status value="active"/>
    <patient>
      <reference value="Patient/example"/>
      <display value="Peter Chalmers"/>
    </patient>
    <encounter>
      <reference value="Encounter/example"/>
      <display value="Inpatient"/>
    </encounter>
    <dateTime value="2014-09-17"/>
    <orderer>
      <reference value="Practitioner/example"/>
      <display value="Dr Adam Careful"/>
    </orderer>
    <allergyIntolerance>
      <reference value="AllergyIntolerance/example"/>
      <display value="Cashew Nuts"/>
    </allergyIntolerance>
    <foodPreferenceModifier>
      <coding>
        <system value="http://hl7.org/fhir/diet"/>
        <code value="dairy-free"/>
      </coding>
    </foodPreferenceModifier>
    <excludeFoodModifier>
      <coding>
        <system value="http://snomed.info/sct"/>
        <version value="20140730"/>
        <code value="227493005"/>
        <display value="Cashew Nut"/>
      </coding>
    </excludeFoodModifier>
    <oralDiet>
      <!-- EH:  list type of diet showing standard code and local codes -->
      <type>
        <coding>
          <system value="http://snomed.info/sct"/>
          <code value="226211001"/>
          <display value="Pureed diet"/>
        </coding>
        <coding>
          <system value="http://goodhealthhospital.org/diet-type-codes"/>
          <code value="1010"/>
          <display value="Pureed diet"/>
        </coding>
        <text value="Pureed diet"/>
      </type>
      <!-- EH example of three times a day starting on 2/10  -->
      <schedule>
        <repeat>
           <boundsPeriod>
             <start value="2015-02-10"/>
           </boundsPeriod>
          <frequency value="3"/>
```

```
          <period value="1"/>
          <periodUnit value="d"/>
        </repeat>
      </schedule>
      <!--EH: this example showing how to code for texture modification -->
      <texture>
        <modifier>
          <coding>
            <system value="http://snomed.info/sct"/>
            <code value="228055009"/>
            <display value="Liquidized food"/>
          </coding>
          <text value="Pureed"/>
        </modifier>
      </texture>
      <fluidConsistencyType>
        <coding>
          <system value="http://snomed.info/sct"/>
          <code value="439021000124105"/>
          <display value="Dietary liquid consistency - nectar thick liquid"/>
        </coding>
        <text value="Nectar thick liquids"/>
      </fluidConsistencyType>
    </oralDiet>
   </NutritionOrder>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/Communication/example"/>
  <resource>
   <Communication xmlns="http://hl7.org/fhir">
     <id value="example"/>
     <text>
       <status value="generated"/>
       <div xmlns="http://www.w3.org/1999/xhtml">Patient has very high serum potassium</div>
     </text>
     <identifier>
       <type>
         <text value="Paging System"/>
       </type>
       <system value="urn:oid:1.3.4.5.6.7"/>
       <value value="2345678901"/>
     </identifier>
     <definition>
    <display value="Hyperkalemia"/>
     </definition>
     <partOf>
    <display value="Serum Potassium Observation"/>
     </partOf>
     <status value="completed"/>
     <category>
       <coding>
         <system value="http://acme.org/messagetypes"/>
         <code value="Alert"/>
       </coding>
       <text value="Alert"/>
     </category>
     <medium>
       <coding>
         <system value="http://hl7.org/fhir/v3/ParticipationMode"/>
         <code value="WRITTEN"/>
         <display value="written"/>
       </coding>
       <text value="written"/>
     </medium>
     <subject>
       <reference value="Patient/example"/>
     </subject>
     <recipient>
       <reference value="Practitioner/example"/>
     </recipient>
     <context>
       <reference value="Encounter/example"/>
     </context>
     <sent value="2014-12-12T18:01:10-08:00"/>
     <received value="2014-12-12T18:01:11-08:00"/>
     <sender>
       <reference value="Device/f001"/>
     </sender>
     <payload>
       <contentString value="Patient 1 has a very high serum potassium value (7.2 mmol/L on 2014-Dec-12 at 5:55 pm)"/>
     </payload>
     <payload>
       <!--Reference to the lab result -->
       <contentReference>
         <display value="Serum Potassium Observation"/>
       </contentReference>
     </payload>
   </Communication>
```

```
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/NutritionOrder/enteralcontinuous"/>
  <resource>
   <NutritionOrder xmlns="http://hl7.org/fhir">
   editor Eric Haas Health eData Inc
   --><!-- Tube feedings given as continuous rate --><id value="enteralcontinuous"/>
    <identifier>
      <system value="http://www.acme.org/nutritionorders"/>
      <value value="123"/>
    </identifier>
    <status value="active"/>
    <patient>
      <reference value="Patient/example"/>
      <display value="Peter Chalmers"/>
    </patient>
    <encounter>
      <reference value="Encounter/example"/>
      <display value="Inpatient"/>
    </encounter>
    <dateTime value="2014-09-17"/>
    <orderer>
      <reference value="Practitioner/example"/>
      <display value="Dr Adam Careful"/>
    </orderer>
    <enteralFormula>
      <baseFormulaType>
        <coding>
          <system value="http://snomed/sct"/>
          <code value="6547210000124112"/>
          <display value="Diabetic specialty enteral formula"/>
        </coding>
      </baseFormulaType>
      <baseFormulaProductName value=" Acme Diabetes Formula"/>

      <caloricDensity>
        <value value="1"/>
        <unit value="calories per milliliter"/>
        <system value="http://unitsofmeasure.org"/>
        <code value="cal/mL"/>
      </caloricDensity>
      <routeofAdministration>
        <coding>
          <system value="http://hl7.org/fhir/v3/RouteOfAdministration"/>
          <code value="NGT"/>
          <display value="Instillation, nasogastric tube"/>
        </coding>
      <!--step up rate of tube feeding by 20 ml/hr every 4 hours -->
      </routeofAdministration>
      <administration>

        <schedule>
        <repeat>
          <boundsPeriod>
            <start value="2014-09-17T07:00:00Z"/>
          </boundsPeriod>
        </repeat>
        </schedule>
        <rateQuantity>
          <value value="60"/>
          <unit value="ml/hr"/>
          <system value="http://unitsofmeasure.org"/>
          <code value="mL/h"/>
        </rateQuantity>
      </administration>
      <administration>
        <schedule>
        <repeat>
          <boundsPeriod>
            <start value="2014-09-17T11:00:00Z"/>
          </boundsPeriod>
        </repeat>
        </schedule>
        <rateQuantity>
          <value value="80"/>
          <unit value="ml/hr"/>
          <system value="http://unitsofmeasure.org"/>
          <code value="mL/h"/>
        </rateQuantity>
      </administration>
      <administration>
        <schedule>
        <repeat>
          <boundsPeriod>
            <start value="2014-09-17T15:00:00Z"/>
          </boundsPeriod>
        </repeat>
        </schedule>
```

```
        <rateQuantity>
          <value value="100"/>
          <unit value="ml/hr"/>
          <system value="http://unitsofmeasure.org"/>
          <code value="mL/h"/>
        </rateQuantity>
      </administration>

      <maxVolumeToDeliver>
        <value value="880"/>
        <unit value="milliliter/day"/>
        <system value="http://unitsofmeasure.org"/>
        <code value="mL/d"/>
      </maxVolumeToDeliver>
      <administrationInstruction value="Hold feedings from 7 pm to 7 am. Add MCT oil to increase calories from 1.0
  cal/mL to 1.5 cal/mL"/>

    </enteralFormula>
   </NutritionOrder>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/List/example"/>
  <resource>
   <List xmlns="http://hl7.org/fhir">
     <id value="example"/>
     <text>
       <status value="generated"/>
       <div xmlns="http://www.w3.org/1999/xhtml">
         <table>
           <thead>
             <tr>
               <th>Condition</th>
               <th>Severity</th>
               <th>Date</th>
               <th>Location</th>
               <th>Status</th>
             </tr>
           </thead>
           <tbody>
             <tr>
               <td>Burnt Ear</td>
               <td>Severe</td>
               <td>24-May 2012</td>
               <td>Left Ear</td>
               <td>deleted</td>
             </tr>
             <tr>
               <td>Asthma</td>
               <td>Mild</td>
               <td>21-Nov 2012</td>
               <td>--</td>
               <td>added</td>
             </tr>
           </tbody>
         </table>
       </div>
     </text>
     <identifier>
       <system value="urn:uuid:a9fcea7c-fcdf-4d17-a5e0-f26dda030b59"/>
       <value value="23974652"/>
     </identifier>
     <status value="current"/>
     <mode value="changes"/>
     <!-- This list doesn't have a code. In actual fact,
       it's a Condition list produced at the end of an encounter
       to a regular primary care practitioner. But the only way
       to know this is to hunt down the place it is used
       and find out -->
     <subject>
       <reference value="Patient/example"/>
     </subject>
     <encounter>
       <reference value="Encounter/example"/>
     </encounter>
     <date value="2012-11-25T22:17:00+11:00"/>
     <source>
       <reference value="Patient/example"/>
     </source>
     <entry>
       <flag>
         <text value="Deleted due to error"/>
       </flag>
       <deleted value="true"/>
       <item>
         <reference value="Condition/example"/>
       </item>
     </entry>
     <entry>
```

```
      <flag>
        <text value="Added"/>
      </flag>
      <date value="2012-11-21"/>
      <item>
        <reference value="Condition/example2"/>
      </item>
    </entry>
   </List>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/ClinicalImpression/example"/>
  <resource>
   <ClinicalImpression xmlns="http://hl7.org/fhir">
    <id value="example"/>
    <identifier>
         <value value="12345"/>
    </identifier>
    <status value="completed"/>
    <description value="This 26 yo male patient is brought into ER by ambulance after being involved in a motor
  vehicle accident"/>
    <subject>
      <reference value="Patient/example"/>
    </subject>
    <context>
      <reference value="Encounter/example"/>
    </context>
    <effectivePeriod>
        <start value="2014-12-06T20:00:00+11:00"/>
        <end value="2014-12-06T22:33:00+11:00"/>
    </effectivePeriod>
    <date value="2014-12-06T22:33:00+11:00"/>
    <assessor>
      <reference value="Practitioner/example"/>
    </assessor>
    <problem>
      <display value="MVA"/> <!-- todo: reference to condition with snomed code 418399005 : Motor vehicle accident
  (event) -->
    </problem>
    <investigation>
      <code>
        <text value="Initial Examination"/>
      </code>
      <!-- todo: turn all of these into observations -->
      <item>
        <display value="deep laceration of the scalp (left temporo-occipital)"/>
      </item>
      <item>
        <display value="decreased level of consciousness"/>
      </item>
      <item>
        <display value="disoriented to time and place"/>
      </item>
      <item>
        <display value="restless"/>
      </item>
    </investigation>
    <summary value="provisional diagnoses of laceration of head and traumatic brain injury (TBI)"/>
    <finding>
      <itemCodeableConcept>
        <coding>
          <system value="http://hl7.org/fhir/sid/icd-9"/>
          <code value="850.0"/>
        </coding>
      </itemCodeableConcept>
    </finding>
   </ClinicalImpression>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/NutritionOrder/proteinsupplement"/>
  <resource>
   <NutritionOrder xmlns="http://hl7.org/fhir">
   edited by Eric Haas Health eData Inc
   --><!-- EH: This example has no food exclusions or preferences --><!-- EH: use supplement element --><id
  value="proteinsupplement"/>
    <identifier>
      <system value="http://goodhealthhospital.org/nutrition-requests"/>
      <value value="123"/>
    </identifier>
    <status value="active"/>
    <patient>
      <reference value="Patient/example"/>
      <display value="Peter Chalmers"/>
    </patient>
    <encounter>
      <reference value="Encounter/example"/>
      <display value="Inpatient"/>
```

```
    </encounter>
    <dateTime value="2014-09-17"/>
    <orderer>
      <reference value="Practitioner/example"/>
      <display value="Dr Adam Careful"/>
    </orderer>
    <supplement>
      <type>
        <coding>
          <system value="http://snomed.info/sct"/>
          <code value="442991000124104"/>
          <display value="Adult high protein formula"/>
        </coding>
        <coding>
          <system value="http://goodhealthhospital.org/supplement-type-codes"/>
          <code value="1000"/>
          <display value="High Protein Powder"/>
        </coding>
        <text value="High Protein Powder"/>
      </type>
      <productName value="Beneprotein"/>
      <schedule>
        <repeat>
           <boundsPeriod>
             <start value="2015-02-10"/>
           </boundsPeriod>
          <when value="C"/>
        </repeat>
      </schedule>
      <quantity>
        <value value="1"/>
        <unit value="scoop"/>
        <system value="http://unitsofmeasure.org"/>
        <code value="{scoop}"/>
      </quantity>
      <!-- EH:  use the .instructions element to list diet  supplement order-->
      <instruction value="Beneprotein 1 scoop TID with meal.starting on  2015-02-10"/>
    </supplement>
   </NutritionOrder>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/ChargeItem/example"/>
  <resource>
   <?xml-model href="../../publish/chargeitem.sch" type="application/xml"
  schematypens="http://purl.oclc.org/dsdl/schematron"?><ChargeItem xmlns="http://hl7.org/fhir">
    <id value="example"/>
    <text>
      <status value="generated"/>
      <div xmlns="http://www.w3.org/1999/xhtml">Example of ChargeItem Usage in Context of the German EBM Billing code
  system</div>
    </text>
    <identifier>
      <system value="http://myHospital.org/ChargeItems"/>
      <value value="654321"/>
    </identifier>
    <definition value="http://www.kbv.de/tools/ebm/html/01520_2904360860826220813632.html"/>
    <status value="billable"/>

    <code>
      <coding>
        <code value="01510"/>
        <display value="Zusatzpauschale für Beobachtung nach diagnostischer Koronarangiografie"/>
      </coding>
    </code>
    <subject>
      <reference value="Patient/example"/>
    </subject>
    <context>
      <reference value="Encounter/example"/>
    </context>
    <occurrencePeriod>
      <start value="2017-01-25T08:00:00+01:00"/>
      <end value="2017-01-25T12:35:00+01:00"/>
    </occurrencePeriod>
    <participant>
      <role>
        <coding>
          <system value="http://snomed.info/sct"/>
          <code value="17561000"/>
          <display value="Cardiologist"/>
        </coding>

      </role>
      <actor>
        <reference value="Practitioner/example"/>
      </actor>
    </participant>
    <participant>
      <role>
```

```
        <coding>
          <system value="http://snomed.info/sct"/>
          <code value="224542009"/>
          <display value="Coronary Care Nurse"/>
        </coding>
      </role>
      <actor>
        <reference value="Practitioner/example"/>
      </actor>
    </participant>
    <performingOrganization>
      <identifier>
        <system value="http://myhospital/NamingSystem/departments"/>
        <value value="CARD_INTERMEDIATE_CARE"/>
      </identifier>
    </performingOrganization>
    <requestingOrganization>
      <identifier>
        <system value="http://myhospital/NamingSystem/departments"/>
        <value value="CARD_U1"/>
      </identifier>
    </requestingOrganization>
    <quantity>
      <value value="1"/>
    </quantity>
    <factorOverride value="0.8"/>
    <priceOverride>
        <value value="40"/>
           <unit value="EUR"/>
           <system value="urn:iso:std:iso:4217"/>
           <code value="EUR"/>
    </priceOverride>
    <overrideReason value="Patient is Cardiologist's golf buddy, so he gets a 20% discount!"/>
    <enterer>
      <reference value="Practitioner/example"/>
    </enterer>
    <enteredDate value="2017-01-25T23:55:04+01:00"/>
    <reason>
        <coding>
          <system value="http://hl7.org/fhir/sid/icd-10"/>
          <code value="123456"/>
          <display value="DIAG-1"/>
        </coding>
    </reason>
    <service>
      <reference value="Procedure/example"/>
    </service>
    <account>
      <reference value="Account/example"/>
    </account>
    <note>
      <authorReference>
        <reference value="Practitioner/example"/>
      </authorReference>
      <time value="2017-01-25T23:55:04+01:00"/>
      <text value="The code is only applicable for periods longer than 4h"/>
    </note>

   </ChargeItem>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/NutritionOrder/energysupplement"/>
  <resource>
   <NutritionOrder xmlns="http://hl7.org/fhir">
   edited by Eric Haas Health eData Inc
   --><!--EH  Show all the elements in this examples for preference, and intolerance --><!-- EH: use supplement
  element --><id value="energysupplement"/>
    <identifier>
      <system value="http://goodhealthhospital.org/nutrition-requests"/>
      <value value="123"/>
    </identifier>
    <status value="active"/>
    <patient>
      <reference value="Patient/example"/>
      <display value="Peter Chalmers"/>
    </patient>
    <encounter>
      <reference value="Encounter/example"/>
      <display value="Inpatient"/>
    </encounter>
    <dateTime value="2014-09-17"/>
    <orderer>
      <reference value="Practitioner/example"/>
      <display value="Dr Adam Careful"/>
    </orderer>
    <allergyIntolerance>
      <reference value="AllergyIntolerance/example"/>
      <display value="Cashew Nuts"/>
```

```
    </allergyIntolerance>
    <foodPreferenceModifier>
      <coding>
        <system value="http://hl7.org/fhir/diet"/>
        <code value="kosher"/>
      </coding>
    </foodPreferenceModifier>
    <excludeFoodModifier>
      <coding>
        <system value="http://snomed.info/sct"/>
        <version value="20140730"/>
        <code value="227493005"/>
        <display value="Cashew Nut"/>
      </coding>
    </excludeFoodModifier>
    <supplement>
      <type>
        <coding>
          <system value="http://snomed.info/sct"/>
          <code value="442971000124100"/>
          <display value="Adult high energy formula"/>
        </coding>
        <coding>
          <system value="http://goodhealthhospital.org/supplement-type-codes"/>
          <code value="1010"/>
          <display value="Adult high energy drink"/>
        </coding>
        <text value="Adult high energy drink"/>
      </type>
      <productName value="Ensure"/>
      <!--EH:  1 8 oz can tid using coded data -->
      <schedule>
        <repeat>
           <boundsPeriod>
             <start value="2015-02-10"/>
           </boundsPeriod>
          <frequency value="3"/>
          <period value="1"/>
          <periodUnit value="d"/>
        </repeat>
      </schedule>
      <quantity>
        <value value="1"/>
        <unit value="8 oz can"/>
      </quantity>
    </supplement>
   </NutritionOrder>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/Claim/960150"/>
  <resource>
   <Claim xmlns="http://hl7.org/fhir">
    <id value="960150"/>

    <text>
      <status value="generated"/>
      <div xmlns="http://www.w3.org/1999/xhtml">A human-readable rendering of the Claim</div>
    </text>

    <identifier>
      <system value="http://happyhospital.com/claim"/>
      <value value="9612345"/>
    </identifier>

    <status value="active"/>

    <type>
      <coding>
        <system value="http://hl7.org/fhir/ex-claimtype"/>
        <code value="institutional"/>
      </coding>
    </type>

    <subType>
      <coding>
        <system value="http://hl7.org/fhir/ex-claimsubtype"/>
        <code value="emergency"/>
      </coding>
    </subType>

    <use value="complete"/>

    <patient>
      <reference value="Patient/1"/>
    </patient>

    <created value="2014-08-16"/>
```

```
<enterer>
  <identifier>
    <system value="http://jurisdiction.org/facilities/HOSP1234/users"/>
    <value value="UC1234"/>
  </identifier>
</enterer>

<insurer>
  <reference value="Organization/2"/>
</insurer>

<organization>
  <reference value="Organization/1"/>
</organization>

<priority>
  <coding>
    <code value="normal"/>
  </coding>
</priority>

<payee>
  <type>
    <coding>
      <code value="provider"/>
    </coding>
  </type>
</payee>

<facility>
  <identifier>
    <system value="http://jurisdiction.org/facilities"/>
    <value value="HOSP1234"/>
  </identifier>
</facility>

<careTeam>
  <sequence value="1"/>
  <provider>
    <reference value="Practitioner/example"/>
  </provider>
</careTeam>

<diagnosis>
  <sequence value="1"/>
  <diagnosisCodeableConcept>
    <coding>
      <code value="654456"/>
    </coding>
  </diagnosisCodeableConcept>
</diagnosis>

<insurance>
  <sequence value="1"/>
  <focal value="true"/>
  <coverage>
    <reference value="Coverage/9876B1"/>
  </coverage>
</insurance>

<item>
  <sequence value="1"/>
  <careTeamLinkId value="1"/>
  <service>
    <coding>
      <system value="http://hl7.org/fhir/ex-serviceproduct"/>
      <code value="exam"/>
    </coding>
  </service>
  <servicedDate value="2014-08-16"/>
  <unitPrice>
    <value value="125.00"/>
    <system value="urn:iso:std:iso:4217"/>
    <code value="USD"/>
  </unitPrice>
  <net>
    <value value="125.00"/>
    <system value="urn:iso:std:iso:4217"/>
    <code value="USD"/>
  </net>
  <encounter>
    <reference value="Encounter/example"/>
  </encounter>
</item>

<total>
  <value value="125.00"/>
  <system value="urn:iso:std:iso:4217"/>
  <code value="USD"/>
```

```
    </total>
   </Claim>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/NutritionOrder/diabeticsupplement"/>
  <resource>
   <NutritionOrder xmlns="http://hl7.org/fhir">
   edited by Eric Haas Health eData Inc

   --><!-- EH: use supplement element --><id value="diabeticsupplement"/>
    <identifier>
      <system value="http://goodhealthhospital.org/nutrition-requests"/>
      <value value="123"/>
    </identifier>
    <status value="active"/>
    <patient>
      <reference value="Patient/example"/>
      <display value="Peter Chalmers"/>
    </patient>
    <encounter>
      <reference value="Encounter/example"/>
      <display value="Inpatient"/>
    </encounter>
    <dateTime value="2014-09-17"/>
    <orderer>
      <reference value="Practitioner/example"/>
      <display value="Dr Adam Careful"/>
    </orderer>
    <allergyIntolerance>
      <reference value="AllergyIntolerance/example"/>
      <display value="Cashew Nuts"/>
    </allergyIntolerance>
    <foodPreferenceModifier>
      <coding>
        <system value="http://hl7.org/fhir/diet"/>
        <code value="kosher"/>
      </coding>
    </foodPreferenceModifier>
    <excludeFoodModifier>
      <coding>
        <system value="http://snomed.info/sct"/>
        <version value="20140730"/>
        <code value="227493005"/>
        <display value="Cashew Nut"/>
      </coding>
    </excludeFoodModifier>
    <supplement>
      <type>
        <coding>
          <system value="http://snomed.info/sct"/>
          <code value="443051000124104"/>
          <display value="Adult diabetes specialty formula"/>
        </coding>
        <coding>
          <system value="http://goodhealthhospital.org/supplement-type-codes"/>
          <code value="1010"/>
          <display value="Adult diabetic formula"/>
        </coding>
        <text value="Adult diabetic formula"/>
      </type>
      <productName value="Glucerna"/>
      <!--EH:Glucerna Shake at 3pm and HS-->
      <schedule>
        <repeat>
          <boundsPeriod>
            <start value="2015-02-10T15:00:00Z"/>
          </boundsPeriod>
          <frequency value="1"/>
          <period value="24"/>
          <periodUnit value="h"/>
        </repeat>
      </schedule>
      <schedule>
        <repeat>
          <duration value="1"/>
          <durationUnit value="h"/>
          <when value="HS"/>
        </repeat>
      </schedule>
      <quantity>
        <value value="1"/>
        <unit value="8 oz bottle"/>
      </quantity>
      <!-- EH:  use the .instructions element to list diet  supplement order
      <instruction value="Glucerna Shake at 3pm and HS"/>
   -->
    </supplement></NutritionOrder>
  </resource>
```

```
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/Immunization/example"/>
  <resource>
   <!--Authored by Joginder Madra--><Immunization xmlns="http://hl7.org/fhir">
    <id value="example"/>
    <identifier>
      <system value="urn:ietf:rfc:3986"/>
      <value value="urn:oid:1.3.6.1.4.1.21367.2005.3.7.1234"/>
    </identifier>
    <status value="completed"/>
    <notGiven value="false"/>
    <vaccineCode>
      <coding>
        <system value="urn:oid:1.2.36.1.2001.1005.17"/>
        <code value="FLUVAX"/>
      </coding>
      <text value="Fluvax (Influenza)"/>
    </vaccineCode>
    <patient>
      <reference value="Patient/example"/>
    </patient>
    <encounter>
      <reference value="Encounter/example"/>
    </encounter>
    <date value="2013-01-10"/>
    <primarySource value="true"/>
    <location>
      <reference value="Location/1"/>
    </location>
    <manufacturer>
      <reference value="Organization/hl7"/>
    </manufacturer>
    <lotNumber value="AAJN11K"/>
    <expirationDate value="2015-02-15"/>
    <site>
      <coding>
        <system value="http://hl7.org/fhir/v3/ActSite"/>
        <code value="LA"/>
        <display value="left arm"/>
      </coding>
    </site>
    <route>
      <coding>
        <system value="http://hl7.org/fhir/v3/RouteOfAdministration"/>
        <code value="IM"/>
        <display value="Injection, intramuscular"/>
      </coding>
    </route>
    <doseQuantity>
      <value value="5"/>
      <system value="http://unitsofmeasure.org"/>
      <code value="mg"/>
    </doseQuantity>
    <practitioner>
      <role>
        <coding>
          <system value="http://hl7.org/fhir/v2/0443"/>
          <code value="OP"/>
        </coding>
      </role>
          <actor>
              <reference value="Practitioner/example"/>
          </actor>
    </practitioner>
    <practitioner>
      <role>
        <coding>
          <system value="http://hl7.org/fhir/v2/0443"/>
          <code value="AP"/>
        </coding>
      </role>
          <actor>
              <reference value="Practitioner/example"/>
          </actor>
    </practitioner>
    <note>
      <text value="Notes on adminstration of vaccine"/>
    </note>
    <explanation>
      <reason>
        <coding>
          <system value="http://snomed.info/sct"/>
          <code value="429060002"/>
        </coding>
      </reason>
    </explanation>
    <reaction>
      <date value="2013-01-10"/>
```

```
      <detail>
        <reference value="Observation/example"/>
      </detail>
      <reported value="true"/>
    </reaction>
    <vaccinationProtocol>
      <doseSequence value="1"/>
      <description value="Vaccination Protocol Sequence 1"/>
      <authority>
        <reference value="Organization/hl7"/>
      </authority>
      <series value="Vaccination Series 1"/>
      <seriesDoses value="2"/>
      <targetDisease>
        <coding>
          <system value="http://snomed.info/sct"/>
          <code value="1857005"/>
        </coding>
      </targetDisease>
      <doseStatus>
        <coding>
          <system value="http://hl7.org/fhir/vaccination-protocol-dose-status"/>
          <code value="count"/>
          <display value="Counts"/>
        </coding>
      </doseStatus>
      <doseStatusReason>
        <coding>
          <system value="http://hl7.org/fhir/vaccination-protocol-dose-status-reason"/>
          <code value="coldchbrk"/>
          <display value="Cold chain break"/>
        </coding>
      </doseStatusReason>
    </vaccinationProtocol>
   </Immunization>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/Task/example5"/>
  <resource>
   <Task xmlns="http://hl7.org/fhir">
    <!--This is an example to demonstrate using task for actioning a procedurerequest and to illustrate how to
  populate many of the task elements this is the first in a series of related Task examples:  example1 action lipid
  panel -in progress =>. example2 specimen collection subtask -accepted => example4  specimen collection subtask -
  completed  with output Specimen=>example 5 action lipid panel -in progress specimen collected with output Specimen
  => example 6 action lipid panel - completed with output Specimen and DiagnosticReport -->
    <id value="example5"/>
    <identifier>
      <use value="official"/>
      <system value="http:/goodhealth.org/identifiers"/>
      <value value="20170201-001"/>
    </identifier>
    <basedOn>
      <display value="General Wellness Careplan"/>
    </basedOn>
    <groupIdentifier>
      <use value="official"/>
      <system value="http:/goodhealth.org/accession/identifiers"/>
      <value value="G20170201-001"/>
    </groupIdentifier>
    <status value="in-progress"/>
    <!--fictive business status populated for this example -->
    <businessStatus>
      <text value="specimen received, test in progress"/>
    </businessStatus>
    <intent value="order"/>
    <priority value="routine"/>
    <!--fictive code and description elements populated for this example -->
    <code>
      <text value="Lipid Panel"/>
    </code>
    <description value="Create order for getting specimen, Set up inhouse testing,  generate order for any sendouts
  and submit with specimen"/>
    <!--  Use the lipid procedurerequest example -->
    <focus>
      <reference value="ProcedureRequest/lipid"/>
      <display value="Lipid Panel Request"/>
    </focus>
    <!--  who the task is for
   -->
    <for>
      <reference value="Patient/example"/>
      <display value="Peter James Chalmers"/>
    </for>
    <context>
      <reference value="Encounter/example"/>
      <display value="Example In-Patient Encounter"/>
    </context>
    <!--  Lots of dates -->
```

```
    <executionPeriod>
      <start value="2016-10-31T08:25:05+10:00"/>
    </executionPeriod>
    <authoredOn value="2016-10-31T08:25:05+10:00"/>
    <!--  updated last modified date to track task activity-->
    <lastModified value="2016-10-31T16:45:05+10:00"/>
    <requester>
      <agent>
        <reference value="Practitioner/example"/>
        <display value="Dr Adam Careful"/>
      </agent>
      <onBehalfOf>
        <reference value="Organization/2.16.840.1.113883.19.5"/>
        <display value="Good Health Clinic"/>
      </onBehalfOf>
    </requester>
    <performerType>
      <coding>
        <system value="http://hl7.org/fhir/task-performer-type"/>
        <code value="performer"/>
        <display value="Performer"/>
      </coding>
      <text value="Performer"/>
    </performerType>
    <owner>
      <reference value="Organization/1832473e-2fe0-452d-abe9-3cdb9879522f"/>
      <display value="Clinical Laboratory @ Acme Hospital"/>
    </owner>
    <!--  populate the reason just to demonstrate it use in this example
   -->
    <reason>
      <text value="The Task.reason should only be included if there is no Task.focus or if it differs from the reason
  indicated on the focus"/>
    </reason>
    <note>
      <text value="This is an example to demonstrate using task for actioning a procedurerequest and to illustrate how
   to populate many of the task elements - this is the parent task that will be broken into subtask to grab the
  specimen and a sendout lab test "/>
    </note>
    <!--  add the link to the signature on the task to show how this element is usedemonstrating in this example the
  relevant history element using a contained resource for the purpose of this example only typically would reference
  an external resource

   -->
    <!--  complete the request within 48 hrs
   -->
    <restriction>
      <repetitions value="1"/>
      <period>
        <end value="2016-11-02T09:45:05+10:00"/>
      </period>
    </restriction>
    <!--The expected output of this task is a DiagnosticRequest and will generated once the task is near completion
  -->
      <output>
      <type>
        <text value="collected specimen"/>
      </type>
      <valueReference>
        <reference value="Specimen/101"/>
      </valueReference>
    </output>

   </Task>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/Task/example1"/>
  <resource>
   <Task xmlns="http://hl7.org/fhir">
    <!--This is an example to demonstrate using task for actioning a procedurerequest and to illustrate how to
  populate many of the task elements - this is the parent task that will be broken into subtask to grab the specimen
   example1 action lipid panel -in progress =>. example2 specimen collection subtask -accepted => example4  specimen
  collection subtask - completed  with output Specimen=>example 5 action lipid panel -in progress specimen collected
  => example 6 action lipid panel - completed with output DiagnosticReport -->
    <id value="example1"/>
    <contained>
      <!-- using contained resource for the purpose of this example only typically this would be a separate
  independent resource -->
      <Provenance>
        <id value="signature"/>
        <target>
          <reference value="ProcedureRequest/physiotherapy/_history/1"/>
        </target>
        <recorded value="2016-10-31T08:25:05+10:00"/>
        <agent>
          <role>
            <coding>
              <system value="http://hl7.org/fhir/v3/ParticipationType"/>
```

```
            <code value="AUT"/>
          </coding>
        </role>
        <whoReference>
          <reference value="Practitioner/f202"/>
          <display value="Luigi Maas"/>
        </whoReference>
      </agent>
      <signature>
        <type>
          <system value="urn:iso-astm:E1762-95:2013"/>
          <code value="1.2.840.10065.1.12.1.1"/>
          <display value="Author's Signature"/>
        </type>
        <when value="2016-10-31T08:25:05+10:00"/>
        <whoReference>
          <reference value="Practitioner/example"/>
          <display value="Dr Adam Careful"/>
        </whoReference>
        <contentType value="application/signature+xml"/>
        <blob value="dGhpcyBibG9iIGlzIHNuaXBwZWQ="/>
      </signature>
    </Provenance>
  </contained>
  <identifier>
    <use value="official"/>
    <system value="http:/goodhealth.org/identifiers"/>
    <value value="20170201-001"/>
  </identifier>
  <basedOn>
    <display value="General Wellness Careplan"/>
  </basedOn>
  <groupIdentifier>
    <use value="official"/>
    <system value="http:/goodhealth.org/accession/identifiers"/>
    <value value="G20170201-001"/>
  </groupIdentifier>
  <status value="in-progress"/>
  <!--fictive business status populated for this example -->
  <businessStatus>
    <text value="waiting for specimen"/>
  </businessStatus>
  <intent value="order"/>
  <priority value="routine"/>
  <!--fictive code and description elements populated for this example -->
  <code>
    <text value="Lipid Panel"/>
  </code>
  <description value="Create order for getting specimen, Set up inhouse testing,  generate order for any sendouts
and submit with specimen"/>
  <!--  Use the lipid procedurerequest example -->
  <focus>
    <reference value="ProcedureRequest/lipid"/>
    <display value="Lipid Panel Request"/>
  </focus>
  <!--  who the task is for
 -->
  <for>
    <reference value="Patient/example"/>
    <display value="Peter James Chalmers"/>
  </for>
  <context>
    <reference value="Encounter/example"/>
    <display value="Example In-Patient Encounter"/>
  </context>
  <!--  Lots of dates -->
  <executionPeriod>
    <start value="2016-10-31T08:25:05+10:00"/>
  </executionPeriod>
  <authoredOn value="2016-10-31T08:25:05+10:00"/>
  <lastModified value="2016-10-31T09:45:05+10:00"/>
  <requester>
    <agent>
      <reference value="Practitioner/example"/>
      <display value="Dr Adam Careful"/>
    </agent>
    <onBehalfOf>
      <reference value="Organization/2.16.840.1.113883.19.5"/>
      <display value="Good Health Clinic"/>
    </onBehalfOf>
  </requester>
  <performerType>
    <coding>
      <system value="http://hl7.org/fhir/task-performer-type"/>
      <code value="performer"/>
      <display value="Performer"/>
    </coding>
    <text value="Performer"/>
  </performerType>
```

```
    <owner>
      <reference value="Organization/1832473e-2fe0-452d-abe9-3cdb9879522f"/>
      <display value="Clinical Laboratory @ Acme Hospital"/>
    </owner>
    <!--  populate the reason just to demonstrate it use in this example
   -->
    <reason>
      <text value="The Task.reason should only be included if there is no Task.focus or if it differs from the reason
  indicated on the focus"/>
    </reason>
    <note>
      <text value="This is an example to demonstrate using task for actioning a procedurerequest and to illustrate how
   to populate many of the task elements - this is the parent task that will be broken into subtask to grab the
  specimen and a sendout lab test "/>
    </note>
    <!--  add the link to the signature on the task to show how this element is usedemonstrating in this example the
  relevant history element using a contained resource for the purpose of this example only typically would reference
  an external resource

   -->
    <relevantHistory>
      <reference value="#signature"/>
      <display value="Author's Signature"/>
    </relevantHistory>
    <!--  complete the request within 48 hrs
   -->
    <restriction>
      <repetitions value="1"/>
      <period>
        <end value="2016-11-02T09:45:05+10:00"/>
      </period>
    </restriction>
    <!--The expected output of this task is a DiagnosticRequest and will generated once the task is near completion
  -->
   </Task>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/Observation/example"/>
  <resource>
   <!--
    This is an example of a simple weight measurement.
    --><Observation xmlns="http://hl7.org/fhir">
    <id value="example"/>
    <!--  the mandatory quality flags:  -->
    <status value="final"/>
    <!-- category code is A code that classifies the general type of observation being made. This is used for
  searching, sorting and display purposes.-->
    <category>
      <coding>
        <system value="http://hl7.org/fhir/observation-category"/>
        <code value="vital-signs"/>
        <display value="Vital Signs"/>
      </coding>
    </category>
    <!--
       Observations are often coded in multiple code systems.
         - LOINC provides codes of varying granularity (though not usefully more specific in this particular case) and
   more generic LOINCs  can be mapped to more specific codes as shown here
         - snomed provides a clinically relevant code that is usually less granular than LOINC
         - the source system provides its own code, which may be less or more granular than LOINC
      -->
    <code>
      <!--  LOINC - always recommended to have a LOINC code  -->
      <coding>
        <system value="http://loinc.org"/>
        <code value="29463-7"/> <!--more generic methodless LOINC-->
        <display value="Body Weight"/>
      </coding>
      <coding>
        <system value="http://loinc.org"/>
        <code value="3141-9"/><!--translation is more specific method = measured LOINC-->
        <display value="Body weight Measured"/>
      </coding>
      <!--  SNOMED CT Codes - becoming more common  -->
      <coding>
        <system value="http://snomed.info/sct"/>
        <code value="27113001"/>
        <display value="Body weight"/>
      </coding>
      <!--  Also, a local code specific to the source system  -->
      <coding>
        <system value="http://acme.org/devices/clinical-codes"/>
        <code value="body-weight"/>
        <display value="Body Weight"/>
      </coding>
    </code>
    <subject>
      <reference value="Patient/example"/>
```

```
    </subject>
    <context>
      <reference value="Encounter/example"/>
    </context>
    <effectiveDateTime value="2016-03-28"/>
    <!--  In FHIR, units may be represented twice. Once in the
       agreed human representation, and once in a coded form.
       Both is best, since it's not always possible to infer
       one from the other in code.

       When a computable unit is provided, UCUM (http://unitsofmeasure.org)
       is always preferred, but it doesn't provide notional units (such as
       "tablet"), etc. For these, something else is required (e.g. SNOMED CT)
       -->
    <valueQuantity>
      <value value="185"/>
      <unit value="lbs"/>
      <system value="http://unitsofmeasure.org"/>
      <code value="[lb_av]"/>
    </valueQuantity>
   </Observation>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/ExplanationOfBenefit/EB3500"/>
  <resource>
   <ExplanationOfBenefit xmlns="http://hl7.org/fhir">
    <id value="EB3500"/>

    <text>
      <status value="generated"/>
      <div xmlns="http://www.w3.org/1999/xhtml">A human-readable rendering of the ExplanationOfBenefit</div>
    </text>

    <identifier>
      <system value="http://www.BenefitsInc.com/fhir/explanationofbenefit"/>
      <value value="987654321"/>
    </identifier>

    <status value="active"/>

    <type>
      <coding>
        <system value="http://hl7.org/fhir/ex-claimtype"/>
        <code value="oral"/>
      </coding>
    </type>

    <patient>
      <reference value="Patient/pat1"/>
    </patient>

    <created value="2014-08-16"/>

    <enterer>
      <reference value="Practitioner/1"/>
    </enterer>

    <provider>
      <reference value="Practitioner/1"/>
    </provider>

    <organization>
      <reference value="Organization/2"/>
    </organization>

    <facility>
      <reference value="Location/1"/>
    </facility>

    <claim>
      <reference value="Claim/100150"/>
    </claim>

    <claimResponse>
      <reference value="ClaimResponse/R3500"/>
    </claimResponse>

    <outcome>
      <coding>
        <system value="http://hl7.org/fhir/remittance-outcome"/>
        <code value="complete"/>
      </coding>
    </outcome>

    <disposition value="Claim settled as per contract."/>

    <payee>
      <type>
```

```
    <coding>
      <system value="http://hl7.org/fhir/payeetype"/>
      <code value="provider"/>
    </coding>
  </type>
  <resourceType>
    <coding>
      <system value="http://hl7.org/fhir/resource-type-link"/>
      <code value="organization"/>
    </coding>
  </resourceType>
  <party>
    <reference value="Organization/2"/>
  </party>
</payee>

<careTeam>
  <sequence value="1"/>
  <provider>
    <reference value="Practitioner/example"/>
  </provider>
</careTeam>

<insurance>
  <coverage>
    <reference value="Coverage/9876B1"/>
  </coverage>
</insurance>

<item>
  <sequence value="1"/>
  <careTeamLinkId value="1"/>
  <service>
    <coding>
      <system value="http://hl7.org/fhir/service-uscls"/>
      <code value="1200"/>
    </coding>
  </service>
  <servicedDate value="2014-08-16"/>
  <unitPrice>
    <value value="135.57"/>
    <system value="urn:iso:std:iso:4217"/>
    <code value="USD"/>
  </unitPrice>
  <net>
    <value value="135.57"/>
    <system value="urn:iso:std:iso:4217"/>
    <code value="USD"/>
  </net>
  <encounter>
    <reference value="Encounter/example"/>
  </encounter>
  <adjudication>
    <category>
      <coding>
        <code value="eligible"/>
      </coding>
    </category>
    <amount>
      <value value="120.00"/>
      <system value="urn:iso:std:iso:4217"/>
      <code value="USD"/>
    </amount>
  </adjudication>
  <adjudication>
    <category>
      <coding>
        <code value="eligpercent"/>
      </coding>
    </category>
    <value value="0.80"/>
  </adjudication>
  <adjudication>
    <category>
      <coding>
        <code value="benefit"/>
      </coding>
    </category>
    <amount>
      <value value="96.00"/>
      <system value="urn:iso:std:iso:4217"/>
      <code value="USD"/>
    </amount>
  </adjudication>
</item>

<totalCost>
  <value value="135.57"/>
  <system value="urn:iso:std:iso:4217"/>
```

```
      <code value="USD"/>
    </totalCost>

    <totalBenefit>
      <value value="96.00"/>
      <system value="urn:iso:std:iso:4217"/>
      <code value="USD"/>
    </totalBenefit>

   </ExplanationOfBenefit>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/NutritionOrder/fiberrestricteddiet"/>
  <resource>
   <NutritionOrder xmlns="http://hl7.org/fhir">
   edited by Eric Haas Health eData Inc
   --><!--EH  Show all the elements in this examples for preference, and intolerance --><!-- EH: use oral Diet element
   --><id value="fiberrestricteddiet"/>
    <identifier>
      <system value="http://goodhealthhospital.org/nutrition-requests"/>
      <value value="123"/>
    </identifier>
    <status value="active"/>
    <patient>
      <reference value="Patient/example"/>
      <display value="Peter Chalmers"/>
    </patient>
    <encounter>
      <reference value="Encounter/example"/>
      <display value="Inpatient"/>
    </encounter>
    <dateTime value="2014-09-17"/>
    <orderer>
      <reference value="Practitioner/example"/>
      <display value="Dr Adam Careful"/>
    </orderer>
    <allergyIntolerance>
      <reference value="AllergyIntolerance/example"/>
      <display value="Cashew Nuts"/>
    </allergyIntolerance>
    <foodPreferenceModifier>
      <coding>
        <system value="http://hl7.org/fhir/diet"/>
        <code value="dairy-free"/>
      </coding>
    </foodPreferenceModifier>
    <excludeFoodModifier>
      <coding>
        <system value="http://snomed.info/sct"/>
        <version value="20140730"/>
        <code value="227493005"/>
        <display value="Cashew Nut"/>
      </coding>
    </excludeFoodModifier>
    <oralDiet>
      <!-- EH:  list type of diet showing standard code and local codes -->
      <type>
        <coding>
          <system value="http://snomed.info/sct"/>
          <code value="15108003"/>
          <display value="Restricted fiber diet"/>
        </coding>
        <coding>
          <system value="http://goodhealthhospital.org/diet-type-codes"/>
          <code value="1000"/>
          <display value="Fiber restricted"/>
        </coding>
        <text value="Fiber restricted diet"/>
      </type>
      <type>
        <coding>
          <system value="http://snomed.info/sct"/>
          <code value="16208003"/>
          <display value="Low fat diet"/>
        </coding>
        <coding>
          <system value="http://goodhealthhospital.org/diet-type-codes"/>
          <code value="1100"/>
          <display value="Low Fat"/>
        </coding>
        <text value="Low fat diet"/>
      </type>
      <!-- EH example of three times a day starting on 2/10  -->
      <schedule>
        <repeat>
           <boundsPeriod>
             <start value="2015-02-10"/>
           </boundsPeriod>
```

```
          <frequency value="3"/>
          <period value="1"/>
          <periodUnit value="d"/>
        </repeat>
      </schedule>
      <nutrient>
        <!-- list nutrients and amounts here -->
        <modifier>
          <coding>
            <system value="http://snomed.info/sct"/>
            <code value="256674009"/>
            <display value="Fat"/>
          </coding>
        </modifier>
        <amount>
          <value value="50"/>
          <unit value="grams"/>
          <system value="http://unitsofmeasure.org"/>
          <code value="g"/>
        </amount>
      </nutrient>
    </oralDiet>
   </NutritionOrder>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/Task/example6"/>
  <resource>
   <Task xmlns="http://hl7.org/fhir">
    <!--This is an example to demonstrate using task for actioning a procedurerequest and to illustrate how to
  populate many of the task elements this is the first in a series of related Task examples:  example1 action lipid
  panel -in progress =>. example2 specimen collection subtask -accepted => example4  specimen collection subtask -
  completed  with output Specimen=>example 5 action lipid panel -in progress specimen collected with output Specimen
  => example 6 action lipid panel - completed with output Specimen and DiagnosticReport -->
    <id value="example6"/>
    <identifier>
      <use value="official"/>
      <system value="http:/goodhealth.org/identifiers"/>
      <value value="20170201-001"/>
    </identifier>
    <basedOn>
      <display value="General Wellness Careplan"/>
    </basedOn>
    <groupIdentifier>
      <use value="official"/>
      <system value="http:/goodhealth.org/accession/identifiers"/>
      <value value="G20170201-001"/>
    </groupIdentifier>
    <status value="completed"/>
    <!--fictive business status populated for this example -->
    <businessStatus>
      <text value="test completed and posted"/>
    </businessStatus>
    <intent value="order"/>
    <priority value="routine"/>
    <!--fictive code and description elements populated for this example -->
    <code>
      <text value="Lipid Panel"/>
    </code>
    <description value="Create order for getting specimen, Set up inhouse testing,  generate order for any sendouts
  and submit with specimen"/>
    <!--  Use the lipid procedurerequest example -->
    <focus>
      <reference value="ProcedureRequest/lipid"/>
      <display value="Lipid Panel Request"/>
    </focus>
    <!--  who the task is for
   -->
    <for>
      <reference value="Patient/example"/>
      <display value="Peter James Chalmers"/>
    </for>
    <context>
      <reference value="Encounter/example"/>
      <display value="Example In-Patient Encounter"/>
    </context>
    <!--  Lots of dates -->
    <executionPeriod>
      <start value="2016-10-31T08:25:05+10:00"/>
      <!--added completion date-->
      <end value="2016-10-31T18:45:05+10:00"/>
    </executionPeriod>
    <authoredOn value="2016-10-31T08:25:05+10:00"/>
    <!--  updated last modified date to track task activity-->
    <lastModified value="2016-10-31T18:45:05+10:00"/>
    <requester>
      <agent>
        <reference value="Practitioner/example"/>
        <display value="Dr Adam Careful"/>
```

```
      </agent>
      <onBehalfOf>
        <reference value="Organization/2.16.840.1.113883.19.5"/>
        <display value="Good Health Clinic"/>
      </onBehalfOf>
    </requester>
    <performerType>
      <coding>
        <system value="http://hl7.org/fhir/task-performer-type"/>
        <code value="performer"/>
        <display value="Performer"/>
      </coding>
      <text value="Performer"/>
    </performerType>
    <owner>
      <reference value="Organization/1832473e-2fe0-452d-abe9-3cdb9879522f"/>
      <display value="Clinical Laboratory @ Acme Hospital"/>
    </owner>
    <!--  populate the reason just to demonstrate it use in this example
   -->
    <reason>
      <text value="The Task.reason should only be included if there is no Task.focus or if it differs from the reason
  indicated on the focus"/>
    </reason>
    <note>
      <text value="This is an example to demonstrate using task for actioning a procedurerequest and to illustrate how
   to populate many of the task elements - this is the parent task that will be broken into subtask to grab the
  specimen and a sendout lab test "/>
    </note>
    <!--  add the link to the signature on the task to show how this element is usedemonstrating in this example the
  relevant history element using a contained resource for the purpose of this example only typically would reference
  an external resource

   -->
    <!--  complete the request within 48 hrs
   -->
    <restriction>
      <repetitions value="1"/>
      <period>
        <end value="2016-11-02T09:45:05+10:00"/>
      </period>
    </restriction>
    <!--The expected output of this task is a DiagnosticReport and Specimen-->
    <output>
      <type>
        <text value="DiagnosticReport generated"/>
      </type>
      <valueReference>
        <reference value="DiagnosticReport/lipids"/>
      </valueReference>
    </output>
      <output>
      <type>
        <text value="collected specimen"/>
      </type>
      <valueReference>
        <reference value="Specimen/101"/>
      </valueReference>
    </output>

   </Task>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/RiskAssessment/cardiac"/>
  <resource>
   <RiskAssessment xmlns="http://hl7.org/fhir">
    <id value="cardiac"/>
     <text>
       <status value="additional"/>
       <div xmlns="http://www.w3.org/1999/xhtml">
         <table>
           <tr>
             <td colspan="2">
               <h1>Information about your risk score:</h1>
             </td>
           </tr>
           <tr>
             <td>
               <strong>Age:</strong>
             </td>
             <td>39</td>
           </tr>
           <tr>
             <td>
               <strong>Gender:</strong>
             </td>
             <td>male</td>
           </tr>
```

```
          <tr>
            <td>
              <strong>Total Cholesterol:</strong>
            </td>
            <td>244 mg/dL</td>
          </tr>
          <tr>
            <td>
              <strong>HDL Cholesterol:</strong>
            </td>
            <td>50 mg/dL
            </td>
          </tr>
          <tr>
            <td>
              <strong>Smoker:</strong>
            </td>
            <td>No</td>
          </tr>
          <tr>
            <td>
              <strong>Systolic Blood Pressure:</strong>
            </td>
            <td>107 mm/Hg</td>
          </tr>
          <tr>
            <td>
              <strong>On medication for HBP:</strong>
            </td>
            <td>No</td>
          </tr>
          <tr>
            <td valign="top">
              <b>Risk Score*</b>
            </td>
            <td>2%<br/>Means 2 of 100 people with this level of risk will have a heart attack in the next 10
years.</td>
          </tr>
          <tr>
            <td colspan="2">* Your risk score was calculated using an equation.  Other NCEP products, such as printed
 ATP III materials, use a point system to determine a risk score that is close to the equation score.</td>
          </tr>
          <tr>
            <td colspan="2">To find out what your risk score means and how to lower your risk for a heart attack, go
to<br/>
              <a href="http://www.nhlbi.nih.gov/health/public/heart/chol/hbc_what.htm">"High Blood Cholesterol—What
You Need to Know"</a>.<br/>
              <a href="http://www.nhlbi.nih.gov/health/public/heart/chol/chol_tlc.htm">Your Guide to Lowering
Cholesterol with Therapeutic Lifestyle Changes (TLC)</a><br/>
              <a href="http://www.nhlbi.nih.gov/health/public/heart/other/hhw/index.htm">The Healthy Heart Handbook
for Women</a>
            </td>
          </tr>
        </table>
      </div>
    </text>
    <identifier>
      <use value="official"/>
      <system value="http://example.org"/>
      <value value="risk-assessment-cardiac"/>
    </identifier>
    <status value="final"/>
    <subject>
      <reference value="Patient/pat2"/>
    </subject>
    <context>
      <reference value="Encounter/example"/>
    </context>
    <occurrenceDateTime value="2014-07-19T16:04:00Z"/>
    <performer>
      <display value="http://cvdrisk.nhlbi.nih.gov/#cholesterol"/>
    </performer>
    <basis>
      <reference value="Patient/pat2"/>
    </basis>
    <basis>
      <reference value="DiagnosticReport/lipids"/>
    </basis>
    <basis>
      <reference value="Observation/blood-pressure"/>
    </basis>
    <prediction>
      <outcome>
        <text value="Heart Attack"/>
      </outcome>
      <probabilityDecimal value="0.02"/>
      <whenRange>
        <low>
          <value value="39"/>
```

```
          <unit value="years"/>
          <system value="http://unitsofmeasure.org"/>
          <code value="a"/>
        </low>
        <high>
          <value value="49"/>
          <unit value="years"/>
          <system value="http://unitsofmeasure.org"/>
          <code value="a"/>
        </high>
      </whenRange>
    </prediction>
   </RiskAssessment>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/DiagnosticReport/101"/>
  <resource>
   <DiagnosticReport xmlns="http://hl7.org/fhir">
    <id value="101"/>
    <meta>
       <tag>
         <system value="http://example.org/fhir/CodeSystem/workflow-codes"/>
         <code value="01"/>
         <display value="Needs Review"/>
       </tag>
    </meta>
    <text>
      <status value="generated"/>
      <div xmlns="http://www.w3.org/1999/xhtml">
        <h3>CBC Report for Wile. E. COYOTE (MRN: 23453) issued 3-Mar 2011 11:45</h3>
        <pre>
   Test                  Units       Value       Reference Range
   Haemoglobin           g/L         176         135 - 180
   Red Cell Count        x10*12/L    5.9         4.2 - 6.0
   Haematocrit                       0.55+       0.38 - 0.52
   Mean Cell Volume      fL          99+         80 - 98
   Mean Cell Haemoglobin pg          36+         27 - 35
   Platelet Count        x10*9/L     444         150 - 450
   White Cell Count      x10*9/L     4.6         4.0 - 11.0
   Neutrophils           %           20
   Neutrophils           x10*9/L     0.9---      2.0 - 7.5
   Lymphocytes           %           20
   Lymphocytes           x10*9/L     0.9-        1.1 - 4.0
   Monocytes             %           20
   Monocytes             x10*9/L     0.9         0.2 - 1.0
   Eosinophils           %           20
   Eosinophils           x10*9/L     0.92++      0.04 - 0.40
   Basophils             %           20
   Basophils             x10*9/L     0.92+++     &lt;0.21
        </pre>
        <p>Acme Laboratory, Inc signed: Dr Pete Pathologist</p>
      </div>
    </text>
    <contained>
      <Observation>
        <!--
     all the data items (= Observations) are contained
     in this diagnostic report. It would be equally
       valid - and normal - for them to be separate trackable
       items. However for the purposes of this example, it's
     more convenient to have them here. For more discussion,
     see under "Contained Resources" on the Resource Definitions
     topic page    -->
        <!--    for users steeped in v2, each observation roughly corresponds with an
       OBX, and the Diagnostic Report with an ORU_R01 message    -->
        <id value="r1"/>
        <status value="final"/>
        <code>
          <coding>
            <system value="http://loinc.org"/>
            <code value="718-7"/>
            <display value="Hemoglobin [Mass/volume] in Blood"/>
          </coding>
          <text value="Haemoglobin"/>
        </code>
        <subject>
          <reference value="Patient/pat2"/>
        </subject>
        <performer>
          <reference value="Organization/1832473e-2fe0-452d-abe9-3cdb9879522f"/>
          <display value="Acme Laboratory, Inc"/>
        </performer>
        <valueQuantity>
          <value value="176"/>
          <unit value="g/L"/>
          <system value="http://unitsofmeasure.org"/>
          <code value="g/L"/>
        </valueQuantity>
```

```
    <referenceRange>
      <low>
        <value value="135"/>
        <unit value="g/L"/>
        <system value="http://unitsofmeasure.org"/>
        <code value="g/L"/>
      </low>
      <high>
        <value value="180"/>
        <unit value="g/L"/>
        <system value="http://unitsofmeasure.org"/>
        <code value="g/L"/>
      </high>
    </referenceRange>
  </Observation>
</contained>
<contained>
  <Observation>
    <id value="r2"/>
    <status value="final"/>
    <code>
      <coding>
        <system value="http://loinc.org"/>
        <code value="789-8"/>
        <display value="Erythrocytes [#/volume] in Blood by Automated count"/>
      </coding>
      <text value="Red Cell Count"/>
    </code>
    <subject>
      <reference value="Patient/pat2"/>
    </subject>
    <performer>
      <reference value="Organization/1832473e-2fe0-452d-abe9-3cdb9879522f"/>
      <display value="Acme Laboratory, Inc"/>
    </performer>
    <valueQuantity>
      <value value="5.9"/>
      <unit value="x10*12/L"/>
      <system value="http://unitsofmeasure.org"/>
      <code value="10*12/L"/>
    </valueQuantity>
    <referenceRange>
      <low>
        <value value="4.2"/>
        <unit value="x10*12/L"/>
        <system value="http://unitsofmeasure.org"/>
        <code value="10*12/L"/>
      </low>
      <high>
        <value value="6.0"/>
        <unit value="x10*12/L"/>
        <system value="http://unitsofmeasure.org"/>
        <code value="10*12/L"/>
      </high>
    </referenceRange>
  </Observation>
</contained>
<contained>
  <Observation>
    <id value="r3"/>
    <status value="final"/>
    <code>
      <coding>
        <system value="http://loinc.org"/>
        <code value="4544-3"/>
        <display value="Hematocrit [Volume Fraction] of Blood by Automated count"/>
      </coding>
      <text value="Haematocrit"/>
    </code>
    <subject>
      <reference value="Patient/pat2"/>
    </subject>
    <performer>
      <reference value="Organization/1832473e-2fe0-452d-abe9-3cdb9879522f"/>
      <display value="Acme Laboratory, Inc"/>
    </performer>
    <valueQuantity>
      <value value="55"/>
      <unit value="%"/>
    </valueQuantity>
    <interpretation>
      <coding>
        <system value="http://hl7.org/fhir/v2/0078"/>
        <code value="H"/>
      </coding>
    </interpretation>
    <referenceRange>
      <low>
        <value value="38"/>
```

```
        <unit value="%"/>
      </low>
      <high>
        <value value="52"/>
        <unit value="%"/>
      </high>
    </referenceRange>
  </Observation>
</contained>
<contained>
  <Observation>
    <id value="r4"/>
    <status value="final"/>
    <code>
      <coding>
        <system value="http://loinc.org"/>
        <code value="787-2"/>
        <display value="Erythrocyte mean corpuscular volume [Entitic volume] by Automated count"/>
      </coding>
      <text value="Mean Cell Volume"/>
    </code>
    <subject>
      <reference value="Patient/pat2"/>
    </subject>
    <performer>
      <reference value="Organization/1832473e-2fe0-452d-abe9-3cdb9879522f"/>
      <display value="Acme Laboratory, Inc"/>
    </performer>
    <valueQuantity>
      <value value="99"/>
      <unit value="fL"/>
      <system value="http://unitsofmeasure.org"/>
      <code value="fL"/>
    </valueQuantity>
    <interpretation>
      <coding>
        <system value="http://hl7.org/fhir/v2/0078"/>
        <code value="H"/>
      </coding>
    </interpretation>
    <referenceRange>
      <low>
        <value value="80"/>
        <unit value="fL"/>
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        <code value="fL"/>
      </low>
      <high>
        <value value="98"/>
        <unit value="fL"/>
        <system value="http://unitsofmeasure.org"/>
        <code value="fL"/>
      </high>
    </referenceRange>
  </Observation>
</contained>
<contained>
  <Observation>
    <id value="r5"/>
    <status value="final"/>
    <code>
      <coding>
        <system value="http://loinc.org"/>
        <code value="785-6"/>
        <display value="Erythrocyte mean corpuscular hemoglobin [Entitic mass] by Automated count"/>
      </coding>
      <text value="Mean Cell Haemoglobin"/>
    </code>
    <subject>
      <reference value="Patient/pat2"/>
    </subject>
    <performer>
      <reference value="Organization/1832473e-2fe0-452d-abe9-3cdb9879522f"/>
      <display value="Acme Laboratory, Inc"/>
    </performer>
    <valueQuantity>
      <value value="36"/>
      <unit value="pg"/>
      <system value="http://unitsofmeasure.org"/>
      <code value="pg"/>
    </valueQuantity>
    <interpretation>
      <coding>
        <system value="http://hl7.org/fhir/v2/0078"/>
        <code value="H"/>
      </coding>
    </interpretation>
    <referenceRange>
      <low>
```

```
        <value value="27"/>
        <unit value="pg"/>
        <system value="http://unitsofmeasure.org"/>
        <code value="pg"/>
      </low>
      <high>
        <value value="35"/>
        <unit value="pg"/>
        <system value="http://unitsofmeasure.org"/>
        <code value="pg"/>
      </high>
    </referenceRange>
  </Observation>
</contained>
<contained>
  <Observation>
    <id value="r6"/>
    <status value="final"/>
    <code>
      <coding>
        <system value="http://loinc.org"/>
        <code value="777-3"/>
        <display value="Platelets [#/volume] in Blood by Automated count"/>
      </coding>
      <text value="Platelet Count"/>
    </code>
    <subject>
      <reference value="Patient/pat2"/>
    </subject>
    <performer>
      <reference value="Organization/1832473e-2fe0-452d-abe9-3cdb9879522f"/>
      <display value="Acme Laboratory, Inc"/>
    </performer>
    <valueQuantity>
      <value value="444"/>
      <unit value="x10*9/L"/>
      <system value="http://unitsofmeasure.org"/>
      <code value="10*9/L"/>
    </valueQuantity>
    <referenceRange>
      <low>
        <value value="150"/>
        <unit value="x10*9/L"/>
        <system value="http://unitsofmeasure.org"/>
        <code value="10*9/L"/>
      </low>
      <high>
        <value value="450"/>
        <unit value="x10*9/L"/>
        <system value="http://unitsofmeasure.org"/>
        <code value="10*9/L"/>
      </high>
    </referenceRange>
  </Observation>
</contained>
<contained>
  <Observation>
    <id value="r7"/>
    <status value="final"/>
    <code>
      <coding>
        <system value="http://loinc.org"/>
        <code value="6690-2"/>
        <display value="Leukocytes [#/volume] in Blood by Automated count"/>
      </coding>
      <text value="White Cell Count"/>
    </code>
    <subject>
      <reference value="Patient/pat2"/>
    </subject>
    <performer>
      <reference value="Organization/1832473e-2fe0-452d-abe9-3cdb9879522f"/>
      <display value="Acme Laboratory, Inc"/>
    </performer>
    <valueQuantity>
      <value value="4.6"/>
      <unit value="x10*9/L"/>
      <system value="http://unitsofmeasure.org"/>
      <code value="10*9/L"/>
    </valueQuantity>
    <referenceRange>
      <low>
        <value value="4.0"/>
        <unit value="x10*9/L"/>
        <system value="http://unitsofmeasure.org"/>
        <code value="10*9/L"/>
      </low>
      <high>
        <value value="11.0"/>
```

```
        <unit value="x10*9/L"/>
        <system value="http://unitsofmeasure.org"/>
        <code value="10*9/L"/>
      </high>
    </referenceRange>
  </Observation>
</contained>
<contained>
  <Observation>
    <id value="r8"/>
    <status value="final"/>
    <code>
      <coding>
        <system value="http://loinc.org"/>
        <code value="770-8"/>
        <display value="Neutrophils/100 leukocytes in Blood by Automated count"/>
      </coding>
      <text value="Neutrophils"/>
    </code>
    <subject>
      <reference value="Patient/pat2"/>
    </subject>
    <performer>
      <reference value="Organization/1832473e-2fe0-452d-abe9-3cdb9879522f"/>
      <display value="Acme Laboratory, Inc"/>
    </performer>
    <valueQuantity>
      <value value="20"/>
      <unit value="%"/>
      <system value="http://unitsofmeasure.org"/>
      <code value="%"/>
    </valueQuantity>
  </Observation>
</contained>
<contained>
  <Observation>
    <id value="r9"/>
    <status value="final"/>
    <code>
      <coding>
        <system value="http://loinc.org"/>
        <code value="751-8"/>
        <display value="Neutrophils [#/volume] in Blood by Automated count"/>
      </coding>
      <text value="Neutrophils"/>
    </code>
    <subject>
      <reference value="Patient/pat2"/>
    </subject>
    <performer>
      <reference value="Organization/1832473e-2fe0-452d-abe9-3cdb9879522f"/>
      <display value="Acme Laboratory, Inc"/>
    </performer>
    <valueQuantity>
      <value value="0.9"/>
      <unit value="x10*9/L"/>
      <system value="http://unitsofmeasure.org"/>
      <code value="10*9/L"/>
    </valueQuantity>
    <interpretation>
      <coding>
        <system value="http://hl7.org/fhir/v2/0078"/>
        <code value="LL"/>
      </coding>
    </interpretation>
    <referenceRange>
      <low>
        <value value="2.0"/>
        <unit value="x10*9/L"/>
        <system value="http://unitsofmeasure.org"/>
        <code value="10*9/L"/>
      </low>
      <high>
        <value value="7.5"/>
        <unit value="x10*9/L"/>
        <system value="http://unitsofmeasure.org"/>
        <code value="10*9/L"/>
      </high>
    </referenceRange>
  </Observation>
</contained>
<contained>
  <Observation>
    <id value="r10"/>
    <status value="final"/>
    <code>
      <coding>
        <system value="http://loinc.org"/>
        <code value="736-9"/>
```

```
        <display value="Lymphocytes/100 leukocytes in Blood by Automated count"/>
      </coding>
      <text value="Lymphocytes"/>
    </code>
    <subject>
      <reference value="Patient/pat2"/>
    </subject>
    <performer>
      <reference value="Organization/1832473e-2fe0-452d-abe9-3cdb9879522f"/>
      <display value="Acme Laboratory, Inc"/>
    </performer>
    <valueQuantity>
      <value value="20"/>
      <unit value="%"/>
      <system value="http://unitsofmeasure.org"/>
      <code value="%"/>
    </valueQuantity>
  </Observation>
</contained>
<contained>
  <Observation>
    <id value="r11"/>
    <status value="final"/>
    <code>
      <coding>
        <system value="http://loinc.org"/>
        <code value="731-0"/>
        <display value="Lymphocytes [#/volume] in Blood by Automated count"/>
      </coding>
      <text value="Lymphocytes"/>
    </code>
    <subject>
      <reference value="Patient/pat2"/>
    </subject>
    <performer>
      <reference value="Organization/1832473e-2fe0-452d-abe9-3cdb9879522f"/>
      <display value="Acme Laboratory, Inc"/>
    </performer>
    <valueQuantity>
      <value value="0.9"/>
      <unit value="x10*9/L"/>
      <system value="http://unitsofmeasure.org"/>
      <code value="10*9/L"/>
    </valueQuantity>
    <interpretation>
      <coding>
        <system value="http://hl7.org/fhir/v2/0078"/>
        <code value="L"/>
      </coding>
    </interpretation>
    <referenceRange>
      <low>
        <value value="1.1"/>
        <unit value="x10*9/L"/>
        <system value="http://unitsofmeasure.org"/>
        <code value="10*9/L"/>
      </low>
      <high>
        <value value="4.0"/>
        <unit value="x10*9/L"/>
        <system value="http://unitsofmeasure.org"/>
        <code value="10*9/L"/>
      </high>
    </referenceRange>
  </Observation>
</contained>
<contained>
  <Observation>
    <id value="r12"/>
    <status value="final"/>
    <code>
      <coding>
        <system value="http://loinc.org"/>
        <code value="5905-5"/>
        <display value="Monocytes/100 leukocytes in Blood by Automated count"/>
      </coding>
      <text value="Monocytes"/>
    </code>
    <subject>
      <reference value="Patient/pat2"/>
    </subject>
    <performer>
      <reference value="Organization/1832473e-2fe0-452d-abe9-3cdb9879522f"/>
      <display value="Acme Laboratory, Inc"/>
    </performer>
    <valueQuantity>
      <value value="20"/>
      <unit value="%"/>
      <system value="http://unitsofmeasure.org"/>
```

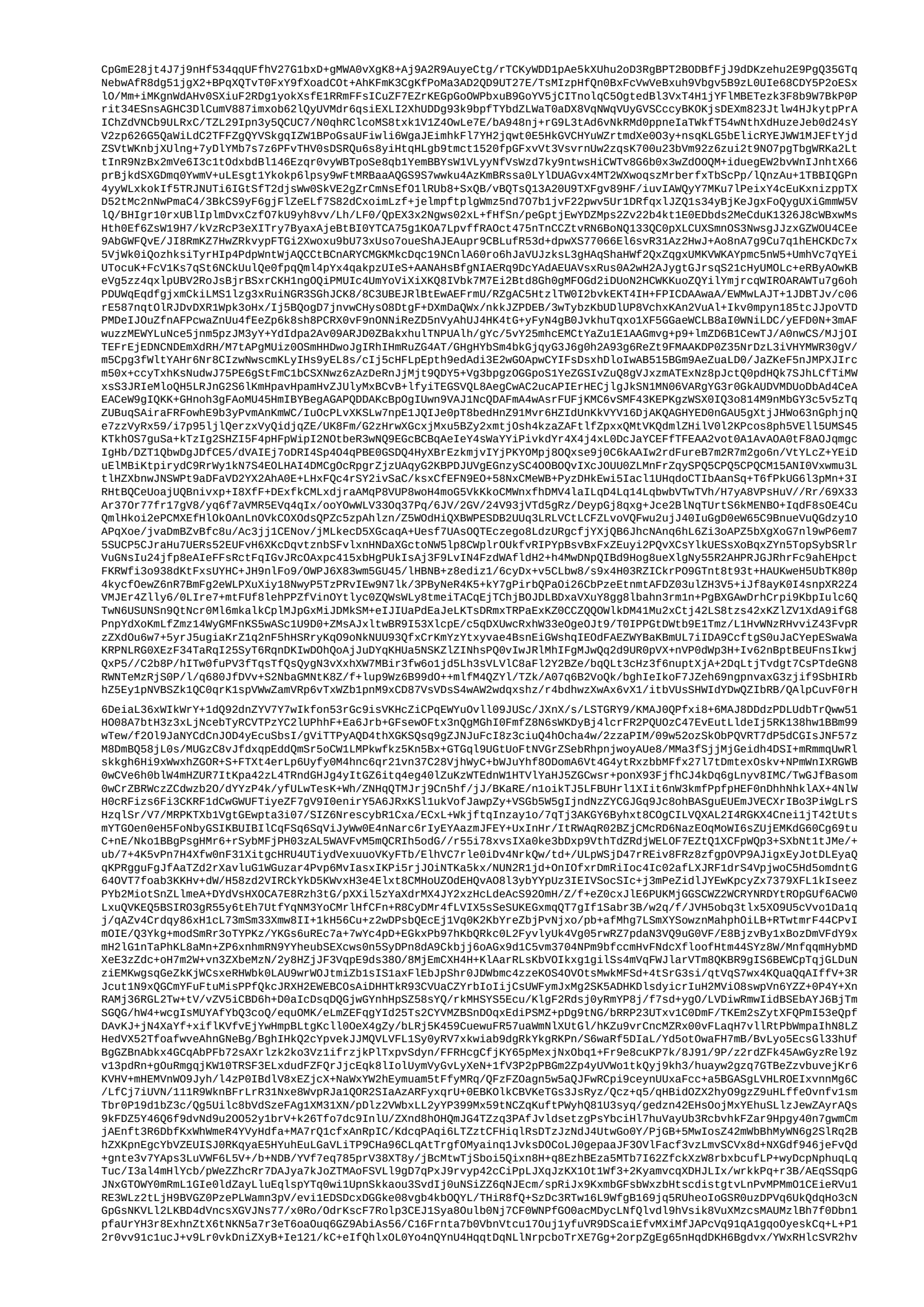
```
      <code value="%"/>
    </valueQuantity>
  </Observation>
</contained>
<contained>
  <Observation>
    <id value="r13"/>
    <status value="final"/>
    <code>
      <coding>
        <system value="http://loinc.org"/>
        <code value="742-7"/>
        <display value="Monocytes [#/volume] in Blood by Automated count"/>
      </coding>
      <text value="Monocytes"/>
    </code>
    <subject>
      <reference value="Patient/pat2"/>
    </subject>
    <performer>
      <reference value="Organization/1832473e-2fe0-452d-abe9-3cdb9879522f"/>
      <display value="Acme Laboratory, Inc"/>
    </performer>
    <valueQuantity>
      <value value="0.9"/>
      <unit value="x10*9/L"/>
      <system value="http://unitsofmeasure.org"/>
      <code value="10*9/L"/>
    </valueQuantity>
    <referenceRange>
      <low>
        <value value="0.2"/>
        <unit value="x10*9/L"/>
        <system value="http://unitsofmeasure.org"/>
        <code value="10*9/L"/>
      </low>
      <high>
        <value value="1.0"/>
        <unit value="x10*9/L"/>
        <system value="http://unitsofmeasure.org"/>
        <code value="10*9/L"/>
      </high>
    </referenceRange>
  </Observation>
</contained>
<contained>
  <Observation>
    <id value="r14"/>
    <status value="final"/>
    <code>
      <coding>
        <system value="http://loinc.org"/>
        <code value="713-8"/>
        <display value="Eosinophils/100 leukocytes in Blood by Automated count"/>
      </coding>
      <text value="Eosinophils"/>
    </code>
    <subject>
      <reference value="Patient/pat2"/>
    </subject>
    <performer>
      <reference value="Organization/1832473e-2fe0-452d-abe9-3cdb9879522f"/>
      <display value="Acme Laboratory, Inc"/>
    </performer>
    <valueQuantity>
      <value value="20"/>
      <unit value="%"/>
      <system value="http://unitsofmeasure.org"/>
      <code value="%"/>
    </valueQuantity>
  </Observation>
</contained>
<contained>
  <Observation>
    <id value="r15"/>
    <status value="final"/>
    <code>
      <coding>
        <system value="http://loinc.org"/>
        <code value="711-2"/>
        <display value="Eosinophils [#/volume] in Blood by Automated count"/>
      </coding>
      <text value="Eosinophils"/>
    </code>
    <subject>
      <reference value="Patient/pat2"/>
    </subject>
    <performer>
      <reference value="Organization/1832473e-2fe0-452d-abe9-3cdb9879522f"/>
```

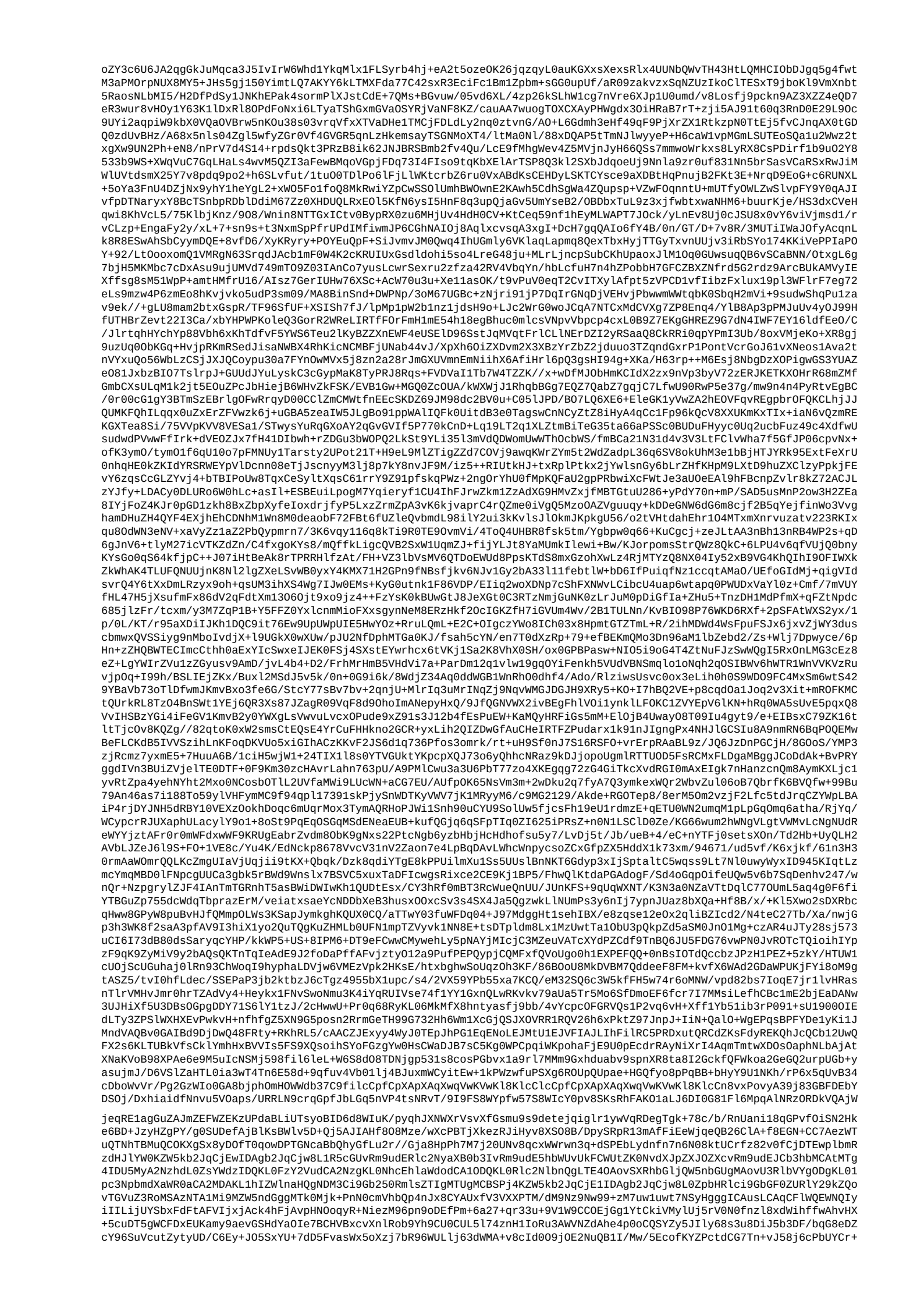
```
      <display value="Acme Laboratory, Inc"/>
    </performer>
    <valueQuantity>
      <value value="0.92"/>
      <unit value="x10*9/L"/>
      <system value="http://unitsofmeasure.org"/>
      <code value="10*9/L"/>
    </valueQuantity>
    <interpretation>
      <coding>
        <system value="http://hl7.org/fhir/v2/0078"/>
        <code value="HH"/>
      </coding>
    </interpretation>
    <referenceRange>
      <low>
        <value value="0.04"/>
        <unit value="x10*9/L"/>
        <system value="http://unitsofmeasure.org"/>
        <code value="10*9/L"/>
      </low>
      <high>
        <value value="0.40"/>
        <unit value="x10*9/L"/>
        <system value="http://unitsofmeasure.org"/>
        <code value="10*9/L"/>
      </high>
    </referenceRange>
  </Observation>
</contained>
<contained>
  <Observation>
    <id value="r16"/>
    <status value="final"/>
    <code>
      <coding>
        <system value="http://loinc.org"/>
        <code value="706-2"/>
        <display value="Basophils/100 leukocytes in Blood by Automated count"/>
      </coding>
      <text value="Basophils"/>
    </code>
    <subject>
      <reference value="Patient/pat2"/>
    </subject>
    <performer>
      <reference value="Organization/1832473e-2fe0-452d-abe9-3cdb9879522f"/>
      <display value="Acme Laboratory, Inc"/>
    </performer>
    <valueQuantity>
      <value value="20"/>
      <unit value="%"/>
      <system value="http://unitsofmeasure.org"/>
      <code value="%"/>
    </valueQuantity>
  </Observation>
</contained>
<contained>
  <Observation>
    <id value="r17"/>
    <status value="final"/>
    <code>
      <coding>
        <system value="http://loinc.org"/>
        <code value="704-7"/>
        <display value="Basophils [#/volume] in Blood by Automated count"/>
      </coding>
      <text value="Basophils"/>
    </code>
    <subject>
      <reference value="Patient/pat2"/>
    </subject>
    <performer>
      <reference value="Organization/1832473e-2fe0-452d-abe9-3cdb9879522f"/>
      <display value="Acme Laboratory, Inc"/>
    </performer>
    <valueQuantity>
      <value value="0.92"/>
      <unit value="x10*9/L"/>
      <system value="http://unitsofmeasure.org"/>
      <code value="10*9/L"/>
    </valueQuantity>
    <referenceRange>
      <high>
        <value value="0.21"/>
        <unit value="x10*9/L"/>
        <system value="http://unitsofmeasure.org"/>
        <code value="10*9/L"/>
      </high>
```

```
    </referenceRange>
  </Observation>
</contained>
<identifier>
  <system value="http://acme.com/lab/reports"/>
  <value value="5234342"/>
</identifier>
<status value="final"/>
<category>
  <coding>
    <system value="http://hl7.org/fhir/v2/0074"/>
    <code value="HM"/>
  </coding>
</category>
<!--    first, various administrative/context stuff    -->
<code>
  <coding>
    <system value="http://loinc.org"/>
    <code value="58410-2"/>
    <display value="Complete blood count (hemogram) panel - Blood by Automated count"/>
  </coding>
  <coding>
    <code value="CBC"/>
    <display value="MASTER FULL BLOOD COUNT"/>
  </coding>
  <text value="Complete Blood Count"/>
</code>
<subject>
  <reference value="Patient/pat2"/>
</subject>
<context>
  <reference value="Encounter/example"/>
</context>
<effectiveDateTime value="2011-03-04T08:30:00+11:00"/>
<!--    all this report is final    -->
<issued value="2011-03-04T11:45:33+11:00"/>
<performer>
  <actor>
    <reference value="Organization/1832473e-2fe0-452d-abe9-3cdb9879522f"/>
    <display value="Acme Laboratory, Inc"/>
  </actor>
</performer>
<!--    now the atomic results    -->
<result>
  <reference value="#r1"/>
</result>
<result>
  <reference value="#r2"/>
</result>
<result>
  <reference value="#r3"/>
</result>
<result>
  <reference value="#r4"/>
</result>
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  <reference value="#r5"/>
</result>
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</result>
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  <reference value="#r7"/>
</result>
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</result>
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  <reference value="#r13"/>
</result>
<result>
  <reference value="#r14"/>
</result>
<result>
  <reference value="#r15"/>
</result>
<result>
```

```
    <reference value="#r16"/>
  </result>
  <result>
    <reference value="#r17"/>
  </result>
  <!--    finally, here's a pdf representation of the same report.
     A consuming application could choose to display either the
  html version above, or the pdf version - they both need to convey the
  same information    -->
  <presentedForm>
    <contentType value="application/pdf"/>
    <language value="en-AU"/>
    <data value="JVBERi0xLjQKJcfsj6IKNSAwIG9iago8PC9MZW5ndGggNiAwIFIvRmlsdGVyIC9GbGF0ZURlY29kZT4+CnN0cmVhbQp4nO1aWW8
URxAW2MviXcs32AYfY2OzM4Zp990zr5GiSFFeQCvlIeSJBPIQI8H/f0j3HF01UPbaZn3hYCHVVldVV1V/XX1Mf044EzLh4a8l3p8MPg8U54l1wjLrkpO
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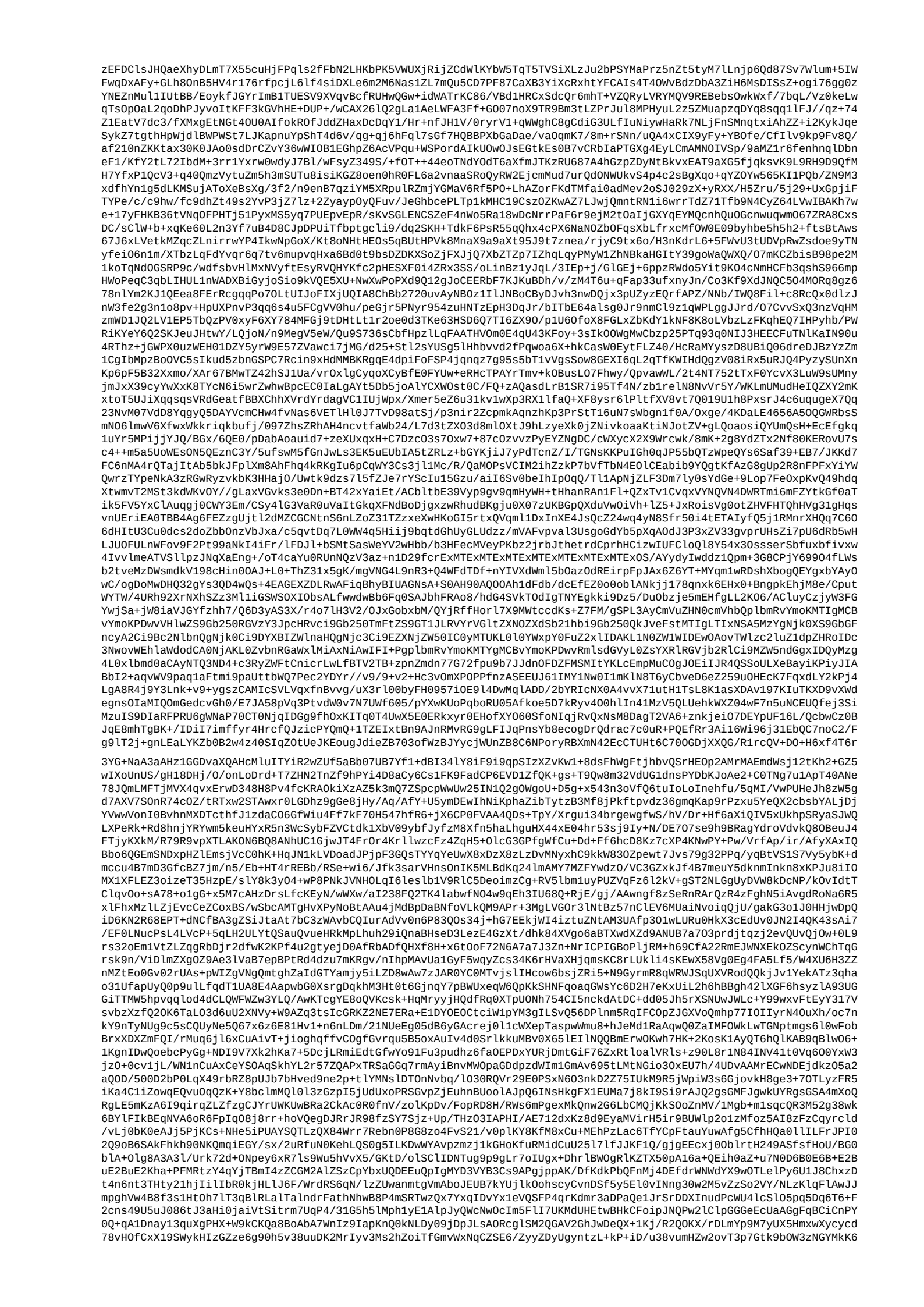




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   <ProcedureRequest xmlns="http://hl7.org/fhir" xmlns:xhtml="http://www.w3.org/1999/xhtml">
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    <contained>
      <Observation>
        <id value="fasting"/>
        <!--    the mandatory quality flag:    -->
        <status value="final"/>
        <code>
          <coding>
            <system value="http://loinc.org"/>
            <code value="49541-6"/>
            <display value="Fasting status - Reported"/>
          </coding>
        </code>
        <subject>
          <reference value="Patient/example"/>
        </subject>
        <valueCodeableConcept>
          <coding>
            <system value="http://hl7.org/fhir/v2/0136"/>
            <code value="Y"/>
            <display value="Yes"/>
          </coding>
        </valueCodeableConcept>
      </Observation>
    </contained>
    <contained>
      <Specimen>
        <id value="serum"/>
        <identifier>
          <system value="http://acme.org/specimens"/>
          <value value="20150107-0012"/>
        </identifier>
        <type>
          <coding>
            <system value="http://snomed.info/sct"/>
            <code value="119364003"/>
            <display value="Serum sample"/>
          </coding>
        </type>
        <subject>
          <reference value="Patient/example"/>
        </subject>
        <collection>
          <collectedDateTime value="2015-08-16T06:40:17Z"/>
        </collection>
      </Specimen>
    </contained>
    <identifier>
      <type>
        <coding>
          <system value="http://hl7.org/fhir/identifier-type"/>
          <code value="PLAC"/>
        </coding>
        <text value="Placer"/>
      </type>
      <system value="urn:oid:1.3.4.5.6.7"/>
      <value value="2345234234234"/>
    </identifier>
    <status value="active"/>
    <intent value="original-order"/>
    <code>
      <coding>
        <system value="http://acme.org/tests"/>
```

```
        <code value="LIPID"/>
       </coding>
       <text value="Lipid Panel"/>
     </code>
     <subject>
       <reference value="Patient/example"/>
     </subject>
     <context>
       <reference value="Encounter/example"/>
     </context>
     <occurrenceDateTime value="2013-05-02T16:16:00-07:00"/>
     <requester>
       <agent>
         <reference value="Practitioner/example"/>
       </agent>
       <onBehalfOf>
         <reference value="Organization/f001"/>
       </onBehalfOf>
     </requester>
     <performer>
       <reference value="Practitioner/f202"/>
     </performer>
     <reasonCode>
       <coding>
         <system value="http://hl7.org/fhir/sid/icd-9"/>
         <code value="V173"/>
         <display value="Fam hx-ischem heart dis"/>
       </coding>
     </reasonCode>
     <supportingInfo>
       <reference value="#fasting"/>
       <display value="Fasting status"/>
     </supportingInfo>
     <specimen>
       <reference value="#serum"/>
       <display value="Serum specimen"/>
     </specimen>
     <note>
       <text value="patient is afraid of needles"/>
     </note>
    </ProcedureRequest>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/Media/xray"/>
  <resource>
   <Media xmlns="http://hl7.org/fhir">
     <id value="xray"/>

     <text>
       <status value="generated"/>
       <div xmlns="http://www.w3.org/1999/xhtml">Xray of left hand for Patient Henry Levin (MRN 12345)
  2016-03-15</div>
     </text>
     <basedOn>
       <identifier>
         <system value="http://someclinic.org/fhir/NamingSystem/imaging-orders"/>
         <value value="111222"/>
         <assigner>
           <display value="XYZ Medical Clinic"/>
         </assigner>
       </identifier>
     </basedOn>
     <type value="photo"/>

     <subtype>
       <coding>
         <system value="http://snomed.info/sct"/>
         <code value="39714003"/>
         <display value="Skeletal X-ray of wrist and hand"/>
       </coding>
     </subtype>

     <subject>
       <reference value="Patient/example"/>
     </subject>
     <context>
       <reference value="Encounter/example"/>
     </context>
     <occurrenceDateTime value="2016-03-15"/>
     <bodySite>
       <coding>
         <system value="http://snomed.info.sct"/>
         <code value="85151006"/>
         <display value="Structure of left hand (body structure)"/>
       </coding>
     </bodySite>
     <height value="432"/>
     <width value="640"/>
```

```
      <content id="a1">
        <contentType value="image/jpeg"/>
        <url value="http://someimagingcenter.org/fhir/Binary/A12345"/>
        <!-- the date this image was created -->
        <creation value="2016-03-15"/>
      </content>
    </Media>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/CommunicationRequest/fm-solicit"/>
  <resource>
   <CommunicationRequest xmlns="http://hl7.org/fhir">
     <id value="fm-solicit"/>
     <text>
       <status value="generated"/>
       <div xmlns="http://www.w3.org/1999/xhtml">Request for Accident Report</div>
     </text>

     <!-- insert contents here -->

       <contained>
       <Organization>
         <id value="provider"/>
         <identifier>
           <system value="http://www.jurisdiction.com/provideroffices"/>
           <value value="3456"/>
         </identifier>
       </Organization>
     </contained>

     <contained>
       <Organization>
         <id value="payor"/>
         <identifier>
           <system value="http://www.jurisdiction.com/insurer"/>
           <value value="123456"/>
         </identifier>
       </Organization>
     </contained>

     <contained>
       <Practitioner>
         <id value="requester"/>
         <identifier>
            <value value="6789"/>
         </identifier>
       </Practitioner>
     </contained>
     <!-- body of the resource -->

     <identifier>
       <system value="http://www.jurisdiction.com/insurer/123456"/>
       <value value="ABC123"/> <!-- this is the value to which the response will refer -->
     </identifier>
      <basedOn>
       <display value="EligibilityRequest"/>
     </basedOn>
      <replaces>
       <display value="prior CommunicationRequest"/>
     </replaces>
     <groupIdentifier>
          <value value="12345"/>
     </groupIdentifier>
     <status value="active"/>

     <category>
        <coding>
          <system value="http://acme.org/messagetypes"/>
          <code value="SolicitedAttachmentRequest"/>
        </coding>
     </category>
     <priority value="routine"/>
     <medium>
       <coding>
         <system value="http://hl7.org/fhir/v3/ParticipationMode"/>
         <code value="WRITTEN"/>
         <display value="written"/>
       </coding>
       <text value="written"/>
     </medium>
     <recipient>
         <reference value="#provider"/>
     </recipient>
     <context>
       <reference value="Encounter/example"/>
     </context>
     <payload>
       <contentString value="Please provide the accident report and any associated pictures to support your Claim#
```

```
  DEF5647."/>
     </payload>
     <occurrenceDateTime value="2016-06-10T11:01:10-08:00"/>
     <authoredOn value="2016-06-10T11:01:10-08:00"/>
     <sender>
        <reference value="#payor"/>
      </sender>
      <requester>
       <agent>
       <reference value="#requester"/>
       </agent>
      </requester>

   </CommunicationRequest>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/QuestionnaireResponse/3141"/>
  <resource>
   <QuestionnaireResponse xmlns="http://hl7.org/fhir">
     <id value="3141"/>
     <text>
       <status value="generated"/>
       <div xmlns="http://www.w3.org/1999/xhtml">
         <pre>
               Comorbidity? YES
                 Cardial Comorbidity? YES

                   Angina? YES
                   MI? NO
                 Vascular Comorbidity?
                   (no answers)
                 ...
               Histopathology
                 Abdominal
                   pT category: 1a
                 ...
             </pre>
       </div>
     </text>
     <contained>
       <Patient>
         <id value="patsub"/>
         <identifier>
           <system value="http://cancer.questionnaire.org/systems/id/patientnr"/>
           <value value="A34442332"/>
         </identifier>
         <identifier>
           <type>
             <text value="Dutch BSN"/>
           </type>
           <system value="urn:oid:2.16.840.1.113883.2.4.6.3"/>
           <value value="188912345"/>
         </identifier>
         <gender value="male"/>
         <birthDate value="1972-11-30"/>
       </Patient>
     </contained>
     <contained>
       <ReferralRequest>
         <id value="order"/>
         <status value="unknown"/>
         <intent value="order"/>
         <subject>
           <reference value="#patsub"/>
         </subject>
         <requester>
           <agent>
             <reference value="Practitioner/example"/>
           </agent>
         </requester>
       </ReferralRequest>
     </contained>
     <contained>
       <Practitioner>
         <id value="questauth"/>
         <identifier>
           <type>
             <text value="AUMC, Den Helder"/>
           </type>
           <system value="http://cancer.questionnaire.org/systems/id/org"/>
           <value value="AUMC"/>
         </identifier>
       </Practitioner>
     </contained>
     <!--   TODO: decide how to do this
     <contained>

       <Observation>
         <id value="obs.pt-category"/>
```

```
    <status value="final"/>
    <code>
      <coding>
        <system value="http://snomed.info/sct"/>
        <code value="53786006"/>
        <display value="pT1 category"/>
      </coding>
    </code>
    <subject>
      <reference value="#patsub"/>
    </subject>
    <performer>
      <reference value="#questauth"/>
    </performer>
    <valueCodeableConcept>
      <coding>
        <system value="http://snomed.info/sct"/>
        <code value="443357004"/>
        <display value="pT1a category"/>
      </coding>
    </valueCodeableConcept>
  </Observation>
</contained> -->
<identifier>
  <system value="http://example.org/fhir/NamingSystem/questionnaire-ids"/>
  <value value="Q12349876"/>
</identifier>
<basedOn>
  <reference value="#order"/>
</basedOn>
<parent>
  <reference value="Procedure/f201"/>
</parent>
<status value="completed"/>
<subject>
  <reference value="#patsub"/>
</subject>
<context>
  <reference value="Encounter/example"/>
</context>
<authored value="2013-02-19T14:15:00-05:00"/>
<author>
  <reference value="#questauth"/>
</author>
<item>
  <!--   COMORBIDITY   -->
  <!--   First main section of the form, questions about comorbidity   -->
  <!--   section contains one question: whether there is comorbidity   -->
  <linkId value="1"/>
  <item>
    <linkId value="1.1"/>
    <answer>
      <valueCoding>
        <system value="http://cancer.questionnaire.org/system/code/yesno"/>
        <code value="1"/>
        <display value="Yes"/>
      </valueCoding>
      <item>
        <!--   COMORBIDITY/CARDIAL   -->
        <!--   Subsection about specific comorbidity: cardial   -->
        <linkId value="1.1.1"/>
        <item>
          <linkId value="1.1.1.1"/>
          <answer>
            <valueCoding>
              <system value="http://cancer.questionnaire.org/system/code/yesno"/>
              <code value="1"/>
            </valueCoding>
          </answer>
        </item>
        <item>
          <!--   This answer carries both the questionnaire-specific name and an equivalent SNOMED CT code   -->
          <linkId value="1.1.1.2"/>
          <answer>
            <valueCoding>
              <system value="http://cancer.questionnaire.org/system/code/yesno"/>
              <code value="1"/>
            </valueCoding>
          </answer>
        </item>
        <item>
          <linkId value="1.1.1.3"/>
          <answer>
            <valueCoding>
              <system value="http://cancer.questionnaire.org/system/code/yesno"/>
              <code value="0"/>
            </valueCoding>
          </answer>
        </item>
```

```
          </item>
        </answer>
      </item>
    </item>
    <!--   HISTOPATHOLOGY   -->
    <!--   TODO: decide how to do this
    <item>

      <item>
        <item>
         <dataResource>
            <reference value="#obs.pt-category"/>
          </dataResource>
        </item>
      </item>
    </item>   -->
   </QuestionnaireResponse>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/GuidanceResponse/example"/>
  <resource>
   <GuidanceResponse xmlns="http://hl7.org/fhir">
    <id value="example"/>
    <!-- This example demonstrates the response to an $evaluate operation using the GuidanceResponse resource. -->
    <contained>
      <Parameters>
        <id value="outputParameters1"/>
        <parameter>
          <name value="score"/>
          <valueDecimal value="7"/>
        </parameter>
        <parameter>
          <name value="item-assessed"/>
     <valueCodeableConcept>
       <coding>
       <system value="http://www.ama-assn.org/go/cpt"/>
       <code value="70450"/>
       <display value="CT, head, wo iv contrast"/>
       </coding>
     </valueCodeableConcept>
        </parameter>
        <parameter>
          <name value="device"/>
          <valueReference>
            <reference value="Device/software"/>
          </valueReference>
        </parameter>
        <parameter>
          <name value="guideline-followed"/>
          <valueUri value="http://someguidelineprovider.org/radiology-appropriateness-guidelines.html"/>
        </parameter>
      </Parameters>
    </contained>
    <requestId value="guidanceRequest1"/>
    <identifier>
      <system value="http://example.org"/>
      <value value="guidanceResponse1"/>
    </identifier>
    <module>
      <reference value="ServiceDefinition/example"/>
    </module>
    <status value="success"/>
    <subject>
      <reference value="Patient/example"/>
    </subject>
    <context>
      <reference value="Encounter/example"/>
    </context>
    <occurrenceDateTime value="2017-03-10T16:02:00Z"/>
    <performer>
      <reference value="Device/software"/>
    </performer>
    <reasonCodeableConcept>
      <text value="Guideline Appropriate Ordering Assessment"/>
    </reasonCodeableConcept>
    <outputParameters>
      <reference value="#outputParameters1"/>
    </outputParameters>
   </GuidanceResponse>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/NutritionOrder/cardiacdiet"/>
  <resource>
   <?xml-stylesheet type="text/xsl" href="file://Mac/Home/Documents/FHIRBuild/reorder-nodes-
  template.xslt"?><NutritionOrder xmlns="http://hl7.org/fhir">
   edited by Eric Haas Health eData Inc
   -->
```

```
 <!--EH  Show all the elements in this examples for preference, and intolerance --><!-- EH: use oral Diet element
-->
 <id value="cardiacdiet"/>
  <identifier>
    <system value="http://goodhealthhospital.org/nutrition-requests"/>
    <value value="123"/>
  </identifier>
  <status value="active"/>
  <patient>
    <reference value="Patient/example"/>
    <display value="Peter Chalmers"/>
  </patient>
  <encounter>
    <reference value="Encounter/example"/>
    <display value="Inpatient"/>
  </encounter>
  <dateTime value="2014-09-17"/>
  <orderer>
    <reference value="Practitioner/example"/>
    <display value="Dr Adam Careful"/>
  </orderer>
  <allergyIntolerance>
    <reference value="AllergyIntolerance/example"/>
    <display value="Cashew Nuts"/>
  </allergyIntolerance>
  <foodPreferenceModifier>
    <coding>
      <system value="http://hl7.org/fhir/diet"/>
      <code value="dairy-free"/>
    </coding>
  </foodPreferenceModifier>
  <excludeFoodModifier>
    <coding>
      <system value="http://snomed.info/sct"/>
      <version value="20140730"/>
      <code value="227493005"/>
      <display value="Cashew Nut"/>
    </coding>
  </excludeFoodModifier>
  <oralDiet>
    <!-- EH:  list type of diet showing standard code and local codes -->
    <type>
      <coding>
        <system value="http://snomed.info/sct"/>
        <code value="386619000"/>
        <display value="Low sodium diet"/>
      </coding>
      <coding>
        <system value="http://goodhealthhospital.org/diet-type-codes"/>
        <code value="1040"/>
        <display value="Low Sodium Diet"/>
      </coding>
      <text value="Low sodium diet"/>
    </type>
    <type>
      <coding>
        <system value="http://snomed.info/sct"/>
        <code value="226208002"/>
        <display value="Fluid restricted diet"/>
      </coding>
      <coding>
        <system value="http://goodhealthhospital.org/diet-type-codes"/>
        <code value="1040"/>
        <display value="Fluid restricted diet"/>
      </coding>
      <text value="Fluid restricted diet"/>
    </type>
    <!-- list nutrients and amounts here -->
    <nutrient>
      <modifier>
        <coding>
          <system value="http://snomed.info/sct"/>
          <code value="39972003"/>
          <display value="Sodium"/>
        </coding>
      </modifier>
      <amount>
        <value value="2"/>
        <unit value="grams"/>
        <system value="http://unitsofmeasure.org"/>
        <code value="g"/>
      </amount>
    </nutrient>
    <nutrient>
      <modifier>
        <coding>
          <system value="http://snomed.info/sct"/>
          <code value="33463005"/>
          <display value="Fluid"/>
```

```
          </coding>
        </modifier>
        <amount>
          <value value="1500"/>
          <unit value="milliliter"/>
          <system value="http://unitsofmeasure.org"/>
          <code value="mL"/>
        </amount>
      </nutrient>
      <!--instructions as free text -->
      <instruction value="Starting on 2/10 breakfast, maximum 400 ml fluids per meal"/>
    </oralDiet>
   </NutritionOrder>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/CareTeam/example"/>
  <resource>
   <CareTeam xmlns="http://hl7.org/fhir">
    <id value="example"/>

    <text>
      <status value="generated"/>
      <div xmlns="http://www.w3.org/1999/xhtml">Care Team</div>
    </text>

    <contained>
      <Practitioner>
        <id value="pr1"/>
        <name>
          <family value="Dietician"/>
          <given value="Dorothy"/>
        </name>
      </Practitioner>
    </contained>
    <identifier>
         <value value="12345"/>
    </identifier>
    <status value="active"/>
    <category>
       <coding>
         <system value="http://hl7.org/fhir/care-team-category"/>
         <code value="encounter"/>
       </coding>
    </category>
    <name value="Peter James Charlmers Care Plan for Inpatient Encounter"/>
    <subject>
      <reference value="Patient/example"/>
      <display value="Peter James Chalmers"/>
    </subject>
    <context>
   <reference value="Encounter/example"/>
    </context>
    <period>
      <end value="2013-01-01"/>
    </period>

    <participant>
      <role>
        <text value="responsiblePerson"/>
      </role>
      <member>
        <reference value="Patient/example"/>
        <display value="Peter James Chalmers"/>
      </member>
    </participant>

    <participant>
      <role>
        <text value="adviser"/>
      </role>
      <member>
        <reference value="#pr1"/>
        <display value="Dorothy Dietition"/>
      </member>
      <onBehalfOf>
        <reference value="Organization/f001"/>
      </onBehalfOf>
      <period>
        <end value="2013-01-01"/>
      </period>
    </participant>
    <managingOrganization>
           <reference value="Organization/f001"/>
       </managingOrganization>
   </CareTeam>
  </resource>
</entry>
<entry>
```

```
<fullUrl value="http://hl7.org/fhir/RequestGroup/kdn5-example"/>
<resource>
 <RequestGroup xmlns="http://hl7.org/fhir">
   <id value="kdn5-example"/>
   <text>
     <status value="generated"/>
     <div xmlns="http://www.w3.org/1999/xhtml">Administer gemcitabine and carboplatin.</div>
   </text>
   <contained>
     <MedicationRequest>
       <id value="1111"/>
       <intent value="proposal"/>
       <medicationCodeableConcept>
         <coding>
           <system value="http://www.nlm.nih.gov/research/umls/rxnorm"/>
           <code value="12574"/>
           <display value="gemcitabine"/>
         </coding>
         <text value="gemcitabine"/>
       </medicationCodeableConcept>
       <subject>
         <reference value="Patient/example"/>
       </subject>
       <dosageInstruction>
         <text value="1250 mg/m² IV over 30 minutes"/>
         <timing>
           <repeat>
             <duration value="30"/>
             <durationUnit value="min"/>
           </repeat>
         </timing>
         <route>
           <text value="IV"/>
         </route>
         <doseQuantity>
           <value value="1250"/>
           <unit value="mg/m²"/>
         </doseQuantity>
       </dosageInstruction>
     </MedicationRequest>
   </contained>
   <contained>
     <MedicationRequest>
       <id value="2222"/>
       <intent value="proposal"/>
       <medicationCodeableConcept>
         <coding>
           <system value="http://www.nlm.nih.gov/research/umls/rxnorm"/>
           <code value="40048"/>
           <display value="Carboplatin"/>
         </coding>
         <text value="CARBOplatin"/>
       </medicationCodeableConcept>
       <subject>
         <reference value="Patient/example"/>
       </subject>
       <dosageInstruction>
         <text value="AUC 5 IV over 30 minutes"/>
         <timing>
           <repeat>
             <duration value="30"/>
             <durationUnit value="min"/>
           </repeat>
         </timing>
         <route>
           <text value="IV"/>
         </route>
         <doseQuantity>
           <!-- AUC as dose unit or as an extension? -->
           <extension url="http://example.org/fhir/AUC-dose">
             <valueInteger value="5"/>
           </extension>
           <!-- <value value="5" />
           <unit value="AUC" />
           <system value="http://snomed.info/sct" />
           <code value="413573002" /> -->
         </doseQuantity>
       </dosageInstruction>
     </MedicationRequest>
   </contained>
   <identifier>
     <value value="requestgroup-kdn5"/>
   </identifier>
   <definition>
     <reference value="PlanDefinition/KDN5"/>
   </definition>
   <status value="draft"/>
   <intent value="plan"/>
   <priority value="routine"/>
```

```
    <subject>
      <reference value="Patient/example"/>
    </subject>
    <context>
      <reference value="Encounter/example"/>
    </context>
    <authoredOn value="2017-03-06T17:31:00Z"/>
    <author>
      <reference value="Practitioner/1"/>
    </author>

    <action>
      <!-- list of regimen options. in this case only one -->
      <selectionBehavior value="exactly-one"/>
      <action>
        <!-- list of regimen parts. in this case only one -->
        <selectionBehavior value="all"/>
        <action>
          <!-- list of cycle definitions. in this case only one -->
          <groupingBehavior value="sentence-group"/>
          <selectionBehavior value="exactly-one"/>
          <action id="cycle-definition-1">
            <textEquivalent value="21-day cycle for 6 cycles"/>

            <timingTiming>
              <repeat>
                <count value="6"/>
                <duration value="21"/>
                <durationUnit value="d"/>
              </repeat>
            </timingTiming>

            <!-- list of regimen entry references for this regimen part -->
            <!-- references would duplicated in each cycle definition option
            (see codesystem-action-grouping-behavior.html#sentence-group)-->
            <action id="action-1">
              <extension url="http://hl7.org/fhir/StructureDefinition/timing-daysOfCycle">
                <extension url="day">
                  <valueInteger value="1"/>
                </extension>
                <extension url="day">
                  <valueInteger value="8"/>
                </extension>
              </extension>
              <textEquivalent value="Gemcitabine 1250 mg/m² IV over 30 minutes on days 1 and 8"/>
              <resource>
                <reference value="#1111"/>
              </resource>
            </action>
            <action id="action-2">
              <extension url="http://hl7.org/fhir/StructureDefinition/timing-daysOfCycle">
                <extension url="day">
                  <valueInteger value="1"/>
                </extension>
              </extension>
              <textEquivalent value="CARBOplatin AUC 5 IV over 30 minutes on Day 1"/>
              <relatedAction>
                <actionId value="action-1"/>
                <relationship value="concurrent-with-start"/>
       </relatedAction>
              <resource>
                <reference value="#2222"/>
              </resource>
            </action> <!-- end of regimen entries -->
          </action> <!-- end of cycle definitions -->
        </action> <!-- end of regimen parts -->
      </action> <!-- end of regimen options -->
    </action>
   </RequestGroup>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/CarePlan/gpvisit"/>
  <resource>
   <CarePlan xmlns="http://hl7.org/fhir">
     <id value="gpvisit"/>
     <text>
       <status value="additional"/>
       <div xmlns="http://www.w3.org/1999/xhtml">
         <p>  Represents the flow of a patient within a practice. The plan is created when
           they arrive and represents the 'care' of the patient over the course of that encounter.
           They first see the nurse for basic observations (BP, pulse, temp) then the doctor for
           the consultation and finally the nurse again for a tetanus immunization. As the plan is
           updated (e.g. a new activity added), different versions of the plan exist, and workflow timings
           for reporting can be gained by examining the plan history. This example is the version after
           seeing the doctor, and waiting for the nurse.The plan can either be created 'ad hoc' and modified as
           the parient progresses, or start with a standard template (which can, of course, be altered to suit the
  patient.</p>
       </div>
```

```
  </text>
  <contained>
    <!-- This is the reason for the encounter. It is referenced by the concern -->
    <Condition>
      <id value="p1"/>
      <clinicalStatus value="active"/>
   <verificationStatus value="confirmed"/>
      <code>
        <!-- Could coded if we wanted to...-->
        <text value="Overseas encounter"/>
      </code>
      <subject>
        <reference value="Patient/100"/>
        <display value="Peter James Chalmers"/>
      </subject>
    </Condition>
  </contained>

  <contained>
    <CareTeam>
        <id value="careteam"/>
        <participant id="part1">

     <role>

       <coding>

       <system value="http://example.org/local"/>

       <code value="nur"/>

       </coding>

   <text value="nurse"/>

     </role>

     <member>

   <reference value="Practitioner/13"/>

   <display value="Nurse Nancy"/>

     </member>

 </participant>

 <participant id="part2">

     <role>

   <coding>

       <system value="http://example.org/local"/>

       <code value="doc"/>

         </coding>

   <text value="doctor"/>

      </role>

      <member>

   <reference value="Practitioner/14"/>

   <display value="Doctor Dave"/>

      </member>

 </participant>

    </CareTeam>
  </contained>


<contained>
    <Goal>
      <id value="goal"/>
      <status value="planned"/>
   <description>
        <text value="Complete consultation"/>
      </description>
   <subject>
        <reference value="Patient/100"/>
        <display value="Peter James Chalmers"/>
      </subject>
    </Goal>
```

```
    </contained>
    <status value="active"/>
    <intent value="plan"/>
    <subject>
      <reference value="Patient/100"/>
      <display value="Peter James Chalmers"/>
    </subject>
    <period>
      <!-- This is the time the plan started - i.e. when they arrived-->
      <start value="2013-01-01T10:30:00+00:00"/>
      <!-- No end yet as the encounter is still in progress.-->
    </period>
    <careTeam>
      <reference value="#careteam"/>
    </careTeam>
    <addresses>
      <reference value="#p1"/>
      <display value="obesity"/>
    </addresses>
    <goal>
      <reference value="#goal"/>
    </goal>
    <activity>
      <outcomeReference>
     <!-- This is a link to the nurse encounter. The assumption is that all contacts with practitioners
     are modelled as separate encounters. Ideally, there will be  a 'master/parent' encounter that ties them
  together.
      If there is a single encounter, then all participants will be linked to that encounter.-->
      <reference value="Encounter/example"/>
      </outcomeReference>
      <detail>
        <!-- This activity is for the initial nurse encounter where vitals are taken. It has been completed.-->
        <category><coding><system value="http://hl7.org/fhir/care-plan-activity-category"/><code
  value="encounter"/></coding></category>
        <code>
          <coding>
            <system value="http://example.org/local"/>
            <code value="nursecon"/>
          </coding>
          <text value="Nurse Consultation"/>
        </code>
        <status value="completed"/>
        <prohibited value="false"/>
        <scheduledPeriod>
          <!-- the nurse saw the patient between 10:38 and 10:50 -->
          <start value="2013-01-01T10:38:00+00:00"/>
          <end value="2013-01-01T10:50:00+00:00"/>
        </scheduledPeriod>
        <performer>
          <!-- refer to the participant (the nurse) in this resource-->
          <reference value="Practitioner/13"/>
          <display value="Nurse Nancy"/>
        </performer>
      </detail>
    </activity>
    <activity>
      <detail>
        <!-- This activity is for the encounter with the doctor. It is scheduled, but not yet started,
                so there is no timing[x]  or actionTaken element yet -->
        <category><coding><system value="http://hl7.org/fhir/care-plan-activity-category"/><code
  value="encounter"/></coding></category>
        <code>
          <coding>
            <system value="http://example.org/local"/>
            <code value="doccon"/>
          </coding>
          <text value="Doctor Consultation"/>
        </code>
        <status value="scheduled"/>
        <prohibited value="false"/>
        <!-- The status is 'scheduled' so the doctor knows the patient is waiting.  -->
        <performer>
          <!-- refer to the participant (the nurse) in this resource-->
          <reference value="Practitioner/14"/>
          <display value="Doctor Dave"/>
        </performer>
      </detail>
    </activity>
   </CarePlan>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/NutritionOrder/renaldiet"/>
  <resource>
   <NutritionOrder xmlns="http://hl7.org/fhir">
   edited by Eric Haas Health eData Inc
   --><!-- EH: use oral Diet element --><id value="renaldiet"/>
    <identifier>
      <system value="http://goodhealthhospital.org/nutrition-requests"/>
```

```
  <value value="123"/>
</identifier>
<status value="active"/>
<patient>
  <reference value="Patient/example"/>
  <display value="Peter Chalmers"/>
</patient>
<encounter>
  <reference value="Encounter/example"/>
  <display value="Inpatient"/>
</encounter>
<dateTime value="2014-09-17"/>
<orderer>
  <reference value="Practitioner/example"/>
  <display value="Dr Adam Careful"/>
</orderer>
<allergyIntolerance>
  <!--EH  Show all the elements in this examples for preference, and intolerance -->
  <reference value="AllergyIntolerance/example"/>
  <display value="Cashew Nuts"/>
</allergyIntolerance>
<foodPreferenceModifier>
  <coding>
    <system value="http://hl7.org/fhir/diet"/>
    <code value="dairy-free"/>
  </coding>
</foodPreferenceModifier>
<excludeFoodModifier>
  <coding>
    <system value="http://snomed.info/sct"/>
    <version value="20140730"/>
    <code value="227493005"/>
    <display value="Cashew Nut"/>
  </coding>
</excludeFoodModifier>
<oralDiet>
  <type>
    <coding>
      <system value="http://snomed.info/sct"/>
      <code value="33489005"/>
      <display value="Renal diet"/>
    </coding>
    <coding>
      <system value="http://goodhealthhospital.org/diet-type-codes"/>
      <code value="1050"/>
      <display value="Renal"/>
    </coding>
    <text value="Renal Diet"/>
  </type>
  <!-- EH example of three times a day starting on 2/10 -->
  <schedule>
    <repeat>
       <boundsPeriod>
         <start value="2015-02-10"/>
       </boundsPeriod>
      <frequency value="3"/>
      <period value="1"/>
      <periodUnit value="d"/>
    </repeat>
  </schedule>
  <nutrient>
    <!-- list nutrients and amounts here -->
    <modifier>
      <coding>
        <system value="http://snomed.info/sct"/>
        <code value="39972003"/>
        <display value="Sodium"/>
      </coding>
    </modifier>
    <amount>
      <value value="2"/>
      <unit value="grams"/>
      <system value="http://unitsofmeasure.org"/>
      <code value="g"/>
    </amount>
  </nutrient>
  <nutrient>
    <modifier>
      <coding>
        <system value="http://snomed.info/sct"/>
        <code value="30820000"/>
        <display value="Phosphorus"/>
      </coding>
    </modifier>
    <amount>
      <value value="1000"/>
      <unit value="milligrams"/>
      <system value="http://unitsofmeasure.org"/>
      <code value="mg"/>
```

```
        </amount>
       </nutrient>
       <nutrient>
         <modifier>
           <coding>
             <system value="http://snomed.info/sct"/>
             <code value="88480006"/>
             <display value="Potassium"/>
           </coding>
         </modifier>
         <amount>
           <value value="2"/>
           <unit value="grams"/>
           <system value="http://unitsofmeasure.org"/>
           <code value="g"/>
         </amount>
       </nutrient>
     </oralDiet>
   </NutritionOrder>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/CommunicationRequest/example"/>
  <resource>
   <CommunicationRequest xmlns="http://hl7.org/fhir">
     <id value="example"/>
     <text>
       <status value="generated"/>
       <div xmlns="http://www.w3.org/1999/xhtml">To be filled out at a later time</div>
     </text>
     <status value="active"/>
     <!-- insert contents here -->
     <subject>
       <reference value="Patient/example"/>
     </subject>
     <context>
    <reference value="Encounter/example"/>
     </context>
   </CommunicationRequest>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/RequestGroup/example"/>
  <resource>
   <RequestGroup xmlns="http://hl7.org/fhir">
     <id value="example"/>
     <text>
       <status value="generated"/>
       <div xmlns="http://www.w3.org/1999/xhtml">Example RequestGroup illustrating related actions to administer
  medications in sequence with time delay.</div>
     </text>
     <contained>
       <MedicationRequest>
         <id value="medicationrequest-1"/>
         <intent value="proposal"/>
         <medicationCodeableConcept>
           <text value="Medication 1"/>
         </medicationCodeableConcept>
         <subject>
           <reference value="Patient/example"/>
         </subject>
       </MedicationRequest>
     </contained>
     <contained>
       <MedicationRequest>
         <id value="medicationrequest-2"/>
         <intent value="proposal"/>
         <medicationCodeableConcept>
           <text value="Medication 2"/>
         </medicationCodeableConcept>
         <subject>
           <reference value="Patient/example"/>
         </subject>
       </MedicationRequest>
     </contained>
     <identifier>
       <value value="requestgroup-1"/>
     </identifier>
     <groupIdentifier>
       <system value="http://example.org/treatment-group"/>
       <value value="00001"/>
     </groupIdentifier>
     <status value="draft"/>
     <intent value="plan"/>
     <priority value="routine"/>
     <subject>
       <reference value="Patient/example"/>
     </subject>
     <context>
```

```
        <reference value="Encounter/example"/>
       </context>
       <authoredOn value="2017-03-06T17:31:00Z"/>
       <author>
         <reference value="Practitioner/1"/>
       </author>
       <reasonCodeableConcept>
         <text value="Treatment"/>
       </reasonCodeableConcept>
       <note>
         <text value="Additional notes about the request group"/>
       </note>
       <action>
         <label value="1"/>
         <title value="Administer Medications"/>
         <description value="Administer medications at the appropriate time"/>
         <textEquivalent value="Administer medication 1, followed an hour later by medication 2"/>
         <timingDateTime value="2017-03-06T19:00:00Z"/>
         <participant>
           <reference value="Practitioner/1"/>
         </participant>
         <groupingBehavior value="logical-group"/>
         <selectionBehavior value="all"/>
         <requiredBehavior value="must"/>
         <precheckBehavior value="yes"/>
         <cardinalityBehavior value="single"/>
         <action id="medication-action-1">
           <description value="Administer medication 1"/>
           <type>
          <code value="create"/>
           </type>
           <resource>
             <reference value="#medicationrequest-1"/>
           </resource>
         </action>
         <action id="medication-action-2">
           <description value="Administer medication 2"/>
           <relatedAction>
             <actionId value="medication-action-1"/>
             <relationship value="after-end"/>
             <offsetDuration>
               <value value="1"/>
               <unit value="h"/>
             </offsetDuration>
           </relatedAction>
           <type>
         <code value="create"/>
           </type>
           <resource>
             <reference value="#medicationrequest-2"/>
           </resource>
         </action>
       </action>
     </RequestGroup>
   </resource>
 </entry>
 <entry>
   <fullUrl value="http://hl7.org/fhir/NutritionOrder/infantenteral"/>
   <resource>
    <NutritionOrder xmlns="http://hl7.org/fhir">
    edited by Eric Haas Health eData Inc
    --><id value="infantenteral"/>
     <identifier>
       <system value="http://www.acme.org/nutritionorders"/>
       <value value="123"/>
     </identifier>
     <status value="active"/>
     <patient>
       <reference value="Patient/example"/>
       <display value="Peter Chalmers"/>
     </patient>
     <encounter>
       <reference value="Encounter/example"/>
       <display value="Inpatient"/>
     </encounter>
     <dateTime value="2014-09-17"/>
     <orderer>
       <reference value="Practitioner/example"/>
       <display value="Dr Adam Careful"/>
     </orderer>
     <enteralFormula>
       <baseFormulaType>
         <coding>
           <system value="http://snomed.info/sct"/>
           <code value="412414007"/>
           <display value="infant formula + iron"/>
         </coding>
       </baseFormulaType>
       <baseFormulaProductName value="Acme Infant Formula + Iron"/>
```

```
      <additiveType>
        <coding>
          <system value="http://hl7.org/fhir/entformula-additive"/>
          <code value="carbohydrate"/>
          <display value="Carbohydrate"/>
        </coding>
      </additiveType>
      <additiveProductName value="Acme High Carbohydrate Additive"/>
      <caloricDensity>
        <value value="20"/>
        <unit value="calories per ounce"/>
        <system value="http://unitsofmeasure.org"/>
        <code value="cal/[foz_us]"/>
      </caloricDensity>
      <routeofAdministration>
        <coding>
          <system value="http://hl7.org/fhir/v3/RouteOfAdministration"/>
          <code value="PO"/>
          <display value="Swallow, oral"/>
          <userSelected value="true"/>
        </coding>
      </routeofAdministration>
      <administration>
      <!-- scheduled 4 oz every 3hrs not to exceed 32 oz/day-->
      <schedule>
        <repeat>
        <boundsPeriod>
          <start value="2014-09-17"/>
        </boundsPeriod>
          <frequency value="1"/>
          <period value="3"/>
          <periodUnit value="h"/>
        </repeat>
      </schedule>
      <quantity>
        <value value="4"/>
        <unit value="ounces"/>
        <system value="http://unitsofmeasure.org"/>
        <code value="[foz_us]"/>
      </quantity>
      </administration>
      <maxVolumeToDeliver>
        <value value="32"/>
        <unit value="ounces"/>
        <system value="http://unitsofmeasure.org"/>
        <code value="[foz_us]"/>
      </maxVolumeToDeliver>
      <!-- EH infant feeding instruction and additional instructions -->
      <administrationInstruction value="Add high calorie high carbohydrate additive to increase cal/oz from 24 cal/oz
  to 27 cal/oz."/>

    </enteralFormula>
   </NutritionOrder>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/ImagingStudy/example-xr"/>
  <resource>
   <ImagingStudy xmlns="http://hl7.org/fhir">
    <id value="example-xr"/>
    <text>
      <status value="generated"/>
      <div xmlns="http://www.w3.org/1999/xhtml">XR Wrist 3+ Views. John Smith (MRN: 09236). Accession: W12342398.
  Performed: 2017-01-01. 1 series, 2 images.</div>
    </text>
    <uid value="urn:oid:2.16.124.113543.6003.1154777499.30246.19789.3503430046"/>
    <accession>
      <use value="usual"/>
      <type>
        <coding>
          <system value="http://hl7.org/fhir/v2/0203"/>
          <code value="ACSN"/>
        </coding>
      </type>
      <value value="W12342398"/>
      <assigner>
        <reference value="Organization/dicom-organization"/>
      </assigner>
    </accession>
    <identifier>
      <use value="secondary"/>
      <value value="55551234"/>
      <assigner>
        <reference value="Organization/dicom-organization"/>
      </assigner>
    </identifier>
    <availability value="ONLINE"/>
    <modalityList>
      <system value="http://dicom.nema.org/resources/ontology/DCM"/>
```

```
      <code value="DX"/>
    </modalityList>
    <patient>
      <reference value="Patient/dicom"/>
    </patient>
    <context>
      <reference value="Encounter/example"/>
    </context>
    <started value="2017-01-01T11:01:20+03:00"/>
    <basedOn>
      <reference value="ProcedureRequest/example"/>
    </basedOn>
    <referrer>
      <reference value="Practitioner/example"/>
    </referrer>
    <interpreter>
      <reference value="Practitioner/example"/>
    </interpreter>
    <endpoint>
      <reference value="Endpoint/example-wadors"/>
    </endpoint>
    <numberOfSeries value="1"/>
    <numberOfInstances value="2"/>
    <procedureReference>
      <reference value="Procedure/example"/>
    </procedureReference>
    <procedureCode>
      <coding>
        <system value="http://www.radlex.org"/>
        <code value="RPID2589"/>
        <display value="XR Wrist 3+ Views"/>
      </coding>
      <text value="XR Wrist 3+ Views"/>
    </procedureCode>
    <reason>
      <coding>
        <system value="http://snomed.info/sct"/>
        <code value="357009"/>
        <display value="Closed fracture of trapezoidal bone of wrist"/>
      </coding>
    </reason>
    <description value="XR Wrist 3+ Views"/>
    <series>
      <uid value="urn:oid:2.16.124.113543.6003.1154777499.30246.19789.3503430045.1"/>
      <number value="3"/>
      <modality>
        <system value="http://dicom.nema.org/resources/ontology/DCM"/>
        <code value="DX"/>
      </modality>
      <description value="XR Wrist 3+ Views"/>
      <numberOfInstances value="2"/>
      <availability value="ONLINE"/>
      <endpoint>
        <reference value="Endpoint/example-wadors"/>
      </endpoint>
      <bodySite>
        <!--   Added body site to support duplicate checking example   -->
        <system value="http://snomed.info/sct"/>
        <code value="T-15460"/>
        <display value="Wrist Joint"/>
      </bodySite>
      <laterality>
        <system value="http://snomed.info/sct"/>
        <code value="419161000"/>
        <display value="Unilateral left"/>
      </laterality>
      <started value="2011-01-01T11:01:20+03:00"/>
      <performer>
        <reference value="Practitioner/example"/>
      </performer>
      <instance>
        <uid value="urn:oid:2.16.124.113543.6003.1154777499.30246.19789.3503430045.1.1"/>
        <number value="1"/>
        <sopClass value="urn:oid:1.2.840.10008.5.1.4.1.1.2"/>
        <title value="PA VIEW"/>
      </instance>
      <instance>
        <uid value="urn:oid:2.16.124.113543.6003.1154777499.30246.19789.3503430045.1.2"/>
        <number value="2"/>
        <sopClass value="urn:oid:1.2.840.10008.5.1.4.1.1.2"/>
        <title value="LL VIEW"/>
      </instance>
    </series>
   </ImagingStudy>
  </resource>
</entry>
<entry>
  <fullUrl value="http://hl7.org/fhir/NutritionOrder/enteralbolus"/>
  <resource>
```

```
<NutritionOrder xmlns="http://hl7.org/fhir">
Tube feedings are given in boluses several times a day   edited by Eric Haas Health eData Inc
--><!--EH  Show all the elements in this examples for preference, and intolerance --><id value="enteralbolus"/>
 <identifier>
   <system value="http://www.acme.org/nutritionorders"/>
   <value value="123"/>
 </identifier>
 <status value="active"/>
 <patient>
   <reference value="Patient/example"/>
   <display value="Peter Chalmers"/>
 </patient>
 <encounter>
   <reference value="Encounter/example"/>
   <display value="Inpatient"/>
 </encounter>
 <dateTime value="2014-09-17"/>
 <orderer>
   <reference value="Practitioner/example"/>
   <display value="Dr Adam Careful"/>
 </orderer>
 <allergyIntolerance>
   <reference value="AllergyIntolerance/example"/>
   <display value="Cashew Nuts"/>
 </allergyIntolerance>
 <foodPreferenceModifier>
   <coding>
     <system value="http://hl7.org/fhir/diet"/>
     <code value="dairy-free"/>
   </coding>
 </foodPreferenceModifier>
 <excludeFoodModifier>
   <coding>
     <system value="http://snomed.info/sct"/>
     <version value="20140730"/>
     <code value="227493005"/>
     <display value="Cashew Nut"/>
   </coding>
 </excludeFoodModifier>
 <enteralFormula>

   <baseFormulaType>
     <coding>
       <!--EH:  bolus tube feeding instruction in text-->
       <system value="http://usextension/snomed.info/sct"/>
       <!--EH  temporary system value for us extension -->
       <code value="659311000124118"/>
       <display value="Adult high protein formula"/>
     </coding>
   </baseFormulaType>
   <baseFormulaProductName value="Acme High Protein Formula"/>

   <additiveType>
     <coding>
       <system value="http://hl7.org/fhir/entformula-additive"/>
       <code value="lipid"/>
       <display value="Lipid"/>
     </coding>
   </additiveType>
   <additiveProductName value="Acme Lipid Additive"/>
   <caloricDensity>
     <value value="1.5"/>
     <unit value="calories per milliliter"/>
     <system value="http://unitsofmeasure.org"/>
     <code value="cal/mL"/>
   </caloricDensity>
   <routeofAdministration>
     <coding>
       <system value="http://hl7.org/fhir/v3/RouteOfAdministration"/>
       <code value="GT"/>
       <display value="Instillation, gastrostomy tube"/>
     </coding>
   </routeofAdministration>
   <administration>
     <schedule>
       <repeat>
       <boundsPeriod>
         <start value="2014-09-17T16:00:00Z"/>
       </boundsPeriod>
         <frequency value="1"/>
         <period value="4"/>
         <periodUnit value="h"/>
       </repeat>
     </schedule>
     <quantity>
       <value value="240"/>
       <unit value="milliliters"/>
       <system value="http://unitsofmeasure.org"/>
       <code value="mL"/>
```

```
          </quantity>
        </administration>
        <maxVolumeToDeliver>
          <value value="1440"/>
          <unit value="milliliter/day"/>
          <system value="http://unitsofmeasure.org"/>
          <code value="mL/d"/>
        </maxVolumeToDeliver>
        <administrationInstruction value="240 mls every 4hrs "/>
      </enteralFormula>
     </NutritionOrder>
    </resource>
  </entry>
</Bundle>
```

8.11.14 Resource Encounter - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/encounter-version-maps.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.11.14 Resource Encounter - Detailed Descriptions

Patient Administration Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Encounter, Patient, Practitioner, RelatedPerson

R2 : R3 Converson maps for Encounter.

Functional status for this map: 10 tests that all execute ok. 1 fail round-trip testing and 2 r3 resources are invalid (2 errors). (see documentation)

8.11.14.1 R2 to R3

```
map "http://hl7.org/fhir/StructureMap/Encounter2to3" = "R2 to R3 Conversions for Encounter"

conceptmap "ActEncounterCode" {
  prefix s = "http://hl7.org/fhir/encounter-class"
  prefix t = "http://hl7.org/fhir/v3/ActCode"

  s:inpatient = t:IMP
  s:outpatient  = t:AMB
  s:ambulatory  = t:AMB
  s:emergency = t:EMER
  s:home  = t:HH
  s:field = t:FLD
  s:daytime = t:SS
  s:virtual = t:VR
  s:other = s:other
}


uses "http://hl7.org/fhir/DSTU2/StructureDefinition/Encounter" alias EncounterR2 as source
uses "http://hl7.org/fhir/StructureDefinition/Encounter" alias Encounter as target

imports "http://hl7.org/fhir/StructureMap/*2to3"

group for type+types  Encounter extends DomainResource
  input src : EncounterR2 as source
  input tgt : Encounter as target

  "Encounter-identifier" : for src.identifier make tgt.identifier
  "Encounter-status" : for src.status make tgt.status
  "Encounter-statusHistory" : for src.statusHistory as vs0 make tgt.statusHistory as vt0 then encStatusHistory(vs0, vt0)
  "Encounter-class" : for src.class as v make tgt.class = translate(v, "#ActEncounterCode", "Coding")
  "Encounter-type" : for src.type make tgt.type
  "Encounter-priority" : for src.priority make tgt.priority
  "Encounter-patient" : for src.patient make tgt.subject
  "Encounter-episodeOfCare" : for src.episodeOfCare make tgt.episodeOfCare
  "Encounter-incomingReferral" : for src.incomingReferral make tgt.incomingReferral
  "Encounter-participant" : for src.participant as vs0 make tgt.participant as vt0 then encParticipant(vs0, vt0)
  "Encounter-appointment" : for src.appointment make tgt.appointment
  "Encounter-period" : for src.period make tgt.period
  "Encounter-length" : for src.length as vs make tgt.length as vt then QuantityToDuration(vs, vt)
  "Encounter-reason" : for src.reason make tgt.reason
  "Encounter-indication" : for src.indication as v make tgt.diagnosis as d, d.condition = v
  "Encounter-hospitalization" : for src.hospitalization as vs0 make tgt.hospitalization as vt0 then
    encHospitaization(vs0, vt0, tgt)
  "Encounter-location" : for src.location as vs0 make tgt.location as vt0 then encLocation(vs0, vt0)
  "Encounter-serviceProvider" : for src.serviceProvider make tgt.serviceProvider
  "Encounter-partOf" : for src.partOf make tgt.partOf
endgroup

group encStatusHistory extends BackboneElement
  input src as source
  input tgt as target

  "Encounter.statusHistory-status" : for src.status make tgt.status
  "Encounter.statusHistory-period" : for src.period make tgt.period
```

```
endgroup

group encParticipant extends BackboneElement
  input src as source
  input tgt as target

  "Encounter.participant-type" : for src.type make tgt.type
  "Encounter.participant-period" : for src.period make tgt.period
  "Encounter.participant-individual" : for src.individual make tgt.individual
endgroup

group encHospitaization extends BackboneElement
  input src as source
  input tgt as target
  input res as target

  "Encounter.hospitalization-preAdmissionIdentifier" : for src.preAdmissionIdentifier make tgt.preAdmissionIdentifier
  "Encounter.hospitalization-origin" : for src.origin make tgt.origin
  "Encounter.hospitalization-admitSource" : for src.admitSource make tgt.admitSource
  "Encounter.hospitalization-admittingDiagnosis" : for src.admittingDiagnosis as v make tgt.diagnosis as d, d.role =
    cc("http://hl7.org/fhir/diagnosis-role", "AD"), d.condition = v
  "Encounter.hospitalization-reAdmission" : for src.reAdmission make tgt.reAdmission
  "Encounter.hospitalization-dietPreference" : for src.dietPreference make tgt.dietPreference
  "Encounter.hospitalization-specialCourtesy" : for src.specialCourtesy make tgt.specialCourtesy
  "Encounter.hospitalization-specialArrangement" : for src.specialArrangement make tgt.specialArrangement
  "Encounter.hospitalization-destination" : for src.destination make tgt.destination
  "Encounter.hospitalization-dischargeDisposition" : for src.dischargeDisposition make tgt.dischargeDisposition
  "Encounter.hospitalization-dischargeDiagnosis" : for src.dischargeDiagnosis as v make tgt.diagnosis as d, d.role =
    cc("http://hl7.org/fhir/diagnosis-role", "DD"), d.condition = v

endgroup

group encLocation extends BackboneElement
  input src as source
  input tgt as target

  "Encounter.location-location" : for src.location make tgt.location
  "Encounter.location-status" : for src.status make tgt.status
  "Encounter.location-period" : for src.period make tgt.period
endgroup
```

8.11.14.2 R3 to R2

```
map "http://hl7.org/fhir/StructureMap/Encounter3to2" = "R3 to R2 Conversion for Encounter"

conceptmap "ActEncounterCode" {
  prefix s = "http://hl7.org/fhir/encounter-class"
  prefix t = "http://hl7.org/fhir/v3/ActCode"

  t:IMP = s:inpatient
  t:AMB = s:outpatient
  t:AMB = s:ambulatory
  t:EMER  = s:emergency
  t:HH  = s:home
  t:FLD = s:field
  t:SS  = s:daytime
  t:VR  = s:virtual
}


uses "http://hl7.org/fhir/StructureDefinition/Encounter" alias Encounter as source
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/Encounter" alias EncounterR2 as target

imports "http://hl7.org/fhir/StructureMap/*3to2"

group for type+types  Encounter extends DomainResource
  input src : Encounter as source
  input tgt : EncounterR2 as target

  "Encounter-identifier" : for src.identifier make tgt.identifier
  "Encounter-status" : for src.status check status != 'entered-in-error' make tgt.status
  "Encounter-statusHistory" : for src.statusHistory as vs0 make tgt.statusHistory as vt0 then encStatusHistory(vs0, vt0)
  "Encounter-class" : for src.class as v make tgt.class = translate(v, "#ActEncounterCode", "code")
  "Encounter-type" : for src.type make tgt.type
  "Encounter-priority" : for src.priority make tgt.priority
  "Encounter-patient" : for src.subject make tgt.patient
  "Encounter-episodeOfCare" : for src.episodeOfCare make tgt.episodeOfCare
  "Encounter-incomingReferral" : for src.incomingReferral make tgt.incomingReferral
  "Encounter-participant" : for src.participant as vs0 make tgt.participant as vt0 then encParticipant(vs0, vt0)
  "Encounter-appointment" : for src.appointment make tgt.appointment
  "Encounter-period" : for src.period make tgt.period
  "Encounter-length" : for src.length as vs make tgt.length as vt then DurationToQuantity(vs, vt)
  "Encounter-reason" : for src.reason make tgt.reason
  "Encounter-indication" : for src.diagnosis as d then encDiagnosis(d, tgt)
  "Encounter-hospitalization" : for src.hospitalization as vs0 make tgt.hospitalization as vt0 then
    encHospitaization(vs0, vt0)
  "Encounter-location" : for src.location as vs0 make tgt.location as vt0 then encLocation(vs0, vt0)
  "Encounter-serviceProvider" : for src.serviceProvider make tgt.serviceProvider
  "Encounter-partOf" : for src.partOf make tgt.partOf
endgroup
```

```
group encDiagnosis extends BackboneElement
  input src as source
  input tgt as target

  "Encounter.indication-condition1" : for src.condition as v where src.role.coding.code.contains("AD").not() and
    role.coding.code.contains("DD").not() make tgt.indication = v
  "Encounter.indication-condition2" : for src.condition as v where src.role.coding.code.contains("AD") make
    tgt.hospitalization as h, h.admittingDiagnosis = v
  "Encounter.indication-condition3" : for src.condition as v where src.role.coding.code.contains("DD") make
    tgt.hospitalization as h, h.dischargeDiagnosis = v
endgroup

group encStatusHistory extends BackboneElement
  input src as source
  input tgt as target

  "Encounter.statusHistory-status" : for src.status make tgt.status
  "Encounter.statusHistory-period" : for src.period make tgt.period
endgroup

group encParticipant extends BackboneElement
  input src as source
  input tgt as target

  "Encounter.participant-type" : for src.type make tgt.type
  "Encounter.participant-period" : for src.period make tgt.period
  "Encounter.participant-individual" : for src.individual make tgt.individual
endgroup

group encHospitaization extends BackboneElement
  input src as source
  input tgt as target

  "Encounter.hospitalization-preAdmissionIdentifier" : for src.preAdmissionIdentifier make tgt.preAdmissionIdentifier
  "Encounter.hospitalization-origin" : for src.origin make tgt.origin
  "Encounter.hospitalization-admitSource" : for src.admitSource make tgt.admitSource
  "Encounter.hospitalization-reAdmission" : for src.reAdmission make tgt.reAdmission
  "Encounter.hospitalization-dietPreference" : for src.dietPreference make tgt.dietPreference
  "Encounter.hospitalization-specialCourtesy" : for src.specialCourtesy make tgt.specialCourtesy
  "Encounter.hospitalization-specialArrangement" : for src.specialArrangement make tgt.specialArrangement
  "Encounter.hospitalization-destination" : for src.destination make tgt.destination
  "Encounter.hospitalization-dischargeDisposition" : for src.dischargeDisposition make tgt.dischargeDisposition

endgroup

group encLocation extends BackboneElement
  input src as source
  input tgt as target

  "Encounter.location-location" : for src.location make tgt.location
  "Encounter.location-status" : for src.status make tgt.status
  "Encounter.location-period" : for src.period make tgt.period
endgroup
```

8.11.14.3 R3 Validation Errors

encounter-example | ERROR @ Encounter.extension: The extension http://hl7.org/fhir/StructureDefinition/encounter-relatedCondition is unknown, and not allowed here (src = InstanceValidator)

f202 | ERROR @ Encounter.length: There SHALL be a code if there is a value and it SHALL be an expression of time. If system is present, it SHALL be UCUM. [code.exists() implies ((system = %ucum) and value.exists())] (src = InstanceValidator)

8.12 Resource Account - Content

URL: https://hl7.org/fhir/STU3/account.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

Patient Administration Work Group | Maturity Level: 2 | Trial Use | Compartments: Device, Patient, Practitioner

A financial tool for tracking value accrued for a particular purpose. In the healthcare field, used to track charges for a patient, cost centers, etc.

8.12.1 Scope and Usage

Todo

This resource is referenced by ChargeItem, Encounter and EpisodeOfCare

8.12.2 Resource Content

* Structure

* UML

* XML

* JSON

* Turtle

* R2 Diff

* All

Structure

Name | Flags | Card. | Type | Description & Constraints

Account | | | DomainResource | Tracks balance, charges, for patient or cost center Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension

identifier | ? | 0..* | Identifier | Account number

status | ?!? | 0..1 | code | active | inactive | entered-in-error AccountStatus (Required)

type | ? | 0..1 | CodeableConcept | E.g. patient, expense, depreciation Account Types (Example)

name | ? | 0..1 | string | Human-readable label

subject | ? | 0..1 | Reference(Patient | Device | Practitioner | Location | HealthcareService | Organization) | What is account tied to?

period | ? | 0..1 | Period | Transaction window

active | ? | 0..1 | Period | Time window that transactions may be posted to this account

balance | | 0..1 | Money | How much is in account?

coverage | ? | 0..* | BackboneElement | The party(s) that are responsible for covering the payment of this account, and what order should they be applied to the account

coverage | ? | 1..1 | Reference(Coverage) | The party(s) that are responsible for covering the payment of this account

priority | ? | 0..1 | positiveInt | The priority of the coverage in the context of this account

owner | ? | 0..1 | Reference(Organization) | Who is responsible?

description | ? | 0..1 | string | Explanation of purpose/use

guarantor | | 0..* | BackboneElement | Responsible for the account

party | | 1..1 | Reference(Patient | RelatedPerson | Organization) | Responsible entity

onHold | | 0..1 | boolean | Credit or other hold applied

period | | 0..1 | Period | Guarrantee account during

Documentation for this format

UML Diagram (Legend)

XML Template

```
<Account xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Account number --></identifier>
 <status value="[code]"/><!-- 0..1 active | inactive | entered-in-error -->
 <type><!-- 0..1 CodeableConcept E.g. patient, expense, depreciation --></type>
 <name value="[string]"/><!-- 0..1 Human-readable label -->
 <subject><!-- 0..1 Reference(Patient|Device|Practitioner|Location|
   HealthcareService|Organization) What is account tied to? --></subject>
 <period><!-- 0..1 Period Transaction window --></period>
 <active><!-- 0..1 Period Time window that transactions may be posted to this account --></active>
 <balance><!-- 0..1 Money How much is in account? --></balance>
 <coverage>  <!-- 0..* The party(s) that are responsible for covering the payment of this account, and what order should
     they be applied to the account -->
  <coverage><!-- 1..1 Reference(Coverage) The party(s) that are responsible for covering the payment of this account
    --></coverage>
  <priority value="[positiveInt]"/><!-- 0..1 The priority of the coverage in the context of this account -->
 </coverage>
 <owner><!-- 0..1 Reference(Organization) Who is responsible? --></owner>
 <description value="[string]"/><!-- 0..1 Explanation of purpose/use -->
 <guarantor>  <!-- 0..* Responsible for the account -->
  <party><!-- 1..1 Reference(Patient|RelatedPerson|Organization) Responsible entity --></party>
  <onHold value="[boolean]"/><!-- 0..1 Credit or other hold applied -->
  <period><!-- 0..1 Period Guarrantee account during --></period>
 </guarantor>
</Account>
```

JSON Template

```
{
  "resourceType" : "Account",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Account number
  "status" : "<code>", // active | inactive | entered-in-error
  "type" : { CodeableConcept }, // E.g. patient, expense, depreciation
  "name" : "<string>", // Human-readable label
  "subject" : { Reference(Patient|Device|Practitioner|Location|
   HealthcareService|Organization) }, // What is account tied to?
  "period" : { Period }, // Transaction window
  "active" : { Period }, // Time window that transactions may be posted to this account
  "balance" : { Money }, // How much is in account?
  "coverage" : [{ // The party(s) that are responsible for covering the payment of this account, and what order should
    they be applied to the account
    "coverage" : { Reference(Coverage) }, // R!  The party(s) that are responsible for covering the payment of this
    account
    "priority" : "<positiveInt>" // The priority of the coverage in the context of this account
  }],
  "owner" : { Reference(Organization) }, // Who is responsible?
  "description" : "<string>", // Explanation of purpose/use
  "guarantor" : [{ // Responsible for the account
    "party" : { Reference(Patient|RelatedPerson|Organization) }, // R!  Responsible entity
```

```
    "onHold" : <boolean>, // Credit or other hold applied
    "period" : { Period } // Guarrantee account during
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Account;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Account.identifier [ Identifier ], ... ; # 0..* Account number
  fhir:Account.status [ code ]; # 0..1 active | inactive | entered-in-error
  fhir:Account.type [ CodeableConcept ]; # 0..1 E.g. patient, expense, depreciation
  fhir:Account.name [ string ]; # 0..1 Human-readable label
  fhir:Account.subject [ Reference(Patient|Device|Practitioner|Location|HealthcareService|Organization) ]; # 0..1 What
    is account tied to?
  fhir:Account.period [ Period ]; # 0..1 Transaction window
  fhir:Account.active [ Period ]; # 0..1 Time window that transactions may be posted to this account
  fhir:Account.balance [ Money ]; # 0..1 How much is in account?
  fhir:Account.coverage [ # 0..* The party(s) that are responsible for covering the payment of this account, and what
    order should they be applied to the account
    fhir:Account.coverage.coverage [ Reference(Coverage) ]; # 1..1 The party(s) that are responsible for covering the
    payment of this account
    fhir:Account.coverage.priority [ positiveInt ]; # 0..1 The priority of the coverage in the context of this account
  ], ...;
  fhir:Account.owner [ Reference(Organization) ]; # 0..1 Who is responsible?
  fhir:Account.description [ string ]; # 0..1 Explanation of purpose/use
  fhir:Account.guarantor [ # 0..* Responsible for the account
    fhir:Account.guarantor.party [ Reference(Patient|RelatedPerson|Organization) ]; # 1..1 Responsible entity
    fhir:Account.guarantor.onHold [ boolean ]; # 0..1 Credit or other hold applied
    fhir:Account.guarantor.period [ Period ]; # 0..1 Guarrantee account during
  ], ...;
]
```

Changes since DSTU2

Account
Account.status | Add Binding http://hl7.org/fhir/ValueSet/account-status (required)
Account.period | Added Element
Account.active | Renamed from activePeriod to active
Account.balance | Type changed from Quantity{http://hl7.org/fhir/StructureDefinition/Money} to Money
Account.coverage | Added Element
Account.coverage.coverage | Added Element
Account.coverage.priority | Added Element
Account.guarantor | Added Element
Account.guarantor.party | Added Element
Account.guarantor.onHold | Added Element
Account.guarantor.period | Added Element
Account.currency | deleted
Account.coveragePeriod | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R2 <--> R3 Conversion Maps (status = 1 test that all execute ok. All tests pass round-trip testing and all r3 resources are valid.).

Structure

Name | Flags | Card. | Type | Description & Constraints
Account | | | DomainResource | Tracks balance, charges, for patient or cost center Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | ? | 0..* | Identifier | Account number
status | ?!? | 0..1 | code | active | inactive | entered-in-error AccountStatus (Required)
type | ? | 0..1 | CodeableConcept | E.g. patient, expense, depreciation Account Types (Example)
name | ? | 0..1 | string | Human-readable label
subject | ? | 0..1 | Reference(Patient | Device | Practitioner | Location | HealthcareService | Organization) | What is account tied to?
period | ? | 0..1 | Period | Transaction window
active | ? | 0..1 | Period | Time window that transactions may be posted to this account
balance | | 0..1 | Money | How much is in account?
coverage | ? | 0..* | BackboneElement | The party(s) that are responsible for covering the payment of this account, and what order should they be applied to the account
coverage | ? | 1..1 | Reference(Coverage) | The party(s) that are responsible for covering the payment of this account
priority | ? | 0..1 | positiveInt | The priority of the coverage in the context of this account
owner | ? | 0..1 | Reference(Organization) | Who is responsible?
description | ? | 0..1 | string | Explanation of purpose/use
guarantor | | 0..* | BackboneElement | Responsible for the account
party | | 1..1 | Reference(Patient | RelatedPerson | Organization) | Responsible entity
onHold | | 0..1 | boolean | Credit or other hold applied

period | | 0..1 | Period | Guarrantee account during
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Account xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Account number --></identifier>
 <status value="[code]"/><!-- 0..1 active | inactive | entered-in-error -->
 <type><!-- 0..1 CodeableConcept E.g. patient, expense, depreciation --></type>
 <name value="[string]"/><!-- 0..1 Human-readable label -->
 <subject><!-- 0..1 Reference(Patient|Device|Practitioner|Location|
   HealthcareService|Organization) What is account tied to? --></subject>
 <period><!-- 0..1 Period Transaction window --></period>
 <active><!-- 0..1 Period Time window that transactions may be posted to this account --></active>
 <balance><!-- 0..1 Money How much is in account? --></balance>
 <coverage>  <!-- 0..* The party(s) that are responsible for covering the payment of this account, and what order should
     they be applied to the account -->
  <coverage><!-- 1..1 Reference(Coverage) The party(s) that are responsible for covering the payment of this account
    --></coverage>
  <priority value="[positiveInt]"/><!-- 0..1 The priority of the coverage in the context of this account -->
 </coverage>
 <owner><!-- 0..1 Reference(Organization) Who is responsible? --></owner>
 <description value="[string]"/><!-- 0..1 Explanation of purpose/use -->
 <guarantor>  <!-- 0..* Responsible for the account -->
  <party><!-- 1..1 Reference(Patient|RelatedPerson|Organization) Responsible entity --></party>
  <onHold value="[boolean]"/><!-- 0..1 Credit or other hold applied -->
  <period><!-- 0..1 Period Guarrantee account during --></period>
 </guarantor>
</Account>
```

JSON Template

```
{
  "resourceType" : "Account",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Account number
  "status" : "<code>", // active | inactive | entered-in-error
  "type" : { CodeableConcept }, // E.g. patient, expense, depreciation
  "name" : "<string>", // Human-readable label
  "subject" : { Reference(Patient|Device|Practitioner|Location|
   HealthcareService|Organization) }, // What is account tied to?
  "period" : { Period }, // Transaction window
  "active" : { Period }, // Time window that transactions may be posted to this account
  "balance" : { Money }, // How much is in account?
  "coverage" : [{ // The party(s) that are responsible for covering the payment of this account, and what order should
    they be applied to the account
    "coverage" : { Reference(Coverage) }, // R!  The party(s) that are responsible for covering the payment of this
    account
    "priority" : "<positiveInt>" // The priority of the coverage in the context of this account
  }],
  "owner" : { Reference(Organization) }, // Who is responsible?
  "description" : "<string>", // Explanation of purpose/use
  "guarantor" : [{ // Responsible for the account
    "party" : { Reference(Patient|RelatedPerson|Organization) }, // R!  Responsible entity
    "onHold" : <boolean>, // Credit or other hold applied
    "period" : { Period } // Guarrantee account during
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Account;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Account.identifier [ Identifier ], ... ; # 0..* Account number
  fhir:Account.status [ code ]; # 0..1 active | inactive | entered-in-error
  fhir:Account.type [ CodeableConcept ]; # 0..1 E.g. patient, expense, depreciation
  fhir:Account.name [ string ]; # 0..1 Human-readable label
  fhir:Account.subject [ Reference(Patient|Device|Practitioner|Location|HealthcareService|Organization) ]; # 0..1 What
    is account tied to?
  fhir:Account.period [ Period ]; # 0..1 Transaction window
  fhir:Account.active [ Period ]; # 0..1 Time window that transactions may be posted to this account
  fhir:Account.balance [ Money ]; # 0..1 How much is in account?
  fhir:Account.coverage [ # 0..* The party(s) that are responsible for covering the payment of this account, and what
    order should they be applied to the account
    fhir:Account.coverage.coverage [ Reference(Coverage) ]; # 1..1 The party(s) that are responsible for covering the
    payment of this account
    fhir:Account.coverage.priority [ positiveInt ]; # 0..1 The priority of the coverage in the context of this account
  ], ...;
  fhir:Account.owner [ Reference(Organization) ]; # 0..1 Who is responsible?
  fhir:Account.description [ string ]; # 0..1 Explanation of purpose/use
  fhir:Account.guarantor [ # 0..* Responsible for the account
```

```
    fhir:Account.guarantor.party [ Reference(Patient|RelatedPerson|Organization) ]; # 1..1 Responsible entity
    fhir:Account.guarantor.onHold [ boolean ]; # 0..1 Credit or other hold applied
    fhir:Account.guarantor.period [ Period ]; # 0..1 Guarrantee account during
  ], ...;
]
```

Changes since DSTU2

Account
Account.status | Add Binding http://hl7.org/fhir/ValueSet/account-status (required)
Account.period | Added Element
Account.active | Renamed from activePeriod to active
Account.balance | Type changed from Quantity{http://hl7.org/fhir/StructureDefinition/Money} to Money
Account.coverage | Added Element
Account.coverage.coverage | Added Element
Account.coverage.priority | Added Element
Account.guarantor | Added Element
Account.guarantor.party | Added Element
Account.guarantor.onHold | Added Element
Account.guarantor.period | Added Element
Account.currency | deleted
Account.coveragePeriod | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R2 <--> R3 Conversion Maps (status = 1 test that all execute ok. All tests pass round-trip testing and all r3 resources are valid.).

Alternate definitions: Master Definition (XML, JSON), XML Schema/Schematron (for) + JSON Schema, ShEx (for Turtle)

8.12.2.1 Terminology Bindings

Path | Definition | Type | Reference
Account.status | Indicates whether the account is available to be used. | Required | AccountStatus
Account.type | The usage type of this account, permits categorization of accounts. | Example | Account Types

8.12.3 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
balance | quantity | How much is in account? | Account.balance
identifier | token | Account number | Account.identifier
name | string | Human-readable label | Account.name
owner | reference | Who is responsible? | Account.owner (Organization)
patient | reference | What is account tied to? | Account.subject (Patient)
period | date | Transaction window | Account.period
status | token | active | inactive | entered-in-error | Account.status
subject | reference | What is account tied to? | Account.subject (Practitioner, Organization, Device, Patient, HealthcareService, Location)
type | token | E.g. patient, expense, depreciation | Account.type

8.12.4 Resource Account - Examples

URL: https://hl7.org/fhir/STU3/account-examples.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.12.4 Resource Account - Examples

Patient Administration Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Device, Patient, Practitioner

Example List:

General Account example | example | XML | JSON | Turtle
Example with a Guarantor | ewg | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

8.12.5 Resource Account - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/account-definitions.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.12.5 Resource Account - Detailed Descriptions

Patient Administration Work Group | Maturity Level: 2 | Trial Use | Compartments: Device, Patient, Practitioner

Detailed Descriptions for the elements in the Account resource.

Account
Definition | A financial tool for tracking value accrued for a particular purpose. In the healthcare field, used to track charges for a patient, cost centers, etc.
Control | 1..1
Alternate Names | Cost center; Record
Account.identifier
Definition | Unique identifier used to reference the account. May or may not be intended for human use (e.g. credit card number).
Note | This is a business identifer, not a resource identifier (see discussion)
Control | 0..*
Type | Identifier
Summary | true
Account.status
Definition | Indicates whether the account is presently used/usable or not.
Control | 0..1
Terminology Binding | AccountStatus (Required)
Type | code
Is Modifier | true
Summary | true
Comments | This element is labeled as a modifier because the status contains the codes inactive and entered-in-error that mark the Account as not currently valid.
Account.type
Definition | Categorizes the account for reporting and searching purposes.
Control | 0..1
Terminology Binding | Account Types (Example)
Type | CodeableConcept
Summary | true
Account.name
Definition | Name used for the account when displaying it to humans in reports, etc.
Control | 0..1
Type | string
Alternate Names | Title
Summary | true
Account.subject
Definition | Identifies the patient, device, practitioner, location or other object the account is associated with.
Control | 0..1
Type | Reference(Patient | Device | Practitioner | Location | HealthcareService | Organization)
Alternate Names | target
Summary | true
Comments | Accounts can be applied to non patients for tracking other non patient related activities, such as group services (patients not tracked, and costs charged to another body), or may not be allocated.
Account.period
Definition | Identifies the period of time the account applies to; e.g. accounts created per fiscal year, quarter, etc.
Control | 0..1
Type | Period
Summary | true
Comments | It is possible for transactions relevant to a coverage period to be posted to the account before or after the coverage period, however the services that are being charged for must be within this period.
Account.active
Definition | Indicates the period of time over which the account is allowed to have transactions posted to it. This period may be different to the coveragePeriod which is the duration of time that services may occur.
Control | 0..1
Type | Period
Requirements | Accounts are able to be active only during a specific period, and the dates to which charges should be able to be applied.
Summary | true
Account.balance
Definition | Represents the sum of all credits less all debits associated with the account. Might be positive, zero or negative.
Control | 0..1
Type | Money
Account.coverage
Definition | The party(s) that are responsible for covering the payment of this account, and what order should they be applied to the account.
Control | 0..*
Summary | true
Comments | Typically this may be some form of insurance, internal charges, or self-pay. Local or jurisdicational business rules may determine which coverage covers which types of billable items charged to the account, and in which order. Where the order is important, a local/jurisdicational extension may be defined to specify the order for the type of charge.
Account.coverage.coverage
Definition | The party(s) that are responsible for payment (or part of) of charges applied to this account (including self-pay). A coverage may only be resposible for specific types of charges, and the sequence of the coverages in the account could be important when processing billing.
Control | 1..1
Type | Reference(Coverage)

Summary | true
Account.coverage.priority
Definition | The priority of the coverage in the context of this account.
Control | 0..1
Type | positiveInt
Alternate Names | coordination of benefit priority
Summary | true
Comments | It is common in some jurisdictions for there to be multiple coverages allocated to an account, and a sequence is required to order the settling of the account (often with insurance claiming).
Account.owner
Definition | Indicates the organization, department, etc. with responsibility for the account.
Control | 0..1
Type | Reference(Organization)
Summary | true
Account.description
Definition | Provides additional information about what the account tracks and how it is used.
Control | 0..1
Type | string
Summary | true
Account.guarantor
Definition | Parties financially responsible for the account.
Control | 0..*
Account.guarantor.party
Definition | The entity who is responsible.
Control | 1..1
Type | Reference(Patient | RelatedPerson | Organization)
Account.guarantor.onHold
Definition | A guarantor may be placed on credit hold or otherwise have their role temporarily suspended.
Control | 0..1
Type | boolean
Account.guarantor.period
Definition | The timeframe during which the guarantor accepts responsibility for the account.
Control | 0..1
Type | Period

8.12.6 Resource Account - Mappings

URL: https://hl7.org/fhir/STU3/account-mappings.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.12.6 Resource Account - Mappings

Patient Administration Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Device, Patient, Practitioner

Mappings:

RIM Mapping

Mappings for the account resource.

8.12.6.1 Mappings for RIM Mapping (http://hl7.org/v3)

Account | Account
identifier | .id
status | .statusCode
type | .code
name | .title
subject | .subject
period | .coverage.insurancePolicy.effectiveTime
active | .effectiveTime
balance | .balanceAmt
coverage | .coverage
coverage | .coverage.insurancePolicy.author
priority | .coverage.sequenceNumber
owner | .holder
description | .subjectOf.observationEvent[code="annotation"].value
guarantor | .holder.guarantorRole
party | .holder.guarantorRole[guarantorPerson|guarantorOrganization]
onHold | .holder.guarantorRole.statusCode
period | .holder.guarantorRole.effectiveTime

8.12.7 Resource Account - Design Notes

URL: https://hl7.org/fhir/STU3/account-explanations.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.12.7 Resource Account - Design Notes

to do

8.12.8 Resource Account - Extensions & Profiles

URL: https://hl7.org/fhir/STU3/account-profiles.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.12.8 Resource Account - Extensions & Profiles

Patient Administration Work Group | Maturity Level: N/A | Ballot Status: Informative

This table lists profiles and extensions for the Account resource. For background information, see Profiling Resources and Extensibility. Additional profiles and resources may be found in published Implementation Guides , or in the Conformance resource registry .

8.12.8.1 Profiles

No Profiles defined for this resource

8.12.8.2 Extensions

No Extensions defined for this resource

Extensions that reference this resource:

No Extensions refer to this resource

8.12.8.3 Search Extensions

No Search Extensions defined for this resource

8.12.9 Resource Account - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/account-version-maps.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.12.9 Resource Account - Detailed Descriptions

Patient Administration Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Device, Patient, Practitioner

R2 : R3 Converson maps for Account.

Functional status for this map: 1 test that all execute ok. All tests pass round-trip testing and all r3 resources are valid. (see documentation)

8.12.9.1 R2 to R3

```
map "http://hl7.org/fhir/StructureMap/Account2to3" = "R2 to R3 Conversions for Account"

uses "http://hl7.org/fhir/DSTU2/StructureDefinition/Account" alias AccountR2 as source
uses "http://hl7.org/fhir/StructureDefinition/Account" alias Account as target

imports "http://hl7.org/fhir/StructureMap/*2to3"

group for type+types  Account extends DomainResource
  input src : AccountR2 as source
  input tgt : Account as target

  "Account-identifier" : for src.identifier make tgt.identifier
  "Account-name" : for src.name make tgt.name
  "Account-type" : for src.type make tgt.type
  "Account-status" : for src.status make tgt.status
  "Account-active" : for src.activePeriod make tgt.active
  "Account-balance" : for src.balance make tgt.balance
//  "Account-coveragePeriod" : for src.coveragePeriod make tgt.coveragePeriod - moved into coverage resource
  "Account-subject" : for src.subject make tgt.subject
  "Account-owner" : for src.owner make tgt.owner
  "Account-description" : for src.description make tgt.description

endgroup
```

8.12.9.2 R3 to R2

```
map "http://hl7.org/fhir/StructureMap/Account3to2" = "R3 to R2 Conversion for Account"

uses "http://hl7.org/fhir/StructureDefinition/Account" alias Account as source
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/Account" alias AccountR2 as target
```

```
imports "http://hl7.org/fhir/StructureMap/*3to2"

group for type+types  Account extends DomainResource
  input src : Account as source
  input tgt : AccountR2 as target

  "Account-identifier" : for src.identifier make tgt.identifier
  "Account-name" : for src.name make tgt.name
  "Account-type" : for src.type make tgt.type
  "Account-status" : for src.status make tgt.status
  "Account-activePeriod" : for src.active make tgt.activePeriod
  "Account-balance" : for src.balance make tgt.balance
//  "Account-coveragePeriod" : for src.coveragePeriod make tgt.coveragePeriod - moved into coverage resource
  "Account-subject" : for src.subject make tgt.subject
  "Account-owner" : for src.owner make tgt.owner
  "Account-description" : for src.description make tgt.description

endgroup
```

8.12.9.3 R3 Validation Errors

8.13 Resource Flag - Content

URL: https://hl7.org/fhir/STU3/flag.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

Patient Care Work Group | Maturity Level: 1 | Trial Use | Compartments: Device, Patient, Practitioner

Prospective warnings of potential issues when providing care to the patient.

8.13.1 Scope and Usage

A flag is a warning or notification of some sort presented to the user - who may be a clinician or some other person involve in patient care. It usually represents something of sufficient significance to be warrant a special display of some sort - rather than just a note in the resource. A flag has a subject representing the resource that will trigger its display. This subject can be of different types, as described in the examples below:

* A note that a patient has an overdue account, which the provider may wish to discuss with them - in case of hardship for example (subject = Patient)

* An outbreak of Ebola in a particular region (subject=Location) so that all patients from that region have a higher risk of having that condition

* A particular provider is unavailable for referrals over a given period (subject = Practitioner)

* A patient who is enrolled in a clinical trial (subject=Group)

* Special guidance or caveats to be aware of when following a protocol (subject=PlanDefinition)

* Warnings about using a drug in a formulary requires special approval (subject=Medication)

* etc.

A flag is typically presented as a label in a prominent location in the record to notify the clinician of the potential issues, though it may also appear in other contexts; e.g. notes applicable to a radiology technician, or to a clinician performing a home visit. For patients, the information in the flag will often be derived from the record, and therefore, for a thorough and careful clinician, who has the time to review the notes will be redundant. However, given the volume of information frequently found in patients' records and the potentially serious consequences of losing sight of some facts, this redundancy is deemed appropriate. As well, some flags may reflect information not captured by any other resource in the record. (E.g. "Patient has large dog at home")

In line with its purpose, a flag is concise, highlighting a small set of high-priority issues among the much larger set of data in the chart. Readers who want more detail should consult the chart or other source of information. Caution should be exercised in creating Flag instances. If entries are created for information that could be gleaned in a sufficiently timely fashion by reviewing the patient record, the flag list will itself become overwhelming and will cease to serve its intended purpose.

Flags are expected to persist in a record for some period of time and are, at most, targeted to particular types of practitioners or to practitioners in particular system.

Examples of Patient related issues that might appear in flags:

* Risks to the patient (functional risk of falls, spousal restraining order, latex allergy)

* Patient's needs for special accommodations (hard of hearing, need for easy-open caps)

* Risks to providers (dog in house, patient may bite, infection control precautions)

* Administrative concerns (incomplete information, pre-payment required due to credit risk)

Examples of issues that should not appear only in flags:

* Potential allergy or drug interaction to planned therapy (use DetectedIssue)

* Known adverse reaction to a substance (use AllergyIntolerance)

Note that we include "latex allergy" in the "in scope" list, and "allergy" in the "not in scope" list. The Flag resource is not designed to replace the normal order checking process, and one should not expect to see all allergies in Flags. However, if there is an activity that might occur prior to careful evaluation of the record (e.g. donning of latex gloves) and that activity might pose a risk to the patient, that is the sort of eventuality the Flag is intended to support.

Specific guidelines about what type of information is appropriate to expose using Flag, as well as what categories of individuals should see particular flags, will vary by interoperability community.

8.13.2 Boundaries and Relationships

Flags may highlight a highly condensed view of information found in the AllergyIntolerance, Condition, Observation, Procedure and possibly other resources. A common extension allows the linkage of a Flag to the supporting detail resource. The purpose of these other resources is to provide detailed clinical information. The purpose of a Flag is to alert practitioners to information that is important to influence their interaction with a Patient prior to detailed review of the record.

Flags are not used to convey information to a specific individual or organization (e.g. an abnormal lab result reported to the ordering clinician, reporting of an adverse reaction to a regulatory authority). These are handled using the CommunicationRequest and the Communication resources.

Flags are not raised as a result of a reported or proposed action (e.g. drug-drug interactions, duplicate therapy warnings). These would be handled using DetectedIssue.

8.13.3 Resource Content

* Structure

* UML

* XML

* JSON

* Turtle

* R2 Diff

* All

Structure

Name | Flags | Card. | Type | Description & Constraints
Flag | | | DomainResource | Key information to flag to healthcare providers Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | ? | 0..* | Identifier | Business identifier
status | ?!? | 1..1 | code | active | inactive | entered-in-error FlagStatus (Required)
category | ? | 0..1 | CodeableConcept | Clinical, administrative, etc. Flag Category (Example)
code | ? | 1..1 | CodeableConcept | Coded or textual message to display to user Flag Code (Example)
subject | ? | 1..1 | Reference(Patient | Location | Group | Organization | Practitioner | PlanDefinition | Medication | Procedure) | Who/What is flag about?
period | ? | 0..1 | Period | Time period when flag is active
encounter | ? | 0..1 | Reference(Encounter) | Alert relevant during encounter
author | ? | 0..1 | Reference(Device | Organization | Patient | Practitioner) | Flag creator
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Flag xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Business identifier --></identifier>
 <status value="[code]"/><!-- 1..1 active | inactive | entered-in-error -->
 <category><!-- 0..1 CodeableConcept Clinical, administrative, etc. --></category>
 <code><!-- 1..1 CodeableConcept Coded or textual message to display to user --></code>
 <subject><!-- 1..1 Reference(Patient|Location|Group|Organization|Practitioner|
   PlanDefinition|Medication|Procedure) Who/What is flag about? --></subject>
 <period><!-- 0..1 Period Time period when flag is active --></period>
 <encounter><!-- 0..1 Reference(Encounter) Alert relevant during encounter --></encounter>
 <author><!-- 0..1 Reference(Device|Organization|Patient|Practitioner) Flag creator --></author>
</Flag>
```

JSON Template

```
{
  "resourceType" : "Flag",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Business identifier
  "status" : "<code>", // R!  active | inactive | entered-in-error
  "category" : { CodeableConcept }, // Clinical, administrative, etc.
  "code" : { CodeableConcept }, // R!  Coded or textual message to display to user
  "subject" : { Reference(Patient|Location|Group|Organization|Practitioner|
   PlanDefinition|Medication|Procedure) }, // R!  Who/What is flag about?
  "period" : { Period }, // Time period when flag is active
  "encounter" : { Reference(Encounter) }, // Alert relevant during encounter
  "author" : { Reference(Device|Organization|Patient|Practitioner) } // Flag creator
```

```
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Flag;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Flag.identifier [ Identifier ], ... ; # 0..* Business identifier
  fhir:Flag.status [ code ]; # 1..1 active | inactive | entered-in-error
  fhir:Flag.category [ CodeableConcept ]; # 0..1 Clinical, administrative, etc.
  fhir:Flag.code [ CodeableConcept ]; # 1..1 Coded or textual message to display to user
  fhir:Flag.subject [ Reference(Patient|Location|Group|Organization|Practitioner|PlanDefinition|Medication|
  Procedure) ]; # 1..1 Who/What is flag about?
  fhir:Flag.period [ Period ]; # 0..1 Time period when flag is active
  fhir:Flag.encounter [ Reference(Encounter) ]; # 0..1 Alert relevant during encounter
  fhir:Flag.author [ Reference(Device|Organization|Patient|Practitioner) ]; # 0..1 Flag creator
]
```

Changes since DSTU2

Flag

Flag.subject | Add Reference(PlanDefinition), Add Reference(Medication), Add Reference(Procedure)

See the Full Difference for further information

This analysis is available as XML or JSON.

See R2 <--> R3 Conversion Maps (status = 3 tests that all execute ok. All tests pass round-trip testing and all r3 resources are valid.).

Structure

Name | Flags | Card. | Type | Description & Constraints

Flag | | | DomainResource | Key information to flag to healthcare providers Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension

identifier | ? | 0..* | Identifier | Business identifier

status | ?!? | 1..1 | code | active | inactive | entered-in-error FlagStatus (Required)

category | ? | 0..1 | CodeableConcept | Clinical, administrative, etc. Flag Category (Example)

code | ? | 1..1 | CodeableConcept | Coded or textual message to display to user Flag Code (Example)

subject | ? | 1..1 | Reference(Patient | Location | Group | Organization | Practitioner | PlanDefinition | Medication | Procedure) | Who/What is flag about?

period | ? | 0..1 | Period | Time period when flag is active

encounter | ? | 0..1 | Reference(Encounter) | Alert relevant during encounter

author | ? | 0..1 | Reference(Device | Organization | Patient | Practitioner) | Flag creator

Documentation for this format

UML Diagram (Legend)

XML Template

```
<Flag xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Business identifier --></identifier>
 <status value="[code]"/><!-- 1..1 active | inactive | entered-in-error -->
 <category><!-- 0..1 CodeableConcept Clinical, administrative, etc. --></category>
 <code><!-- 1..1 CodeableConcept Coded or textual message to display to user --></code>
 <subject><!-- 1..1 Reference(Patient|Location|Group|Organization|Practitioner|
   PlanDefinition|Medication|Procedure) Who/What is flag about? --></subject>
 <period><!-- 0..1 Period Time period when flag is active --></period>
 <encounter><!-- 0..1 Reference(Encounter) Alert relevant during encounter --></encounter>
 <author><!-- 0..1 Reference(Device|Organization|Patient|Practitioner) Flag creator --></author>
</Flag>
```

JSON Template

```
{
  "resourceType" : "Flag",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Business identifier
  "status" : "<code>", // R!  active | inactive | entered-in-error
  "category" : { CodeableConcept }, // Clinical, administrative, etc.
  "code" : { CodeableConcept }, // R!  Coded or textual message to display to user
  "subject" : { Reference(Patient|Location|Group|Organization|Practitioner|
   PlanDefinition|Medication|Procedure) }, // R!  Who/What is flag about?
  "period" : { Period }, // Time period when flag is active
  "encounter" : { Reference(Encounter) }, // Alert relevant during encounter
  "author" : { Reference(Device|Organization|Patient|Practitioner) } // Flag creator
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Flag;
```

```
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Flag.identifier [ Identifier ], ... ; # 0..* Business identifier
  fhir:Flag.status [ code ]; # 1..1 active | inactive | entered-in-error
  fhir:Flag.category [ CodeableConcept ]; # 0..1 Clinical, administrative, etc.
  fhir:Flag.code [ CodeableConcept ]; # 1..1 Coded or textual message to display to user
  fhir:Flag.subject [ Reference(Patient|Location|Group|Organization|Practitioner|PlanDefinition|Medication|
  Procedure) ]; # 1..1 Who/What is flag about?
  fhir:Flag.period [ Period ]; # 0..1 Time period when flag is active
  fhir:Flag.encounter [ Reference(Encounter) ]; # 0..1 Alert relevant during encounter
  fhir:Flag.author [ Reference(Device|Organization|Patient|Practitioner) ]; # 0..1 Flag creator
]
```

Changes since DSTU2

Flag

Flag.subject | Add Reference(PlanDefinition), Add Reference(Medication), Add Reference(Procedure)

See the Full Difference for further information

This analysis is available as XML or JSON.

See R2 <--> R3 Conversion Maps (status = 3 tests that all execute ok. All tests pass round-trip testing and all r3 resources are valid.).

Alternate definitions: Master Definition (XML, JSON), XML Schema/Schematron (for) + JSON Schema, ShEx (for Turtle)

8.13.3.1 Terminology Bindings

Path | Definition | Type | Reference

Flag.status | Indicates whether this flag is active and needs to be displayed to a user, or whether it is no longer needed or entered in error. | Required | FlagStatus

Flag.category | A general category for flags for filtering/display purposes. | Example | Flag Category

Flag.code | Detail codes identifying specific flagged issues. | Example | Flag Code

8.13.4 Notes

The Flag resource is sometimes used as "patient notes" and MAY be used to warn of issues such as:

* Issues that impact on the patient's ability to receive/respond to care the care provision process itself (e.g., poor language comprehension, low compliance expected)

* Issues that impact on the ability to provide care (e.g., patient has a big dog at home)

* Financial matters (e.g., patient is a bad debtor)

8.13.5 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common

author | reference | Flag creator | Flag.author (Practitioner, Organization, Device, Patient)

date | date | Time period when flag is active | Flag.period | 18 Resources

encounter | reference | Alert relevant during encounter | Flag.encounter (Encounter) | 12 Resources

identifier | token | Business identifier | Flag.identifier

patient | reference | The identity of a subject to list flags for | Flag.subject (Patient) | 31 Resources

subject | reference | The identity of a subject to list flags for | Flag.subject (Practitioner, Group, Organization, Medication, Patient, PlanDefinition, Procedure, Location)

8.13.6 Resource Flag - Examples

URL: https://hl7.org/fhir/STU3/flag-examples.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.13.6 Resource Flag - Examples

Patient Care Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Device, Patient, Practitioner

Example List:

Example of flag | example | XML | JSON | Turtle

Encounter Specific Alert | example-encounter | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

8.13.7 Resource Flag - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/flag-definitions.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.13.7 Resource Flag - Detailed Descriptions

Patient Care Work Group | Maturity Level: 1 | Trial Use | Compartments: Device, Patient, Practitioner

Detailed Descriptions for the elements in the Flag resource.

Flag
Definition | Prospective warnings of potential issues when providing care to the patient.
Control | 1..1
Alternate Names | Barriers to Care; Alert; Warning
Flag.identifier
Definition | Identifier assigned to the flag for external use (outside the FHIR environment).
Note | This is a business identifer, not a resource identifier (see discussion)
Control | 0..*
Type | Identifier
Summary | true
Flag.status
Definition | Supports basic workflow.
Control | 1..1
Terminology Binding | FlagStatus (Required)
Type | code
Is Modifier | true
Summary | true
Comments | This element is labeled as a modifier because the status contains codes that mark the resource as not currently valid.
Flag.category
Definition | Allows an flag to be divided into different categories like clinical, administrative etc. Intended to be used as a means of filtering which flags are displayed to particular user or in a given context.
Control | 0..1
Terminology Binding | Flag Category (Example)
Type | CodeableConcept
Summary | true
Flag.code
Definition | The coded value or textual component of the flag to display to the user.
Control | 1..1
Terminology Binding | Flag Code (Example)
Type | CodeableConcept
Summary | true
Comments | If non coded, use CodeableConcept.text. This element should always be included in the narrative.
Flag.subject
Definition | The patient, location, group , organization , or practitioner, etc. this is about record this flag is associated with.
Control | 1..1
Type | Reference(Patient | Location | Group | Organization | Practitioner | PlanDefinition | Medication | Procedure)
Summary | true
Flag.period
Definition | The period of time from the activation of the flag to inactivation of the flag. If the flag is active, the end of the period should be unspecified.
Control | 0..1
Type | Period
Summary | true
Flag.encounter
Definition | This alert is only relevant during the encounter.
Control | 0..1
Type | Reference(Encounter)
Summary | true
Comments | If both Flag.encounter and Flag.period are valued, then Flag.period.start shall not be before Encounter.period.start and Flag.period.end shall not be after Encounter.period.end.
Flag.author
Definition | The person, organization or device that created the flag.
Control | 0..1
Type | Reference(Device | Organization | Patient | Practitioner)
Summary | true

8.13.8 Resource Flag - Mappings

URL: https://hl7.org/fhir/STU3/flag-mappings.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.13.8 Resource Flag - Mappings

Patient Care Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Device, Patient, Practitioner

Mappings:

RIM Mapping

Mappings for the flag resource.

8.13.8.1 Mappings for RIM Mapping (http://hl7.org/v3)

Flag | Observation[classCode=ISSUE, moodCode=EVN]
identifier | .id
status | .status
category | .code
code | .value
subject | .participation[typeCode=SBJ].role[classCode=PAT | LOCE | ASSIGNED]
period
encounter
author | .participation[typeCode=AUT].role

8.13.9 Resource Flag - Design Notes

URL: https://hl7.org/fhir/STU3/flag-explanations.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.13.9 Resource Flag - Design Notes

to do

8.13.10 Resource Flag - Extensions & Profiles

URL: https://hl7.org/fhir/STU3/flag-profiles.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.13.10 Resource Flag - Extensions & Profiles

Patient Care Work Group | Maturity Level: N/A | Ballot Status: Informative

This table lists profiles and extensions for the Flag resource. For background information, see Profiling Resources and Extensibility. Additional profiles and resources may be found in published Implementation Guides , or in the Conformance resource registry .

8.13.10.1 Profiles

No Profiles defined for this resource

8.13.10.2 Extensions

flag-detail | detail | for Flag HL7 Extensions
flag-priority | priority | for Flag HL7 Extensions

Extensions that reference this resource:

No Extensions refer to this resource

8.13.10.3 Search Extensions

No Search Extensions defined for this resource

8.13.11 Flag HL7 Extensions

URL: https://hl7.org/fhir/STU3/flag-extensions.html

Defines common extensions used with or related to the Flag resource

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.13.11 Flag HL7 Extensions

(No assigned work group) | Maturity Level: N/A | Ballot Status: Informative

8.13.11.1 Content

Extensions:
flag-detail | detail : Points to the Observation, AllergyIntolerance or other record that provides additional supporting information about this flag.
flag-priority | priority : A code that identifies the priority of the alert, for example the Alert Priority flags column in IHE PCD TF 2 Table B.8-4.

8.13.12 Resource Flag - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/flag-version-maps.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.13.12 Resource Flag - Detailed Descriptions

Patient Care Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Device, Patient, Practitioner

R2 : R3 Converson maps for Flag.

Functional status for this map: 3 tests that all execute ok. All tests pass round-trip testing and all r3 resources are valid. (see documentation)

8.13.12.1 R2 to R3

```
map "http://hl7.org/fhir/StructureMap/Flag2to3" = "R2 to R3 Conversions for Flag"

uses "http://hl7.org/fhir/DSTU2/StructureDefinition/Flag" alias FlagR2 as source
uses "http://hl7.org/fhir/StructureDefinition/Flag" alias Flag as target

imports "http://hl7.org/fhir/StructureMap/*2to3"

group for type+types  Flag extends DomainResource
  input src : FlagR2 as source
  input tgt : Flag as target

  "Flag-identifier" : for src.identifier make tgt.identifier
  "Flag-category" : for src.category make tgt.category
  "Flag-status" : for src.status make tgt.status
  "Flag-period" : for src.period make tgt.period
  "Flag-subject" : for src.subject make tgt.subject
  "Flag-encounter" : for src.encounter make tgt.encounter
  "Flag-author" : for src.author make tgt.author
  "Flag-code" : for src.code make tgt.code

endgroup
```

8.13.12.2 R3 to R2

```
map "http://hl7.org/fhir/StructureMap/Flag3to2" = "R3 to R2 Conversion for Flag"

uses "http://hl7.org/fhir/StructureDefinition/Flag" alias Flag as source
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/Flag" alias FlagR2 as target

imports "http://hl7.org/fhir/StructureMap/*3to2"

group for type+types  Flag extends DomainResource
  input src : Flag as source
  input tgt : FlagR2 as target

  "Flag-identifier" : for src.identifier make tgt.identifier
  "Flag-category" : for src.category make tgt.category
  "Flag-status" : for src.status make tgt.status
  "Flag-period" : for src.period make tgt.period
  "Flag-subject" : for src.subject make tgt.subject
  "Flag-encounter" : for src.encounter make tgt.encounter
  "Flag-author" : for src.author make tgt.author
  "Flag-code" : for src.code make tgt.code

endgroup
```

8.13.12.3 R3 Validation Errors

8.14 Resource Device - Content

URL: https://hl7.org/fhir/STU3/device.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

Orders and Observations Work Group | Maturity Level: 2 | Trial Use | Compartments: Device

This resource identifies an instance or a type of a manufactured item that is used in the provision of healthcare without being substantially changed through that activity. The device may be a medical or non-medical device. Medical devices include durable (reusable) medical equipment, implantable devices, as well as disposable equipment used for diagnostic, treatment, and research for healthcare and public health. Non-medical devices may include items such as a machine, cellphone, computer, application, etc.

8.14.1 Scope and Usage

This resource is an administrative resource that tracks individual device types or instances of a device and their location. It is referenced by other resource for recording which device performed an action such as a procedure or an observation. It is also referenced when prescribing and dispensing devices for patient use or for ordering supplies. It is used to record and transmit Unique Device Identifer (UDI) information about a device such as a patient's implant.

8.14.2 Boundaries and Relationships

These are the device related resources

* Device (this resource)

* DeviceMetric - Describes a measurement, calculation or setting capability of a medical device.

* DeviceComponent that the DeviceMetric is part of. This can be a DeviceComponent of any kind like a VirtualMedicalDevice (VMD), a MedicalDeviceSystem (MDS) , or a Channel

In FHIR, the "Device" is the "administrative" resource for the device (it does not change much and has manufacturer information etc.), whereas the DeviceComponent and DeviceMetric (which is really a kind of DeviceComponent) model the physical part, including operation status and is much more volatile. The physical composition of a Device is done by the DeviceComponents pointing to their "parent" component using DeviceComponent.parent, e.g. channel to VMD and VMD to the MDS. All components point to the "logical" Device they belong to, using DeviceComponent.source. Similarly, if for example you want to express which logical device your observation came from, you can just use Device, but if you need to be more specific observation can point to DeviceMetric which in turn points to the "logical" Device it belongs to, using DeviceMetric.source.

Devices differ from medications because they are not "used up" - they remain active in a patient in an ongoing fashion. However, the specific boundary between medications and devices is defined at the implementation level and this standard does not enforce a boundary with the exception of devices that are implanted in a patient. The Medication resource should not be used to represent implanted devices.

This resource is referenced by Account, AdverseEvent, Appointment, AppointmentResponse, AuditEvent, ChargeItem, Claim, Communication, CommunicationRequest, Composition, Consent, Contract, DetectedIssue, DeviceComponent, DeviceMetric, DeviceRequest, DeviceUseStatement, DiagnosticReport, DocumentManifest, DocumentReference, ExplanationOfBenefit, Flag, Group, GuidanceResponse, ImagingManifest, List, Media, MedicationAdministration, MedicationDispense, MedicationRequest, MessageHeader, Observation, Procedure, ProcedureRequest, Provenance, QuestionnaireResponse, ReferralRequest, RequestGroup, RiskAssessment, Schedule, Sequence, Specimen, SupplyDelivery, SupplyRequest and Task

8.14.3 Resource Content

* Structure

* UML

* XML

* JSON

* Turtle

* R2 Diff

* All

Structure

Name | Flags | Card. | Type | Description & Constraints
Device | | | DomainResource | Item used in healthcare Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | | 0..* | Identifier | Instance identifier
udi | ? | 0..1 | BackboneElement | Unique Device Identifier (UDI) Barcode string
deviceIdentifier | ? | 0..1 | string | Mandatory fixed portion of UDI
name | ? | 0..1 | string | Device Name as appears on UDI label
jurisdiction | | 0..1 | uri | Regional UDI authority
carrierHRF | ? | 0..1 | string | UDI Human Readable Barcode String
carrierAIDC | ? | 0..1 | base64Binary | UDI Machine Readable Barcode String
issuer | | 0..1 | uri | UDI Issuing Organization
entryType | | 0..1 | code | barcode | rfid | manual + UDIEntryType (Required)
status | ?!? | 0..1 | code | active | inactive | entered-in-error | unknown FHIRDeviceStatus (Required)
type | | 0..1 | CodeableConcept | What kind of device this is FHIR Device Types (Example)
lotNumber | | 0..1 | string | Lot number of manufacture
manufacturer | | 0..1 | string | Name of device manufacturer
manufactureDate | | 0..1 | dateTime | Date when the device was made
expirationDate | | 0..1 | dateTime | Date and time of expiry of this device (if applicable)
model | | 0..1 | string | Model id assigned by the manufacturer
version | | 0..1 | string | Version number (i.e. software)
patient | | 0..1 | Reference(Patient) | Patient to whom Device is affixed
owner | | 0..1 | Reference(Organization) | Organization responsible for device
contact | | 0..* | ContactPoint | Details for human/organization for support
location | | 0..1 | Reference(Location) | Where the resource is found
url | | 0..1 | uri | Network address to contact device
note | | 0..* | Annotation | Device notes and comments
safety | ? | 0..* | CodeableConcept | Safety Characteristics of Device DeviceSafety (Example)
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Device xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Instance identifier --></identifier>
 <udi>  <!-- 0..1 Unique Device Identifier (UDI) Barcode string -->
  <deviceIdentifier value="[string]"/><!-- 0..1 Mandatory fixed portion of UDI -->
  <name value="[string]"/><!-- 0..1 Device Name as appears on UDI label -->
  <jurisdiction value="[uri]"/><!-- 0..1 Regional UDI authority -->
  <carrierHRF value="[string]"/><!-- 0..1 UDI Human Readable Barcode String -->
  <carrierAIDC value="[base64Binary]"/><!-- 0..1 UDI Machine Readable Barcode String -->
  <issuer value="[uri]"/><!-- 0..1 UDI Issuing Organization -->
  <entryType value="[code]"/><!-- 0..1 barcode | rfid | manual + -->
 </udi>
 <status value="[code]"/><!-- 0..1 active | inactive | entered-in-error | unknown -->
 <type><!-- 0..1 CodeableConcept What kind of device this is --></type>
 <lotNumber value="[string]"/><!-- 0..1 Lot number of manufacture -->
 <manufacturer value="[string]"/><!-- 0..1 Name of device manufacturer -->
 <manufactureDate value="[dateTime]"/><!-- 0..1 Date when the device was made -->
 <expirationDate value="[dateTime]"/><!-- 0..1 Date and time of expiry of this device (if applicable) -->
 <model value="[string]"/><!-- 0..1 Model id assigned by the manufacturer -->
 <version value="[string]"/><!-- 0..1 Version number (i.e. software) -->
 <patient><!-- 0..1 Reference(Patient) Patient to whom Device is affixed --></patient>
 <owner><!-- 0..1 Reference(Organization) Organization responsible for device --></owner>
 <contact><!-- 0..* ContactPoint Details for human/organization for support --></contact>
 <location><!-- 0..1 Reference(Location) Where the resource is found --></location>
 <url value="[uri]"/><!-- 0..1 Network address to contact device -->
 <note><!-- 0..* Annotation Device notes and comments --></note>
 <safety><!-- 0..* CodeableConcept Safety Characteristics of Device --></safety>
</Device>
```

JSON Template

```
{
  "resourceType" : "Device",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Instance identifier
  "udi" : { // Unique Device Identifier (UDI) Barcode string
    "deviceIdentifier" : "<string>", // Mandatory fixed portion of UDI
    "name" : "<string>", // Device Name as appears on UDI label
    "jurisdiction" : "<uri>", // Regional UDI authority
    "carrierHRF" : "<string>", // UDI Human Readable Barcode String
    "carrierAIDC" : "<base64Binary>", // UDI Machine Readable Barcode String
    "issuer" : "<uri>", // UDI Issuing Organization
    "entryType" : "<code>" // barcode | rfid | manual +
  },
  "status" : "<code>", // active | inactive | entered-in-error | unknown
  "type" : { CodeableConcept }, // What kind of device this is
  "lotNumber" : "<string>", // Lot number of manufacture
  "manufacturer" : "<string>", // Name of device manufacturer
  "manufactureDate" : "<dateTime>", // Date when the device was made
  "expirationDate" : "<dateTime>", // Date and time of expiry of this device (if applicable)
  "model" : "<string>", // Model id assigned by the manufacturer
  "version" : "<string>", // Version number (i.e. software)
  "patient" : { Reference(Patient) }, // Patient to whom Device is affixed
  "owner" : { Reference(Organization) }, // Organization responsible for device
  "contact" : [{ ContactPoint }], // Details for human/organization for support
  "location" : { Reference(Location) }, // Where the resource is found
  "url" : "<uri>", // Network address to contact device
  "note" : [{ Annotation }], // Device notes and comments
  "safety" : [{ CodeableConcept }] // Safety Characteristics of Device
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Device;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Device.identifier [ Identifier ], ... ; # 0..* Instance identifier
  fhir:Device.udi [ # 0..1 Unique Device Identifier (UDI) Barcode string
    fhir:Device.udi.deviceIdentifier [ string ]; # 0..1 Mandatory fixed portion of UDI
    fhir:Device.udi.name [ string ]; # 0..1 Device Name as appears on UDI label
    fhir:Device.udi.jurisdiction [ uri ]; # 0..1 Regional UDI authority
    fhir:Device.udi.carrierHRF [ string ]; # 0..1 UDI Human Readable Barcode String
    fhir:Device.udi.carrierAIDC [ base64Binary ]; # 0..1 UDI Machine Readable Barcode String
    fhir:Device.udi.issuer [ uri ]; # 0..1 UDI Issuing Organization
    fhir:Device.udi.entryType [ code ]; # 0..1 barcode | rfid | manual +
  ];
  fhir:Device.status [ code ]; # 0..1 active | inactive | entered-in-error | unknown
  fhir:Device.type [ CodeableConcept ]; # 0..1 What kind of device this is
  fhir:Device.lotNumber [ string ]; # 0..1 Lot number of manufacture
  fhir:Device.manufacturer [ string ]; # 0..1 Name of device manufacturer
  fhir:Device.manufactureDate [ dateTime ]; # 0..1 Date when the device was made
  fhir:Device.expirationDate [ dateTime ]; # 0..1 Date and time of expiry of this device (if applicable)
```

```
  fhir:Device.model [ string ]; # 0..1 Model id assigned by the manufacturer
  fhir:Device.version [ string ]; # 0..1 Version number (i.e. software)
  fhir:Device.patient [ Reference(Patient) ]; # 0..1 Patient to whom Device is affixed
  fhir:Device.owner [ Reference(Organization) ]; # 0..1 Organization responsible for device
  fhir:Device.contact [ ContactPoint ], ... ; # 0..* Details for human/organization for support
  fhir:Device.location [ Reference(Location) ]; # 0..1 Where the resource is found
  fhir:Device.url [ uri ]; # 0..1 Network address to contact device
  fhir:Device.note [ Annotation ], ... ; # 0..* Device notes and comments
  fhir:Device.safety [ CodeableConcept ], ... ; # 0..* Safety Characteristics of Device
]
```

Changes since DSTU2

Device
Device.udi | Type changed from string to BackboneElement
Device.udi.deviceIdentifier | Added Element
Device.udi.name | Added Element
Device.udi.jurisdiction | Added Element
Device.udi.carrierHRF | Added Element
Device.udi.carrierAIDC | Added Element
Device.udi.issuer | Added Element
Device.udi.entryType | Added Element
Device.status | Change value set from http://hl7.org/fhir/ValueSet/devicestatus to http://hl7.org/fhir/ValueSet/device-status
Device.type | Min Cardinality changed from 1 to 0
Device.expirationDate | Renamed from expiry to expirationDate
Device.safety | Added Element

See the Full Difference for further information

This analysis is available as XML or JSON.

See R2 <--> R3 Conversion Maps (status = 5 tests that all execute ok. 2 fail round-trip testing and all r3 resources are valid.).

Structure

Name | Flags | Card. | Type | Description & Constraints
Device | | | DomainResource | Item used in healthcare Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | | 0..* | Identifier | Instance identifier
udi | ? | 0..1 | BackboneElement | Unique Device Identifier (UDI) Barcode string
deviceIdentifier | ? | 0..1 | string | Mandatory fixed portion of UDI
name | ? | 0..1 | string | Device Name as appears on UDI label
jurisdiction | | 0..1 | uri | Regional UDI authority
carrierHRF | ? | 0..1 | string | UDI Human Readable Barcode String
carrierAIDC | ? | 0..1 | base64Binary | UDI Machine Readable Barcode String
issuer | | 0..1 | uri | UDI Issuing Organization
entryType | | 0..1 | code | barcode | rfid | manual + UDIEntryType (Required)
status | ?!? | 0..1 | code | active | inactive | entered-in-error | unknown FHIRDeviceStatus (Required)
type | | 0..1 | CodeableConcept | What kind of device this is FHIR Device Types (Example)
lotNumber | | 0..1 | string | Lot number of manufacture
manufacturer | | 0..1 | string | Name of device manufacturer
manufactureDate | | 0..1 | dateTime | Date when the device was made
expirationDate | | 0..1 | dateTime | Date and time of expiry of this device (if applicable)
model | | 0..1 | string | Model id assigned by the manufacturer
version | | 0..1 | string | Version number (i.e. software)
patient | | 0..1 | Reference(Patient) | Patient to whom Device is affixed
owner | | 0..1 | Reference(Organization) | Organization responsible for device
contact | | 0..* | ContactPoint | Details for human/organization for support
location | | 0..1 | Reference(Location) | Where the resource is found
url | | 0..1 | uri | Network address to contact device
note | | 0..* | Annotation | Device notes and comments
safety | ? | 0..* | CodeableConcept | Safety Characteristics of Device DeviceSafety (Example)
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Device xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Instance identifier --></identifier>
 <udi>  <!-- 0..1 Unique Device Identifier (UDI) Barcode string -->
  <deviceIdentifier value="[string]"/><!-- 0..1 Mandatory fixed portion of UDI -->
  <name value="[string]"/><!-- 0..1 Device Name as appears on UDI label -->
  <jurisdiction value="[uri]"/><!-- 0..1 Regional UDI authority -->
  <carrierHRF value="[string]"/><!-- 0..1 UDI Human Readable Barcode String -->
  <carrierAIDC value="[base64Binary]"/><!-- 0..1 UDI Machine Readable Barcode String -->
  <issuer value="[uri]"/><!-- 0..1 UDI Issuing Organization -->
  <entryType value="[code]"/><!-- 0..1 barcode | rfid | manual + -->
 </udi>
 <status value="[code]"/><!-- 0..1 active | inactive | entered-in-error | unknown -->
 <type><!-- 0..1 CodeableConcept What kind of device this is --></type>
```

```
 <lotNumber value="[string]"/><!-- 0..1 Lot number of manufacture -->
 <manufacturer value="[string]"/><!-- 0..1 Name of device manufacturer -->
 <manufactureDate value="[dateTime]"/><!-- 0..1 Date when the device was made -->
 <expirationDate value="[dateTime]"/><!-- 0..1 Date and time of expiry of this device (if applicable) -->
 <model value="[string]"/><!-- 0..1 Model id assigned by the manufacturer -->
 <version value="[string]"/><!-- 0..1 Version number (i.e. software) -->
 <patient><!-- 0..1 Reference(Patient) Patient to whom Device is affixed --></patient>
 <owner><!-- 0..1 Reference(Organization) Organization responsible for device --></owner>
 <contact><!-- 0..* ContactPoint Details for human/organization for support --></contact>
 <location><!-- 0..1 Reference(Location) Where the resource is found --></location>
 <url value="[uri]"/><!-- 0..1 Network address to contact device -->
 <note><!-- 0..* Annotation Device notes and comments --></note>
 <safety><!-- 0..* CodeableConcept Safety Characteristics of Device --></safety>
</Device>
```

JSON Template

```
{
  "resourceType" : "Device",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Instance identifier
  "udi" : { // Unique Device Identifier (UDI) Barcode string
    "deviceIdentifier" : "<string>", // Mandatory fixed portion of UDI
    "name" : "<string>", // Device Name as appears on UDI label
    "jurisdiction" : "<uri>", // Regional UDI authority
    "carrierHRF" : "<string>", // UDI Human Readable Barcode String
    "carrierAIDC" : "<base64Binary>", // UDI Machine Readable Barcode String
    "issuer" : "<uri>", // UDI Issuing Organization
    "entryType" : "<code>" // barcode | rfid | manual +
  },
  "status" : "<code>", // active | inactive | entered-in-error | unknown
  "type" : { CodeableConcept }, // What kind of device this is
  "lotNumber" : "<string>", // Lot number of manufacture
  "manufacturer" : "<string>", // Name of device manufacturer
  "manufactureDate" : "<dateTime>", // Date when the device was made
  "expirationDate" : "<dateTime>", // Date and time of expiry of this device (if applicable)
  "model" : "<string>", // Model id assigned by the manufacturer
  "version" : "<string>", // Version number (i.e. software)
  "patient" : { Reference(Patient) }, // Patient to whom Device is affixed
  "owner" : { Reference(Organization) }, // Organization responsible for device
  "contact" : [{ ContactPoint }], // Details for human/organization for support
  "location" : { Reference(Location) }, // Where the resource is found
  "url" : "<uri>", // Network address to contact device
  "note" : [{ Annotation }], // Device notes and comments
  "safety" : [{ CodeableConcept }] // Safety Characteristics of Device
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Device;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Device.identifier [ Identifier ], ... ; # 0..* Instance identifier
  fhir:Device.udi [ # 0..1 Unique Device Identifier (UDI) Barcode string
    fhir:Device.udi.deviceIdentifier [ string ]; # 0..1 Mandatory fixed portion of UDI
    fhir:Device.udi.name [ string ]; # 0..1 Device Name as appears on UDI label
    fhir:Device.udi.jurisdiction [ uri ]; # 0..1 Regional UDI authority
    fhir:Device.udi.carrierHRF [ string ]; # 0..1 UDI Human Readable Barcode String
    fhir:Device.udi.carrierAIDC [ base64Binary ]; # 0..1 UDI Machine Readable Barcode String
    fhir:Device.udi.issuer [ uri ]; # 0..1 UDI Issuing Organization
    fhir:Device.udi.entryType [ code ]; # 0..1 barcode | rfid | manual +
  ];
  fhir:Device.status [ code ]; # 0..1 active | inactive | entered-in-error | unknown
  fhir:Device.type [ CodeableConcept ]; # 0..1 What kind of device this is
  fhir:Device.lotNumber [ string ]; # 0..1 Lot number of manufacture
  fhir:Device.manufacturer [ string ]; # 0..1 Name of device manufacturer
  fhir:Device.manufactureDate [ dateTime ]; # 0..1 Date when the device was made
  fhir:Device.expirationDate [ dateTime ]; # 0..1 Date and time of expiry of this device (if applicable)
  fhir:Device.model [ string ]; # 0..1 Model id assigned by the manufacturer
  fhir:Device.version [ string ]; # 0..1 Version number (i.e. software)
  fhir:Device.patient [ Reference(Patient) ]; # 0..1 Patient to whom Device is affixed
  fhir:Device.owner [ Reference(Organization) ]; # 0..1 Organization responsible for device
  fhir:Device.contact [ ContactPoint ], ... ; # 0..* Details for human/organization for support
  fhir:Device.location [ Reference(Location) ]; # 0..1 Where the resource is found
  fhir:Device.url [ uri ]; # 0..1 Network address to contact device
  fhir:Device.note [ Annotation ], ... ; # 0..* Device notes and comments
  fhir:Device.safety [ CodeableConcept ], ... ; # 0..* Safety Characteristics of Device
]
```

Changes since DSTU2

Device

Device.udi | Type changed from string to BackboneElement

Device.udi.deviceIdentifier | Added Element

Device.udi.name | Added Element

Device.udi.jurisdiction | Added Element
Device.udi.carrierHRF | Added Element
Device.udi.carrierAIDC | Added Element
Device.udi.issuer | Added Element
Device.udi.entryType | Added Element
Device.status | Change value set from http://hl7.org/fhir/ValueSet/devicestatus to http://hl7.org/fhir/ValueSet/device-status
Device.type | Min Cardinality changed from 1 to 0
Device.expirationDate | Renamed from expiry to expirationDate
Device.safety | Added Element

See the Full Difference for further information

This analysis is available as XML or JSON.

See R2 <--> R3 Conversion Maps (status = 5 tests that all execute ok. 2 fail round-trip testing and all r3 resources are valid.).

Alternate definitions: Master Definition (XML, JSON), XML Schema/Schematron (for) + JSON Schema, ShEx (for Turtle)

8.14.3.1 Terminology Bindings

Path | Definition | Type | Reference
Device.udi.entryType | Codes to identify how UDI data was entered | Required | UDIEntryType
Device.status | The availability status of the device. | Required | FHIRDeviceStatus
Device.type | Codes to identify medical devices | Example | FHIR Device Types
Device.safety | Codes used to identify medical devices safety characterics. These codes are derived in part from the [United States Food and Drug Administration recommendations](http://www.fda.gov/downloads/medicaldevices/deviceregulationandguidance/guidancedocuments/ucm107708.pdf) and are provided here as a suggestive example. | Example | DeviceSafety

8.14.3.2 Notes

8.14.3.2.1 Device Types

The example binding used for the device element is from SNOMED CT. However, there are many other sources of possible codes for device type including:

* Global Medical Device Nomenclature (GMDN®)

* Rosetta Terminology Mapping (RTM)

Note that there may be translations of type code using one or more of these and other vocabularies

8.14.3.2.2 Device Identifiers

Nearly all devices are assigned one or more identifiers, which are usually printed or affixed to the device using either barcodes or RFIDs. The identifiers can come from the manufacturer (often called the "serial number," "reference number," or "catalog number"), various institution and registries. Any of these identifiers assigned to the device can and should be recorded in the device resource. The different identifiers are differentiated by their use, label, and system values.

8.14.3.2.3 Unique Device Identifier (UDI)

The International Medical Device Regulators Forum IMDRF UDI Working Group published UDI System for Medical Devices (Version 2.0) , the base specification for Unique Device Identifiers (UDI). The United States Food and Drug Administration has produced an implementation guide for Unique Device Identifiers (UDI) which implements the IMDRF specification and other jurisdictions may produce similar IMDRF implementation guides as well. The full UDI string that represents the barcode as printed on the packaging of the device or Automatic Identification and Data Capture (AIDC) representation is called the "UDI carrier". The UDI has 2 components*:

* Device identifier (DI)**, which is the actual identification component

* Production identifier(s)(PI) which provide the means to track a device through its manufacture, distribution and use.

*non-UDI elements may also appear within the UDI carrier. **a "GTIN" (sometimes also called an EAN number) is a code developed by GS1 for the kind of device not an identifier for the device. A GTIN may appear on its own or it may appear in a UDI string as the DI component.

The DI of the UDI may be stored in a jurisdictional repository and used as the primary key to access other device information. For example, in the United States, the DI of the UDI is submitted in a device record to the Global Unique Device Identification Database (GUDID) . The UDI may identify an instance of a device uniquely (when the PI includes a serial number), or it may just identify the type of the device. The UDI is parsed into its constituent parts (DI, PI and other elements) by parsing rules developed by each Issuing Agency standard. Where the device has an assigned UDI, the other details carried in the resource (e.g., lot, expiration date, etc.) SHALL be consistent with the information encoded in the UDI string or registered in the local repository.

Best practice guidelines for transmitting UDI data using the Device resource dictate transmitting both the UDI Carrier and all components found within the UDI as described in Device UDI Mapping. Several examples are provided for futher guidance.

8.14.4 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common

device-name | string | A server defined search that may match any of the string fields in the Device.udi.name or Device.type.coding.display or Device.type.text | Device.udi.name | Device.type.text | Device.type.coding.display
identifier | token | Instance id from manufacturer, owner, and others | Device.identifier
location | reference | A location, where the resource is found | Device.location (Location)
manufacturer | string | The manufacturer of the device | Device.manufacturer
model | string | The model of the device | Device.model
organization | reference | The organization responsible for the device | Device.owner (Organization)
patient | reference | Patient information, if the resource is affixed to a person | Device.patient (Patient)
status | token | active | inactive | entered-in-error | unknown | Device.status
type | token | The type of the device | Device.type
udi-carrier | string | UDI Barcode (RFID or other technology) string either in HRF format or AIDC format converted to base64 string. | Device.udi.carrierHRF | Device.udi.carrierAIDC
udi-di | string | The udi Device Identifier (DI) | Device.udi.deviceIdentifier
url | uri | Network address to contact device | Device.url

8.14.5 Resource Device - Examples

URL: https://hl7.org/fhir/STU3/device-examples.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.14.5 Resource Device - Examples

Orders and Observations Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Device

Example List:

General Device Example | example | XML | JSON | Turtle
Example for IHE-PCD example | ihe-pcd | XML | JSON | Turtle
Example of device | f001 | XML | JSON | Turtle
Example of a pacemaker device | example-pacemaker | XML | JSON | Turtle
Example of an EHR software installation | software | XML | JSON | Turtle
Implantable device/UDI example GS-1 Format | example-udi1 | XML | JSON | Turtle
Implantable device/UDI example ICCBBA Format | example-udi2 | XML | JSON | Turtle
Implantable device/UDI example HIBBC Format | example-udi3 | XML | JSON | Turtle
Blood Bag UDI example ICCBBA Format | example-udi4 | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

8.14.6 Resource Device - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/device-definitions.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.14.6 Resource Device - Detailed Descriptions

Orders and Observations Work Group | Maturity Level: 2 | Trial Use | Compartments: Device

Detailed Descriptions for the elements in the Device resource.

Device
Definition | This resource identifies an instance or a type of a manufactured item that is used in the provision of healthcare without being substantially changed through that activity. The device may be a medical or non-medical device. Medical devices include durable (reusable) medical equipment, implantable devices, as well as disposable equipment used for diagnostic, treatment, and research for healthcare and public health. Non-medical devices may include items such as a machine, cellphone, computer, application, etc.
Control | 1..1
Requirements | Allows institutions to track their devices.
Device.identifier
Definition | Unique instance identifiers assigned to a device by manufacturers other organizations or owners.
Note | This is a business identifer, not a resource identifier (see discussion)
Control | 0..*
Type | Identifier
Alternate Names | Serial Number
Comments | The barcode string from a barcode present on a device label or package may identify the instance, include names given to the device in local usage, or may identify the type of device. If the identifier identifies the type of device, Device.type element should be used. For UDI, this element corresponds to the variable portion of the UDI that identifies the serial number of a specific device. See UDI mappings for a complete mapping of UDI parts to Device.
Device.udi
Definition | Unique device identifier (UDI) assigned to device label or package.
Control | 0..1
Summary | true
Comments | UDI may identify an unique instance of a device, or it may only identify the type of the device. See UDI mappings for a complete mapping of UDI parts to Device.
Device.udi.deviceIdentifier

Definition | The device identifier (DI) is a mandatory, fixed portion of a UDI that identifies the labeler and the specific version or model of a device.
Control | 0..1
Type | string
Alternate Names | DI
Summary | true
Device.udi.name
Definition | Name of device as used in labeling or catalog.
Control | 0..1
Type | string
Alternate Names | Brand Name
Summary | true
Comments | Use the Device.type for a generic type or kind of device name. Note the GUDID lists the name as the 'Brand Name'.
Device.udi.jurisdiction
Definition | The identity of the authoritative source for UDI generation within a jurisdiction. All UDIs are globally unique within a single namespace. with the appropriate repository uri as the system. For example, UDIs of devices managed in the U.S. by the FDA, the value is http://hl7.org/fhir/NamingSystem/fda-udi.
Control | 0..1
Type | uri
Requirements | Allows a recipient of a UDI to know which database will contain the UDI-associated metadata.
Device.udi.carrierHRF
Definition | The full UDI carrier as the human readable form (HRF) representation of the barcode string as printed on the packaging of the device.
Control | 0..1
Type | string
Alternate Names | Human Readable Form
Summary | true
Comments | If separate barcodes for DI and PI are present, concatenate the string with DI first and in order of human readable expression on label.
Device.udi.carrierAIDC
Definition | The full UDI carrier of the Automatic Identification and Data Capture (AIDC) technology representation of the barcode string as printed on the packaging of the device - E.g a barcode or RFID. Because of limitations on character sets in XML and the need to round-trip JSON data through XML, AIDC Formats SHALL be base64 encoded.
Control | 0..1
Type | base64Binary
Alternate Names | Automatic Identification and Data Capture
Summary | true
Comments | The AIDC form of UDIs should be scanned or otherwise used for the identification of the device whenever possible to minimize errors in records resulting from manual transcriptions. If separate barcodes for DI and PI are present, concatenate the string with DI first and in order of human readable expression on label.
Device.udi.issuer
Definition | Organization that is charged with issuing UDIs for devices. For example, the US FDA issuers include : 1) GS1: http://hl7.org/fhir/NamingSystem/gs1-di, 2) HIBCC: http://hl7.org/fhir/NamingSystem/hibcc-dI, 3) ICCBBA for blood containers: http://hl7.org/fhir/NamingSystem/iccbba-blood-di, 4) ICCBA for other devices: http://hl7.org/fhir/NamingSystem/iccbba-other-di.
Control | 0..1
Type | uri
Alternate Names | Barcode System
Device.udi.entryType
Definition | A coded entry to indicate how the data was entered.
Control | 0..1
Terminology Binding | UDIEntryType (Required)
Type | code
Requirements | Supports a way to distinguish hand entered from machine read data.
Device.status
Definition | Status of the Device availability.
Control | 0..1
Terminology Binding | FHIRDeviceStatus (Required)
Type | code
Is Modifier | true
Summary | true
Comments | This element is labeled as a modifier because the status contains the codes inactive and entered-in-error that mark the device (record)as not currently valid.
Device.type
Definition | Code or identifier to identify a kind of device.
Control | 0..1
Terminology Binding | FHIR Device Types (Example)
Type | CodeableConcept
Device.lotNumber
Definition | Lot number assigned by the manufacturer.
Control | 0..1
Type | string
Device.manufacturer
Definition | A name of the manufacturer.
Control | 0..1
Type | string
Device.manufactureDate

Definition | The date and time when the device was manufactured.
Control | 0..1
Type | dateTime
Device.expirationDate
Definition | The date and time beyond which this device is no longer valid or should not be used (if applicable).
Control | 0..1
Type | dateTime
Device.model
Definition | The "model" is an identifier assigned by the manufacturer to identify the product by its type. This number is shared by the all devices sold as the same type.
Control | 0..1
Type | string
Device.version
Definition | The version of the device, if the device has multiple releases under the same model, or if the device is software or carries firmware.
Note | This is a business versionId, not a resource version id (see discussion)
Control | 0..1
Type | string
Device.patient
Definition | Patient information, If the device is affixed to a person.
Control | 0..1
Type | Reference(Patient)
Requirements | If the device is implanted in a patient, then need to associate the device to the patient.
Device.owner
Definition | An organization that is responsible for the provision and ongoing maintenance of the device.
Control | 0..1
Type | Reference(Organization)
Device.contact
Definition | Contact details for an organization or a particular human that is responsible for the device.
Control | 0..*
Type | ContactPoint
Comments | used for troubleshooting etc.
Device.location
Definition | The place where the device can be found.
Control | 0..1
Type | Reference(Location)
Requirements | Device.location can be used to track device location.
Device.url
Definition | A network address on which the device may be contacted directly.
Control | 0..1
Type | uri
Comments | If the device is running a FHIR server, the network address should be the Base URL from which a conformance statement may be retrieved.
Device.note
Definition | Descriptive information, usage information or implantation information that is not captured in an existing element.
Control | 0..*
Type | Annotation
Device.safety
Definition | Provides additional safety characteristics about a medical device. For example devices containing latex.
Control | 0..*
Terminology Binding | DeviceSafety (Example)
Type | CodeableConcept
Summary | true

8.14.7 Resource Device - Mappings

URL: https://hl7.org/fhir/STU3/device-mappings.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.14.7 Resource Device - Mappings

Orders and Observations Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Device

Mappings:

RIM Mapping

UDI Mapping

Mappings for the device resource.

8.14.7.1 Mappings for RIM Mapping (http://hl7.org/v3)

Device | Device
identifier | .id
udi | .id and .code

deviceIdentifier | Role.id.extension
name | .displayName
jurisdiction | Role.scoper
carrierHRF | Role.id.extension
carrierAIDC | Role.id.extension
issuer | Role.id.root
entryType | NA
status | .statusCode
type | .code
lotNumber | .lotNumberText
manufacturer | .playedRole[typeCode=MANU].scoper.name
manufactureDate | .existenceTime.low
expirationDate | .expirationTime
model | .playedRole[typeCode=MANU].code
version | .softwareName (included as part)
patient | .playedRole[typeCode=USED].scoper.playedRole[typeCode=PAT]
owner | .playedRole[typeCode=OWN].scoper
contact | .scopedRole[typeCode=CON].player
location | .playedRole[typeCode=LOCE].scoper
url | .telecom
note | .text
safety | NA

8.14.7.2 Mappings for UDI Mapping (http://www.fda.gov/MedicalDevices/DeviceRegulationandGuidance/UniqueDeviceIdentification/default.htm)

UDI is a unique numeric or alphanumeric code that consists of two parts: (1) a device identifier (DI), a mandatory, fixed portion of a UDI that identifies the labeler and the specific version or model of a device, and (2) a production identifier (PI), a conditional, variable portion of a UDI that identifies one or more of the following when included on the label of a device: (2a) the lot or batch number within which a device was manufactured; (2b) the serial number of a specific device; (2c) the expiration date of a specific device; (2d) the date a specific device was manufactured; (2e) the distinct identification code required by §1271.290(c) for a human cell, tissue, or cellular and tissue-based product (HCT/P) regulated as a device.

Device
identifier | The serial number which is a component of the production identifier (PI), a conditional, variable portion of a UDI. The identifier.type code should be set to “SNO”(Serial Number) and the system left empty.
udi
deviceIdentifier | The device identifier (DI), a mandatory, fixed portion of a UDI that identifies the labeler and the specific version or model of a device.
name | Propietary/Trade Name of the medical Device as used in the labeling or catalog
jurisdiction
carrierHRF | A unique device identifier (UDI) on a device label in plain text
carrierAIDC | A unique device identifier (UDI) on a device label a form that uses automatic identification and data capture (AIDC) technology.
issuer | All UDIs are to be issued under a system operated by an Jurisdiction-accredited issuing agency. GS1 DIs: http://hl7.org/fhir/NamingSystem/gs1 HIBCC DIs: http://hl7.org/fhir/NamingSystem/hibcc ICCBBA DIs for blood containers: http://hl7.org/fhir/NamingSystem/iccbba-blood ICCBA DIs for other devices: http://hl7.org/fhir/NamingSystem/iccbba-other
entryType
status
type | Type description using GMDN or other code system
lotNumber | The lot or batch number within which a device was manufactured - which is a component of the production identifier (PI), a conditional, variable portion of a UDI.
manufacturer | N/A
manufactureDate | The date a specific device was manufactured - which is a component of the production identifier (PI), a conditional, variable portion of a UDI. For FHIR, The datetime syntax must converted to YYYY-MM-DD[THH:MM:SS]. If hour is present, the minutes and seconds should both be set to “00”.
expirationDate | the expiration date of a specific device - which is a component of the production identifier (PI), a conditional, variable portion of a UDI. For FHIR, The datetime syntax must converted to YYYY-MM-DD[THH:MM:SS]. If hour is present, the minutes and seconds should both be set to “00”.
model
version
patient
owner
contact
location
url
note
safety

8.14.8 Resource Device - Design Notes

URL: https://hl7.org/fhir/STU3/device-explanations.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.14.8 Resource Device - Design Notes

to do

8.14.9 Resource Device - Extensions & Profiles

URL: https://hl7.org/fhir/STU3/device-profiles.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.14.9 Resource Device - Extensions & Profiles

Orders and Observations Work Group | Maturity Level: N/A | Ballot Status: Informative

This table lists profiles and extensions for the Device resource. For background information, see Profiling Resources and Extensibility. Additional profiles and resources may be found in published Implementation Guides , or in the Conformance resource registry .

8.14.9.1 Profiles

No Profiles defined for this resource

8.14.9.2 Extensions

device-din | din | for Device HL7 Extensions
device-implant-status | implant-status | for Device HL7 Extensions

Extensions that reference this resource:

communicationrequest-orderedBy | orderedBy | for CommunicationRequest HL7 Extensions
questionnaireresponse-author | author | for Core extensions for QuestionnaireResponse
task-candidateList | candidateList | for Task HL7 Extensions

8.14.9.3 Search Extensions

No Search Extensions defined for this resource

8.14.10 Device HL7 Extensions

URL: https://hl7.org/fhir/STU3/device-extensions.html

Defines common extensions used with or related to the Device resource

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.14.10 Device HL7 Extensions

(No assigned work group) | Maturity Level: N/A | Ballot Status: Informative

8.14.10.1 Content

Extensions:

device-din | din : The distinct identification code required by §1271.290(c) for a human cell, tissue, or cellular and tissue-based product (HCT/P) regulated as a device - which is a conditional, variable portion of the production identifier (PI) of a Unique device Identifier (UDI) . If the content is from an ICCBBA -encoded UDI, the URI to use for the system is: http://hl7.org/fhir/NamingSystem/iccbba-din. GS1 and HIBCC formatted UDIs do not currently convey the DIN element. *Note to balloters: The Orders and Observations Workgroup has not yet determined whether the Device resource will include Devices of Human (or other) Origin (tissue, blood, etc.) and so the need for Donation Identification number is currently being met through this extension. If it is determined a separate resource is to be used for Devices of Human (or other) Origin (tissue, blood, etc.) then this extension may be deprecated in the future. Feedback on this issue is welcome.

device-implant-status | implant-status : Codes to represent the functional status of a device implanted in a patient. Both overall device status and an implant status need to be considered. The implant status should only be used when the device status is active.

8.14.10.2 Search Parameters

Search parameters defined by this package. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Paths | Source
din | token | The donation identification number (DIN) | | XML / JSON

8.14.10.3 Device-extensions-Device-din

URL: https://hl7.org/fhir/STU3/device-extensions-Device-din.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.14.10.3 Device-extensions-Device-din

Search Parameter din (type: token) defined as part of the device-extensions Profile.

The donation identification number (DIN)

Other Details:

* Full URL : http://hl7.org/fhir/SearchParameter/device-extensions-Device-din

* Paths :

* Targets :

8.14.11 Resource Device - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/device-version-maps.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.14.11 Resource Device - Detailed Descriptions

Orders and Observations Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Device

R2 : R3 Converson maps for Device.

Functional status for this map: 5 tests that all execute ok. 2 fail round-trip testing and all r3 resources are valid. (see documentation)

8.14.11.1 R2 to R3

```
map "http://hl7.org/fhir/StructureMap/Device2to3" = "R2 to R3 Conversions for Device"

conceptmap "DeviceStatus" {
  prefix s = "http://hl7.org/fhir/devicestatus"
  prefix t = "http://hl7.org/fhir/devicestatus"

  s:available  = t:active
  s:inactive = t:"not-available"
  s:"entered-in-error" = t:"entered-in-error"
}


uses "http://hl7.org/fhir/DSTU2/StructureDefinition/Device" alias DeviceR2 as source
uses "http://hl7.org/fhir/StructureDefinition/Device" alias Device as target

imports "http://hl7.org/fhir/StructureMap/*2to3"

group for type+types  Device extends DomainResource
  input src : DeviceR2 as source
  input tgt : Device as target

  "Device-identifier" : for src.identifier make tgt.identifier
  "Device-type" : for src.type make tgt.type
  "Device-note" : for src.note make tgt.note
  "Device-status" : for src.status as v make tgt.status = translate(v, "#DeviceStatus", "code")
  "Device-manufacturer" : for src.manufacturer make tgt.manufacturer
  "Device-model" : for src.model make tgt.model
  "Device-version" : for src.version make tgt.version
  "Device-manufactureDate" : for src.manufactureDate make tgt.manufactureDate
  "Device-expirationDate" : for src.expiry make tgt.expirationDate
  "Device-udi" : for src.udi as u make tgt.udi as t, t.deviceIdentifier = u
  "Device-lotNumber" : for src.lotNumber make tgt.lotNumber
  "Device-owner" : for src.owner make tgt.owner
  "Device-location" : for src.location make tgt.location
  "Device-patient" : for src.patient make tgt.patient
  "Device-contact" : for src.contact make tgt.contact
  "Device-url" : for src.url make tgt.url
endgroup
```

8.14.11.2 R3 to R2

```
map "http://hl7.org/fhir/StructureMap/Device3to2" = "R3 to R2 Conversion for Device"

conceptmap "DeviceStatus" {
  prefix s = "http://hl7.org/fhir/devicestatus"
  prefix t = "http://hl7.org/fhir/devicestatus"

  s:available  = t:active
  s:"not-available" = t:inactive
  s:"entered-in-error" = t:"entered-in-error"
}

uses "http://hl7.org/fhir/StructureDefinition/Device" alias Device as source
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/Device" alias DeviceR2 as target

imports "http://hl7.org/fhir/StructureMap/*3to2"

group for type+types  Device extends DomainResource
  input src : Device as source
  input tgt : DeviceR2 as target

  "Device-identifier" : for src.identifier make tgt.identifier
```

```
  "Device-type" : for src.type make tgt.type
  "Device-note" : for src.note make tgt.note
  "Device-status" : for src.status as v make tgt.status = translate(v, "#DeviceStatus", "code")
  "Device-manufacturer" : for src.manufacturer make tgt.manufacturer
  "Device-model" : for src.model make tgt.model
  "Device-version" : for src.version make tgt.version
  "Device-manufactureDate" : for src.manufactureDate make tgt.manufactureDate
  "Device-expirationDate" : for src.expirationDate make tgt.expiry
  "Device-udi" : for src.udi as u then udi(u, tgt)
  "Device-lotNumber" : for src.lotNumber make tgt.lotNumber
  "Device-owner" : for src.owner make tgt.owner
  "Device-location" : for src.location make tgt.location
  "Device-patient" : for src.patient make tgt.patient
  "Device-contact" : for src.contact make tgt.contact
  "Device-url" : for src.url make tgt.url
endgroup

group udi
  input src as source
  input tgt as target

  "udi" : for src.deviceIdentifier make tgt.udi
endgroup
```

8.14.11.3 R3 Validation Errors

8.15 Resource DeviceComponent - Content

URL: https://hl7.org/fhir/STU3/devicecomponent.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

Health Care Devices Work Group | Maturity Level: 1 | Trial Use | Compartments: Device

The characteristics, operational status and capabilities of a medical-related component of a medical device.

8.15.1 Scope and Usage

The DeviceComponent resource is used to describe the characteristics, operational status and capabilities of a medical-related component of a medical device. It can be a physical component that is integrated inside the device, a removable physical component, or a non-physical component that allows physiological measurement data and its derived data to be grouped in a hierarchical information organization.

Note:

For the initial scope, this DeviceComponent resource is only applicable to describe a single node in the containment tree that is produced by the context scanner in any medical device that implements or derives from the ISO/IEEE 11073 standard and that does not represent a metric. Examples for such a node are MDS, VMD, or Channel.

8.15.2 Boundaries and Relationships

The DeviceComponent allows us to change the configuration of the device without having to change the device resource instance. The life-cycle of the configuration may be completely different than the one of the device itself.

There are several related resources

* Device - Used by the MedicalDeviceSystem profile

* Patient - Used by the MedicalDeviceSystem profile

* Location - Used by the MedicalDeviceSystem profile

* DeviceComponent

8.15.3 Background and Context

8.15.3.1 Structure of a DeviceComponent Resource

A Context Scanner object of a medical device that implements or derives from ISO/IEEE 11073 standard is responsible for observing device configuration changes. After instantiation, the Context Scanner object is responsible for announcing the object instances in the device's MDIB, a hierarchical containment (MDS->VMD->Channel->Metric). The DeviceComponent resource can be used to describe the characteristics, operational status and capabilities of a medical-related component of a medical device. It can be a physical component that is integrated inside the device, a removable physical component, or a non-physical component that allows physiological measurement data and its derived data to be grouped in a hierarchical information organization. Devices are conceptualized using the following main structure:

* MedicalDeviceSystem - An actual device that external systems communicate with. In 11073, this is known as a MDS.

* VirtualMedicalDevice - A medical-related subsystem of a medical device. It can either be a physical hardware piece or a pure software plugin component of a medical device. In 11073, this is known as a VMD.

* Channel - A non-physical component that allows physiological measurement data and its derived data to be grouped in a hierarchical information organization.

Very simple devices may have only a single virtual device with a single channel and one metric, while complex devices may have multiple items at every level.

This resource is referenced by devicemetric

8.15.4 Resource Content

* Structure

* UML

* XML

* JSON

* Turtle

* R2 Diff

* All

Structure

Name | Flags | Card. | Type | Description & Constraints

DeviceComponent | ? | | DomainResource | An instance of a medical-related component of a medical device Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension

identifier | ? | 1..1 | Identifier | Instance id assigned by the software stack

type | ? | 1..1 | CodeableConcept | What kind of component it is FHIR Device Types (Preferred)

lastSystemChange | ? | 0..1 | instant | Recent system change timestamp

source | ? | 0..1 | Reference(Device) | Top-level device resource link

parent | ? | 0..1 | Reference(DeviceComponent) | Parent resource link

operationalStatus | ? | 0..* | CodeableConcept | Current operational status of the component, for example On, Off or Standby DeviceComponentOperationalStatus (Extensible)

parameterGroup | ? | 0..1 | CodeableConcept | Current supported parameter group DeviceComponentParameterGroup (Extensible)

measurementPrinciple | ? | 0..1 | code | other | chemical | electrical | impedance | nuclear | optical | thermal | biological | mechanical | acoustical | manual+ MeasmntPrinciple (Required)

productionSpecification | ? | 0..* | BackboneElement | Specification details such as Component Revisions, or Serial Numbers

specType | ? | 0..1 | CodeableConcept | Type or kind of production specification, for example serial number or software revision DeviceSpecificationSpecType (Extensible)

componentId | ? | 0..1 | Identifier | Internal component unique identification

productionSpec | ? | 0..1 | string | A printable string defining the component

languageCode | ? | 0..1 | CodeableConcept | Language code for the human-readable text strings produced by the device Common Languages (Extensible but limited to All Languages)

Documentation for this format

UML Diagram (Legend)

XML Template

```
<DeviceComponent xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 1..1 Identifier Instance id assigned by the software stack --></identifier>
 <type><!-- 1..1 CodeableConcept What kind of component it is --></type>
 <lastSystemChange value="[instant]"/><!-- 0..1 Recent system change timestamp -->
 <source><!-- 0..1 Reference(Device) Top-level device resource link --></source>
 <parent><!-- 0..1 Reference(DeviceComponent) Parent resource link --></parent>
 <operationalStatus><!-- 0..* CodeableConcept Current operational status of the component, for example On, Off or
    Standby --></operationalStatus>
 <parameterGroup><!-- 0..1 CodeableConcept Current supported parameter group --></parameterGroup>
 <measurementPrinciple value="[code]"/><!-- 0..1 other | chemical | electrical | impedance | nuclear | optical | thermal
     | biological | mechanical | acoustical | manual+ -->
 <productionSpecification>  <!-- 0..* Specification details such as Component Revisions, or Serial Numbers -->
  <specType><!-- 0..1 CodeableConcept Type or kind of production specification, for example serial number or software
    revision --></specType>
  <componentId><!-- 0..1 Identifier Internal component unique identification --></componentId>
  <productionSpec value="[string]"/><!-- 0..1 A printable string defining the component -->
 </productionSpecification>
 <languageCode><!-- 0..1 CodeableConcept Language code for the human-readable text strings produced by the device
    --></languageCode>
</DeviceComponent>
```

JSON Template

```
{
  "resourceType" : "DeviceComponent",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : { Identifier }, // R!  Instance id assigned by the software stack
  "type" : { CodeableConcept }, // R!  What kind of component it is
  "lastSystemChange" : "<instant>", // Recent system change timestamp
  "source" : { Reference(Device) }, // Top-level device resource link
  "parent" : { Reference(DeviceComponent) }, // Parent resource link
  "operationalStatus" : [{ CodeableConcept }], // Current operational status of the component, for example On, Off or
    Standby
  "parameterGroup" : { CodeableConcept }, // Current supported parameter group
  "measurementPrinciple" : "<code>", // other | chemical | electrical | impedance | nuclear | optical | thermal |
    biological | mechanical | acoustical | manual+
  "productionSpecification" : [{ // Specification details such as Component Revisions, or Serial Numbers
```

```
    "specType" : { CodeableConcept }, // Type or kind of production specification, for example serial number or software
     revision
    "componentId" : { Identifier }, // Internal component unique identification
    "productionSpec" : "<string>" // A printable string defining the component
  }],
  "languageCode" : { CodeableConcept } // Language code for the human-readable text strings produced by the device
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:DeviceComponent;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:DeviceComponent.identifier [ Identifier ]; # 1..1 Instance id assigned by the software stack
  fhir:DeviceComponent.type [ CodeableConcept ]; # 1..1 What kind of component it is
  fhir:DeviceComponent.lastSystemChange [ instant ]; # 0..1 Recent system change timestamp
  fhir:DeviceComponent.source [ Reference(Device) ]; # 0..1 Top-level device resource link
  fhir:DeviceComponent.parent [ Reference(DeviceComponent) ]; # 0..1 Parent resource link
  fhir:DeviceComponent.operationalStatus [ CodeableConcept ], ... ; # 0..* Current operational status of the component,
    for example On, Off or Standby
  fhir:DeviceComponent.parameterGroup [ CodeableConcept ]; # 0..1 Current supported parameter group
  fhir:DeviceComponent.measurementPrinciple [ code ]; # 0..1 other | chemical | electrical | impedance | nuclear |
    optical | thermal | biological | mechanical | acoustical | manual+
  fhir:DeviceComponent.productionSpecification [ # 0..* Specification details such as Component Revisions, or Serial
    Numbers
    fhir:DeviceComponent.productionSpecification.specType [ CodeableConcept ]; # 0..1 Type or kind of production
    specification, for example serial number or software revision
    fhir:DeviceComponent.productionSpecification.componentId [ Identifier ]; # 0..1 Internal component unique
    identification
    fhir:DeviceComponent.productionSpecification.productionSpec [ string ]; # 0..1 A printable string defining the
    component
  ], ...;
  fhir:DeviceComponent.languageCode [ CodeableConcept ]; # 0..1 Language code for the human-readable text strings
    produced by the device
]
```

Changes since DSTU2

DeviceComponent

DeviceComponent.lastSystemChange | Min Cardinality changed from 1 to 0

DeviceComponent.operationalStatus | Add Binding http://hl7.org/fhir/ValueSet/operational-status (extensible)

DeviceComponent.parameterGroup | Add Binding http://hl7.org/fhir/ValueSet/parameter-group (extensible)

DeviceComponent.productionSpecification.specType | Add Binding http://hl7.org/fhir/ValueSet/specification-type (extensible)

DeviceComponent.languageCode | Change binding strength from required to extensible Change value set from http://tools.ietf.org/html/bcp47 to http://hl7.org/fhir/ValueSet/languages

See the Full Difference for further information

This analysis is available as XML or JSON.

See R2 <--> R3 Conversion Maps (status = 2 tests that all execute ok. All tests pass round-trip testing and all r3 resources are valid.).

Structure

Name | Flags | Card. | Type | Description & Constraints

DeviceComponent | ? | | DomainResource | An instance of a medical-related component of a medical device Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension

identifier | ? | 1..1 | Identifier | Instance id assigned by the software stack

type | ? | 1..1 | CodeableConcept | What kind of component it is FHIR Device Types (Preferred)

lastSystemChange | ? | 0..1 | instant | Recent system change timestamp

source | ? | 0..1 | Reference(Device) | Top-level device resource link

parent | ? | 0..1 | Reference(DeviceComponent) | Parent resource link

operationalStatus | ? | 0..* | CodeableConcept | Current operational status of the component, for example On, Off or Standby DeviceComponentOperationalStatus (Extensible)

parameterGroup | ? | 0..1 | CodeableConcept | Current supported parameter group DeviceComponentParameterGroup (Extensible)

measurementPrinciple | ? | 0..1 | code | other | chemical | electrical | impedance | nuclear | optical | thermal | biological | mechanical | acoustical | manual+ MeasmntPrinciple (Required)

productionSpecification | ? | 0..* | BackboneElement | Specification details such as Component Revisions, or Serial Numbers

specType | ? | 0..1 | CodeableConcept | Type or kind of production specification, for example serial number or software revision DeviceSpecificationSpecType (Extensible)

componentId | ? | 0..1 | Identifier | Internal component unique identification

productionSpec | ? | 0..1 | string | A printable string defining the component

languageCode | ? | 0..1 | CodeableConcept | Language code for the human-readable text strings produced by the device Common Languages (Extensible but limited to All Languages)

Documentation for this format

UML Diagram (Legend)

XML Template

```
<DeviceComponent xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
```

```
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 1..1 Identifier Instance id assigned by the software stack --></identifier>
 <type><!-- 1..1 CodeableConcept What kind of component it is --></type>
 <lastSystemChange value="[instant]"/><!-- 0..1 Recent system change timestamp -->
 <source><!-- 0..1 Reference(Device) Top-level device resource link --></source>
 <parent><!-- 0..1 Reference(DeviceComponent) Parent resource link --></parent>
 <operationalStatus><!-- 0..* CodeableConcept Current operational status of the component, for example On, Off or
    Standby --></operationalStatus>
 <parameterGroup><!-- 0..1 CodeableConcept Current supported parameter group --></parameterGroup>
 <measurementPrinciple value="[code]"/><!-- 0..1 other | chemical | electrical | impedance | nuclear | optical | thermal
     | biological | mechanical | acoustical | manual+ -->
 <productionSpecification>  <!-- 0..* Specification details such as Component Revisions, or Serial Numbers -->
  <specType><!-- 0..1 CodeableConcept Type or kind of production specification, for example serial number or software
    revision --></specType>
  <componentId><!-- 0..1 Identifier Internal component unique identification --></componentId>
  <productionSpec value="[string]"/><!-- 0..1 A printable string defining the component -->
 </productionSpecification>
 <languageCode><!-- 0..1 CodeableConcept Language code for the human-readable text strings produced by the device
    --></languageCode>
</DeviceComponent>
```

JSON Template

```
{
  "resourceType" : "DeviceComponent",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : { Identifier }, // R!  Instance id assigned by the software stack
  "type" : { CodeableConcept }, // R!  What kind of component it is
  "lastSystemChange" : "<instant>", // Recent system change timestamp
  "source" : { Reference(Device) }, // Top-level device resource link
  "parent" : { Reference(DeviceComponent) }, // Parent resource link
  "operationalStatus" : [{ CodeableConcept }], // Current operational status of the component, for example On, Off or
    Standby
  "parameterGroup" : { CodeableConcept }, // Current supported parameter group
  "measurementPrinciple" : "<code>", // other | chemical | electrical | impedance | nuclear | optical | thermal |
    biological | mechanical | acoustical | manual+
  "productionSpecification" : [{ // Specification details such as Component Revisions, or Serial Numbers
    "specType" : { CodeableConcept }, // Type or kind of production specification, for example serial number or software
     revision
    "componentId" : { Identifier }, // Internal component unique identification
    "productionSpec" : "<string>" // A printable string defining the component
  }],
  "languageCode" : { CodeableConcept } // Language code for the human-readable text strings produced by the device
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:DeviceComponent;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:DeviceComponent.identifier [ Identifier ]; # 1..1 Instance id assigned by the software stack
  fhir:DeviceComponent.type [ CodeableConcept ]; # 1..1 What kind of component it is
  fhir:DeviceComponent.lastSystemChange [ instant ]; # 0..1 Recent system change timestamp
  fhir:DeviceComponent.source [ Reference(Device) ]; # 0..1 Top-level device resource link
  fhir:DeviceComponent.parent [ Reference(DeviceComponent) ]; # 0..1 Parent resource link
  fhir:DeviceComponent.operationalStatus [ CodeableConcept ], ... ; # 0..* Current operational status of the component,
    for example On, Off or Standby
  fhir:DeviceComponent.parameterGroup [ CodeableConcept ]; # 0..1 Current supported parameter group
  fhir:DeviceComponent.measurementPrinciple [ code ]; # 0..1 other | chemical | electrical | impedance | nuclear |
    optical | thermal | biological | mechanical | acoustical | manual+
  fhir:DeviceComponent.productionSpecification [ # 0..* Specification details such as Component Revisions, or Serial
    Numbers
    fhir:DeviceComponent.productionSpecification.specType [ CodeableConcept ]; # 0..1 Type or kind of production
    specification, for example serial number or software revision
    fhir:DeviceComponent.productionSpecification.componentId [ Identifier ]; # 0..1 Internal component unique
    identification
    fhir:DeviceComponent.productionSpecification.productionSpec [ string ]; # 0..1 A printable string defining the
    component
  ], ...;
  fhir:DeviceComponent.languageCode [ CodeableConcept ]; # 0..1 Language code for the human-readable text strings
    produced by the device
]
```

Changes since DSTU2

DeviceComponent

DeviceComponent.lastSystemChange | Min Cardinality changed from 1 to 0

DeviceComponent.operationalStatus | Add Binding http://hl7.org/fhir/ValueSet/operational-status (extensible)

DeviceComponent.parameterGroup | Add Binding http://hl7.org/fhir/ValueSet/parameter-group (extensible)

DeviceComponent.productionSpecification.specType | Add Binding http://hl7.org/fhir/ValueSet/specification-type (extensible)

DeviceComponent.languageCode | Change binding strength from required to extensible Change value set from http://tools.ietf.org/html/bcp47 to http://hl7.org/fhir/ValueSet/languages

See the Full Difference for further information

This analysis is available as XML or JSON.

See R2 <--> R3 Conversion Maps (status = 2 tests that all execute ok. All tests pass round-trip testing and all r3 resources are valid.).

Alternate definitions: Master Definition (XML, JSON), XML Schema/Schematron (for) + JSON Schema, ShEx (for Turtle)

8.15.4.1 Terminology Bindings

Path | Definition | Type | Reference
DeviceComponent.type | Describes the type of the component. | Preferred | FHIR Device Types
DeviceComponent.operationalStatus | Codes representing the current status of the device - on, off, suspended, etc. | Extensible | DeviceComponentOperationalStatus
DeviceComponent.parameterGroup | Codes identifying groupings of parameters; e.g. Cardiovascular. | Extensible | DeviceComponentParameterGroup
DeviceComponent.measurementPrinciple | Different measurement principle supported by the device. | Required | MeasmntPrinciple
DeviceComponent.productionSpecification.specType | Codes for device specification types such as serial number, part number, hardware revision, software revision, etc. | Extensible | DeviceSpecificationSpecType
DeviceComponent.languageCode | A human language. | Extensible, but limited to All Languages | Common Languages

8.15.5 Notes:

* The correct codes for the medical device system (MDS), virtual medical device (VMD), and Channel will be specified in the profiles for specific types of device component. Generally, these codes are registered in the RTM Management service , but this is not required. See Terminology Systems for the correct representation of these codes in a Coding data type.

8.15.6 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
identifier | token | The identifier of the component | DeviceComponent.identifier
parent | reference | The parent DeviceComponent resource | DeviceComponent.parent (DeviceComponent)
source | reference | The device source | DeviceComponent.source (Device)
type | token | The device component type | DeviceComponent.type

8.15.7 Resource DeviceComponent - Examples

URL: https://hl7.org/fhir/STU3/devicecomponent-examples.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.15.7 Resource DeviceComponent - Examples

Health Care Devices Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Device

Example List:

General DeviceComponent example | example | XML | JSON | Turtle
Example for DeviceComponent includes production specification information | example-prodspec | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

8.15.8 Resource DeviceComponent - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/devicecomponent-definitions.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.15.8 Resource DeviceComponent - Detailed Descriptions

Health Care Devices Work Group | Maturity Level: 1 | Trial Use | Compartments: Device

Detailed Descriptions for the elements in the DeviceComponent resource.

DeviceComponent
Definition | The characteristics, operational status and capabilities of a medical-related component of a medical device.
Control | 1..1
Summary | true
Comments | For the initial scope, this DeviceComponent resource is only applicable to describe a single node in the containment tree that is produced by the context scanner in any medical device that implements or derives from the ISO/IEEE 11073 standard and that does not represent a metric. Examples for such a node are MDS, VMD, or Channel.
DeviceComponent.identifier
Definition | The locally assigned unique identification by the software. For example: handle ID.
Note | This is a business identifer, not a resource identifier (see discussion)
Control | 1..1
Type | Identifier

Summary | true
DeviceComponent.type
Definition | The component type as defined in the object-oriented or metric nomenclature partition.
Control | 1..1
Terminology Binding | FHIR Device Types (Preferred)
Type | CodeableConcept
Summary | true
Comments | DeviceComponent.type can be referred to either GDMN (nomenclature system supported by FDA Global UDI Database) or preferable RTMMS coding system.
DeviceComponent.lastSystemChange
Definition | The timestamp for the most recent system change which includes device configuration or setting change.
Control | 0..1
Type | instant
Summary | true
DeviceComponent.source
Definition | The link to the source Device that contains administrative device information such as manufacture, serial number, etc.
Control | 0..1
Type | Reference(Device)
Summary | true
DeviceComponent.parent
Definition | The link to the parent resource. For example: Channel is linked to its VMD parent.
Control | 0..1
Type | Reference(DeviceComponent)
Summary | true
DeviceComponent.operationalStatus
Definition | The current operational status of the device. For example: On, Off, Standby, etc.
Control | 0..*
Terminology Binding | DeviceComponentOperationalStatus (Extensible)
Type | CodeableConcept
Summary | true
Comments | OperationalStatus for the MDS, VMD, or Channel will be bound to a specific ValueSet that is defined in its profile.
DeviceComponent.parameterGroup
Definition | The parameter group supported by the current device component that is based on some nomenclature, e.g. cardiovascular.
Control | 0..1
Terminology Binding | DeviceComponentParameterGroup (Extensible)
Type | CodeableConcept
Summary | true
DeviceComponent.measurementPrinciple
Definition | The physical principle of the measurement. For example: thermal, chemical, acoustical, etc.
Control | 0..1
Terminology Binding | MeasmntPrinciple (Required)
Type | code
Summary | true
DeviceComponent.productionSpecification
Definition | The production specification such as component revision, serial number, etc.
Control | 0..*
Summary | true
DeviceComponent.productionSpecification.specType
Definition | The specification type, such as, serial number, part number, hardware revision, software revision, etc.
Control | 0..1
Terminology Binding | DeviceSpecificationSpecType (Extensible)
Type | CodeableConcept
Summary | true
DeviceComponent.productionSpecification.componentId
Definition | The internal component unique identification. This is a provision for manufacture specific standard components using a private OID. 11073-10101 has a partition for private OID semantic that the manufacturer can make use of.
Control | 0..1
Type | Identifier
Summary | true
DeviceComponent.productionSpecification.productionSpec
Definition | The printable string defining the component.
Control | 0..1
Type | string
Summary | true
DeviceComponent.languageCode
Definition | The language code for the human-readable text string produced by the device. This language code will follow the IETF language tag. Example: en-US.
Control | 0..1
Terminology Binding | Common Languages (Extensible but limited to ??)
Type | CodeableConcept
Summary | true

8.15.9 Resource DeviceComponent - Mappings

URL: https://hl7.org/fhir/STU3/devicecomponent-mappings.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.15.9 Resource DeviceComponent - Mappings

Health Care Devices Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Device

Mappings for the devicecomponent resource.

8.15.10 Resource DeviceComponent - Design Notes

URL: https://hl7.org/fhir/STU3/devicecomponent-explanations.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.15.10 Resource DeviceComponent - Design Notes

to do

8.15.11 Resource DeviceComponent - Extensions & Profiles

URL: https://hl7.org/fhir/STU3/devicecomponent-profiles.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.15.11 Resource DeviceComponent - Extensions & Profiles

Health Care Devices Work Group | Maturity Level: N/A | Ballot Status: Informative

This table lists profiles and extensions for the DeviceComponent resource. For background information, see Profiling Resources and Extensibility. Additional profiles and resources may be found in published Implementation Guides , or in the Conformance resource registry .

8.15.11.1 Profiles

No Profiles defined for this resource

8.15.11.2 Extensions

No Extensions defined for this resource

Extensions that reference this resource:

No Extensions refer to this resource

8.15.11.3 Search Extensions

No Search Extensions defined for this resource

8.15.12 Resource DeviceComponent - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/devicecomponent-version-maps.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.15.12 Resource DeviceComponent - Detailed Descriptions

Health Care Devices Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Device

R2 : R3 Converson maps for DeviceComponent.

Functional status for this map: 2 tests that all execute ok. All tests pass round-trip testing and all r3 resources are valid. (see documentation)

8.15.12.1 R2 to R3

```
map "http://hl7.org/fhir/StructureMap/DeviceComponent2to3" = "R2 to R3 Conversions for DeviceComponent"

uses "http://hl7.org/fhir/DSTU2/StructureDefinition/DeviceComponent" alias DeviceComponentR2 as source
uses "http://hl7.org/fhir/StructureDefinition/DeviceComponent" alias DeviceComponent as target

imports "http://hl7.org/fhir/StructureMap/*2to3"

group for type+types  DeviceComponent extends DomainResource
  input src : DeviceComponentR2 as source
  input tgt : DeviceComponent as target

  "DeviceComponent-type" : for src.type make tgt.type
  "DeviceComponent-identifier" : for src.identifier make tgt.identifier
```

```
  "DeviceComponent-lastSystemChange" : for src.lastSystemChange make tgt.lastSystemChange
  "DeviceComponent-source" : for src.source make tgt.source
  "DeviceComponent-parent" : for src.parent make tgt.parent
  "DeviceComponent-operationalStatus" : for src.operationalStatus make tgt.operationalStatus
  "DeviceComponent-parameterGroup" : for src.parameterGroup make tgt.parameterGroup
  "DeviceComponent-measurementPrinciple" : for src.measurementPrinciple make tgt.measurementPrinciple
  "DeviceComponent-productionSpecification" : for src.productionSpecification as vs0 make tgt.productionSpecification as
     vt0 then dvcProductionSpecification(vs0, vt0)
  "DeviceComponent-languageCode" : for src.languageCode make tgt.languageCode
endgroup

group dvcProductionSpecification extends BackboneElement
  input src as source
  input tgt as target

  "DeviceComponent.productionSpecification-specType" : for src.specType make tgt.specType
  "DeviceComponent.productionSpecification-componentId" : for src.componentId make tgt.componentId
  "DeviceComponent.productionSpecification-productionSpec" : for src.productionSpec make tgt.productionSpec
endgroup
```

8.15.12.2 R3 to R2

```
map "http://hl7.org/fhir/StructureMap/DeviceComponent3to2" = "R3 to R2 Conversion for DeviceComponent"

uses "http://hl7.org/fhir/StructureDefinition/DeviceComponent" alias DeviceComponent as source
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/DeviceComponent" alias DeviceComponentR2 as target

imports "http://hl7.org/fhir/StructureMap/*3to2"

group for type+types  DeviceComponent extends DomainResource
  input src : DeviceComponent as source
  input tgt : DeviceComponentR2 as target

  "DeviceComponent-type" : for src.type make tgt.type
  "DeviceComponent-identifier" : for src.identifier make tgt.identifier
  "DeviceComponent-lastSystemChange" : for src.lastSystemChange make tgt.lastSystemChange
  "DeviceComponent-source" : for src.source make tgt.source
  "DeviceComponent-parent" : for src.parent make tgt.parent
  "DeviceComponent-operationalStatus" : for src.operationalStatus make tgt.operationalStatus
  "DeviceComponent-parameterGroup" : for src.parameterGroup make tgt.parameterGroup
  "DeviceComponent-measurementPrinciple" : for src.measurementPrinciple make tgt.measurementPrinciple
  "DeviceComponent-productionSpecification" : for src.productionSpecification as vs0 make tgt.productionSpecification as
     vt0 then dvcProductionSpecification(vs0, vt0)
  "DeviceComponent-languageCode" : for src.languageCode make tgt.languageCode
endgroup

group dvcProductionSpecification extends BackboneElement
  input src as source
  input tgt as target

  "DeviceComponent.productionSpecification-specType" : for src.specType make tgt.specType
  "DeviceComponent.productionSpecification-componentId" : for src.componentId make tgt.componentId
  "DeviceComponent.productionSpecification-productionSpec" : for src.productionSpec make tgt.productionSpec
endgroup
```

8.15.12.3 R3 Validation Errors

8.16 Resource DeviceMetric - Content

URL: https://hl7.org/fhir/STU3/devicemetric.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

Health Care Devices Work Group | Maturity Level: 1 | Trial Use | Compartments: Device

Describes a measurement, calculation or setting capability of a medical device.

8.16.1 Scope and Usage

The DeviceMetric resource describes mandatory static properties that characterize a direct or derived, quantitative or qualitative biosignal measurement, setting, or calculation produced by a medical device. The DeviceMetric resource can also be used to describe the non-static but highly relevant properties to the metric such as metric status, metric last calibration time and type, measurement mode, color, reference link to the parent DeviceComponent to where it belongs, and any capabilities that the metric offers (for example: setting the metric label).

Note:

For the initial scope, this DeviceMetric resource is only applicable to describe a single metric node in the containment tree that is produced by the context scanner in any medical device that implements or derives from the ISO/IEEE 11073 standard.

8.16.2 Boundaries and Relationships

There are two related resources

* Device - The physical device that this DeviceMetric belongs to.

* DeviceComponent - The DeviceComponent that this DeviceMetric is part of. This can be a DeviceComponent of any kind like a VirtualMedicalDevice, a MedicalDeviceSystem, or a Channel.

This resource is referenced by Media and Observation

8.16.3 Resource Content

* Structure

* UML

* XML

* JSON

* Turtle

* R2 Diff

* All

Structure

Name | Flags | Card. | Type | Description & Constraints
DeviceMetric | ? | | DomainResource | Measurement, calculation or setting capability of a medical device Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | ? | 1..1 | Identifier | Unique identifier of this DeviceMetric
type | ? | 1..1 | CodeableConcept | Identity of metric, for example Heart Rate or PEEP Setting Device Metric and Component Types (Preferred)
unit | ? | 0..1 | CodeableConcept | Unit of Measure for the Metric Device Metric and Component Types (Preferred)
source | ? | 0..1 | Reference(Device) | Describes the link to the source Device
parent | ? | 0..1 | Reference(DeviceComponent) | Describes the link to the parent DeviceComponent
operationalStatus | ? | 0..1 | code | on | off | standby | entered-in-error DeviceMetricOperationalStatus (Required)
color | ? | 0..1 | code | black | red | green | yellow | blue | magenta | cyan | white DeviceMetricColor (Required)
category | ? | 1..1 | code | measurement | setting | calculation | unspecified DeviceMetricCategory (Required)
measurementPeriod | ? | 0..1 | Timing | Describes the measurement repetition time
calibration | ? | 0..* | BackboneElement | Describes the calibrations that have been performed or that are required to be performed
type | ? | 0..1 | code | unspecified | offset | gain | two-point DeviceMetricCalibrationType (Required)
state | ? | 0..1 | code | not-calibrated | calibration-required | calibrated | unspecified DeviceMetricCalibrationState (Required)
time | ? | 0..1 | instant | Describes the time last calibration has been performed
Documentation for this format

UML Diagram (Legend)

XML Template

```
<DeviceMetric xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 1..1 Identifier Unique identifier of this DeviceMetric --></identifier>
 <type><!-- 1..1 CodeableConcept Identity of metric, for example Heart Rate or PEEP Setting --></type>
 <unit><!-- 0..1 CodeableConcept Unit of Measure for the Metric --></unit>
 <source><!-- 0..1 Reference(Device) Describes the link to the source Device --></source>
 <parent><!-- 0..1 Reference(DeviceComponent) Describes the link to the parent DeviceComponent --></parent>
 <operationalStatus value="[code]"/><!-- 0..1 on | off | standby | entered-in-error -->
 <color value="[code]"/><!-- 0..1 black | red | green | yellow | blue | magenta | cyan | white -->
 <category value="[code]"/><!-- 1..1 measurement | setting | calculation | unspecified -->
 <measurementPeriod><!-- 0..1 Timing Describes the measurement repetition time --></measurementPeriod>
 <calibration>  <!-- 0..* Describes the calibrations that have been performed or that are required to be performed -->
  <type value="[code]"/><!-- 0..1 unspecified | offset | gain | two-point -->
  <state value="[code]"/><!-- 0..1 not-calibrated | calibration-required | calibrated | unspecified -->
  <time value="[instant]"/><!-- 0..1 Describes the time last calibration has been performed -->
 </calibration>
</DeviceMetric>
```

JSON Template

```
{
  "resourceType" : "DeviceMetric",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : { Identifier }, // R!  Unique identifier of this DeviceMetric
  "type" : { CodeableConcept }, // R!  Identity of metric, for example Heart Rate or PEEP Setting
  "unit" : { CodeableConcept }, // Unit of Measure for the Metric
  "source" : { Reference(Device) }, // Describes the link to the source Device
  "parent" : { Reference(DeviceComponent) }, // Describes the link to the parent DeviceComponent
  "operationalStatus" : "<code>", // on | off | standby | entered-in-error
  "color" : "<code>", // black | red | green | yellow | blue | magenta | cyan | white
  "category" : "<code>", // R!  measurement | setting | calculation | unspecified
  "measurementPeriod" : { Timing }, // Describes the measurement repetition time
  "calibration" : [{ // Describes the calibrations that have been performed or that are required to be performed
    "type" : "<code>", // unspecified | offset | gain | two-point
    "state" : "<code>", // not-calibrated | calibration-required | calibrated | unspecified
    "time" : "<instant>" // Describes the time last calibration has been performed
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[ a fhir:DeviceMetric;
```

```
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:DeviceMetric.identifier [ Identifier ]; # 1..1 Unique identifier of this DeviceMetric
  fhir:DeviceMetric.type [ CodeableConcept ]; # 1..1 Identity of metric, for example Heart Rate or PEEP Setting
  fhir:DeviceMetric.unit [ CodeableConcept ]; # 0..1 Unit of Measure for the Metric
  fhir:DeviceMetric.source [ Reference(Device) ]; # 0..1 Describes the link to the source Device
  fhir:DeviceMetric.parent [ Reference(DeviceComponent) ]; # 0..1 Describes the link to the parent DeviceComponent
  fhir:DeviceMetric.operationalStatus [ code ]; # 0..1 on | off | standby | entered-in-error
  fhir:DeviceMetric.color [ code ]; # 0..1 black | red | green | yellow | blue | magenta | cyan | white
  fhir:DeviceMetric.category [ code ]; # 1..1 measurement | setting | calculation | unspecified
  fhir:DeviceMetric.measurementPeriod [ Timing ]; # 0..1 Describes the measurement repetition time
  fhir:DeviceMetric.calibration [ # 0..* Describes the calibrations that have been performed or that are required to be
    performed
    fhir:DeviceMetric.calibration.type [ code ]; # 0..1 unspecified | offset | gain | two-point
    fhir:DeviceMetric.calibration.state [ code ]; # 0..1 not-calibrated | calibration-required | calibrated |
    unspecified
    fhir:DeviceMetric.calibration.time [ instant ]; # 0..1 Describes the time last calibration has been performed
  ], ...;
]
```

Changes since DSTU2

DeviceMetric

See the Full Difference for further information

This analysis is available as XML or JSON.

See R2 <--> R3 Conversion Maps (status = 1 test that all execute ok. All tests pass round-trip testing and all r3 resources are valid.).

Structure

Name | Flags | Card. | Type | Description & Constraints

DeviceMetric | ? | | DomainResource | Measurement, calculation or setting capability of a medical device Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension

identifier | ? | 1..1 | Identifier | Unique identifier of this DeviceMetric

type | ? | 1..1 | CodeableConcept | Identity of metric, for example Heart Rate or PEEP Setting Device Metric and Component Types (Preferred)

unit | ? | 0..1 | CodeableConcept | Unit of Measure for the Metric Device Metric and Component Types (Preferred)

source | ? | 0..1 | Reference(Device) | Describes the link to the source Device

parent | ? | 0..1 | Reference(DeviceComponent) | Describes the link to the parent DeviceComponent

operationalStatus | ? | 0..1 | code | on | off | standby | entered-in-error DeviceMetricOperationalStatus (Required)

color | ? | 0..1 | code | black | red | green | yellow | blue | magenta | cyan | white DeviceMetricColor (Required)

category | ? | 1..1 | code | measurement | setting | calculation | unspecified DeviceMetricCategory (Required)

measurementPeriod | ? | 0..1 | Timing | Describes the measurement repetition time

calibration | ? | 0..* | BackboneElement | Describes the calibrations that have been performed or that are required to be performed

type | ? | 0..1 | code | unspecified | offset | gain | two-point DeviceMetricCalibrationType (Required)

state | ? | 0..1 | code | not-calibrated | calibration-required | calibrated | unspecified DeviceMetricCalibrationState (Required)

time | ? | 0..1 | instant | Describes the time last calibration has been performed

Documentation for this format

UML Diagram (Legend)

XML Template

```
<DeviceMetric xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 1..1 Identifier Unique identifier of this DeviceMetric --></identifier>
 <type><!-- 1..1 CodeableConcept Identity of metric, for example Heart Rate or PEEP Setting --></type>
 <unit><!-- 0..1 CodeableConcept Unit of Measure for the Metric --></unit>
 <source><!-- 0..1 Reference(Device) Describes the link to the source Device --></source>
 <parent><!-- 0..1 Reference(DeviceComponent) Describes the link to the parent DeviceComponent --></parent>
 <operationalStatus value="[code]"/><!-- 0..1 on | off | standby | entered-in-error -->
 <color value="[code]"/><!-- 0..1 black | red | green | yellow | blue | magenta | cyan | white -->
 <category value="[code]"/><!-- 1..1 measurement | setting | calculation | unspecified -->
 <measurementPeriod><!-- 0..1 Timing Describes the measurement repetition time --></measurementPeriod>
 <calibration>  <!-- 0..* Describes the calibrations that have been performed or that are required to be performed -->
  <type value="[code]"/><!-- 0..1 unspecified | offset | gain | two-point -->
  <state value="[code]"/><!-- 0..1 not-calibrated | calibration-required | calibrated | unspecified -->
  <time value="[instant]"/><!-- 0..1 Describes the time last calibration has been performed -->
 </calibration>
</DeviceMetric>
```

JSON Template

```
{
  "resourceType" : "DeviceMetric",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : { Identifier }, // R!  Unique identifier of this DeviceMetric
  "type" : { CodeableConcept }, // R!  Identity of metric, for example Heart Rate or PEEP Setting
  "unit" : { CodeableConcept }, // Unit of Measure for the Metric
  "source" : { Reference(Device) }, // Describes the link to the source Device
  "parent" : { Reference(DeviceComponent) }, // Describes the link to the parent DeviceComponent
  "operationalStatus" : "<code>", // on | off | standby | entered-in-error
  "color" : "<code>", // black | red | green | yellow | blue | magenta | cyan | white
```

```
  "category" : "<code>", // R!  measurement | setting | calculation | unspecified
  "measurementPeriod" : { Timing }, // Describes the measurement repetition time
  "calibration" : [{ // Describes the calibrations that have been performed or that are required to be performed
    "type" : "<code>", // unspecified | offset | gain | two-point
    "state" : "<code>", // not-calibrated | calibration-required | calibrated | unspecified
    "time" : "<instant>" // Describes the time last calibration has been performed
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:DeviceMetric;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:DeviceMetric.identifier [ Identifier ]; # 1..1 Unique identifier of this DeviceMetric
  fhir:DeviceMetric.type [ CodeableConcept ]; # 1..1 Identity of metric, for example Heart Rate or PEEP Setting
  fhir:DeviceMetric.unit [ CodeableConcept ]; # 0..1 Unit of Measure for the Metric
  fhir:DeviceMetric.source [ Reference(Device) ]; # 0..1 Describes the link to the source Device
  fhir:DeviceMetric.parent [ Reference(DeviceComponent) ]; # 0..1 Describes the link to the parent DeviceComponent
  fhir:DeviceMetric.operationalStatus [ code ]; # 0..1 on | off | standby | entered-in-error
  fhir:DeviceMetric.color [ code ]; # 0..1 black | red | green | yellow | blue | magenta | cyan | white
  fhir:DeviceMetric.category [ code ]; # 1..1 measurement | setting | calculation | unspecified
  fhir:DeviceMetric.measurementPeriod [ Timing ]; # 0..1 Describes the measurement repetition time
  fhir:DeviceMetric.calibration [ # 0..* Describes the calibrations that have been performed or that are required to be
    performed
    fhir:DeviceMetric.calibration.type [ code ]; # 0..1 unspecified | offset | gain | two-point
    fhir:DeviceMetric.calibration.state [ code ]; # 0..1 not-calibrated | calibration-required | calibrated |
    unspecified
    fhir:DeviceMetric.calibration.time [ instant ]; # 0..1 Describes the time last calibration has been performed
  ], ...;
]
```

Changes since DSTU2

DeviceMetric

See the Full Difference for further information

This analysis is available as XML or JSON.

See R2 <--> R3 Conversion Maps (status = 1 test that all execute ok. All tests pass round-trip testing and all r3 resources are valid.).

Alternate definitions: Master Definition (XML, JSON), XML Schema/Schematron (for) + JSON Schema, ShEx (for Turtle)

8.16.3.1 Terminology Bindings

Path | Definition | Type | Reference
DeviceMetric.type | Describes the metric type. | Preferred | Device Metric and Component Types
DeviceMetric.unit | Describes the unit of the metric. | Preferred | Device Metric and Component Types
DeviceMetric.operationalStatus | Describes the operational status of the DeviceMetric. | Required | DeviceMetricOperationalStatus
DeviceMetric.color | Describes the typical color of representation. | Required | DeviceMetricColor
DeviceMetric.category | Describes the category of the metric. | Required | DeviceMetricCategory
DeviceMetric.calibration.type | Describes the type of a metric calibration. | Required | DeviceMetricCalibrationType
DeviceMetric.calibration.state | Describes the state of a metric calibration. | Required | DeviceMetricCalibrationState

8.16.4 Notes:

* The correct codes for the metric types are registered in the RTM Management service , but this is not required. See Terminology Systems for the correct representation of these codes in a Coding data type.

8.16.5 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
category | token | The category of the metric | DeviceMetric.category
identifier | token | The identifier of the metric | DeviceMetric.identifier
parent | reference | The parent DeviceMetric resource | DeviceMetric.parent (DeviceComponent)
source | reference | The device resource | DeviceMetric.source (Device)
type | token | The component type | DeviceMetric.type

8.16.6 Resource DeviceMetric - Examples

URL: https://hl7.org/fhir/STU3/devicemetric-examples.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.16.6 Resource DeviceMetric - Examples

Health Care Devices Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Device

Example List:

General DeviceMetric example | example | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

8.16.7 Resource DeviceMetric - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/devicemetric-definitions.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.16.7 Resource DeviceMetric - Detailed Descriptions

Health Care Devices Work Group | Maturity Level: 1 | Trial Use | Compartments: Device

Detailed Descriptions for the elements in the DeviceMetric resource.

DeviceMetric
Definition | Describes a measurement, calculation or setting capability of a medical device.
Control | 1..1
Summary | true
Comments | For the initial scope, this DeviceMetric resource is only applicable to describe a single metric node in the containment tree that is produced by the context scanner in any medical device that implements or derives from the ISO/IEEE 11073 standard.
DeviceMetric.identifier
Definition | Describes the unique identification of this metric that has been assigned by the device or gateway software. For example: handle ID. It should be noted that in order to make the identifier unique, the system element of the identifier should be set to the unique identifier of the device.
Note | This is a business identifer, not a resource identifier (see discussion)
Control | 1..1
Type | Identifier
Summary | true
DeviceMetric.type
Definition | Describes the type of the metric. For example: Heart Rate, PEEP Setting, etc.
Control | 1..1
Terminology Binding | Device Metric and Component Types (Preferred)
Type | CodeableConcept
Summary | true
Comments | DeviceMetric.type can be referred to either IEEE 11073-10101 or LOINC.
DeviceMetric.unit
Definition | Describes the unit that an observed value determined for this metric will have. For example: Percent, Seconds, etc.
Control | 0..1
Terminology Binding | Device Metric and Component Types (Preferred)
Type | CodeableConcept
Summary | true
Comments | DeviceMetric.unit can refer to either UCUM or preferable a RTMMS coding system.
DeviceMetric.source
Definition | Describes the link to the Device that this DeviceMetric belongs to and that contains administrative device information such as manufacturer, serial number, etc.
Control | 0..1
Type | Reference(Device)
Summary | true
DeviceMetric.parent
Definition | Describes the link to the DeviceComponent that this DeviceMetric belongs to and that provide information about the location of this DeviceMetric in the containment structure of the parent Device. An example would be a DeviceComponent that represents a Channel. This reference can be used by a client application to distinguish DeviceMetrics that have the same type, but should be interpreted based on their containment location.
Control | 0..1
Type | Reference(DeviceComponent)
Summary | true
DeviceMetric.operationalStatus
Definition | Indicates current operational state of the device. For example: On, Off, Standby, etc.
Control | 0..1
Terminology Binding | DeviceMetricOperationalStatus (Required)
Type | code
Summary | true
DeviceMetric.color
Definition | Describes the color representation for the metric. This is often used to aid clinicians to track and identify parameter types by color. In practice, consider a Patient Monitor that has ECG/HR and Pleth for example; the parameters are displayed in different characteristic colors, such as HR-blue, BP-green, and PR and SpO2- magenta.
Control | 0..1
Terminology Binding | DeviceMetricColor (Required)
Type | code
Summary | true

DeviceMetric.category
Definition | Indicates the category of the observation generation process. A DeviceMetric can be for example a setting, measurement, or calculation.
Control | 1..1
Terminology Binding | DeviceMetricCategory (Required)
Type | code
Summary | true
DeviceMetric.measurementPeriod
Definition | Describes the measurement repetition time. This is not necessarily the same as the update period. The measurement repetition time can range from milliseconds up to hours. An example for a measurement repetition time in the range of milliseconds is the sampling rate of an ECG. An example for a measurement repetition time in the range of hours is a NIBP that is triggered automatically every hour. The update period may be different than the measurement repetition time, if the device does not update the published observed value with the same frequency as it was measured.
Control | 0..1
Type | Timing
Summary | true
DeviceMetric.calibration
Definition | Describes the calibrations that have been performed or that are required to be performed.
Control | 0..*
Summary | true
DeviceMetric.calibration.type
Definition | Describes the type of the calibration method.
Control | 0..1
Terminology Binding | DeviceMetricCalibrationType (Required)
Type | code
Summary | true
DeviceMetric.calibration.state
Definition | Describes the state of the calibration.
Control | 0..1
Terminology Binding | DeviceMetricCalibrationState (Required)
Type | code
Summary | true
DeviceMetric.calibration.time
Definition | Describes the time last calibration has been performed.
Control | 0..1
Type | instant
Summary | true

8.16.8 Resource DeviceMetric - Mappings

URL: https://hl7.org/fhir/STU3/devicemetric-mappings.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.16.8 Resource DeviceMetric - Mappings

Health Care Devices Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Device

Mappings for the devicemetric resource.

8.16.9 Resource DeviceMetric - Design Notes

URL: https://hl7.org/fhir/STU3/devicemetric-explanations.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.16.9 Resource DeviceMetric - Design Notes

to do

8.16.10 Resource DeviceMetric - Extensions & Profiles

URL: https://hl7.org/fhir/STU3/devicemetric-profiles.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.16.10 Resource DeviceMetric - Extensions & Profiles

Health Care Devices Work Group | Maturity Level: N/A | Ballot Status: Informative

This table lists profiles and extensions for the DeviceMetric resource. For background information, see Profiling Resources and Extensibility. Additional profiles and resources may be found in published Implementation Guides , or in the Conformance resource registry .

8.16.10.1 Profiles

No Profiles defined for this resource

8.16.10.2 Extensions

No Extensions defined for this resource

Extensions that reference this resource:

No Extensions refer to this resource

8.16.10.3 Search Extensions

No Search Extensions defined for this resource

8.16.11 Resource DeviceMetric - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/devicemetric-version-maps.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.16.11 Resource DeviceMetric - Detailed Descriptions

Health Care Devices Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Device

R2 : R3 Converson maps for DeviceMetric.

Functional status for this map: 1 test that all execute ok. All tests pass round-trip testing and all r3 resources are valid. (see documentation)

8.16.11.1 R2 to R3

```
map "http://hl7.org/fhir/StructureMap/DeviceMetric2to3" = "R2 to R3 Conversions for DeviceMetric"

uses "http://hl7.org/fhir/DSTU2/StructureDefinition/DeviceMetric" alias DeviceMetricR2 as source
uses "http://hl7.org/fhir/StructureDefinition/DeviceMetric" alias DeviceMetric as target

imports "http://hl7.org/fhir/StructureMap/*2to3"

group for type+types  DeviceMetric extends DomainResource
  input src : DeviceMetricR2 as source
  input tgt : DeviceMetric as target

  "DeviceMetric-type" : for src.type make tgt.type
  "DeviceMetric-identifier" : for src.identifier make tgt.identifier
  "DeviceMetric-unit" : for src.unit make tgt.unit
  "DeviceMetric-source" : for src.source make tgt.source
  "DeviceMetric-parent" : for src.parent make tgt.parent
  "DeviceMetric-operationalStatus" : for src.operationalStatus make tgt.operationalStatus
  "DeviceMetric-color" : for src.color make tgt.color
  "DeviceMetric-category" : for src.category make tgt.category
  "DeviceMetric-measurementPeriod" : for src.measurementPeriod make tgt.measurementPeriod
  "DeviceMetric-calibration" : for src.calibration as vs0 make tgt.calibration as vt0 then dvmCalibration(vs0, vt0)
endgroup

group dvmCalibration extends BackboneElement
  input src as source
  input tgt as target

  "DeviceMetric.calibration-type" : for src.type make tgt.type
  "DeviceMetric.calibration-state" : for src.state make tgt.state
  "DeviceMetric.calibration-time" : for src.time make tgt.time
endgroup
```

8.16.11.2 R3 to R2

```
map "http://hl7.org/fhir/StructureMap/DeviceMetric3to2" = "R3 to R2 Conversion for DeviceMetric"

uses "http://hl7.org/fhir/StructureDefinition/DeviceMetric" alias DeviceMetric as source
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/DeviceMetric" alias DeviceMetricR2 as target

imports "http://hl7.org/fhir/StructureMap/*3to2"

group for type+types  DeviceMetric extends DomainResource
  input src : DeviceMetric as source
  input tgt : DeviceMetricR2 as target

  "DeviceMetric-type" : for src.type make tgt.type
  "DeviceMetric-identifier" : for src.identifier make tgt.identifier
  "DeviceMetric-unit" : for src.unit make tgt.unit
  "DeviceMetric-source" : for src.source make tgt.source
  "DeviceMetric-parent" : for src.parent make tgt.parent
  "DeviceMetric-operationalStatus" : for src.operationalStatus make tgt.operationalStatus
  "DeviceMetric-color" : for src.color make tgt.color
  "DeviceMetric-category" : for src.category make tgt.category
  "DeviceMetric-measurementPeriod" : for src.measurementPeriod make tgt.measurementPeriod
  "DeviceMetric-calibration" : for src.calibration as vs0 make tgt.calibration as vt0 then dvmCalibration(vs0, vt0)
endgroup

group dvmCalibration extends BackboneElement
```

```
  input src as source
  input tgt as target

  "DeviceMetric.calibration-type" : for src.type make tgt.type
  "DeviceMetric.calibration-state" : for src.state make tgt.state
  "DeviceMetric.calibration-time" : for src.time make tgt.time
endgroup
```

8.16.11.3 R3 Validation Errors

8.17 Resource Substance - Content

URL: https://hl7.org/fhir/STU3/substance.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

Orders and Observations Work Group | Maturity Level: 2 | Trial Use | Compartments: Not linked to any defined compartments

A homogeneous material with a definite composition.

8.17.1 Scope and Usage

This resource allows for a material to be represented. The resource can be used to represent either a kind of a substance - e.g. a formulation commonly used for treating patients, or it can be used to describe a particular package of a known substance (e.g. bottle, jar, packet).

The composition of the material can be specified in terms of a mix of other materials, including with precise amounts if required.

8.17.2 Boundaries and Relationships

A medication is a substance that is packaged and used as an administered medication. The medication resource uses the substance resource to represent the actual ingredients of a medication.

This resource is referenced by ActivityDefinition, AdverseEvent, CarePlan, Contract, Group, Medication, Procedure, Specimen, SupplyDelivery and SupplyRequest

8.17.3 Resource Content

* Structure

* UML

* XML

* JSON

* Turtle

* R2 Diff

* All

Structure

Name | Flags | Card. | Type | Description & Constraints
Substance | | | DomainResource | A homogeneous material with a definite composition Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | ? | 0..* | Identifier | Unique identifier
status | ? | 0..1 | code | active | inactive | entered-in-error FHIRSubstanceStatus (Required)
category | ? | 0..* | CodeableConcept | What class/type of substance this is Substance Category Codes (Extensible)
code | ? | 1..1 | CodeableConcept | What substance this is Substance Code (Example)
description | ? | 0..1 | string | Textual description of the substance, comments
instance | ? | 0..* | BackboneElement | If this describes a specific package/container of the substance
identifier | ? | 0..1 | Identifier | Identifier of the package/container
expiry | ? | 0..1 | dateTime | When no longer valid to use
quantity | ? | 0..1 | SimpleQuantity | Amount of substance in the package
ingredient | ? | 0..* | BackboneElement | Composition information about the substance
quantity | ? | 0..1 | Ratio | Optional amount (concentration)
substance[x] | ? | 1..1 | | A component of the substance Substance Code (Example)
substanceCodeableConcept | | | CodeableConcept
substanceReference | | | Reference(Substance)
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Substance xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Unique identifier --></identifier>
 <status value="[code]"/><!-- 0..1 active | inactive | entered-in-error -->
 <category><!-- 0..* CodeableConcept What class/type of substance this is --></category>
 <code><!-- 1..1 CodeableConcept What substance this is --></code>
```

```
 <description value="[string]"/><!-- 0..1 Textual description of the substance, comments -->
 <instance>  <!-- 0..* If this describes a specific package/container of the substance -->
  <identifier><!-- 0..1 Identifier Identifier of the package/container --></identifier>
  <expiry value="[dateTime]"/><!-- 0..1 When no longer valid to use -->
  <quantity><!-- 0..1 Quantity(SimpleQuantity) Amount of substance in the package --></quantity>
 </instance>
 <ingredient>  <!-- 0..* Composition information about the substance -->
  <quantity><!-- 0..1 Ratio Optional amount (concentration) --></quantity>
  <substance[x]><!-- 1..1 CodeableConcept|Reference(Substance) A component of the substance --></substance[x]>
 </ingredient>
</Substance>
```

JSON Template

```
{
  "resourceType" : "Substance",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Unique identifier
  "status" : "<code>", // active | inactive | entered-in-error
  "category" : [{ CodeableConcept }], // What class/type of substance this is
  "code" : { CodeableConcept }, // R!  What substance this is
  "description" : "<string>", // Textual description of the substance, comments
  "instance" : [{ // If this describes a specific package/container of the substance
    "identifier" : { Identifier }, // Identifier of the package/container
    "expiry" : "<dateTime>", // When no longer valid to use
    "quantity" : { Quantity(SimpleQuantity) } // Amount of substance in the package
  }],
  "ingredient" : [{ // Composition information about the substance
    "quantity" : { Ratio }, // Optional amount (concentration)
    // substance[x]: A component of the substance. One of these 2:
    "substanceCodeableConcept" : { CodeableConcept }
    "substanceReference" : { Reference(Substance) }
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Substance;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Substance.identifier [ Identifier ], ... ; # 0..* Unique identifier
  fhir:Substance.status [ code ]; # 0..1 active | inactive | entered-in-error
  fhir:Substance.category [ CodeableConcept ], ... ; # 0..* What class/type of substance this is
  fhir:Substance.code [ CodeableConcept ]; # 1..1 What substance this is
  fhir:Substance.description [ string ]; # 0..1 Textual description of the substance, comments
  fhir:Substance.instance [ # 0..* If this describes a specific package/container of the substance
    fhir:Substance.instance.identifier [ Identifier ]; # 0..1 Identifier of the package/container
    fhir:Substance.instance.expiry [ dateTime ]; # 0..1 When no longer valid to use
    fhir:Substance.instance.quantity [ Quantity(SimpleQuantity) ]; # 0..1 Amount of substance in the package
  ], ...;
  fhir:Substance.ingredient [ # 0..* Composition information about the substance
    fhir:Substance.ingredient.quantity [ Ratio ]; # 0..1 Optional amount (concentration)
    # Substance.ingredient.substance[x] : 1..1 A component of the substance. One of these 2
      fhir:Substance.ingredient.substanceCodeableConcept [ CodeableConcept ]
      fhir:Substance.ingredient.substanceReference [ Reference(Substance) ]
  ], ...;
]
```

Changes since DSTU2

Substance

Substance.status | Added Element

Substance.ingredient.substance[x] | Renamed from substance to substance[x] Add CodeableConcept

See the Full Difference for further information

This analysis is available as XML or JSON.

See R2 <--> R3 Conversion Maps (status = 6 tests that all execute ok. All tests pass round-trip testing and all r3 resources are valid.).

Structure

Name | Flags | Card. | Type | Description & Constraints

Substance | | | DomainResource | A homogeneous material with a definite composition Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension

identifier | ? | 0..* | Identifier | Unique identifier

status | ? | 0..1 | code | active | inactive | entered-in-error FHIRSubstanceStatus (Required)

category | ? | 0..* | CodeableConcept | What class/type of substance this is Substance Category Codes (Extensible)

code | ? | 1..1 | CodeableConcept | What substance this is Substance Code (Example)

description | ? | 0..1 | string | Textual description of the substance, comments

instance | ? | 0..* | BackboneElement | If this describes a specific package/container of the substance

identifier | ? | 0..1 | Identifier | Identifier of the package/container

expiry | ? | 0..1 | dateTime | When no longer valid to use

quantity | ? | 0..1 | SimpleQuantity | Amount of substance in the package
ingredient | ? | 0..* | BackboneElement | Composition information about the substance
quantity | ? | 0..1 | Ratio | Optional amount (concentration)
substance[x] | ? | 1..1 | | A component of the substance Substance Code (Example)
substanceCodeableConcept | | | CodeableConcept
substanceReference | | | Reference(Substance)
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Substance xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Unique identifier --></identifier>
 <status value="[code]"/><!-- 0..1 active | inactive | entered-in-error -->
 <category><!-- 0..* CodeableConcept What class/type of substance this is --></category>
 <code><!-- 1..1 CodeableConcept What substance this is --></code>
 <description value="[string]"/><!-- 0..1 Textual description of the substance, comments -->
 <instance>  <!-- 0..* If this describes a specific package/container of the substance -->
  <identifier><!-- 0..1 Identifier Identifier of the package/container --></identifier>
  <expiry value="[dateTime]"/><!-- 0..1 When no longer valid to use -->
  <quantity><!-- 0..1 Quantity(SimpleQuantity) Amount of substance in the package --></quantity>
 </instance>
 <ingredient>  <!-- 0..* Composition information about the substance -->
  <quantity><!-- 0..1 Ratio Optional amount (concentration) --></quantity>
  <substance[x]><!-- 1..1 CodeableConcept|Reference(Substance) A component of the substance --></substance[x]>
 </ingredient>
</Substance>
```

JSON Template

```
{
  "resourceType" : "Substance",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Unique identifier
  "status" : "<code>", // active | inactive | entered-in-error
  "category" : [{ CodeableConcept }], // What class/type of substance this is
  "code" : { CodeableConcept }, // R!  What substance this is
  "description" : "<string>", // Textual description of the substance, comments
  "instance" : [{ // If this describes a specific package/container of the substance
    "identifier" : { Identifier }, // Identifier of the package/container
    "expiry" : "<dateTime>", // When no longer valid to use
    "quantity" : { Quantity(SimpleQuantity) } // Amount of substance in the package
  }],
  "ingredient" : [{ // Composition information about the substance
    "quantity" : { Ratio }, // Optional amount (concentration)
    // substance[x]: A component of the substance. One of these 2:
    "substanceCodeableConcept" : { CodeableConcept }
    "substanceReference" : { Reference(Substance) }
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Substance;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Substance.identifier [ Identifier ], ... ; # 0..* Unique identifier
  fhir:Substance.status [ code ]; # 0..1 active | inactive | entered-in-error
  fhir:Substance.category [ CodeableConcept ], ... ; # 0..* What class/type of substance this is
  fhir:Substance.code [ CodeableConcept ]; # 1..1 What substance this is
  fhir:Substance.description [ string ]; # 0..1 Textual description of the substance, comments
  fhir:Substance.instance [ # 0..* If this describes a specific package/container of the substance
    fhir:Substance.instance.identifier [ Identifier ]; # 0..1 Identifier of the package/container
    fhir:Substance.instance.expiry [ dateTime ]; # 0..1 When no longer valid to use
    fhir:Substance.instance.quantity [ Quantity(SimpleQuantity) ]; # 0..1 Amount of substance in the package
  ], ...;
  fhir:Substance.ingredient [ # 0..* Composition information about the substance
    fhir:Substance.ingredient.quantity [ Ratio ]; # 0..1 Optional amount (concentration)
    # Substance.ingredient.substance[x] : 1..1 A component of the substance. One of these 2
      fhir:Substance.ingredient.substanceCodeableConcept [ CodeableConcept ]
      fhir:Substance.ingredient.substanceReference [ Reference(Substance) ]
  ], ...;
]
```

Changes since DSTU2

Substance
Substance.status | Added Element
Substance.ingredient.substance[x] | Renamed from substance to substance[x] Add CodeableConcept

See the Full Difference for further information

This analysis is available as XML or JSON.

See R2 <--> R3 Conversion Maps (status = 6 tests that all execute ok. All tests pass round-trip testing and all r3 resources are valid.).

Alternate definitions: Master Definition (XML, JSON), XML Schema/Schematron (for) + JSON Schema, ShEx (for Turtle)

8.17.3.1 Terminology Bindings

Path | Definition | Type | Reference
Substance.status | A code to indicate if the substance is actively used | Required | FHIRSubstanceStatus
Substance.category | Category or classification of substance | Extensible | Substance Category Codes
Substance.code | Substance codes | Example | Substance Code
Substance.ingredient.substance[x] | Substance Ingredient codes | Example | Substance Code

8.17.4 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
category | token | The category of the substance | Substance.category
code | token | The code of the substance or ingredient | Substance.code | Substance.ingredient.substance.as(CodeableConcept)
container-identifier | token | Identifier of the package/container | Substance.instance.identifier
expiry | date | Expiry date of package or container of substance | Substance.instance.expiry
identifier | token | Unique identifier for the substance | Substance.identifier
quantity | quantity | Amount of substance in the package | Substance.instance.quantity
status | token | active | inactive | entered-in-error | Substance.status
substance-reference | reference | A component of the substance | Substance.ingredient.substance.as(Reference) (Substance)

8.17.5 Resource Substance - Examples

URL: https://hl7.org/fhir/STU3/substance-examples.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.17.5 Resource Substance - Examples

Orders and Observations Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Not linked to any defined compartments

Example List:

General Example | example | XML | JSON | Turtle
Dust Mite Example | f201 | XML | JSON | Turtle
Staphylococcus Example | f202 | XML | JSON | Turtle
Potassium Example | f203 | XML | JSON | Turtle
Silver Nitrate Product Example | f204 | XML | JSON | Turtle
Amoxicillin-Clavulanate Example | f205 | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

8.17.6 Resource Substance - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/substance-definitions.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.17.6 Resource Substance - Detailed Descriptions

Orders and Observations Work Group | Maturity Level: 2 | Trial Use | Compartments: Not linked to any defined compartments

Detailed Descriptions for the elements in the Substance resource.

Substance
Definition | A homogeneous material with a definite composition.
Control | 1..1
Substance.identifier
Definition | Unique identifier for the substance.
Note | This is a business identifer, not a resource identifier (see discussion)
Control | 0..*
Type | Identifier
Summary | true
Comments | This identifier is associated with the kind of substance in contrast to the Substance.instance.identifier which is associated with the package/container.
Substance.status
Definition | A code to indicate if the substance is actively used.
Control | 0..1

Terminology Binding | FHIRSubstanceStatus (Required)
Type | code
Summary | true
Substance.category
Definition | A code that classifies the general type of substance. This is used for searching, sorting and display purposes.
Control | 0..*
Terminology Binding | Substance Category Codes (Extensible)
Type | CodeableConcept
Summary | true
Comments | The level of granularity is defined by the category concepts in the value set. More fine-grained filtering can be performed using the metadata and/or terminology hierarchy in Substance.code.
Substance.code
Definition | A code (or set of codes) that identify this substance.
Control | 1..1
Terminology Binding | Substance Code (Example)
Type | CodeableConcept
Summary | true
Comments | This could be a reference to an externally defined code. It could also be a locally assigned code (e.g. a formulary), optionally with translations to the standard drug codes.
Substance.description
Definition | A description of the substance - its appearance, handling requirements, and other usage notes.
Control | 0..1
Type | string
Summary | true
Substance.instance
Definition | Substance may be used to describe a kind of substance, or a specific package/container of the substance: an instance.
Control | 0..*
Meaning if Missing | If this element is not present, then the substance resource describes a kind of substance
Summary | true
Substance.instance.identifier
Definition | Identifier associated with the package/container (usually a label affixed directly).
Note | This is a business identifer, not a resource identifier (see discussion)
Control | 0..1
Type | Identifier
Summary | true
Substance.instance.expiry
Definition | When the substance is no longer valid to use. For some substances, a single arbitrary date is used for expiry.
Control | 0..1
Type | dateTime
Summary | true
Substance.instance.quantity
Definition | The amount of the substance.
Control | 0..1
Type | SimpleQuantity
Summary | true
Substance.ingredient
Definition | A substance can be composed of other substances.
Control | 0..*
Summary | true
Substance.ingredient.quantity
Definition | The amount of the ingredient in the substance - a concentration ratio.
Control | 0..1
Type | Ratio
Summary | true
Substance.ingredient.substance[x]
Definition | Another substance that is a component of this substance.
Control | 1..1
Terminology Binding | Substance Code (Example)
Type | CodeableConcept|Reference(Substance)
[x] Note | See Choice of Data Types for further information about how to use [x]
Summary | true

8.17.7 Resource Substance - Mappings

URL: https://hl7.org/fhir/STU3/substance-mappings.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.17.7 Resource Substance - Mappings

Orders and Observations Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Not linked to any defined compartments

Mappings:

RIM Mapping

Ontological RIM Mapping

Mappings for the substance resource.

8.17.7.1 Mappings for RIM Mapping (http://hl7.org/v3)

Substance | Material
identifier | .id
status | .statusCode
category | .code
code | .code
description | .player.desc
instance | .player.determinerCode
identifier | .id
expiry | .expiryTime
quantity | .player.quantity
ingredient | .player.scopesRole[classCode=INGR]
quantity | .quantity
substance[x] | .player

8.17.7.2 Mappings for Ontological RIM Mapping (http://hl7.org/orim)

Substance | fhir:Substance rdfs:subClassOf rim:Entity, [a owl:Restriction; owl:onProperty rim:Act.classCode; owl:allValuesFrom cs:EntityClass\#MAT].
identifier
status
category
code | fhir:Substance.code rdfs:subPropertyOf rim:Entity.code
description
instance
identifier
expiry
quantity
ingredient
quantity
substance[x]

8.17.8 Resource Substance - Design Notes

URL: https://hl7.org/fhir/STU3/substance-explanations.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.17.8 Resource Substance - Design Notes

to do

8.17.9 Resource Substance - Extensions & Profiles

URL: https://hl7.org/fhir/STU3/substance-profiles.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.17.9 Resource Substance - Extensions & Profiles

Orders and Observations Work Group | Maturity Level: N/A | Ballot Status: Informative

This table lists profiles and extensions for the Substance resource. For background information, see Profiling Resources and Extensibility. Additional profiles and resources may be found in published Implementation Guides , or in the Conformance resource registry .

8.17.9.1 Profiles

No Profiles defined for this resource

8.17.9.2 Extensions

No Extensions defined for this resource

Extensions that reference this resource:

No Extensions refer to this resource

8.17.9.3 Search Extensions

No Search Extensions defined for this resource

8.17.10 Resource Substance - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/substance-version-maps.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.17.10 Resource Substance - Detailed Descriptions

Orders and Observations Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Not linked to any defined compartments

R2 : R3 Converson maps for Substance.

Functional status for this map: 6 tests that all execute ok. All tests pass round-trip testing and all r3 resources are valid. (see documentation)

8.17.10.1 R2 to R3

```
map "http://hl7.org/fhir/StructureMap/Substance2to3" = "R2 to R3 Conversions for Substance"

uses "http://hl7.org/fhir/DSTU2/StructureDefinition/Substance" alias SubstanceR2 as source
uses "http://hl7.org/fhir/StructureDefinition/Substance" alias Substance as target

imports "http://hl7.org/fhir/StructureMap/*2to3"

group for type+types  Substance extends DomainResource
  input src : SubstanceR2 as source
  input tgt : Substance as target

  "Substance-identifier" : for src.identifier make tgt.identifier
  "Substance-category" : for src.category make tgt.category
  "Substance-code" : for src.code make tgt.code
  "Substance-description" : for src.description make tgt.description
  "Substance-instance" : for src.instance as vs0 make tgt.instance as vt0 then sbsInstance(vs0, vt0)
  "Substance-ingredient" : for src.ingredient as vs0 make tgt.ingredient as vt0 then sbsIngredient(vs0, vt0)
endgroup


group sbsInstance extends BackboneElement
  input src : Substance as source
  input tgt : SubstanceR2 as target

  "Substance.instance-identifier" : for src.identifier make tgt.identifier
  "Substance.instance-expiry" : for src.expiry make tgt.expiry
  "Substance.instance-quantity" : for src.quantity make tgt.quantity
endgroup

group sbsIngredient extends BackboneElement
  input src : Substance as source
  input tgt : SubstanceR2 as target

  "Substance.ingredient-quantity" : for src.quantity make tgt.quantity
  "Substance.ingredient-substance" : for src.substance make tgt.substance
endgroup
```

8.17.10.2 R3 to R2

```
map "http://hl7.org/fhir/StructureMap/Substance3to2" = "R3 to R2 Conversion for Substance"

uses "http://hl7.org/fhir/StructureDefinition/Substance" alias Substance as source
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/Substance" alias SubstanceR2 as target

imports "http://hl7.org/fhir/StructureMap/*3to2"

group for type+types  Substance extends DomainResource
  input src : Substance as source
  input tgt : SubstanceR2 as target

  "Substance-identifier" : for src.identifier make tgt.identifier
  "Substance-category" : for src.category make tgt.category
  "Substance-code" : for src.code make tgt.code
  "Substance-description" : for src.description make tgt.description
  "Substance-instance" : for src.instance as vs0 make tgt.instance as vt0 then sbsInstance(vs0, vt0)
  "Substance-ingredient" : for src.ingredient as vs0 make tgt.ingredient as vt0 then sbsIngredient(vs0, vt0)
endgroup


group sbsInstance extends BackboneElement
  input src : Substance as source
  input tgt : SubstanceR2 as target

  "Substance.instance-identifier" : for src.identifier make tgt.identifier
  "Substance.instance-expiry" : for src.expiry make tgt.expiry
  "Substance.instance-quantity" : for src.quantity make tgt.quantity
endgroup

group sbsIngredient extends BackboneElement
  input src : Substance as source
  input tgt : SubstanceR2 as target
```

```
  "Substance.ingredient-quantity" : for src.quantity make tgt.quantity
  "Substance.ingredient-substance" : for src.substance where $this is "Reference" make tgt.substance
endgroup
```

8.17.10.3 R3 Validation Errors

8.18 Resource Person - Content

URL: https://hl7.org/fhir/STU3/person.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

Patient Administration Work Group | Maturity Level: 2 | Trial Use | Compartments: Patient, Practitioner, RelatedPerson

Demographics and administrative information about a person independent of a specific health-related context.

8.18.1 Scope and Usage

An individual has identity outside of a healthcare setting. The Person resource is used to capture this information and to relate the person as an individual to other resources that do have a health-related context.

For example, while a patient resource may be created and maintained by each organization providing care for that person as a patient, a person resource provides a mechanism for linking patient resources across different organizations and their unique patient identity domains.

This resource is not referenced by any other resources.

This resource is referenced by requestgroup

8.18.2 Resource Content

* Structure

* UML

* XML

* JSON

* Turtle

* R2 Diff

* All

Structure

Name | Flags | Card. | Type | Description & Constraints
Person | | | DomainResource | A generic person record Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | | 0..* | Identifier | A human identifier for this person
name | ? | 0..* | HumanName | A name associated with the person
telecom | ? | 0..* | ContactPoint | A contact detail for the person
gender | ? | 0..1 | code | male | female | other | unknown AdministrativeGender (Required)
birthDate | ? | 0..1 | date | The date on which the person was born
address | | 0..* | Address | One or more addresses for the person
photo | | 0..1 | Attachment | Image of the person
managingOrganization | ? | 0..1 | Reference(Organization) | The organization that is the custodian of the person record
active | ?!? | 0..1 | boolean | This person's record is in active use
link | | 0..* | BackboneElement | Link to a resource that concerns the same actual person
target | | 1..1 | Reference(Patient | Practitioner | RelatedPerson | Person) | The resource to which this actual person is associated
assurance | | 0..1 | code | level1 | level2 | level3 | level4 IdentityAssuranceLevel (Required)
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Person xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier A human identifier for this person --></identifier>
 <name><!-- 0..* HumanName A name associated with the person --></name>
 <telecom><!-- 0..* ContactPoint A contact detail for the person --></telecom>
 <gender value="[code]"/><!-- 0..1 male | female | other | unknown -->
 <birthDate value="[date]"/><!-- 0..1 The date on which the person was born -->
 <address><!-- 0..* Address One or more addresses for the person --></address>
 <photo><!-- 0..1 Attachment Image of the person --></photo>
 <managingOrganization><!-- 0..1 Reference(Organization) The organization that is the custodian of the person record
    --></managingOrganization>
 <active value="[boolean]"/><!-- 0..1 This person's record is in active use -->
 <link>  <!-- 0..* Link to a resource that concerns the same actual person -->
  <target><!-- 1..1 Reference(Patient|Practitioner|RelatedPerson|Person) The resource to which this actual person is
    associated --></target>
  <assurance value="[code]"/><!-- 0..1 level1 | level2 | level3 | level4 -->
 </link>
</Person>
```

JSON Template

```
{
  "resourceType" : "Person",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // A human identifier for this person
  "name" : [{ HumanName }], // A name associated with the person
  "telecom" : [{ ContactPoint }], // A contact detail for the person
  "gender" : "<code>", // male | female | other | unknown
  "birthDate" : "<date>", // The date on which the person was born
  "address" : [{ Address }], // One or more addresses for the person
  "photo" : { Attachment }, // Image of the person
  "managingOrganization" : { Reference(Organization) }, // The organization that is the custodian of the person record
  "active" : <boolean>, // This person's record is in active use
  "link" : [{ // Link to a resource that concerns the same actual person
    "target" : { Reference(Patient|Practitioner|RelatedPerson|Person) }, // R!  The resource to which this actual person
     is associated
    "assurance" : "<code>" // level1 | level2 | level3 | level4
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Person;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Person.identifier [ Identifier ], ... ; # 0..* A human identifier for this person
  fhir:Person.name [ HumanName ], ... ; # 0..* A name associated with the person
  fhir:Person.telecom [ ContactPoint ], ... ; # 0..* A contact detail for the person
  fhir:Person.gender [ code ]; # 0..1 male | female | other | unknown
  fhir:Person.birthDate [ date ]; # 0..1 The date on which the person was born
  fhir:Person.address [ Address ], ... ; # 0..* One or more addresses for the person
  fhir:Person.photo [ Attachment ]; # 0..1 Image of the person
  fhir:Person.managingOrganization [ Reference(Organization) ]; # 0..1 The organization that is the custodian of the
    person record
  fhir:Person.active [ boolean ]; # 0..1 This person's record is in active use
  fhir:Person.link [ # 0..* Link to a resource that concerns the same actual person
    fhir:Person.link.target [ Reference(Patient|Practitioner|RelatedPerson|Person) ]; # 1..1 The resource to which this
    actual person is associated
    fhir:Person.link.assurance [ code ]; # 0..1 level1 | level2 | level3 | level4
  ], ...;
]
```

Changes since DSTU2

Person

See the Full Difference for further information

This analysis is available as XML or JSON.

See R2 <--> R3 Conversion Maps (status = 5 tests that all execute ok. All tests pass round-trip testing and all r3 resources are valid.).

Structure

Name | Flags | Card. | Type | Description & Constraints

Person | | | DomainResource | A generic person record Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension

identifier | | 0..* | Identifier | A human identifier for this person

name | ? | 0..* | HumanName | A name associated with the person

telecom | ? | 0..* | ContactPoint | A contact detail for the person

gender | ? | 0..1 | code | male | female | other | unknown AdministrativeGender (Required)

birthDate | ? | 0..1 | date | The date on which the person was born

address | | 0..* | Address | One or more addresses for the person

photo | | 0..1 | Attachment | Image of the person

managingOrganization | ? | 0..1 | Reference(Organization) | The organization that is the custodian of the person record

active | ?!? | 0..1 | boolean | This person's record is in active use

link | | 0..* | BackboneElement | Link to a resource that concerns the same actual person

target | | 1..1 | Reference(Patient | Practitioner | RelatedPerson | Person) | The resource to which this actual person is associated

assurance | | 0..1 | code | level1 | level2 | level3 | level4 IdentityAssuranceLevel (Required)

Documentation for this format

UML Diagram (Legend)

XML Template

```
<Person xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier A human identifier for this person --></identifier>
 <name><!-- 0..* HumanName A name associated with the person --></name>
 <telecom><!-- 0..* ContactPoint A contact detail for the person --></telecom>
 <gender value="[code]"/><!-- 0..1 male | female | other | unknown -->
```

```
 <birthDate value="[date]"/><!-- 0..1 The date on which the person was born -->
 <address><!-- 0..* Address One or more addresses for the person --></address>
 <photo><!-- 0..1 Attachment Image of the person --></photo>
 <managingOrganization><!-- 0..1 Reference(Organization) The organization that is the custodian of the person record
    --></managingOrganization>
 <active value="[boolean]"/><!-- 0..1 This person's record is in active use -->
 <link>  <!-- 0..* Link to a resource that concerns the same actual person -->
  <target><!-- 1..1 Reference(Patient|Practitioner|RelatedPerson|Person) The resource to which this actual person is
    associated --></target>
  <assurance value="[code]"/><!-- 0..1 level1 | level2 | level3 | level4 -->
 </link>
</Person>
```

JSON Template

```
{
  "resourceType" : "Person",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // A human identifier for this person
  "name" : [{ HumanName }], // A name associated with the person
  "telecom" : [{ ContactPoint }], // A contact detail for the person
  "gender" : "<code>", // male | female | other | unknown
  "birthDate" : "<date>", // The date on which the person was born
  "address" : [{ Address }], // One or more addresses for the person
  "photo" : { Attachment }, // Image of the person
  "managingOrganization" : { Reference(Organization) }, // The organization that is the custodian of the person record
  "active" : <boolean>, // This person's record is in active use
  "link" : [{ // Link to a resource that concerns the same actual person
    "target" : { Reference(Patient|Practitioner|RelatedPerson|Person) }, // R!  The resource to which this actual person
     is associated
    "assurance" : "<code>" // level1 | level2 | level3 | level4
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Person;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Person.identifier [ Identifier ], ... ; # 0..* A human identifier for this person
  fhir:Person.name [ HumanName ], ... ; # 0..* A name associated with the person
  fhir:Person.telecom [ ContactPoint ], ... ; # 0..* A contact detail for the person
  fhir:Person.gender [ code ]; # 0..1 male | female | other | unknown
  fhir:Person.birthDate [ date ]; # 0..1 The date on which the person was born
  fhir:Person.address [ Address ], ... ; # 0..* One or more addresses for the person
  fhir:Person.photo [ Attachment ]; # 0..1 Image of the person
  fhir:Person.managingOrganization [ Reference(Organization) ]; # 0..1 The organization that is the custodian of the
    person record
  fhir:Person.active [ boolean ]; # 0..1 This person's record is in active use
  fhir:Person.link [ # 0..* Link to a resource that concerns the same actual person
    fhir:Person.link.target [ Reference(Patient|Practitioner|RelatedPerson|Person) ]; # 1..1 The resource to which this
    actual person is associated
    fhir:Person.link.assurance [ code ]; # 0..1 level1 | level2 | level3 | level4
  ], ...;
]
```

Changes since DSTU2

Person

See the Full Difference for further information

This analysis is available as XML or JSON.

See R2 <--> R3 Conversion Maps (status = 5 tests that all execute ok. All tests pass round-trip testing and all r3 resources are valid.).

Alternate definitions: Master Definition (XML, JSON), XML Schema/Schematron (for) + JSON Schema, ShEx (for Turtle)

8.18.2.1 Terminology Bindings

Path | Definition | Type | Reference

Person.gender | The gender of a person used for administrative purposes. | Required | AdministrativeGender

Person.link.assurance | The level of confidence that this link represents the same actual person, based on NIST Authentication Levels. | Required | IdentityAssuranceLevel

8.18.3 De-normalized Data

Most of the properties of the Person resource are replicated within the other resources that they are shared with. This is intentional and highlights that the "disconnectedness" of the resources.

Not many systems actually implement a shared Person record, and as such the values DO become out of sync with each other. The inclusion of this resource does permit a capability for systems to identify other instances of this actual person's data via a centralized registry that can assist in keeping things up to date.

8.18.4 Person and Linking

The link element is used to relate resources under a common person record. This element supports two primary scenarios where other resources refer to the same person resource.

The link element cannot be used to link to RelatedPerson entries. So we can utilize a Person resource to relate these elements together identifying them as the same individual.

8.18.5 Cross-Domain Patient Directory

In a data sharing network, finding the location of patient records across different systems is a necessary pre-requisite for accessing external patient data. Using the link element, systems associate patient resources from different organizations. The assuranceLevel associated with the link provides a way for a system to qualify its confidence in the asserted link. For example, a relationship from the person to a patient using a probabilistic matching algorithm may be represented using a link with an assurance level of level1, while a relationship established using a government-issued photo ID may be created with an assurance level of level3.

8.18.6 Cross-Domain Provider Directory

Similarly, providers working in multiple healthcare service settings may be linked across different organizations using the link element. The various practitioner resources can be related using a common person resource with a link for each of the practitioner resources located in other organizations.

8.18.7 Client Portal

Client Portals provide consumer access to a window of their data locked up in healthcare systems. In many cases these systems are externally integrated and do not have access to the legacy CIS/PAS systems.

In contrast to most systems where a user has access to multiple patients depending on their roles and permissions, a client portal provides a consumer with direct access to their data (with permissions applied). This can also include not just their data (via the Person.link Patient entries), but also to information that they have entered, or were involved with (via the Person.link RelatedPerson entries), such as observations that they entered. With appropriate permissions/consent applied, the user could have access to other Patient records linked via a RelatedPerson.

8.18.8 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
address | string | A server defined search that may match any of the string fields in the Address, including line, city, state, country, postalCode, and/or text | Person.address | 3 Resources
address-city | string | A city specified in an address | Person.address.city | 3 Resources
address-country | string | A country specified in an address | Person.address.country | 3 Resources
address-postalcode | string | A postal code specified in an address | Person.address.postalCode | 3 Resources
address-state | string | A state specified in an address | Person.address.state | 3 Resources
address-use | token | A use code specified in an address | Person.address.use | 3 Resources
birthdate | date | The person's date of birth | Person.birthDate | 2 Resources
email | token | A value in an email contact | Person.telecom.where(system='email') | 4 Resources
gender | token | The gender of the person | Person.gender | 3 Resources
identifier | token | A person Identifier | Person.identifier
link | reference | Any link has this Patient, Person, RelatedPerson or Practitioner reference | Person.link.target (Practitioner, Patient, Person, RelatedPerson)
name | string | A server defined search that may match any of the string fields in the HumanName, including family, give, prefix, suffix, suffix, and/or text | Person.name
organization | reference | The organization at which this person record is being managed | Person.managingOrganization (Organization)
patient | reference | The Person links to this Patient | Person.link.target (Patient)
phone | token | A value in a phone contact | Person.telecom.where(system='phone') | 4 Resources
phonetic | string | A portion of name using some kind of phonetic matching algorithm | Person.name | 3 Resources
practitioner | reference | The Person links to this Practitioner | Person.link.target (Practitioner)
relatedperson | reference | The Person links to this RelatedPerson | Person.link.target (RelatedPerson)
telecom | token | The value in any kind of contact | Person.telecom | 4 Resources

8.18.9 Resource Person - Examples

URL: https://hl7.org/fhir/STU3/person-examples.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.18.9 Resource Person - Examples

Patient Administration Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Patient, Practitioner, RelatedPerson

Example List:

General Person Example | example | XML | JSON | Turtle

Grahame's person resource | grahame | XML | JSON | Turtle
Patient Portal Example | pp | XML | JSON | Turtle
Provider Directory Example | pd | XML | JSON | Turtle
Same person from RelatedPerson | f002 | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

8.18.10 Resource Person - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/person-definitions.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.18.10 Resource Person - Detailed Descriptions

Patient Administration Work Group | Maturity Level: 2 | Trial Use | Compartments: Patient, Practitioner, RelatedPerson

Detailed Descriptions for the elements in the Person resource.

Person
Definition | Demographics and administrative information about a person independent of a specific health-related context.
Control | 1..1
Requirements | Need to track persons potentially across multiple roles.
Comments | The Person resource does justice to person registries that keep track of persons regardless of their role. The Person resource is also a primary resource to point to for people acting in a particular role such as SubjectofCare, Practitioner, and Agent. Very few attributes are specific to any role and so Person is kept lean. Most attributes are expected to be tied to the role the Person plays rather than the Person himself. Examples of that are Guardian (SubjectofCare), ContactParty (SubjectOfCare, Practitioner), and multipleBirthInd (SubjectofCare).
Person.identifier
Definition | Identifier for a person within a particular scope.
Note | This is a business identifer, not a resource identifier (see discussion)
Control | 0..*
Type | Identifier
Requirements | People are known by a variety of ids. Some institutions maintain several, and most collect identifiers for exchange with other organizations concerning the person. Examples are national person identifier and local identifier.
Person.name
Definition | A name associated with the person.
Control | 0..*
Type | HumanName
Requirements | Need to be able to track the person by multiple names. Examples are your official name and a partner name.
Summary | true
Comments | Person may have multiple names with different uses or applicable periods.
Person.telecom
Definition | A contact detail for the person, e.g. a telephone number or an email address.
Control | 0..*
Type | ContactPoint
Requirements | People have (primary) ways to contact them in some way such as phone, email.
Summary | true
Comments | Person may have multiple ways to be contacted with different uses or applicable periods. May need to have options for contacting the person urgently, and also to help with identification.
Person.gender
Definition | Administrative Gender.
Control | 0..1
Terminology Binding | AdministrativeGender (Required)
Type | code
Requirements | Needed for identification of the person, in combination with (at least) name and birth date. Gender of person drives many clinical processes.
Summary | true
Comments | The gender may not match the biological sex as determined by genetics, or the individual's preferred identification. Note that for both humans and particularly animals, there are other legitimate possibilities than M and F, though the vast majority of systems and contexts only support M and F.
Person.birthDate
Definition | The birth date for the person.
Control | 0..1
Type | date
Requirements | Age of person drives many clinical processes, and is often used in performing identification of the person. Times are not included so as to not confuse things with potential timezone issues.
Summary | true
Comments | At least an estimated year should be provided as a guess if the real DOB is unknown.
Person.address
Definition | One or more addresses for the person.
Control | 0..*
Type | Address
Requirements | May need to keep track of person's addresses for contacting, billing or reporting requirements and also to help with identification.
Comments | Person may have multiple addresses with different uses or applicable periods.

Person.photo
Definition | An image that can be displayed as a thumbnail of the person to enhance the identification of the individual.
Control | 0..1
Type | Attachment
Person.managingOrganization
Definition | The organization that is the custodian of the person record.
Control | 0..1
Type | Reference(Organization)
Requirements | Need to know who recognizes this person record, manages and updates it.
Summary | true
Person.active
Definition | Whether this person's record is in active use.
Control | 0..1
Type | boolean
Is Modifier | true
Requirements | Need to be able to mark a person record as not to be used because it was created in error.
Summary | true
Person.link
Definition | Link to a resource that concerns the same actual person.
Control | 0..*
Person.link.target
Definition | The resource to which this actual person is associated.
Control | 1..1
Type | Reference(Patient | Practitioner | RelatedPerson | Person)
Person.link.assurance
Definition | Level of assurance that this link is actually associated with the target resource.
Control | 0..1
Terminology Binding | IdentityAssuranceLevel (Required)
Type | code

8.18.11 Resource Person - Mappings

URL: https://hl7.org/fhir/STU3/person-mappings.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.18.11 Resource Person - Mappings

Patient Administration Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Patient, Practitioner, RelatedPerson

Mappings:

RIM Mapping

HL7 v2 Mapping

Mappings for the person resource.

8.18.11.1 Mappings for RIM Mapping (http://hl7.org/v3)

Person | Person(classCode="PSN" and determinerCode="INST" and quantity="1")
identifier | .plays:Role(classCode='IDENT').id
name | ./name
telecom | ./telecom
gender | player[classCode=PSN|ANM and determinerCode=INSTANCE]/administrativeGender
birthDate | ./birthTime
address | ./addr
photo | player[classCode=PSN|ANM and determinerCode=INSTANCE]/desc
managingOrganization | scoper
active | statusCode
link | outboundLink
target | id
assurance | priorityNumber?

8.18.11.2 Mappings for HL7 v2 Mapping (http://hl7.org/v2)

Person
identifier | PID-3
name | PID-5, PID-9
telecom | PID-13, PID-14
gender | PID-8
birthDate | PID-7
address | PID-11
photo | OBX-5 - needs a profile
managingOrganization
active

link
target
assurance

8.18.12 Resource Person - Design Notes

URL: https://hl7.org/fhir/STU3/person-explanations.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.18.12 Resource Person - Design Notes

to do

8.18.13 Resource Person - Extensions & Profiles

URL: https://hl7.org/fhir/STU3/person-profiles.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.18.13 Resource Person - Extensions & Profiles

Patient Administration Work Group | Maturity Level: N/A | Ballot Status: Informative

This table lists profiles and extensions for the Person resource. For background information, see Profiling Resources and Extensibility. Additional profiles and resources may be found in published Implementation Guides , or in the Conformance resource registry .

8.18.13.1 Profiles

No Profiles defined for this resource

8.18.13.2 Extensions

No Extensions defined for this resource

Extensions that reference this resource:

cqif-guidanceinitiatingPerson | initiatingPerson | for CQIF Guidance Extensions
cqif-guidancereceivingPerson | receivingPerson | for CQIF Guidance Extensions

8.18.13.3 Search Extensions

No Search Extensions defined for this resource

8.18.14 Resource Person - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/person-version-maps.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.18.14 Resource Person - Detailed Descriptions

Patient Administration Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Patient, Practitioner, RelatedPerson

R2 : R3 Converson maps for Person.

Functional status for this map: 5 tests that all execute ok. All tests pass round-trip testing and all r3 resources are valid. (see documentation)

8.18.14.1 R2 to R3

```
map "http://hl7.org/fhir/StructureMap/Person2to3" = "R2 to R3 Conversions for Person"

uses "http://hl7.org/fhir/DSTU2/StructureDefinition/Person" alias PersonR2 as source
uses "http://hl7.org/fhir/StructureDefinition/Person" alias Person as target

imports "http://hl7.org/fhir/StructureMap/*2to3"

group for type+types  Person extends DomainResource
  input src : PersonR2 as source
  input tgt : Person as target

  "Person-identifier" : for src.identifier make tgt.identifier
  "Person-name" : for src.name make tgt.name
  "Person-telecom" : for src.telecom make tgt.telecom
  "Person-gender" : for src.gender make tgt.gender
  "Person-birthDate" : for src.birthDate make tgt.birthDate
  "Person-address" : for src.address make tgt.address
  "Person-photo" : for src.photo make tgt.photo
  "Person-managingOrganization" : for src.managingOrganization make tgt.managingOrganization
  "Person-active" : for src.active make tgt.active
```

```
  "Person-link" : for src.link as vs0 make tgt.link as vt0 then psnLink(vs0, vt0)
endgroup

group psnLink extends BackboneElement
  input src as source
  input tgt as target

  "Person.link-target" : for src.target make tgt.target
  "Person.link-assurance" : for src.assurance make tgt.assurance
endgroup
```

8.18.14.2 R3 to R2

```
map "http://hl7.org/fhir/StructureMap/Person3to2" = "R3 to R2 Conversion for Person"

uses "http://hl7.org/fhir/StructureDefinition/Person" alias Person as source
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/Person" alias PersonR2 as target

imports "http://hl7.org/fhir/StructureMap/*3to2"

group for type+types  Person extends DomainResource
  input src : Person as source
  input tgt : PersonR2 as target

  "Person-identifier" : for src.identifier make tgt.identifier
  "Person-name" : for src.name make tgt.name
  "Person-telecom" : for src.telecom make tgt.telecom
  "Person-gender" : for src.gender make tgt.gender
  "Person-birthDate" : for src.birthDate make tgt.birthDate
  "Person-address" : for src.address make tgt.address
  "Person-photo" : for src.photo make tgt.photo
  "Person-managingOrganization" : for src.managingOrganization make tgt.managingOrganization
  "Person-active" : for src.active make tgt.active
  "Person-link" : for src.link as vs0 make tgt.link as vt0 then psnLink(vs0, vt0)
endgroup

group psnLink extends BackboneElement
  input src as source
  input tgt as target

  "Person.link-target" : for src.target make tgt.target
  "Person.link-assurance" : for src.assurance make tgt.assurance
endgroup
```

8.18.14.3 R3 Validation Errors

8.19 Resource Linkage - Content

URL: https://hl7.org/fhir/STU3/linkage.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

Patient Care Work Group | Maturity Level: 0 | Draft | Compartments: Practitioner

Identifies two or more records (resource instances) that are referring to the same real-world "occurrence".

8.19.1 Scope and Usage

This resource allows the assertion of linkages between multiple resource instances (generally of the same type) that are referring to the same underlying business objects. For example, multiple Condition records that refer to the same underlying problem/issue for a Patient; multiple AllergyIntolerance records that refer to the same reaction susceptibility; multiple Patient, Practioner and/or RelatedPerson records that refer to the same human being or animal.

FHIR supports a process for sharing electronic records. It is common for multiple records to exist that deal with the same real-world phenomenon. This can result from information being captured by different systems, information being captured within a single system by different users (either deliberately to represent distinct perspectives or accidentally when a new record is created rather than updating an existing record). These multiple records may be referred to as "duplicate" records, but in practice they aren't often actually "duplicate" in that the data represented (and the history of the records) will be at least somewhat distinct.

In some cases, the solution upon identifying duplicates is to deprecate one of the records (e.g. by changing the status to "entered in error") and move all relevant information to the surviving record. In other cases, the resource may support the ability to establish a linkage directly between the resources. However, in some cases, both records may need to survive or there's a desire to have both resources continue to be maintained, perhaps because the resources live on different servers or have different 'owners'. The Linkage resource is intended to satisfy this use-case.

8.19.2 Boundaries and Relationships

The Person resource should be used to link a person independent of a specific health-related context. Linkage can be used for all other use cases.

8.19.3 Resource Content

* Structure
* UML
* XML
* JSON
* Turtle
* R2 Diff
* All

Structure

Name | Flags | Card. | Type | Description & Constraints
Linkage | ?I | | DomainResource | Links records for 'same' item + Must have at least two items Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
active | ? | 0..1 | boolean | Whether this linkage assertion is active or not
author | ? | 0..1 | Reference(Practitioner | Organization) | Who is responsible for linkages
item | ?I | 1..* | BackboneElement | Item to be linked
type | ? | 1..1 | code | source | alternate | historical LinkageType (Required)
resource | ? | 1..1 | Reference() | Resource being linked
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Linkage xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <active value="[boolean]"/><!-- 0..1 Whether this linkage assertion is active or not -->
 <author><!-- 0..1 Reference(Practitioner|Organization) Who is responsible for linkages --></author>
 <item>  <!--  1..* Item to be linked -->
  <type value="[code]"/><!-- 1..1 source | alternate | historical -->
  <resource><!-- 1..1 Reference Resource being linked --></resource>
 </item>
</Linkage>
```

JSON Template

```
{
  "resourceType" : "Linkage",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "active" : <boolean>, // Whether this linkage assertion is active or not
  "author" : { Reference(Practitioner|Organization) }, // Who is responsible for linkages
  "item" : [{ // C? R!  Item to be linked
    "type" : "<code>", // R!  source | alternate | historical
    "resource" : { Reference } // R!  Resource being linked
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Linkage;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Linkage.active [ boolean ]; # 0..1 Whether this linkage assertion is active or not
  fhir:Linkage.author [ Reference(Practitioner|Organization) ]; # 0..1 Who is responsible for linkages
  fhir:Linkage.item [ # 1..* Item to be linked
    fhir:Linkage.item.type [ code ]; # 1..1 source | alternate | historical
    fhir:Linkage.item.resource [ Reference ]; # 1..1 Resource being linked
  ], ...;
]
```

Changes since DSTU2

This resource did not exist in Release 2

This analysis is available as XML or JSON.

Structure

Name | Flags | Card. | Type | Description & Constraints
Linkage | ?I | | DomainResource | Links records for 'same' item + Must have at least two items Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
active | ? | 0..1 | boolean | Whether this linkage assertion is active or not
author | ? | 0..1 | Reference(Practitioner | Organization) | Who is responsible for linkages
item | ?I | 1..* | BackboneElement | Item to be linked
type | ? | 1..1 | code | source | alternate | historical LinkageType (Required)
resource | ? | 1..1 | Reference() | Resource being linked
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Linkage xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <active value="[boolean]"/><!-- 0..1 Whether this linkage assertion is active or not -->
 <author><!-- 0..1 Reference(Practitioner|Organization) Who is responsible for linkages --></author>
 <item>  <!--  1..* Item to be linked -->
  <type value="[code]"/><!-- 1..1 source | alternate | historical -->
  <resource><!-- 1..1 Reference Resource being linked --></resource>
 </item>
</Linkage>
```

JSON Template

```
{
  "resourceType" : "Linkage",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "active" : <boolean>, // Whether this linkage assertion is active or not
  "author" : { Reference(Practitioner|Organization) }, // Who is responsible for linkages
  "item" : [{ // C? R!  Item to be linked
    "type" : "<code>", // R!  source | alternate | historical
    "resource" : { Reference } // R!  Resource being linked
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Linkage;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Linkage.active [ boolean ]; # 0..1 Whether this linkage assertion is active or not
  fhir:Linkage.author [ Reference(Practitioner|Organization) ]; # 0..1 Who is responsible for linkages
  fhir:Linkage.item [ # 1..* Item to be linked
    fhir:Linkage.item.type [ code ]; # 1..1 source | alternate | historical
    fhir:Linkage.item.resource [ Reference ]; # 1..1 Resource being linked
  ], ...;
]
```

Changes since DSTU2

This resource did not exist in Release 2

This analysis is available as XML or JSON.

Alternate definitions: Master Definition (XML, JSON), XML Schema/Schematron (for) + JSON Schema, ShEx (for Turtle)

8.19.3.1 Terminology Bindings

Path | Definition | Type | Reference
Linkage.item.type | Used to distinguish different roles a resource can play within a set of linked resources | Required | LinkageType

8.19.3.2 Constraints

* lnk-1: Must have at least two items (expression : item.count()>1)

8.19.4 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
author | reference | Author of the Linkage | Linkage.author (Practitioner, Organization)
item | reference | Matches on any item in the Linkage | Linkage.item.resource
source | reference | Matches on any item in the Linkage with a type of 'source' | Linkage.item.resource

8.19.5 Resource Linkage - Examples

URL: https://hl7.org/fhir/STU3/linkage-examples.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.19.5 Resource Linkage - Examples

Patient Care Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Practitioner

Example List:

Example of linkage | example | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

8.19.6 Resource Linkage - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/linkage-definitions.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.19.6 Resource Linkage - Detailed Descriptions

Patient Care Work Group | Maturity Level: 0 | Draft | Compartments: Practitioner

Detailed Descriptions for the elements in the Linkage resource.

Linkage
Definition | Identifies two or more records (resource instances) that are referring to the same real-world "occurrence".
Control | 1..1
Summary | true
Invariants | Defined on this element lnk-1: Must have at least two items (expression : item.count()>1, xpath: count(f:item)>1)
Linkage.active
Definition | Indicates whether the asserted set of linkages are considered to be "in effect".
Control | 0..1
Type | boolean
Default Value | true
Requirements | Need to be able to mark a linkage record as not to be used because it was created in error.
Summary | true
Comments | If false, any asserted linkages should not be considered current/relevant/applicable.
Linkage.author
Definition | Identifies the user or organization responsible for asserting the linkages and who establishes the context for evaluating the nature of each linkage.
Control | 0..1
Type | Reference(Practitioner | Organization)
Requirements | "Primary" for the same set of linked resources may be different from different user's/organization's perspectives.
Summary | true
Linkage.item
Definition | Identifies one of the records that is considered to refer to the same real-world occurrence as well as how the items hould be evaluated within the collection of linked items.
Control | 1..*
Summary | true
Invariants | Affect this element lnk-1: Must have at least two items (expression : item.count()>1, xpath: count(f:item)>1)
Linkage.item.type
Definition | Distinguishes which item is "source of truth" (if any) and which items are no longer considered to be current representations.
Control | 1..1
Terminology Binding | LinkageType (Required)
Type | code
Summary | true
Linkage.item.resource
Definition | The resource instance being linked as part of the group.
Control | 1..1
Type | Reference
Summary | true

8.19.7 Resource Linkage - Mappings

URL: https://hl7.org/fhir/STU3/linkage-mappings.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.19.7 Resource Linkage - Mappings

Patient Care Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Practitioner

Mappings:

RIM Mapping

Mappings for the linkage resource.

8.19.7.1 Mappings for RIM Mapping (http://hl7.org/v3)

Linkage | N/A - RIM doesn't know how to do this
active
author
item
type
resource

8.19.8 Resource Linkage - Design Notes

URL: https://hl7.org/fhir/STU3/linkage-explanations.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.19.8 Resource Linkage - Design Notes

to do

8.19.9 Resource Linkage - Extensions & Profiles

URL: https://hl7.org/fhir/STU3/linkage-profiles.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.19.9 Resource Linkage - Extensions & Profiles

Patient Care Work Group | Maturity Level: N/A | Ballot Status: Informative

This table lists profiles and extensions for the Linkage resource. For background information, see Profiling Resources and Extensibility. Additional profiles and resources may be found in published Implementation Guides , or in the Conformance resource registry .

8.19.9.1 Profiles

No Profiles defined for this resource

8.19.9.2 Extensions

No Extensions defined for this resource

Extensions that reference this resource:

No Extensions refer to this resource

8.19.9.3 Search Extensions

No Search Extensions defined for this resource

8.20 Resource Communication - Content

URL: https://hl7.org/fhir/STU3/communication.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

Patient Care Work Group | Maturity Level: 2 | Trial Use | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

An occurrence of information being transmitted; e.g. an alert that was sent to a responsible provider, a public health agency was notified about a reportable condition.

8.20.1 Scope and Usage

Communication is one of the event resources in the FHIR workflow specification.

This resource is a record of a communication. A communication is a conveyance of information from one entity, a sender, to another entity, a receiver. The sender and receivers may be patients, practitioners, related persons, organizations, or devices. Communication use cases include:

* A reminder or alert delivered to a responsible provider

* A recorded notification from the nurse that a patient's temperature exceeds a value

* A notification to a public health agency of a patient presenting with a communicable disease reportable to the public health agency

* Patient educational material sent by a provider to a patient

Non-patient specific communication use cases may include:

* A nurse call from a hall bathroom

* Advisory for battery service from a pump

8.20.2 Boundaries and Relationships

This resource is a record of a communication that has occurred. It does not represent the actual flow of communication. While AuditEvent can track electronic disclosures of information, it cannot track conversations, phone calls, letters and other interactions that are not system-to-system. And even for system-to-system communications, the specific end recipients may not be known. As well, AuditEvents are not considered to be "part" of the patient record, while Communication instances are. The Communication resource is not used as a general audit mechanism to track every disclosure of every record. Rather, it is used when a clinician or other user wants to ensure a record of a particular communication is itself maintained as part of the reviewable health record.

Flag resources represent a continuous ongoing "communication" alerting anyone dealing with the patient of certain precautions to take or issues to be aware of. The flags are continuously present as an ongoing reminder. This is distinct from Communication where there is a specific intended sender and receiver and the information is delivered only once.

Communication and Encounter

The Communication is about the transfer of information (which may or may not occur as part of an encounter), while Encounter is about the coming together (in person or virtually) of a Patient with a Practitioner. Communication does not deal with the duration of a call, it represents the fact that information was transferred at a particular point in time.

The phone calls involving the Patient should be handled using Encounter. Phone calls not involving the patient (e.g. between practitioners or practitioner to relative) that are tracked for billing or other purposes can use Communication to represent the information transferred, but are not ideal to represent the call itself. A better mechanism for handling such calls will be explored in a future release.

8.20.3 Resource Content

* Structure

* UML

* XML

* JSON

* Turtle

* R2 Diff

* All

Structure

Name | Flags | Card. | Type | Description & Constraints
Communication | I | | DomainResource | A record of information transmitted from a sender to a receiver + Not Done Reason can only be specified if NotDone is "true" Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | ? | 0..* | Identifier | Unique identifier
definition | ? | 0..* | Reference(PlanDefinition | ActivityDefinition) | Instantiates protocol or definition
basedOn | ? | 0..* | Reference(Any) | Request fulfilled by this communication
partOf | ? | 0..* | Reference(Any) | Part of this action
status | ?!? | 1..1 | code | preparation | in-progress | suspended | aborted | completed | entered-in-error EventStatus (Required)
notDone | ?!? | 0..1 | boolean | Communication did not occur
notDoneReason | ?I | 0..1 | CodeableConcept | Why communication did not occur CommunicationNotDoneReason (Example)
category | | 0..* | CodeableConcept | Message category CommunicationCategory (Example)
medium | | 0..* | CodeableConcept | A channel of communication v3 Code System ParticipationMode (Example)
subject | ? | 0..1 | Reference(Patient | Group) | Focus of message
recipient | | 0..* | Reference(Device | Organization | Patient | Practitioner | RelatedPerson | Group) | Message recipient
topic | | 0..* | Reference(Any) | Focal resources
context | ? | 0..1 | Reference(Encounter | EpisodeOfCare) | Encounter or episode leading to message
sent | | 0..1 | dateTime | When sent
received | | 0..1 | dateTime | When received
sender | | 0..1 | Reference(Device | Organization | Patient | Practitioner | RelatedPerson) | Message sender
reasonCode | ? | 0..* | CodeableConcept | Indication for message SNOMED CT Clinical Findings (Example)
reasonReference | ? | 0..* | Reference(Condition | Observation) | Why was communication done?
payload | | 0..* | BackboneElement | Message payload
content[x] | | 1..1 | | Message part content
contentString | | | string
contentAttachment | | | Attachment
contentReference | | | Reference(Any)
note | | 0..* | Annotation | Comments made about the communication
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Communication xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Unique identifier --></identifier>
 <definition><!-- 0..* Reference(PlanDefinition|ActivityDefinition) Instantiates protocol or definition --></definition>
 <basedOn><!-- 0..* Reference(Any) Request fulfilled by this communication --></basedOn>
 <partOf><!-- 0..* Reference(Any) Part of this action --></partOf>
 <status value="[code]"/><!-- 1..1 preparation | in-progress | suspended | aborted | completed | entered-in-error -->
 <notDone value="[boolean]"/><!-- 0..1 Communication did not occur -->
 <notDoneReason><!--  0..1 CodeableConcept Why communication did not occur --></notDoneReason>
 <category><!-- 0..* CodeableConcept Message category --></category>
 <medium><!-- 0..* CodeableConcept A channel of communication --></medium>
 <subject><!-- 0..1 Reference(Patient|Group) Focus of message --></subject>
 <recipient><!-- 0..* Reference(Device|Organization|Patient|Practitioner|
   RelatedPerson|Group) Message recipient --></recipient>
 <topic><!-- 0..* Reference(Any) Focal resources --></topic>
 <context><!-- 0..1 Reference(Encounter|EpisodeOfCare) Encounter or episode leading to message --></context>
 <sent value="[dateTime]"/><!-- 0..1 When sent -->
```

```
 <received value="[dateTime]"/><!-- 0..1 When received -->
 <sender><!-- 0..1 Reference(Device|Organization|Patient|Practitioner|
   RelatedPerson) Message sender --></sender>
 <reasonCode><!-- 0..* CodeableConcept Indication for message --></reasonCode>
 <reasonReference><!-- 0..* Reference(Condition|Observation) Why was communication done? --></reasonReference>
 <payload>  <!-- 0..* Message payload -->
  <content[x]><!-- 1..1 string|Attachment|Reference(Any) Message part content --></content[x]>
 </payload>
 <note><!-- 0..* Annotation Comments made about the communication --></note>
</Communication>
```

JSON Template

```
{
  "resourceType" : "Communication",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Unique identifier
  "definition" : [{ Reference(PlanDefinition|ActivityDefinition) }], // Instantiates protocol or definition
  "basedOn" : [{ Reference(Any) }], // Request fulfilled by this communication
  "partOf" : [{ Reference(Any) }], // Part of this action
  "status" : "<code>", // R!  preparation | in-progress | suspended | aborted | completed | entered-in-error
  "notDone" : <boolean>, // Communication did not occur
  "notDoneReason" : { CodeableConcept }, // C? Why communication did not occur
  "category" : [{ CodeableConcept }], // Message category
  "medium" : [{ CodeableConcept }], // A channel of communication
  "subject" : { Reference(Patient|Group) }, // Focus of message
  "recipient" : [{ Reference(Device|Organization|Patient|Practitioner|
   RelatedPerson|Group) }], // Message recipient
  "topic" : [{ Reference(Any) }], // Focal resources
  "context" : { Reference(Encounter|EpisodeOfCare) }, // Encounter or episode leading to message
  "sent" : "<dateTime>", // When sent
  "received" : "<dateTime>", // When received
  "sender" : { Reference(Device|Organization|Patient|Practitioner|
   RelatedPerson) }, // Message sender
  "reasonCode" : [{ CodeableConcept }], // Indication for message
  "reasonReference" : [{ Reference(Condition|Observation) }], // Why was communication done?
  "payload" : [{ // Message payload
    // content[x]: Message part content. One of these 3:
    "contentString" : "<string>"
    "contentAttachment" : { Attachment }
    "contentReference" : { Reference(Any) }
  }],
  "note" : [{ Annotation }] // Comments made about the communication
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Communication;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Communication.identifier [ Identifier ], ... ; # 0..* Unique identifier
  fhir:Communication.definition [ Reference(PlanDefinition|ActivityDefinition) ], ... ; # 0..* Instantiates protocol or
    definition
  fhir:Communication.basedOn [ Reference(Any) ], ... ; # 0..* Request fulfilled by this communication
  fhir:Communication.partOf [ Reference(Any) ], ... ; # 0..* Part of this action
  fhir:Communication.status [ code ]; # 1..1 preparation | in-progress | suspended | aborted | completed | entered-in-
    error
  fhir:Communication.notDone [ boolean ]; # 0..1 Communication did not occur
  fhir:Communication.notDoneReason [ CodeableConcept ]; # 0..1 Why communication did not occur
  fhir:Communication.category [ CodeableConcept ], ... ; # 0..* Message category
  fhir:Communication.medium [ CodeableConcept ], ... ; # 0..* A channel of communication
  fhir:Communication.subject [ Reference(Patient|Group) ]; # 0..1 Focus of message
  fhir:Communication.recipient [ Reference(Device|Organization|Patient|Practitioner|RelatedPerson|Group) ], ... ; # 0..*
     Message recipient
  fhir:Communication.topic [ Reference(Any) ], ... ; # 0..* Focal resources
  fhir:Communication.context [ Reference(Encounter|EpisodeOfCare) ]; # 0..1 Encounter or episode leading to message
  fhir:Communication.sent [ dateTime ]; # 0..1 When sent
  fhir:Communication.received [ dateTime ]; # 0..1 When received
  fhir:Communication.sender [ Reference(Device|Organization|Patient|Practitioner|RelatedPerson) ]; # 0..1 Message sender
  fhir:Communication.reasonCode [ CodeableConcept ], ... ; # 0..* Indication for message
  fhir:Communication.reasonReference [ Reference(Condition|Observation) ], ... ; # 0..* Why was communication done?
  fhir:Communication.payload [ # 0..* Message payload
    # Communication.payload.content[x] : 1..1 Message part content. One of these 3
      fhir:Communication.payload.contentString [ string ]
      fhir:Communication.payload.contentAttachment [ Attachment ]
      fhir:Communication.payload.contentReference [ Reference(Any) ]
  ], ...;
  fhir:Communication.note [ Annotation ], ... ; # 0..* Comments made about the communication
]
```

Changes since DSTU2

Communication

Communication.definition | Added Element

Communication.basedOn | Renamed from requestDetail to basedOn Max Cardinality changed from 1 to * Type changed from Reference(CommunicationRequest) to Reference(Resource)
Communication.partOf | Added Element
Communication.status | Min Cardinality changed from 0 to 1 Change value set from http://hl7.org/fhir/ValueSet/communication-status to http://hl7.org/fhir/ValueSet/event-status
Communication.notDone | Added Element
Communication.notDoneReason | Added Element
Communication.category | Max Cardinality changed from 1 to *
Communication.subject | Add Reference(Group)
Communication.topic | Added Element
Communication.context | Renamed from encounter to context Add Reference(EpisodeOfCare)
Communication.reasonCode | Renamed from reason to reasonCode
Communication.reasonReference | Added Element
Communication.note | Added Element

See the Full Difference for further information

This analysis is available as XML or JSON.

See R2 <--> R3 Conversion Maps (status = 2 tests that all execute ok. All tests pass round-trip testing and all r3 resources are valid.).

Structure

Name | Flags | Card. | Type | Description & Constraints
Communication | I | | DomainResource | A record of information transmitted from a sender to a receiver + Not Done Reason can only be specified if NotDone is "true" Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | ? | 0..* | Identifier | Unique identifier
definition | ? | 0..* | Reference(PlanDefinition | ActivityDefinition) | Instantiates protocol or definition
basedOn | ? | 0..* | Reference(Any) | Request fulfilled by this communication
partOf | ? | 0..* | Reference(Any) | Part of this action
status | ?!? | 1..1 | code | preparation | in-progress | suspended | aborted | completed | entered-in-error EventStatus (Required)
notDone | ?!? | 0..1 | boolean | Communication did not occur
notDoneReason | ?I | 0..1 | CodeableConcept | Why communication did not occur CommunicationNotDoneReason (Example)
category | | 0..* | CodeableConcept | Message category CommunicationCategory (Example)
medium | | 0..* | CodeableConcept | A channel of communication v3 Code System ParticipationMode (Example)
subject | ? | 0..1 | Reference(Patient | Group) | Focus of message
recipient | | 0..* | Reference(Device | Organization | Patient | Practitioner | RelatedPerson | Group) | Message recipient
topic | | 0..* | Reference(Any) | Focal resources
context | ? | 0..1 | Reference(Encounter | EpisodeOfCare) | Encounter or episode leading to message
sent | | 0..1 | dateTime | When sent
received | | 0..1 | dateTime | When received
sender | | 0..1 | Reference(Device | Organization | Patient | Practitioner | RelatedPerson) | Message sender
reasonCode | ? | 0..* | CodeableConcept | Indication for message SNOMED CT Clinical Findings (Example)
reasonReference | ? | 0..* | Reference(Condition | Observation) | Why was communication done?
payload | | 0..* | BackboneElement | Message payload
content[x] | | 1..1 | | Message part content
contentString | | | string
contentAttachment | | | Attachment
contentReference | | | Reference(Any)
note | | 0..* | Annotation | Comments made about the communication
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Communication xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Unique identifier --></identifier>
 <definition><!-- 0..* Reference(PlanDefinition|ActivityDefinition) Instantiates protocol or definition --></definition>
 <basedOn><!-- 0..* Reference(Any) Request fulfilled by this communication --></basedOn>
 <partOf><!-- 0..* Reference(Any) Part of this action --></partOf>
 <status value="[code]"/><!-- 1..1 preparation | in-progress | suspended | aborted | completed | entered-in-error -->
 <notDone value="[boolean]"/><!-- 0..1 Communication did not occur -->
 <notDoneReason><!--  0..1 CodeableConcept Why communication did not occur --></notDoneReason>
 <category><!-- 0..* CodeableConcept Message category --></category>
 <medium><!-- 0..* CodeableConcept A channel of communication --></medium>
 <subject><!-- 0..1 Reference(Patient|Group) Focus of message --></subject>
 <recipient><!-- 0..* Reference(Device|Organization|Patient|Practitioner|
   RelatedPerson|Group) Message recipient --></recipient>
 <topic><!-- 0..* Reference(Any) Focal resources --></topic>
 <context><!-- 0..1 Reference(Encounter|EpisodeOfCare) Encounter or episode leading to message --></context>
 <sent value="[dateTime]"/><!-- 0..1 When sent -->
 <received value="[dateTime]"/><!-- 0..1 When received -->
 <sender><!-- 0..1 Reference(Device|Organization|Patient|Practitioner|
   RelatedPerson) Message sender --></sender>
 <reasonCode><!-- 0..* CodeableConcept Indication for message --></reasonCode>
 <reasonReference><!-- 0..* Reference(Condition|Observation) Why was communication done? --></reasonReference>
 <payload>  <!-- 0..* Message payload -->
  <content[x]><!-- 1..1 string|Attachment|Reference(Any) Message part content --></content[x]>
```

```
 </payload>
 <note><!-- 0..* Annotation Comments made about the communication --></note>
</Communication>
```

JSON Template

```
{
  "resourceType" : "Communication",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Unique identifier
  "definition" : [{ Reference(PlanDefinition|ActivityDefinition) }], // Instantiates protocol or definition
  "basedOn" : [{ Reference(Any) }], // Request fulfilled by this communication
  "partOf" : [{ Reference(Any) }], // Part of this action
  "status" : "<code>", // R!  preparation | in-progress | suspended | aborted | completed | entered-in-error
  "notDone" : <boolean>, // Communication did not occur
  "notDoneReason" : { CodeableConcept }, // C? Why communication did not occur
  "category" : [{ CodeableConcept }], // Message category
  "medium" : [{ CodeableConcept }], // A channel of communication
  "subject" : { Reference(Patient|Group) }, // Focus of message
  "recipient" : [{ Reference(Device|Organization|Patient|Practitioner|
   RelatedPerson|Group) }], // Message recipient
  "topic" : [{ Reference(Any) }], // Focal resources
  "context" : { Reference(Encounter|EpisodeOfCare) }, // Encounter or episode leading to message
  "sent" : "<dateTime>", // When sent
  "received" : "<dateTime>", // When received
  "sender" : { Reference(Device|Organization|Patient|Practitioner|
   RelatedPerson) }, // Message sender
  "reasonCode" : [{ CodeableConcept }], // Indication for message
  "reasonReference" : [{ Reference(Condition|Observation) }], // Why was communication done?
  "payload" : [{ // Message payload
    // content[x]: Message part content. One of these 3:
    "contentString" : "<string>"
    "contentAttachment" : { Attachment }
    "contentReference" : { Reference(Any) }
  }],
  "note" : [{ Annotation }] // Comments made about the communication
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Communication;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Communication.identifier [ Identifier ], ... ; # 0..* Unique identifier
  fhir:Communication.definition [ Reference(PlanDefinition|ActivityDefinition) ], ... ; # 0..* Instantiates protocol or
    definition
  fhir:Communication.basedOn [ Reference(Any) ], ... ; # 0..* Request fulfilled by this communication
  fhir:Communication.partOf [ Reference(Any) ], ... ; # 0..* Part of this action
  fhir:Communication.status [ code ]; # 1..1 preparation | in-progress | suspended | aborted | completed | entered-in-
    error
  fhir:Communication.notDone [ boolean ]; # 0..1 Communication did not occur
  fhir:Communication.notDoneReason [ CodeableConcept ]; # 0..1 Why communication did not occur
  fhir:Communication.category [ CodeableConcept ], ... ; # 0..* Message category
  fhir:Communication.medium [ CodeableConcept ], ... ; # 0..* A channel of communication
  fhir:Communication.subject [ Reference(Patient|Group) ]; # 0..1 Focus of message
  fhir:Communication.recipient [ Reference(Device|Organization|Patient|Practitioner|RelatedPerson|Group) ], ... ; # 0..*
     Message recipient
  fhir:Communication.topic [ Reference(Any) ], ... ; # 0..* Focal resources
  fhir:Communication.context [ Reference(Encounter|EpisodeOfCare) ]; # 0..1 Encounter or episode leading to message
  fhir:Communication.sent [ dateTime ]; # 0..1 When sent
  fhir:Communication.received [ dateTime ]; # 0..1 When received
  fhir:Communication.sender [ Reference(Device|Organization|Patient|Practitioner|RelatedPerson) ]; # 0..1 Message sender
  fhir:Communication.reasonCode [ CodeableConcept ], ... ; # 0..* Indication for message
  fhir:Communication.reasonReference [ Reference(Condition|Observation) ], ... ; # 0..* Why was communication done?
  fhir:Communication.payload [ # 0..* Message payload
    # Communication.payload.content[x] : 1..1 Message part content. One of these 3
      fhir:Communication.payload.contentString [ string ]
      fhir:Communication.payload.contentAttachment [ Attachment ]
      fhir:Communication.payload.contentReference [ Reference(Any) ]
  ], ...;
  fhir:Communication.note [ Annotation ], ... ; # 0..* Comments made about the communication
]
```

Changes since DSTU2

Communication

Communication.definition | Added Element

Communication.basedOn | Renamed from requestDetail to basedOn Max Cardinality changed from 1 to * Type changed from Reference(CommunicationRequest) to Reference(Resource)

Communication.partOf | Added Element

Communication.status | Min Cardinality changed from 0 to 1 Change value set from http://hl7.org/fhir/ValueSet/communication-status to http://hl7.org/fhir/ValueSet/event-status

Communication.notDone | Added Element

Communication.notDoneReason | Added Element

Communication.category | Max Cardinality changed from 1 to *
Communication.subject | Add Reference(Group)
Communication.topic | Added Element
Communication.context | Renamed from encounter to context Add Reference(EpisodeOfCare)
Communication.reasonCode | Renamed from reason to reasonCode
Communication.reasonReference | Added Element
Communication.note | Added Element

See the Full Difference for further information

This analysis is available as XML or JSON.

See R2 <--> R3 Conversion Maps (status = 2 tests that all execute ok. All tests pass round-trip testing and all r3 resources are valid.).

Alternate definitions: Master Definition (XML, JSON), XML Schema/Schematron (for) + JSON Schema, ShEx (for Turtle)

8.20.3.1 Terminology Bindings

Path | Definition | Type | Reference
Communication.status | The status of the communication. | Required | EventStatus
Communication.notDoneReason | Codes for the reason why a communication was not done. | Example | CommunicationNotDoneReason
Communication.category | Codes for general categories of communications such as alerts, instruction, etc. | Example | CommunicationCategory
Communication.medium | Codes for communication mediums such as phone, fax, email, in person, etc. | Example | v3 Code System ParticipationMode
Communication.reasonCode | Codes for describing reasons for the occurrence of a communication. | Example | SNOMED CT Clinical Findings

8.20.3.2 Constraints

* com-1: Not Done Reason can only be specified if NotDone is "true" (expression : notDone or notDoneReason.exists().not())

Notes to reviewers:

At this time, the code bindings are placeholders to be fleshed out upon further review by the community.

8.20.3.3 Communication.sender and Communication.recepient

Communication.sender and Communication.recipient allow Patient|Practitioner|RelatedPerson - but it is not unusual to have a communication target - even a defined one - where it is unknown what kind of role the person is playing.

If the communication is to or from an individual, whose role is not known (practitioner, patient or related person), - for example, only email address is captured in the system; then RelatedPerson should be used by default.

8.20.4 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
based-on | reference | Request fulfilled by this communication | Communication.basedOn (Any)
category | token | Message category | Communication.category
context | reference | Encounter or episode leading to message | Communication.context (EpisodeOfCare, Encounter)
definition | reference | Instantiates protocol or definition | Communication.definition (PlanDefinition, ActivityDefinition)
encounter | reference | Encounter leading to message | Communication.context (Encounter)
identifier | token | Unique identifier | Communication.identifier
medium | token | A channel of communication | Communication.medium
part-of | reference | Part of this action | Communication.partOf (Any)
patient | reference | Focus of message | Communication.subject (Patient)
received | date | When received | Communication.received
recipient | reference | Message recipient | Communication.recipient (Practitioner, Group, Organization, Device, Patient, RelatedPerson)
sender | reference | Message sender | Communication.sender (Practitioner, Organization, Device, Patient, RelatedPerson)
sent | date | When sent | Communication.sent
status | token | preparation | in-progress | suspended | aborted | completed | entered-in-error | Communication.status
subject | reference | Focus of message | Communication.subject (Group, Patient)

8.20.5 Resource Communication - Examples

URL: https://hl7.org/fhir/STU3/communication-examples.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.20.5 Resource Communication - Examples

Patient Care Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

An introduction to this page as well as proper examples are to be submitted shortly.

Example List:

An example of an alert message | example | XML | JSON | Turtle
An example of a response to a request for additional information | fm-solicited | XML | JSON | Turtle
An example of an unsolicited attachment - provision of additional information | fm-attachment | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

8.20.6 Resource Communication - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/communication-definitions.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.20.6 Resource Communication - Detailed Descriptions

Patient Care Work Group | Maturity Level: 2 | Trial Use | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

Detailed Descriptions for the elements in the Communication resource.

Communication
Definition | An occurrence of information being transmitted; e.g. an alert that was sent to a responsible provider, a public health agency was notified about a reportable condition.
Control | 1..1
Invariants | Defined on this element com-1: Not Done Reason can only be specified if NotDone is "true" (expression : notDone or notDoneReason.exists().not(), xpath: f:notDone/@value=true() or not(exists(f:notDoneReason)))
Communication.identifier
Definition | Identifiers associated with this Communication that are defined by business processes and/ or used to refer to it when a direct URL reference to the resource itself is not appropriate (e.g. in CDA documents, or in written / printed documentation).
Note | This is a business identifer, not a resource identifier (see discussion)
Control | 0..*
Type | Identifier
Summary | true
Communication.definition
Definition | A protocol, guideline, or other definition that was adhered to in whole or in part by this communication event.
Control | 0..*
Type | Reference(PlanDefinition | ActivityDefinition)
Summary | true
Communication.basedOn
Definition | An order, proposal or plan fulfilled in whole or in part by this Communication.
Control | 0..*
Type | Reference(Any)
Alternate Names | fulfills
Summary | true
Comments | This must point to some sort of a 'Request' resource, such as CarePlan, CommunicationRequest, ReferralRequest, MedicationRequest, etc.
Communication.partOf
Definition | Part of this action.
Control | 0..*
Type | Reference(Any)
Alternate Names | container
Summary | true
Communication.status
Definition | The status of the transmission.
Control | 1..1
Terminology Binding | EventStatus (Required)
Type | code
Is Modifier | true
Summary | true
Comments | This element is labeled as a modifier because the status contains the codes aborted and entered-in-error that mark the communication as not currently valid.
Communication.notDone
Definition | If true, indicates that the described communication event did not actually occur.
Control | 0..1
Type | boolean
Is Modifier | true
Requirements | Creating a Communication where notDone is true is intended for situations where there's a need for a specific statement in the record about something not being done. If the need is merely to indicate that a request wasn't fulfilled, that should be handled using Task.
Summary | true
Comments | This element is labeled as a modifier because it marks the communication as a communication that did not occur. The more attributes are populated, the more constrained the negated statement is.
Communication.notDoneReason
Definition | Describes why the communication event did not occur in coded and/or textual form.
Control | 0..1
Terminology Binding | CommunicationNotDoneReason (Example)
Type | CodeableConcept
Summary | true
Invariants | Affect this element com-1: Not Done Reason can only be specified if NotDone is "true" (expression : notDone or notDoneReason.exists().not(), xpath: f:notDone/@value=true() or not(exists(f:notDoneReason)))

Communication.category
Definition | The type of message conveyed such as alert, notification, reminder, instruction, etc.
Control | 0..*
Terminology Binding | CommunicationCategory (Example)
Type | CodeableConcept
Comments | There may be multiple axes of categorization and one communication may serve multiple purposes.
Communication.medium
Definition | A channel that was used for this communication (e.g. email, fax).
Control | 0..*
Terminology Binding | v3 Code System ParticipationMode (Example)
Type | CodeableConcept
Communication.subject
Definition | The patient or group that was the focus of this communication.
Control | 0..1
Type | Reference(Patient | Group)
Alternate Names | patient
Summary | true
Communication.recipient
Definition | The entity (e.g. person, organization, clinical information system, or device) which was the target of the communication. If receipts need to be tracked by individual, a separate resource instance will need to be created for each recipient. Multiple recipient communications are intended where either a receipt(s) is not tracked (e.g. a mass mail-out) or is captured in aggregate (all emails confirmed received by a particular time).
Control | 0..*
Type | Reference(Device | Organization | Patient | Practitioner | RelatedPerson | Group)
Communication.topic
Definition | The resources which were responsible for or related to producing this communication.
Control | 0..*
Type | Reference(Any)
Communication.context
Definition | The encounter within which the communication was sent.
Control | 0..1
Type | Reference(Encounter | EpisodeOfCare)
Alternate Names | encounter
Summary | true
Communication.sent
Definition | The time when this communication was sent.
Control | 0..1
Type | dateTime
Communication.received
Definition | The time when this communication arrived at the destination.
Control | 0..1
Type | dateTime
Communication.sender
Definition | The entity (e.g. person, organization, clinical information system, or device) which was the source of the communication.
Control | 0..1
Type | Reference(Device | Organization | Patient | Practitioner | RelatedPerson)
Communication.reasonCode
Definition | The reason or justification for the communication.
Control | 0..*
Terminology Binding | SNOMED CT Clinical Findings (Example)
Type | CodeableConcept
Summary | true
Comments | Textual reasons can be caprued using reasonCode.text.
Communication.reasonReference
Definition | Indicates another resource whose existence justifies this communication.
Control | 0..*
Type | Reference(Condition | Observation)
Summary | true
Communication.payload
Definition | Text, attachment(s), or resource(s) that was communicated to the recipient.
Control | 0..*
Communication.payload.content[x]
Definition | A communicated content (or for multi-part communications, one portion of the communication).
Control | 1..1
Type | string|Attachment|Reference(Any)
[x] Note | See Choice of Data Types for further information about how to use [x]
Communication.note
Definition | Additional notes or commentary about the communication by the sender, receiver or other interested parties.
Control | 0..*
Type | Annotation

8.20.7 Resource Communication - Mappings

URL: https://hl7.org/fhir/STU3/communication-mappings.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.20.7 Resource Communication - Mappings

Patient Care Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

Mappings:

RIM Mapping

Workflow Mapping

Mappings for the communication resource.

8.20.7.1 Mappings for RIM Mapping (http://hl7.org/v3)

Communication
identifier
definition
basedOn
partOf
status
notDone
notDoneReason
category
medium
subject
recipient
topic
context
sent
received
sender
reasonCode | .reasonCode
reasonReference | .outboundRelationship[typeCode=RSON].target
payload
content[x]
note

8.20.7.2 Mappings for Workflow Mapping (http://hl7.org/fhir/workflow)

Communication | Event
identifier | Event.identifier
definition | Event.definition
basedOn | Event.basedOn
partOf | Event.partOf
status | Event.status
notDone | Event.notDone
notDoneReason | Event.notDoneReason
category | Event.code
medium
subject | Event.subject
recipient
topic
context | Event.context
sent | Event.occurrenceDateTime (Invariant: maps to either sent OR received)
received | Event.occurrenceDateTime (Invariant: maps to either sent OR received)
sender | Event.performer.actor
reasonCode | Event.reasonCodableConcept
reasonReference | Event.reasonReference
payload
content[x]
note | Event.note

8.20.8 Resource Communication - Design Notes

URL: https://hl7.org/fhir/STU3/communication-explanations.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.20.8 Resource Communication - Design Notes

to do

8.20.9 Resource Communication - Extensions & Profiles

URL: https://hl7.org/fhir/STU3/communication-profiles.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.20.9 Resource Communication - Extensions & Profiles

Patient Care Work Group | Maturity Level: N/A | Ballot Status: Informative

This table lists profiles and extensions for the Communication resource. For background information, see Profiling Resources and Extensibility. Additional profiles and resources may be found in published Implementation Guides , or in the Conformance resource registry .

8.20.9.1 Profiles

No Profiles defined for this resource

8.20.9.2 Extensions

communication-media | media | for Communication HL7 Extensions
communication-reasonNotPerformed | reasonNotPerformed | for Communication HL7 Extensions

Extensions that reference this resource:

No Extensions refer to this resource

8.20.9.3 Search Extensions

No Search Extensions defined for this resource

8.20.10 Communication HL7 Extensions

URL: https://hl7.org/fhir/STU3/communication-extensions.html

Defines common extensions used with or related to the Communication resource

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.20.10 Communication HL7 Extensions

(No assigned work group) | Maturity Level: N/A | Ballot Status: Informative

8.20.10.1 Content

Extensions:

communication-reasonNotPerformed | reasonNotPerformed : Explanation for rejection or failure of the communication. Applicable only if the status is suspended, rejected, or failed.

communication-media | media : It contains enriched media representation of the alert message, such as a voice recording. This may be used, for example for compliance with jurisdictional accessibility requirements, literacy issues, or translations of the unstructured text content in other languages.

8.20.11 Resource Communication - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/communication-version-maps.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.20.11 Resource Communication - Detailed Descriptions

Patient Care Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

R2 : R3 Converson maps for Communication.

Functional status for this map: 2 tests that all execute ok. All tests pass round-trip testing and all r3 resources are valid. (see documentation)

8.20.11.1 R2 to R3

```
map "http://hl7.org/fhir/StructureMap/Communication2to3" = "R2 to R3 Conversions for Communication"

uses "http://hl7.org/fhir/DSTU2/StructureDefinition/Communication" alias CommunicationR2 as source
uses "http://hl7.org/fhir/StructureDefinition/Communication" alias Communication as target

imports "http://hl7.org/fhir/StructureMap/primitives2to3"
imports "http://hl7.org/fhir/StructureMap/DomainResource2to3"
imports "http://hl7.org/fhir/StructureMap/Attachment2to3"
imports "http://hl7.org/fhir/StructureMap/Identifier2to3"
imports "http://hl7.org/fhir/StructureMap/Reference2to3"
imports "http://hl7.org/fhir/StructureMap/CodeableConcept2to3"

group for type+types  Communication extends DomainResource
```

```
  input src : CommunicationR2 as source
  input tgt : Communication as target

  "Communication-identifier" : for src.identifier make tgt.identifier
  "Communication-category" : for src.category make tgt.category
  "Communication-sender" : for src.sender make tgt.sender
  "Communication-recipient" : for src.recipient make tgt.recipient
  "Communication-payload" : for src.payload as vs0 make tgt.payload as vt0 then cmmPayload(vs0, vt0)
  "Communication-medium" : for src.medium make tgt.medium
  "Communication-status" : for src.status make tgt.status
  "Communication-encounter" : for src.encounter make tgt.context
  "Communication-sent" : for src.sent make tgt.sent
  "Communication-received" : for src.received make tgt.received
  "Communication-reason" : for src.reason make tgt.reasonCode
  "Communication-subject" : for src.subject make tgt.subject
  "Communication-requestDetail" : for src.requestDetail make tgt.basedOn
endgroup

group cmmPayload
  input src as source
  input tgt as target

  "Communication.payload-content" : for src.content make tgt.content
endgroup
```

8.20.11.2 R3 to R2

```
map "http://hl7.org/fhir/StructureMap/Communication3to2" = "R3 to R2 Conversion for Communication"

uses "http://hl7.org/fhir/StructureDefinition/Communication" alias Communication as source
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/Communication" alias CommunicationR2 as target

imports "http://hl7.org/fhir/StructureMap/primitives3to2"
imports "http://hl7.org/fhir/StructureMap/DomainResource3to2"
imports "http://hl7.org/fhir/StructureMap/Attachment3to2"
imports "http://hl7.org/fhir/StructureMap/Identifier3to2"
imports "http://hl7.org/fhir/StructureMap/Reference3to2"
imports "http://hl7.org/fhir/StructureMap/CodeableConcept3to2"

group for type+types  Communication extends DomainResource
  input src : Communication as source
  input tgt : CommunicationR2 as target

  "Communication-identifier" : for src.identifier make tgt.identifier
  "Communication-category" : for src.category make tgt.category
  "Communication-sender" : for src.sender make tgt.sender
  "Communication-recipient" : for src.recipient make tgt.recipient
  "Communication-payload" : for src.payload as vs0 make tgt.payload as vt0 then cmmPayload(vs0, vt0)
  "Communication-medium" : for src.medium make tgt.medium
  "Communication-status" : for src.status make tgt.status
  "Communication-encounter" : for src.context make tgt.encounter
  "Communication-sent" : for src.sent make tgt.sent
  "Communication-received" : for src.received make tgt.received
  "Communication-reason" : for src.reasonCode make tgt.reason
  "Communication-subject" : for src.subject make tgt.subject
  "Communication-requestDetail" : for src.basedOn make tgt.requestDetail
endgroup

group cmmPayload
  input src as source
  input tgt as target

  "Communication.payload-content" : for src.content make tgt.content
endgroup
```

8.20.11.3 R3 Validation Errors

8.21 Resource CommunicationRequest - Content

URL: https://hl7.org/fhir/STU3/communicationrequest.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

Patient Care Work Group | Maturity Level: 2 | Trial Use | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

A request to convey information; e.g. the CDS system proposes that an alert be sent to a responsible provider, the CDS system proposes that the public health agency be notified about a reportable condition.

8.21.1 Scope and Usage

CommunicationRequest is one of the request resources in the FHIR workflow specification.

This resource is a record of a request for a communication to be performed. A communication is a conveyance of information from one entity, a sender, to another entity, a receiver. The sender and receivers may be patients, practitioners, related persons, organizations, and devices. Uses of communication request include:

* A computer-based decision-support system requesting a reminder or alert be delivered to a responsible provider

* A physician requesting notification from the nurse if a patient's temperature exceeds a value

* A monitoring system or a provider requesting a staff member or department to notify a public health agency of a patient presenting with a communicable disease reportable to the public health agency

* A computer-based decision-support system proposes to send educational material to a patient

8.21.2 Boundaries and Relationships

This resource is a record of a request. It does not represent the actual flow of communication.

The use of CommunicationRequest excludes requests for referrals which are covered by the ReferralRequest resource. It also excludes requests for therapy or counseling which would be handled by the ProcedureRequest resource. The performance of a CommunicationRequest may result in a Communication resource.

This resource is referenced by CarePlan, ClaimResponse and ProcessResponse

8.21.3 Resource Content

* Structure

* UML

* XML

* JSON

* Turtle

* R2 Diff

* All

Structure

Name | Flags | Card. | Type | Description & Constraints
CommunicationRequest | | | DomainResource | A request for information to be sent to a receiver Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | ? | 0..* | Identifier | Unique identifier
basedOn | ? | 0..* | Reference(Any) | Fulfills plan or proposal
replaces | ? | 0..* | Reference(CommunicationRequest) | Request(s) replaced by this request
groupIdentifier | ? | 0..1 | Identifier | Composite request this is part of
status | ?!? | 1..1 | code | draft | active | suspended | cancelled | completed | entered-in-error | unknown RequestStatus (Required)
category | | 0..* | CodeableConcept | Message category CommunicationCategory (Example)
priority | ? | 0..1 | code | Message urgency RequestPriority (Required)
medium | | 0..* | CodeableConcept | A channel of communication v3 Code System ParticipationMode (Example)
subject | | 0..1 | Reference(Patient | Group) | Focus of message
recipient | | 0..* | Reference(Device | Organization | Patient | Practitioner | RelatedPerson | Group | CareTeam) | Message recipient
topic | | 0..* | Reference(Any) | Focal resources
context | ? | 0..1 | Reference(Encounter | EpisodeOfCare) | Encounter or episode leading to message
payload | | 0..* | BackboneElement | Message payload
content[x] | | 1..1 | | Message part content
contentString | | | string
contentAttachment | | | Attachment
contentReference | | | Reference(Any)
occurrence[x] | ? | 0..1 | | When scheduled
occurrenceDateTime | | | dateTime
occurrencePeriod | | | Period
authoredOn | ? | 0..1 | dateTime | When request transitioned to being actionable
sender | | 0..1 | Reference(Device | Organization | Patient | Practitioner | RelatedPerson) | Message sender
requester | ?I | 0..1 | BackboneElement | Who/what is requesting service + onBehalfOf can only be specified if agent is practitioner or device
agent | ? | 1..1 | Reference(Practitioner | Organization | Patient | RelatedPerson | Device) | Individual making the request
onBehalfOf | ?I | 0..1 | Reference(Organization) | Organization agent is acting for
reasonCode | ? | 0..* | CodeableConcept | Why is communication needed? v3 Code System ActReason (Example)
reasonReference | ? | 0..* | Reference(Condition | Observation) | Why is communication needed?
note | | 0..* | Annotation | Comments made about communication request
Documentation for this format

UML Diagram (Legend)

XML Template

```
<CommunicationRequest xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Unique identifier --></identifier>
 <basedOn><!-- 0..* Reference(Any) Fulfills plan or proposal --></basedOn>
 <replaces><!-- 0..* Reference(CommunicationRequest) Request(s) replaced by this request --></replaces>
 <groupIdentifier><!-- 0..1 Identifier Composite request this is part of --></groupIdentifier>
 <status value="[code]"/><!-- 1..1 draft | active | suspended | cancelled | completed | entered-in-error | unknown -->
 <category><!-- 0..* CodeableConcept Message category --></category>
 <priority value="[code]"/><!-- 0..1 Message urgency -->
 <medium><!-- 0..* CodeableConcept A channel of communication --></medium>
 <subject><!-- 0..1 Reference(Patient|Group) Focus of message --></subject>
 <recipient><!-- 0..* Reference(Device|Organization|Patient|Practitioner|
```

```
   RelatedPerson|Group|CareTeam) Message recipient --></recipient>
 <topic><!-- 0..* Reference(Any) Focal resources --></topic>
 <context><!-- 0..1 Reference(Encounter|EpisodeOfCare) Encounter or episode leading to message --></context>
 <payload>  <!-- 0..* Message payload -->
  <content[x]><!-- 1..1 string|Attachment|Reference(Any) Message part content --></content[x]>
 </payload>
 <occurrence[x]><!-- 0..1 dateTime|Period When scheduled --></occurrence[x]>
 <authoredOn value="[dateTime]"/><!-- 0..1 When request transitioned to being actionable -->
 <sender><!-- 0..1 Reference(Device|Organization|Patient|Practitioner|
   RelatedPerson) Message sender --></sender>
 <requester>  <!-- 0..1 Who/what is requesting service -->
  <agent><!-- 1..1 Reference(Practitioner|Organization|Patient|RelatedPerson|
    Device) Individual making the request --></agent>
  <onBehalfOf><!--  0..1 Reference(Organization) Organization agent is acting for --></onBehalfOf>
 </requester>
 <reasonCode><!-- 0..* CodeableConcept Why is communication needed? --></reasonCode>
 <reasonReference><!-- 0..* Reference(Condition|Observation) Why is communication needed? --></reasonReference>
 <note><!-- 0..* Annotation Comments made about communication request --></note>
</CommunicationRequest>
```

JSON Template

```
{
  "resourceType" : "CommunicationRequest",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Unique identifier
  "basedOn" : [{ Reference(Any) }], // Fulfills plan or proposal
  "replaces" : [{ Reference(CommunicationRequest) }], // Request(s) replaced by this request
  "groupIdentifier" : { Identifier }, // Composite request this is part of
  "status" : "<code>", // R!  draft | active | suspended | cancelled | completed | entered-in-error | unknown
  "category" : [{ CodeableConcept }], // Message category
  "priority" : "<code>", // Message urgency
  "medium" : [{ CodeableConcept }], // A channel of communication
  "subject" : { Reference(Patient|Group) }, // Focus of message
  "recipient" : [{ Reference(Device|Organization|Patient|Practitioner|
   RelatedPerson|Group|CareTeam) }], // Message recipient
  "topic" : [{ Reference(Any) }], // Focal resources
  "context" : { Reference(Encounter|EpisodeOfCare) }, // Encounter or episode leading to message
  "payload" : [{ // Message payload
    // content[x]: Message part content. One of these 3:
    "contentString" : "<string>"
    "contentAttachment" : { Attachment }
    "contentReference" : { Reference(Any) }
  }],
  // occurrence[x]: When scheduled. One of these 2:
  "occurrenceDateTime" : "<dateTime>",
  "occurrencePeriod" : { Period },
  "authoredOn" : "<dateTime>", // When request transitioned to being actionable
  "sender" : { Reference(Device|Organization|Patient|Practitioner|
   RelatedPerson) }, // Message sender
  "requester" : { // Who/what is requesting service
    "agent" : { Reference(Practitioner|Organization|Patient|RelatedPerson|
    Device) }, // R!  Individual making the request
    "onBehalfOf" : { Reference(Organization) } // C? Organization agent is acting for
  },
  "reasonCode" : [{ CodeableConcept }], // Why is communication needed?
  "reasonReference" : [{ Reference(Condition|Observation) }], // Why is communication needed?
  "note" : [{ Annotation }] // Comments made about communication request
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:CommunicationRequest;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:CommunicationRequest.identifier [ Identifier ], ... ; # 0..* Unique identifier
  fhir:CommunicationRequest.basedOn [ Reference(Any) ], ... ; # 0..* Fulfills plan or proposal
  fhir:CommunicationRequest.replaces [ Reference(CommunicationRequest) ], ... ; # 0..* Request(s) replaced by this
    request
  fhir:CommunicationRequest.groupIdentifier [ Identifier ]; # 0..1 Composite request this is part of
  fhir:CommunicationRequest.status [ code ]; # 1..1 draft | active | suspended | cancelled | completed | entered-in-
    error | unknown
  fhir:CommunicationRequest.category [ CodeableConcept ], ... ; # 0..* Message category
  fhir:CommunicationRequest.priority [ code ]; # 0..1 Message urgency
  fhir:CommunicationRequest.medium [ CodeableConcept ], ... ; # 0..* A channel of communication
  fhir:CommunicationRequest.subject [ Reference(Patient|Group) ]; # 0..1 Focus of message
  fhir:CommunicationRequest.recipient [ Reference(Device|Organization|Patient|Practitioner|RelatedPerson|Group|CareTeam)
     ], ... ; # 0..* Message recipient
  fhir:CommunicationRequest.topic [ Reference(Any) ], ... ; # 0..* Focal resources
  fhir:CommunicationRequest.context [ Reference(Encounter|EpisodeOfCare) ]; # 0..1 Encounter or episode leading to
    message
  fhir:CommunicationRequest.payload [ # 0..* Message payload
    # CommunicationRequest.payload.content[x] : 1..1 Message part content. One of these 3
      fhir:CommunicationRequest.payload.contentString [ string ]
      fhir:CommunicationRequest.payload.contentAttachment [ Attachment ]
```

```
      fhir:CommunicationRequest.payload.contentReference [ Reference(Any) ]
  ], ...;
  # CommunicationRequest.occurrence[x] : 0..1 When scheduled. One of these 2
    fhir:CommunicationRequest.occurrenceDateTime [ dateTime ]
    fhir:CommunicationRequest.occurrencePeriod [ Period ]
  fhir:CommunicationRequest.authoredOn [ dateTime ]; # 0..1 When request transitioned to being actionable
  fhir:CommunicationRequest.sender [ Reference(Device|Organization|Patient|Practitioner|RelatedPerson) ]; # 0..1 Message
     sender
  fhir:CommunicationRequest.requester [ # 0..1 Who/what is requesting service
    fhir:CommunicationRequest.requester.agent [ Reference(Practitioner|Organization|Patient|RelatedPerson|Device) ]; #
    1..1 Individual making the request
    fhir:CommunicationRequest.requester.onBehalfOf [ Reference(Organization) ]; # 0..1 Organization agent is acting for
  ];
  fhir:CommunicationRequest.reasonCode [ CodeableConcept ], ... ; # 0..* Why is communication needed?
  fhir:CommunicationRequest.reasonReference [ Reference(Condition|Observation) ], ... ; # 0..* Why is communication
    needed?
  fhir:CommunicationRequest.note [ Annotation ], ... ; # 0..* Comments made about communication request
]
```

Changes since DSTU2

CommunicationRequest
CommunicationRequest.basedOn | Added Element
CommunicationRequest.replaces | Added Element
CommunicationRequest.groupIdentifier | Added Element
CommunicationRequest.status | Min Cardinality changed from 0 to 1 Change value set from http://hl7.org/fhir/ValueSet/communication-request-status to http://hl7.org/fhir/ValueSet/request-status
CommunicationRequest.category | Max Cardinality changed from 1 to *
CommunicationRequest.priority | Type changed from CodeableConcept to code Add Binding http://hl7.org/fhir/ValueSet/request-priority (required)
CommunicationRequest.subject | Add Reference(Group)
CommunicationRequest.recipient | Add Reference(Group), Add Reference(CareTeam)
CommunicationRequest.topic | Added Element
CommunicationRequest.context | Renamed from encounter to context Add Reference(EpisodeOfCare)
CommunicationRequest.occurrence[x] | Added Element
CommunicationRequest.authoredOn | Renamed from requestedOn to authoredOn
CommunicationRequest.requester | Remove Reference(Practitioner), Remove Reference(Patient), Remove Reference(RelatedPerson)
CommunicationRequest.requester.agent | Added Element
CommunicationRequest.requester.onBehalfOf | Added Element
CommunicationRequest.reasonCode | Renamed from reason to reasonCode
CommunicationRequest.reasonReference | Added Element
CommunicationRequest.note | Added Element
CommunicationRequest.scheduled[x] | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R2 <--> R3 Conversion Maps (status = 2 tests that all execute ok. 1 fail round-trip testing and 1 r3 resources are invalid (1 errors).).

Structure

Name | Flags | Card. | Type | Description & Constraints
CommunicationRequest | | | DomainResource | A request for information to be sent to a receiver Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | ? | 0..* | Identifier | Unique identifier
basedOn | ? | 0..* | Reference(Any) | Fulfills plan or proposal
replaces | ? | 0..* | Reference(CommunicationRequest) | Request(s) replaced by this request
groupIdentifier | ? | 0..1 | Identifier | Composite request this is part of
status | ?!? | 1..1 | code | draft | active | suspended | cancelled | completed | entered-in-error | unknown RequestStatus (Required)
category | | 0..* | CodeableConcept | Message category CommunicationCategory (Example)
priority | ? | 0..1 | code | Message urgency RequestPriority (Required)
medium | | 0..* | CodeableConcept | A channel of communication v3 Code System ParticipationMode (Example)
subject | | 0..1 | Reference(Patient | Group) | Focus of message
recipient | | 0..* | Reference(Device | Organization | Patient | Practitioner | RelatedPerson | Group | CareTeam) | Message recipient
topic | | 0..* | Reference(Any) | Focal resources
context | ? | 0..1 | Reference(Encounter | EpisodeOfCare) | Encounter or episode leading to message
payload | | 0..* | BackboneElement | Message payload
content[x] | | 1..1 | | Message part content
contentString | | | string
contentAttachment | | | Attachment
contentReference | | | Reference(Any)
occurrence[x] | ? | 0..1 | | When scheduled
occurrenceDateTime | | | dateTime
occurrencePeriod | | | Period
authoredOn | ? | 0..1 | dateTime | When request transitioned to being actionable
sender | | 0..1 | Reference(Device | Organization | Patient | Practitioner | RelatedPerson) | Message sender
requester | ?I | 0..1 | BackboneElement | Who/what is requesting service + onBehalfOf can only be specified if agent is practitioner or device
agent | ? | 1..1 | Reference(Practitioner | Organization | Patient | RelatedPerson | Device) | Individual making the request
onBehalfOf | ?I | 0..1 | Reference(Organization) | Organization agent is acting for

reasonCode | ? | 0..* | CodeableConcept | Why is communication needed? v3 Code System ActReason (Example)
reasonReference | ? | 0..* | Reference(Condition | Observation) | Why is communication needed?
note | | 0..* | Annotation | Comments made about communication request
Documentation for this format

UML Diagram (Legend)

XML Template

```
<CommunicationRequest xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Unique identifier --></identifier>
 <basedOn><!-- 0..* Reference(Any) Fulfills plan or proposal --></basedOn>
 <replaces><!-- 0..* Reference(CommunicationRequest) Request(s) replaced by this request --></replaces>
 <groupIdentifier><!-- 0..1 Identifier Composite request this is part of --></groupIdentifier>
 <status value="[code]"/><!-- 1..1 draft | active | suspended | cancelled | completed | entered-in-error | unknown -->
 <category><!-- 0..* CodeableConcept Message category --></category>
 <priority value="[code]"/><!-- 0..1 Message urgency -->
 <medium><!-- 0..* CodeableConcept A channel of communication --></medium>
 <subject><!-- 0..1 Reference(Patient|Group) Focus of message --></subject>
 <recipient><!-- 0..* Reference(Device|Organization|Patient|Practitioner|
   RelatedPerson|Group|CareTeam) Message recipient --></recipient>
 <topic><!-- 0..* Reference(Any) Focal resources --></topic>
 <context><!-- 0..1 Reference(Encounter|EpisodeOfCare) Encounter or episode leading to message --></context>
 <payload>  <!-- 0..* Message payload -->
  <content[x]><!-- 1..1 string|Attachment|Reference(Any) Message part content --></content[x]>
 </payload>
 <occurrence[x]><!-- 0..1 dateTime|Period When scheduled --></occurrence[x]>
 <authoredOn value="[dateTime]"/><!-- 0..1 When request transitioned to being actionable -->
 <sender><!-- 0..1 Reference(Device|Organization|Patient|Practitioner|
   RelatedPerson) Message sender --></sender>
 <requester>  <!-- 0..1 Who/what is requesting service -->
  <agent><!-- 1..1 Reference(Practitioner|Organization|Patient|RelatedPerson|
    Device) Individual making the request --></agent>
  <onBehalfOf><!--  0..1 Reference(Organization) Organization agent is acting for --></onBehalfOf>
 </requester>
 <reasonCode><!-- 0..* CodeableConcept Why is communication needed? --></reasonCode>
 <reasonReference><!-- 0..* Reference(Condition|Observation) Why is communication needed? --></reasonReference>
 <note><!-- 0..* Annotation Comments made about communication request --></note>
</CommunicationRequest>
```

JSON Template

```
{
  "resourceType" : "CommunicationRequest",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Unique identifier
  "basedOn" : [{ Reference(Any) }], // Fulfills plan or proposal
  "replaces" : [{ Reference(CommunicationRequest) }], // Request(s) replaced by this request
  "groupIdentifier" : { Identifier }, // Composite request this is part of
  "status" : "<code>", // R!  draft | active | suspended | cancelled | completed | entered-in-error | unknown
  "category" : [{ CodeableConcept }], // Message category
  "priority" : "<code>", // Message urgency
  "medium" : [{ CodeableConcept }], // A channel of communication
  "subject" : { Reference(Patient|Group) }, // Focus of message
  "recipient" : [{ Reference(Device|Organization|Patient|Practitioner|
   RelatedPerson|Group|CareTeam) }], // Message recipient
  "topic" : [{ Reference(Any) }], // Focal resources
  "context" : { Reference(Encounter|EpisodeOfCare) }, // Encounter or episode leading to message
  "payload" : [{ // Message payload
    // content[x]: Message part content. One of these 3:
    "contentString" : "<string>"
    "contentAttachment" : { Attachment }
    "contentReference" : { Reference(Any) }
  }],
  // occurrence[x]: When scheduled. One of these 2:
  "occurrenceDateTime" : "<dateTime>",
  "occurrencePeriod" : { Period },
  "authoredOn" : "<dateTime>", // When request transitioned to being actionable
  "sender" : { Reference(Device|Organization|Patient|Practitioner|
   RelatedPerson) }, // Message sender
  "requester" : { // Who/what is requesting service
    "agent" : { Reference(Practitioner|Organization|Patient|RelatedPerson|
    Device) }, // R!  Individual making the request
    "onBehalfOf" : { Reference(Organization) } // C? Organization agent is acting for
  },
  "reasonCode" : [{ CodeableConcept }], // Why is communication needed?
  "reasonReference" : [{ Reference(Condition|Observation) }], // Why is communication needed?
  "note" : [{ Annotation }] // Comments made about communication request
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:CommunicationRequest;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
```

```
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:CommunicationRequest.identifier [ Identifier ], ... ; # 0..* Unique identifier
  fhir:CommunicationRequest.basedOn [ Reference(Any) ], ... ; # 0..* Fulfills plan or proposal
  fhir:CommunicationRequest.replaces [ Reference(CommunicationRequest) ], ... ; # 0..* Request(s) replaced by this
    request
  fhir:CommunicationRequest.groupIdentifier [ Identifier ]; # 0..1 Composite request this is part of
  fhir:CommunicationRequest.status [ code ]; # 1..1 draft | active | suspended | cancelled | completed | entered-in-
    error | unknown
  fhir:CommunicationRequest.category [ CodeableConcept ], ... ; # 0..* Message category
  fhir:CommunicationRequest.priority [ code ]; # 0..1 Message urgency
  fhir:CommunicationRequest.medium [ CodeableConcept ], ... ; # 0..* A channel of communication
  fhir:CommunicationRequest.subject [ Reference(Patient|Group) ]; # 0..1 Focus of message
  fhir:CommunicationRequest.recipient [ Reference(Device|Organization|Patient|Practitioner|RelatedPerson|Group|CareTeam)
     ], ... ; # 0..* Message recipient
  fhir:CommunicationRequest.topic [ Reference(Any) ], ... ; # 0..* Focal resources
  fhir:CommunicationRequest.context [ Reference(Encounter|EpisodeOfCare) ]; # 0..1 Encounter or episode leading to
    message
  fhir:CommunicationRequest.payload [ # 0..* Message payload
    # CommunicationRequest.payload.content[x] : 1..1 Message part content. One of these 3
      fhir:CommunicationRequest.payload.contentString [ string ]
      fhir:CommunicationRequest.payload.contentAttachment [ Attachment ]
      fhir:CommunicationRequest.payload.contentReference [ Reference(Any) ]
  ], ...;
  # CommunicationRequest.occurrence[x] : 0..1 When scheduled. One of these 2
    fhir:CommunicationRequest.occurrenceDateTime [ dateTime ]
    fhir:CommunicationRequest.occurrencePeriod [ Period ]
  fhir:CommunicationRequest.authoredOn [ dateTime ]; # 0..1 When request transitioned to being actionable
  fhir:CommunicationRequest.sender [ Reference(Device|Organization|Patient|Practitioner|RelatedPerson) ]; # 0..1 Message
     sender
  fhir:CommunicationRequest.requester [ # 0..1 Who/what is requesting service
    fhir:CommunicationRequest.requester.agent [ Reference(Practitioner|Organization|Patient|RelatedPerson|Device) ]; #
    1..1 Individual making the request
    fhir:CommunicationRequest.requester.onBehalfOf [ Reference(Organization) ]; # 0..1 Organization agent is acting for
  ];
  fhir:CommunicationRequest.reasonCode [ CodeableConcept ], ... ; # 0..* Why is communication needed?
  fhir:CommunicationRequest.reasonReference [ Reference(Condition|Observation) ], ... ; # 0..* Why is communication
    needed?
  fhir:CommunicationRequest.note [ Annotation ], ... ; # 0..* Comments made about communication request
]
```

Changes since DSTU2

CommunicationRequest

CommunicationRequest.basedOn | Added Element

CommunicationRequest.replaces | Added Element

CommunicationRequest.groupIdentifier | Added Element

CommunicationRequest.status | Min Cardinality changed from 0 to 1 Change value set from http://hl7.org/fhir/ValueSet/communication-request-status to http://hl7.org/fhir/ValueSet/request-status

CommunicationRequest.category | Max Cardinality changed from 1 to *

CommunicationRequest.priority | Type changed from CodeableConcept to code Add Binding http://hl7.org/fhir/ValueSet/request-priority (required)

CommunicationRequest.subject | Add Reference(Group)

CommunicationRequest.recipient | Add Reference(Group), Add Reference(CareTeam)

CommunicationRequest.topic | Added Element

CommunicationRequest.context | Renamed from encounter to context Add Reference(EpisodeOfCare)

CommunicationRequest.occurrence[x] | Added Element

CommunicationRequest.authoredOn | Renamed from requestedOn to authoredOn

CommunicationRequest.requester | Remove Reference(Practitioner), Remove Reference(Patient), Remove Reference(RelatedPerson)

CommunicationRequest.requester.agent | Added Element

CommunicationRequest.requester.onBehalfOf | Added Element

CommunicationRequest.reasonCode | Renamed from reason to reasonCode

CommunicationRequest.reasonReference | Added Element

CommunicationRequest.note | Added Element

CommunicationRequest.scheduled[x] | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R2 <--> R3 Conversion Maps (status = 2 tests that all execute ok. 1 fail round-trip testing and 1 r3 resources are invalid (1 errors).).

Alternate definitions: Master Definition (XML, JSON), XML Schema/Schematron (for) + JSON Schema, ShEx (for Turtle)

8.21.3.1 Terminology Bindings

Path | Definition | Type | Reference

CommunicationRequest.status | The status of the communication request. | Required | RequestStatus

CommunicationRequest.category | Codes for general categories of communications such as alerts, instruction, etc. | Example | CommunicationCategory

CommunicationRequest.priority | Codes indicating the relative importance of a communication request. | Required | RequestPriority

CommunicationRequest.medium | Codes for communication mediums such as phone, fax, email, in person, etc. | Example | v3 Code System ParticipationMode

CommunicationRequest.reasonCode | Codes for describing reasons for the occurrence of a communication. | Example | v3 Code System ActReason

8.21.3.2 Constraints

* cmr-1: On CommunicationRequest.requester: onBehalfOf can only be specified if agent is practitioner or device (expression on CommunicationRequest.requester: (agent.resolve() is Practitioner) or (agent.resolve() is Device) or onBehalfOf.exists().not())

Notes to reviewers:

At this time, the code bindings are placeholders to be fleshed out upon further review by the community.

8.21.3.3 CommunicationRequest.sender and CommunicationRequest.recepient

CommunicationRequest.sender and CommunicationRequest.recipient allow Patient|Practitioner|RelatedPerson - but it is not unusual to have a communication target - even a defined one - where it is unknown what kind of role the person is playing.

If the communication request is to or from an individual, whose role is not known (practitioner, patient or related person), - for example, only email address is captured in the system; then RelatedPerson should be used by default.

8.21.4 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
authored | date | When request transitioned to being actionable | CommunicationRequest.authoredOn
based-on | reference | Fulfills plan or proposal | CommunicationRequest.basedOn (Any)
category | token | Message category | CommunicationRequest.category
context | reference | Encounter or episode leading to message | CommunicationRequest.context (EpisodeOfCare, Encounter)
encounter | reference | Encounter leading to message | CommunicationRequest.context (Encounter)
group-identifier | token | Composite request this is part of | CommunicationRequest.groupIdentifier
identifier | token | Unique identifier | CommunicationRequest.identifier
medium | token | A channel of communication | CommunicationRequest.medium
occurrence | date | When scheduled | CommunicationRequest.occurrence.as(DateTime)
patient | reference | Focus of message | CommunicationRequest.subject (Patient)
priority | token | Message urgency | CommunicationRequest.priority
recipient | reference | Message recipient | CommunicationRequest.recipient (Practitioner, Group, Organization, CareTeam, Device, Patient, RelatedPerson)
replaces | reference | Request(s) replaced by this request | CommunicationRequest.replaces (CommunicationRequest)
requester | reference | Individual making the request | CommunicationRequest.requester.agent (Practitioner, Organization, Device, Patient, RelatedPerson)
sender | reference | Message sender | CommunicationRequest.sender (Practitioner, Organization, Device, Patient, RelatedPerson)
status | token | draft | active | suspended | cancelled | completed | entered-in-error | unknown | CommunicationRequest.status
subject | reference | Focus of message | CommunicationRequest.subject (Group, Patient)

8.21.5 Resource CommunicationRequest - Examples

URL: https://hl7.org/fhir/STU3/communicationrequest-examples.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.21.5 Resource CommunicationRequest - Examples

Patient Care Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

An introduction to this page as well as proper examples are to be submitted shortly.

Example List:

An example of an alert message | example | XML | JSON | Turtle
An example of a request for additional information | fm-solicit | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

8.21.6 Resource CommunicationRequest - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/communicationrequest-definitions.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.21.6 Resource CommunicationRequest - Detailed Descriptions

Patient Care Work Group | Maturity Level: 2 | Trial Use | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

Detailed Descriptions for the elements in the CommunicationRequest resource.

CommunicationRequest
Definition | A request to convey information; e.g. the CDS system proposes that an alert be sent to a responsible provider, the CDS system proposes that the public health agency be notified about a reportable condition.
Control | 1..1

CommunicationRequest.identifier
Definition | A unique ID of this request for reference purposes. It must be provided if user wants it returned as part of any output, otherwise it will be autogenerated, if needed, by CDS system. Does not need to be the actual ID of the source system.
Note | This is a business identifer, not a resource identifier (see discussion)
Control | 0..*
Type | Identifier
Summary | true
CommunicationRequest.basedOn
Definition | A plan or proposal that is fulfilled in whole or in part by this request.
Control | 0..*
Type | Reference(Any)
Requirements | Allows tracing of authorization for the request and tracking whether proposals/recommendations were acted upon.
Alternate Names | fulfills
Summary | true
CommunicationRequest.replaces
Definition | Completed or terminated request(s) whose function is taken by this new request.
Control | 0..*
Type | Reference(CommunicationRequest)
Requirements | Allows tracing the continuation of a therapy or administrative process instantiated through multiple requests.
Alternate Names | supersedes; prior
Summary | true
Comments | The replacement could be because the initial request was immediately rejected (due to an issue) or because the previous request was completed, but the need for the action described by the request remains ongoing.
CommunicationRequest.groupIdentifier
Definition | A shared identifier common to all requests that were authorized more or less simultaneously by a single author, representing the identifier of the requisition, prescription or similar form.
Control | 0..1
Type | Identifier
Alternate Names | grouperId
Summary | true
Comments | Requests are linked either by a "basedOn" relationship (i.e. one request is fulfilling another) or by having a common requisition. Requests that are part of the same requisition are generally treated independently from the perspective of changing their state or maintaining them after initial creation.
CommunicationRequest.status
Definition | The status of the proposal or order.
Control | 1..1
Terminology Binding | RequestStatus (Required)
Type | code
Is Modifier | true
Summary | true
Comments | This element is labeled as a modifier because the status contains the codes cancelled and entered-in-error that mark the request as not currently valid.
CommunicationRequest.category
Definition | The type of message to be sent such as alert, notification, reminder, instruction, etc.
Control | 0..*
Terminology Binding | CommunicationCategory (Example)
Type | CodeableConcept
Comments | There may be multiple axes of categorization and one communication request may serve multiple purposes.
CommunicationRequest.priority
Definition | Characterizes how quickly the proposed act must be initiated. Includes concepts such as stat, urgent, routine.
Control | 0..1
Terminology Binding | RequestPriority (Required)
Type | code
Meaning if Missing | If missing, this task should be performed with normal priority
Summary | true
CommunicationRequest.medium
Definition | A channel that was used for this communication (e.g. email, fax).
Control | 0..*
Terminology Binding | v3 Code System ParticipationMode (Example)
Type | CodeableConcept
CommunicationRequest.subject
Definition | The patient or group that is the focus of this communication request.
Control | 0..1
Type | Reference(Patient | Group)
Alternate Names | patient
CommunicationRequest.recipient
Definition | The entity (e.g. person, organization, clinical information system, device, group, or care team) which is the intended target of the communication.
Control | 0..*
Type | Reference(Device | Organization | Patient | Practitioner | RelatedPerson | Group | CareTeam)
CommunicationRequest.topic
Definition | The resources which were related to producing this communication request.
Control | 0..*

Type | Reference(Any)
CommunicationRequest.context
Definition | The encounter or episode of care within which the communication request was created.
Control | 0..1
Type | Reference(Encounter | EpisodeOfCare)
Alternate Names | encounter
Summary | true
CommunicationRequest.payload
Definition | Text, attachment(s), or resource(s) to be communicated to the recipient.
Control | 0..*
CommunicationRequest.payload.content[x]
Definition | The communicated content (or for multi-part communications, one portion of the communication).
Control | 1..1
Type | string|Attachment|Reference(Any)
[x] Note | See Choice of Data Types for further information about how to use [x]
CommunicationRequest.occurrence[x]
Definition | The time when this communication is to occur.
Control | 0..1
Type | dateTime|Period
[x] Note | See Choice of Data Types for further information about how to use [x]
Alternate Names | timing
Summary | true
CommunicationRequest.authoredOn
Definition | For draft requests, indicates the date of initial creation. For requests with other statuses, indicates the date of activation.
Control | 0..1
Type | dateTime
Alternate Names | createdOn
Summary | true
CommunicationRequest.sender
Definition | The entity (e.g. person, organization, clinical information system, or device) which is to be the source of the communication.
Control | 0..1
Type | Reference(Device | Organization | Patient | Practitioner | RelatedPerson)
CommunicationRequest.requester
Definition | The individual who initiated the request and has responsibility for its activation.
Control | 0..1
Alternate Names | author
Summary | true
Invariants | Defined on this element cmr-1: onBehalfOf can only be specified if agent is practitioner or device (expression : (agent.resolve() is Practitioner) or (agent.resolve() is Device) or onBehalfOf.exists().not(), xpath: contains(f:agent/f:reference/@value, '/Practitioner/') or contains(f:agent/f:reference/@value, '/Device/') or not(exists(f:onBehalfOf)))
CommunicationRequest.requester.agent
Definition | The device, practitioner, etc. who initiated the request.
Control | 1..1
Type | Reference(Practitioner | Organization | Patient | RelatedPerson | Device)
Summary | true
CommunicationRequest.requester.onBehalfOf
Definition | The organization the device or practitioner was acting on behalf of.
Control | 0..1
Type | Reference(Organization)
Requirements | Practitioners and Devices can be associated with multiple organizations. This element indicates which organization they were acting on behalf of when authoring the request.
Summary | true
Invariants | Affect this element cmr-1: onBehalfOf can only be specified if agent is practitioner or device (expression : (agent.resolve() is Practitioner) or (agent.resolve() is Device) or onBehalfOf.exists().not(), xpath: contains(f:agent/f:reference/@value, '/Practitioner/') or contains(f:agent/f:reference/@value, '/Device/') or not(exists(f:onBehalfOf)))
CommunicationRequest.reasonCode
Definition | Describes why the request is being made in coded or textual form.
Control | 0..*
Terminology Binding | v3 Code System ActReason (Example)
Type | CodeableConcept
Summary | true
Comments | Textual reasons can be caprued using reasonCode.text.
CommunicationRequest.reasonReference
Definition | Indicates another resource whose existence justifies this request.
Control | 0..*
Type | Reference(Condition | Observation)
Summary | true
CommunicationRequest.note
Definition | Comments made about the request by the requester, sender, recipient, subject or other participants.
Control | 0..*
Type | Annotation

8.21.7 Resource CommunicationRequest - Mappings

URL: https://hl7.org/fhir/STU3/communicationrequest-mappings.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.21.7 Resource CommunicationRequest - Mappings

Patient Care Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

Mappings:

RIM Mapping

Workflow Mapping

Mappings for the communicationrequest resource.

8.21.7.1 Mappings for RIM Mapping (http://hl7.org/v3)

CommunicationRequest
identifier
basedOn | .outboundRelationship[typeCode=FLFS].target
replaces | .outboundRelationship[typeCode=RPLC].target
groupIdentifier | .inboundRelationship(typeCode=COMP].source[moodCode=INT].identifier
status
category
priority
medium
subject
recipient
topic
context
payload
content[x]
occurrence[x]
authoredOn
sender
requester | .participation[typeCode=AUT].role
agent | .player
onBehalfOf | .scoper
reasonCode | .reasonCode
reasonReference | .outboundRelationship[typeCode=RSON].target
note | .inboundRelationship(typeCode=SUBJ].source[classCode=ANNGEN, moodCode=EVN].value[xsi:type=ST]

8.21.7.2 Mappings for Workflow Mapping (http://hl7.org/fhir/workflow)

CommunicationRequest | Request
identifier | Request.identifier
basedOn | Request.basedOn
replaces | Request.replaces
groupIdentifier | Request.groupIdentifier
status | Request.status
category | Request.code
priority | Request.priority
medium
subject | Request.subject
recipient | Request.performer
topic | Request.supportingInfo
context | Request.context
payload
content[x] | Request.supportingInfo IF Reference, Request.note if string; No mapping if Attachment
occurrence[x] | Request.occurrence
authoredOn | Request.authoredOn
sender | Request.requester.agent
requester | Request.requester
agent | Request.requester.agent
onBehalfOf | Request.requester.onBehalfOf
reasonCode | Request.reasonCode
reasonReference | Request.reasonReference
note | Request.note

8.21.8 Resource CommunicationRequest - Design Notes

URL: https://hl7.org/fhir/STU3/communicationrequest-explanations.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.21.8 Resource CommunicationRequest - Design Notes

to do

8.21.9 Resource CommunicationRequest - Extensions & Profiles

URL: https://hl7.org/fhir/STU3/communicationrequest-profiles.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.21.9 Resource CommunicationRequest - Extensions & Profiles

Patient Care Work Group | Maturity Level: N/A | Ballot Status: Informative

This table lists profiles and extensions for the CommunicationRequest resource. For background information, see Profiling Resources and Extensibility. Additional profiles and resources may be found in published Implementation Guides , or in the Conformance resource registry .

8.21.9.1 Profiles

No Profiles defined for this resource

8.21.9.2 Extensions

communicationrequest-definition | definition | for CommunicationRequest HL7 Extensions
communicationrequest-orderedBy | orderedBy | for CommunicationRequest HL7 Extensions
communicationrequest-reasonRejected | reasonRejected | for CommunicationRequest HL7 Extensions
communicationrequest-relevantHistory | relevantHistory | for CommunicationRequest HL7 Extensions
communicationrequest-supportingInfo | supportingInfo | for CommunicationRequest HL7 Extensions

Extensions that reference this resource:

No Extensions refer to this resource

8.21.9.3 Search Extensions

No Search Extensions defined for this resource

8.21.10 CommunicationRequest HL7 Extensions

URL: https://hl7.org/fhir/STU3/communicationrequest-extensions.html

Defines common extensions used with or related to the CommunicationRequest resource

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.21.10 CommunicationRequest HL7 Extensions

(No assigned work group) | Maturity Level: N/A | Ballot Status: Informative

8.21.10.1 Content

Extensions:
communicationrequest-orderedBy | orderedBy : Agent that ordered the communication.
communicationrequest-reasonRejected | reasonRejected : The reason associated with the rejection of this communication request by the receiver. Applicable only if the status is rejected.
communicationrequest-supportingInfo | supportingInfo : Information that may be needed by/relevant to the performer in their execution of this request.
communicationrequest-relevantHistory | relevantHistory : Links to Provenance records for past versions of this resource or fulfilling request or event resources that identify key state transitions or updates that are likely to be relevant to a user looking at the current version of the resource.
communicationrequest-definition | definition : A protocol, guideline, or other definition that is adhered to in whole or in part by this request.

8.21.11 Resource CommunicationRequest - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/communicationrequest-version-maps.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.21.11 Resource CommunicationRequest - Detailed Descriptions

Patient Care Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

R2 : R3 Converson maps for CommunicationRequest.

Functional status for this map: 2 tests that all execute ok. 1 fail round-trip testing and 1 r3 resources are invalid (1 errors). (see documentation)

8.21.11.1 R2 to R3

```
map "http://hl7.org/fhir/StructureMap/CommunicationRequest2to3" = "R2 to R3 Conversions for CommunicationRequest"


conceptmap "CommunicationRequestStatus" {
  prefix s = "http://hl7.org/fhir/communication-request-status"
  prefix t = "http://hl7.org/fhir/communication-request-status"

  s:proposed = t:draft
  s:planned = t:draft
  s:requested = t:active
  s:received = t:active
  s:accepted = t:active
  s:"in-progress" = t:active
  s:completed = t:completed
  s:suspended = t:suspended
  s:rejected = t:cancelled
  s:failed = t:cancelled
}


uses "http://hl7.org/fhir/DSTU2/StructureDefinition/CommunicationRequest" alias CommunicationRequestR2 as source
uses "http://hl7.org/fhir/StructureDefinition/CommunicationRequest" alias CommunicationRequest as target

imports "http://hl7.org/fhir/StructureMap/*2to3"


group for type+types CommunicationRequest extends DomainResource
  input src : CommunicationRequestR2 as source
  input tgt : CommunicationRequest as target

  "CommunicationRequest-identifier" : for src.identifier make tgt.identifier
  "CommunicationRequest-category" : for src.category make tgt.category
  "CommunicationRequest-sender" : for src.sender make tgt.sender
  "CommunicationRequest-recipient" : for src.recipient make tgt.recipient
  "CommunicationRequest-payload" : for src.payload as vs0 make tgt.payload as vt0 then cmrPayload(vs0, vt0)
  "CommunicationRequest-medium" : for src.medium make tgt.medium
  "CommunicationRequest-requester" : for src.requester make tgt.requester
  "CommunicationRequest-status" : for src.status as v make tgt.status = translate(v, "#CommunicationRequestStatus",
    "code")
  "CommunicationRequest-context" : for src.encounter make tgt.context
  "CommunicationRequest-scheduled" : for src.scheduled make tgt.occurrence
  "CommunicationRequest-reason" : for src.reason make tgt.reasonCode
  "CommunicationRequest-requestedOn" : for src.requestedOn make tgt.authoredOn
  "CommunicationRequest-subject" : for src.subject make tgt.subject
  "CommunicationRequest-priority" : for src.priority make tgt.priority
endgroup

group cmrPayload
  input src as source
  input tgt as target

  "CommunicationRequest.payload-content" : for src.content make tgt.content
endgroup
```

8.21.11.2 R3 to R2

```
map "http://hl7.org/fhir/StructureMap/CommunicationRequest3to2" = "R3 to R2 Conversion for CommunicationRequest"

uses "http://hl7.org/fhir/StructureDefinition/CommunicationRequest" alias CommunicationRequest as source
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/CommunicationRequest" alias CommunicationRequestR2 as target

imports "http://hl7.org/fhir/StructureMap/*3to2"

group for type+types  CommunicationRequest extends DomainResource
  input src : CommunicationRequest as source
  input tgt : CommunicationRequestR2 as target

  "CommunicationRequest-identifier" : for src.identifier make tgt.identifier
  "CommunicationRequest-category" : for src.category make tgt.category
  "CommunicationRequest-sender" : for src.sender make tgt.sender
  "CommunicationRequest-recipient" : for src.recipient make tgt.recipient
  "CommunicationRequest-payload" : for src.payload as vs0 make tgt.payload as vt0 then cmrPayload(vs0, vt0)
  "CommunicationRequest-medium" : for src.medium make tgt.medium
  "CommunicationRequest-requester" : for src.requester make tgt.requester
  "CommunicationRequest-status" : for src.status make tgt.status
  "CommunicationRequest-context" : for src.context make tgt.encounter
  "CommunicationRequest-scheduled" : for src.occurrence make tgt.scheduled
  "CommunicationRequest-reason" : for src.reasonCode make tgt.reason
  "CommunicationRequest-requestedOn" : for src.authoredOn make tgt.requestedOn
  "CommunicationRequest-subject" : for src.subject make tgt.subject
  "CommunicationRequest-priority" : for src.priority make tgt.priority
endgroup

group cmrPayload
  input src as source
  input tgt as target

  "CommunicationRequest.payload-content" : for vs0.content make vt0.content
```

```
endgroup
```

8.21.11.3 R3 Validation Errors

example | ERROR @ CommunicationRequest: Profile http://hl7.org/fhir/StructureDefinition/CommunicationRequest, Element 'CommunicationRequest.status': minimum required = 1, but only found 0 (src = InstanceValidator)

8.22 Resource ResearchStudy - Content

URL: https://hl7.org/fhir/STU3/researchstudy.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

Regulated Clinical Research Information Management Work Group | Maturity Level: 0 | Draft | Compartments: Practitioner

A process where a researcher or organization plans and then executes a series of steps intended to increase the field of healthcare-related knowledge. This includes studies of safety, efficacy, comparative effectiveness and other information about medications, devices, therapies and other interventional and investigative techniques. A ResearchStudy involves the gathering of information about human or animal subjects.

8.22.1 Scope and Usage

This resource supports the HL7 mission to create and promote HL7 standards by developing RCRIM standards to improve or enhance information management during clinical research and regulatory evaluation of the safety, efficacy and quality of therapeutic products and procedures worldwide.

This resource is referenced by AdverseEvent and ResearchSubject

8.22.2 Resource Content

* Structure
* UML
* XML
* JSON
* Turtle
* R2 Diff
* All

Structure

Name | Flags | Card. | Type | Description & Constraints
ResearchStudy | | | DomainResource | Investigation to increase healthcare-related patient-independent knowledge Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | ? | 0..* | Identifier | Business Identifier for study
title | ? | 0..1 | string | Name for this study
protocol | ? | 0..* | Reference(PlanDefinition) | Steps followed in executing study
partOf | ? | 0..* | Reference(ResearchStudy) | Part of larger study
status | ?!? | 1..1 | code | draft | in-progress | suspended | stopped | completed | entered-in-error ResearchStudyStatus (Required)
category | ? | 0..* | CodeableConcept | Classifications for the study
focus | ? | 0..* | CodeableConcept | Drugs, devices, conditions, etc. under study
contact | ? | 0..* | ContactDetail | Contact details for the study
relatedArtifact | | 0..* | RelatedArtifact | References and dependencies
keyword | ? | 0..* | CodeableConcept | Used to search for the study
jurisdiction | ? | 0..* | CodeableConcept | Geographic region(s) for study Jurisdiction ValueSet (Extensible)
description | | 0..1 | markdown | What this is study doing
enrollment | ? | 0..* | Reference(Group) | Inclusion & exclusion criteria
period | ? | 0..1 | Period | When the study began and ended
sponsor | ? | 0..1 | Reference(Organization) | Organization responsible for the study
principalInvestigator | ? | 0..1 | Reference(Practitioner) | The individual responsible for the study
site | ? | 0..* | Reference(Location) | Location involved in study execution
reasonStopped | ? | 0..1 | CodeableConcept | Reason for terminating study early
note | | 0..* | Annotation | Comments made about the event
arm | | 0..* | BackboneElement | Defined path through the study for a subject
name | | 1..1 | string | Label for study arm
code | | 0..1 | CodeableConcept | Categorization of study arm
description | | 0..1 | string | Short explanation of study path

Documentation for this format

UML Diagram (Legend)

XML Template

```
<ResearchStudy xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Business Identifier for study --></identifier>
 <title value="[string]"/><!-- 0..1 Name for this study -->
```

```
 <protocol><!-- 0..* Reference(PlanDefinition) Steps followed in executing study --></protocol>
 <partOf><!-- 0..* Reference(ResearchStudy) Part of larger study --></partOf>
 <status value="[code]"/><!-- 1..1 draft | in-progress | suspended | stopped | completed | entered-in-error -->
 <category><!-- 0..* CodeableConcept Classifications for the study --></category>
 <focus><!-- 0..* CodeableConcept Drugs, devices, conditions, etc. under study --></focus>
 <contact><!-- 0..* ContactDetail Contact details for the study --></contact>
 <relatedArtifact><!-- 0..* RelatedArtifact References and dependencies --></relatedArtifact>
 <keyword><!-- 0..* CodeableConcept Used to search for the study --></keyword>
 <jurisdiction><!-- 0..* CodeableConcept Geographic region(s) for study --></jurisdiction>
 <description value="[markdown]"/><!-- 0..1 What this is study doing -->
 <enrollment><!-- 0..* Reference(Group) Inclusion & exclusion criteria --></enrollment>
 <period><!-- 0..1 Period When the study began and ended --></period>
 <sponsor><!-- 0..1 Reference(Organization) Organization responsible for the study --></sponsor>
 <principalInvestigator><!-- 0..1 Reference(Practitioner) The individual responsible for the study
    --></principalInvestigator>
 <site><!-- 0..* Reference(Location) Location involved in study execution --></site>
 <reasonStopped><!-- 0..1 CodeableConcept Reason for terminating study early --></reasonStopped>
 <note><!-- 0..* Annotation Comments made about the event --></note>
 <arm>  <!-- 0..* Defined path through the study for a subject -->
  <name value="[string]"/><!-- 1..1 Label for study arm -->
  <code><!-- 0..1 CodeableConcept Categorization of study arm --></code>
  <description value="[string]"/><!-- 0..1 Short explanation of study path -->
 </arm>
</ResearchStudy>
```

JSON Template

```
{
  "resourceType" : "ResearchStudy",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Business Identifier for study
  "title" : "<string>", // Name for this study
  "protocol" : [{ Reference(PlanDefinition) }], // Steps followed in executing study
  "partOf" : [{ Reference(ResearchStudy) }], // Part of larger study
  "status" : "<code>", // R!  draft | in-progress | suspended | stopped | completed | entered-in-error
  "category" : [{ CodeableConcept }], // Classifications for the study
  "focus" : [{ CodeableConcept }], // Drugs, devices, conditions, etc. under study
  "contact" : [{ ContactDetail }], // Contact details for the study
  "relatedArtifact" : [{ RelatedArtifact }], // References and dependencies
  "keyword" : [{ CodeableConcept }], // Used to search for the study
  "jurisdiction" : [{ CodeableConcept }], // Geographic region(s) for study
  "description" : "<markdown>", // What this is study doing
  "enrollment" : [{ Reference(Group) }], // Inclusion & exclusion criteria
  "period" : { Period }, // When the study began and ended
  "sponsor" : { Reference(Organization) }, // Organization responsible for the study
  "principalInvestigator" : { Reference(Practitioner) }, // The individual responsible for the study
  "site" : [{ Reference(Location) }], // Location involved in study execution
  "reasonStopped" : { CodeableConcept }, // Reason for terminating study early
  "note" : [{ Annotation }], // Comments made about the event
  "arm" : [{ // Defined path through the study for a subject
    "name" : "<string>", // R!  Label for study arm
    "code" : { CodeableConcept }, // Categorization of study arm
    "description" : "<string>" // Short explanation of study path
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:ResearchStudy;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:ResearchStudy.identifier [ Identifier ], ... ; # 0..* Business Identifier for study
  fhir:ResearchStudy.title [ string ]; # 0..1 Name for this study
  fhir:ResearchStudy.protocol [ Reference(PlanDefinition) ], ... ; # 0..* Steps followed in executing study
  fhir:ResearchStudy.partOf [ Reference(ResearchStudy) ], ... ; # 0..* Part of larger study
  fhir:ResearchStudy.status [ code ]; # 1..1 draft | in-progress | suspended | stopped | completed | entered-in-error
  fhir:ResearchStudy.category [ CodeableConcept ], ... ; # 0..* Classifications for the study
  fhir:ResearchStudy.focus [ CodeableConcept ], ... ; # 0..* Drugs, devices, conditions, etc. under study
  fhir:ResearchStudy.contact [ ContactDetail ], ... ; # 0..* Contact details for the study
  fhir:ResearchStudy.relatedArtifact [ RelatedArtifact ], ... ; # 0..* References and dependencies
  fhir:ResearchStudy.keyword [ CodeableConcept ], ... ; # 0..* Used to search for the study
  fhir:ResearchStudy.jurisdiction [ CodeableConcept ], ... ; # 0..* Geographic region(s) for study
  fhir:ResearchStudy.description [ markdown ]; # 0..1 What this is study doing
  fhir:ResearchStudy.enrollment [ Reference(Group) ], ... ; # 0..* Inclusion & exclusion criteria
  fhir:ResearchStudy.period [ Period ]; # 0..1 When the study began and ended
  fhir:ResearchStudy.sponsor [ Reference(Organization) ]; # 0..1 Organization responsible for the study
  fhir:ResearchStudy.principalInvestigator [ Reference(Practitioner) ]; # 0..1 The individual responsible for the study
  fhir:ResearchStudy.site [ Reference(Location) ], ... ; # 0..* Location involved in study execution
  fhir:ResearchStudy.reasonStopped [ CodeableConcept ]; # 0..1 Reason for terminating study early
  fhir:ResearchStudy.note [ Annotation ], ... ; # 0..* Comments made about the event
  fhir:ResearchStudy.arm [ # 0..* Defined path through the study for a subject
    fhir:ResearchStudy.arm.name [ string ]; # 1..1 Label for study arm
    fhir:ResearchStudy.arm.code [ CodeableConcept ]; # 0..1 Categorization of study arm
    fhir:ResearchStudy.arm.description [ string ]; # 0..1 Short explanation of study path
  ], ...;
```

]

Changes since DSTU2

This resource did not exist in Release 2

This analysis is available as XML or JSON.

Structure

Name | Flags | Card. | Type | Description & Constraints
ResearchStudy | | | DomainResource | Investigation to increase healthcare-related patient-independent knowledge Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | ? | 0..* | Identifier | Business Identifier for study
title | ? | 0..1 | string | Name for this study
protocol | ? | 0..* | Reference(PlanDefinition) | Steps followed in executing study
partOf | ? | 0..* | Reference(ResearchStudy) | Part of larger study
status | ?!? | 1..1 | code | draft | in-progress | suspended | stopped | completed | entered-in-error ResearchStudyStatus (Required)
category | ? | 0..* | CodeableConcept | Classifications for the study
focus | ? | 0..* | CodeableConcept | Drugs, devices, conditions, etc. under study
contact | ? | 0..* | ContactDetail | Contact details for the study
relatedArtifact | | 0..* | RelatedArtifact | References and dependencies
keyword | ? | 0..* | CodeableConcept | Used to search for the study
jurisdiction | ? | 0..* | CodeableConcept | Geographic region(s) for study Jurisdiction ValueSet (Extensible)
description | | 0..1 | markdown | What this is study doing
enrollment | ? | 0..* | Reference(Group) | Inclusion & exclusion criteria
period | ? | 0..1 | Period | When the study began and ended
sponsor | ? | 0..1 | Reference(Organization) | Organization responsible for the study
principalInvestigator | ? | 0..1 | Reference(Practitioner) | The individual responsible for the study
site | ? | 0..* | Reference(Location) | Location involved in study execution
reasonStopped | ? | 0..1 | CodeableConcept | Reason for terminating study early
note | | 0..* | Annotation | Comments made about the event
arm | | 0..* | BackboneElement | Defined path through the study for a subject
name | | 1..1 | string | Label for study arm
code | | 0..1 | CodeableConcept | Categorization of study arm
description | | 0..1 | string | Short explanation of study path
Documentation for this format

UML Diagram (Legend)

XML Template

```
<ResearchStudy xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Business Identifier for study --></identifier>
 <title value="[string]"/><!-- 0..1 Name for this study -->
 <protocol><!-- 0..* Reference(PlanDefinition) Steps followed in executing study --></protocol>
 <partOf><!-- 0..* Reference(ResearchStudy) Part of larger study --></partOf>
 <status value="[code]"/><!-- 1..1 draft | in-progress | suspended | stopped | completed | entered-in-error -->
 <category><!-- 0..* CodeableConcept Classifications for the study --></category>
 <focus><!-- 0..* CodeableConcept Drugs, devices, conditions, etc. under study --></focus>
 <contact><!-- 0..* ContactDetail Contact details for the study --></contact>
 <relatedArtifact><!-- 0..* RelatedArtifact References and dependencies --></relatedArtifact>
 <keyword><!-- 0..* CodeableConcept Used to search for the study --></keyword>
 <jurisdiction><!-- 0..* CodeableConcept Geographic region(s) for study --></jurisdiction>
 <description value="[markdown]"/><!-- 0..1 What this is study doing -->
 <enrollment><!-- 0..* Reference(Group) Inclusion & exclusion criteria --></enrollment>
 <period><!-- 0..1 Period When the study began and ended --></period>
 <sponsor><!-- 0..1 Reference(Organization) Organization responsible for the study --></sponsor>
 <principalInvestigator><!-- 0..1 Reference(Practitioner) The individual responsible for the study
    --></principalInvestigator>
 <site><!-- 0..* Reference(Location) Location involved in study execution --></site>
 <reasonStopped><!-- 0..1 CodeableConcept Reason for terminating study early --></reasonStopped>
 <note><!-- 0..* Annotation Comments made about the event --></note>
 <arm>  <!-- 0..* Defined path through the study for a subject -->
  <name value="[string]"/><!-- 1..1 Label for study arm -->
  <code><!-- 0..1 CodeableConcept Categorization of study arm --></code>
  <description value="[string]"/><!-- 0..1 Short explanation of study path -->
 </arm>
</ResearchStudy>
```

JSON Template

```
{
  "resourceType" : "ResearchStudy",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Business Identifier for study
  "title" : "<string>", // Name for this study
  "protocol" : [{ Reference(PlanDefinition) }], // Steps followed in executing study
  "partOf" : [{ Reference(ResearchStudy) }], // Part of larger study
  "status" : "<code>", // R!  draft | in-progress | suspended | stopped | completed | entered-in-error
  "category" : [{ CodeableConcept }], // Classifications for the study
  "focus" : [{ CodeableConcept }], // Drugs, devices, conditions, etc. under study
  "contact" : [{ ContactDetail }], // Contact details for the study
  "relatedArtifact" : [{ RelatedArtifact }], // References and dependencies
```

```
  "keyword" : [{ CodeableConcept }], // Used to search for the study
  "jurisdiction" : [{ CodeableConcept }], // Geographic region(s) for study
  "description" : "<markdown>", // What this is study doing
  "enrollment" : [{ Reference(Group) }], // Inclusion & exclusion criteria
  "period" : { Period }, // When the study began and ended
  "sponsor" : { Reference(Organization) }, // Organization responsible for the study
  "principalInvestigator" : { Reference(Practitioner) }, // The individual responsible for the study
  "site" : [{ Reference(Location) }], // Location involved in study execution
  "reasonStopped" : { CodeableConcept }, // Reason for terminating study early
  "note" : [{ Annotation }], // Comments made about the event
  "arm" : [{ // Defined path through the study for a subject
    "name" : "<string>", // R!  Label for study arm
    "code" : { CodeableConcept }, // Categorization of study arm
    "description" : "<string>" // Short explanation of study path
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:ResearchStudy;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:ResearchStudy.identifier [ Identifier ], ... ; # 0..* Business Identifier for study
  fhir:ResearchStudy.title [ string ]; # 0..1 Name for this study
  fhir:ResearchStudy.protocol [ Reference(PlanDefinition) ], ... ; # 0..* Steps followed in executing study
  fhir:ResearchStudy.partOf [ Reference(ResearchStudy) ], ... ; # 0..* Part of larger study
  fhir:ResearchStudy.status [ code ]; # 1..1 draft | in-progress | suspended | stopped | completed | entered-in-error
  fhir:ResearchStudy.category [ CodeableConcept ], ... ; # 0..* Classifications for the study
  fhir:ResearchStudy.focus [ CodeableConcept ], ... ; # 0..* Drugs, devices, conditions, etc. under study
  fhir:ResearchStudy.contact [ ContactDetail ], ... ; # 0..* Contact details for the study
  fhir:ResearchStudy.relatedArtifact [ RelatedArtifact ], ... ; # 0..* References and dependencies
  fhir:ResearchStudy.keyword [ CodeableConcept ], ... ; # 0..* Used to search for the study
  fhir:ResearchStudy.jurisdiction [ CodeableConcept ], ... ; # 0..* Geographic region(s) for study
  fhir:ResearchStudy.description [ markdown ]; # 0..1 What this is study doing
  fhir:ResearchStudy.enrollment [ Reference(Group) ], ... ; # 0..* Inclusion & exclusion criteria
  fhir:ResearchStudy.period [ Period ]; # 0..1 When the study began and ended
  fhir:ResearchStudy.sponsor [ Reference(Organization) ]; # 0..1 Organization responsible for the study
  fhir:ResearchStudy.principalInvestigator [ Reference(Practitioner) ]; # 0..1 The individual responsible for the study
  fhir:ResearchStudy.site [ Reference(Location) ], ... ; # 0..* Location involved in study execution
  fhir:ResearchStudy.reasonStopped [ CodeableConcept ]; # 0..1 Reason for terminating study early
  fhir:ResearchStudy.note [ Annotation ], ... ; # 0..* Comments made about the event
  fhir:ResearchStudy.arm [ # 0..* Defined path through the study for a subject
    fhir:ResearchStudy.arm.name [ string ]; # 1..1 Label for study arm
    fhir:ResearchStudy.arm.code [ CodeableConcept ]; # 0..1 Categorization of study arm
    fhir:ResearchStudy.arm.description [ string ]; # 0..1 Short explanation of study path
  ], ...;
]
```

Changes since DSTU2

This resource did not exist in Release 2

This analysis is available as XML or JSON.

Alternate definitions: Master Definition (XML, JSON), XML Schema/Schematron (for) + JSON Schema, ShEx (for Turtle)

8.22.2.1 Terminology Bindings

Path | Definition | Type | Reference
ResearchStudy.status | Codes that convey the current status of the research study | Required | ResearchStudyStatus
ResearchStudy.category | Codes that describe the type of research study. E.g. Study phase, Interventional/Observational, blinding type, etc. | Unknown | No details provided yet
ResearchStudy.focus | Codes for medications, devices, conditions and other interventions | Unknown | No details provided yet
ResearchStudy.keyword | Words associated with the study that may be useful in discovery | Unknown | No details provided yet
ResearchStudy.jurisdiction | Countries and regions within which this artifact is targeted for use | Extensible | Jurisdiction ValueSet

8.22.3 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
category | token | Classifications for the study | ResearchStudy.category
date | date | When the study began and ended | ResearchStudy.period
focus | token | Drugs, devices, conditions, etc. under study | ResearchStudy.focus
identifier | token | Business Identifier for study | ResearchStudy.identifier
jurisdiction | token | Geographic region(s) for study | ResearchStudy.jurisdiction
keyword | token | Used to search for the study | ResearchStudy.keyword
partof | reference | Part of larger study | ResearchStudy.partOf (ResearchStudy)
principalinvestigator | reference | The individual responsible for the study | ResearchStudy.principalInvestigator (Practitioner)
protocol | reference | Steps followed in executing study | ResearchStudy.protocol (PlanDefinition)
site | reference | Location involved in study execution | ResearchStudy.site (Location)

sponsor | reference | Organization responsible for the study | ResearchStudy.sponsor (Organization)
status | token | draft | in-progress | suspended | stopped | completed | entered-in-error | ResearchStudy.status
title | string | Name for this study | ResearchStudy.title

8.22.4 Resource ResearchStudy - Examples

URL: https://hl7.org/fhir/STU3/researchstudy-examples.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.22.4 Resource ResearchStudy - Examples

Regulated Clinical Research Information Management Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Practitioner

Example List:

Placeholder example | example | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

8.22.5 Resource ResearchStudy - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/researchstudy-definitions.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.22.5 Resource ResearchStudy - Detailed Descriptions

Regulated Clinical Research Information Management Work Group | Maturity Level: 0 | Draft | Compartments: Practitioner

Detailed Descriptions for the elements in the ResearchStudy resource.

ResearchStudy
Definition | A process where a researcher or organization plans and then executes a series of steps intended to increase the field of healthcare-related knowledge. This includes studies of safety, efficacy, comparative effectiveness and other information about medications, devices, therapies and other interventional and investigative techniques. A ResearchStudy involves the gathering of information about human or animal subjects.
Control | 1..1
Alternate Names | Clinical Trial; Study
Comments | Need to make sure we encompass public health studies.
ResearchStudy.identifier
Definition | Identifiers assigned to this research study by the sponsor or other systems.
Note | This is a business identifer, not a resource identifier (see discussion)
Control | 0..*
Type | Identifier
Requirements | Allows identification of the research study as it is known by various participating systems and in a way that remains consistent across servers.
Summary | true
ResearchStudy.title
Definition | A short, descriptive user-friendly label for the study.
Control | 0..1
Type | string
Summary | true
ResearchStudy.protocol
Definition | The set of steps expected to be performed as part of the execution of the study.
Control | 0..*
Type | Reference(PlanDefinition)
Summary | true
ResearchStudy.partOf
Definition | A larger research study of which this particular study is a component or step.
Control | 0..*
Type | Reference(ResearchStudy)
Requirements | Allows breaking a study into components (e.g. by study site) each with their own PI, status, enrollment, etc.
Alternate Names | container
Summary | true
ResearchStudy.status
Definition | The current state of the study.
Control | 1..1
Terminology Binding | ResearchStudyStatus (Required)
Type | code
Is Modifier | true
Summary | true
Comments | This element is labeled as a modifier because the status contains codes that mark the resource as not currently valid.
To Do | Should this be a common code system for all events?
ResearchStudy.category

Definition | Codes categorizing the type of study such as investigational vs. observational, type of blinding, type of randomization, safety vs. efficacy, etc.
Control | 0..*
Terminology Binding | ResearchStudyCategory:
Type | CodeableConcept
Summary | true
ResearchStudy.focus
Definition | The condition(s), medication(s), food(s), therapy(ies), device(s) or other concerns or interventions that the study is seeking to gain more information about.
Control | 0..*
Terminology Binding | ResearchStudyFocus:
Type | CodeableConcept
Summary | true
ResearchStudy.contact
Definition | Contact details to assist a user in learning more about or engaging with the study.
Control | 0..*
Type | ContactDetail
Summary | true
ResearchStudy.relatedArtifact
Definition | Citations, references and other related documents.
Control | 0..*
Type | RelatedArtifact
ResearchStudy.keyword
Definition | Key terms to aid in searching for or filtering the study.
Control | 0..*
Terminology Binding | ResearchStudyKeyword:
Type | CodeableConcept
Summary | true
ResearchStudy.jurisdiction
Definition | Indicates a country, state or other region where the study is taking place.
Control | 0..*
Terminology Binding | Jurisdiction ValueSet (Extensible)
Type | CodeableConcept
Summary | true
ResearchStudy.description
Definition | A full description of how the study is being conducted.
Control | 0..1
Type | markdown
ResearchStudy.enrollment
Definition | Reference to a Group that defines the criteria for and quantity of subjects participating in the study. E.g. " 200 female Europeans between the ages of 20 and 45 with early onset diabetes".
Control | 0..*
Type | Reference(Group)
Alternate Names | patient
Summary | true
Comments | The Group referenced should not generally enumerate specific subjects. Subjects will be linked to the study using the ResearchSubject resource.
To Do | .participation[typeCode=SBJ].role.
ResearchStudy.period
Definition | Identifies the start date and the expected (or actual, depending on status) end date for the study.
Control | 0..1
Type | Period
Requirements | [The list of types may be constrained as appropriate for the type of event].
Alternate Names | timing
Summary | true
To Do | .effectiveTime.
ResearchStudy.sponsor
Definition | The organization responsible for the execution of the study.
Control | 0..1
Type | Reference(Organization)
Summary | true
To Do | .participation[typeCode=PRF].role[scoper.determinerCode=INSTANCE].
ResearchStudy.principalInvestigator
Definition | Indicates the individual who has primary oversite of the execution of the study.
Control | 0..1
Type | Reference(Practitioner)
Summary | true
ResearchStudy.site
Definition | Clinic, hospital or other healthcare location that is participating in the study.
Control | 0..*
Type | Reference(Location)
Summary | true
ResearchStudy.reasonStopped

Definition | A description and/or code explaining the premature termination of the study.
Control | 0..1
Type | CodeableConcept
Summary | true
ResearchStudy.note
Definition | Comments made about the event by the performer, subject or other participants.
Control | 0..*
Type | Annotation
To Do | .inboundRelationship(typeCode=SUBJ].source[classCode=ANNGEN, moodCode=EVN].value[xsi:type=ST].
ResearchStudy.arm
Definition | Describes an expected sequence of events for one of the participants of a study. E.g. Exposure to drug A, wash-out, exposure to drug B, wash-out, follow-up.
Control | 0..*
ResearchStudy.arm.name
Definition | Unique, human-readable label for this arm of the study.
Control | 1..1
Type | string
ResearchStudy.arm.code
Definition | Categorization of study arm, e.g. experimental, active comparator, placebo comparater.
Control | 0..1
Type | CodeableConcept
ResearchStudy.arm.description
Definition | A succinct description of the path through the study that would be followed by a subject adhering to this arm.
Control | 0..1
Type | string

8.22.6 Resource ResearchStudy - Mappings

URL: https://hl7.org/fhir/STU3/researchstudy-mappings.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.22.6 Resource ResearchStudy - Mappings

Regulated Clinical Research Information Management Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Practitioner

Mappings:

RIM Mapping

HL7 v2 Mapping

ClinicalTrials.gov Mapping

Mappings for the researchstudy resource.

8.22.6.1 Mappings for RIM Mapping (http://hl7.org/v3)

ResearchStudy | Observation[classCode=CLNTRL, moodCode=EVN]
identifier | .identifier
title
protocol | .outboundRelationship[typeCode=DEFN].target
partOf | .inboundRelationship[typeCode=COMP].source[moodCode=EVN]
status | .status
category
focus
contact
relatedArtifact
keyword
jurisdiction
description
enrollment
period
sponsor
principalInvestigator
site
reasonStopped
note
arm
name
code
description

8.22.6.2 Mappings for HL7 v2 Mapping (http://hl7.org/v2)

ResearchStudy | OBX
identifier

title
protocol
partOf
status
category
focus
contact
relatedArtifact
keyword
jurisdiction
description
enrollment
period
sponsor
principalInvestigator
site
reasonStopped
note
arm
name
code
description

8.22.6.3 Mappings for ClinicalTrials.gov Mapping (http://clinicaltrials.gov)

ResearchStudy
identifier | Organization's Unique Protocol ID; Secondary IDs
title | Brief Title
protocol | Interventions; Arm/Intervention Cross-Reference
partOf
status
category | Study Type; Primary Purpose; Study Phase; Interventional Model; Masking; Allocation; study Classification; Observational Study Model; Time Perspective; Biospecimen Retention
focus | Conditions or Focus of Study
contact | Central Contact; Central Contact Backup; Overall Study Officials; Facility Contact & Backup (if Study tied to only one location); Investigators
relatedArtifact | Citations, Links, Available Study Data/Documents
keyword | Keywords
jurisdiction
description | Detailed Description
enrollment | Enrollment; Number of Groups/Cohorts (= number of "enrollment" instances); Groups (Group/Cohort Label = Group.name; Group/Cohort Description = Group.description); Eligibility (Group criteria)
period | Study Start Date (start); Study Completion Date (end)
sponsor | Sponsor
principalInvestigator | Principal Investigator
site | Facility
reasonStopped | Why Study Stopped
note
arm | Arms
name | Arm Label
code | Arm Type
description | Arm Description

8.22.7 Resource ResearchStudy - Design Notes

URL: https://hl7.org/fhir/STU3/researchstudy-explanations.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.22.7 Resource ResearchStudy - Design Notes

to do

8.22.8 Resource ResearchStudy - Extensions & Profiles

URL: https://hl7.org/fhir/STU3/researchstudy-profiles.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.22.8 Resource ResearchStudy - Extensions & Profiles

Regulated Clinical Research Information Management Work Group | Maturity Level: N/A | Ballot Status: Informative

This table lists profiles and extensions for the ResearchStudy resource. For background information, see Profiling Resources and Extensibility. Additional profiles and resources may be found in published Implementation Guides , or in the Conformance resource registry .

8.22.8.1 Profiles

No Profiles defined for this resource

8.22.8.2 Extensions

No Extensions defined for this resource

Extensions that reference this resource:

No Extensions refer to this resource

8.22.8.3 Search Extensions

No Search Extensions defined for this resource

8.23 Resource ResearchSubject - Content

URL: https://hl7.org/fhir/STU3/researchsubject.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

Regulated Clinical Research Information Management Work Group | Maturity Level: 0 | Draft | Compartments: Patient

A process where a researcher or organization plans and then executes a series of steps intended to increase the field of healthcare-related knowledge. This includes studies of safety, efficacy, comparative effectiveness and other information about medications, devices, therapies and other interventional and investigative techniques. A ResearchStudy involves the gathering of information about human or animal subjects.

8.23.1 Scope and Usage

This resource supports the HL7 mission to create and promote HL7 standards by developing RCRIM standards to improve or enhance information management during clinical research and regulatory evaluation of the safety, efficacy and quality of therapeutic products and procedures worldwide.

This resource is referenced by adverseevent

8.23.2 Resource Content

* Structure

* UML

* XML

* JSON

* Turtle

* R2 Diff

* All

Structure

Name | Flags | Card. | Type | Description & Constraints
ResearchSubject | | | DomainResource | Investigation to increase healthcare-related patient-independent knowledge Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | ? | 0..1 | Identifier | Business Identifier for research subject
status | ?!? | 1..1 | code | candidate | enrolled | active | suspended | withdrawn | completed ResearchSubjectStatus (Required)
period | ? | 0..1 | Period | Start and end of participation
study | ? | 1..1 | Reference(ResearchStudy) | Study subject is part of
individual | ? | 1..1 | Reference(Patient) | Who is part of study
assignedArm | | 0..1 | string | What path should be followed
actualArm | | 0..1 | string | What path was followed
consent | | 0..1 | Reference(Consent) | Agreement to participate in study
Documentation for this format

UML Diagram (Legend)

XML Template

```
<ResearchSubject xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..1 Identifier Business Identifier for research subject --></identifier>
 <status value="[code]"/><!-- 1..1 candidate | enrolled | active | suspended | withdrawn | completed -->
 <period><!-- 0..1 Period Start and end of participation --></period>
 <study><!-- 1..1 Reference(ResearchStudy) Study subject is part of --></study>
 <individual><!-- 1..1 Reference(Patient) Who is part of study --></individual>
 <assignedArm value="[string]"/><!-- 0..1 What path should be followed -->
 <actualArm value="[string]"/><!-- 0..1 What path was followed -->
```

```
 <consent><!-- 0..1 Reference(Consent) Agreement to participate in study --></consent>
</ResearchSubject>
```

JSON Template

```
{
  "resourceType" : "ResearchSubject",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : { Identifier }, // Business Identifier for research subject
  "status" : "<code>", // R!  candidate | enrolled | active | suspended | withdrawn | completed
  "period" : { Period }, // Start and end of participation
  "study" : { Reference(ResearchStudy) }, // R!  Study subject is part of
  "individual" : { Reference(Patient) }, // R!  Who is part of study
  "assignedArm" : "<string>", // What path should be followed
  "actualArm" : "<string>", // What path was followed
  "consent" : { Reference(Consent) } // Agreement to participate in study
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:ResearchSubject;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:ResearchSubject.identifier [ Identifier ]; # 0..1 Business Identifier for research subject
  fhir:ResearchSubject.status [ code ]; # 1..1 candidate | enrolled | active | suspended | withdrawn | completed
  fhir:ResearchSubject.period [ Period ]; # 0..1 Start and end of participation
  fhir:ResearchSubject.study [ Reference(ResearchStudy) ]; # 1..1 Study subject is part of
  fhir:ResearchSubject.individual [ Reference(Patient) ]; # 1..1 Who is part of study
  fhir:ResearchSubject.assignedArm [ string ]; # 0..1 What path should be followed
  fhir:ResearchSubject.actualArm [ string ]; # 0..1 What path was followed
  fhir:ResearchSubject.consent [ Reference(Consent) ]; # 0..1 Agreement to participate in study
]
```

Changes since DSTU2

This resource did not exist in Release 2

This analysis is available as XML or JSON.

Structure

Name | Flags | Card. | Type | Description & Constraints

ResearchSubject | | | DomainResource | Investigation to increase healthcare-related patient-independent knowledge Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension

identifier | ? | 0..1 | Identifier | Business Identifier for research subject

status | ?!? | 1..1 | code | candidate | enrolled | active | suspended | withdrawn | completed ResearchSubjectStatus (Required)

period | ? | 0..1 | Period | Start and end of participation

study | ? | 1..1 | Reference(ResearchStudy) | Study subject is part of

individual | ? | 1..1 | Reference(Patient) | Who is part of study

assignedArm | | 0..1 | string | What path should be followed

actualArm | | 0..1 | string | What path was followed

consent | | 0..1 | Reference(Consent) | Agreement to participate in study

Documentation for this format

UML Diagram (Legend)

XML Template

```
<ResearchSubject xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..1 Identifier Business Identifier for research subject --></identifier>
 <status value="[code]"/><!-- 1..1 candidate | enrolled | active | suspended | withdrawn | completed -->
 <period><!-- 0..1 Period Start and end of participation --></period>
 <study><!-- 1..1 Reference(ResearchStudy) Study subject is part of --></study>
 <individual><!-- 1..1 Reference(Patient) Who is part of study --></individual>
 <assignedArm value="[string]"/><!-- 0..1 What path should be followed -->
 <actualArm value="[string]"/><!-- 0..1 What path was followed -->
 <consent><!-- 0..1 Reference(Consent) Agreement to participate in study --></consent>
</ResearchSubject>
```

JSON Template

```
{
  "resourceType" : "ResearchSubject",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : { Identifier }, // Business Identifier for research subject
  "status" : "<code>", // R!  candidate | enrolled | active | suspended | withdrawn | completed
  "period" : { Period }, // Start and end of participation
  "study" : { Reference(ResearchStudy) }, // R!  Study subject is part of
  "individual" : { Reference(Patient) }, // R!  Who is part of study
  "assignedArm" : "<string>", // What path should be followed
  "actualArm" : "<string>", // What path was followed
  "consent" : { Reference(Consent) } // Agreement to participate in study
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:ResearchSubject;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:ResearchSubject.identifier [ Identifier ]; # 0..1 Business Identifier for research subject
  fhir:ResearchSubject.status [ code ]; # 1..1 candidate | enrolled | active | suspended | withdrawn | completed
  fhir:ResearchSubject.period [ Period ]; # 0..1 Start and end of participation
  fhir:ResearchSubject.study [ Reference(ResearchStudy) ]; # 1..1 Study subject is part of
  fhir:ResearchSubject.individual [ Reference(Patient) ]; # 1..1 Who is part of study
  fhir:ResearchSubject.assignedArm [ string ]; # 0..1 What path should be followed
  fhir:ResearchSubject.actualArm [ string ]; # 0..1 What path was followed
  fhir:ResearchSubject.consent [ Reference(Consent) ]; # 0..1 Agreement to participate in study
]
```

Changes since DSTU2

This resource did not exist in Release 2

This analysis is available as XML or JSON.

Alternate definitions: Master Definition (XML, JSON), XML Schema/Schematron (for) + JSON Schema, ShEx (for Turtle)

8.23.2.1 Terminology Bindings

Path | Definition | Type | Reference
ResearchSubject.status | Indicates the progression of a study subject through a study | Required | ResearchSubjectStatus

8.23.3 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
date | date | Start and end of participation | ResearchSubject.period
identifier | token | Business Identifier for research subject | ResearchSubject.identifier
individual | reference | Who is part of study | ResearchSubject.individual (Patient)
patient | reference | Who is part of study | ResearchSubject.individual (Patient)
status | token | candidate | enrolled | active | suspended | withdrawn | completed | ResearchSubject.status

8.23.4 Resource ResearchSubject - Examples

URL: https://hl7.org/fhir/STU3/researchsubject-examples.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.23.4 Resource ResearchSubject - Examples

Regulated Clinical Research Information Management Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Patient

Example List:

Placeholder example | example | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

8.23.5 Resource ResearchSubject - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/researchsubject-definitions.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.23.5 Resource ResearchSubject - Detailed Descriptions

Regulated Clinical Research Information Management Work Group | Maturity Level: 0 | Draft | Compartments: Patient

Detailed Descriptions for the elements in the ResearchSubject resource.

ResearchSubject
Definition | A process where a researcher or organization plans and then executes a series of steps intended to increase the field of healthcare-related knowledge. This includes studies of safety, efficacy, comparative effectiveness and other information about medications, devices, therapies and other interventional and investigative techniques. A ResearchStudy involves the gathering of information about human or animal subjects.
Control | 1..1
Alternate Names | Study Subject
Comments | Need to make sure we encompass public health studies.
ResearchSubject.identifier
Definition | Identifiers assigned to this research study by the sponsor or other systems.

Note | This is a business identifer, not a resource identifier (see discussion)
Control | 0..1
Type | Identifier
Requirements | Allows identification of the research study as it is known by various participating systems and in a way that remains consistent across servers.
Summary | true
ResearchSubject.status
Definition | The current state of the subject.
Control | 1..1
Terminology Binding | ResearchSubjectStatus (Required)
Type | code
Is Modifier | true
Summary | true
Comments | This element is labeled as a modifier because the status contains codes that mark the resource as not currently valid.
To Do | Should this be a common code system for all events?
ResearchSubject.period
Definition | The dates the subject began and ended their participation in the study.
Control | 0..1
Type | Period
Alternate Names | timing
Summary | true
To Do | .effectiveTime.
ResearchSubject.study
Definition | Reference to the study the subject is participating in.
Control | 1..1
Type | Reference(ResearchStudy)
Summary | true
ResearchSubject.individual
Definition | The record of the person or animal who is involved in the study.
Control | 1..1
Type | Reference(Patient)
Summary | true
ResearchSubject.assignedArm
Definition | The name of the arm in the study the subject is expected to follow as part of this study.
Control | 0..1
Type | string
ResearchSubject.actualArm
Definition | The name of the arm in the study the subject actually followed as part of this study.
Control | 0..1
Type | string
ResearchSubject.consent
Definition | A record of the patient's informed agreement to participate in the study.
Control | 0..1
Type | Reference(Consent)

8.23.6 Resource ResearchSubject - Mappings

URL: https://hl7.org/fhir/STU3/researchsubject-mappings.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.23.6 Resource ResearchSubject - Mappings

Regulated Clinical Research Information Management Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Patient

Mappings:

RIM Mapping

HL7 v2 Mapping

Mappings for the researchsubject resource.

8.23.6.1 Mappings for RIM Mapping (http://hl7.org/v3)

ResearchSubject | Observation[classCode=CLNTRL, moodCode=EVN]
identifier | .identifier
status | .status
period
study
individual
assignedArm
actualArm
consent

8.23.6.2 Mappings for HL7 v2 Mapping (http://hl7.org/v2)

ResearchSubject | OBX
identifier
status
period
study
individual
assignedArm
actualArm
consent

8.23.7 Resource ResearchSubject - Design Notes

URL: https://hl7.org/fhir/STU3/researchsubject-explanations.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.23.7 Resource ResearchSubject - Design Notes

to do

8.23.8 Resource ResearchSubject - Extensions & Profiles

URL: https://hl7.org/fhir/STU3/researchsubject-profiles.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.23.8 Resource ResearchSubject - Extensions & Profiles

Regulated Clinical Research Information Management Work Group | Maturity Level: N/A | Ballot Status: Informative

This table lists profiles and extensions for the ResearchSubject resource. For background information, see Profiling Resources and Extensibility. Additional profiles and resources may be found in published Implementation Guides , or in the Conformance resource registry .

8.23.8.1 Profiles

No Profiles defined for this resource

8.23.8.2 Extensions

No Extensions defined for this resource

Extensions that reference this resource:

No Extensions refer to this resource

8.23.8.3 Search Extensions

No Search Extensions defined for this resource

8.24 Resource ChargeItem - Content

URL: https://hl7.org/fhir/STU3/chargeitem.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

Patient Administration Work Group | Maturity Level: 0 | Draft | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

The resource ChargeItem describes the provision of healthcare provider products for a certain patient, therefore referring not only to the product, but containing in addition details of the provision, like date, time, amounts and participating organizations and persons. Main Usage of the ChargeItem is to enable the billing process and internal cost allocation.

8.24.1 Scope and Usage

Tracking Financial information is vital in Patient Administration and Finance systems in most Healthcare Organizations. The resource ChargeItem describes the charge for provision of healthcare provider products for a certain patient, therefore referring not only to the product, but containing in addition details of the provision, like date, time, amounts and participating organizations and persons. Main Usage of the ChargeItem is to enable the billing process and internal cost allocation. They are created as soon as the products are planned or provisioned, references to Encounters and/or Accounts can be maintained in a later process step.

The target of ChargeItem.definition may provide information on the Charge code such as pricing and inclusion/exclusion rules as well as factors that apply under certain conditions. In many cases however this information may been drawn from sources outside of FHIR depending on the distribution format of the code catalogue. The ChargeItem assumes that such information is either implicitly known by the communicating systems or explicitly shared through the ChargeItem.definition. Therefore

explicit pricing information is not shared within the ChargeItem resource. Also, the systems posting the ChargeItems are not expected to apply the rules associated with the charge codes as they may not know the whole context of the patient/encounter to evaluate such rules. It lies within the responsibity of a billing engine, to collect the ChargeItems in the context of an Account or Encounter at a certain point in time (e.g. discharge of the patient) and to evaluate the associated rules resulting in some of the ChargeItems to be set to the status "not billable" in case the rules exclude them from being billed, or to create financial transactions according to base price and factors. Additional references to Encounter/EpisodeOfCare, Patient/Group and Services provide further context to help billing systems determine the appropriate account and establish the clinical/financial context to evaluate the rules associated with the charge codes.

8.24.2 Boundaries and Relationships

This resource is not an actual financial transaction (such as an item on an invoice or any concise monetary amount being transferred from one Account to another), but is the base administrative data that may be used by a billing engine to create the financial transactions based on rules, factors and base prices associated with the charge code.

Unlike the Financial Transaction the ChargeItem primarily describes the provision, whereas the Financial Transaction documents cash flow. Therefore, the Financial Transaction results from ChargeItems created via the subsequent billing- or cost allocation process.

The actual financial transaction resulting from the evaluation of these rules against the clinical and financial context may be represented in formats appropriate to the financial realm. These are considered out of scope for the FHIR Standard, as they are not specific to the healthcare domain. The FHIR Claim resource does contain line items, and this ChargeItem resource provides the source material for the billing engine to create the items on the claim (which may be different due to business rules).

8.24.3 Resource Content

* Structure

* UML

* XML

* JSON

* Turtle

* R2 Diff

* All

Structure

Name | Flags | Card. | Type | Description & Constraints
ChargeItem | | | DomainResource | Item containing charge code(s) associated with the provision of healthcare provider products Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | ? | 0..1 | Identifier | Business Identifier for item
definition | | 0..* | uri | Defining information about the code of this charge item
status | ?!? | 1..1 | code | planned | billable | not-billable | aborted | billed | entered-in-error | unknown ChargeItemStatus (Required)
partOf | | 0..* | Reference(ChargeItem) | Part of referenced ChargeItem
code | ? | 1..1 | CodeableConcept | A code that identifies the charge, like a billing code ChargeItemCode (Example)
subject | ? | 1..1 | Reference(Patient | Group) | Individual service was done for/to
context | ? | 0..1 | Reference(Encounter | EpisodeOfCare) | Encounter / Episode associated with event
occurrence[x] | ? | 0..1 | | When the charged service was applied
occurrenceDateTime | | | dateTime
occurrencePeriod | | | Period
occurrenceTiming | | | Timing
participant | | 0..* | BackboneElement | Who performed charged service
role | | 0..1 | CodeableConcept | What type of performance was done Procedure Performer Role Codes (Example)
actor | | 1..1 | Reference(Practitioner | Organization | Patient | Device | RelatedPerson) | Individual who was performing
performingOrganization | | 0..1 | Reference(Organization) | Organization providing the charged sevice
requestingOrganization | | 0..1 | Reference(Organization) | Organization requesting the charged service
quantity | ? | 0..1 | Quantity | Quantity of which the charge item has been serviced
bodysite | ? | 0..* | CodeableConcept | Anatomical location, if relevant SNOMED CT Body Structures (Example)
factorOverride | | 0..1 | decimal | Factor overriding the associated rules
priceOverride | | 0..1 | Money | Price overriding the associated rules
overrideReason | | 0..1 | string | Reason for overriding the list price/factor
enterer | ? | 0..1 | Reference(Practitioner | Organization | Patient | Device | RelatedPerson) | Individual who was entering
enteredDate | ? | 0..1 | dateTime | Date the charge item was entered
reason | | 0..* | CodeableConcept | Why was the charged service rendered? ICD-10 Codes (Example)
service | | 0..* | Reference(DiagnosticReport | ImagingStudy | Immunization | MedicationAdministration | MedicationDispense | Observation | Procedure | SupplyDelivery) | Which rendered service is being charged?
account | ? | 0..* | Reference(Account) | Account to place this charge
note | | 0..* | Annotation | Comments made about the ChargeItem
supportingInformation | | 0..* | Reference(Any) | Further information supporting the this charge
Documentation for this format

UML Diagram (Legend)

XML Template

```
<ChargeItem xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..1 Identifier Business Identifier for item --></identifier>
 <definition value="[uri]"/><!-- 0..* Defining information about the code of this charge item -->
 <status value="[code]"/><!-- 1..1 planned | billable | not-billable | aborted | billed | entered-in-error | unknown -->
 <partOf><!-- 0..* Reference(ChargeItem) Part of referenced ChargeItem --></partOf>
 <code><!-- 1..1 CodeableConcept A code that identifies the charge, like a billing code --></code>
 <subject><!-- 1..1 Reference(Patient|Group) Individual service was done for/to --></subject>
 <context><!-- 0..1 Reference(Encounter|EpisodeOfCare) Encounter / Episode associated with event --></context>
 <occurrence[x]><!-- 0..1 dateTime|Period|Timing When the charged service was applied --></occurrence[x]>
 <participant>  <!-- 0..* Who performed charged service -->
  <role><!-- 0..1 CodeableConcept What type of performance was done --></role>
  <actor><!-- 1..1 Reference(Practitioner|Organization|Patient|Device|
    RelatedPerson) Individual who was performing --></actor>
 </participant>
 <performingOrganization><!-- 0..1 Reference(Organization) Organization providing the charged sevice
    --></performingOrganization>
 <requestingOrganization><!-- 0..1 Reference(Organization) Organization requesting the charged service
    --></requestingOrganization>
 <quantity><!-- 0..1 Quantity Quantity of which the charge item has been serviced --></quantity>
 <bodysite><!-- 0..* CodeableConcept Anatomical location, if relevant --></bodysite>
 <factorOverride value="[decimal]"/><!-- 0..1 Factor overriding the associated rules -->
 <priceOverride><!-- 0..1 Money Price overriding the associated rules --></priceOverride>
 <overrideReason value="[string]"/><!-- 0..1 Reason for overriding the list price/factor -->
 <enterer><!-- 0..1 Reference(Practitioner|Organization|Patient|Device|
   RelatedPerson) Individual who was entering --></enterer>
 <enteredDate value="[dateTime]"/><!-- 0..1 Date the charge item was entered -->
 <reason><!-- 0..* CodeableConcept Why was the charged  service rendered? --></reason>
 <service><!-- 0..* Reference(DiagnosticReport|ImagingStudy|Immunization|
   MedicationAdministration|MedicationDispense|Observation|Procedure|
   SupplyDelivery) Which rendered service is being charged? --></service>
 <account><!-- 0..* Reference(Account) Account to place this charge --></account>
 <note><!-- 0..* Annotation Comments made about the ChargeItem --></note>
 <supportingInformation><!-- 0..* Reference(Any) Further information supporting the this charge
    --></supportingInformation>
</ChargeItem>
```

JSON Template

```
{
  "resourceType" : "ChargeItem",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : { Identifier }, // Business Identifier for item
  "definition" : ["<uri>"], // Defining information about the code of this charge item
  "status" : "<code>", // R!  planned | billable | not-billable | aborted | billed | entered-in-error | unknown
  "partOf" : [{ Reference(ChargeItem) }], // Part of referenced ChargeItem
  "code" : { CodeableConcept }, // R!  A code that identifies the charge, like a billing code
  "subject" : { Reference(Patient|Group) }, // R!  Individual service was done for/to
  "context" : { Reference(Encounter|EpisodeOfCare) }, // Encounter / Episode associated with event
  // occurrence[x]: When the charged service was applied. One of these 3:
  "occurrenceDateTime" : "<dateTime>",
  "occurrencePeriod" : { Period },
  "occurrenceTiming" : { Timing },
  "participant" : [{ // Who performed charged service
    "role" : { CodeableConcept }, // What type of performance was done
    "actor" : { Reference(Practitioner|Organization|Patient|Device|
    RelatedPerson) } // R!  Individual who was performing
  }],
  "performingOrganization" : { Reference(Organization) }, // Organization providing the charged sevice
  "requestingOrganization" : { Reference(Organization) }, // Organization requesting the charged service
  "quantity" : { Quantity }, // Quantity of which the charge item has been serviced
  "bodysite" : [{ CodeableConcept }], // Anatomical location, if relevant
  "factorOverride" : <decimal>, // Factor overriding the associated rules
  "priceOverride" : { Money }, // Price overriding the associated rules
  "overrideReason" : "<string>", // Reason for overriding the list price/factor
  "enterer" : { Reference(Practitioner|Organization|Patient|Device|
   RelatedPerson) }, // Individual who was entering
  "enteredDate" : "<dateTime>", // Date the charge item was entered
  "reason" : [{ CodeableConcept }], // Why was the charged  service rendered?
  "service" : [{ Reference(DiagnosticReport|ImagingStudy|Immunization|
   MedicationAdministration|MedicationDispense|Observation|Procedure|
   SupplyDelivery) }], // Which rendered service is being charged?
  "account" : [{ Reference(Account) }], // Account to place this charge
  "note" : [{ Annotation }], // Comments made about the ChargeItem
  "supportingInformation" : [{ Reference(Any) }] // Further information supporting the this charge
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:ChargeItem;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
```

```
  fhir:ChargeItem.identifier [ Identifier ]; # 0..1 Business Identifier for item
  fhir:ChargeItem.definition [ uri ], ... ; # 0..* Defining information about the code of this charge item
  fhir:ChargeItem.status [ code ]; # 1..1 planned | billable | not-billable | aborted | billed | entered-in-error |
    unknown
  fhir:ChargeItem.partOf [ Reference(ChargeItem) ], ... ; # 0..* Part of referenced ChargeItem
  fhir:ChargeItem.code [ CodeableConcept ]; # 1..1 A code that identifies the charge, like a billing code
  fhir:ChargeItem.subject [ Reference(Patient|Group) ]; # 1..1 Individual service was done for/to
  fhir:ChargeItem.context [ Reference(Encounter|EpisodeOfCare) ]; # 0..1 Encounter / Episode associated with event
  # ChargeItem.occurrence[x] : 0..1 When the charged service was applied. One of these 3
    fhir:ChargeItem.occurrenceDateTime [ dateTime ]
    fhir:ChargeItem.occurrencePeriod [ Period ]
    fhir:ChargeItem.occurrenceTiming [ Timing ]
  fhir:ChargeItem.participant [ # 0..* Who performed charged service
    fhir:ChargeItem.participant.role [ CodeableConcept ]; # 0..1 What type of performance was done
    fhir:ChargeItem.participant.actor [ Reference(Practitioner|Organization|Patient|Device|RelatedPerson) ]; # 1..1
    Individual who was performing
  ], ...;
  fhir:ChargeItem.performingOrganization [ Reference(Organization) ]; # 0..1 Organization providing the charged sevice
  fhir:ChargeItem.requestingOrganization [ Reference(Organization) ]; # 0..1 Organization requesting the charged service
  fhir:ChargeItem.quantity [ Quantity ]; # 0..1 Quantity of which the charge item has been serviced
  fhir:ChargeItem.bodysite [ CodeableConcept ], ... ; # 0..* Anatomical location, if relevant
  fhir:ChargeItem.factorOverride [ decimal ]; # 0..1 Factor overriding the associated rules
  fhir:ChargeItem.priceOverride [ Money ]; # 0..1 Price overriding the associated rules
  fhir:ChargeItem.overrideReason [ string ]; # 0..1 Reason for overriding the list price/factor
  fhir:ChargeItem.enterer [ Reference(Practitioner|Organization|Patient|Device|RelatedPerson) ]; # 0..1 Individual who
    was entering
  fhir:ChargeItem.enteredDate [ dateTime ]; # 0..1 Date the charge item was entered
  fhir:ChargeItem.reason [ CodeableConcept ], ... ; # 0..* Why was the charged  service rendered?
  fhir:ChargeItem.service [ Reference(DiagnosticReport|ImagingStudy|Immunization|MedicationAdministration|
  MedicationDispense|Observation|Procedure|SupplyDelivery) ], ... ; # 0..* Which rendered service is being charged?
  fhir:ChargeItem.account [ Reference(Account) ], ... ; # 0..* Account to place this charge
  fhir:ChargeItem.note [ Annotation ], ... ; # 0..* Comments made about the ChargeItem
  fhir:ChargeItem.supportingInformation [ Reference(Any) ], ... ; # 0..* Further information supporting the this charge
]
```

Changes since DSTU2

This resource did not exist in Release 2

This analysis is available as XML or JSON.

Structure

Name | Flags | Card. | Type | Description & Constraints

ChargeItem | | | DomainResource | Item containing charge code(s) associated with the provision of healthcare provider products Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension

identifier | ? | 0..1 | Identifier | Business Identifier for item

definition | | 0..* | uri | Defining information about the code of this charge item

status | ?!? | 1..1 | code | planned | billable | not-billable | aborted | billed | entered-in-error | unknown ChargeItemStatus (Required)

partOf | | 0..* | Reference(ChargeItem) | Part of referenced ChargeItem

code | ? | 1..1 | CodeableConcept | A code that identifies the charge, like a billing code ChargeItemCode (Example)

subject | ? | 1..1 | Reference(Patient | Group) | Individual service was done for/to

context | ? | 0..1 | Reference(Encounter | EpisodeOfCare) | Encounter / Episode associated with event

occurrence[x] | ? | 0..1 | | When the charged service was applied

occurrenceDateTime | | | dateTime

occurrencePeriod | | | Period

occurrenceTiming | | | Timing

participant | | 0..* | BackboneElement | Who performed charged service

role | | 0..1 | CodeableConcept | What type of performance was done Procedure Performer Role Codes (Example)

actor | | 1..1 | Reference(Practitioner | Organization | Patient | Device | RelatedPerson) | Individual who was performing

performingOrganization | | 0..1 | Reference(Organization) | Organization providing the charged sevice

requestingOrganization | | 0..1 | Reference(Organization) | Organization requesting the charged service

quantity | ? | 0..1 | Quantity | Quantity of which the charge item has been serviced

bodysite | ? | 0..* | CodeableConcept | Anatomical location, if relevant SNOMED CT Body Structures (Example)

factorOverride | | 0..1 | decimal | Factor overriding the associated rules

priceOverride | | 0..1 | Money | Price overriding the associated rules

overrideReason | | 0..1 | string | Reason for overriding the list price/factor

enterer | ? | 0..1 | Reference(Practitioner | Organization | Patient | Device | RelatedPerson) | Individual who was entering

enteredDate | ? | 0..1 | dateTime | Date the charge item was entered

reason | | 0..* | CodeableConcept | Why was the charged service rendered? ICD-10 Codes (Example)

service | | 0..* | Reference(DiagnosticReport | ImagingStudy | Immunization | MedicationAdministration | MedicationDispense | Observation | Procedure | SupplyDelivery) | Which rendered service is being charged?

account | ? | 0..* | Reference(Account) | Account to place this charge

note | | 0..* | Annotation | Comments made about the ChargeItem

supportingInformation | | 0..* | Reference(Any) | Further information supporting the this charge

Documentation for this format

UML Diagram (Legend)

XML Template

```
<ChargeItem xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..1 Identifier Business Identifier for item --></identifier>
```

```
 <definition value="[uri]"/><!-- 0..* Defining information about the code of this charge item -->
 <status value="[code]"/><!-- 1..1 planned | billable | not-billable | aborted | billed | entered-in-error | unknown -->
 <partOf><!-- 0..* Reference(ChargeItem) Part of referenced ChargeItem --></partOf>
 <code><!-- 1..1 CodeableConcept A code that identifies the charge, like a billing code --></code>
 <subject><!-- 1..1 Reference(Patient|Group) Individual service was done for/to --></subject>
 <context><!-- 0..1 Reference(Encounter|EpisodeOfCare) Encounter / Episode associated with event --></context>
 <occurrence[x]><!-- 0..1 dateTime|Period|Timing When the charged service was applied --></occurrence[x]>
 <participant>  <!-- 0..* Who performed charged service -->
  <role><!-- 0..1 CodeableConcept What type of performance was done --></role>
  <actor><!-- 1..1 Reference(Practitioner|Organization|Patient|Device|
    RelatedPerson) Individual who was performing --></actor>
 </participant>
 <performingOrganization><!-- 0..1 Reference(Organization) Organization providing the charged sevice
    --></performingOrganization>
 <requestingOrganization><!-- 0..1 Reference(Organization) Organization requesting the charged service
    --></requestingOrganization>
 <quantity><!-- 0..1 Quantity Quantity of which the charge item has been serviced --></quantity>
 <bodysite><!-- 0..* CodeableConcept Anatomical location, if relevant --></bodysite>
 <factorOverride value="[decimal]"/><!-- 0..1 Factor overriding the associated rules -->
 <priceOverride><!-- 0..1 Money Price overriding the associated rules --></priceOverride>
 <overrideReason value="[string]"/><!-- 0..1 Reason for overriding the list price/factor -->
 <enterer><!-- 0..1 Reference(Practitioner|Organization|Patient|Device|
   RelatedPerson) Individual who was entering --></enterer>
 <enteredDate value="[dateTime]"/><!-- 0..1 Date the charge item was entered -->
 <reason><!-- 0..* CodeableConcept Why was the charged  service rendered? --></reason>
 <service><!-- 0..* Reference(DiagnosticReport|ImagingStudy|Immunization|
   MedicationAdministration|MedicationDispense|Observation|Procedure|
   SupplyDelivery) Which rendered service is being charged? --></service>
 <account><!-- 0..* Reference(Account) Account to place this charge --></account>
 <note><!-- 0..* Annotation Comments made about the ChargeItem --></note>
 <supportingInformation><!-- 0..* Reference(Any) Further information supporting the this charge
    --></supportingInformation>
</ChargeItem>
```

JSON Template

```
{
  "resourceType" : "ChargeItem",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : { Identifier }, // Business Identifier for item
  "definition" : ["<uri>"], // Defining information about the code of this charge item
  "status" : "<code>", // R!  planned | billable | not-billable | aborted | billed | entered-in-error | unknown
  "partOf" : [{ Reference(ChargeItem) }], // Part of referenced ChargeItem
  "code" : { CodeableConcept }, // R!  A code that identifies the charge, like a billing code
  "subject" : { Reference(Patient|Group) }, // R!  Individual service was done for/to
  "context" : { Reference(Encounter|EpisodeOfCare) }, // Encounter / Episode associated with event
  // occurrence[x]: When the charged service was applied. One of these 3:
  "occurrenceDateTime" : "<dateTime>",
  "occurrencePeriod" : { Period },
  "occurrenceTiming" : { Timing },
  "participant" : [{ // Who performed charged service
    "role" : { CodeableConcept }, // What type of performance was done
    "actor" : { Reference(Practitioner|Organization|Patient|Device|
    RelatedPerson) } // R!  Individual who was performing
  }],
  "performingOrganization" : { Reference(Organization) }, // Organization providing the charged sevice
  "requestingOrganization" : { Reference(Organization) }, // Organization requesting the charged service
  "quantity" : { Quantity }, // Quantity of which the charge item has been serviced
  "bodysite" : [{ CodeableConcept }], // Anatomical location, if relevant
  "factorOverride" : <decimal>, // Factor overriding the associated rules
  "priceOverride" : { Money }, // Price overriding the associated rules
  "overrideReason" : "<string>", // Reason for overriding the list price/factor
  "enterer" : { Reference(Practitioner|Organization|Patient|Device|
   RelatedPerson) }, // Individual who was entering
  "enteredDate" : "<dateTime>", // Date the charge item was entered
  "reason" : [{ CodeableConcept }], // Why was the charged  service rendered?
  "service" : [{ Reference(DiagnosticReport|ImagingStudy|Immunization|
   MedicationAdministration|MedicationDispense|Observation|Procedure|
   SupplyDelivery) }], // Which rendered service is being charged?
  "account" : [{ Reference(Account) }], // Account to place this charge
  "note" : [{ Annotation }], // Comments made about the ChargeItem
  "supportingInformation" : [{ Reference(Any) }] // Further information supporting the this charge
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:ChargeItem;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:ChargeItem.identifier [ Identifier ]; # 0..1 Business Identifier for item
  fhir:ChargeItem.definition [ uri ], ... ; # 0..* Defining information about the code of this charge item
  fhir:ChargeItem.status [ code ]; # 1..1 planned | billable | not-billable | aborted | billed | entered-in-error |
    unknown
  fhir:ChargeItem.partOf [ Reference(ChargeItem) ], ... ; # 0..* Part of referenced ChargeItem
  fhir:ChargeItem.code [ CodeableConcept ]; # 1..1 A code that identifies the charge, like a billing code
```

```
  fhir:ChargeItem.subject [ Reference(Patient|Group) ]; # 1..1 Individual service was done for/to
  fhir:ChargeItem.context [ Reference(Encounter|EpisodeOfCare) ]; # 0..1 Encounter / Episode associated with event
  # ChargeItem.occurrence[x] : 0..1 When the charged service was applied. One of these 3
    fhir:ChargeItem.occurrenceDateTime [ dateTime ]
    fhir:ChargeItem.occurrencePeriod [ Period ]
    fhir:ChargeItem.occurrenceTiming [ Timing ]
  fhir:ChargeItem.participant [ # 0..* Who performed charged service
    fhir:ChargeItem.participant.role [ CodeableConcept ]; # 0..1 What type of performance was done
    fhir:ChargeItem.participant.actor [ Reference(Practitioner|Organization|Patient|Device|RelatedPerson) ]; # 1..1
    Individual who was performing
  ], ...;
  fhir:ChargeItem.performingOrganization [ Reference(Organization) ]; # 0..1 Organization providing the charged sevice
  fhir:ChargeItem.requestingOrganization [ Reference(Organization) ]; # 0..1 Organization requesting the charged service
  fhir:ChargeItem.quantity [ Quantity ]; # 0..1 Quantity of which the charge item has been serviced
  fhir:ChargeItem.bodysite [ CodeableConcept ], ... ; # 0..* Anatomical location, if relevant
  fhir:ChargeItem.factorOverride [ decimal ]; # 0..1 Factor overriding the associated rules
  fhir:ChargeItem.priceOverride [ Money ]; # 0..1 Price overriding the associated rules
  fhir:ChargeItem.overrideReason [ string ]; # 0..1 Reason for overriding the list price/factor
  fhir:ChargeItem.enterer [ Reference(Practitioner|Organization|Patient|Device|RelatedPerson) ]; # 0..1 Individual who
    was entering
  fhir:ChargeItem.enteredDate [ dateTime ]; # 0..1 Date the charge item was entered
  fhir:ChargeItem.reason [ CodeableConcept ], ... ; # 0..* Why was the charged  service rendered?
  fhir:ChargeItem.service [ Reference(DiagnosticReport|ImagingStudy|Immunization|MedicationAdministration|
  MedicationDispense|Observation|Procedure|SupplyDelivery) ], ... ; # 0..* Which rendered service is being charged?
  fhir:ChargeItem.account [ Reference(Account) ], ... ; # 0..* Account to place this charge
  fhir:ChargeItem.note [ Annotation ], ... ; # 0..* Comments made about the ChargeItem
  fhir:ChargeItem.supportingInformation [ Reference(Any) ], ... ; # 0..* Further information supporting the this charge
]
```

Changes since DSTU2

This resource did not exist in Release 2

This analysis is available as XML or JSON.

Alternate definitions: Master Definition (XML, JSON), XML Schema/Schematron (for) + JSON Schema, ShEx (for Turtle)

8.24.3.1 Terminology Bindings

Path | Definition | Type | Reference
ChargeItem.status | Codes identifying the stage lifecycle stage of a ChargeItem | Required | ChargeItemStatus
ChargeItem.code | Example set of codes that can be used for billing purposes | Example | ChargeItemCode
ChargeItem.participant.role | Codes describing the types of functional roles performers can take on when performing events | Example | Procedure Performer Role Codes
ChargeItem.bodysite | Codes describing anatomical locations. May include laterality. | Example | SNOMED CT Body Structures
ChargeItem.reason | Example binding for reason | Example | ICD-10 Codes

8.24.4 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
account | reference | Account to place this charge | ChargeItem.account (Account)
code | token | A code that identifies the charge, like a billing code | ChargeItem.code
context | reference | Encounter / Episode associated with event | ChargeItem.context (EpisodeOfCare, Encounter)
entered-date | date | Date the charge item was entered | ChargeItem.enteredDate
enterer | reference | Individual who was entering | ChargeItem.enterer (Practitioner, Organization, Device, Patient, RelatedPerson)
factor-override | number | Factor overriding the associated rules | ChargeItem.factorOverride
identifier | token | Business Identifier for item | ChargeItem.identifier
occurrence | date | When the charged service was applied | ChargeItem.occurrence
participant-actor | reference | Individual who was performing | ChargeItem.participant.actor (Practitioner, Organization, Device, Patient, RelatedPerson)
participant-role | token | What type of performance was done | ChargeItem.participant.role
patient | reference | Individual service was done for/to | ChargeItem.subject (Patient)
performing-organization | reference | Organization providing the charged sevice | ChargeItem.performingOrganization (Organization)
price-override | quantity | Price overriding the associated rules | ChargeItem.priceOverride
quantity | quantity | Quantity of which the charge item has been serviced | ChargeItem.quantity
requesting-organization | reference | Organization requesting the charged service | ChargeItem.requestingOrganization (Organization)
service | reference | Which rendered service is being charged? | ChargeItem.service (Immunization, MedicationDispense, SupplyDelivery, Observation, DiagnosticReport, ImagingStudy, MedicationAdministration, Procedure)
subject | reference | Individual service was done for/to | ChargeItem.subject (Group, Patient)

8.24.5 Resource ChargeItem - Examples

URL: https://hl7.org/fhir/STU3/chargeitem-examples.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.24.5 Resource ChargeItem - Examples

Patient Administration Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

Example List:

Example of chargeitem | example | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

8.24.6 Resource ChargeItem - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/chargeitem-definitions.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.24.6 Resource ChargeItem - Detailed Descriptions

Patient Administration Work Group | Maturity Level: 0 | Draft | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

Detailed Descriptions for the elements in the ChargeItem resource.

ChargeItem
Definition | The resource ChargeItem describes the provision of healthcare provider products for a certain patient, therefore referring not only to the product, but containing in addition details of the provision, like date, time, amounts and participating organizations and persons. Main Usage of the ChargeItem is to enable the billing process and internal cost allocation.
Control | 1..1
ChargeItem.identifier
Definition | Identifiers assigned to this event performer or other systems.
Note | This is a business identifer, not a resource identifier (see discussion)
Control | 0..1
Type | Identifier
Requirements | Allows identification of the charge Item as it is known by various participating systems and in a way that remains consistent across servers.
Summary | true
ChargeItem.definition
Definition | References the source of pricing information, rules of application for the code this ChargeItem uses.
Control | 0..*
Type | uri
Summary | false
Comments | Target Resource is yet to be defined as a container that holds information about list prices and rules applicable to the charge item code. Once defined, change type back to Reference(Definition Type).
ChargeItem.status
Definition | The current state of the ChargeItem.
Control | 1..1
Terminology Binding | ChargeItemStatus (Required)
Type | code
Is Modifier | true
Summary | true
Comments | Unknown does not represent "other" - one of the defined statuses must apply. Unknown is used when the authoring system is not sure what the current status is. This element is labeled as a modifier because the status contains the code entered-in-error that marks the charge item as not currently valid.
ChargeItem.partOf
Definition | ChargeItems can be grouped to larger ChargeItems covering the whole set.
Control | 0..*
Type | Reference(ChargeItem)
Requirements | E.g. Drug administration as part of a procedure, procedure as part of observation, etc.
Alternate Names | container
Summary | false
ChargeItem.code
Definition | A code that identifies the charge, like a billing code.
Control | 1..1
Terminology Binding | ChargeItemCode (Example)
Type | CodeableConcept
Alternate Names | type
Summary | true
ChargeItem.subject
Definition | The individual or set of individuals the action is being or was performed on.
Control | 1..1
Type | Reference(Patient | Group)
Requirements | Links the event to the Patient context.
Alternate Names | patient
Summary | true
ChargeItem.context
Definition | The encounter or episode of care that establishes the context for this event.
Control | 0..1

Type | Reference(Encounter | EpisodeOfCare)
Requirements | Links the request to the Encounter context.
Alternate Names | encounter
Summary | true
ChargeItem.occurrence[x]
Definition | Date/time(s) or duration when the charged service was applied.
Control | 0..1
Type | dateTime|Period|Timing
[x] Note | See Choice of Data Types for further information about how to use [x]
Alternate Names | timing
Summary | true
Comments | The list of types may be constrained as appropriate for the type of charge item.
ChargeItem.participant
Definition | Indicates who or what performed or participated in the charged service.
Control | 0..*
Summary | false
ChargeItem.participant.role
Definition | Describes the type of performance or participation(e.g. primary surgeon, anaesthesiologiest, etc.).
Control | 0..1
Terminology Binding | Procedure Performer Role Codes (Example)
Type | CodeableConcept
Summary | false
ChargeItem.participant.actor
Definition | The device, practitioner, etc. who performed or participated in the service.
Control | 1..1
Type | Reference(Practitioner | Organization | Patient | Device | RelatedPerson)
Summary | false
ChargeItem.performingOrganization
Definition | The organization requesting the service.
Control | 0..1
Type | Reference(Organization)
Summary | false
Comments | Practitioners and Devices can be associated with multiple organizations. It has to be made clear, on behalf of which Organization the services have been rendered.
ChargeItem.requestingOrganization
Definition | The organization performing the service.
Control | 0..1
Type | Reference(Organization)
Summary | false
Comments | The rendered Service may not be associated with a Request. It has to be made clear, ob which Organization's Request, the services have been rendered. (In many cases, this may just be the Department associated with the Encounter.location).
ChargeItem.quantity
Definition | Quantity of which the charge item has been serviced.
Control | 0..1
Type | Quantity
Summary | true
Comments | In many cases this may just be a value, if the underlying units are implicit in the definition of the charge item code.
ChargeItem.bodysite
Definition | The anatomical location where the related service has been applied.
Control | 0..*
Terminology Binding | SNOMED CT Body Structures (Example)
Type | CodeableConcept
Summary | true
Comments | Only used if not implicit in code found in Condition.code. If the use case requires attributes from the BodySite resource (e.g. to identify and track separately) then use the standard extension body-site-instance. May be a summary code, or a reference to a very precise definition of the location, or both.
ChargeItem.factorOverride
Definition | Factor overriding the factor determined by the rules associated with the code.
Control | 0..1
Type | decimal
Summary | false
Comments | There is no reason to carry the factor in the instance of a ChargeItem unless special circumstances require a manual override. The factors are usually defined by a set of rules in a back catalogue of the billing codes (see ChargeItem.definition). Derived profiles may require a ChargeItem.overrideReason to be provided if either factor or price are manually overriden.
ChargeItem.priceOverride
Definition | Total price of the charge overriding the list price associated with the code.
Control | 0..1
Type | Money
Summary | false
Comments | There is no reason to carry the price in the instance of a ChargeItem unless circumstances require a manual override. The list prices or are usually defined in a back catalogue of the billing codes (see ChargeItem.definition). Derived profiles may require a ChargeItem.overrideReason to be provided if either factor or price are manually overriden.
ChargeItem.overrideReason

Definition | If the list price or the rule based factor associated with the code is overridden, this attribute can capture a text to indicate the reason for this action.
Control | 0..1
Type | string
Summary | false
Comments | Derived Profiles may chose to add invariants requiring this field to be populated if either priceOverride or factorOverride have been filled.
ChargeItem.enterer
Definition | The device, practitioner, etc. who entered the charge item.
Control | 0..1
Type | Reference(Practitioner | Organization | Patient | Device | RelatedPerson)
Summary | true
Comments | The enterer is also te person considered responsible for factor/priceOverrides if applicable.
ChargeItem.enteredDate
Definition | Date the charge item was entered.
Control | 0..1
Type | dateTime
Summary | true
Comments | The actual date when the service associated with the charge has been rendered is captured in occurrence[x].
ChargeItem.reason
Definition | Describes why the event occurred in coded or textual form.
Control | 0..*
Terminology Binding | ICD-10 Codes (Example)
Type | CodeableConcept
Summary | false
Comments | If the application of the charge item requires a reason to be given, it can be captured here. Textual reasons can be captured using reasonCode.text.
ChargeItem.service
Definition | Indicated the rendered service that caused this charge.
Control | 0..*
Type | Reference(DiagnosticReport | ImagingStudy | Immunization | MedicationAdministration | MedicationDispense | Observation | Procedure | SupplyDelivery)
Summary | false
ChargeItem.account
Definition | Account into which this ChargeItems belongs.
Control | 0..*
Type | Reference(Account)
Summary | true
Comments | Systems posting the ChargeItems may not always be able to determine, which accounts the Items need to be places into. It is up to the potprocessing Financial System to apply internal rules to decide based on the Encounter/EpisodeOfCare/Patient/Coverage context and the type of ChargeItem, which Account is appropriate.
ChargeItem.note
Definition | Comments made about the event by the performer, subject or other participants.
Control | 0..*
Type | Annotation
Summary | false
ChargeItem.supportingInformation
Definition | Further information supporting the this charge.
Control | 0..*
Type | Reference(Any)
Summary | false

8.24.7 Resource ChargeItem - Mappings

URL: https://hl7.org/fhir/STU3/chargeitem-mappings.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.24.7 Resource ChargeItem - Mappings

Patient Administration Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

Mappings:

RIM Mapping

HL7 v2 Mapping

Workflow Mapping

Mappings for the chargeitem resource.

8.24.7.1 Mappings for RIM Mapping (http://hl7.org/v3)

ChargeItem | Act[moodCode=EVN]
identifier | .identifier
definition | .outboundRelationship[typeCode=DEFN].target

status | .status
partOf | .inboundRelationship[typeCode=COMP].source[moodCode=EVN]
code | .code
subject | .participation[typeCode=SBJ].role
context | .inboundRelationship(typeCode=COMP].source[classCode<=PCPR, moodCode=EVN]
occurrence[x] | .effectiveTime
participant | .participation[typeCode=PRF].role[scoper.determinerCode=INSTANCE]
role | .participation.functionCode
actor | .player
performingOrganization | .scoper
requestingOrganization | .scoper
quantity
bodysite | .targetBodySiteCode
factorOverride
priceOverride
overrideReason
enterer | .player
enteredDate
reason | .reasonCode
service | .outboundRelationship[typeCode=RSON].target
account
note | .inboundRelationship(typeCode=SUBJ].source[classCode=ANNGEN, moodCode=EVN].value[xsi:type=ST]
supportingInformation

8.24.7.2 Mappings for HL7 v2 Mapping (http://hl7.org/v2)

ChargeItem | FT1
identifier | FT1.2
definition | Varies by domain
status | Varies by domain
partOf | Varies by domain
code | FT1.7
subject | PAT in proximity to FT1 segment
context | PV1 in proximity to FT1 segment
occurrence[x] | Varies by domain
participant | EVN.5 or by domain
role | Varies by domain
actor | EVN.5 or by domain
performingOrganization | N/A
requestingOrganization | N/A
quantity | FT1.10
bodysite
factorOverride | FT1.26
priceOverride | FT1.22
overrideReason
enterer | EVN.5 or by domain
enteredDate
reason | EVN.4 or by domain
service | EVN.4 or by domain
account
note | NTE
supportingInformation

8.24.7.3 Mappings for Workflow Mapping (http://hl7.org/fhir/workflow)

ChargeItem
identifier
definition
status
partOf
code | Event.code
subject | Event.subject
context | Event.context
occurrence[x]
participant
role
actor
performingOrganization
requestingOrganization
quantity
bodysite
factorOverride
priceOverride

overrideReason
enterer
enteredDate
reason | Event.reasonCodeableConcept
service | Event.reasonReference
account
note | Event.note
supportingInformation

8.24.8 Resource ChargeItem - Design Notes

URL: https://hl7.org/fhir/STU3/chargeitem-explanations.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.24.8 Resource ChargeItem - Design Notes

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8.24.9 Resource ChargeItem - Extensions & Profiles

URL: https://hl7.org/fhir/STU3/chargeitem-profiles.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

8.24.9 Resource ChargeItem - Extensions & Profiles

Patient Administration Work Group | Maturity Level: N/A | Ballot Status: Informative

This table lists profiles and extensions for the ChargeItem resource. For background information, see Profiling Resources and Extensibility. Additional profiles and resources may be found in published Implementation Guides , or in the Conformance resource registry .

8.24.9.1 Profiles

No Profiles defined for this resource

8.24.9.2 Extensions

No Extensions defined for this resource

Extensions that reference this resource:

No Extensions refer to this resource

8.24.9.3 Search Extensions

No Search Extensions defined for this resource